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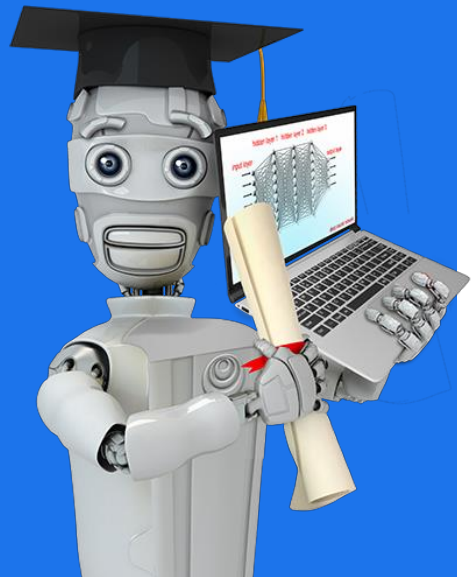
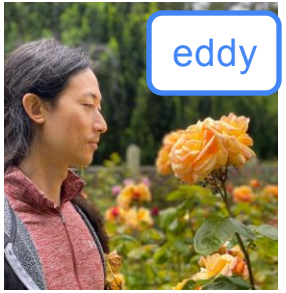


Machine Learning

Welcome!


eddy

aarti

geoff

Ivy

kin

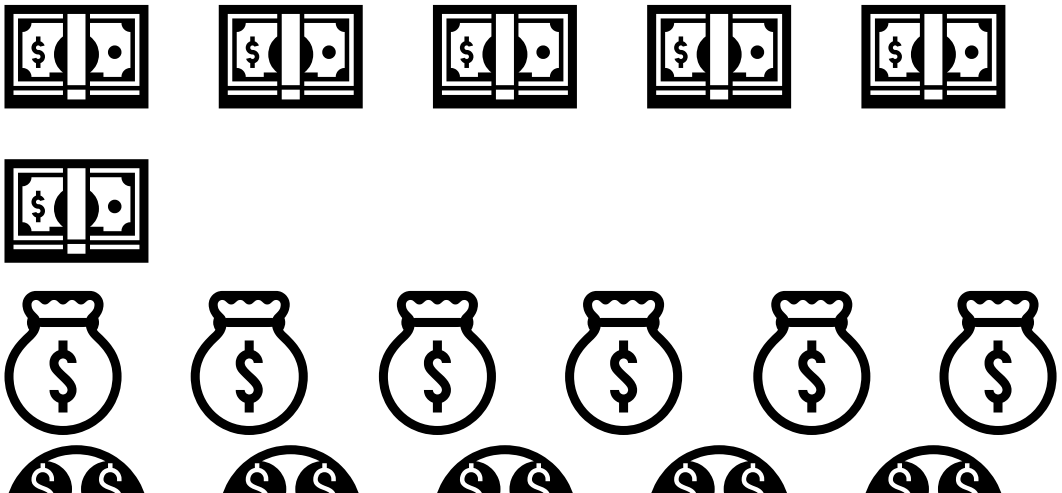
robert

andres

daniel

Re: Urgent Information :) External Spam ×

Congratulations! You've won a million dollars!



Gmail

Compose

Mail

Inbox

Starred

Snoozed

Sent

Drafts 42

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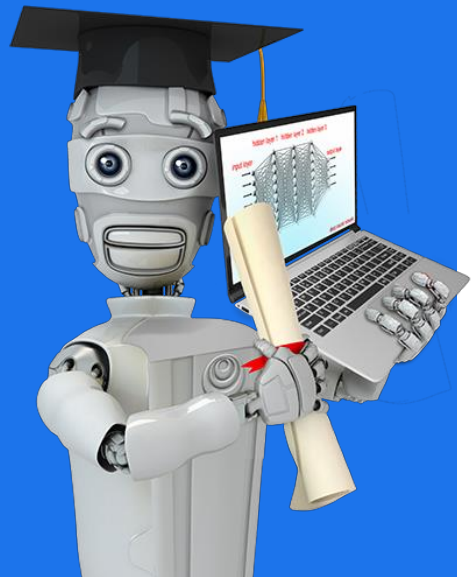

Machine Learning

Applications of Machine Learning

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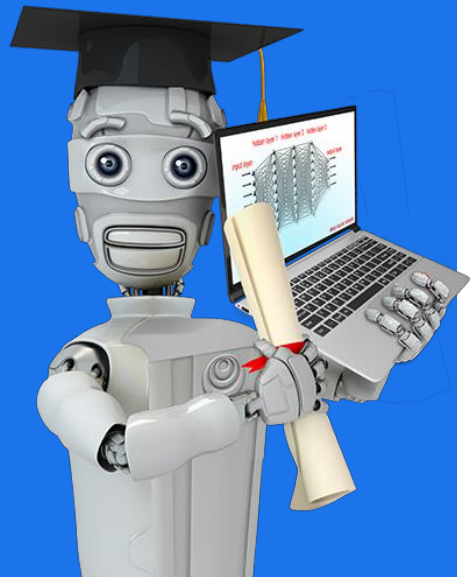


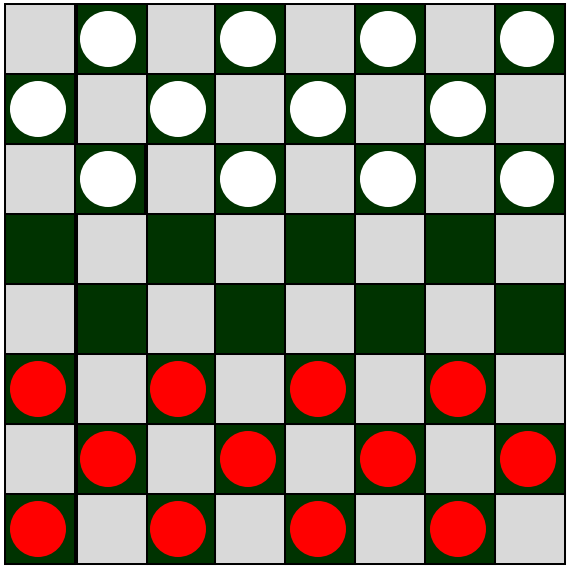
Machine Learning Overview

What is Machine Learning?

Machine learning

"Field of study that gives computers the ability to learn without being explicitly programmed."

Arthur Samuel (1959)



Question

If the checkers program had been allowed to play only ten games (instead of tens of thousands) against itself, a much smaller number of games, how would this have affected its performance?

- Would have made it better
- Would have made it worse

Machine learning algorithms

rapid advancements

used most in real-world applications

- Supervised learning ← course 1, 2
- Unsupervised learning ← course 3
- Recommender systems
- Reinforcement learning

Practical advice for applying learning algorithms

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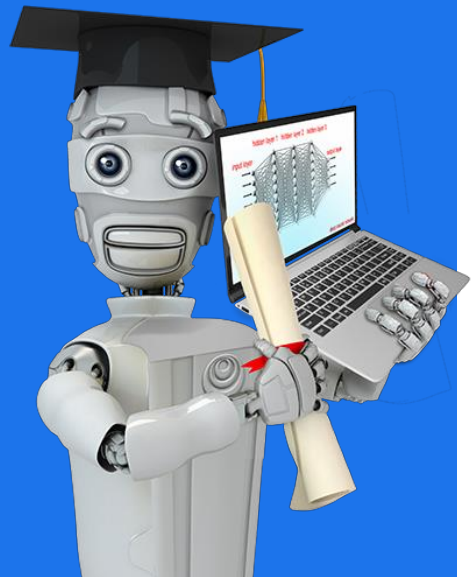
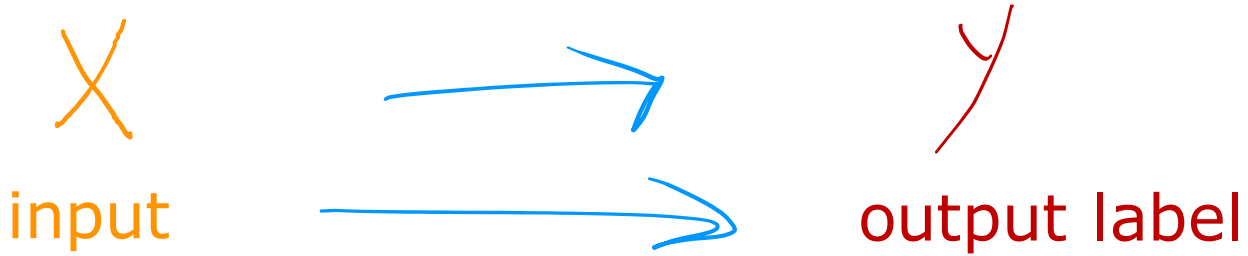


Machine Learning Overview

Supervised Learning Part 1

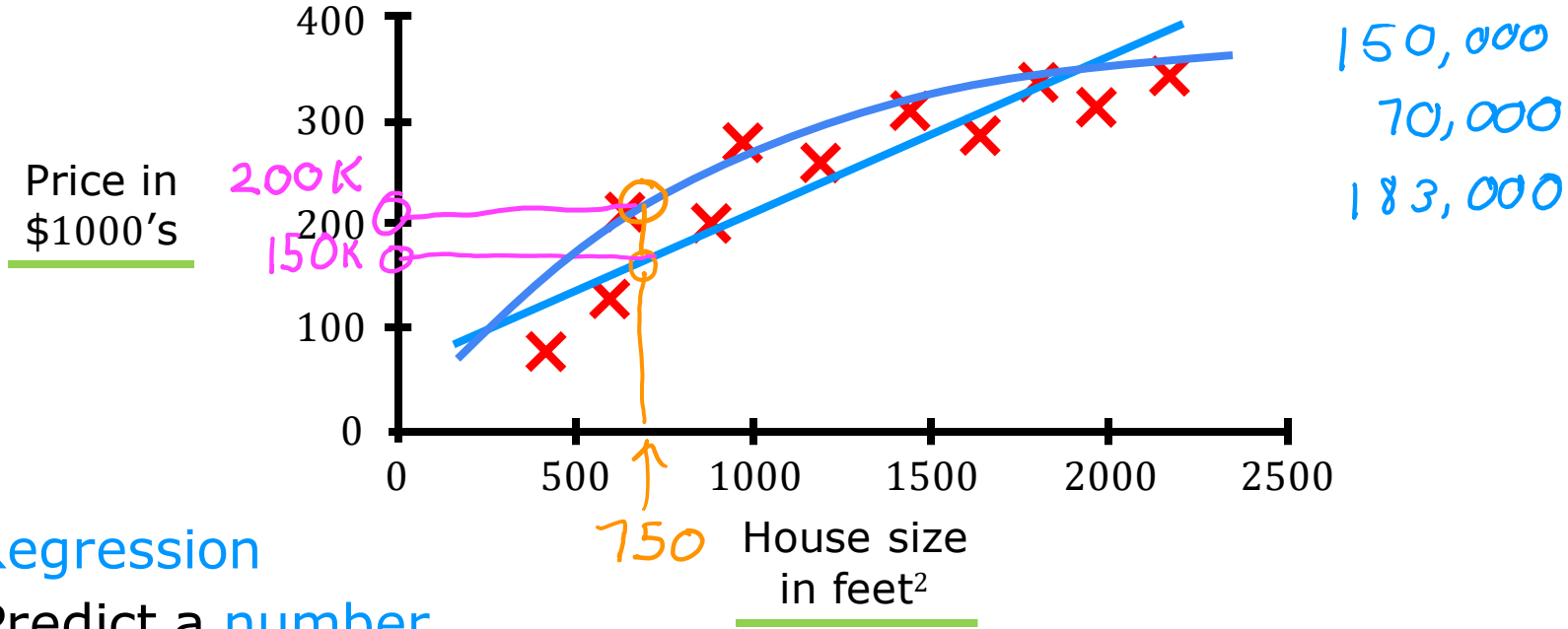
Supervised learning



Learns from being given “right answers”

Input (X)	Output (Y)	Application
email	spam? (0/1)	spam filtering
audio	text transcripts	speech recognition
English	Spanish	machine translation
ad, user info	click? (0/1)	online advertising
image, radar info	position of other cars	self-driving car
image of phone	defect? (0/1)	visual inspection

Regression: Housing price prediction



Regression

Predict a number

infinitely many possible outputs

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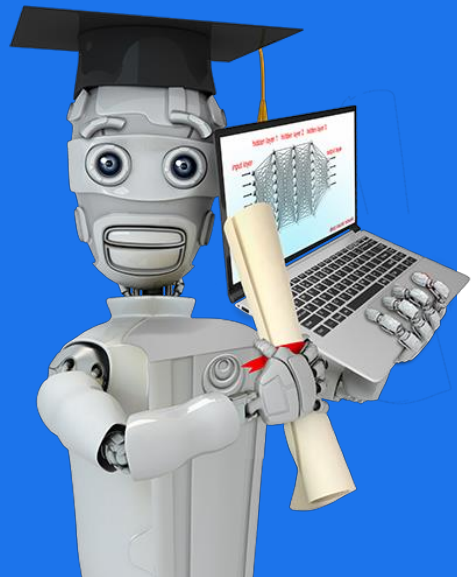
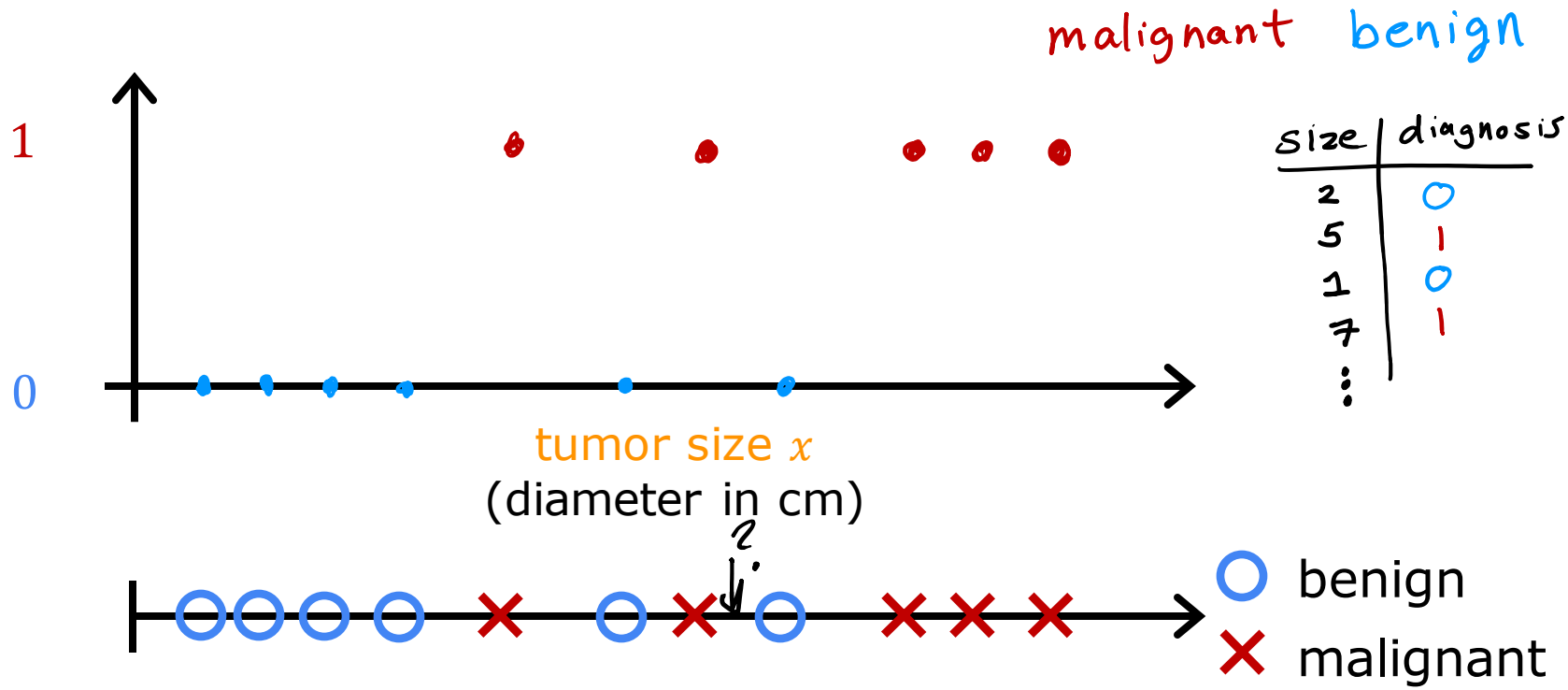


Machine Learning Overview

Supervised Learning Part 2

Classification: Breast cancer detection

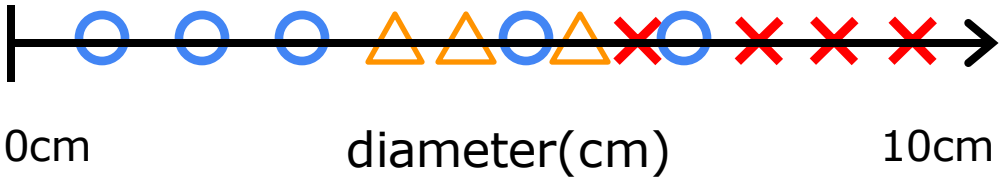


Classification: Breast cancer detection

benign

malignant type1

malignant type 2



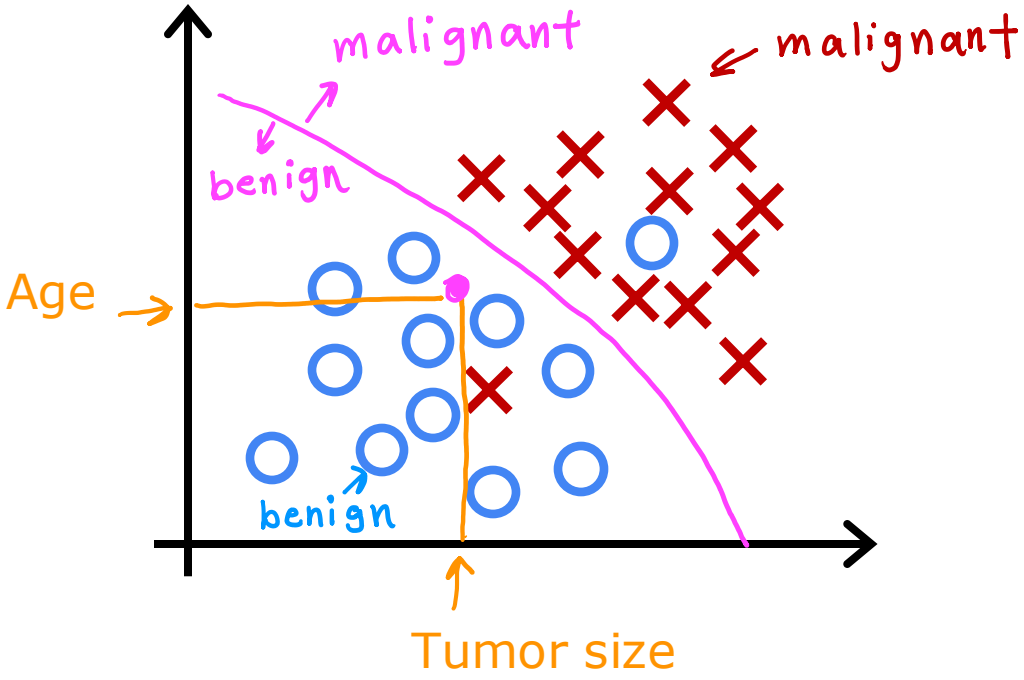
class category

Classification

predict categories cat dog benign malignant 0, 1, 2

small number of possible outputs

Two or more inputs



Supervised learning

Learns from being given "right answers"

Regression

Predict a number

infinitely many possible outputs

Classification

predict categories

small number of possible outputs

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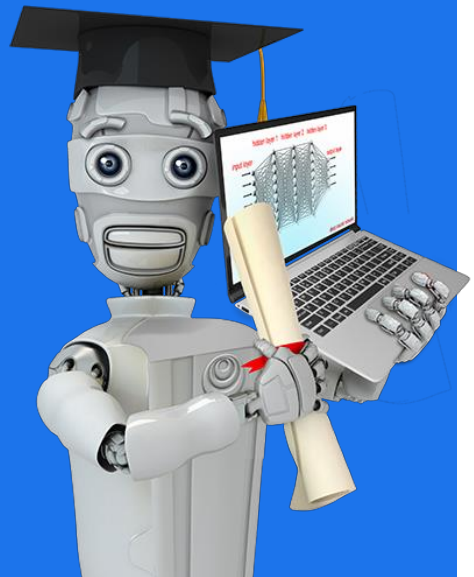
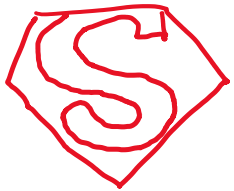


Machine Learning Overview

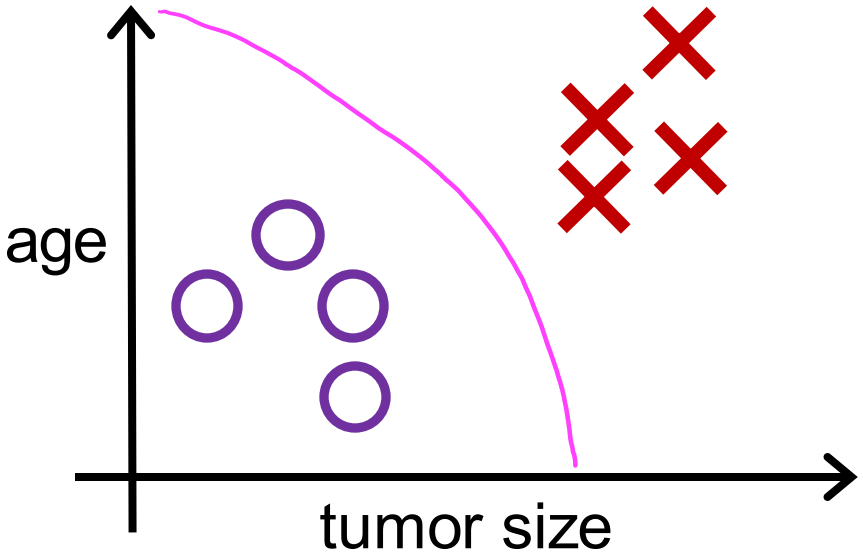
Unsupervised Learning Part 1

Previous: Supervised learning

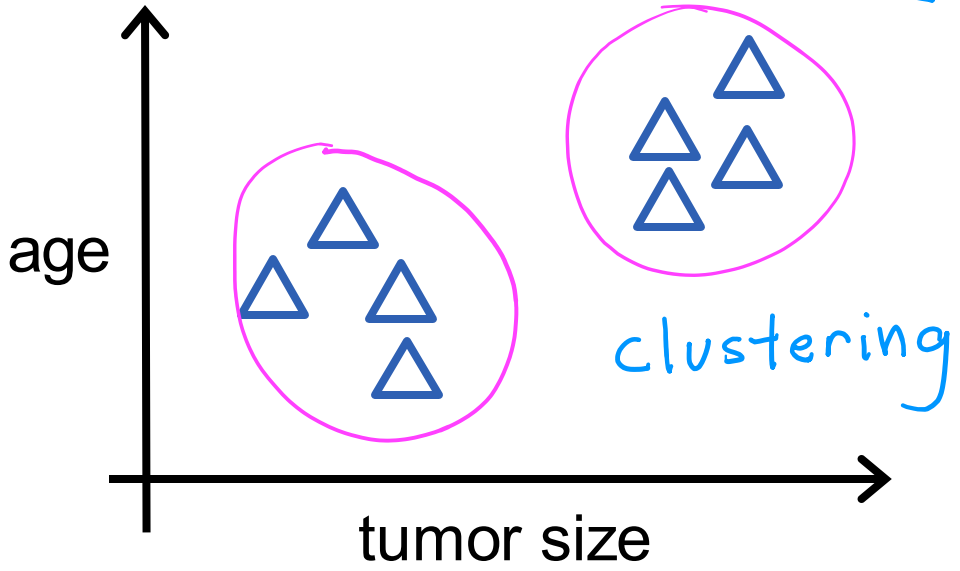
Now: Unsupervised learning



Supervised learning
Learn from data labeled with the "right answers"

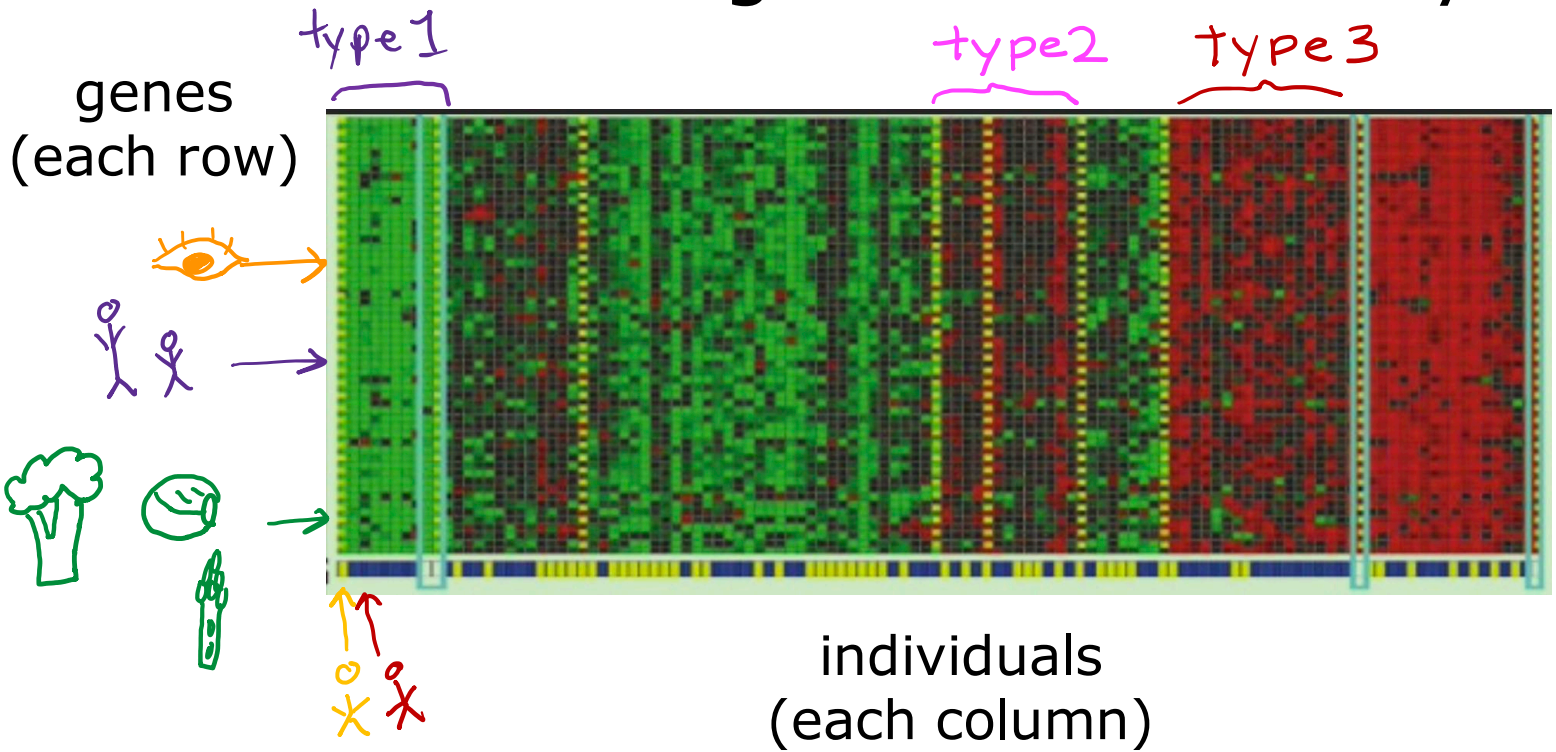


Unsupervised learning
Find something interesting in unlabeled data.

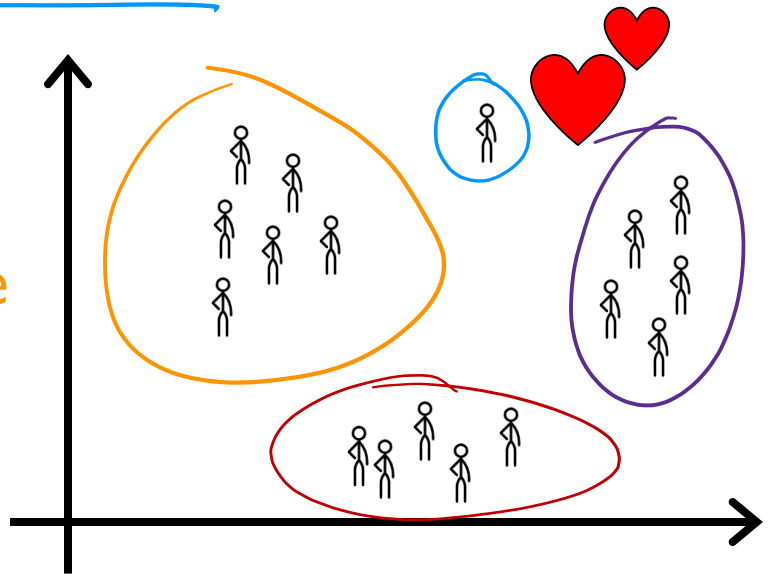


Clustering: Google news

Clustering: DNA microarray



Clustering: Grouping customers



growing skills and knowledge

develop career

stay updated with AI

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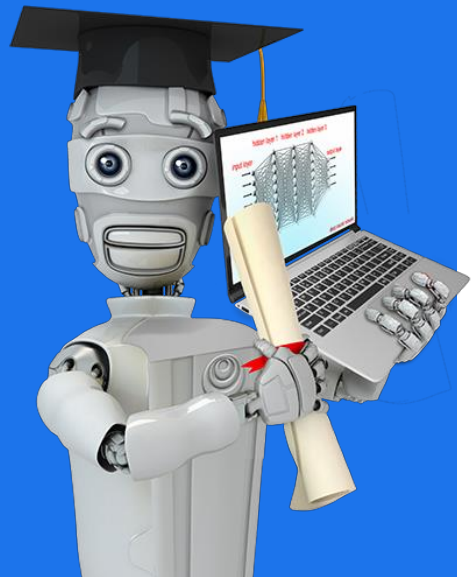

Machine Learning Overview

Unsupervised Learning Part 2

Unsupervised learning

Data only comes with inputs x, but not output labels y.
Algorithm has to find structure in the data.

Clustering
Group similar data points together.

Dimensionality reduction
Compress data using fewer numbers.

Anomaly detection
Find unusual data points.

Question

Of the following examples, which would you address using an unsupervised learning algorithm?

- [ ] Given email labeled as spam/not spam, learn a spam filter.

- [x] Given a set of news articles found on the web, group them into sets of articles about the same story.

- [x] Given a database of customer data, automatically discover market segments and group customers into different market segments.

- [ ] Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not

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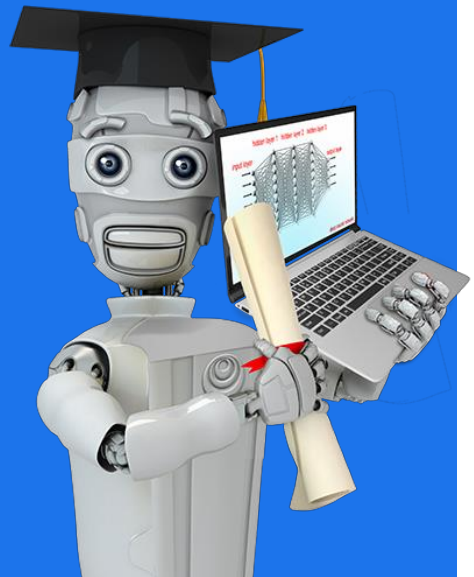

Machine Learning Overview

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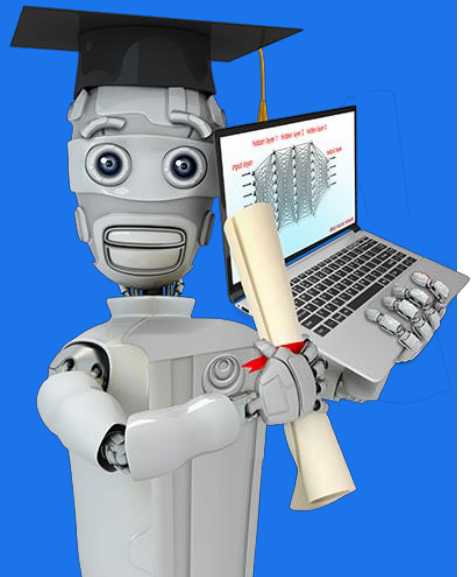
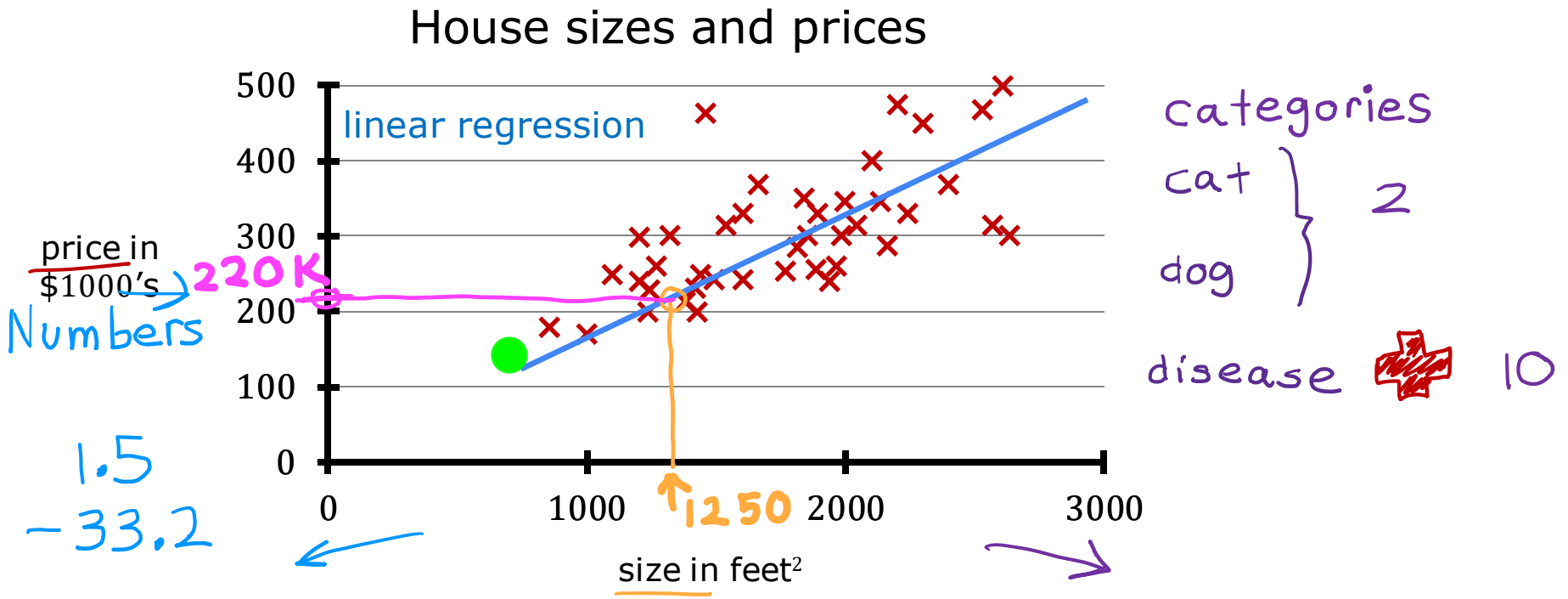


Linear Regression with One Variable

Linear Regression Model Part 1



Regression model
Predicts numbers
Infinitely many possible outputs

Supervised learning model
Data has "right answers"

Classification model
Predicts categories
Small number of possible outputs

House sizes and prices

Data table

size in feet²	price in $1000's
2104	400
1416	232
1534	315
852	178
...	...
3210	870

Terminology

Training set: Data used to train the model

	x size in feet^2	y price in \$1000's
(1)	2104	400
(2)	1416	232
(3)	1534	315
(4)	852	178
…	…	…
(47)	3210	870

$m = 47$

$x^{(1)} = 2104$ $y^{(1)} = 400$

$(x^{(1)}, y^{(1)}) = (2104, 400)$

$x^{(2)} = 1416$ $y^{(2)} \neq x^2$ not exponent

Notation:

x = "input" variable
feature

y = "output" variable
"target" variable

m = number of training examples

(x, y) = single training example

$(x^{(i)}, y^{(i)})$

$(x^{(i)}, y^{(i)})$ = i^{th} training example

index (1^{st}, 2^{nd}, 3^{rd} …)

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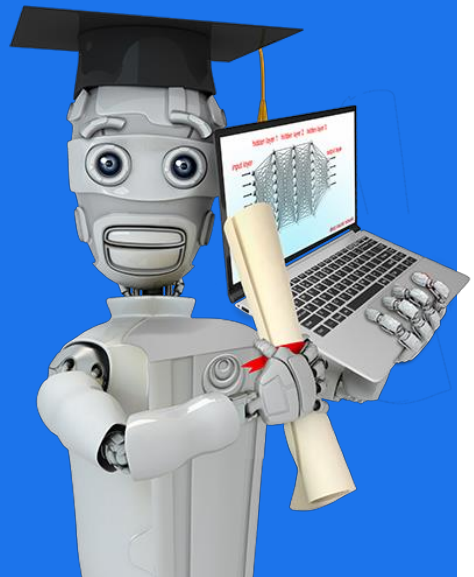
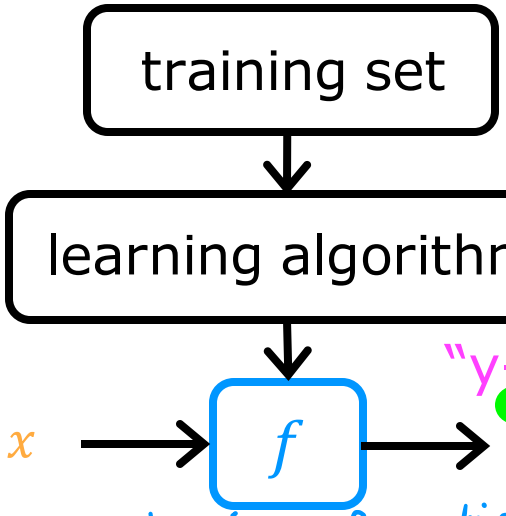


Linear Regression with One Variable

Linear Regression Model Part 2



features
targets

$\hat{y}$

(estimated y)

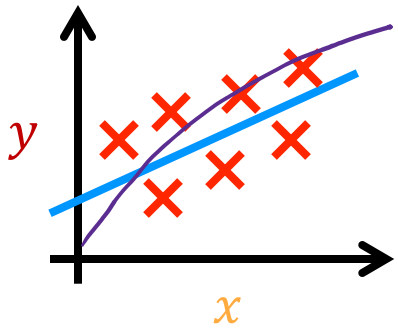
target

size → f → price (estimated)

How to represent f?

$f_{w,b}(x) = wx + b$

$f(x)$



$f_{w,b}(x) = wx + b$

$f(x) = wx + b$

linear

single feature x

Linear regression with one variable.

size

Univariate linear regression.

one variable

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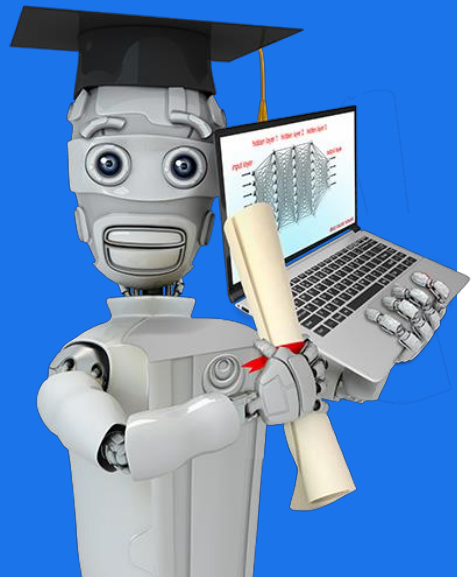


Linear Regression with One Variable

Cost Function

Training set

features | targets

size in feet² (x)	price \$1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

Model: $f_{w,b}(x) = wx + b$

w, b: parameters

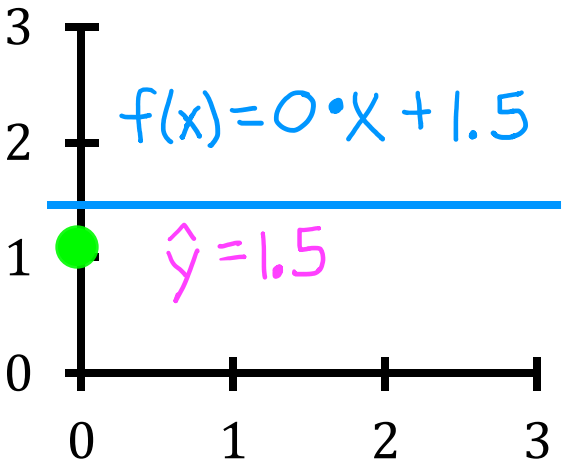
coefficients

weights

What do w, b do?

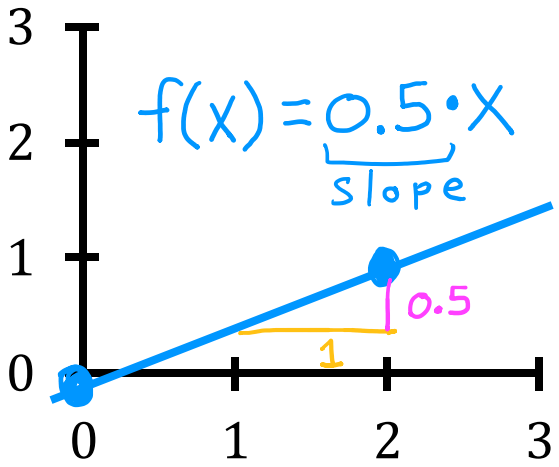
$$f_{w,b}(x) = wx + b$$

$f(x)$



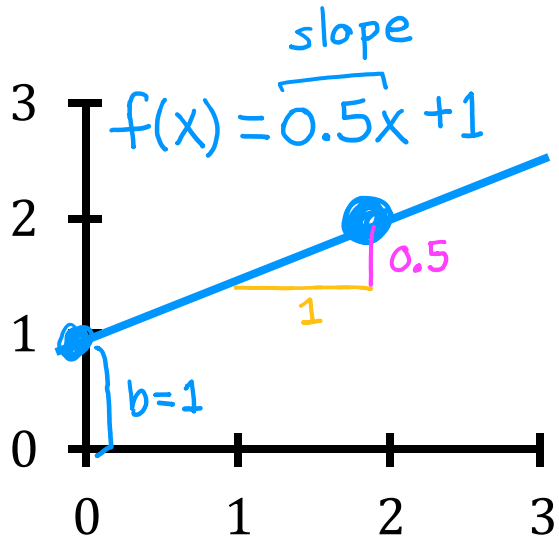
→ $w = 0$

→ $b = 1.5$ (y-intercept)



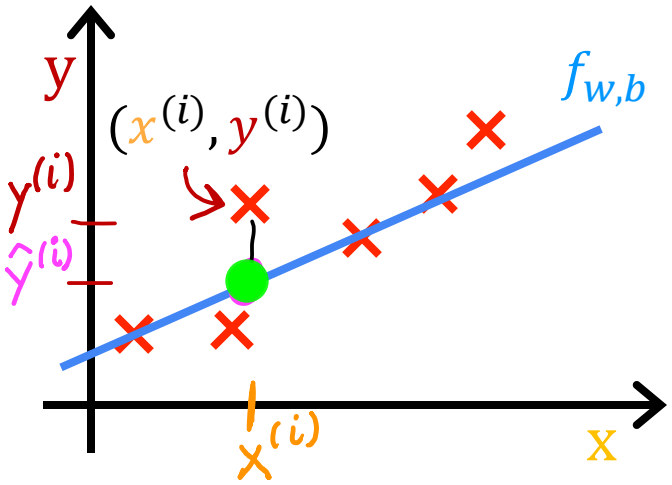
→ $w = 0.5$

→ $b = 0$



→ $w = 0.5$

→ $b = 1$



$$\hat{y}^{(i)} = f_{w,b}(x^{(i)})$$

$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

Cost function: Squared error cost function

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(\hat{y}^{(i)} - y^{(i)}\right)^2$$

$\hat{y}^{(i)}$ − $y^{(i)}$: error

m = number of training examples

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{w,b}(x^{(i)}) - y^{(i)}\right)^2$$

$J(w,b)$: intuition (next!)

Find w, b:

$\hat{y}^{(i)}$ is close to $y^{(i)}$ for all $(x^{(i)}, y^{(i)})$.

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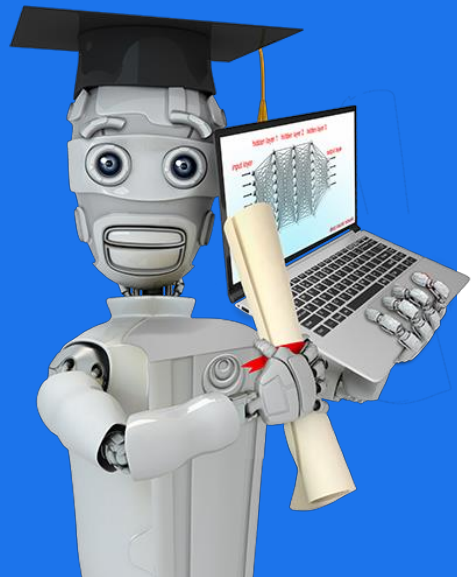


Linear Regression with One Variable

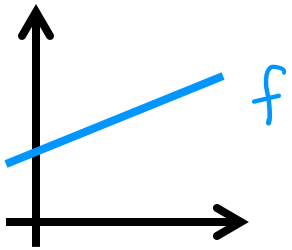
Cost Function Intuition

model:

$$f_{w,b}(x) = wx + b$$

parameters:

w, b



cost function:

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})^2$$

goal:

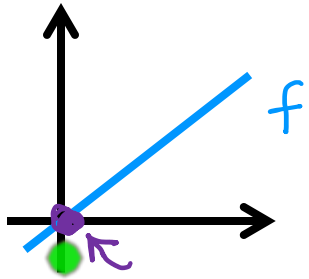
$$\underset{w,b}{\text{minimize}}\, J(w,b)$$

simplified

$$f_w(x) = wx$$

$b = 0$

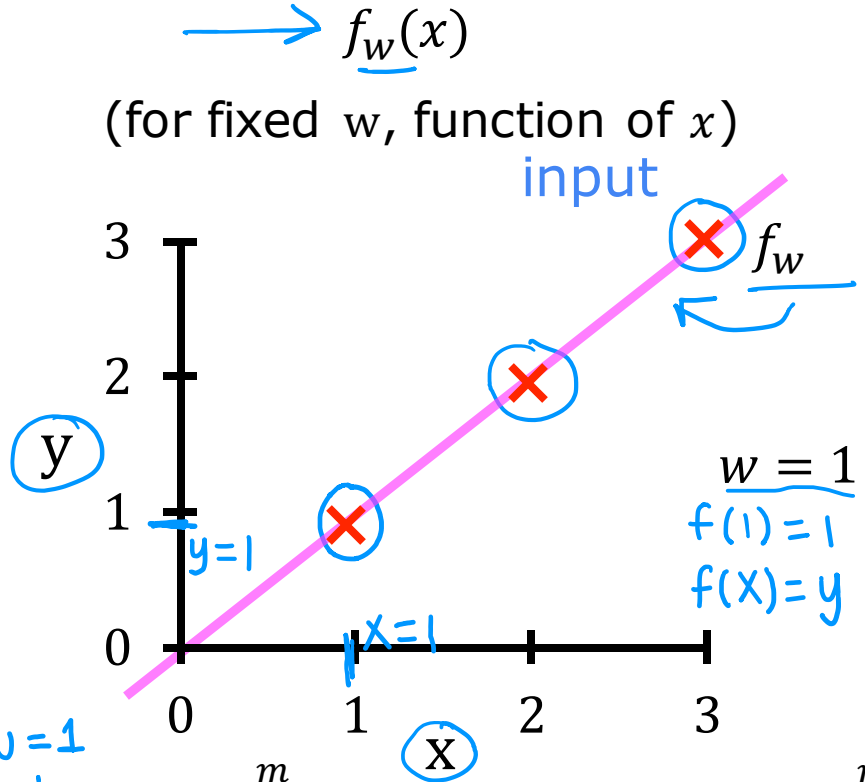
w

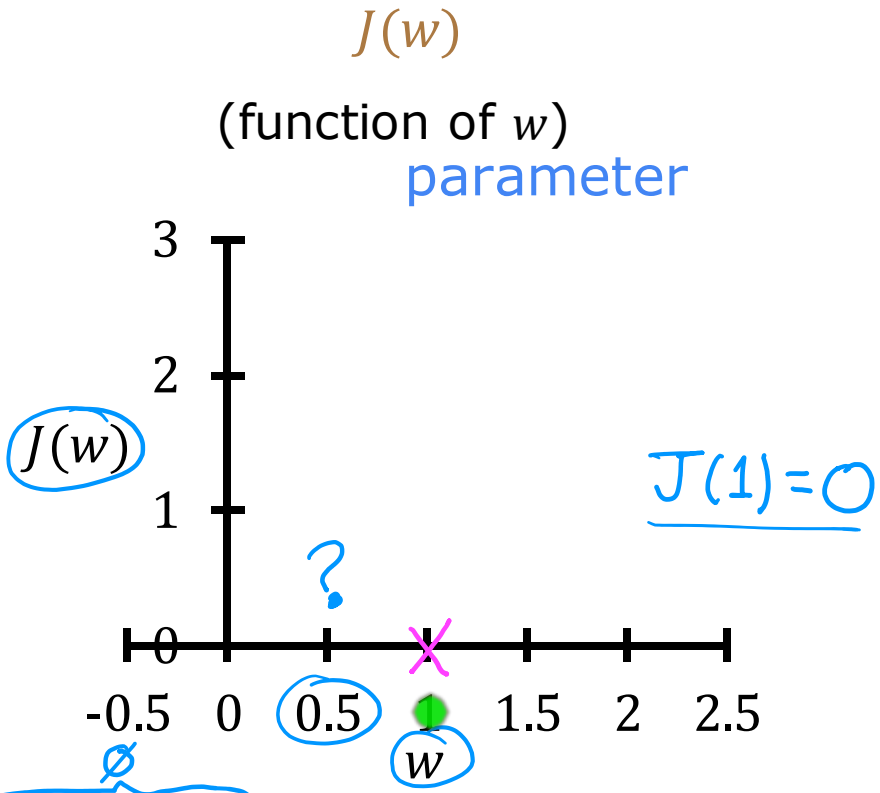


$$J(w) = \frac{1}{2m}\sum_{i=1}^{m}(f_w(x^{(i)}) - y^{(i)})^2$$

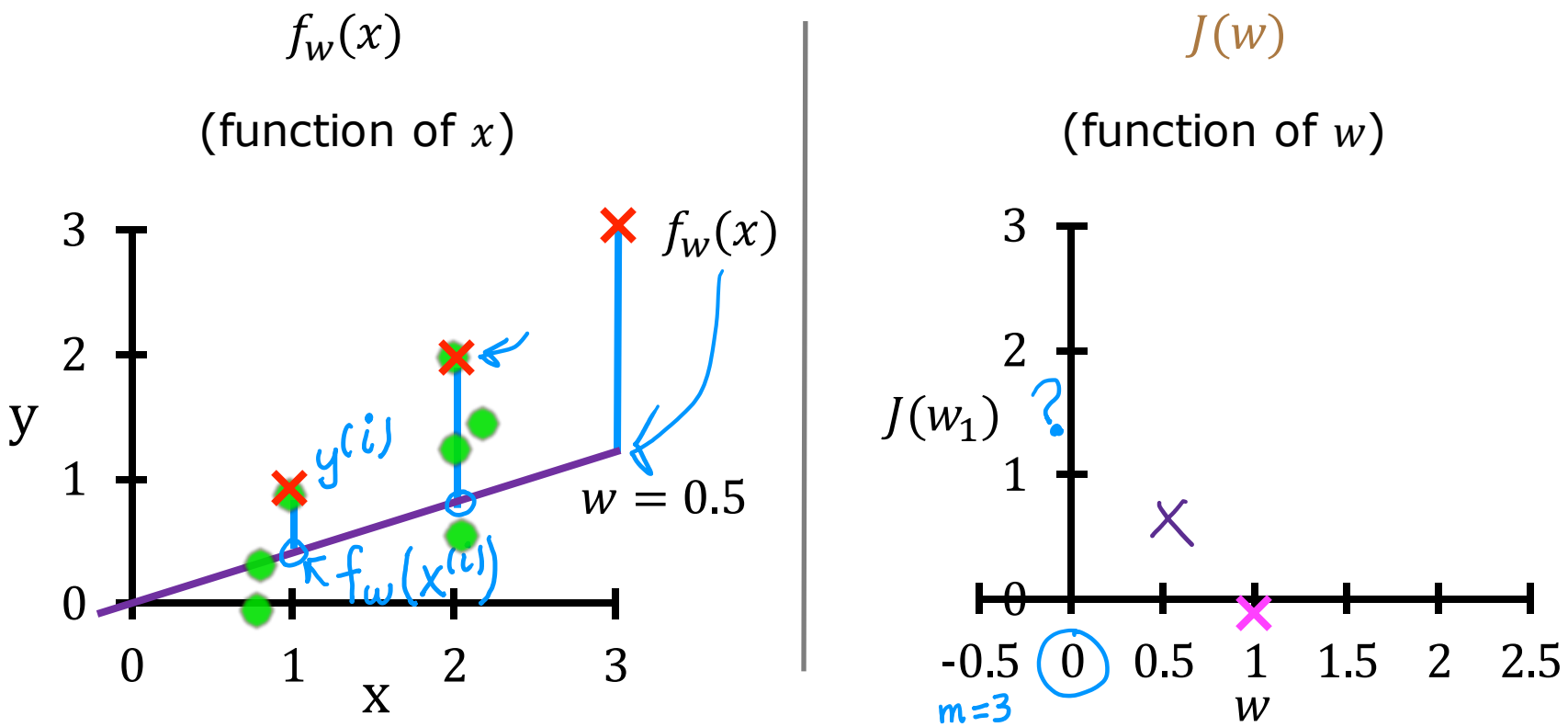
$wx^{(i)}$

$$\underset{w}{\text{minimize}}\, J(w)$$

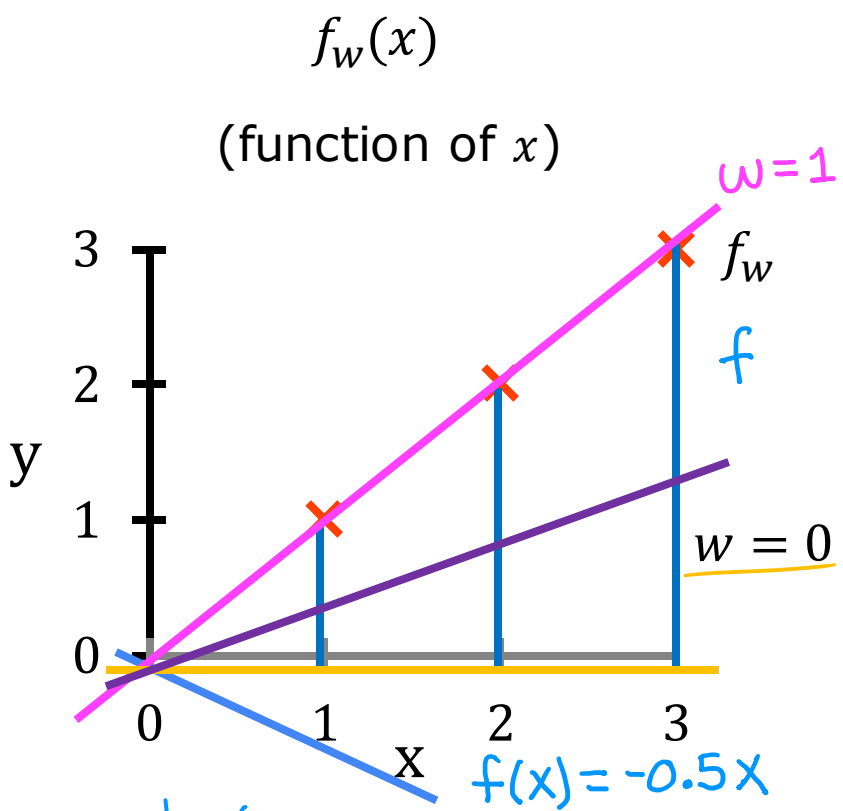




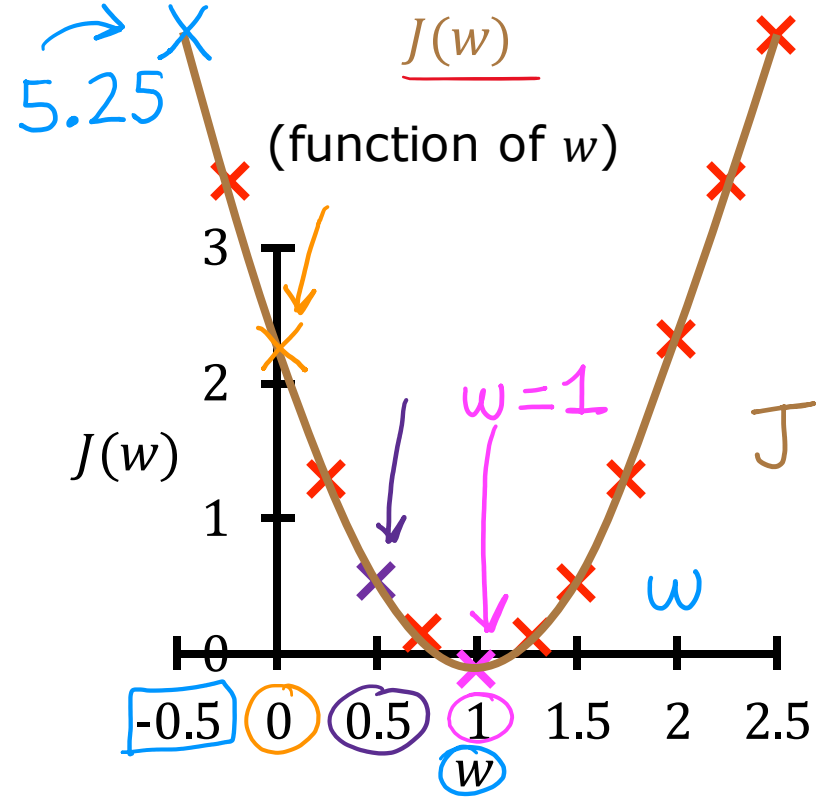
$$J(w) = \frac{1}{2m}\sum_{i=1}^{m}(f_w(x^{(i)}) - y^{(i)})^2 = \frac{1}{2m}\sum_{i=1}^{m}(wx^{(i)} - y^{(i)})^2 = \frac{1}{2m}(0^2 + 0^2 + 0^2) = 0$$



$$J(0.5) = \frac{1}{2m}\left[(0.5-1)^2+(1-2)^2+(1.5-3)^2\right] = \frac{1}{2\times 3}[3.5] = \frac{3.5}{6} \approx 0.58$$



$f(x) = -0.5x$



$$J(0) = \frac{1}{2m}\left(1^2+2^2+3^2\right) = \frac{1}{6}[14] \approx 2.3$$

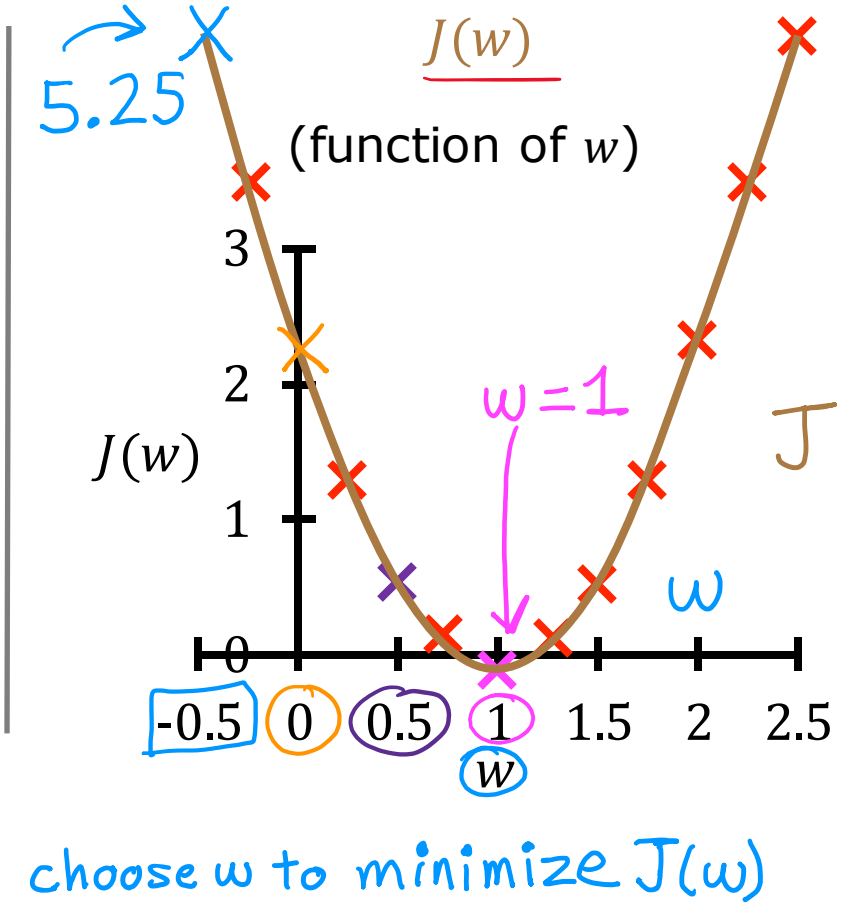
how to choose w?

goal of linear regression:

$$\underset{w}{\text{minimize}}\, J(w)$$

general case:

$$\underset{w,b}{\text{minimize}}\, J(w,b)$$



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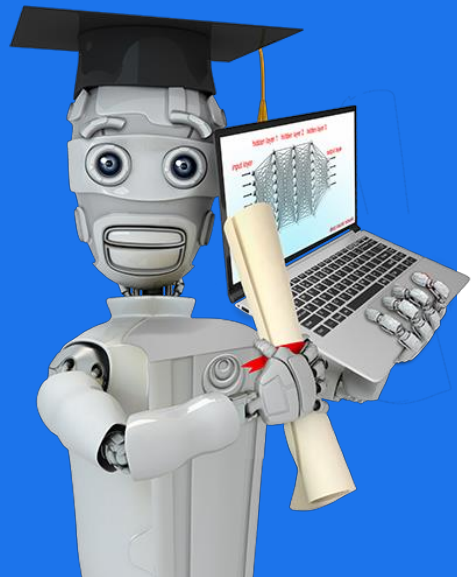


Linear Regression with One Variable

Visualizing the Cost Function

Model $f_{w,b}(x) = wx + b$

~~before: $b = 0$~~

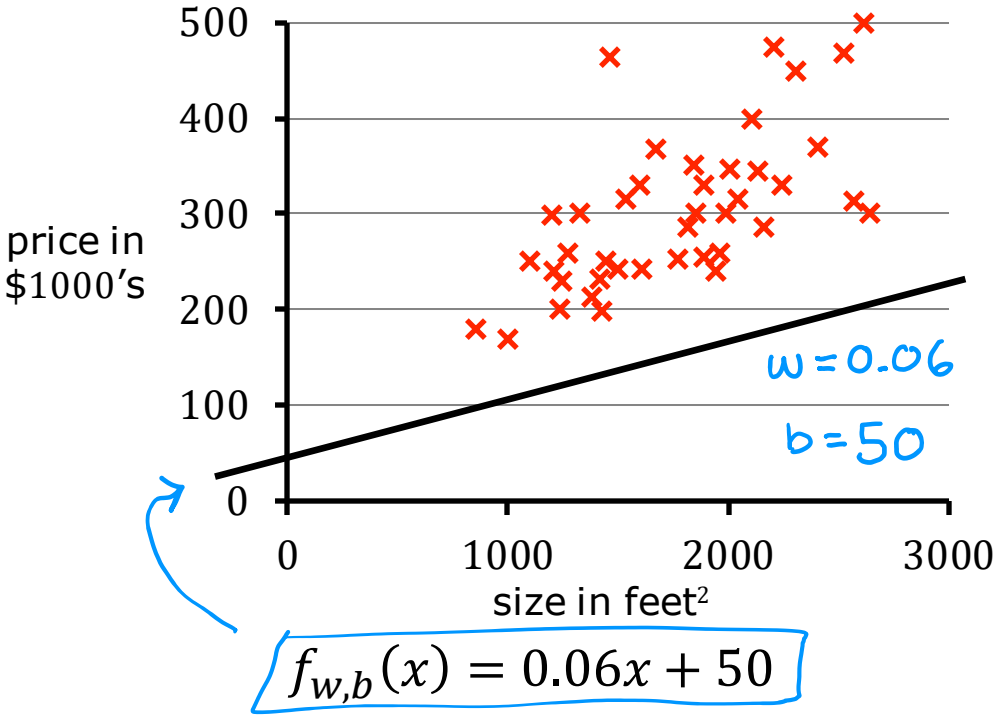
Parameters w, b

Cost Function $J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})^2$

Objective $\underset{w,b}{\text{minimize}}\, J(w,b)$

$f_{w,b}$

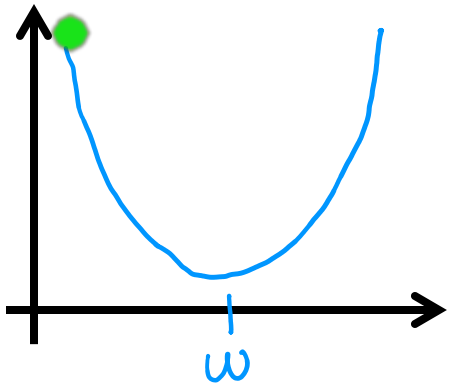
(function of x)



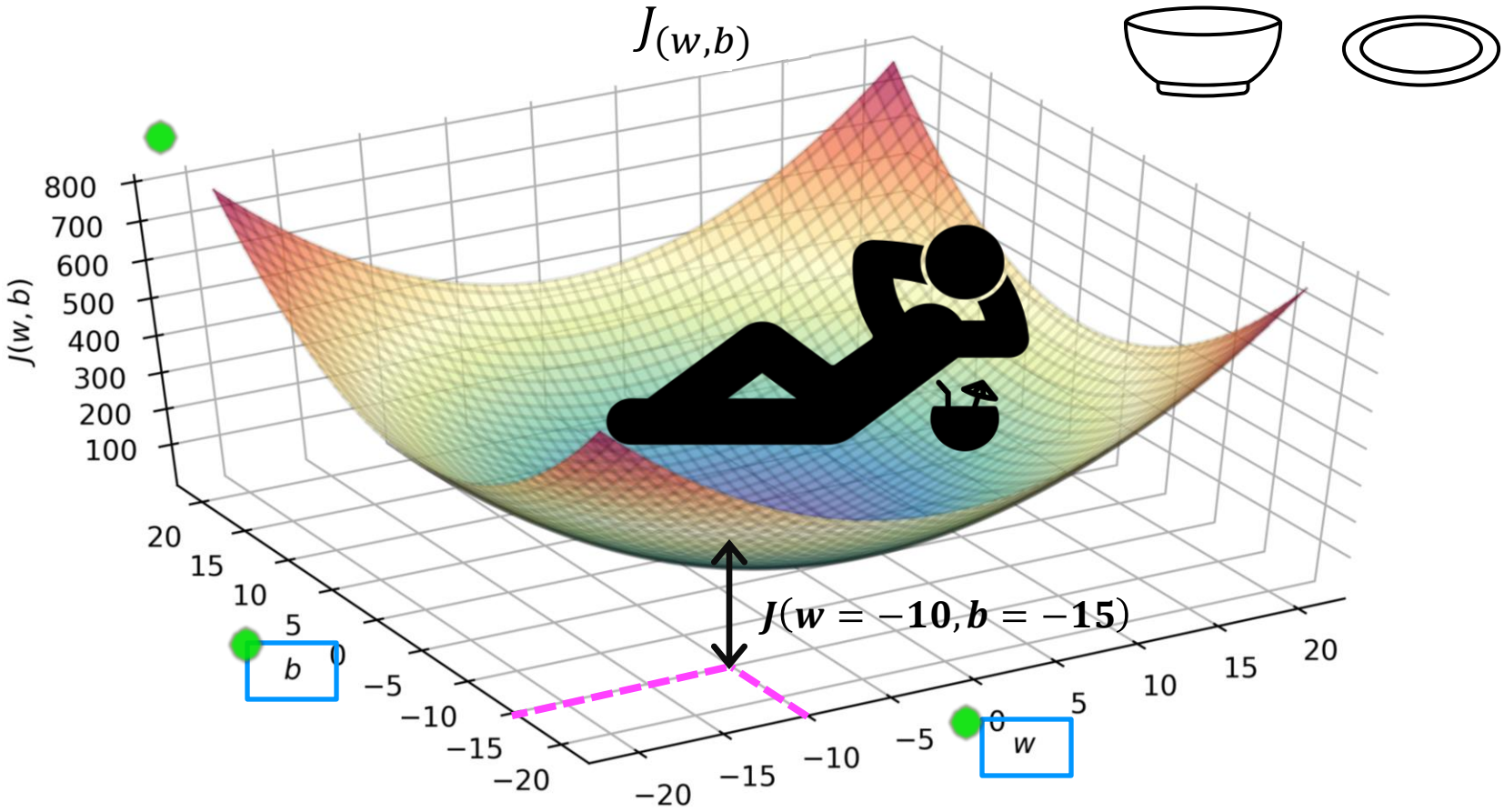
$f_{w,b}(x) = 0.06x + 50$

J

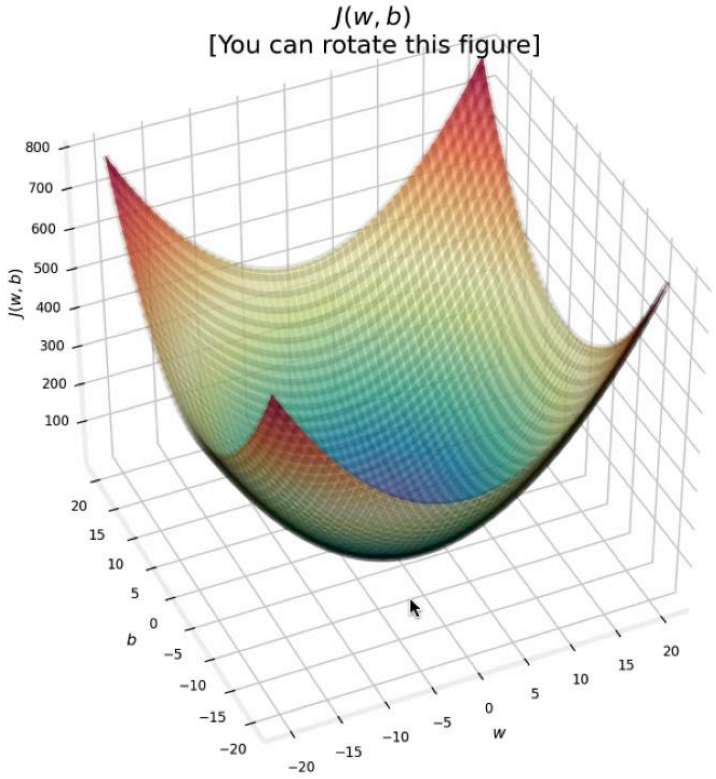
(function of w, b)



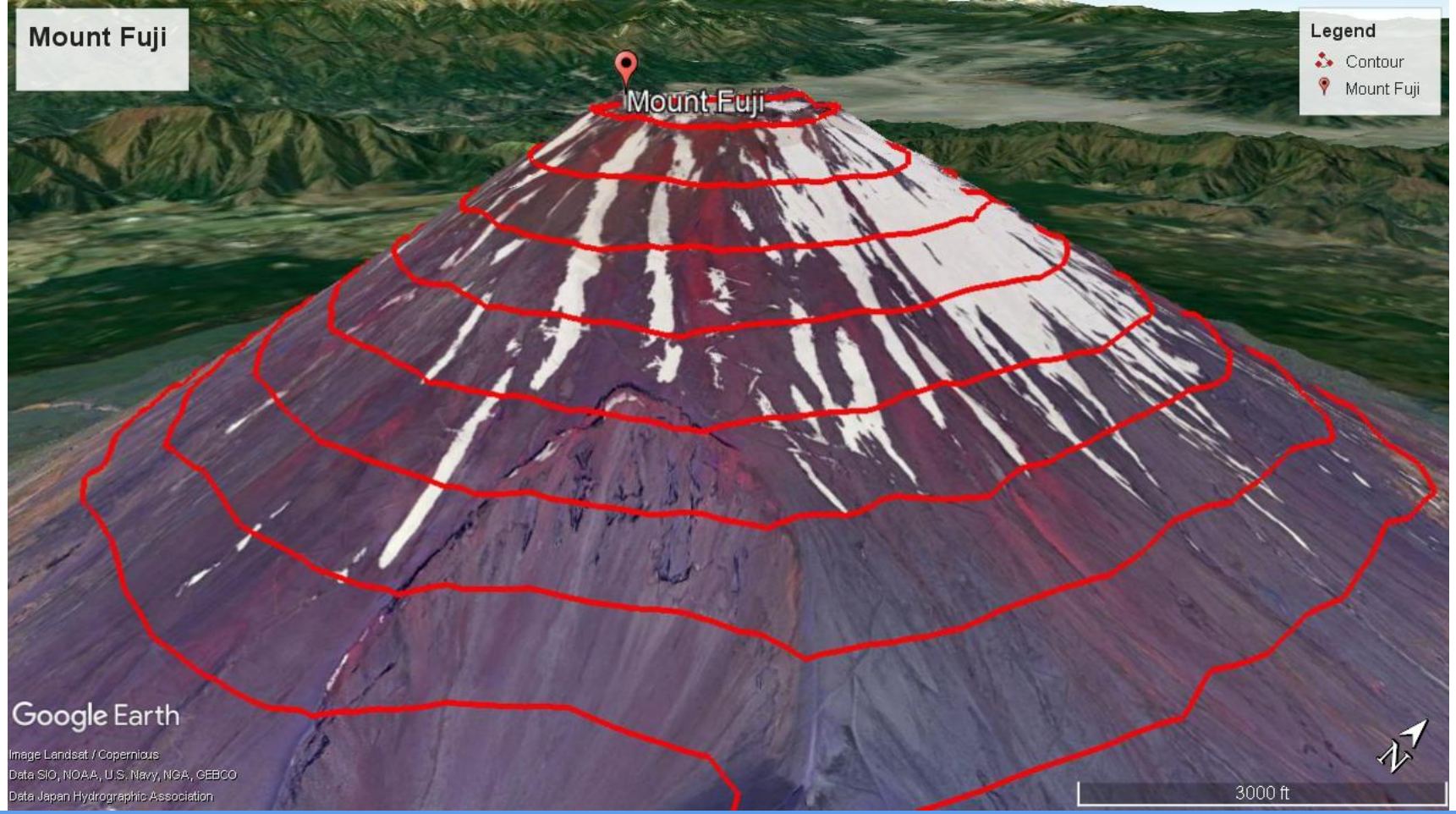
w , b

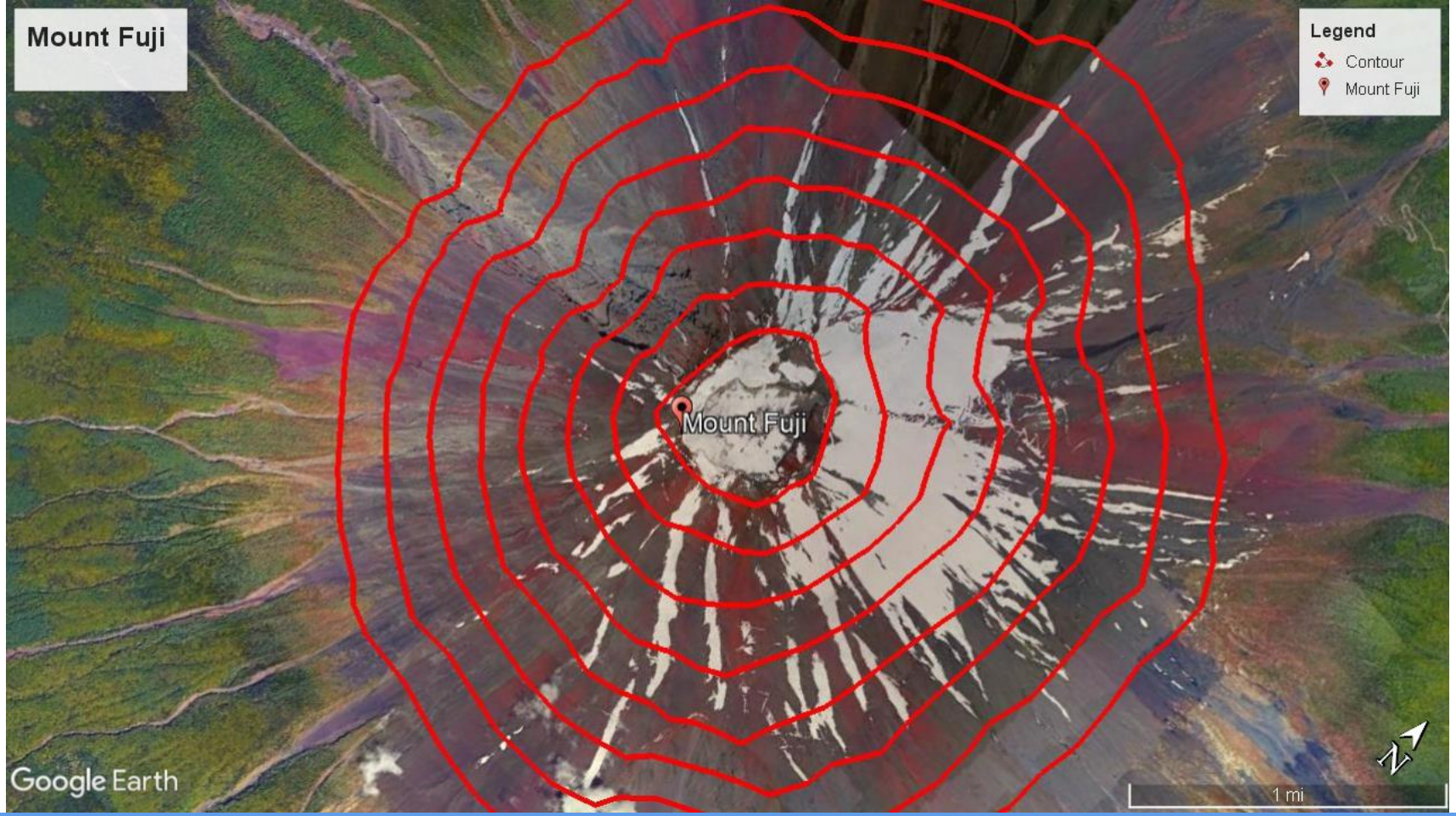

$J_{(w,b)}$
800
700
600
500
400
300
200
100
$J(w,b)$
20
15
10
5
0
b
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
w
$J(w=-10, b=-15)$

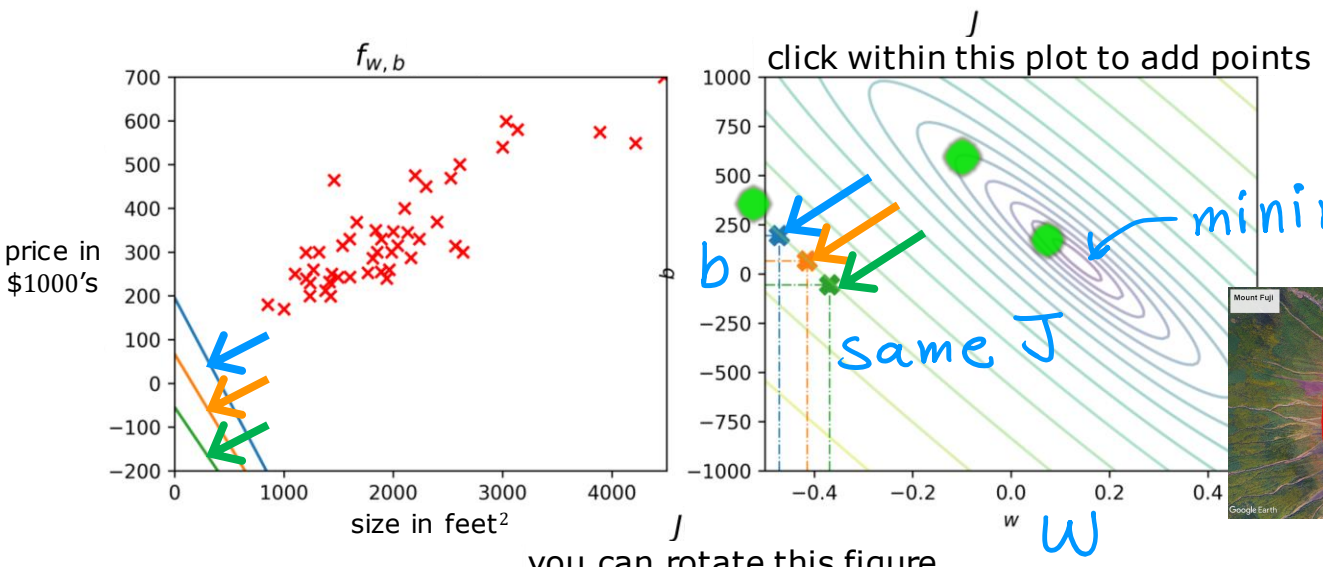
3D surface plot

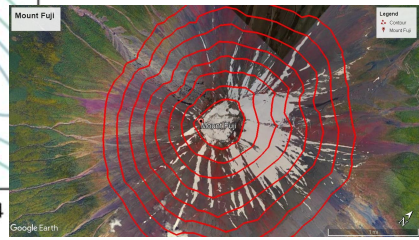


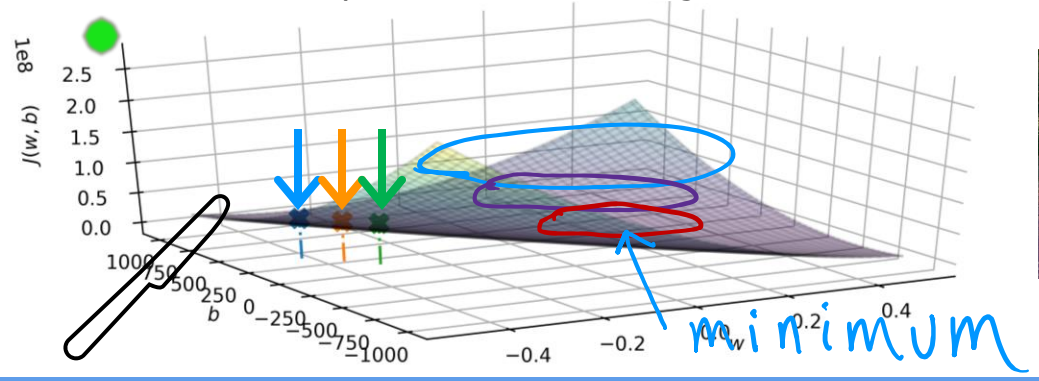
Alternative contour plot

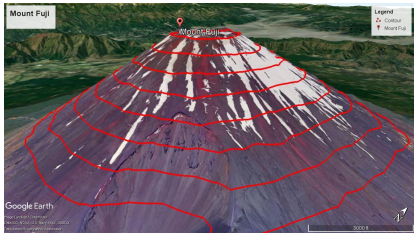

Mount Fuji
Legend
Contour
Mount Fuji
Mount Fuji
Google Earth
Image Landsat / Copernicus
Data SIO, NOAA, U.S. Navy, NGA, GEBCO
Data Japan Hydrographic Association
3000 ft


Mount Fuji
Legend
Contour
Mount Fuji
Mount Fuji
Google Earth
1 mi


$f_{w,b}$
price in $1000's
size in feet^2
J
click within this plot to add points
b
w
minimum
same J


Mount Fuji
Google Earth


J
you can rotate this figure
J(w, b)
1e8
b
w
minimum


Mount Fuji
Google Earth

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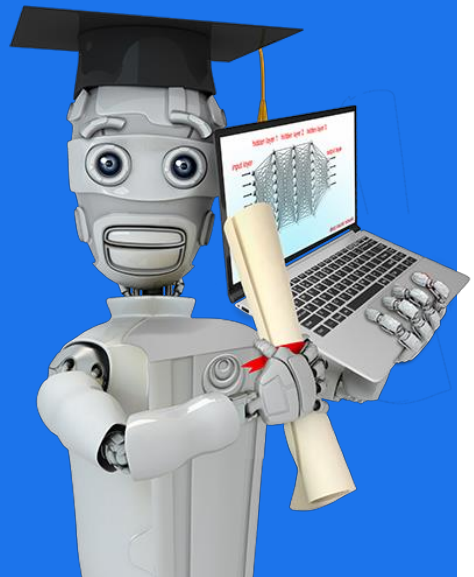
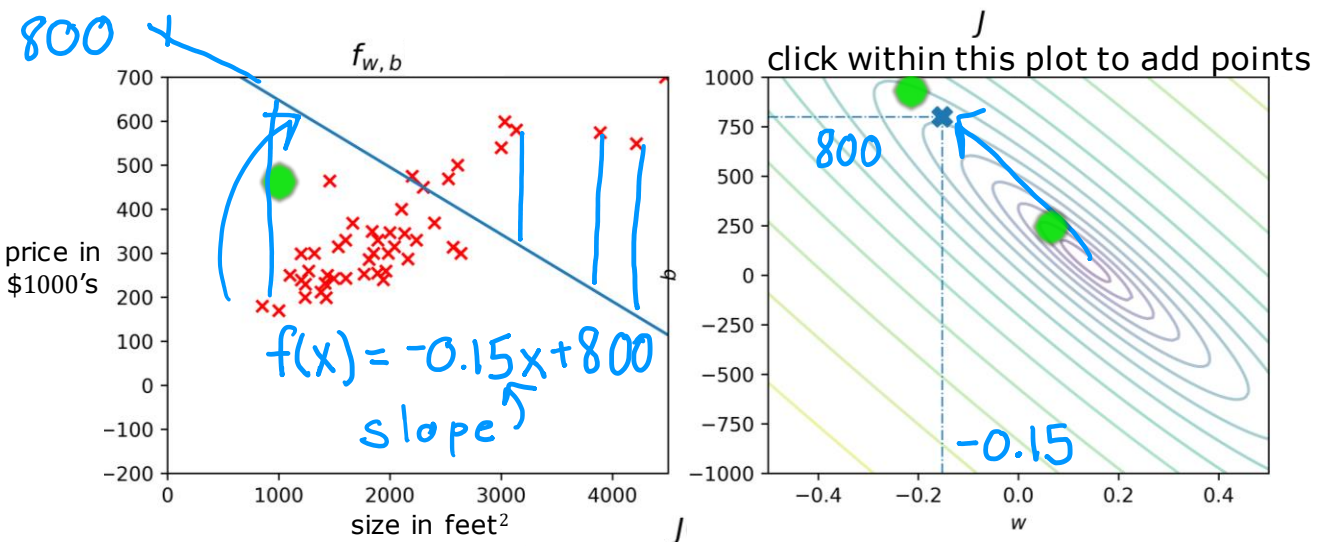


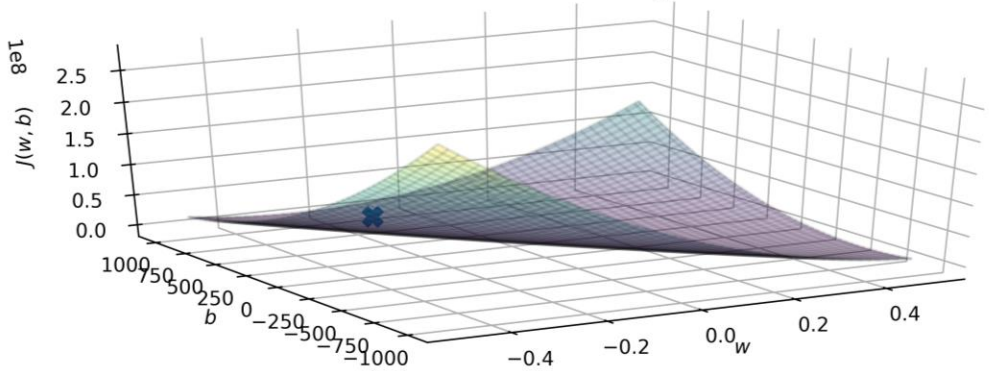
Linear Regression with One Variable

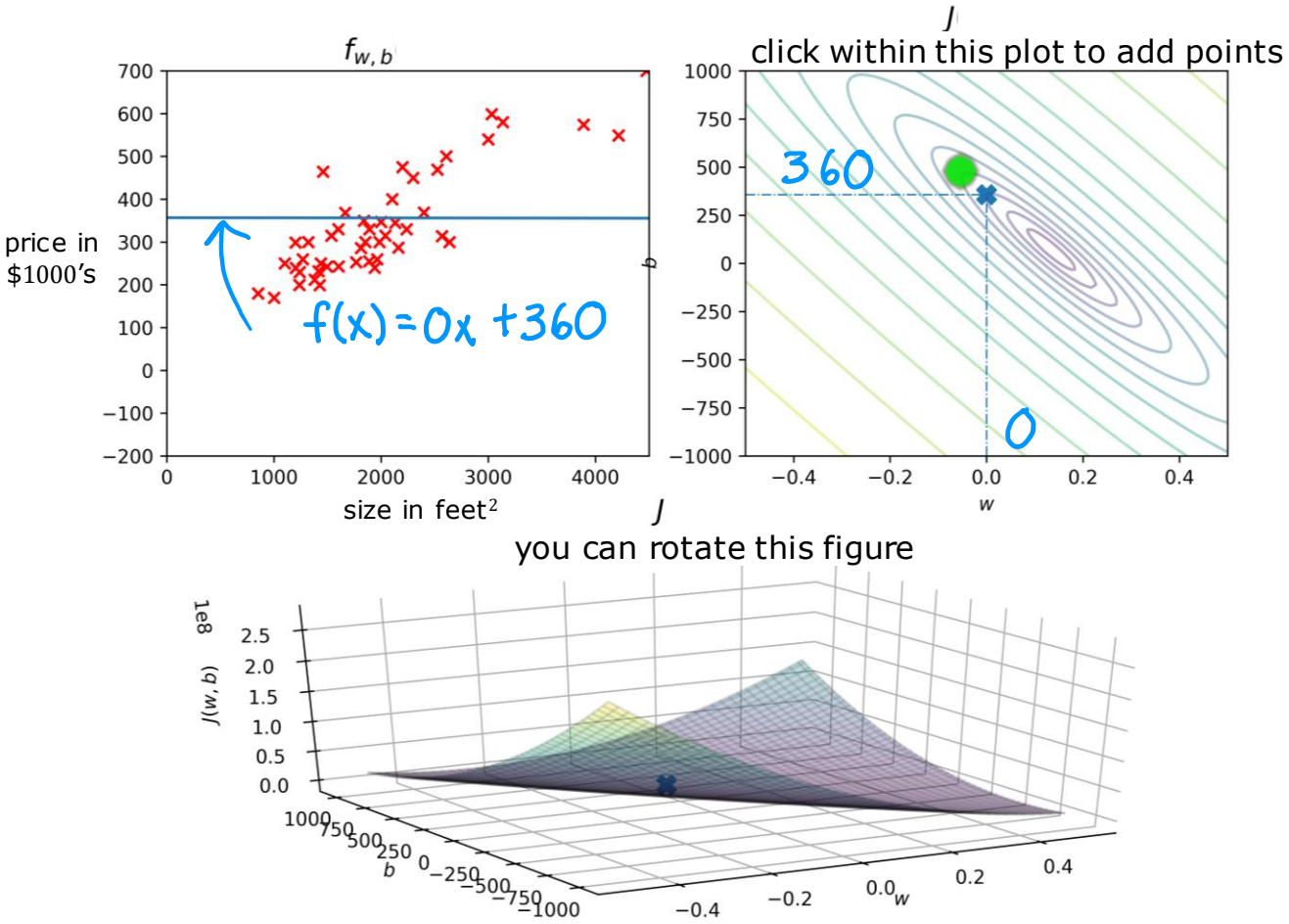
Visualization examples

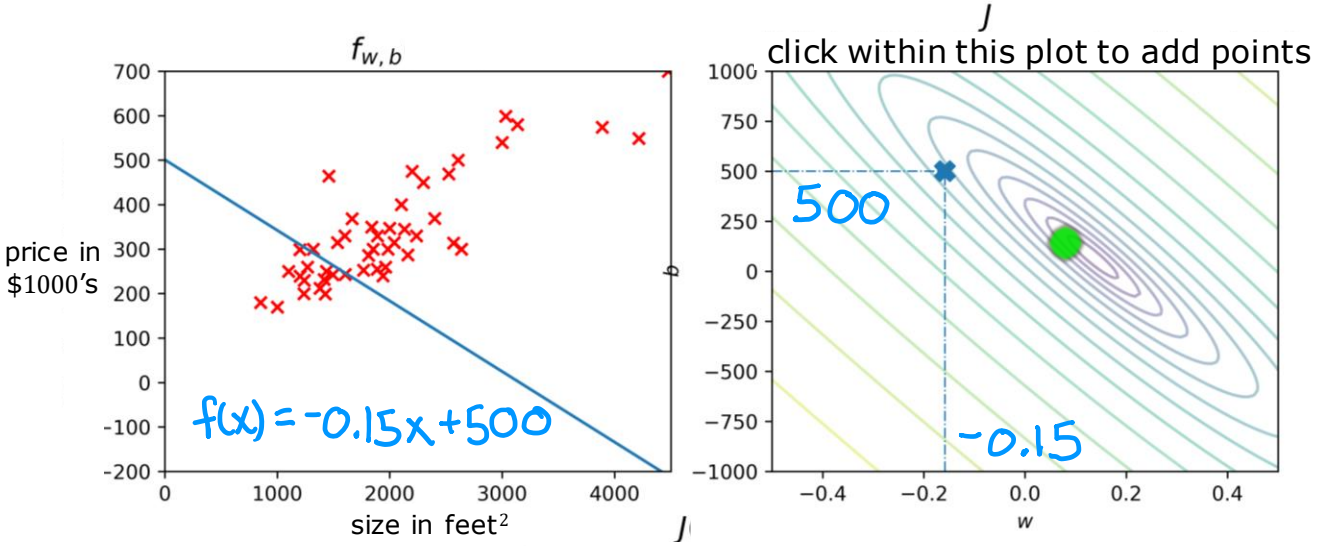


J

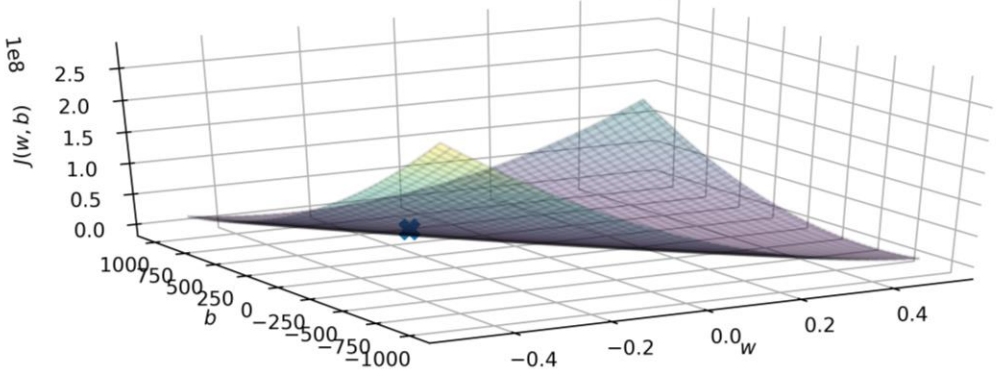
you can rotate this figure

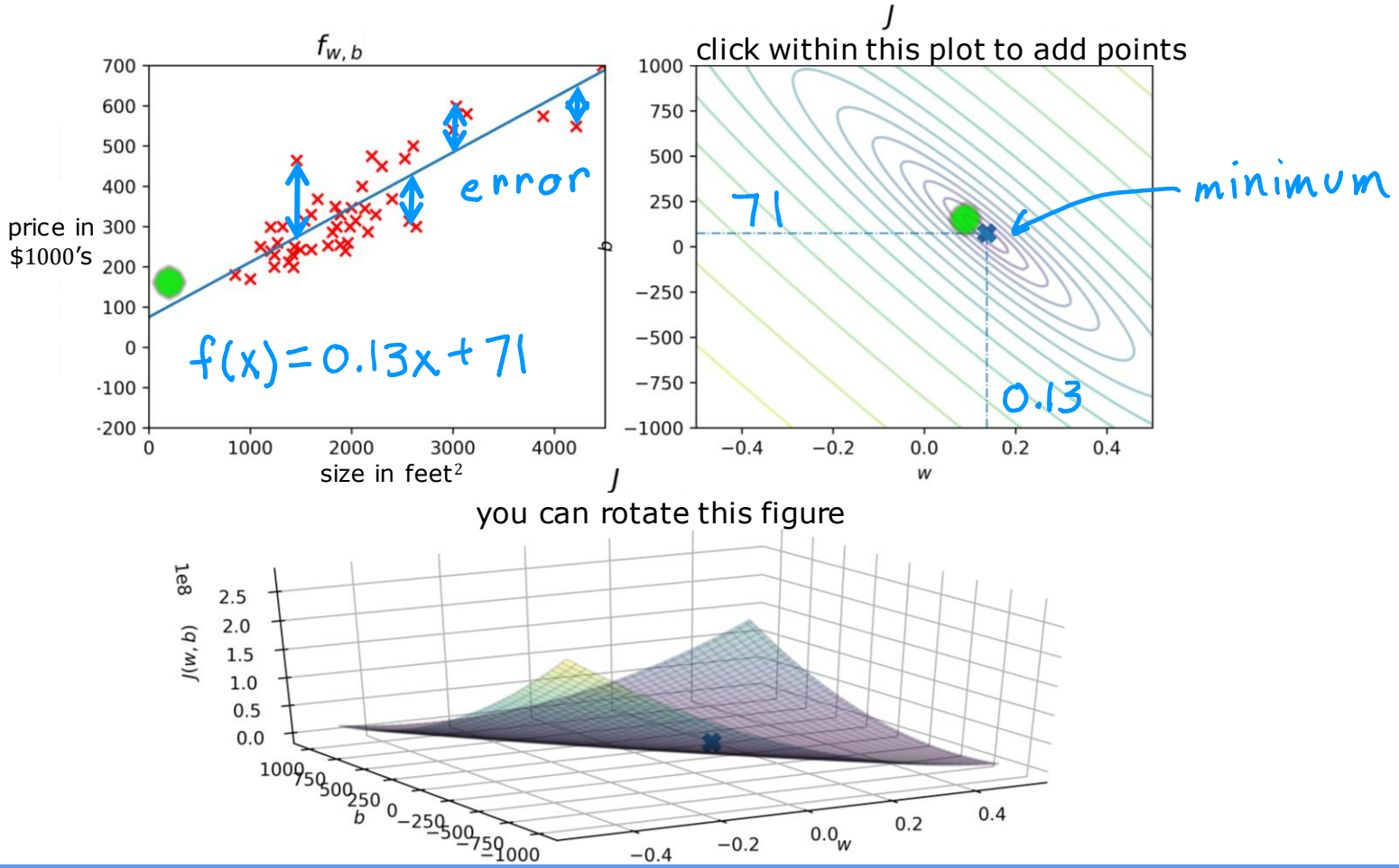



$f_{w,b}$
price in $1000's
size in feet^2
f(x) = 0x + 360
J
click within this plot to add points
360
0
b
w
J
you can rotate this figure
1e8
J(w, b)
b
w



you can rotate this figure




$f_{w,b}$
price in $1000's
size in feet^2
error
$f(x) = 0.13x + 71$
J
click within this plot to add points
71
0.13
minimum
w
b
you can rotate this figure
1e8
J(w, b)

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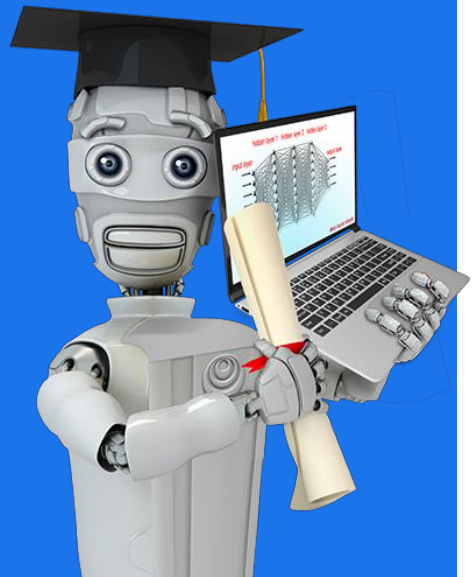


Training Linear Regression

Gradient Descent

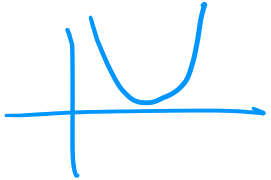
Have some function $J(w,b)$ for linear regression or any function

Want $\min_{w,b} J(w,b)$ $\min_{w_1,\dots,w_n,b} J(w_1,w_2,\dots,w_n,b)$

Outline:

- Start with some w,b (set $w=0, b=0$)
- Keep changing w,b to reduce $J(w,b)$
- Until we settle at or near a minimum

J not always



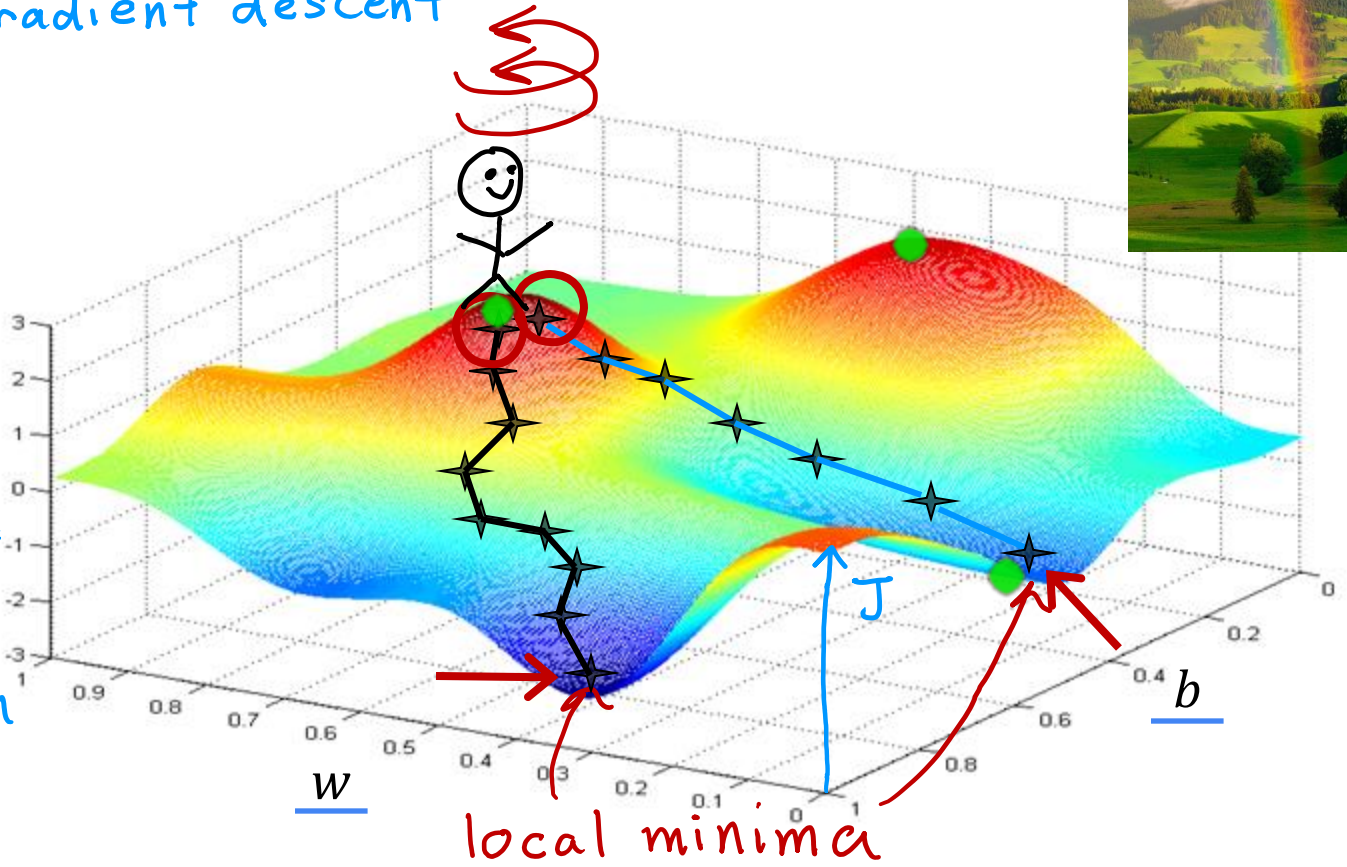
may have >1 minimum

gradient descent

$J(w, b)$

not squared error cost

not linear regression



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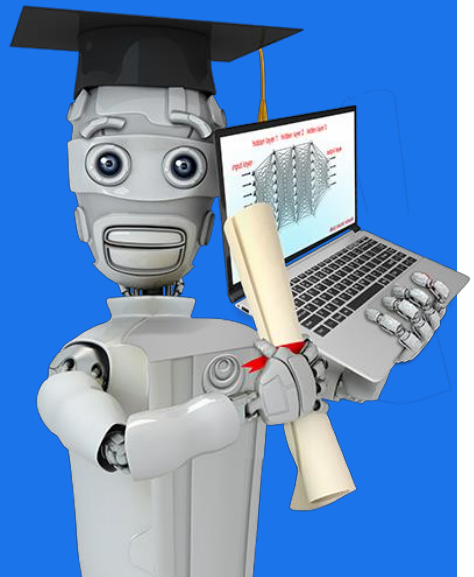

Training Linear Regression

Implementing Gradient Descent

Gradient descent algorithm

Repeat until convergence

$$w = w - \alpha \frac{\partial}{\partial w} J(w,b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w,b)$$

Learning rate

Derivative

Simultaneously update w and b

Assignment

$a = c$

$a = a + 1$

Code

Truth assertion

$a = c$

$a = a + 1$

Math

a==c

Correct: Simultaneous update

$$tmp_w = w - \alpha \frac{\partial}{\partial w} J(w,b)$$

$$tmp_b = b - \alpha \frac{\partial}{\partial b} J(w,b)$$

$$w = tmp_w$$

$$b = tmp_b$$

Incorrect

$$tmp_w = w - \alpha \frac{\partial}{\partial w} J(w,b)$$

$$w = tmp_w$$

$$tmp_b = b - \alpha \frac{\partial}{\partial b} J(w,b)$$

$$b = tmp_b$$

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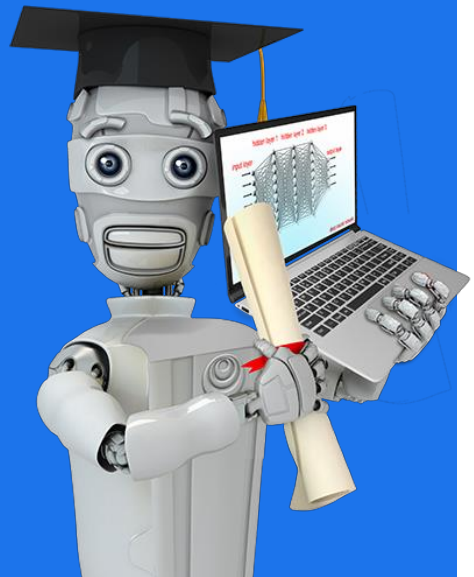


Training Linear Regression

Gradient Descent Intuition

Gradient descent algorithm

repeat until convergence {

learning rate

$$w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

derivative

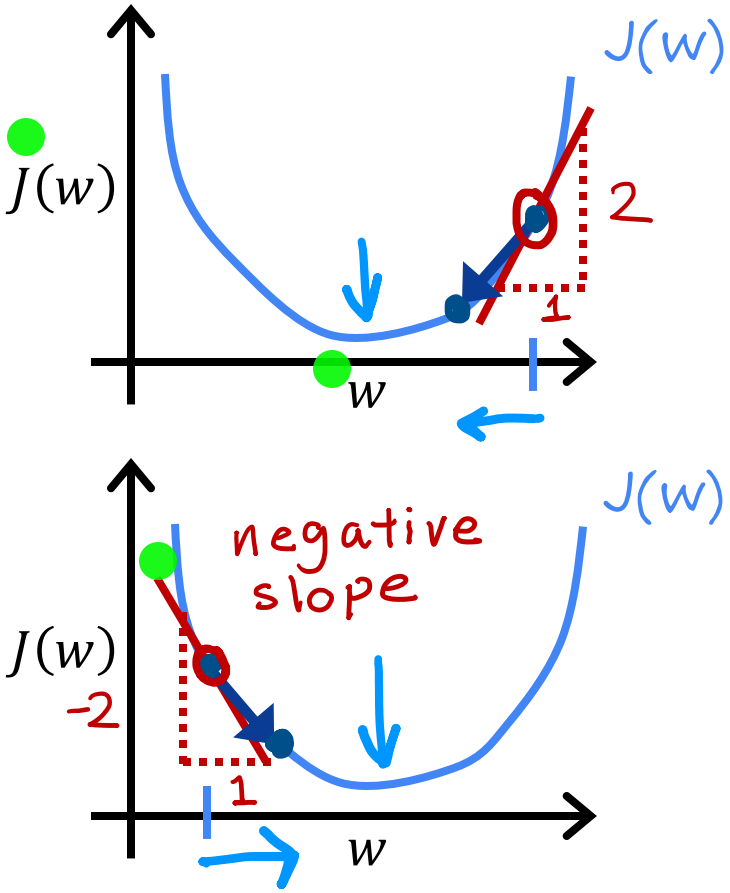
$$b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

}

$J(w)$

$$w = w - \alpha \frac{\partial}{\partial w} J(w)$$

$$\min_{w} J(w)$$



$$w = w - \alpha \underbrace{\frac{d}{dw} J(w)}_{>0}$$

$$w = w - \alpha \cdot (positive\ number)$$

$$\underbrace{\frac{d}{dw} J(w)}_{<0}$$

$$w = w - \alpha \cdot (negative\ number)$$

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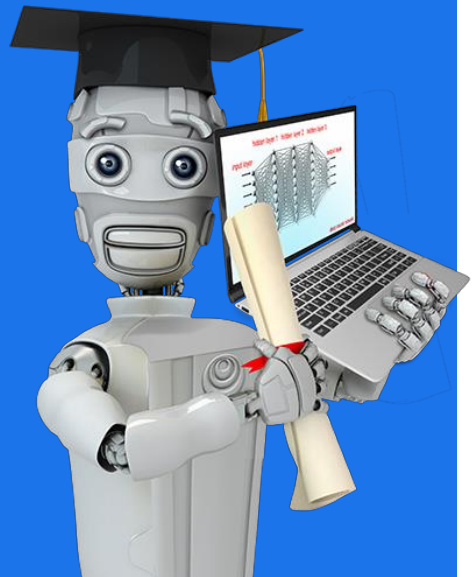


Training Linear Regression

Learning Rate

$$w = w - \alpha \frac{d}{dw} J(w)$$

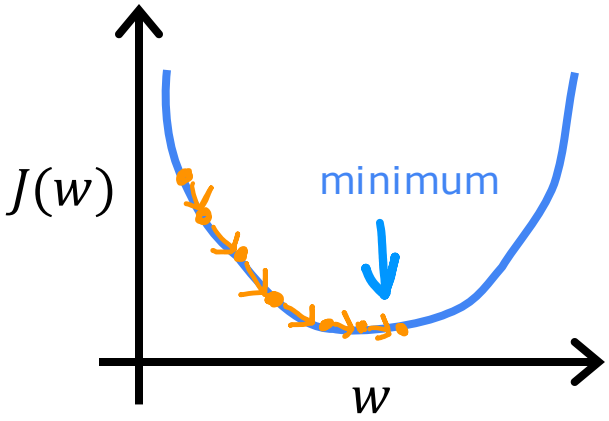
If α is too small…

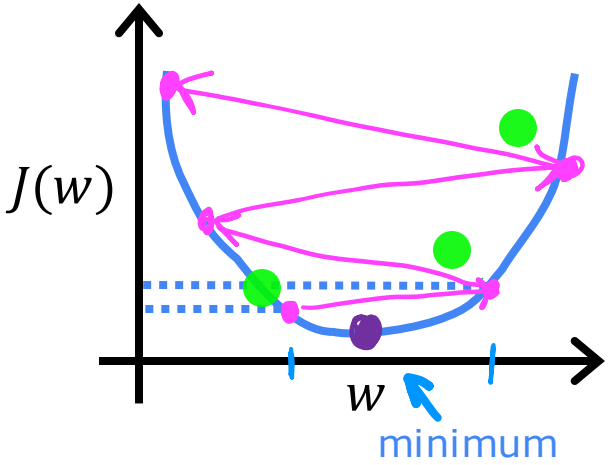
Gradient descent may be slow.

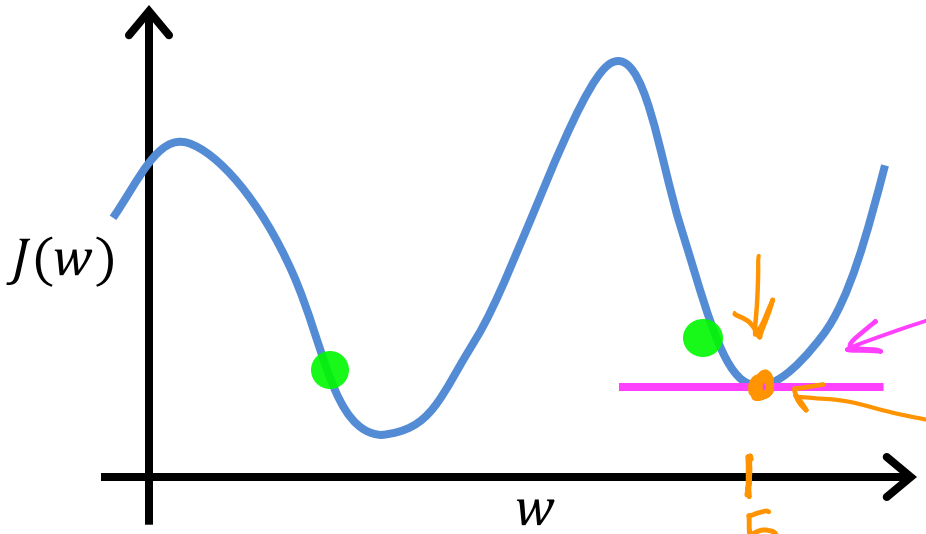
If α is too large…

Gradient descent may:

- Overshoot, never reach minimum
- Fail to converge, diverge







$$w = w - \alpha \frac{d}{dw} J(w)$$

0.1

$=0$

$5 - 0.1 \cdot 0$

$w = w - \alpha \cdot 0$

$w = w$

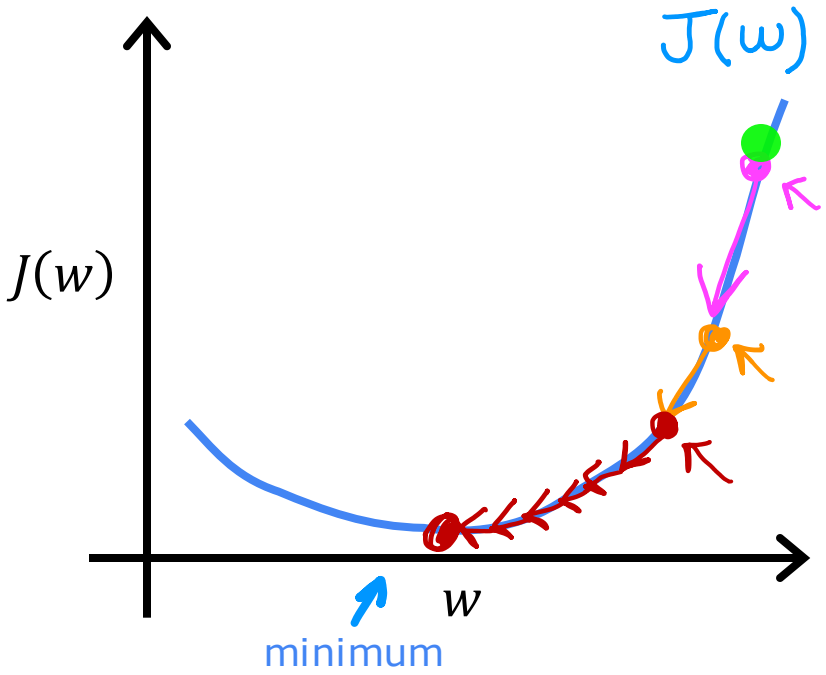
Can reach local minimum with fixed learning rate

$$w = w - \alpha \frac{d}{dw} J(w)$$

smaller

not as large

large



Near a local minimum,
- Derivative becomes smaller
- Update steps become smaller

Can reach minimum without decreasing learning rate α

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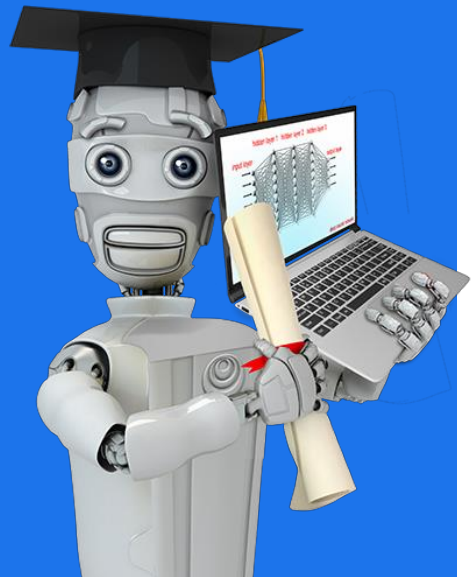


Training Linear Regression

Gradient Descent for Linear Regression

Linear regression model

$$f_{w,b}(x) = wx + b$$

Cost function

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})^2$$

Gradient descent algorithm

repeat until convergence {

$$w = w - \alpha \frac{\partial}{\partial w} J(w,b) \rightarrow \frac{1}{m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})x^{(i)}$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w,b) \rightarrow \frac{1}{m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})$$

}

next slide is optional!

(Optional)

$$\frac{\partial}{\partial w} J(w,b) = \frac{d}{dw} \frac{1}{2m} \sum_{i=1}^{m} \left(f_{w,b}(x^{(i)}) - y^{(i)}\right)^2 = \frac{d}{dw} \frac{1}{2m} \sum_{i=1}^{m} \left(wx^{(i)} + b - y^{(i)}\right)^2$$

$$= \frac{1}{2m} \sum_{i=1}^{m} \left(wx^{(i)} + b - y^{(i)}\right) 2x^{(i)} = \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}(x^{(i)}) - y^{(i)}\right) x^{(i)}$$

$$\frac{\partial}{\partial b} J(w,b) = \frac{d}{db} \frac{1}{2m} \sum_{i=1}^{m} \left(f_{w,b}(x^{(i)}) - y^{(i)}\right)^2 = \frac{d}{db} \frac{1}{2m} \sum_{i=1}^{m} \left(wx^{(i)} + b - y^{(i)}\right)^2$$

$$= \frac{1}{2m} \sum_{i=1}^{m} \left(wx^{(i)} + b - y^{(i)}\right) 2 = \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}(x^{(i)}) - y^{(i)}\right)$$

no $x^{(i)}$

Gradient descent algorithm

$\frac{\partial}{\partial w} J(w,b)$

repeat until convergence {

$$w = w - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}\left(x^{(i)}\right) - y^{(i)} \right) x^{(i)}$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}\left(x^{(i)}\right) - y^{(i)} \right)$$

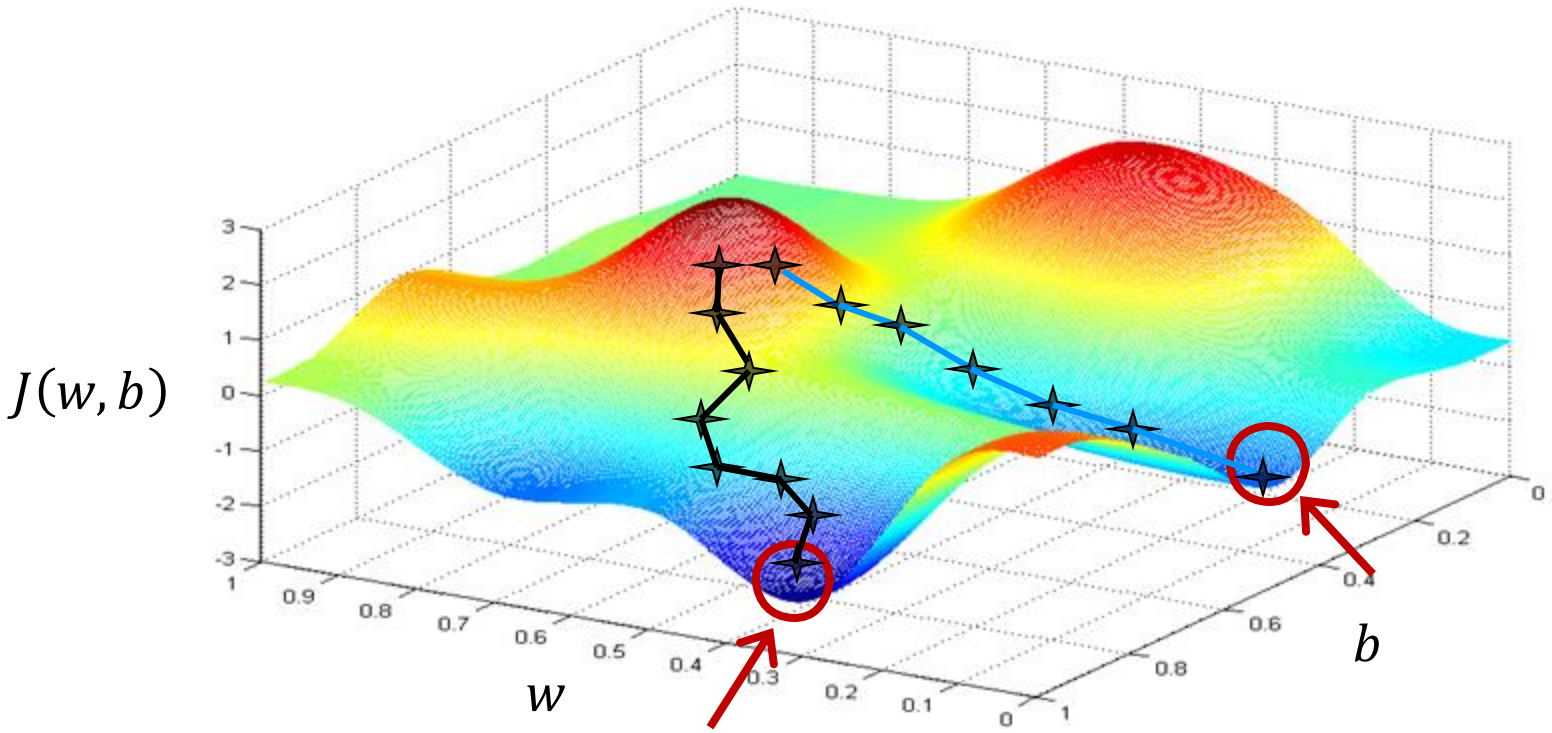
}

Update w and b simultaneously

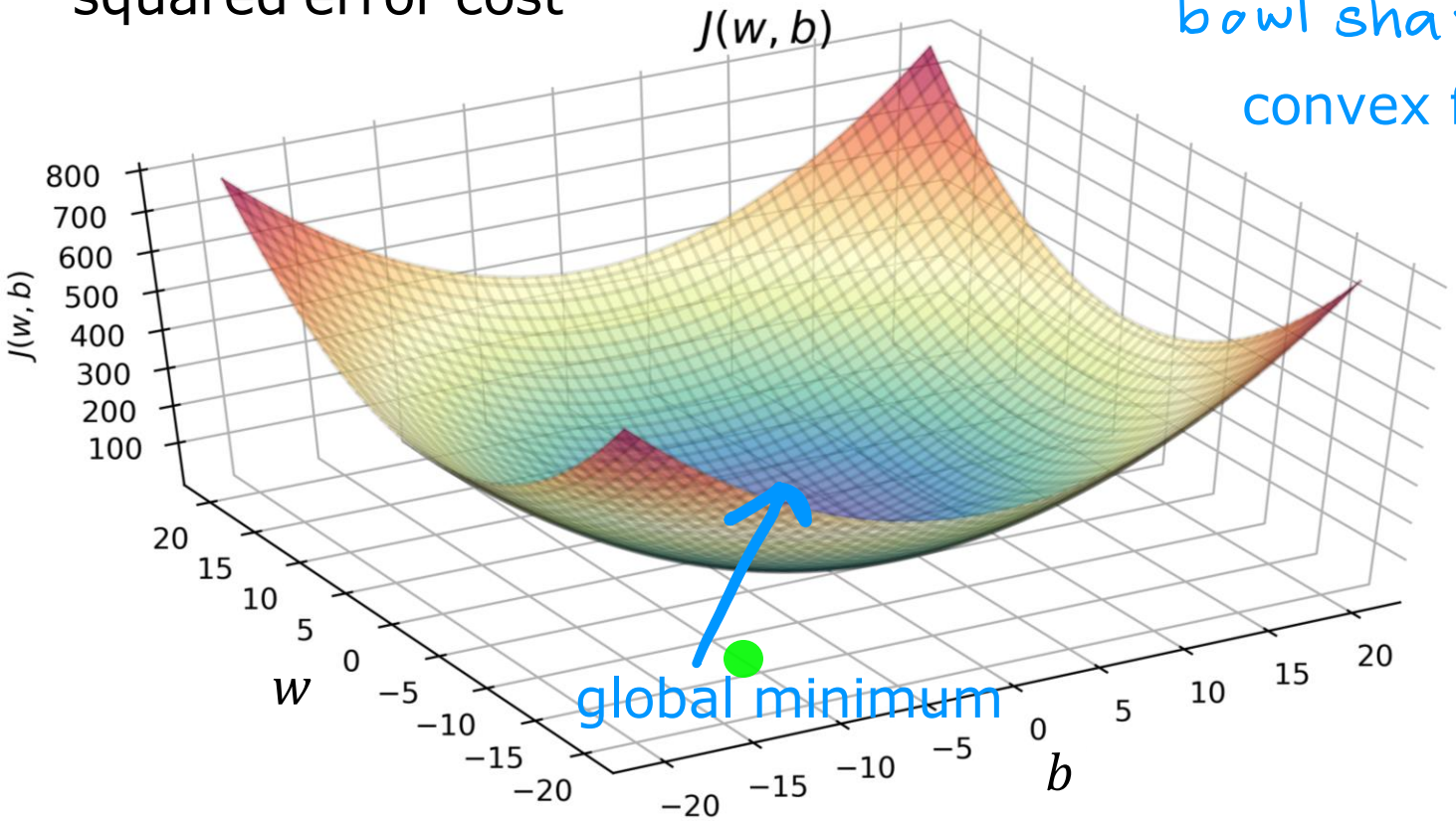
$\frac{\partial}{\partial b} J(w,b)$

$f_{w,b}(x^{(i)}) = wx^{(i)} + b$

More than one local minimum



squared error cost



bowl shape

convex function

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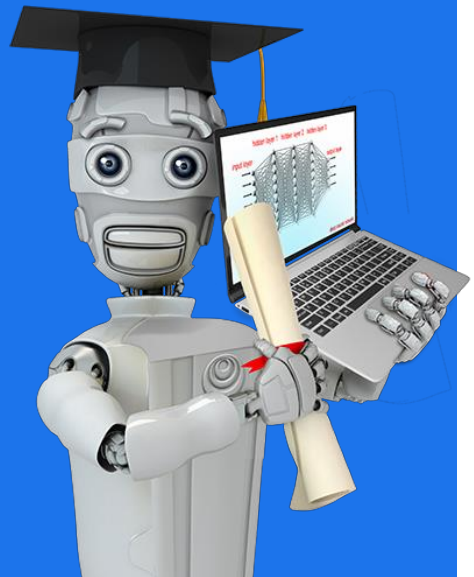
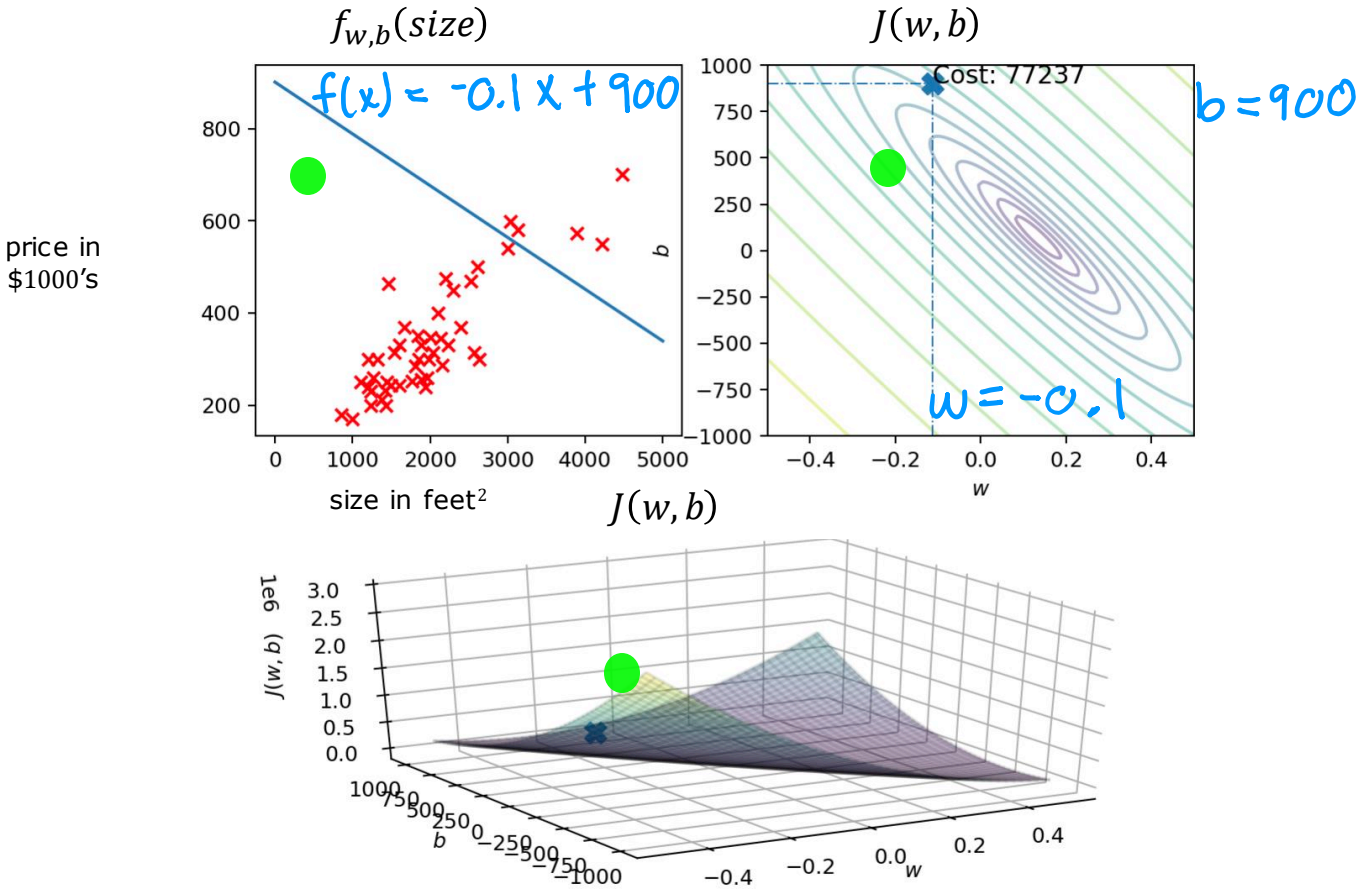
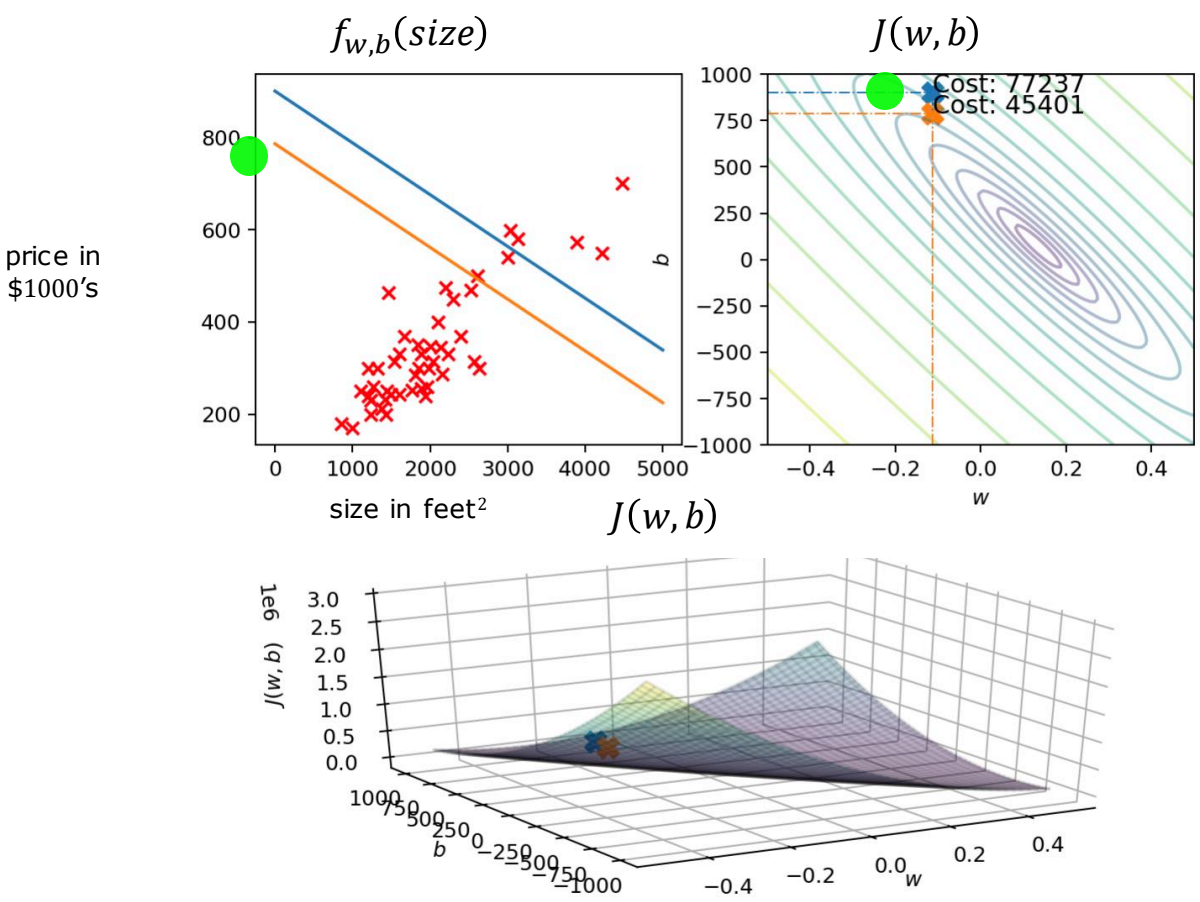
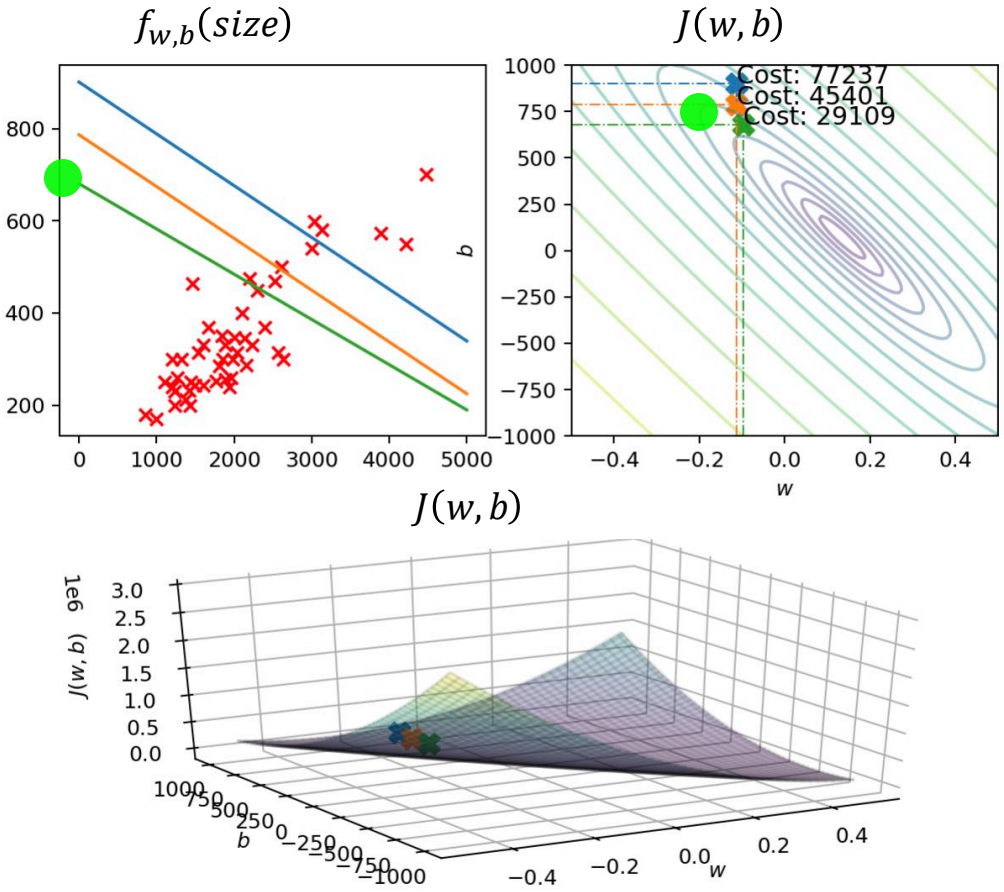


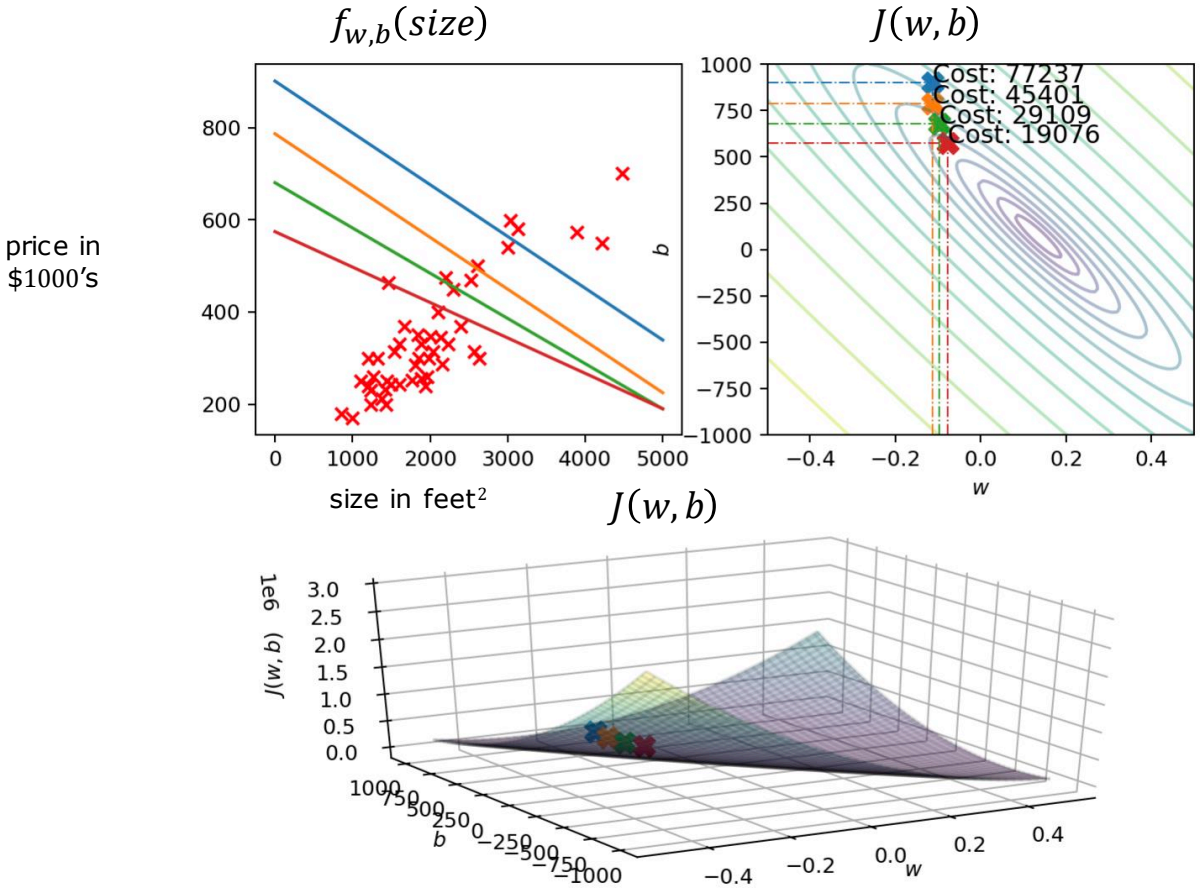
Training Linear Regression

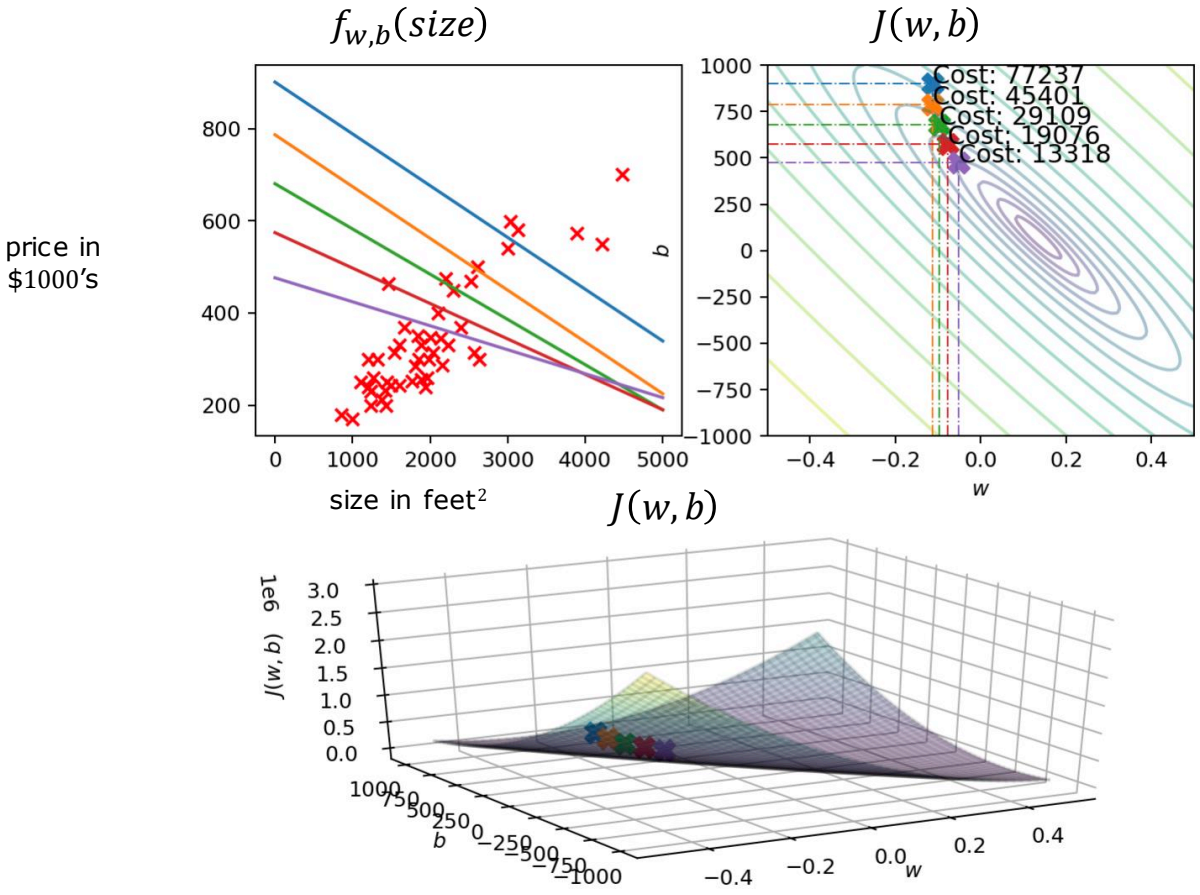
Running Gradient Descent

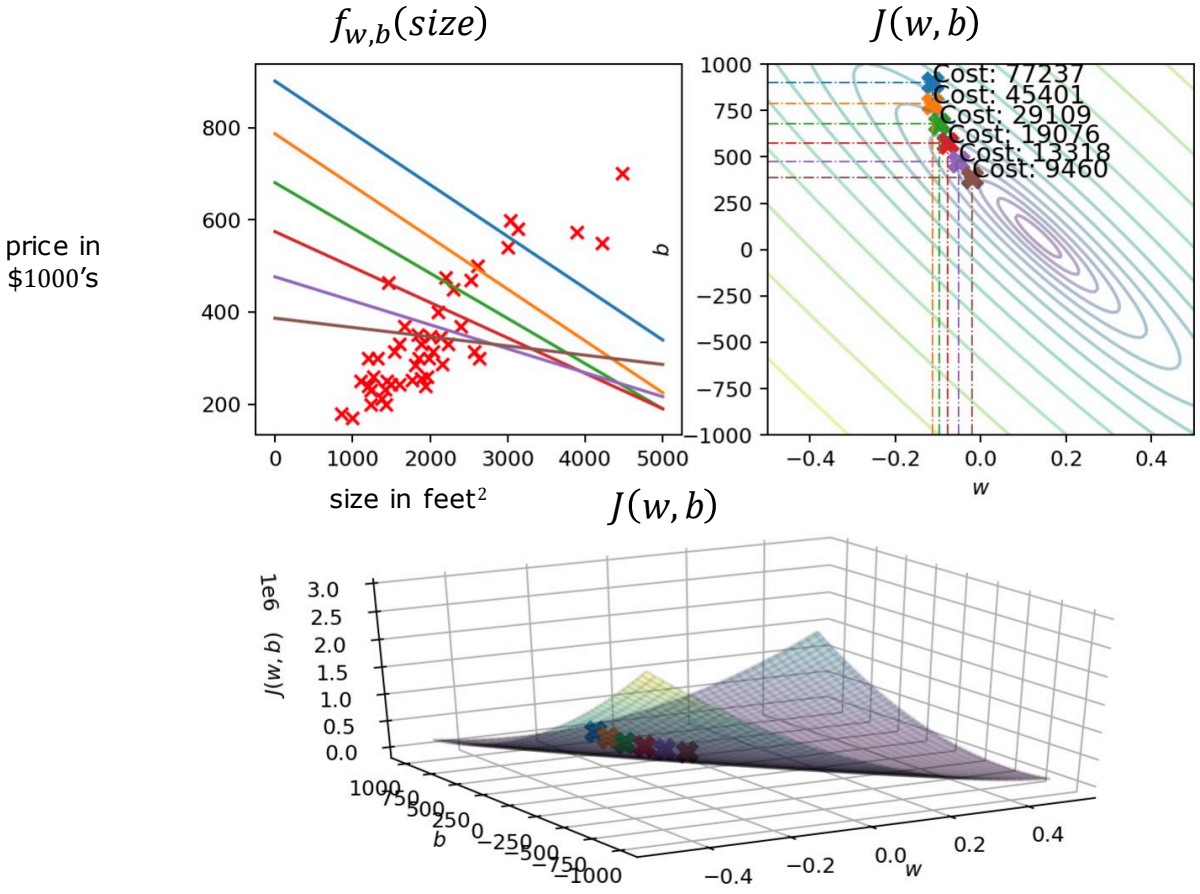

$f_{w,b}(size)$
$J(w,b)$
f(x) = -0.1 x + 900
Cost: 77237
b = 900
w = -0.1
price in $1000's
size in feet²
b
w
$J(w,b)$
$J(w,b)$
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1000
750
500
250
0
−250
−500
−750
−1000
−0.4
−0.2
0.0
0.2
0.4

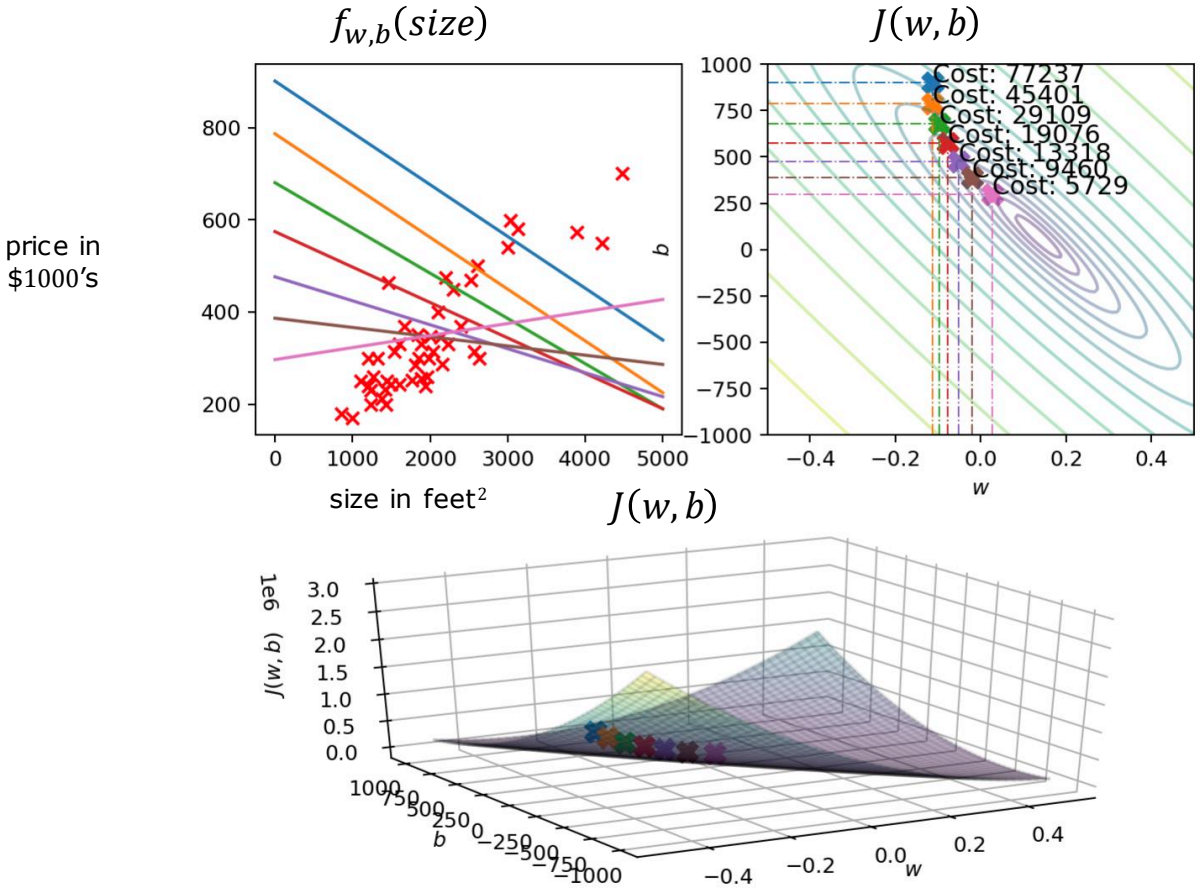

$f_{w,b}(size)$
price in $1000's
size in feet²
$J(w,b)$
Cost: 77237
Cost: 45401
$J(w,b)$

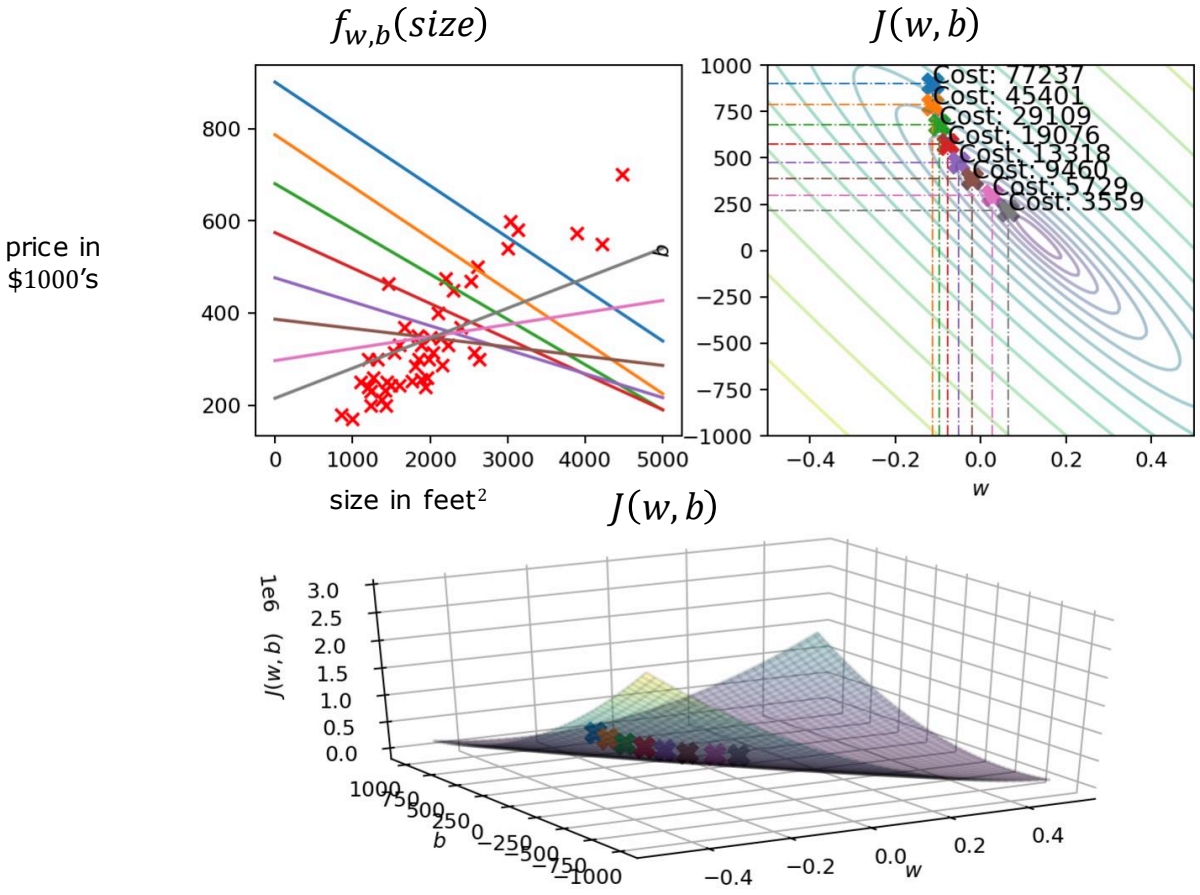

$f_{w,b}(size)$
800
600
400
200
0
1000
2000
3000
4000
5000
b
$J(w,b)$
1000
750
500
250
0
−250
−500
−750
−1000
Cost: 77237
Cost: 45401
Cost: 29109
−0.4
−0.2
0.0
0.2
0.4
w
$J(w,b)$
log
3.0
2.5
2.0
1.5
1.0
0.5
0.0
$J(w,b)$
1000
750
500
250
0
−250
−500
−750
−1000
b
−0.4
−0.2
0.0
0.2
0.4
w

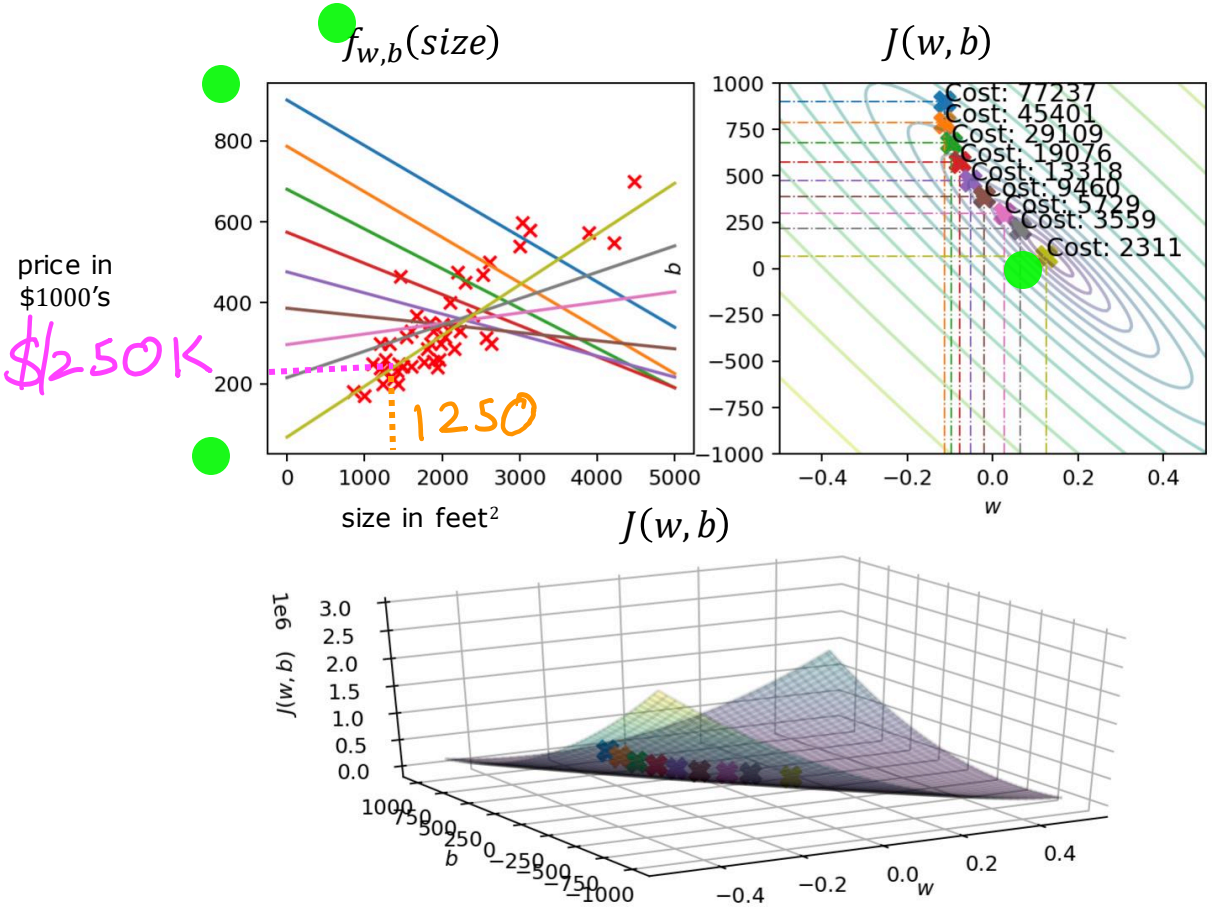

$f_{w,b}(size)$
price in $1000's
size in feet²
$J(w,b)$
Cost: 77237
Cost: 45401
Cost: 29109
Cost: 19076
$J(w,b)$


$f_{w,b}(size)$
price in $1000's
size in feet^2
$J(w,b)$
Cost: 77237
Cost: 45401
Cost: 29109
Cost: 19076
Cost: 13318
w
b
$J(w,b)$
log
3.0
2.5
2.0
1.5
1.0
0.5
0.0


$f_{w,b}(size)$
price in $1000's
size in feet^2
$J(w,b)$
Cost: 77237
Cost: 45401
Cost: 29109
Cost: 19076
Cost: 13318
Cost: 9460
$J(w,b)$


$f_{w,b}(size)$
price in $1000's
size in feet²
$J(w,b)$
Cost: 77237
Cost: 45401
Cost: 29109
Cost: 19076
Cost: 13318
Cost: 9460
Cost: 5729
w
b
$J(w,b)$
log
$J(w,b)$


$f_{w,b}(size)$
price in $1000's
size in feet^2
$J(w,b)$
Cost: 77237
Cost: 45401
Cost: 29109
Cost: 19076
Cost: 13318
Cost: 9460
Cost: 5729
Cost: 3559
w
b
$J(w,b)$


$f_{w,b}(size)$
price in $1000's
$250K
1250
size in feet²
$J(w,b)$
Cost: 77237
Cost: 45401
Cost: 29109
Cost: 19076
Cost: 13318
Cost: 9460
Cost: 5729
Cost: 3559
Cost: 2311
$J(w,b)$
log
$J(w,b)$

"Batch" gradient descent

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THE BATCH

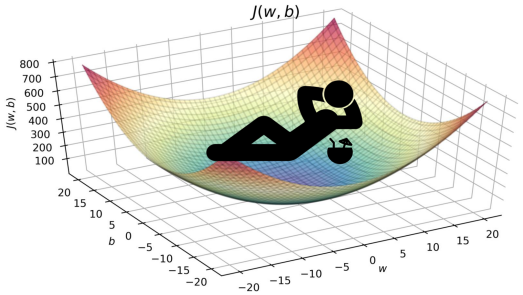
"Batch": Each step of gradient descent uses all the training examples.

other gradient descent: subsets

	x size in feet^2	y price in \$1000's
(1)	2104	400
(2)	1416	232
(3)	1534	315
(4)	852	178
...	...	...
(47)	3210	870

$m = 47$

$$\sum_{i=1}^{m} (f_{w,b}(x^{(i)}) - y^{(i)})^2$$



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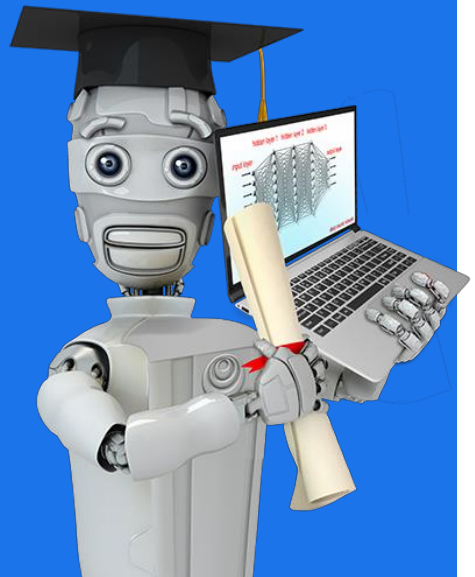

Linear Regression with Multiple Variables

Multiple Features

Multiple features (variables)

one feature →

Size in feet^2 (x)	Price ($) in 1000's (y) ←
2104	400
1416	232
1534	315
852	178
…	…

$$f_{w,b}(x) = wx + b$$

Multiple features (variables)

Size in feet²	Number of bedrooms	Number of floors	Age of home in years	Price ($) in $1000's
x_1	x_2	x_3	x_4	
2104	5	1	45	460
1416	3	2	40	232
1534	3	2	30	315
852	2	1	36	178
...	...	...	...	...

$j = 1 \ldots 4$

$n = 4$

$i = 2$

x_j = j^{th} feature

n = number of features

$\vec{x}^{(i)}$ = features of i^{th} training example

$x_j^{(i)}$ = value of feature j in i^{th} training example

$\vec{x}^{(2)} = [1416 \quad 3 \quad 2 \quad 40]$

$x_3^{(2)} = 2$

Model:

Previously: $f_{w,b}(x) = wx + b$

$$f_{w,b}(X) = w_1 X_1 + w_2 X_2 + w_3 X_3 + w_4 X_4 + b$$

example

$$f_{w,b}(X) = 0.1 X_1 + 4 X_2 + 10 X_3 + -2 X_4 + 80$$

X_1: size; X_2: #bedrooms; X_3: #floors; X_4: years; 80: base price

$$f_{w,b}(x) = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

$$f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

$\vec{w} = [w_1 \; w_2 \; w_3 \ldots w_n]$ parameters of the model

b is a number

vector $\vec{x} = [x_1 \; x_2 \; x_3 \ldots x_n]$

$$f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b = w_1 x_1 + w_2 x_2 + w_3 x_3 + \cdots + w_n x_n + b$$

dot product

multiple linear regression

(not multivariate regression)

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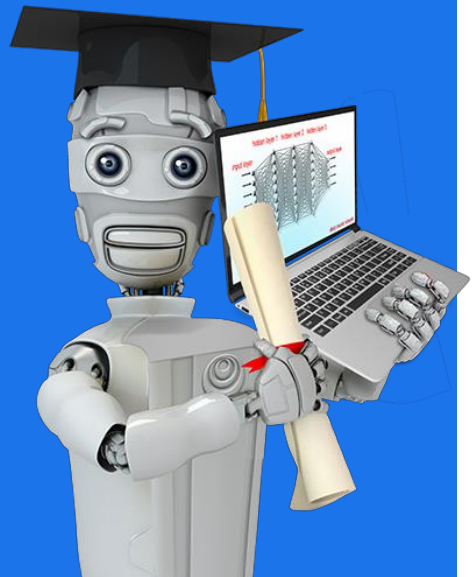

Linear Regression with Multiple Variables

Vectorization Part 1

Parameters and features

$\vec{w} = [w_1 \quad w_2 \quad w_3]$ $n=3$

b is a number

$\vec{x} = [x_1 \quad x_2 \quad x_3]$

linear algebra: count from 1

NumPy

w[0] w[1] w[2]

```
w = np.array([1.0,2.5,-3.3])
b = 4
x = np.array([10,20,30])
```

x[0] x[1] x[2]

code: count from 0

Without vectorization

$n = 100{,}000$

$f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$

```
f = w[0] * x[0] +
    w[1] * x[1] +
    w[2] * x[2] + b
```

Without vectorization

$f_{\vec{w},b}(\vec{x}) = \left(\sum_{j=1}^{n} w_j x_j\right) + b$

$\sum_{j=1}^{n} \rightarrow j = 1 \ldots n$ 1, 2, 3

range(0, n) → j = 0 ... n-1

```
f = 0      range(n)
for j in range(0,n):
    f = f + w[j] * x[j]
f = f + b
```

Vectorization

$f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$

```
f = np.dot(w,x) + b
```

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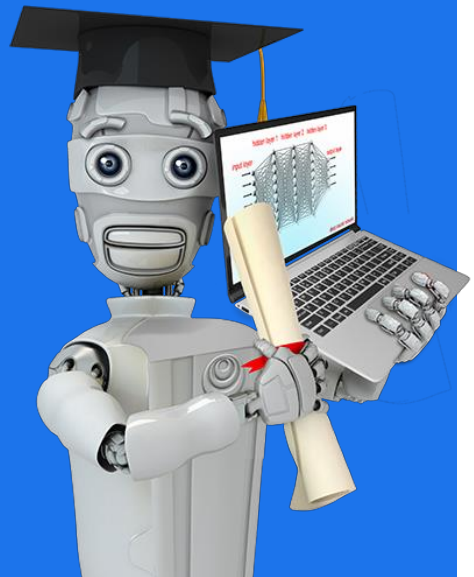


Linear Regression with Multiple Variables

Vectorization Part 2

Without vectorization

```
for j in range(0,16):
    f = f + w[j] * x[j]
```

t_0
```
f + w[0] * x[0]
```

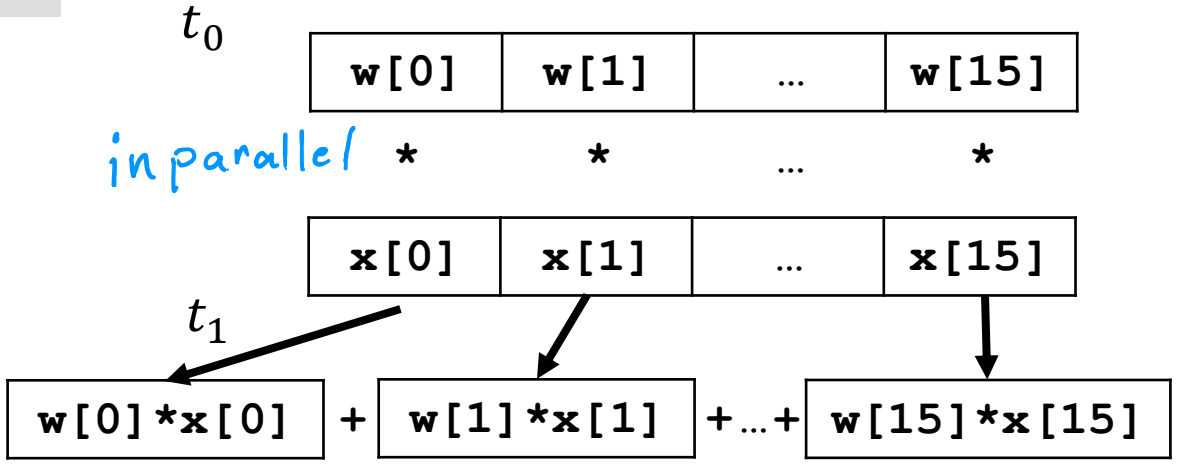
t_1
```
f + w[1] * x[1]
```

...

t_{15}
```
f + w[15] * x[15]
```

Vectorization

```
np.dot(w,x)
```



efficient → scale to large datasets

Gradient descent

$\vec{w} = (w_1 \quad w_2 \quad \cdots \quad w_{16})$ ~~b~~ parameters

derivatives $\vec{d} = (d_1 \quad d_2 \quad \cdots \quad d_{16})$

```
w = np.array([0.5, 1.3, … 3.4])
d = np.array([0.3, 0.2, … 0.4])
```

learning rate α

compute $w_j = w_j - 0.1 d_j$ for $j = 1 \ldots 16$

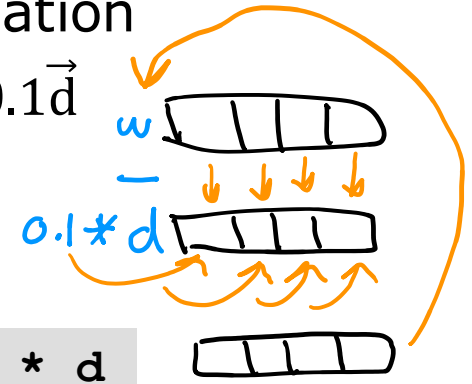
Without vectorization

$$w_1 = w_1 - 0.1 d_1$$
$$w_2 = w_2 - 0.1 d_2$$
$$\vdots$$
$$w_{16} = w_{16} - 0.1 d_{16}$$

```
for j in range(0,16):
    w[j] = w[j] - 0.1 * d[j]
```

With vectorization

$$\vec{w} = \vec{w} - 0.1\vec{d}$$




```
w = w - 0.1 * d
```

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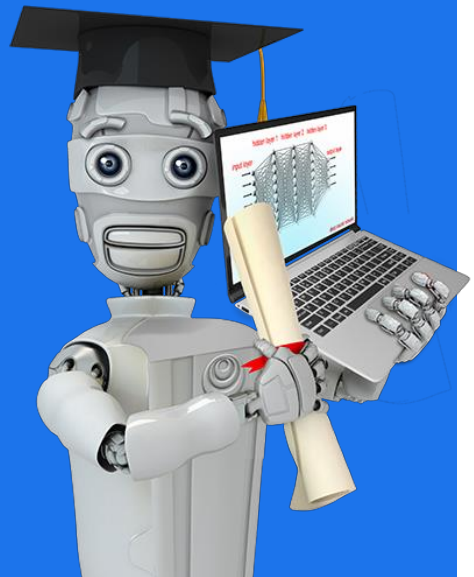

Linear Regression with Multiple Variables

Gradient Descent for Multiple Regression

Previous notation

Parameters $w_1, \cdots, w_n$

b

Model $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + \cdots + w_n x_n + b$

Cost function $J(w_1, \cdots, w_n, b)$

Vector notation

vector of length n

$\vec{w} = [w_1 \quad \cdots \quad w_n]$

b still a number

$f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$

dot product

$J(\vec{w}, b)$

Gradient descent

repeat {

$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(w_1, \cdots, w_n, b)$

$b = b - \alpha \frac{\partial}{\partial b} J(w_1, \cdots, w_n, b)$

}

repeat {

$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$

$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b)$

}

Gradient descent

One feature

repeat {

$$w = w - \alpha \frac{1}{m}\sum_{i=1}^{m}\left(f_{w,b}\left(x^{(i)}\right) - y^{(i)}\right)x^{(i)}$$

$$\frac{\partial}{\partial w}J(w,b)$$

$$b = b - \alpha \frac{1}{m}\sum_{i=1}^{m}\left(f_{w,b}\left(x^{(i)}\right) - y^{(i)}\right)$$

simultaneously update w, b

}

n features ($n \geq 2$)

repeat {

$j = 1$

$$w_1 = w_1 - \alpha \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right) - y^{(i)}\right)x_1^{(i)}$$

$$\frac{\partial}{\partial w_1}J(\vec{w},b)$$

$\vdots$

$j = n$

$$w_n = w_n - \alpha \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right) - y^{(i)}\right)x_n^{(i)}$$

$$b = b - \alpha \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right) - y^{(i)}\right)$$

simultaneously update

w_j (for $j = 1, \cdots, n$) and b

}

An alternative to gradient descent

Normal equation
- Only for linear regression
- Solve for w, b without iterations

Disadvantages
- Doesn't generalize to other learning algorithms.
- Slow when number of features is large (> 10,000)

What you need to know
- Normal equation method may be used in machine learning libraries that implement linear regression.
- Gradient descent is the recommended method for finding parameters w,b

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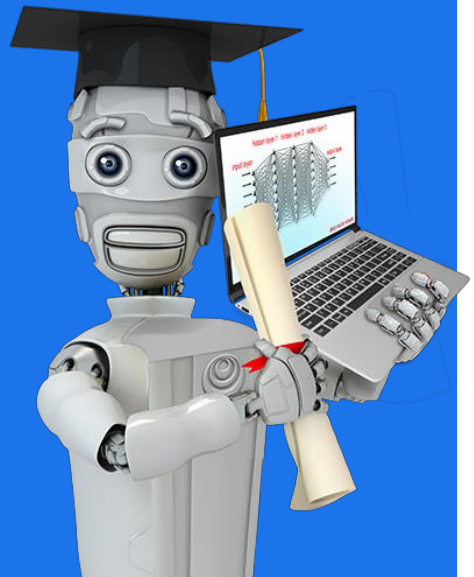


Practical Tips for Linear Regression

Feature Scaling Part 1

Feature and parameter values

$\widehat{price} = w_1 x_1 + w_2 x_2 + b$

size #bedrooms

x_1: size (feet^2)
range: $300 - 2{,}000$

large

x_2: # bedrooms
range: $0 - 5$

small

House: $x_1 = 2000$, $x_2 = 5$, $price = \$500\text{k}$

one training example

size of the parameters w_1, w_2?

$w_1 = 50, \quad w_2 = 0.1, \quad b = 50$

$\widehat{price} = \underbrace{50 * 2000}_{100{,}000\text{k}} + \underbrace{0.1 * 5}_{0.5\text{k}} + \underbrace{50}_{50\text{k}}$

$\widehat{price} = \$100{,}050.5\text{k} = \$100{,}050{,}500$

$w_1 = 0.1, \quad w_2 = 50, \quad b = 50$

small large

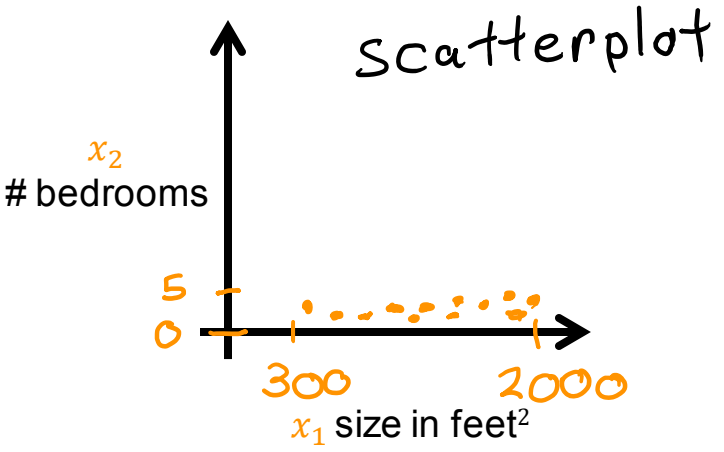
$\widehat{price} = \underbrace{0.1 * 2000k}_{200\text{k}} + \underbrace{50 * 5}_{250\text{k}} + \underbrace{50}_{50\text{k}}$

$\widehat{price} = \$500\text{k}$ more reasonable

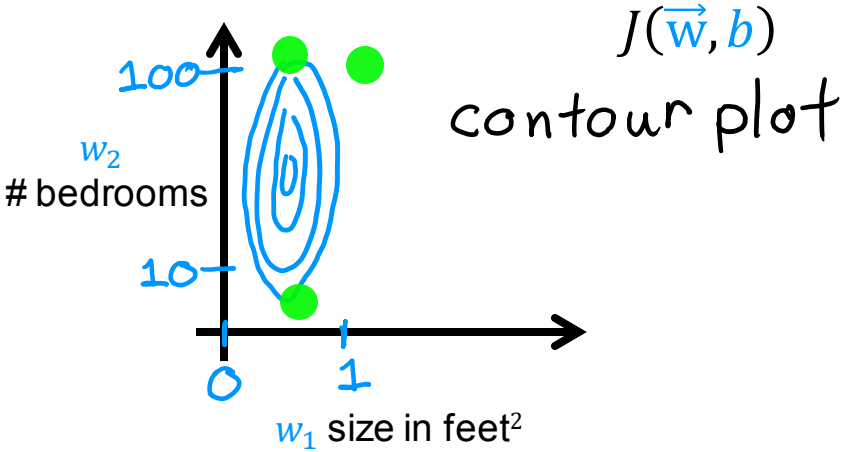
Feature size and parameter size

	size of feature x_j	size of parameter w_j
size in feet^2	⟷	⟷
#bedrooms	⟷	⟷

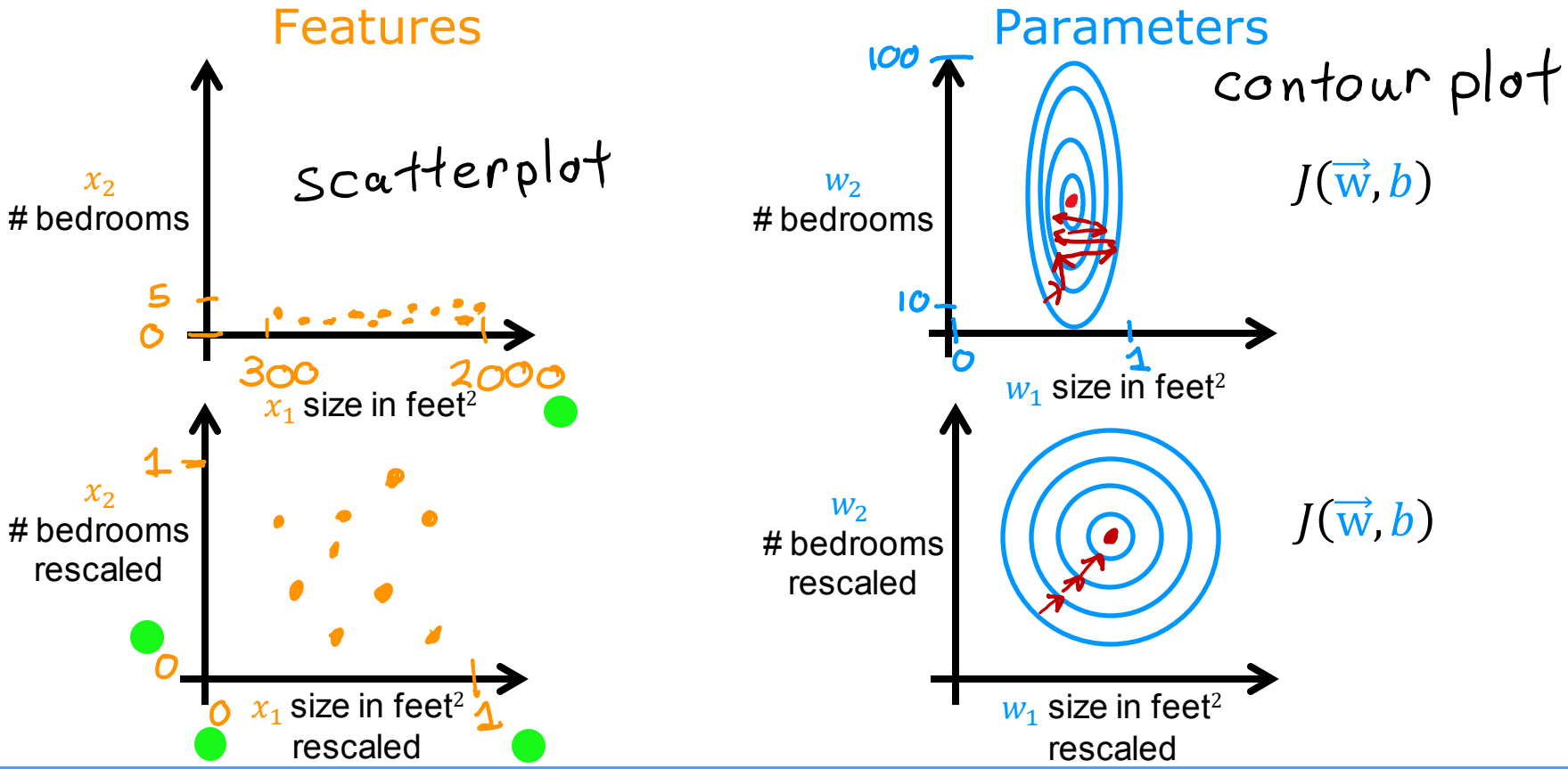
Features



Parameters



Feature size and gradient descent



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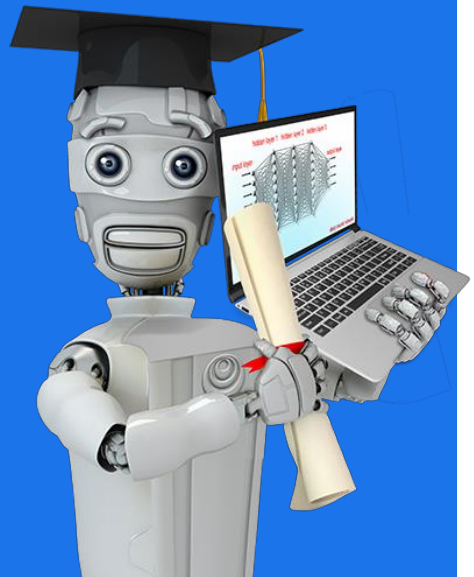
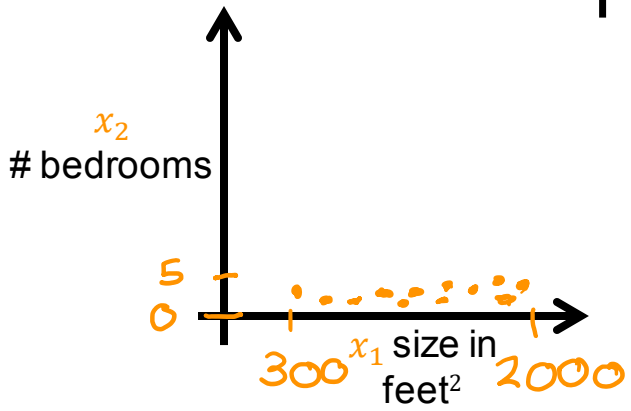
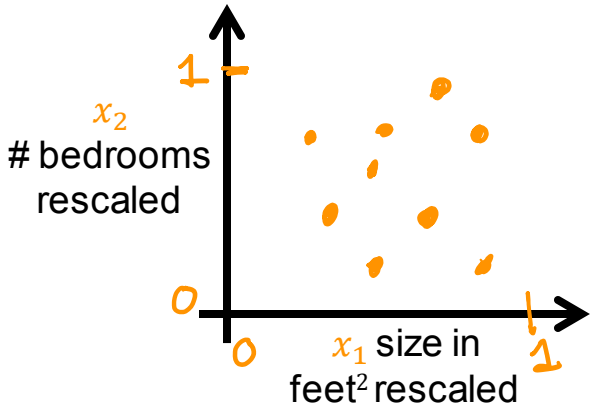


Practical Tips for Linear Regression

Feature Scaling Part 2

Feature scaling





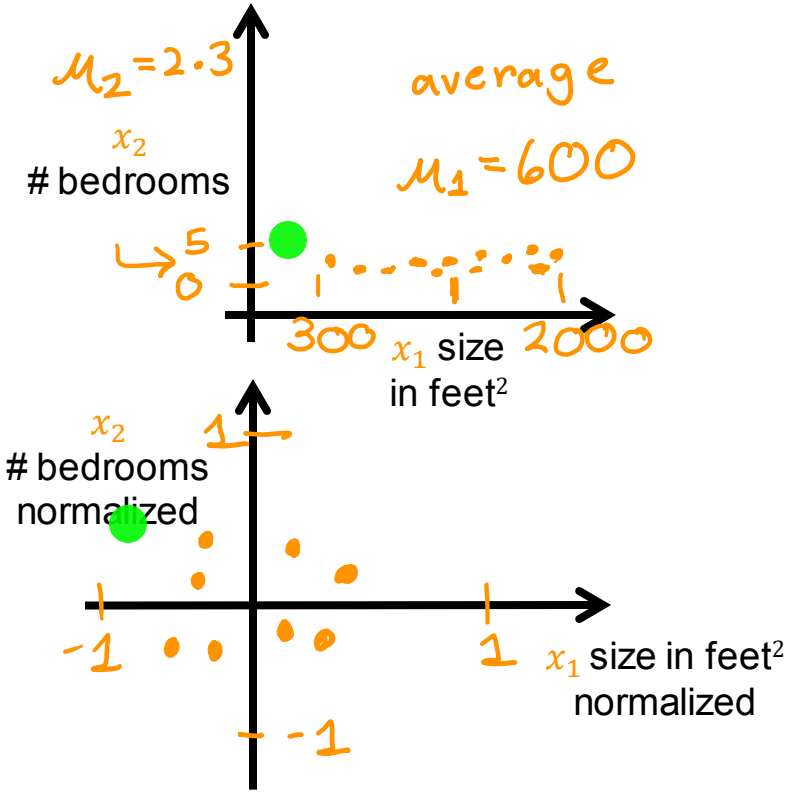
$$300 \le x_1 \le 2000 \qquad 0 \le x_2 \le 5$$

$$x_{1,scaled} = \frac{x_1}{2000} \qquad x_{2,scaled} = \frac{x_2}{5}$$

max max

$$0.15 \le x_{1,scaled} \le 1 \qquad 0 \le x_{2,scaled} \le 1$$

Mean normalization



$$300 \le x_1 \le 2000$$

$$x_1 = \frac{x_1 - \mu_1}{2000 - 300}$$

max-min

$$-0.18 \le x_1 \le 0.82$$

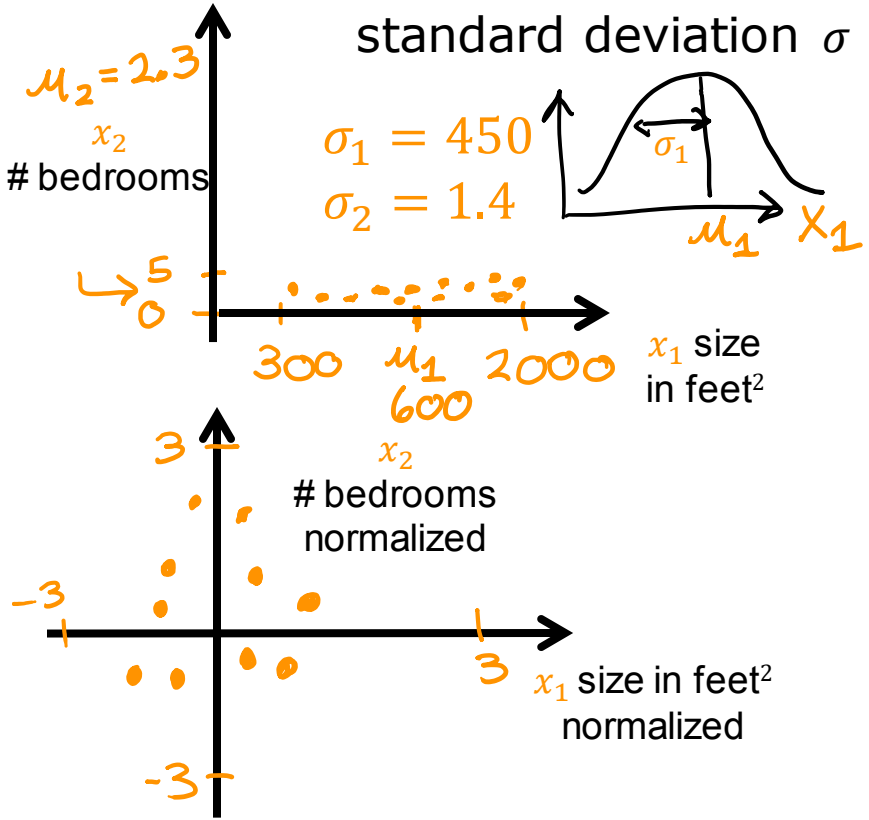
$$0 \le x_2 \le 5$$

$$x_2 = \frac{x_2 - \mu_2}{5 - 0}$$

max-min

$$-0.46 \le x_2 \le 0.54$$

Z-score normalization



$300 \le x_1 \le 2000$ $\quad$ $0 \le x_2 \le 5$

$$x_1 = \frac{x_1 - \mu_1}{\sigma_1} \qquad x_2 = \frac{x_2 - \mu_2}{\sigma_2}$$

$-0.67 \le x_1 \le 3.1$ $\quad$ $-1.6 \le x_2 \le 1.9$

Feature scaling

aim for about $-1 \leq x_j \leq 1$ for each feature x_j

$-3 \leq x_j \leq 3$

$-0.3 \leq x_j \leq 0.3$

} acceptable ranges

$0 \leq x_1 \leq 3$ okay, no rescaling

$-2 \leq x_2 \leq 0.5$ okay, no rescaling

$-100 \leq x_3 \leq 100$ too large → rescale

$-0.001 \leq x_4 \leq 0.001$ too small → rescale

$98.6 \leq x_5 \leq 105$ too large → rescale

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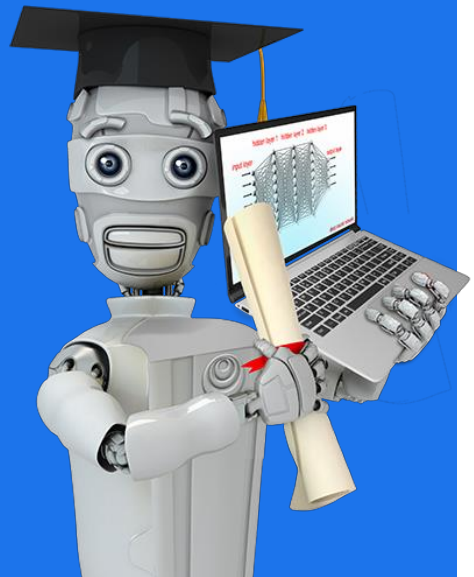


Practical Tips for Linear Regression

Checking Gradient Descent for Convergence

Gradient descent

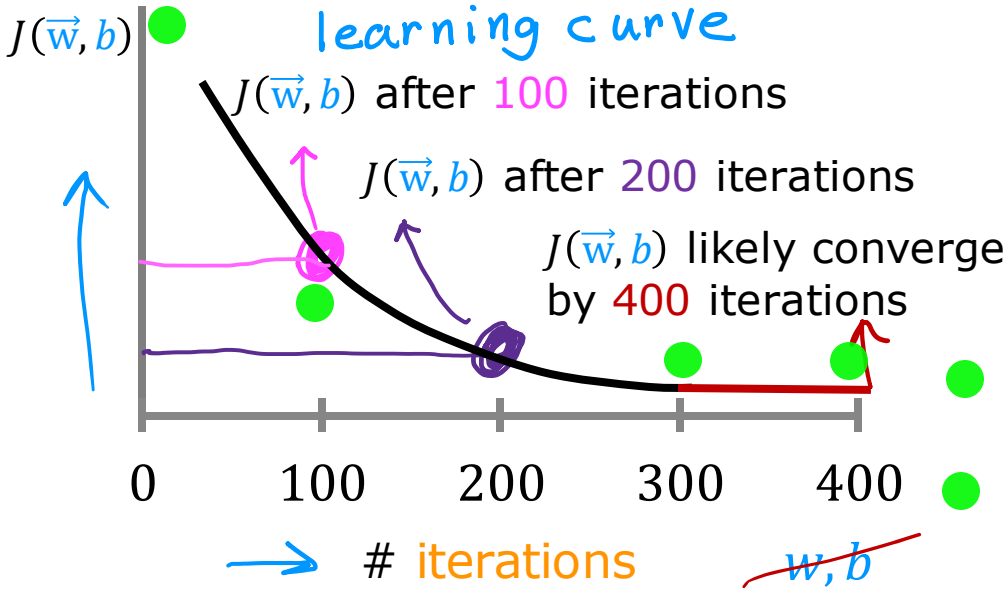
$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b)$$

Make sure gradient descent is working correctly

objective: $\min_{\vec{w},b} J(\vec{w},b)$

$J(\vec{w},b)$ should decrease after every iteration



iterations needed varies 30 1,000 100,000

Automatic convergence test

Let ε "epsilon" be 10^{-3}. 0.001

If $J(\vec{w},b)$ decreases by $\leq \varepsilon$ in one iteration, declare convergence.

(found parameters $\vec{w}, b$ to get close to global minimum)

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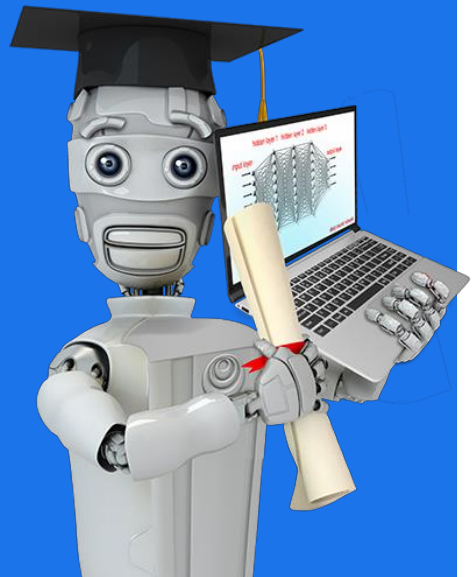
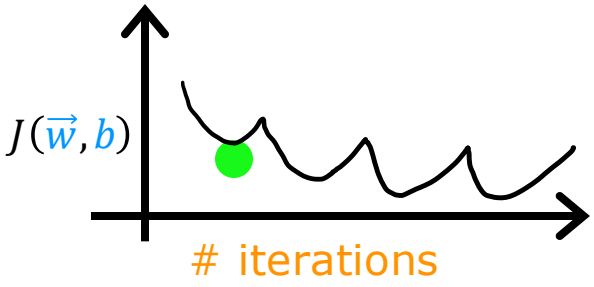
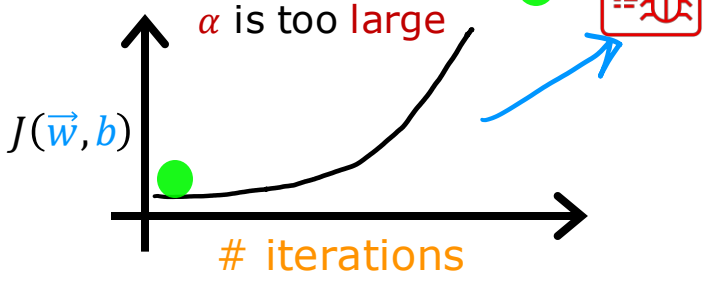


Practical Tips for Linear Regression

Choosing the Learning Rate

Identify problem with gradient descent





or learning rate is too large

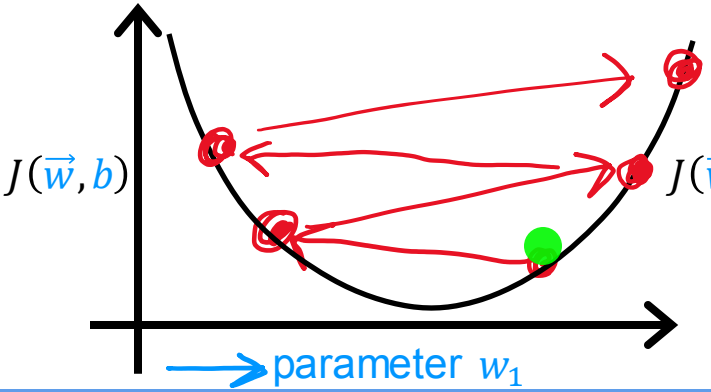
$w_1 = w_1 + \alpha d_1$

use a minus sign

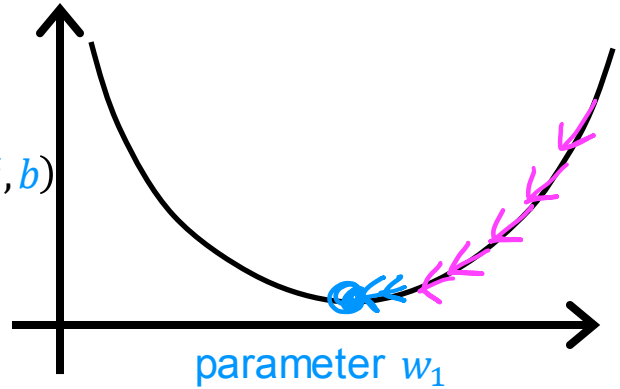
$w_1 = w_1 - \alpha d_1$

Adjust learning rate

α is too big



Use smaller α



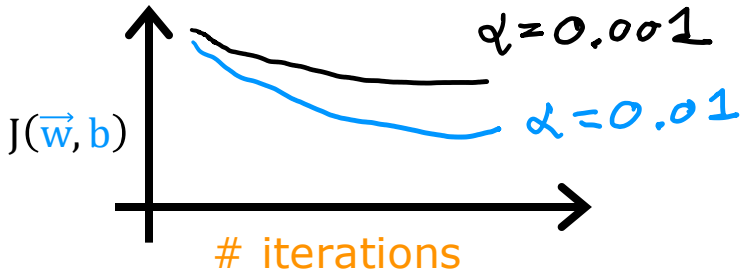
With a small enough α, $J(\vec{w}, b)$ should decrease on every iteration

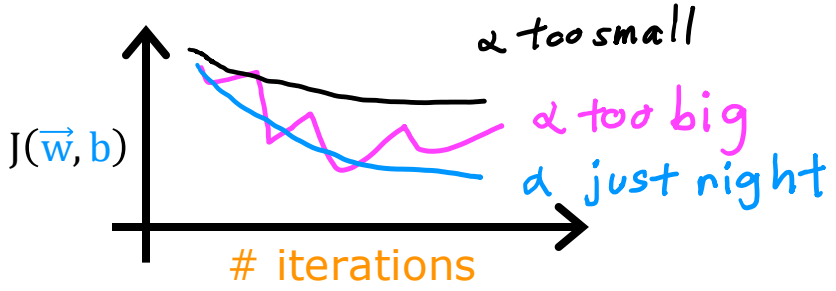
If α is too small, gradient descent takes a lot more iterations to converge

Values of α to try:

... 0.001 0.003 0.01 0.03 0.1 0.3 1 ...

3x ≈3x 3x ≈3x 3x ≈3x





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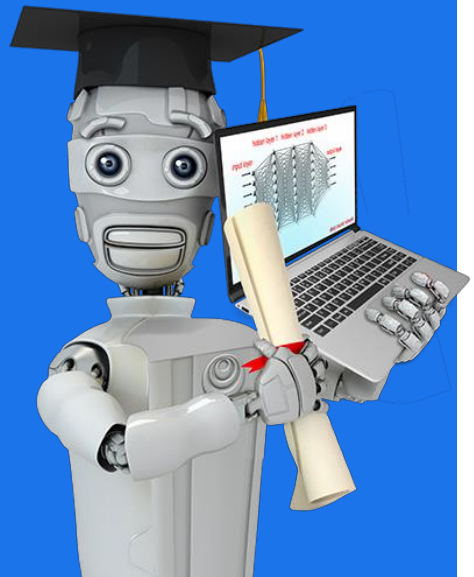


Practical Tips for Linear Regression

Feature Engineering

Feature engineering

$$f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x_2 + b$$

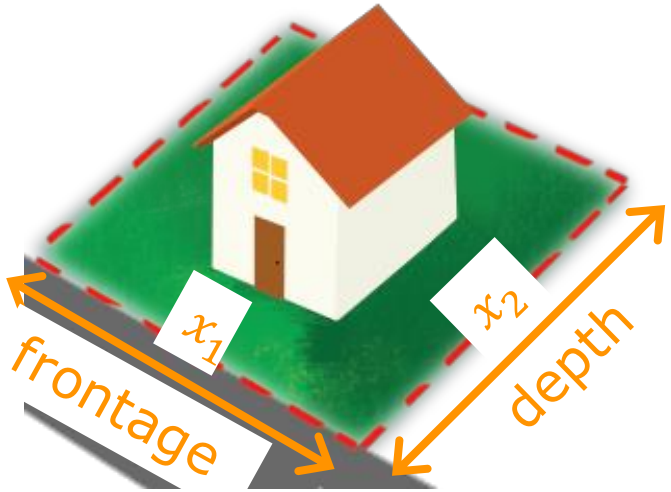
frontage depth

$$area = frontage \times depth$$

$$x_3 = x_1 x_2$$

new feature

$$f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$



Feature engineering:
Using intuition to design new features, by transforming or combining original features.

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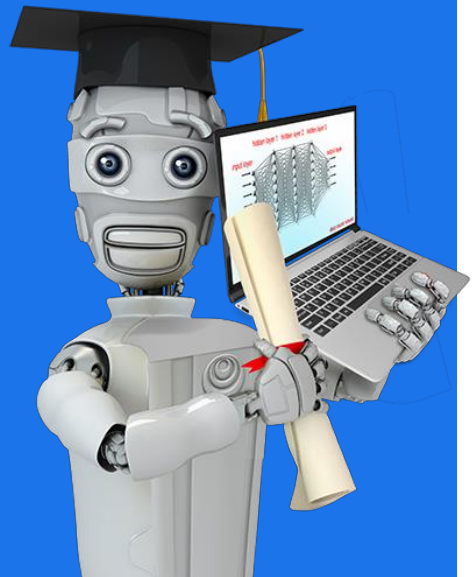
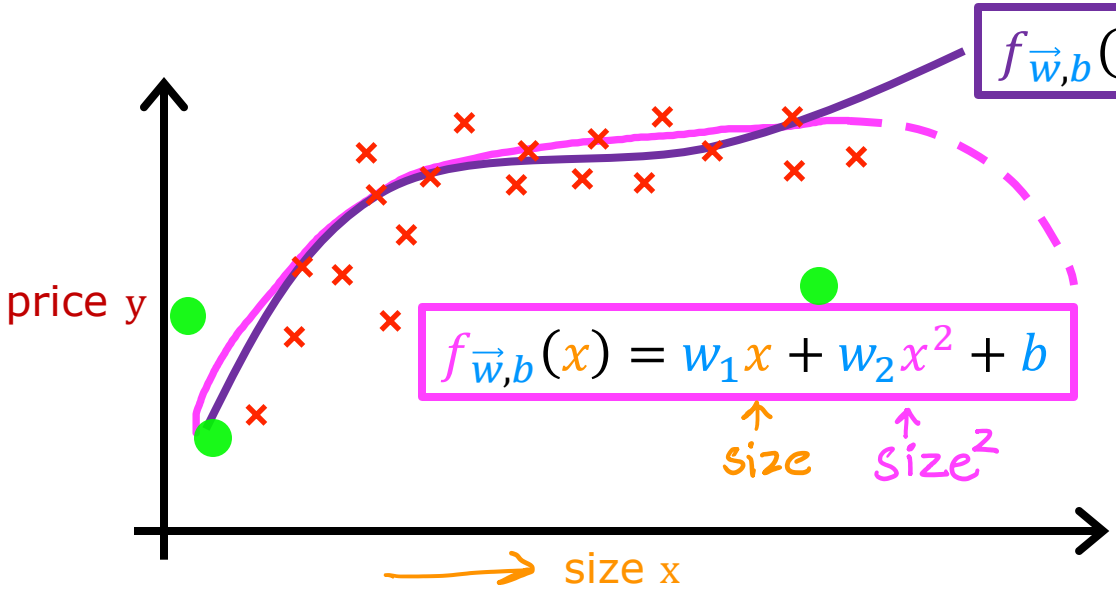


Practical Tips for Linear Regression

Polynomial Regression

Polynomial regression



$$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + b$$

size x $\quad$ x^2

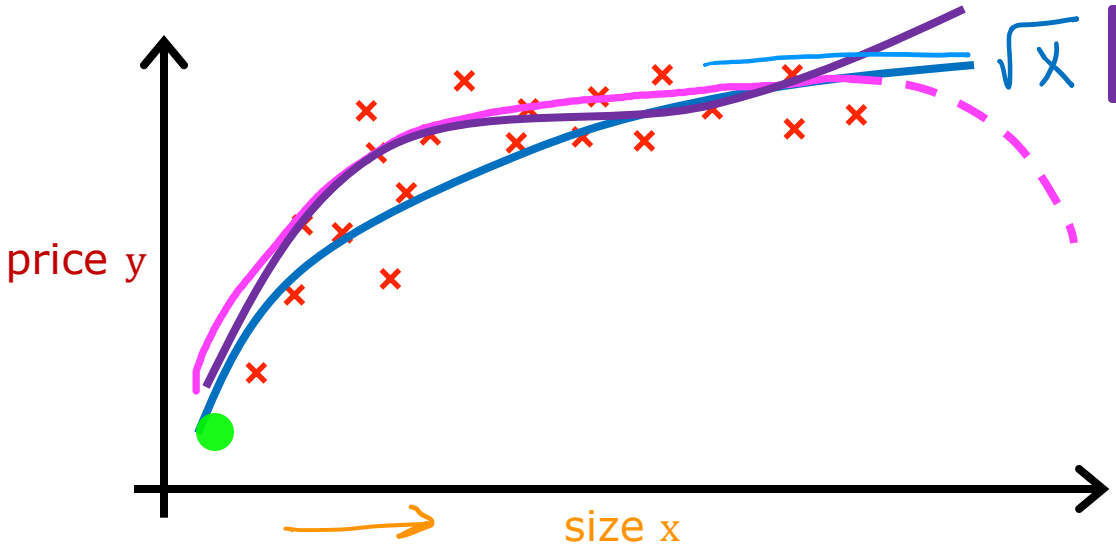
$$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + w_3 x^3 + b$$

size x $\quad$ x^2 $\quad$ x^3

size $1 - 10^3$ $\quad$ $1 - 10^6$ $\quad$ $1 - 10^9$

feature scaling

Choice of features



$$f_{\vec{w},b}(x) = w_1 x + w_2 \sqrt{x} + b$$

size

√size

what features to use?

↳ course 2

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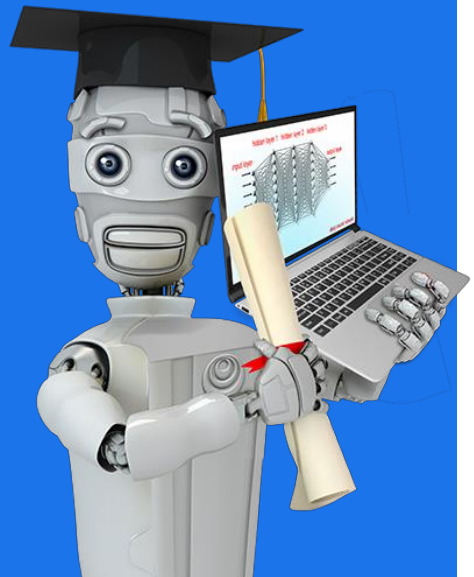


Classification

Motivations

Classification

Question	Answer "y"	
Is this email spam?	no	yes
Is the transaction fraudulent?	no	yes
Is the tumor malignant?	no	yes

y can only be one of two values

"binary classification"

class = category

false true

0 1

useful for classification

"negative class"

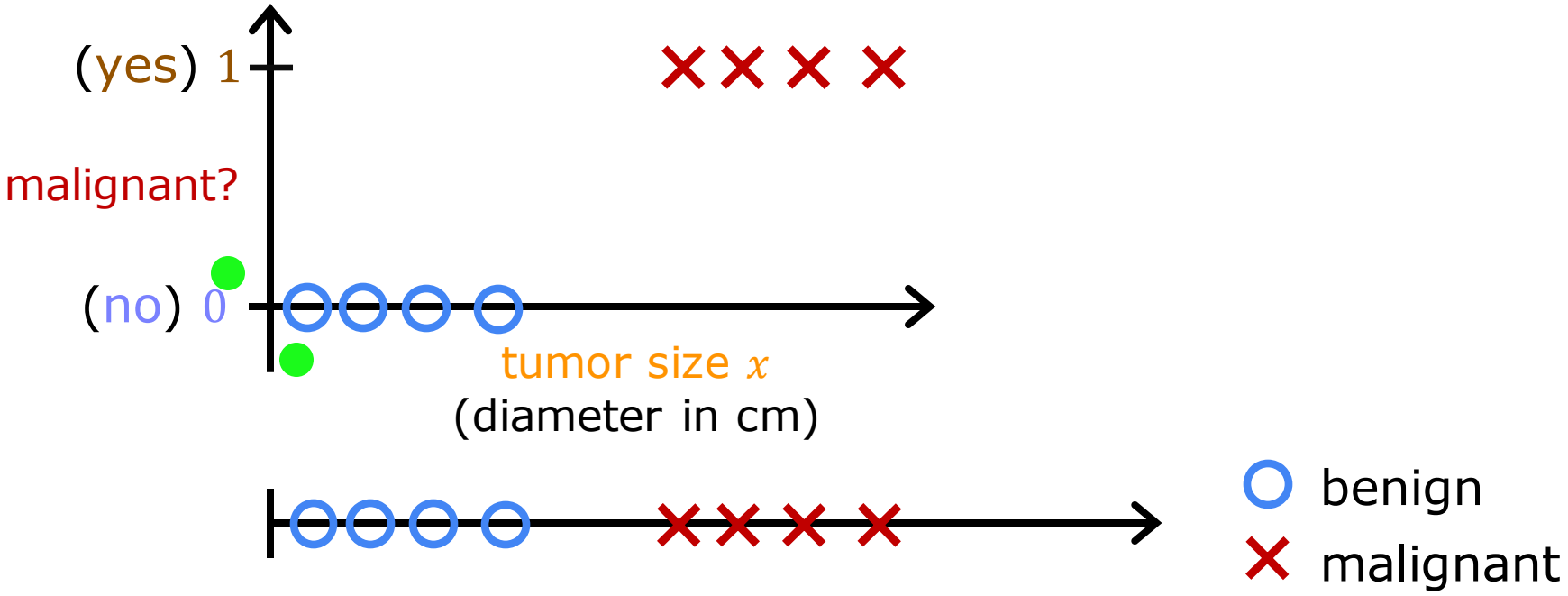
≠ "bad"

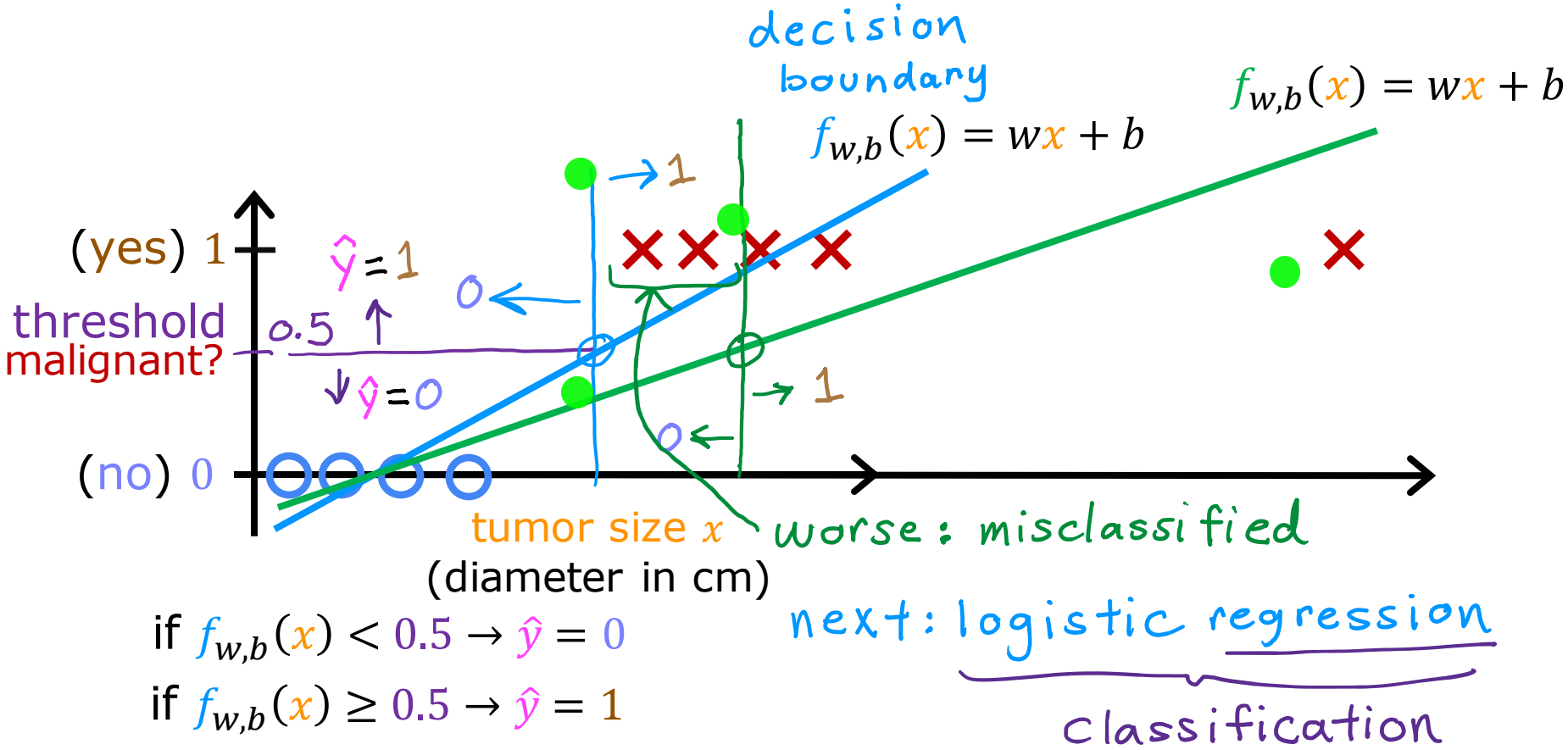
absence

"positive class"

≠ "good"

presence


(yes) 1
malignant?
(no) 0
tumor size x
(diameter in cm)
benign
malignant



if $f_{w,b}(x) < 0.5 \rightarrow \hat{y} = 0$

if $f_{w,b}(x) \geq 0.5 \rightarrow \hat{y} = 1$

next: logistic regression

classification

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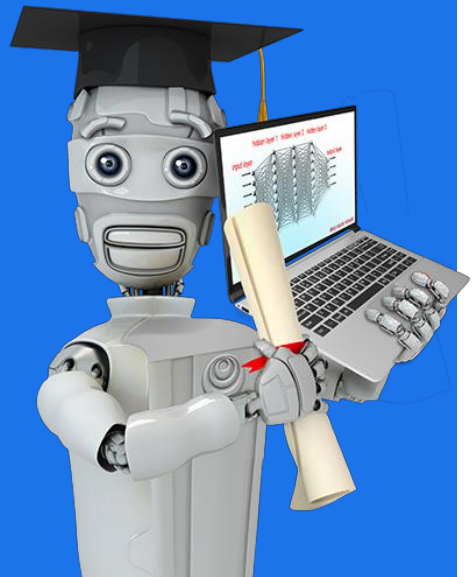
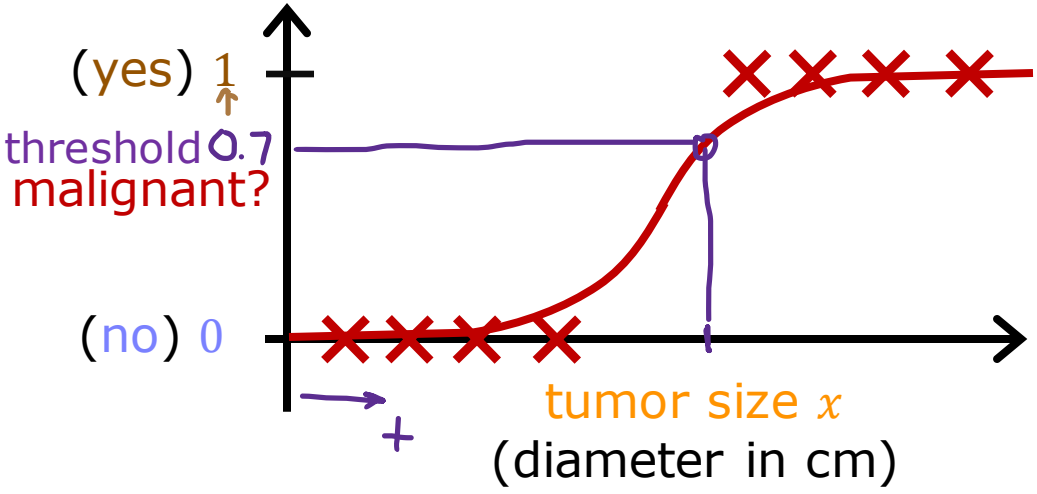
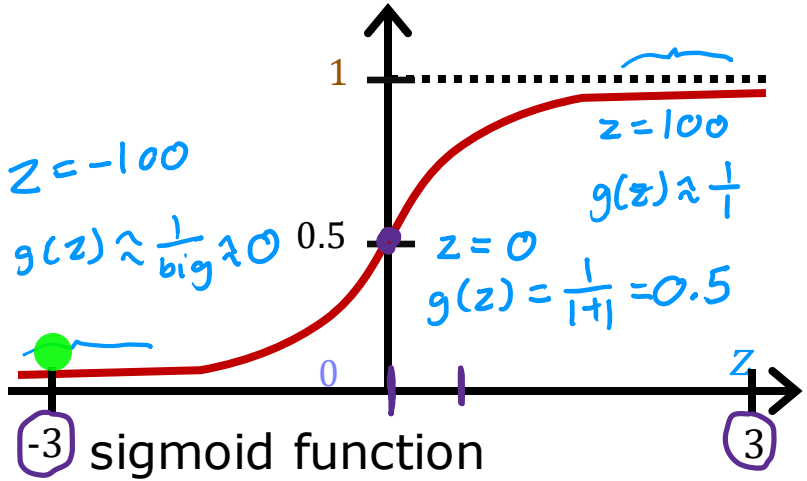


Classification

Logistic Regression



Want outputs between 0 and 1



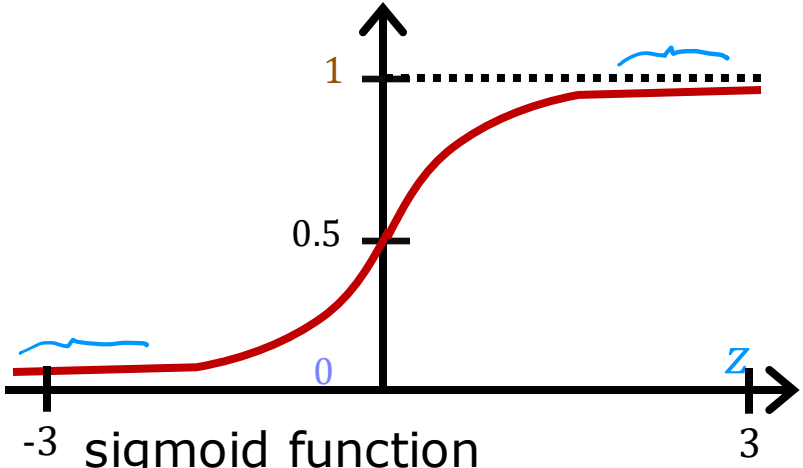
sigmoid function

logistic function

outputs between 0 and 1

$$g(z) = \frac{1}{1+e^{-z}} \qquad 0 < g(z) < 1$$

Want outputs between 0 and 1



sigmoid function

logistic function

outputs between 0 and 1

$$g(z) = \frac{1}{1+e^{-z}} \qquad 0 < g(z) < 1$$

$f_{\vec{w},b}(\vec{x})$

$\vec{w} \cdot \vec{x} + b$

z

$g(z) = \frac{1}{1+e^{-z}}$

$$f_{\vec{w},b}(\vec{x}) = g(\underbrace{\vec{w} \cdot \vec{x} + b}_{z}) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$$

"logistic regression"

$e \approx 2.7$

Interpretation of logistic regression output

$$f_{\vec{w},b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w}\cdot\vec{x}+b)}}$$

"probability" that class is 1

$$f_{\vec{w},b}(\vec{x}) = P(y = 1|\vec{x}; \vec{w}, b)$$

Probability that y is 1, given input $\vec{x}$, parameters $\vec{w}, b$

Example:

x is "tumor size"

y is 0 (not malignant) or 1 (malignant)

$$P(y = 0) + P(y = 1) = 1$$

$f_{\vec{w},b}(\vec{x}) = 0.7$

70% chance that y is 1

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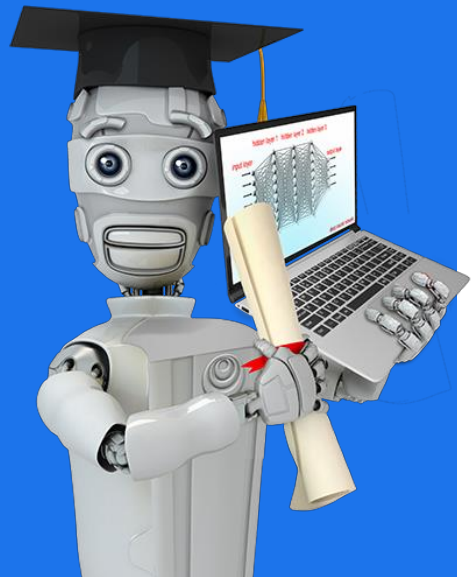
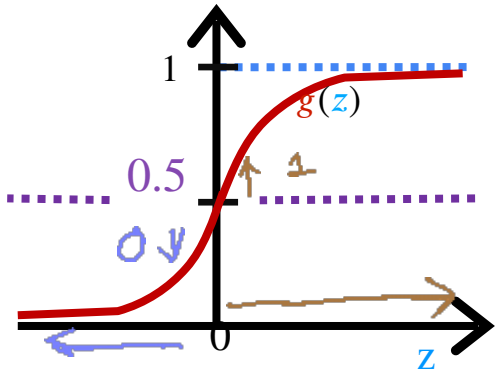


Classification

Decision Boundary



$$f_{\vec{w},b}(\vec{x}) = g(\vec{w} \cdot \vec{x} + b) \qquad z = \vec{w} \cdot \vec{x} + b$$

$$g(z) = \frac{1}{1+e^{-z}}$$

$$= \frac{1}{1+e^{-\left(\vec{w} \cdot \vec{x} + b\right)}}$$

$$= P(y = 1 \mid x; \vec{w}, b)$$

0.7 0.3

0 or 1?

threshold

Is $f_{\vec{w},b}(\vec{x}) \geq 0.5$?

Yes: $\hat{y} = 1$ No: $\hat{y} = 0$

When is $f_{\vec{w},b}(\vec{x}) \geq 0.5$?

$g(z) \geq 0.5$

$z \geq 0$

$\vec{w} \cdot \vec{x} + b \geq 0$

$\hat{y} = 1$

$z < 0$

$\vec{w} \cdot \vec{x} + b < 0$

$\hat{y} = 0$

Decision boundary

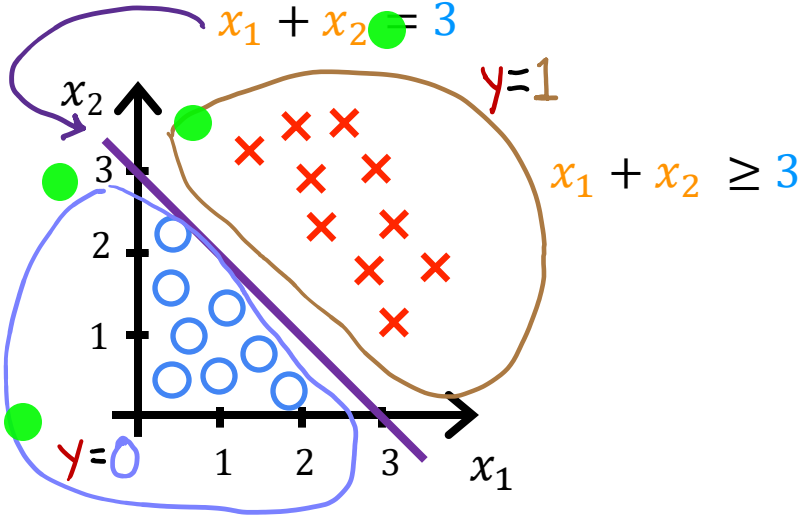
$$f_{\vec{w},b}(\vec{x}) = g(z) = g(w_1 x_1 + w_2 x_2 + b)$$

$w_1 = 1$, $w_2 = 1$, $b = -3$

Decision boundary $z = \vec{w} \cdot \vec{x} + b = 0$

$z = x_1 + x_2 - 3 = 0$

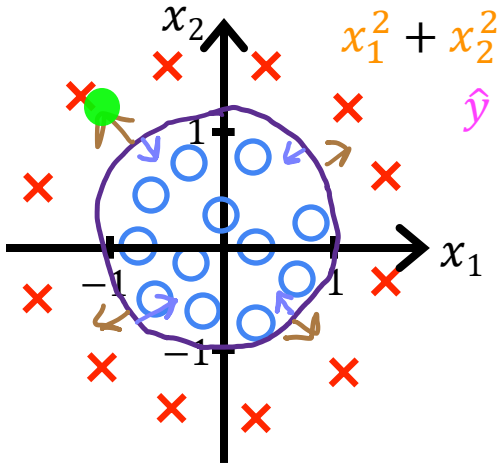
$x_1 + x_2 = 3$



$x_1 + x_2 \geq 3$

$x_1 + x_2 < 3$

Non-linear decision boundaries



$x_1^2 + x_2^2 \geq 1$

$\hat{y} = 1$

$x_1^2 + x_2^2 < 1$

$\hat{y} = 0$

z

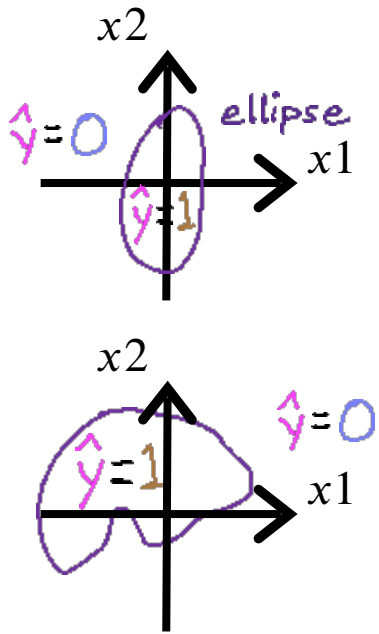
$w_1 x_1^2 + w_2 x_2^2 + b$

1 1 -1

decision boundary $z = x_1^2 + x_2^2 - 1 = 0$

$x_1^2 + x_2^2 = 1$

Non-linear decision boundaries



$$f_{\vec{w},b}\left(\vec{x}\right) = g(z) = g(w_1 x_1 + w_2 x_2 + w_3 x_1^2 + w_4 x_1 x_2 + w_5 x_2^2 + w_6 x_1^3 + \cdots + b)$$

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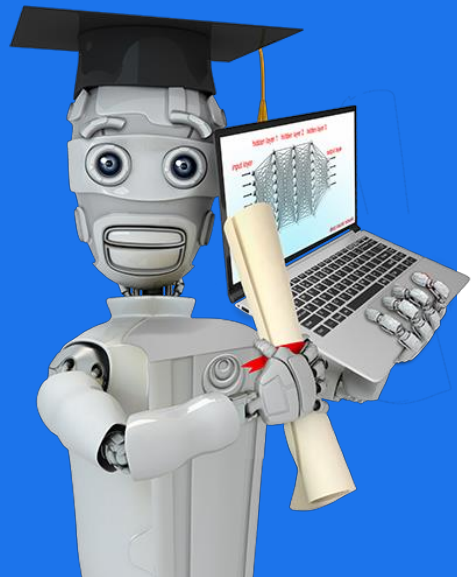


Cost Function

Cost Function for Logistic Regression

Training set

	tumor size (cm) x_1	…	patient's age x_n	malignant? y
$i=1$	10		52	1
	2		73	0
	5		55	0
	12		49	1
$i=m$	…		…	…

$i = 1, \ldots, m$ ← training examples

$j = 1, \ldots, n$ ← features

target y is 0 or 1

$$f_{\vec{w},b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$$

How to choose $\vec{w} = [w_1 \quad w_2 \quad \cdots \quad w_n]$ and b?

Squared error cost

cost

$$J(\vec{w}, b) = \frac{1}{m}\sum_{i=1}^{m}\frac{1}{2}\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right) - y^{(i)}\right)^2$$

average of training set

loss

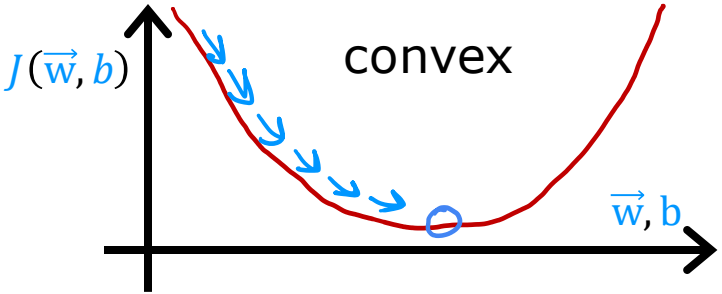
$$L\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right), y^{(i)}\right)$$

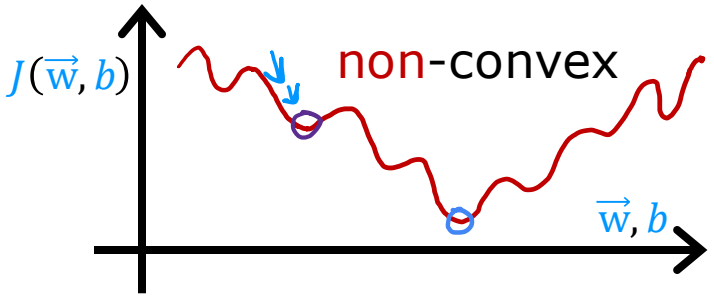
linear regression

$$f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$$

logistic regression

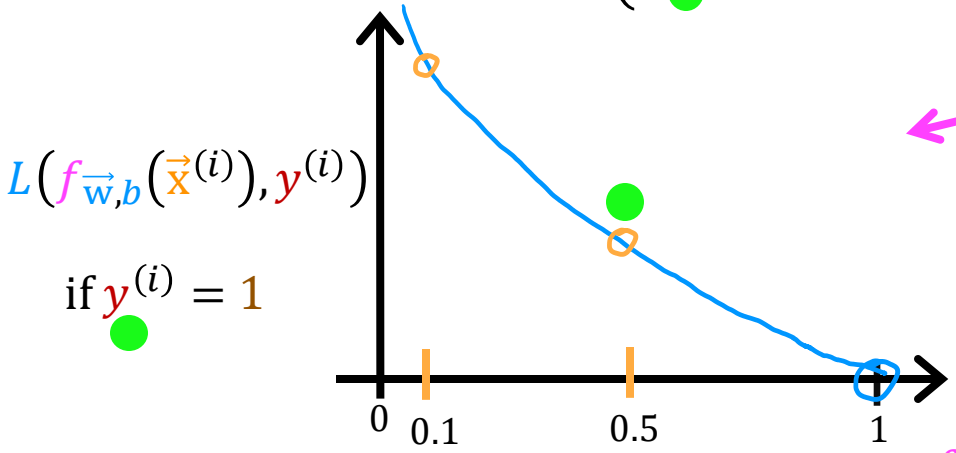
$$f_{\vec{w},b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w}\cdot\vec{x}+b)}}$$

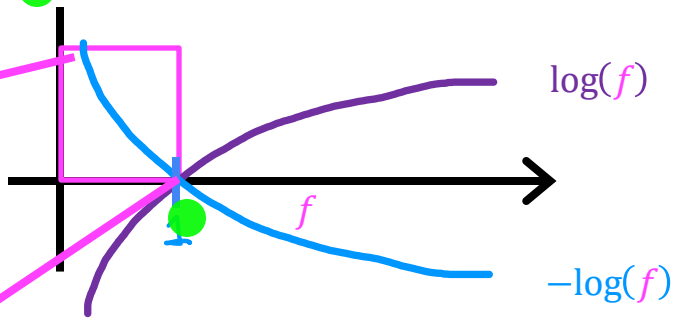




Logistic loss function

$$L\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right), y^{(i)}\right) = \begin{cases} -\log\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) & \text{if } y^{(i)} = 1 \\ -\log\left(1 - f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) & \text{if } y^{(i)} = 0 \end{cases}$$





As $f_{\vec{w},b}\left(\vec{x}^{(i)}\right) \to 1$ then loss $\to 0$

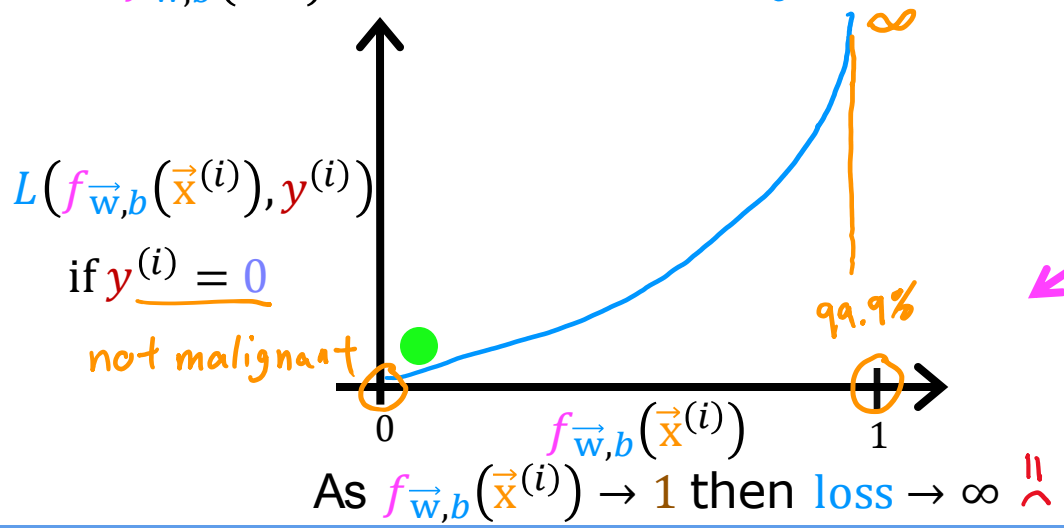
As $f_{\vec{w},b}\left(\vec{x}^{(i)}\right) \to 0$ then loss $\to \infty$

Loss is lowest when $f_{\vec{w},b}\left(\vec{x}^{(i)}\right)$ predicts close to true label $y^{(i)}$.

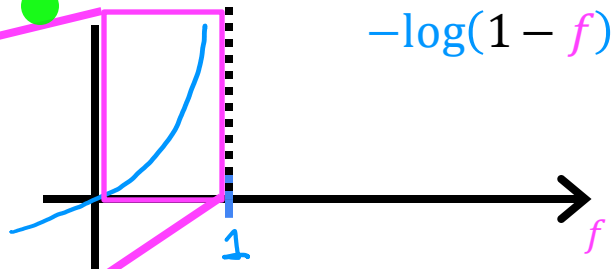
Logistic loss function

$$L\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right), y^{(i)}\right) = \begin{cases} -\log\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) & \text{if } y^{(i)} = 1 \\ -\log\left(1 - f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) & \text{if } y^{(i)} = 0 \end{cases}$$

As $f_{\vec{w},b}\left(\vec{x}^{(i)}\right) \to 0$ then loss $\to 0$



As $f_{\vec{w},b}\left(\vec{x}^{(i)}\right) \to 1$ then loss $\to \infty$



The further prediction $f_{\vec{w},b}\left(\vec{x}^{(i)}\right)$ is from target $y^{(i)}$, the higher the loss.

Cost

cost

$$J(\vec{w}, b) = \frac{1}{m}\sum_{i=1}^{m} L\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right), y^{(i)}\right)$$

loss

$$= \begin{cases} -\log\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) & \text{if } y^{(i)} = 1 \\ -\log\left(1 - f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) & \text{if } y^{(i)} = 0 \end{cases}$$

convex
↳ can reach a global minimum

find w, b that minimize cost J

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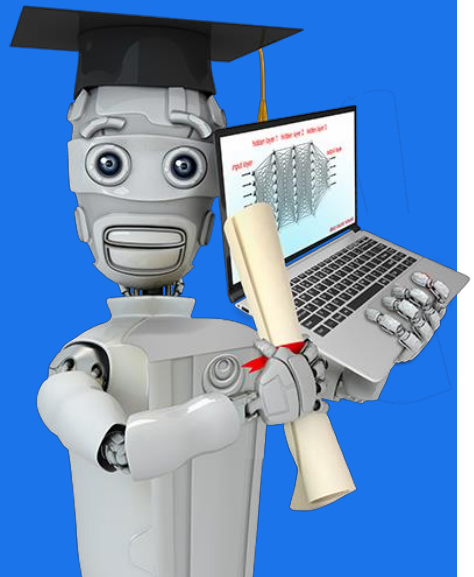

Cost Function

Simplified Cost Function for Logistic Regression

Simplified loss function

$$L\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right), y^{(i)}\right) = \begin{cases} -\log\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) & \text{if } y^{(i)} = 1 \\ -\log\left(1 - f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) & \text{if } y^{(i)} = 0 \end{cases}$$

$$L\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right), y^{(i)}\right) = -y^{(i)}\log\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) - \left(1 - y^{(i)}\right)\log\left(1 - f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right)$$

1

$(1-1)$

0

if $y^{(i)} = 1$:

$$L\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right), y^{(i)}\right) = -1\log\left(f(\vec{x})\right)$$

Simplified loss function

$$L\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right), y^{(i)}\right) = \begin{cases} -\log\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) & \text{if } y^{(i)} = 1 \\ -\log\left(1 - f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) & \text{if } y^{(i)} = 0 \end{cases}$$

$$L\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right), y^{(i)}\right) = -y^{(i)}\log\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) - \left(1 - y^{(i)}\right)\log\left(1 - f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right)$$

$y^{(i)}$ → 0 ; $(1 - y^{(i)})$ → $(1 - 0)$

if $y^{(i)} = 1$:

$$L\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right), y^{(i)}\right) = -1 \log\left(f(\vec{x})\right)$$

if $y^{(i)} = 0$:

$$L\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right), y^{(i)}\right) = -(1-0)\log\left(1 - f(\vec{x})\right)$$

Simplified cost function

loss

$$L\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right), y^{(i)}\right) = -y^{(i)}\log\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) - \left(1 - y^{(i)}\right)\log\left(1 - f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right)$$

convex (single global minimum)

cost

$$J(\vec{w}, b) = \frac{1}{m}\sum_{i=1}^{m}\left[L\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right), y^{(i)}\right)\right]$$

$$= -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) + \left(1 - y^{(i)}\right)\log\left(1 - f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right)\right]$$

maximum likelihood (don't worry about it!)

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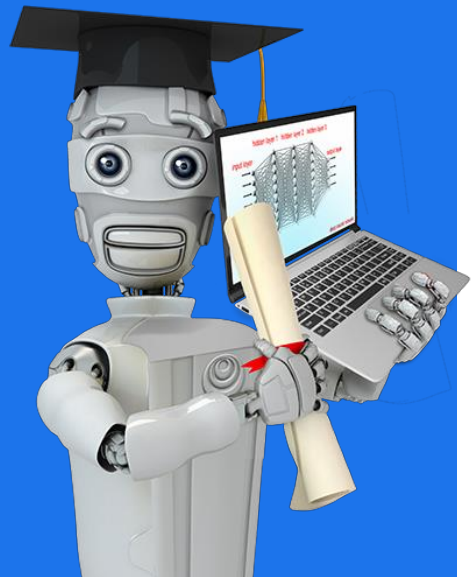

Gradient Descent

Gradient Descent Implementation

Training logistic regression

Find $\vec{w}, b$

Given new $\vec{x}$, output $f_{\vec{w},b}(\vec{x}) = \frac{1}{1+e^{-(\vec{w}\cdot\vec{x}+b)}}$

$$P(y = 1|\vec{x}; \vec{w}, b)$$

Gradient descent

cost

$$J(\vec{w}, b) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(f_{\vec{w},b}(\vec{x}^{(i)})\right) + (1-y^{(i)})\log\left(1-f_{\vec{w},b}(\vec{x}^{(i)})\right)\right]$$

repeat {

$j = 1 \ldots n$

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b)$$

} simultaneous updates

$$\frac{\partial}{\partial w_j} J(\vec{w}, b) = \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)x_j^{(i)}$$

$$\frac{\partial}{\partial b} J(\vec{w}, b) = \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)$$

Gradient descent for logistic regression

repeat {

looks like linear regression!

$$w_j = w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^{m} (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}) x_j^{(i)} \right]$$

$$b = b - \alpha \left[\frac{1}{m} \sum_{i=1}^{m} (f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}) \right]$$

} simultaneous updates

Same concepts:

- Monitor gradient descent (learning curve)
- Vectorized implementation
- Feature scaling

Linear regression $f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$

Logistic regression $f_{\vec{w},b}(\vec{x}) = \frac{1}{1 + e^{(-\vec{w} \cdot \vec{x} + b)}}$

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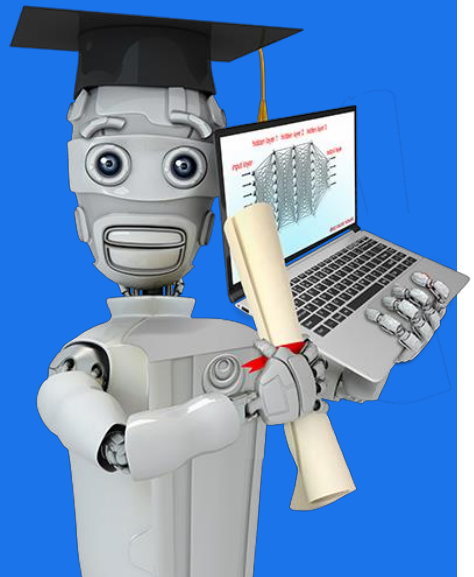
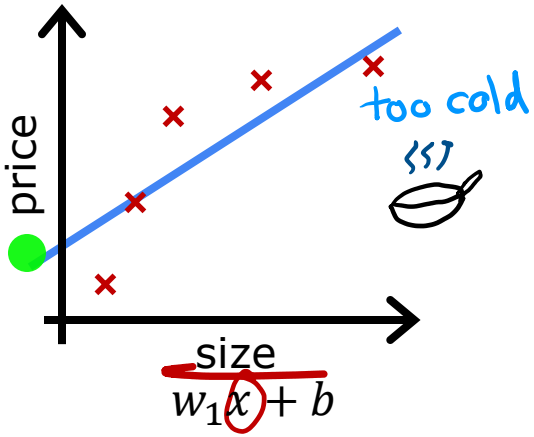


Regularization to Reduce Overfitting

The Problem of Overfitting

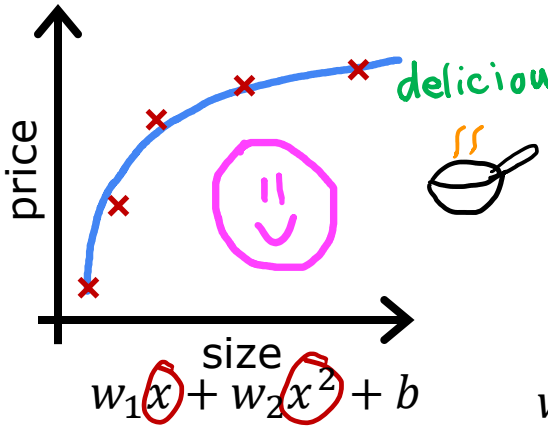
Regression example



underfit

- Does not fit the training set well

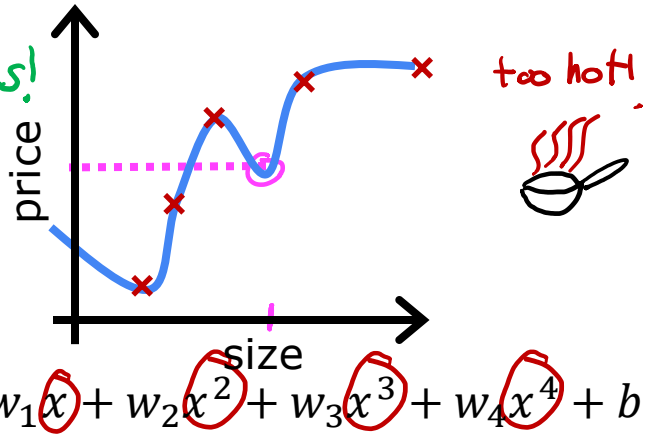
high bias



just right

- Fits training set pretty well

generalization

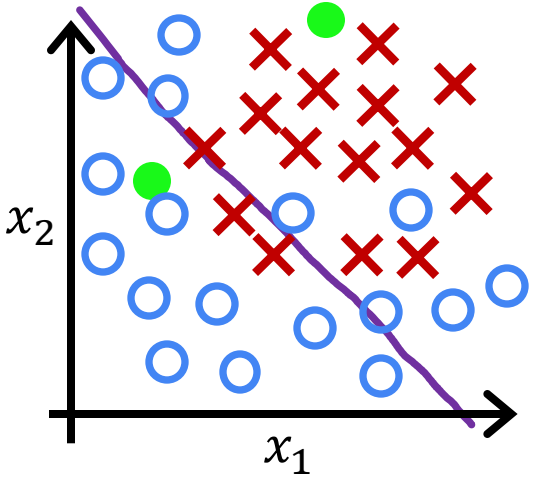


overfit

- Fits the training set extremely well

high variance

Classification

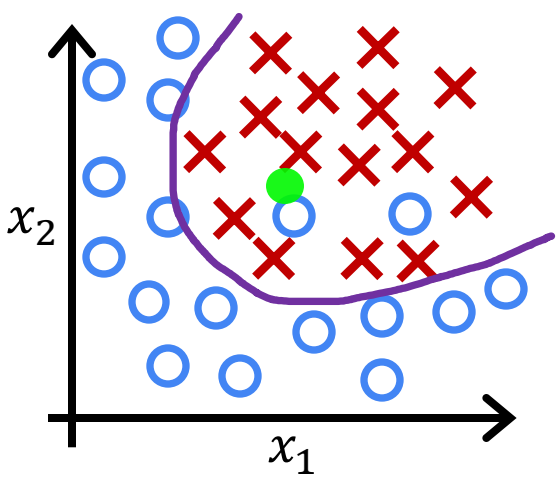


$$z = w_1 x_1 + w_2 x_2 + b$$

$$f_{\vec{w},b}(\vec{x}) = g(z)$$

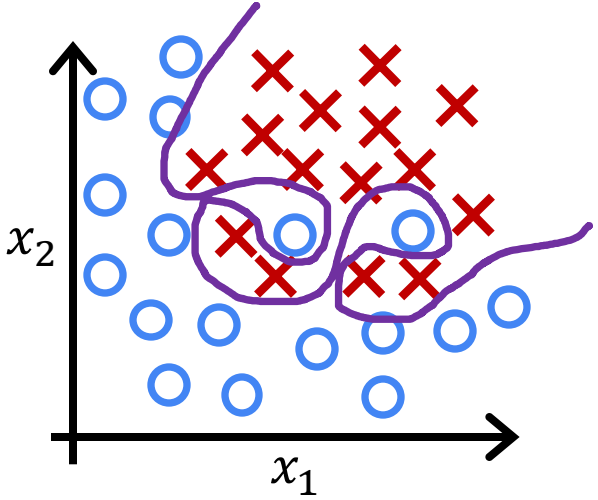
g is the sigmoid function

underfit high bias



$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 + w_4 x_2^2 + w_5 x_1 x_2 + b$$

just right



$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 x_2 + w_4 x_1^2 x_2^2 + w_5 x_1^2 x_2^3 + w_6 x_1^3 x_2 + \cdots + b$$

overfit

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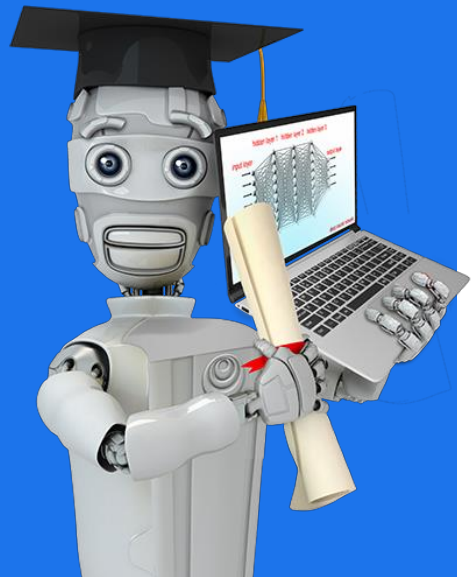
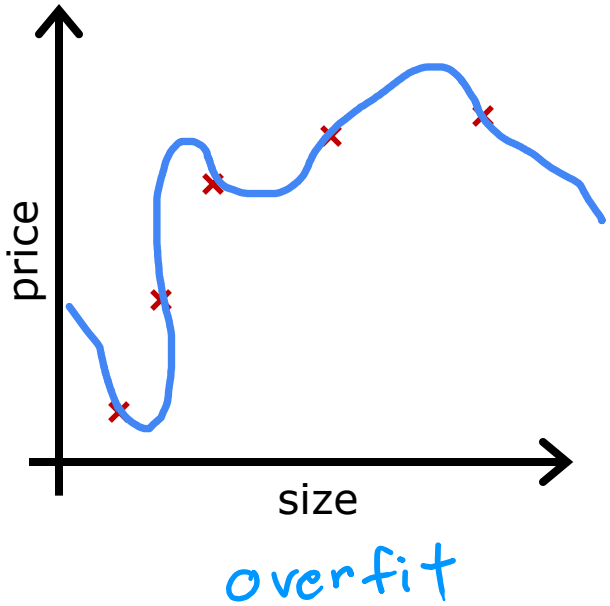
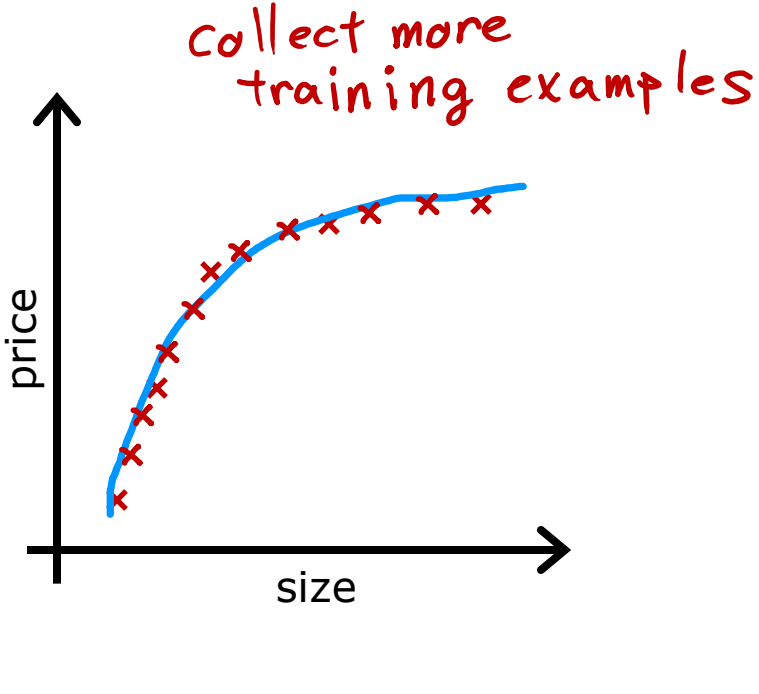


Regularization to Reduce Overfitting

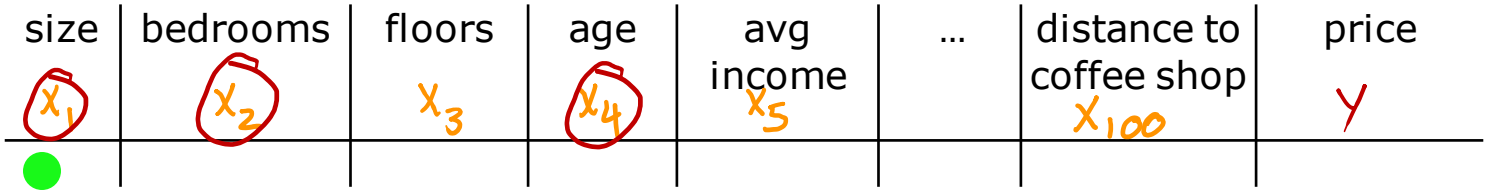
Addressing Overfitting

Collect more training examples





Select features to include/exclude



all features

\+

insufficient data

↓

overfit

selected features

size
bedrooms
age
just right

feature selection

course 2

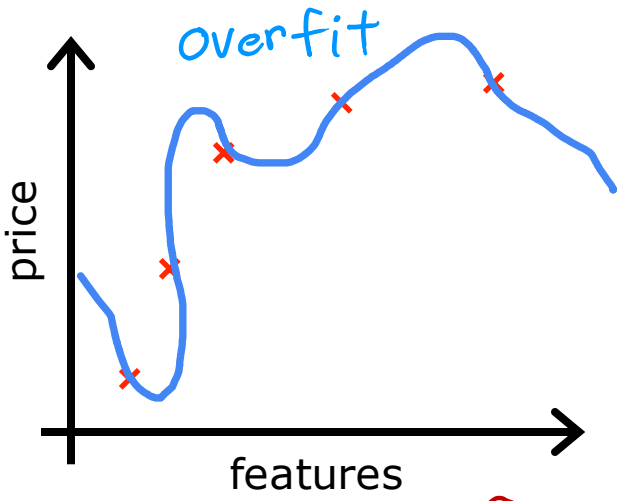
disadvantage

↓

useful features could be lost

Regularization

Reduce the size of parameters w_j

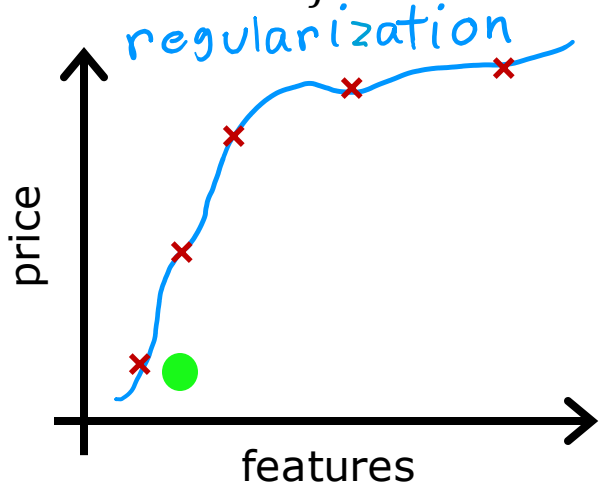


$$f(x) = 28x - 385x^2 + 39x^3 - 174x^4 + 100$$

0

large values for w_j

eliminate feature



$$f(x) = 13x - 0.23x^2 + 0.000014x^3 - 0.0001x^4 + 10$$

small values for w_j

Addressing overfitting

Options

1. Collect more data

2. Select features
   - Feature selection in course 2

3. Reduce size of parameters
   - "Regularization" next videos!

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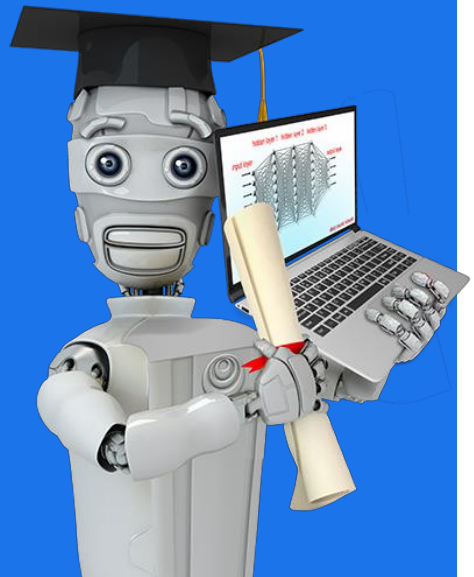
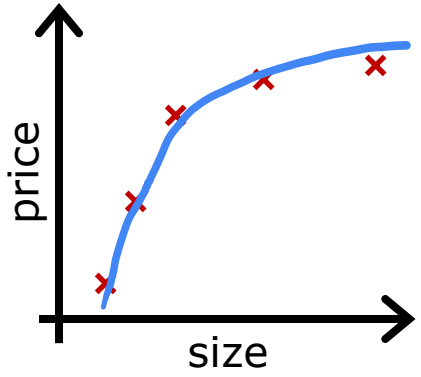


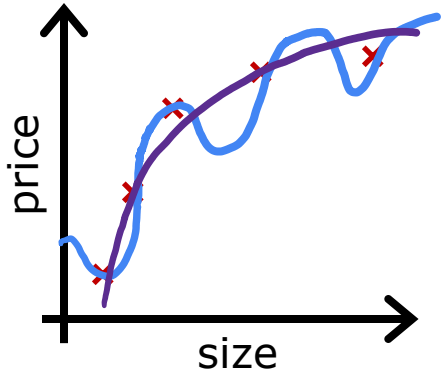
Regularization to Reduce Overfitting

Cost Function with Regularization

Intuition



$w_1 x + w_2 x^2 + b$



$w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b$

$w_3 \approx 0$ $w_4 \approx 0$

make w_3, w_4 really small (≈ 0)

$$\min_{\vec{w},b} \frac{1}{2m} \sum_{i=1}^{m} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)} \right)^2 + 1000 w_3^2 + 1000 w_4^2$$

$w_3 \approx 0.001$ $w_4 \approx 0.002$

Regularization

small values $w_1, w_2, \cdots, w_n, b$

simpler model

less likely to overfit

$w_3 \approx 0$

$w_4 \approx 0$

size	bedrooms	floors	age	avg income	...	distance to coffee shop	price
x_1	x_2	x_3	x_4	x_5		x_{100}	y

$n = 100$

n features

$w_1, w_1, w_2, \cdots, w_{100}, b$

regularization term

$$J(\vec{w}, b) = \frac{1}{2m}\left[\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2 + \frac{\lambda}{2m} b^2\right]$$

can include or exclude b

"lambda"

regularization parameter

$\lambda > 0$

Regularization

mean squared error

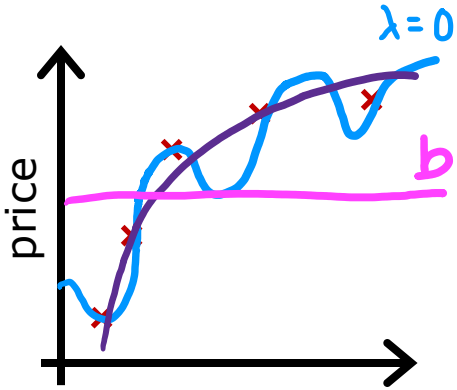
regularization term

$$\min_{\vec{w},b} J(\vec{w},b) = \min_{\vec{w},b} \left[\frac{1}{2m} \sum_{i=1}^{m} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)} \right)^2 + \frac{\lambda}{2m} \sum_{j=1}^{n} w_j^2 \right]$$

fit data

Keep w_j small

λ balances both goals



choose $\lambda = 10^{10}$

$$f_{\vec{w},b}(\vec{x}) = \underset{\approx 0}{w_1 x} + \underset{\approx 0}{w_2 x^2} + \underset{\approx 0}{w_3 x^3} + \underset{\approx 0}{w_4 x^4} + b$$

$f(x) = b$

Choose λ

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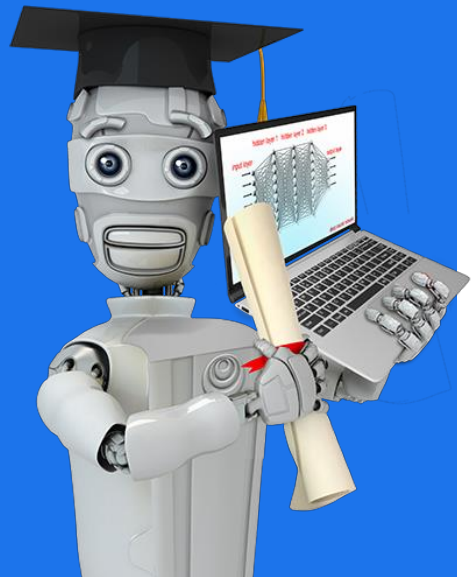

Regularization to Reduce Overfitting

Regularized Linear Regression

Regularized linear regression

$$\min_{\vec{w},b} J(\vec{w},b) = \min_{\vec{w},b}\left[\frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2\right]$$

Gradient descent

repeat {

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w},b) \qquad j = 1,\ldots,n$$

$$= \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right) x_j^{(i)} + \frac{\lambda}{m} w_j$$

$$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w},b)$$

$$= \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)$$

don't have to regularize b

} simultaneous update

Implementing gradient descent

repeat {

- $w_j = w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^{m} \left[\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)} \right) x_j^{(i)} \right] + \frac{\lambda}{m} w_j \right]$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)} \right)$$

} simultaneous update $j = 1 \ldots n$

Implementing gradient descent

repeat {

$$w_j = w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^{m} \left[\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)} \right) x_j^{(i)} \right] + \frac{\lambda}{m} w_j \right]$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)} \right)$$

} simultaneous update $j = 1 \ldots n$

$$w_j = \underbrace{1 w_j - \alpha \frac{\lambda}{m} w_j}_{w_j \left(1 - \alpha \frac{\lambda}{m}\right)} - \underbrace{\alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}(\vec{x}^{(i)}) - y^{(i)} \right) x_j^{(i)}}_{\text{usual update}}$$

shrink w_j

$\alpha \frac{\lambda}{m}$

$0.01 \frac{1}{50} = 0.0002$

$w_j (1 - 0.0002)$

0.9998

How we get the derivative term (optional)

$$\frac{\partial}{\partial w_j} J(\vec{w}, b) = \frac{d}{dw_j}\left[\frac{1}{2m}\sum_{i=1}^{m}\left(\underbrace{f(\vec{x}^{(i)})}_{\vec{w}\cdot\vec{x}^{(i)}+b} - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2\right]$$

$$= \frac{1}{2m}\sum_{i=1}^{m}\left[\left(\vec{w}\cdot\vec{x}^{(i)} + b - y^{(i)}\right) 2x_j^{(i)}\right] + \frac{\lambda}{2m} 2w_j \quad \text{No } \sum_{j=1}^{n}$$

$$= \frac{1}{m}\sum_{i=1}^{m}\left[\left(\underbrace{\vec{w}\cdot\vec{x}^{(i)} + b}_{f(\vec{x})} - y^{(i)}\right) x_j^{(i)}\right] + \frac{\lambda}{m} w_j$$

$$= \frac{1}{m}\sum_{i=1}^{m}\left[\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)x_j^{(i)}\right] + \frac{\lambda}{m} w_j$$

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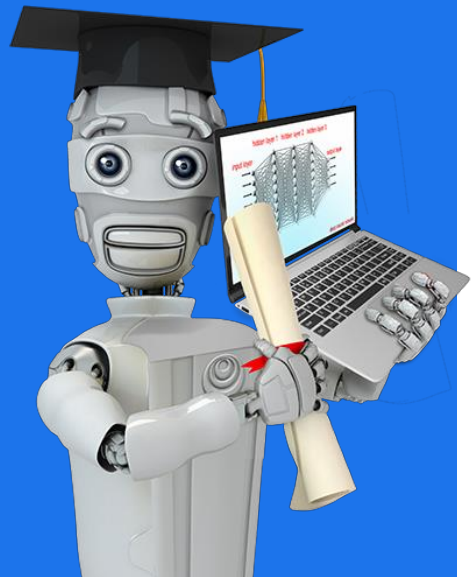
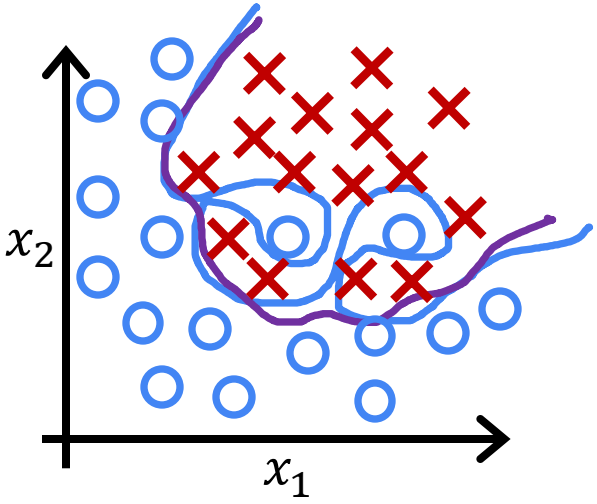


Regularization to Reduce Overfitting

Regularized Logistic Regression

Regularized logistic regression



$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 x_2 + w_4 x_1^2 x_2^2 + w_5 x_1^2 x_2^3 + \cdots + b$$

$$f_{\vec{w},b}(\vec{x}) = \frac{1}{1+e^{-z}}$$

Cost function

$$J(\vec{w},b) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(f_{\vec{w},b}(\vec{x}^{(i)})\right) + (1-y^{(i)})\log\left(1-f_{\vec{w},b}(\vec{x}^{(i)})\right)\right] + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$

$$\min_{\vec{w},b} J(\vec{w},b) \rightarrow w_j \downarrow$$

Regularized logistic regression

$$J(\vec{w}, b) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(f_{\vec{w},b}(\vec{x}^{(i)})\right) + (1-y^{(i)})\log\left(1 - f_{\vec{w},b}(\vec{x}^{(i)})\right)\right] + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$

$\min_{\vec{w},b}$

Gradient descent

repeat {

$$w_j = w_j - \alpha\frac{\partial}{\partial w_j}J(\vec{w}, b)$$

$j = 1 \ldots n$

$$b = b - \alpha\frac{\partial}{\partial b}J(\vec{w}, b)$$

}

Looks same as for linear regression!

$$= \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)x_j^{(i)} + \frac{\lambda}{m}w_j$$

logistic regression

$$= \frac{1}{m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)$$

don't have to regularize b

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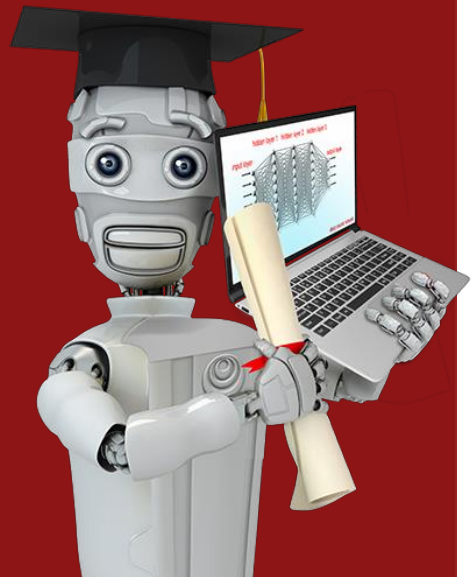

Advanced Learning Algorithms

Welcome!

Advanced learning algorithms

Neural Networks
  inference (prediction)
  training
Practical advice for building machine learning systems
Decision Trees

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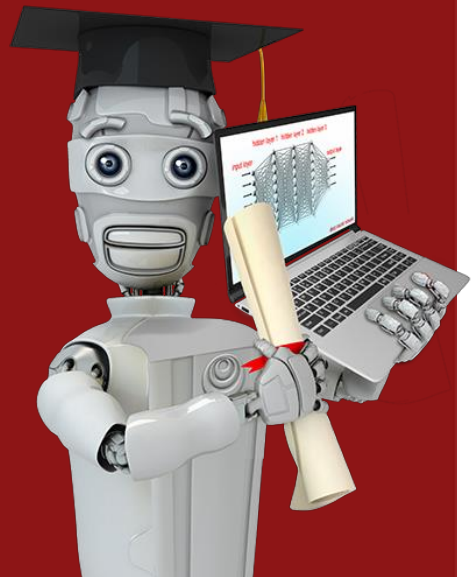


Neural Networks Intuition

Neurons and the brain

Neural networks

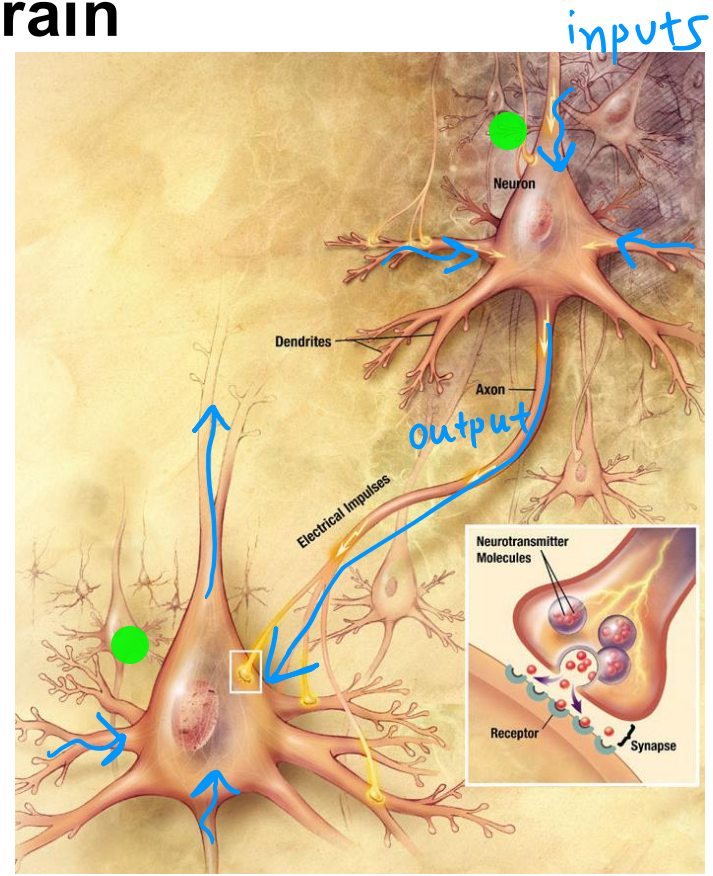
Origins: Algorithms that try to mimic the brain.

Used in the 1980's and early 1990's.
Fell out of favor in the late 1990's.

Resurgence from around 2005.

speech → images → text (NLP) → ...

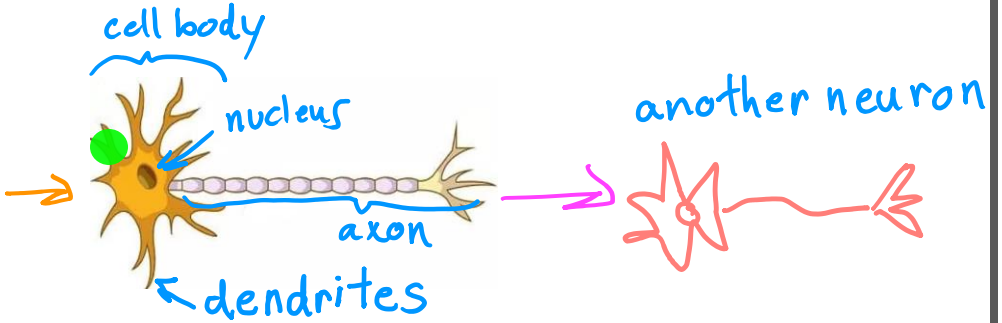
Neurons in the brain



[Credit: US National Institutes of Health, National Institute on Aging]

Biological neuron

inputs outputs



Simplified mathematical model of a neuron

inputs outputs

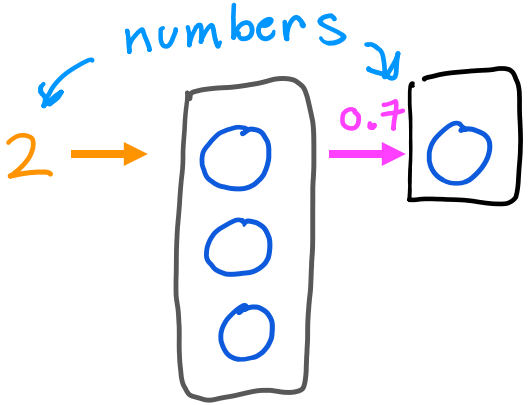
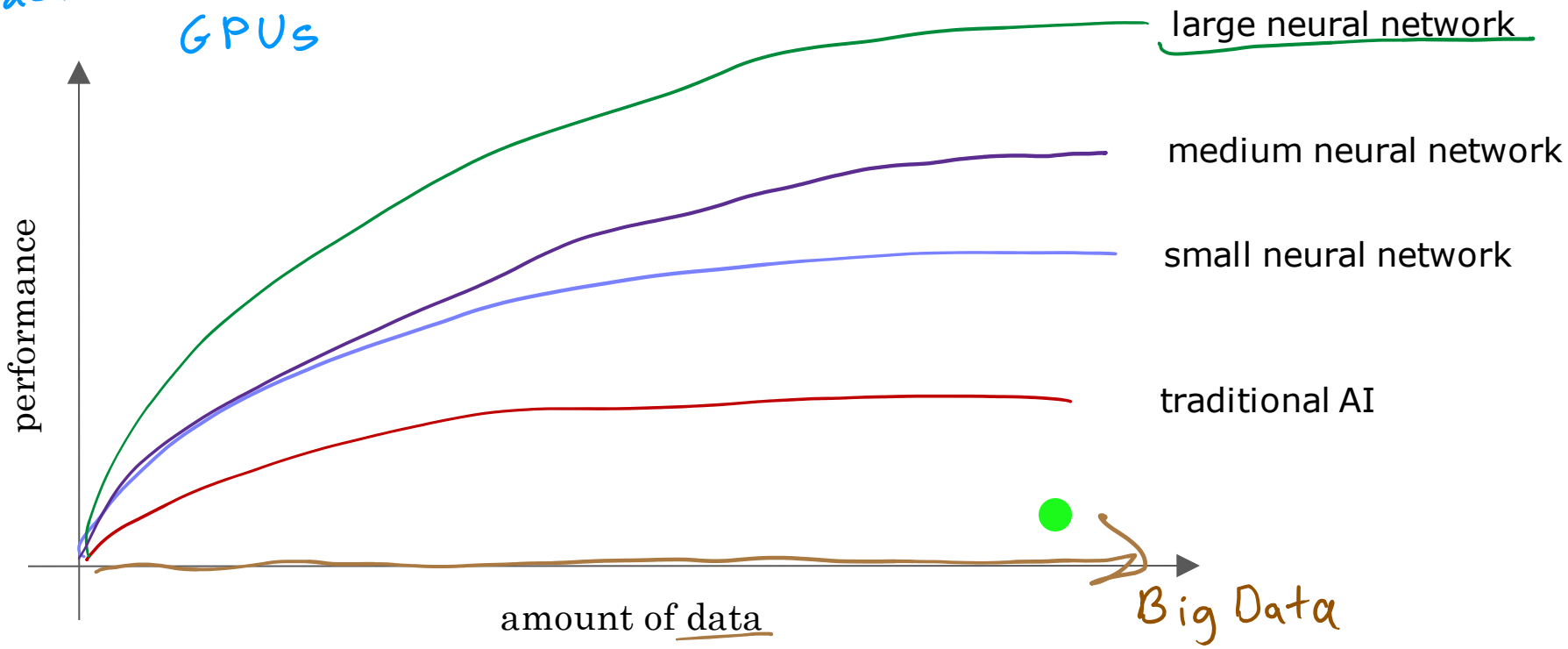


image source: https://biologydictionary.net/sensory-neuron/

Why Now?

Faster computer processors
GPUs



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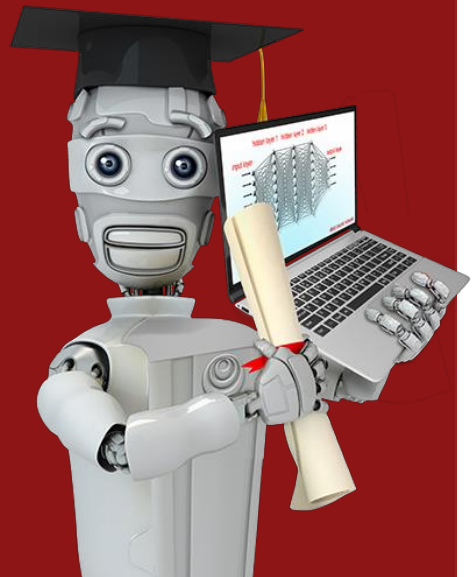
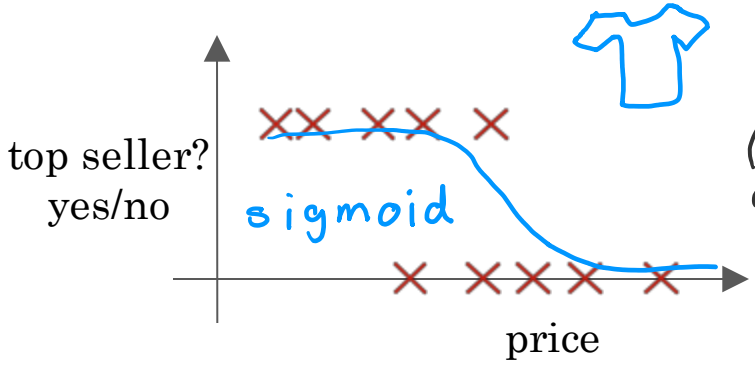


Neural Network Intuition

Demand Prediction

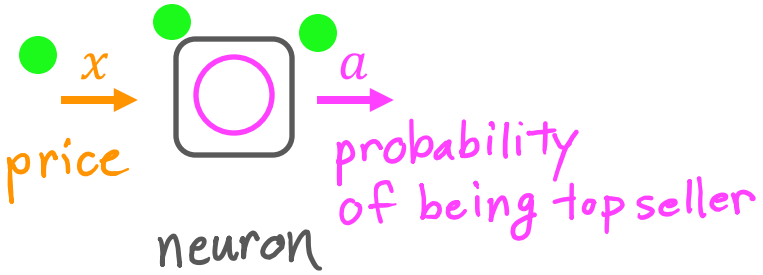
Demand Prediction



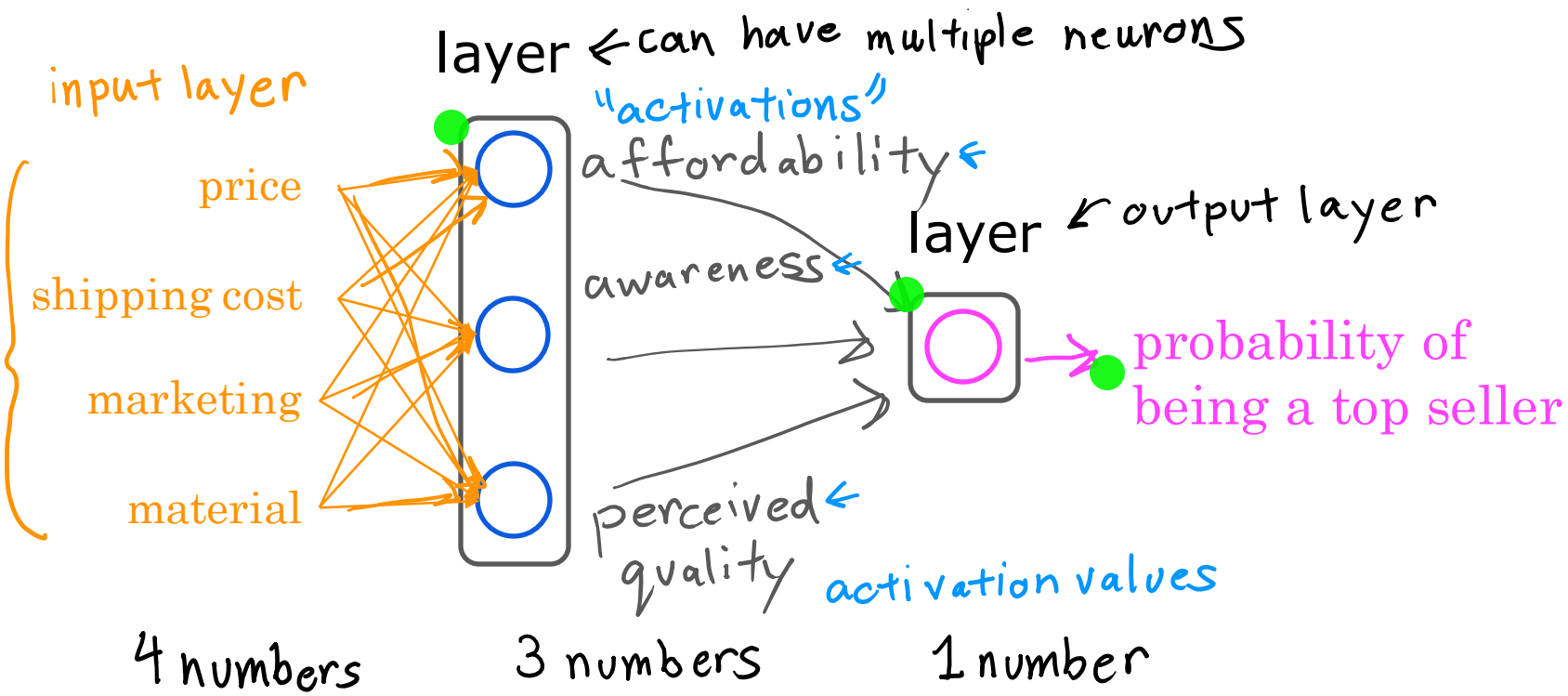
$x = price$ input

$a = f(x) = \frac{1}{1 + e^{-(wx+b)}}$ output

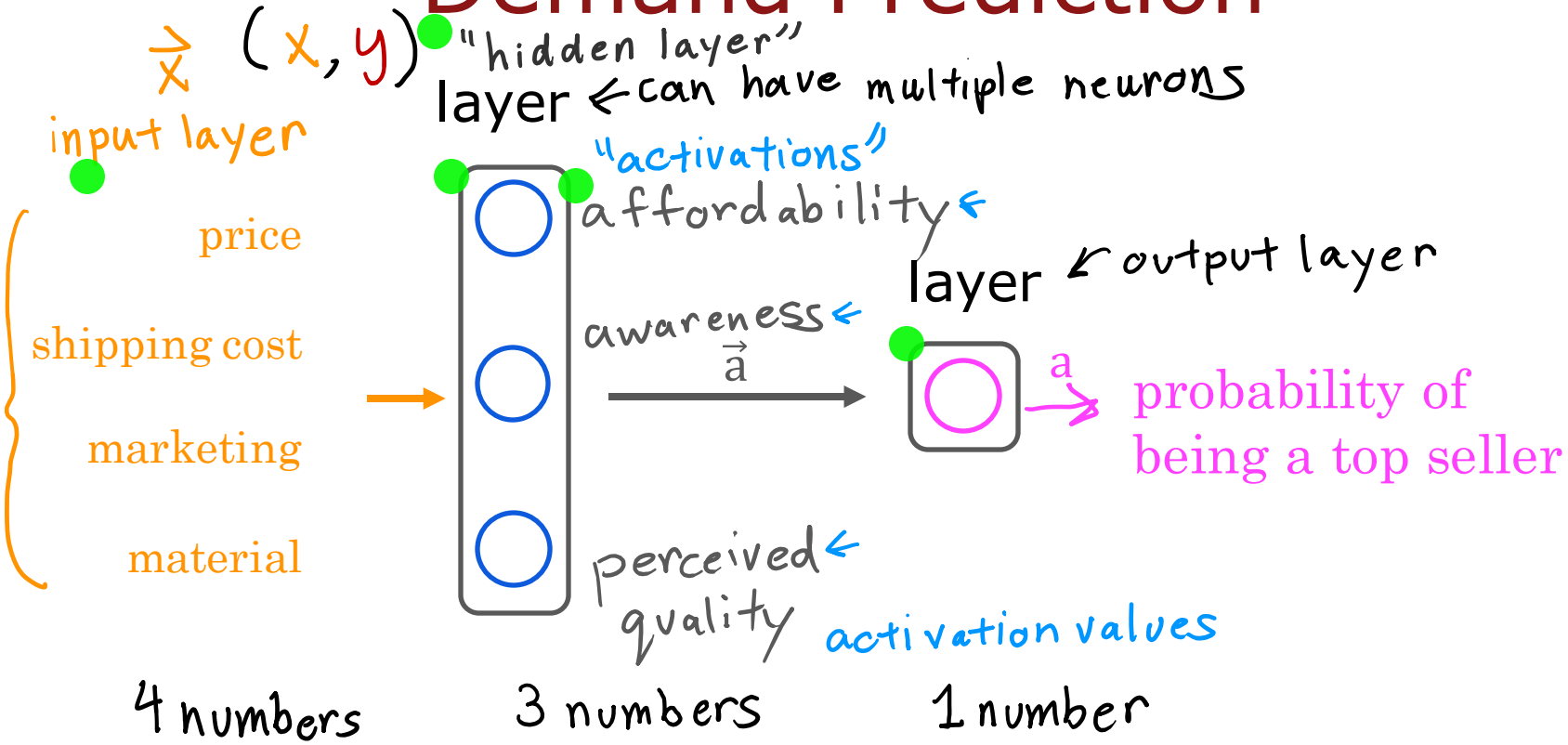
activation



Demand Prediction



Demand Prediction



Demand Prediction

feature engineering

x_1 x_2

$\vec{x}$ (x, y)

input layer

price

shipping cost

marketing

material

4 numbers

"hidden layer"

layer ← can have multiple neurons

"activations"

affordability ←

awareness ←

perceived quality ←

$\vec{a}$

layer ← output layer

a

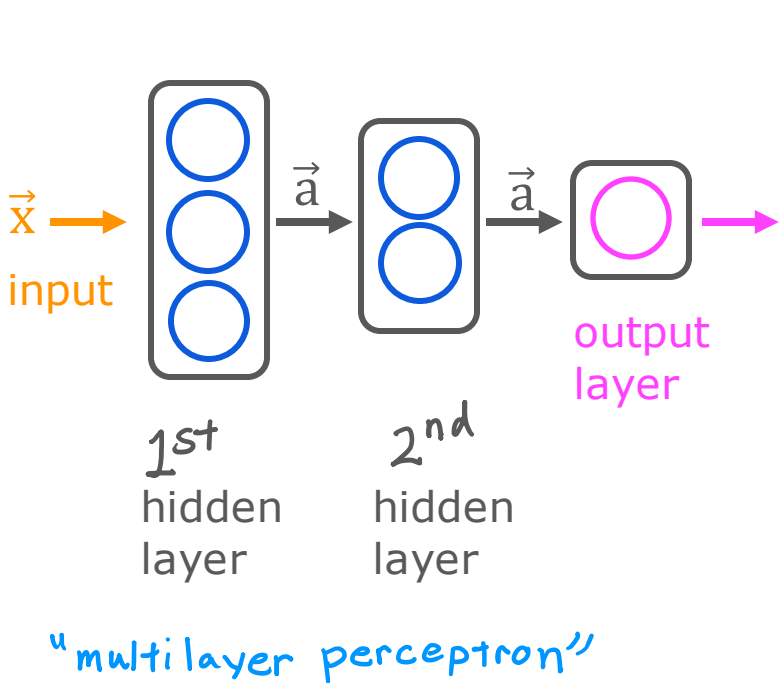
probability of being a top seller

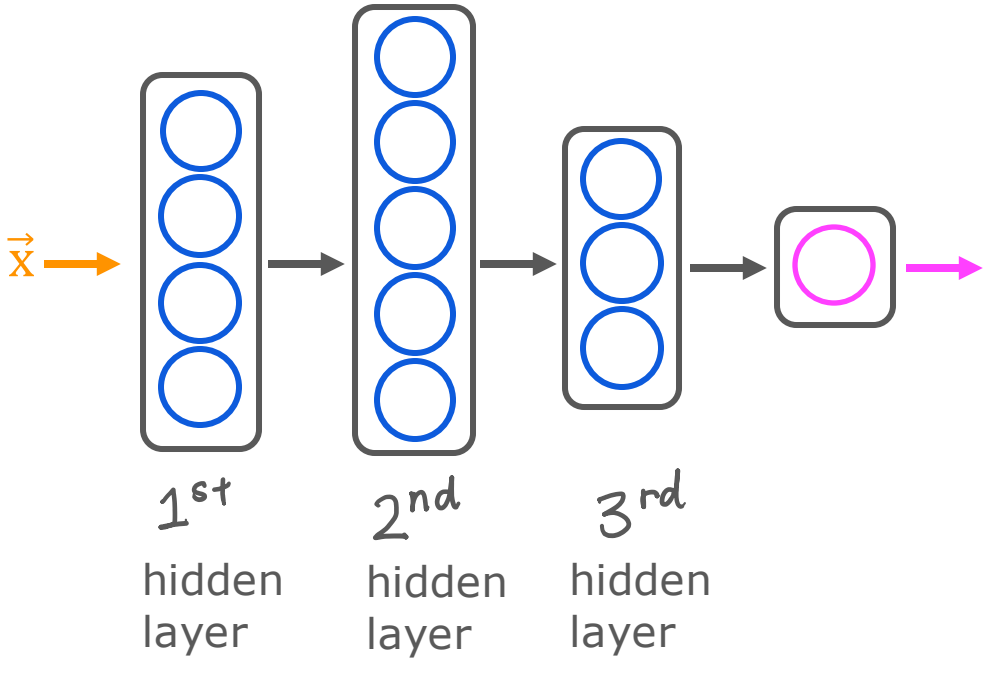
activation values

3 numbers

1 number

Multiple hidden layers





neural network architecture

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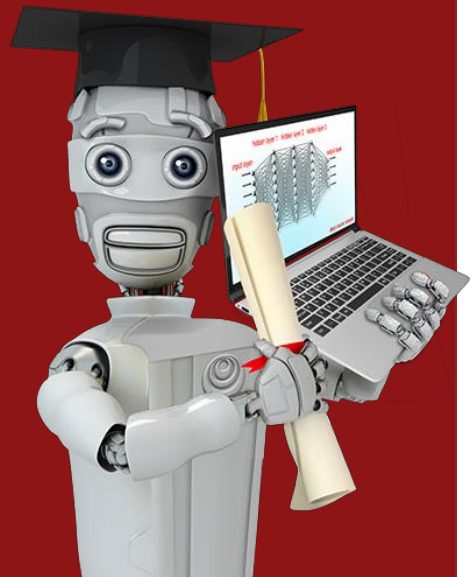


Neural Networks Intuition

Example:
Recognizing Images

Face recognition

1000 pixels

1000 pixels

1000 columns

1000 rows

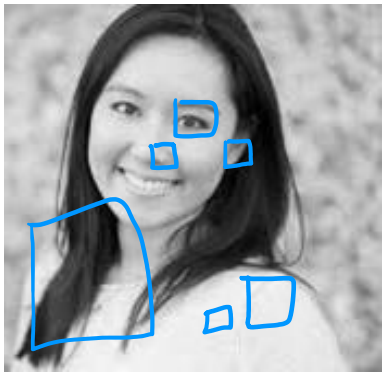
197	185	203	...	...
...	57	64	92	...
⋮				
		...	187	214

$\vec{x} =$

197
185
203
⋮
57
64
92
⋮
187
214

1 million

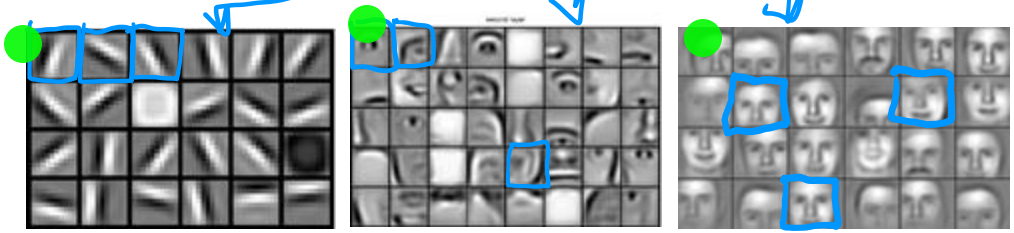
Face recognition



$\vec{x}$
input

output layer

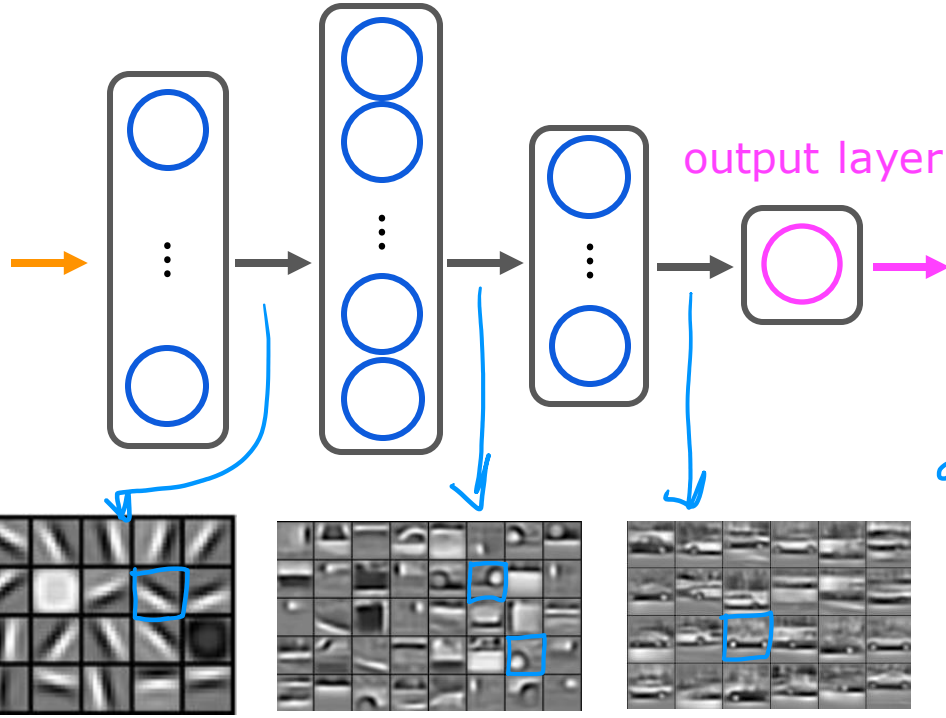
probability of being person 'XYZ'



activations are higher level features

source: Convolutional Deep Belief Networks for Scalable Unsupervised Learning of Hierarchical Representations by Honglak Lee, Roger Grosse, Ranganath Andrew Y. Ng

Car classification



source: Convolutional Deep Belief Networks for Scalable Unsupervised Learning of Hierarchical Representations by Honglak Lee, Roger Grosse, Ranganath Andrew Y. Ng

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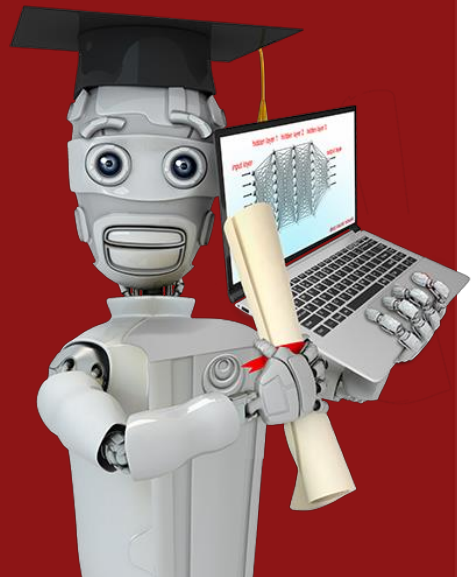
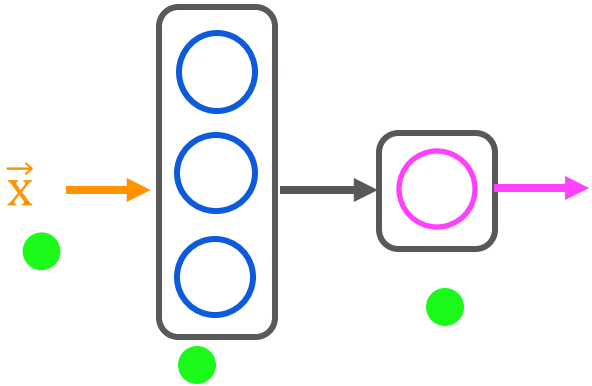


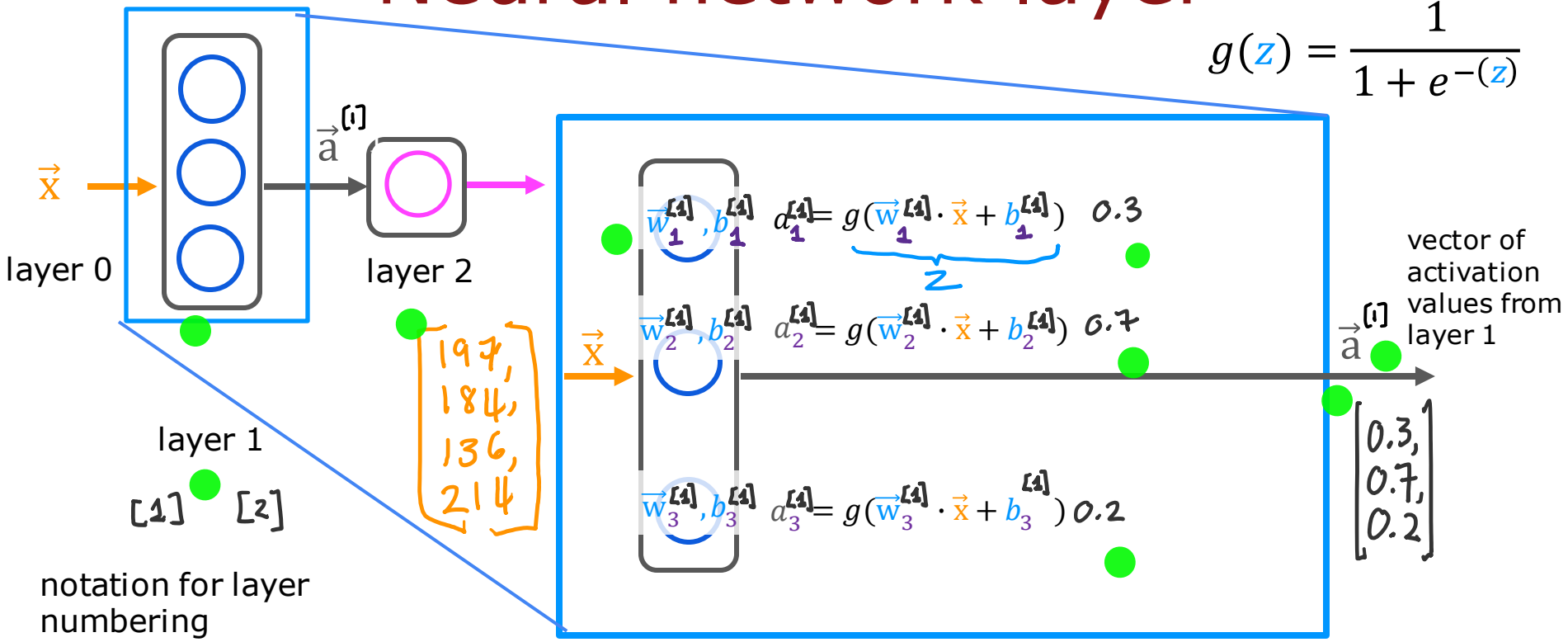
Neural network model

Neural network layer

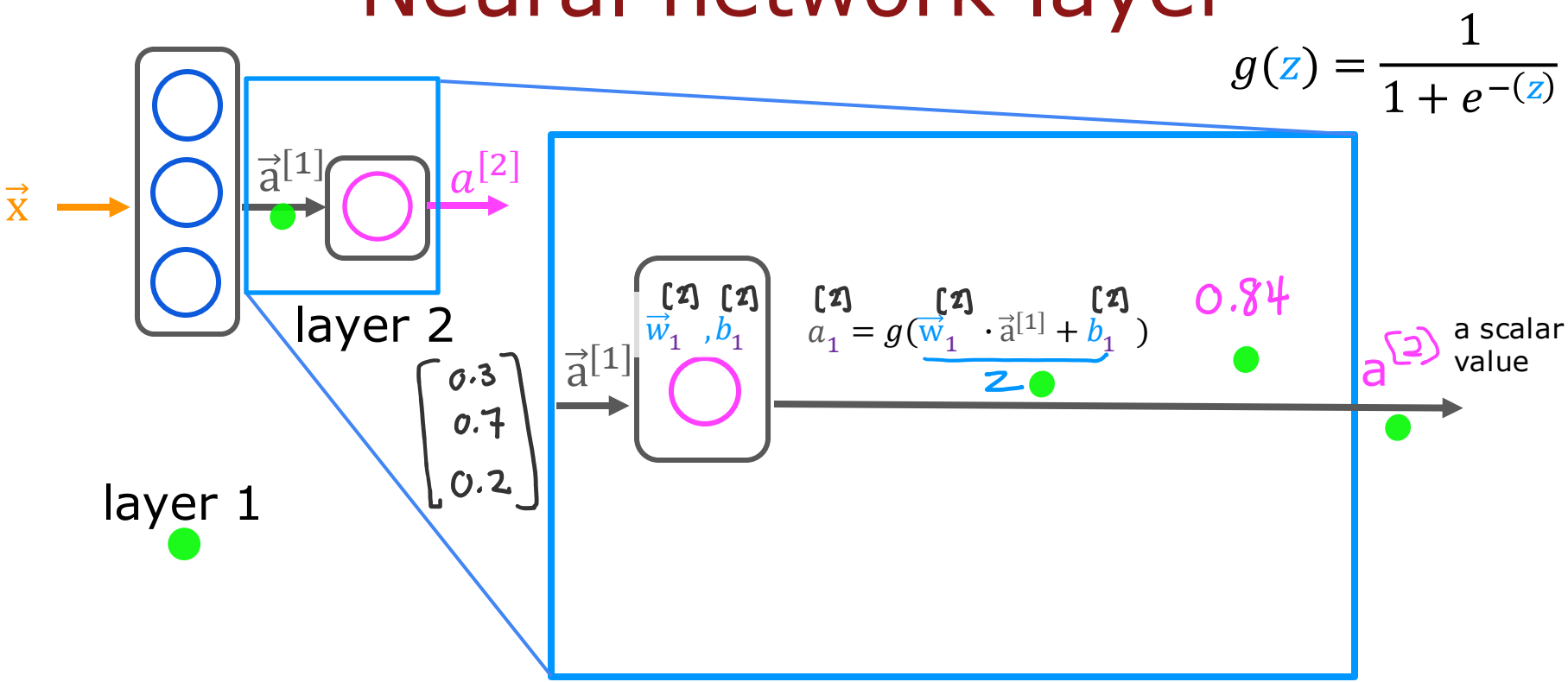
Neural network layer



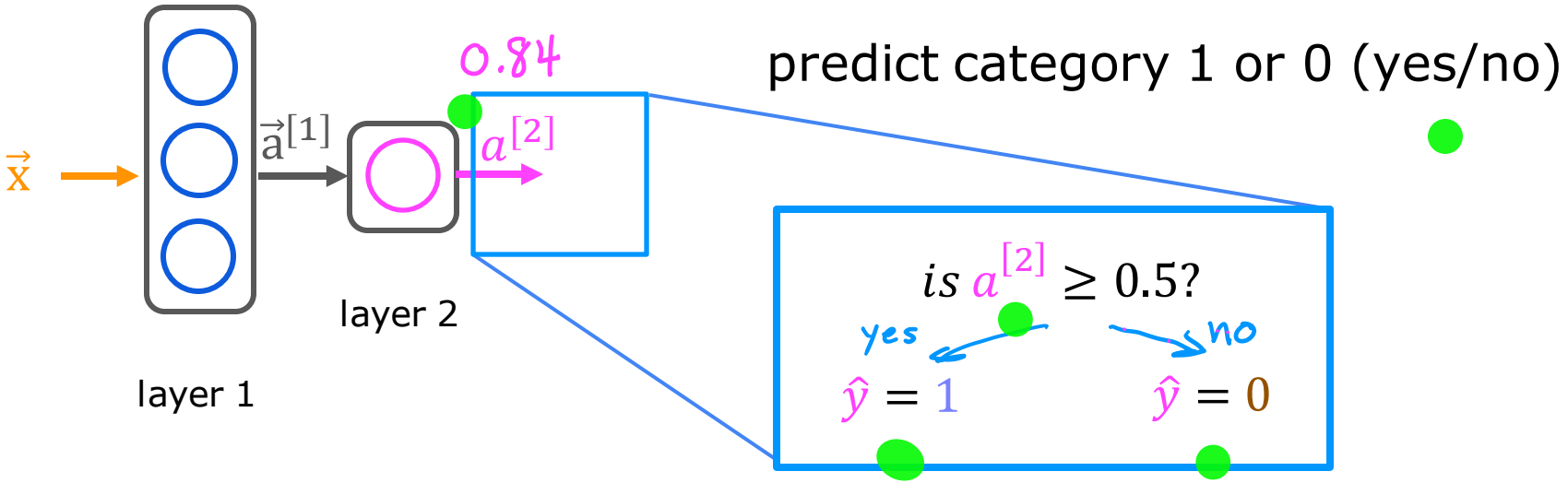
Neural network layer



Neural network layer



Neural network layer



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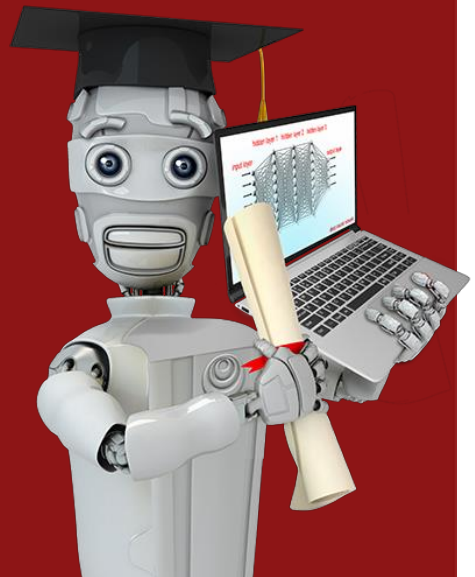
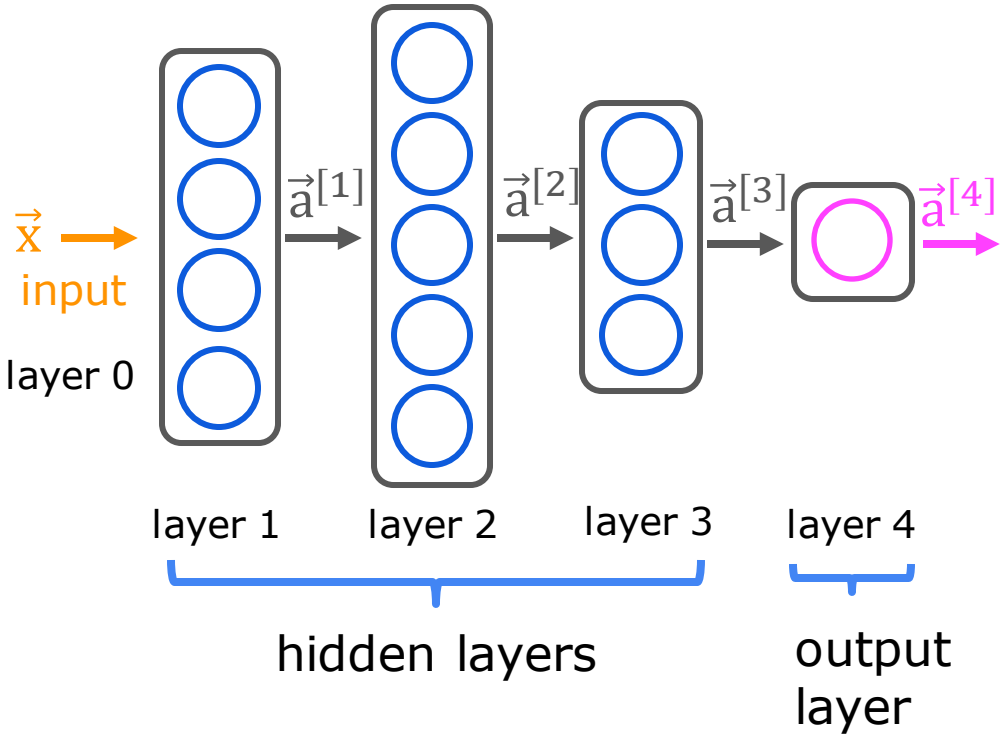


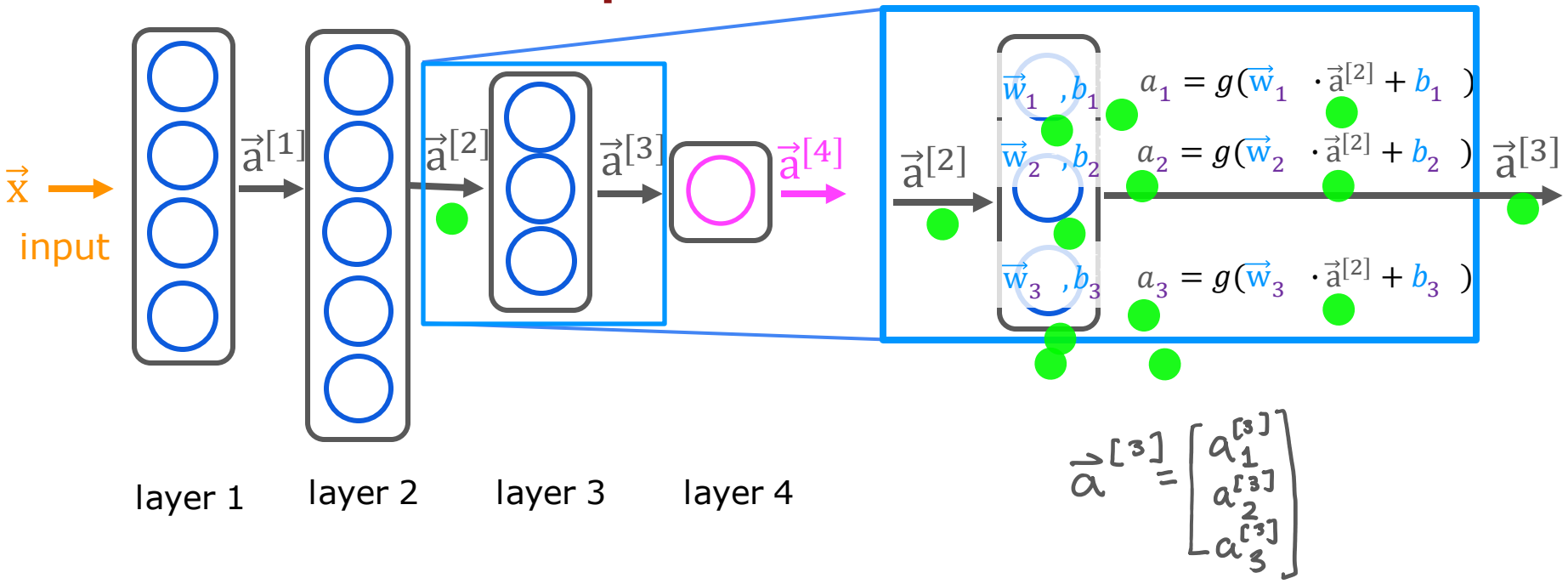
Neural Network Model

More complex neural networks

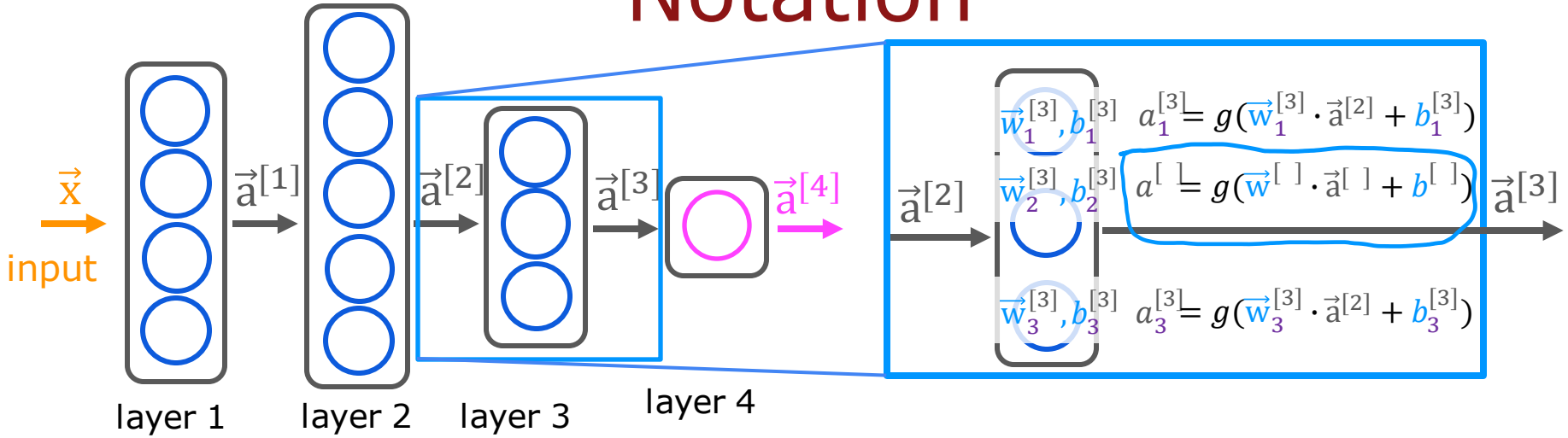
More complex neural network



More complex neural network

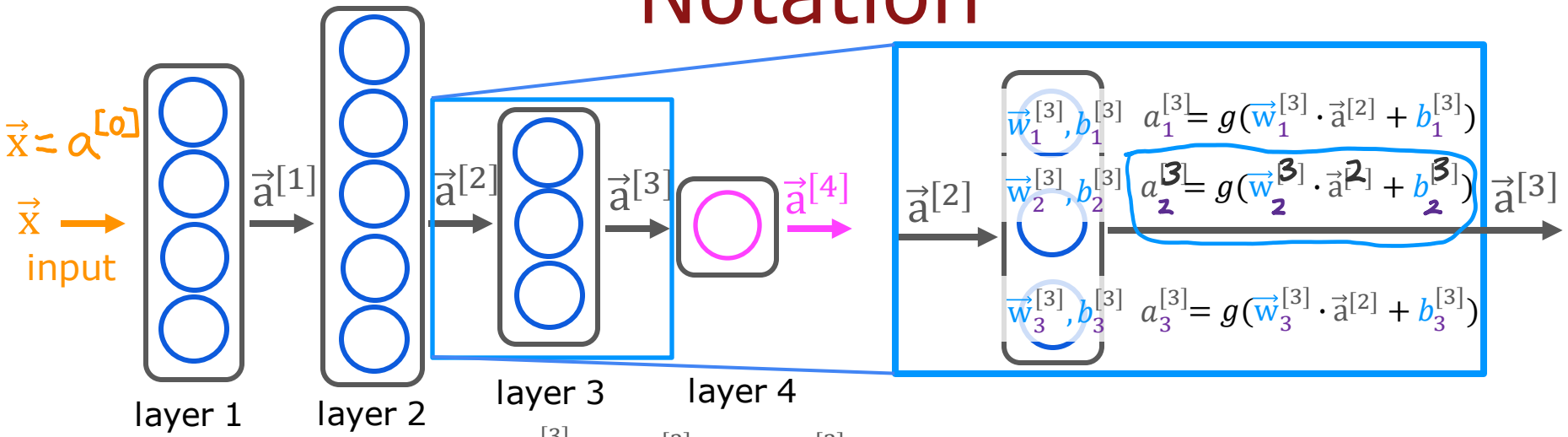


Notation



Question:
Can you fill in the superscripts and subscripts for the second neuron?

Notation



$$a_2^{[3]} = g(\vec{w}_2^{[3]} \cdot \vec{a}^{[2]} + b_2^{[3]})$$

Activation value of layer l, unit(neuron) j

output of layer $l-1$ (previous layer)

$$a_j^{[l]} = g(\vec{w}_j^{[l]} \cdot \vec{a}^{[l-1]} + b_j^{[l]})$$

sigmoid "activation function"

Parameters w & b of layer l, unit j

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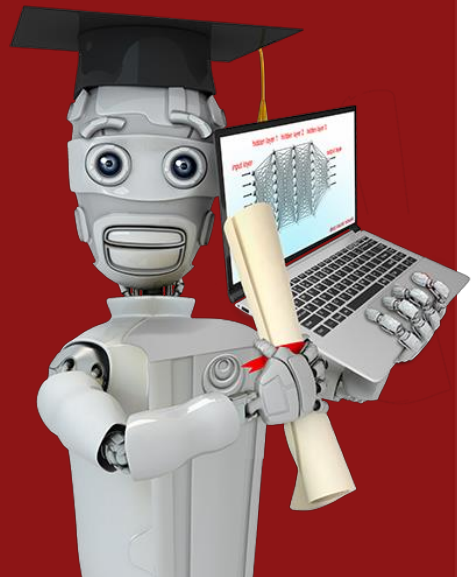
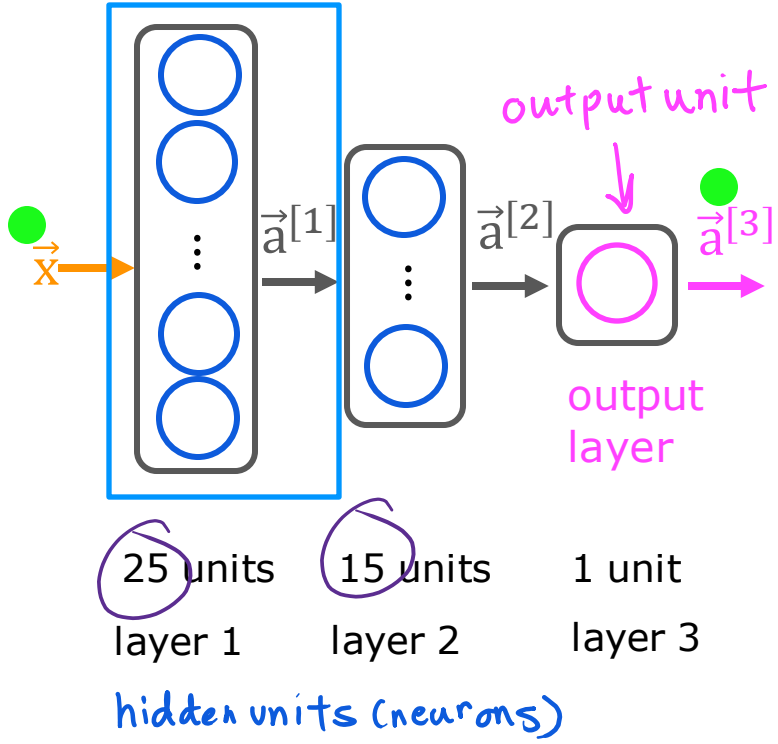
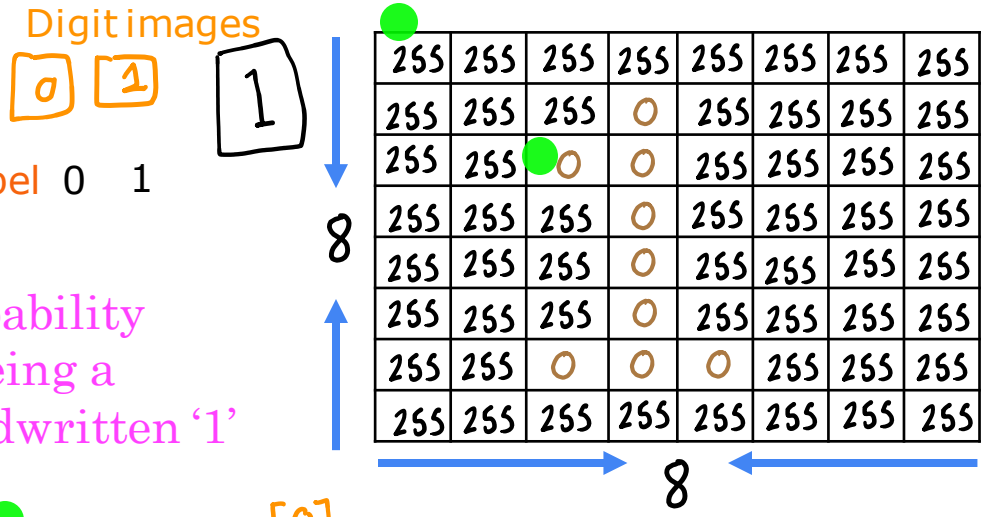


Neural Network Model

Inference: making predictions (forward propagation)

Handwritten digit recognition

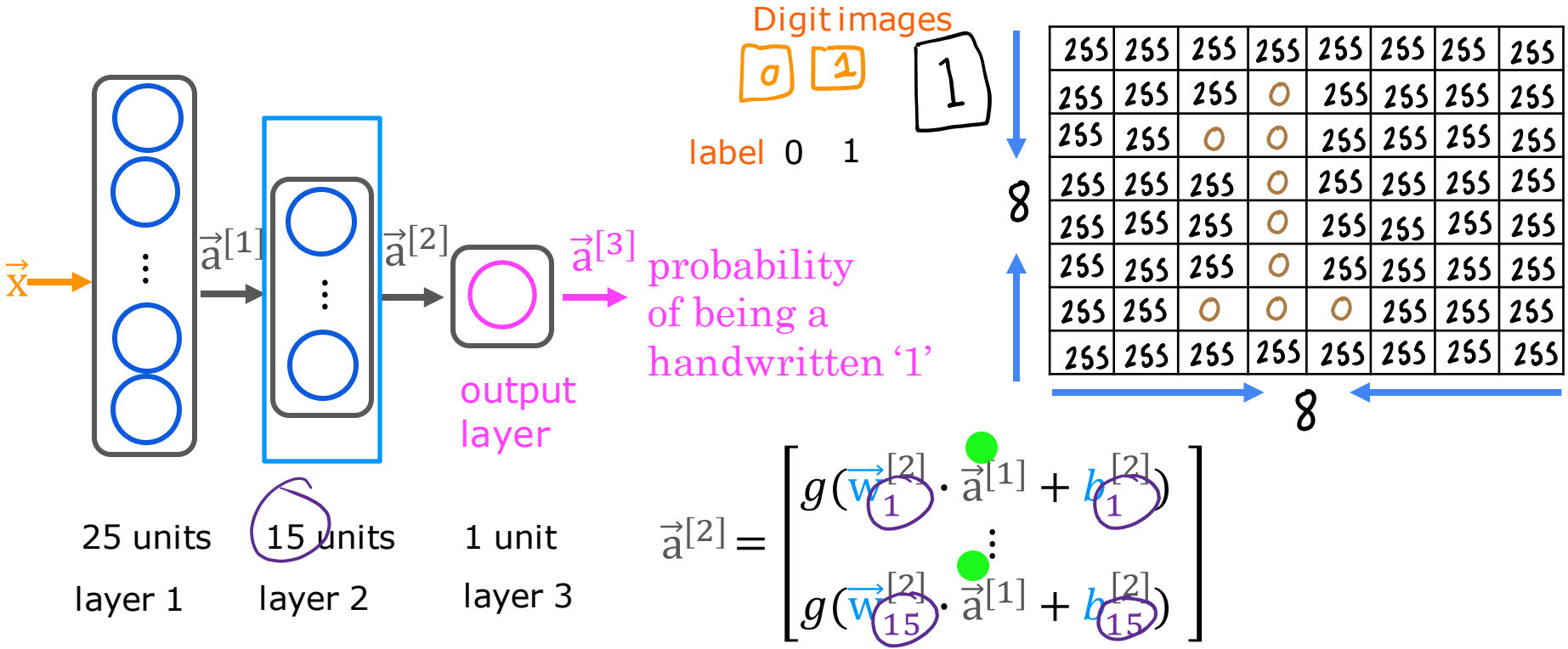




$$a^{[0]}$$

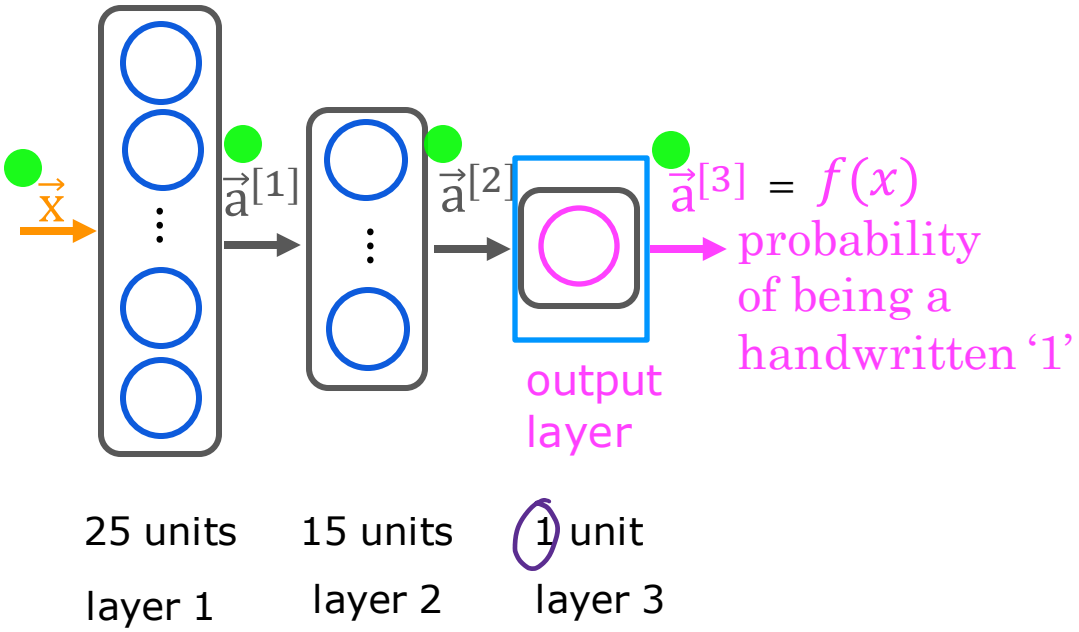
$$\vec{a}^{[1]} = \begin{bmatrix} g(\vec{w}_1^{[1]} \cdot \vec{x} + b_1^{[1]}) \\ \vdots \\ g(\vec{w}_{25}^{[1]} \cdot \vec{x} + b_{25}^{[1]}) \end{bmatrix}$$

Handwritten digit recognition



Handwritten digit recognition

forward propagation



$$\vec{a}^{[3]} = \left[g\left(\vec{w}_1^{[3]} \cdot \vec{a}^{[2]} + b_1^{[3]} \right) \right]$$

$is\ a_1^{[3]} \geq 0.5?$

yes → $\hat{y} = 1$ image is digit 1

no → $\hat{y} = 0$ image isn't digit 1

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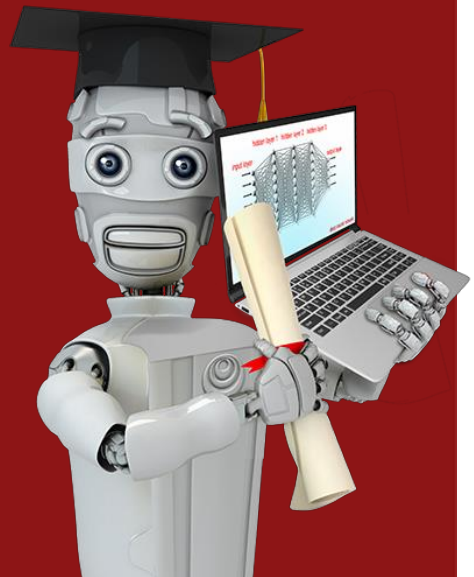
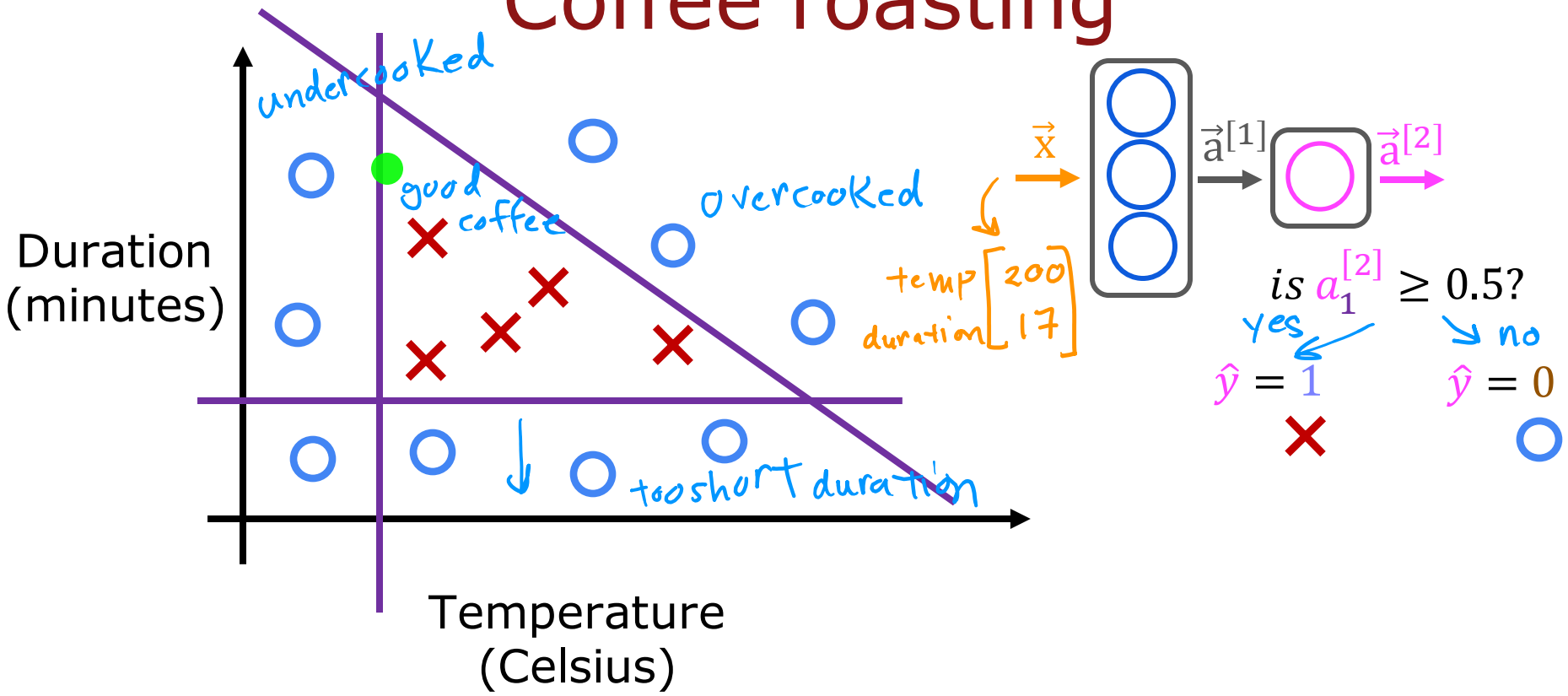
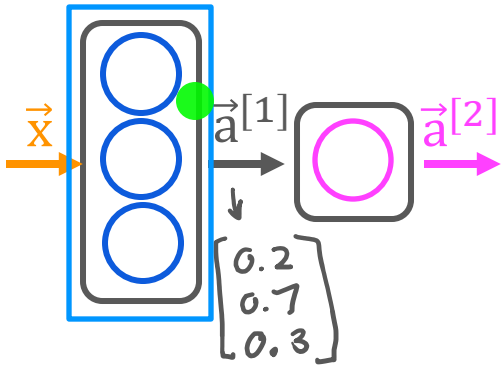


TensorFlow implementation

Inference in Code

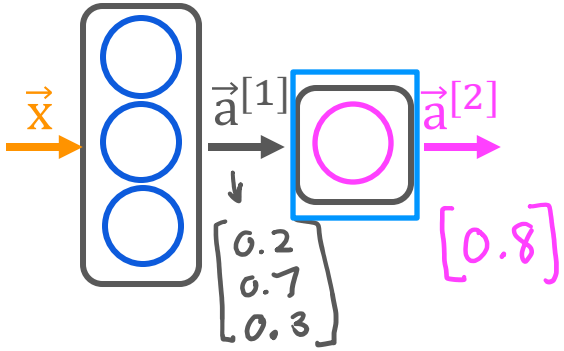
Coffee roasting






```
x = np.array([[200.0, 17.0]])
layer_1 = Dense(units=3, activation='sigmoid')
a1 = layer_1(x)
```

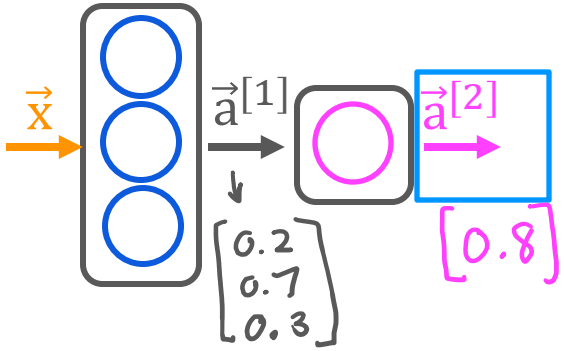
Build the model using TensorFlow

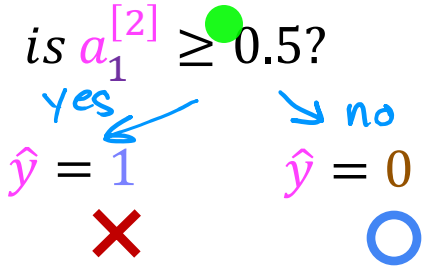



```
x = np.array([[200.0, 17.0]])
layer_1 = Dense(units=3, activation='sigmoid')
a1 = layer_1(x)
```

```
layer_2 = Dense(units=1, activation='sigmoid')
a2 = layer_2(a1)
```

Build the model using TensorFlow



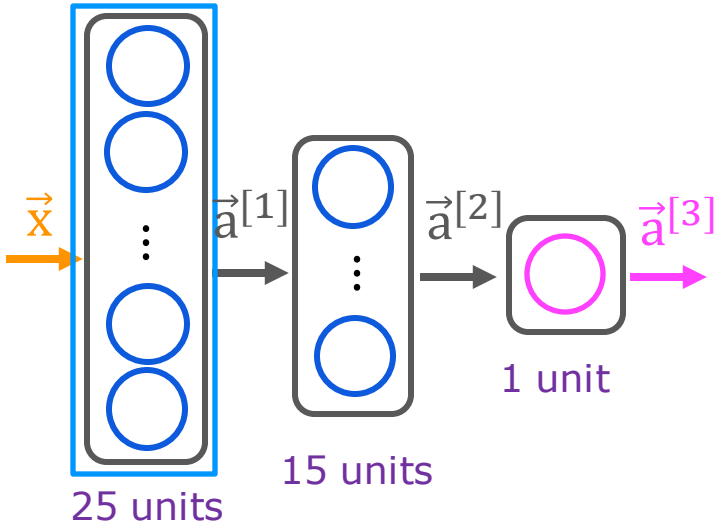



```
x = np.array([[200.0, 17.0]])
layer_1 = Dense(units=3, activation='sigmoid')
a1 = layer_1(x)
```

```
layer_2 = Dense(units=1, activation='sigmoid')
a2 = layer_2(a1)
```

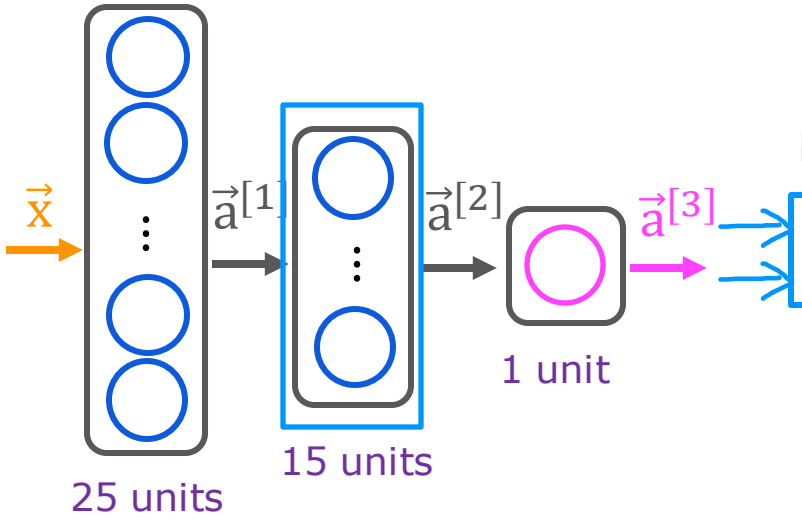
```
if a2 >= 0.5:
  yhat = 1
else:
  yhat = 0
```

Model for digit classification




```
x = np.array([[0.0,...245,...240...0]])
layer_1 = Dense(units=25, activation='sigmoid')
a1 = layer_1(x)
```

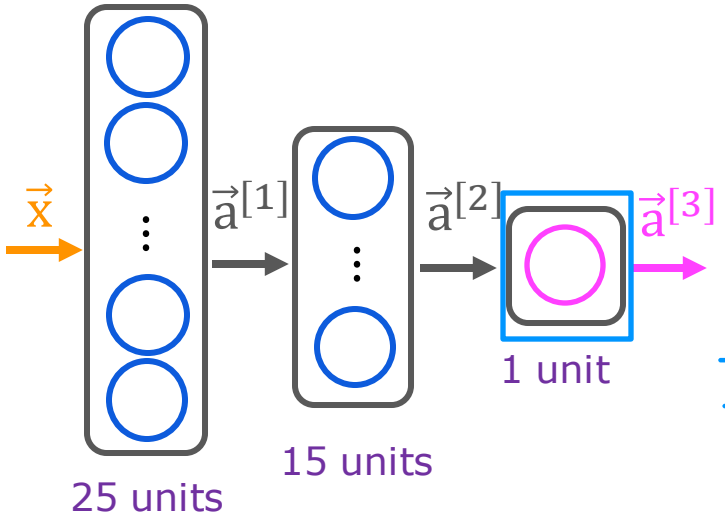
Model for digit classification




```
x = np.array([[0.0,...245,...240...0]])
layer_1 = Dense(units=25, activation='sigmoid')
a1 = layer_1(x)
```

```
layer_2 = Dense(units=15, activation='sigmoid')
a2 = layer_2(a1)
```

Model for digit classification

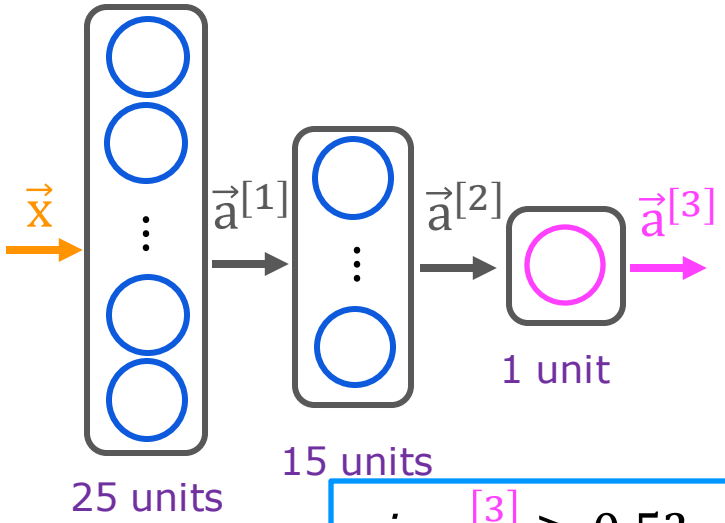



```
x = np.array([[0.0,...245,...240...0]])
layer_1 = Dense(units=25, activation='sigmoid')
a1 = layer_1(x)

layer_2 = Dense(units=15, activation='sigmoid')
a2 = layer_2(a1)

layer_3 = Dense(units=1, activation='sigmoid')
a3 = layer_3(a2)
```

Model for digit classification




```
x = np.array([[0.0,...245,...240...0]])
layer_1 = Dense(units=25, activation='sigmoid')
a1 = layer_1(x)

layer_2 = Dense(units=15, activation='sigmoid')
a2 = layer_2(a1)

layer_3 = Dense(units=1, activation='sigmoid')
a3 = layer_3(a2)
```

$is\ a_1^{[3]} \geq 0.5?$

$\hat{y} = 1$ ✗ $\hat{y} = 0$ ○

```
if a3 >= 0.5:
  yhat = 1
else:
  yhat = 0
```

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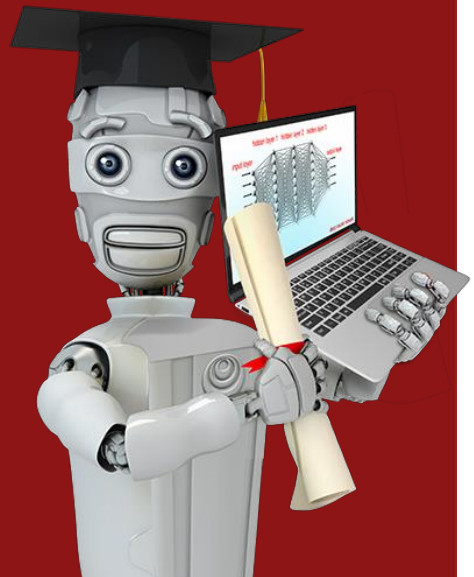


TensorFlow implementation

Data in TensorFlow

Feature vectors

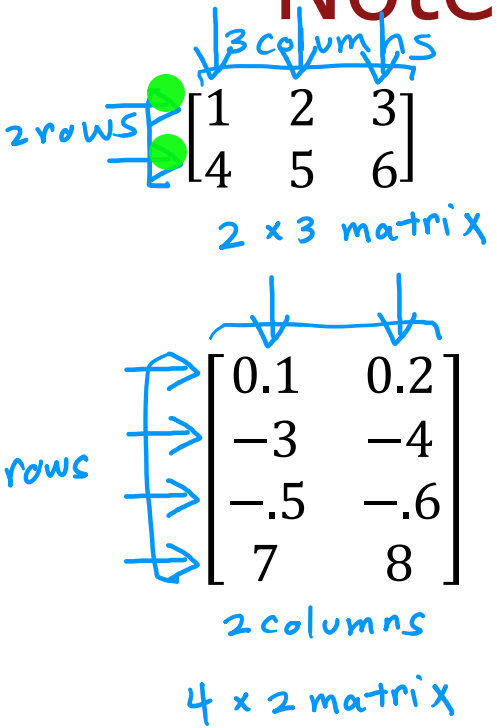
temperature (Celsius)	duration (minutes)	Good coffee? (1/0)
200.0	17.0	1
425.0	18.5	0
…	…	…

```
x = np.array([[200.0, 17.0]])
```

```
[[200.0, 17.0]]
```

Why?

Note about numpy arrays




```
x = np.array([[1, 2, 3],
              [4, 5, 6]])
```

```
[[1, 2, 3],
 [4, 5, 6]]
```

```
x = np.array([[0.1, 0.2],
              [-3.0, -4.0,],
              [-0.5, -0.6,],
              [7.0, 8.0,]])
```

```
[[0.1, 0.2],
 [-3.0, -4.0,],
 [-0.5, -0.6,],
 [7.0, 8.0,]]
```

2D array

2 x 3

4 x 2

1 x 2

2 x 1

Note about numpy arrays

```
x = np.array([[200, 17]])
```

$\begin{bmatrix} 200 & 17 \end{bmatrix}$

1 x 2

```
x = np.array([[200],
              [17]])
```

$\begin{bmatrix} 200 \\ 17 \end{bmatrix}$

2 x 1

```
x = np.array([200,17])
```

1D
"Vector"

Feature vectors

temperature (Celsius)	duration (minutes)	Good coffee? (1/0)
200.0	17.0	1
425.0	18.5	0
…	…	…

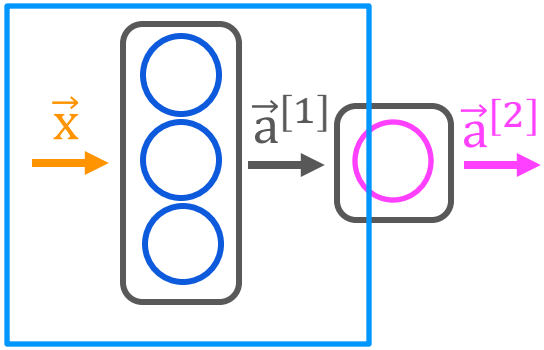
```
x = np.array([[200.0, 17.0]])
```

```
[[200.0, 17.0]]
```

1 x 2

$$\begin{bmatrix} 200.0 & 17.0 \end{bmatrix}$$

Activation vector




```
x = np.array([[200.0, 17.0]])
layer_1 = Dense(units=3, activation='sigmoid')
a1 = layer_1(x)
```

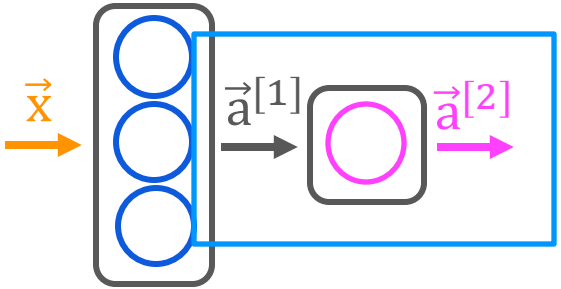
[0.2, 0.7, 0.3] 1 x 3 matrix

```
tf.Tensor([[0.2 0.7 0.3]], shape=(1, 3), dtype=float32)
```

```
a1.numpy()
```

```
array([[1.4661001, 1.125196 , 3.2159438]], dtype=float32)
```

Activation vector




```
layer_2 = Dense(units=1, activation='sigmoid')
a2 = layer_2(a1)
```

[[0.8]]

1 x 1

```
tf.Tensor([[0.8]], shape=(1, 1), dtype=float32)
```

```
a2.numpy()
```

```
array([[0.8]], dtype=float32)
```

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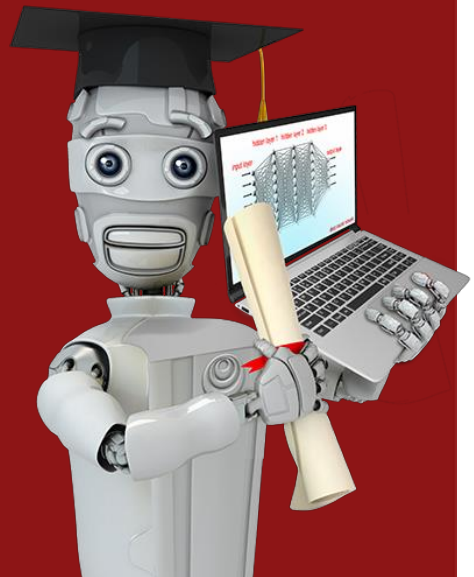
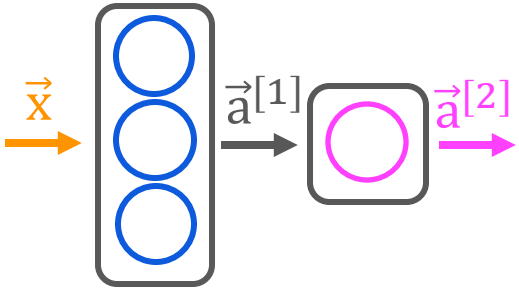


TensorFlow implementation

Building a neural network

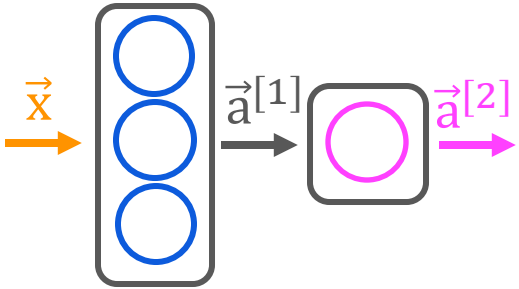
What you saw earlier




```
x = np.array([[200.0, 17.0]])
layer_1 = Dense(units=3, activation="sigmoid")
a1 = layer_1(x)

layer_2 = Dense(units=1, activation="sigmoid")
a2 = layer_2(a1)
```

Building a neural network architecture



		y
200	17	1
120	5	0
425	20	0
212	18	1

layer1

```
layer_1 = Dense(units=3, activation="sigmoid")
layer_2 = Dense(units=1, activation="sigmoid")
model = Sequential([layer_1, layer_2])
```

layer2

```
x = np.array([[200.0, 17.0],
              [120.0, 5.0],
              [425.0, 20.0],
              [212.0, 18.0]])
```

4 x 2

targets

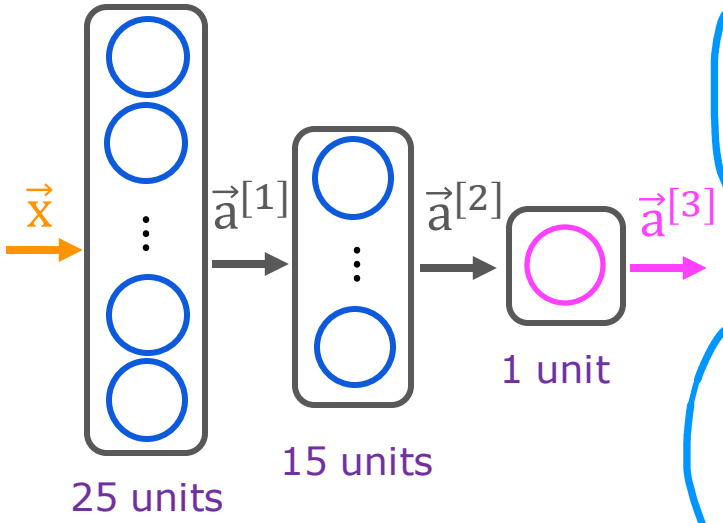
```
y = np.array([1,0,0,1])

model.compile(...)
model.fit(x,y)

model.predict(x_new)
```

← more about this next week!

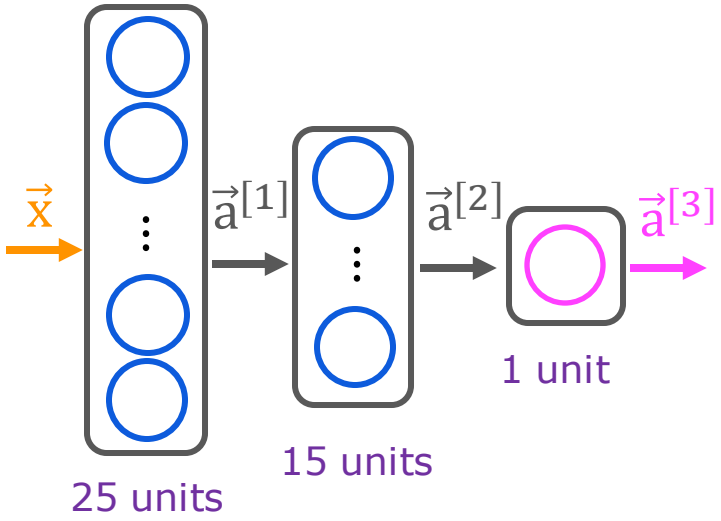
Digit classification model




```
layer_1 = Dense(units=25, activation="sigmoid")
layer_2 = Dense(units=15, activation="sigmoid")
layer_3 = Dense(units=1, activation="sigmoid")
model = Sequential([layer_1, layer_2, layer_3])
model.compile(...)
x = np.array([[0..., 245, ..., 17],
              [0..., 200, ..., 184]])
y = np.array([1,0])
model.fit(x,y)
model.predict(x_new)
```

more about this next week!

Digit classification model




```
model = Sequential([
    Dense(units=25, activation="sigmoid"),
    Dense(units=15, activation="sigmoid"),
    Dense(units=1, activation="sigmoid")])

model.compile(...)

x = np.array([[0..., 245, ..., 17],
              [0..., 200, ..., 184])
y = np.array([1,0])

model.fit(x,y)

model.predict(x_new)
```

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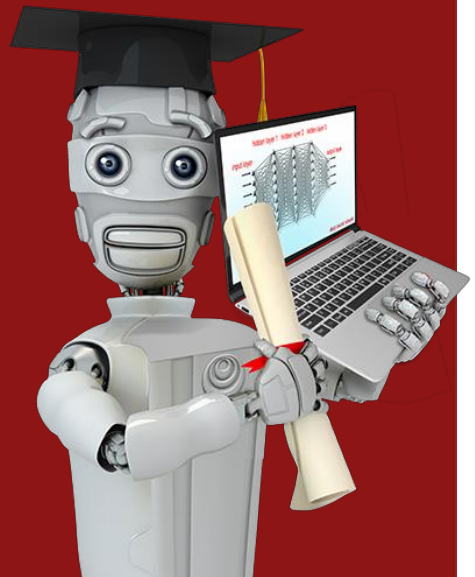
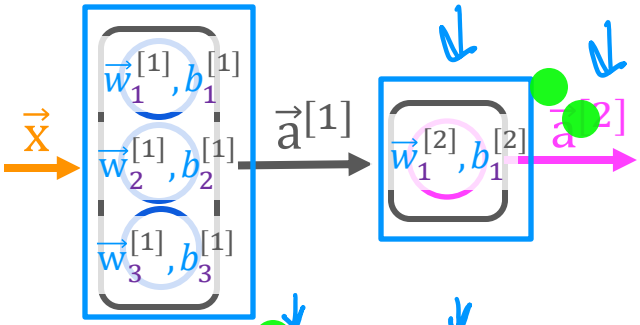


Neural network implementation in Python

Forward prop in a single layer

forward prop (coffee roasting model)



$a_1^{[2]} = g(\vec{w}_1^{[2]} \cdot \vec{a}^{[1]} + b_1^{[2]})$

```
w2_1 = np.array([-7, 8])
b2_1 = np.array([3])
z2_1 = np.dot(w2_1,a1)+b2_1
a2_1 = sigmoid(z1_1)
```

$w_1^{[2]}$ w2_1

```
x = np.array([200, 17])
```

1D arrays

$a_1^{[1]} = g(\vec{w}_1^{[1]} \cdot \vec{x} + b_1^{[1]})$ $a_2^{[1]} = g(\vec{w}_2^{[1]} \cdot \vec{x} + b_2^{[1]})$ $a_3^{[1]} = g(\vec{w}_3^{[1]} \cdot \vec{x} + b_3^{[1]})$

```
w1_1 = np.array([1, 2])
b1_1 = np.array([-1])
z1_1 = np.dot(w1_1,x)+ b
a1_1 = sigmoid(z1_1)
```

```
w1_2 = np.array([-3, 4])
b1_2 = np.array([1])
z1_2 = np.dot(w1_2,x)+b
a1_2 = sigmoid(z1_2)
```

```
w1_3 = np.array([5, -6])
b1_3 = np.array([2])
z1_3 = np.dot(w1_3,x)+b
a1_3 = sigmoid(z1_3)
```

```
a1 = np.array([a1_1, a1_2, a1_3])
```

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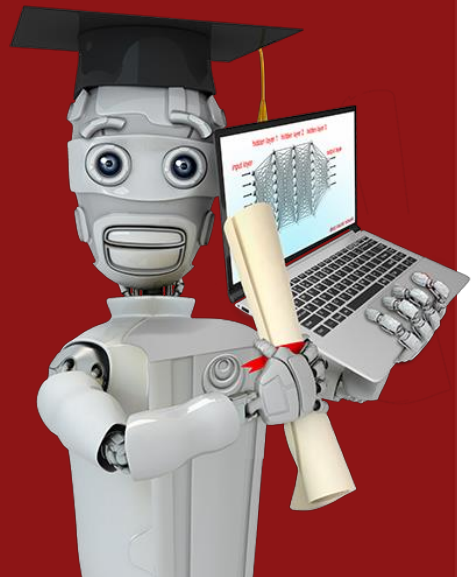
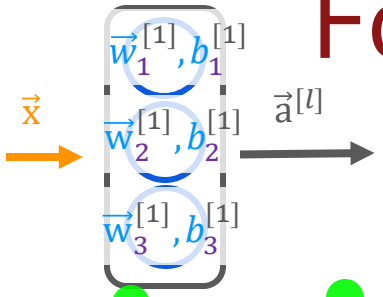


Neural network implemenation in Python

General implementation of forward propagation

Forward prop in NumPy



$\vec{w}_1^{[1]} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\vec{w}_2^{[1]} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$ $\vec{w}_3^{[1]} = \begin{bmatrix} 5 \\ -6 \end{bmatrix}$

```
W = np.array([
     [1,  -3,  5]
     [2,   4, -6]])
```
2 by 3

$b_1^{[l]} = -1$ $b_2^{[l]} = 1$ $b_3^{[l]} = 2$

```
b = np.array([-1, 1, 2])
```

$\vec{a}^{[0]} = \vec{x}$

```
a_in = np.array([-2, 4])
```

```
def dense(a_in,W,b, g):
    units = W.shape[1]
    a_out = np.zeros(units)
    for j in range(units):
        w = W[:,j]
        z = np.dot(w,a_in) + b[j]
        a_out[j] = g(z)
    return a_out
```
3 [0,0,0] 0,1,2

```
def sequential(x):
    a1 = dense(x,W1,b1,g)
    a2 = dense(a1,W2,b2,g)
    a3 = dense(a2,W3,b3,g)
    a4 = dense(a3,W4,b4,g)
    f_x = a4
    return f_x
```
$a^{[1]}$ $a^{[2]}$

capital W refers to a matrix

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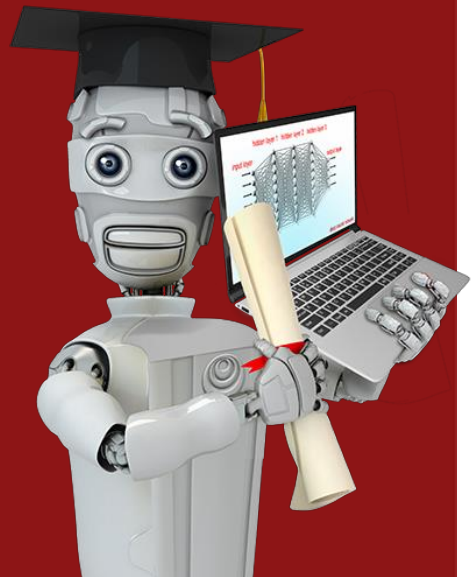
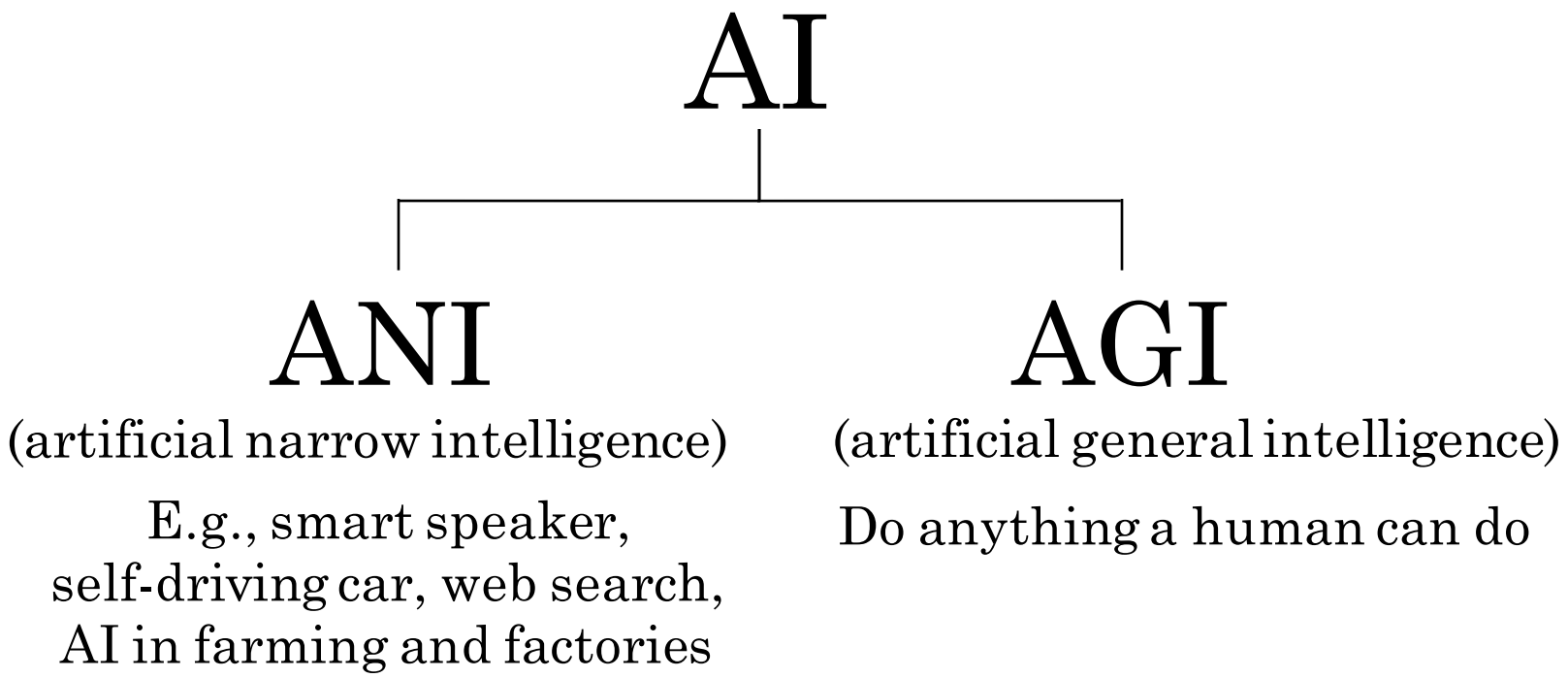


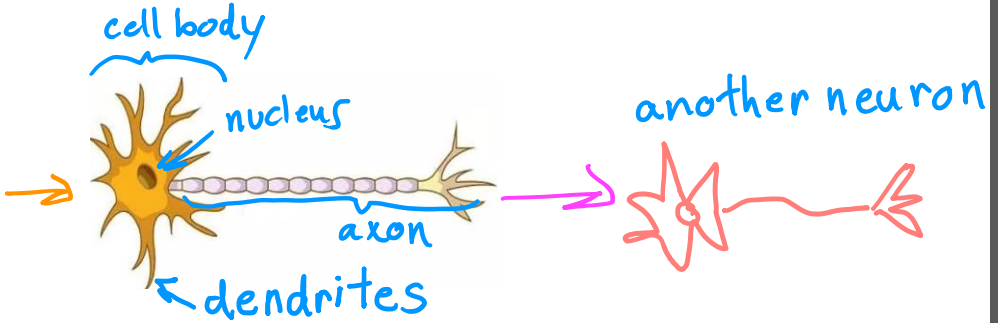
Speculations on artificial general intelligence (AGI)

Is there a path to AGI?


AI
ANI
(artificial narrow intelligence)
E.g., smart speaker,
self-driving car, web search,
AI in farming and factories
AGI
(artificial general intelligence)
Do anything a human can do

Biological neuron

inputs outputs



Simplified mathematical model of a neuron

inputs outputs

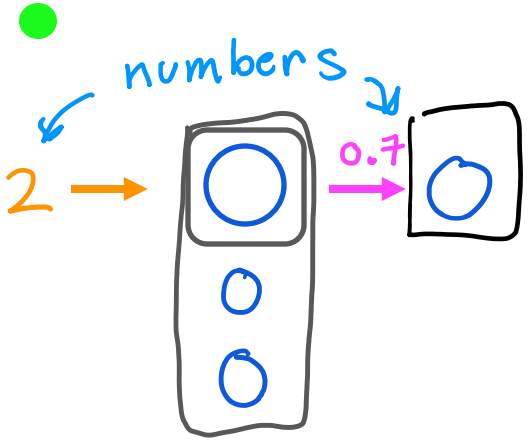
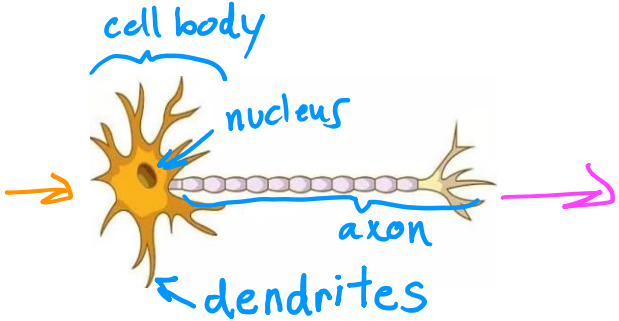


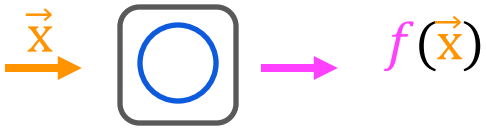
image source: https://biologydictionary.net/sensory-neuron/

Neural network and the brain

Can we mimic the human brain?

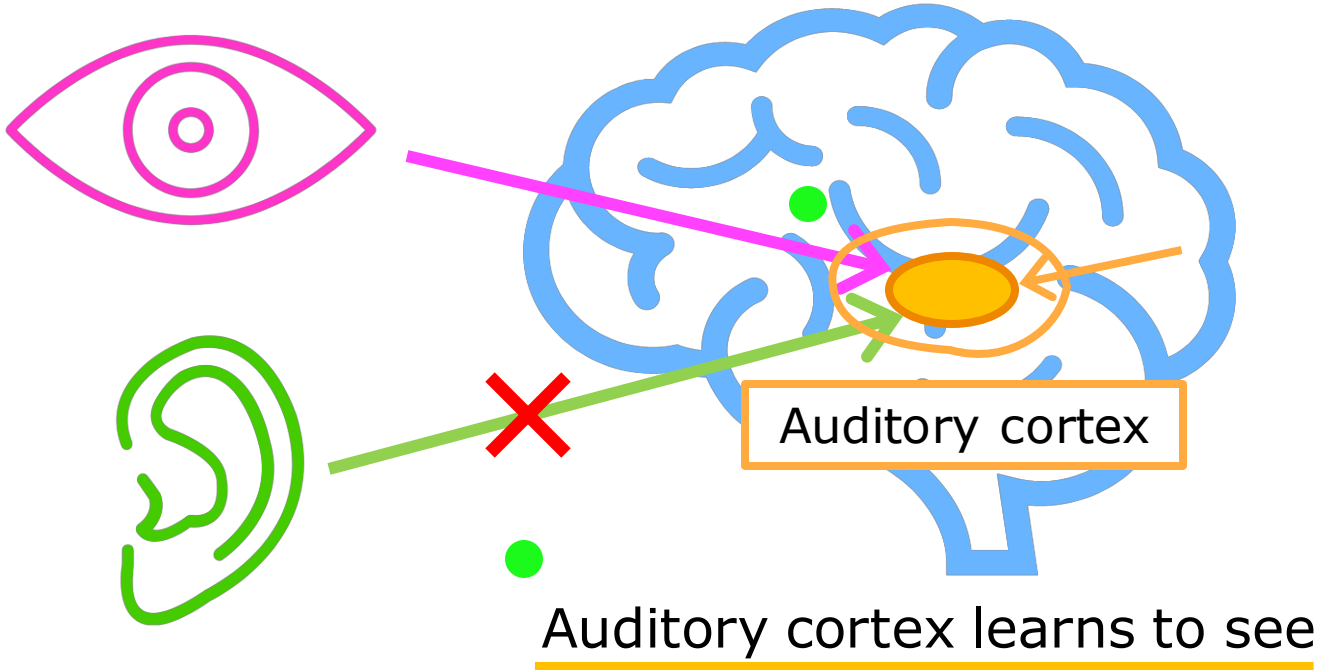


vs



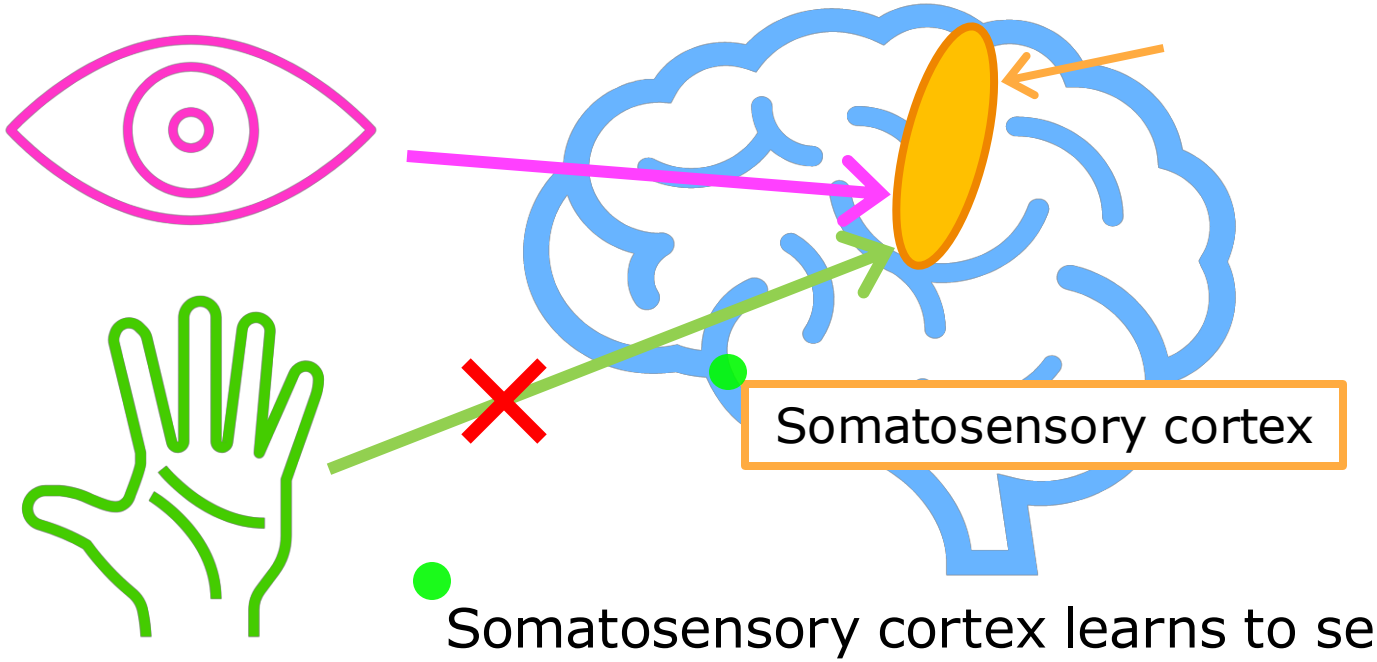
We have (almost) no idea how the brain works

The ”one learning algorithm” hypothesis



[Roe et al., 1992]

The ”one learning algorithm” hypothesis



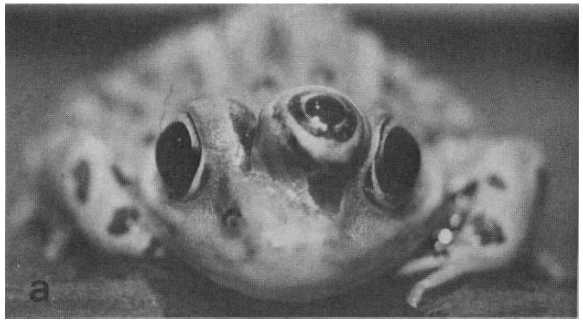
[Metin & Frost, 1989]

Sensor representations in the brain

Seeing with your tongue

Human echolocation (sonar)

Haptic belt: Direction sense



Implanting a 3rd eye

[BrainPort; Welsh & Blasch, 1997; Nagel et al., 2005; Constantine-Paton & Law, 2009]

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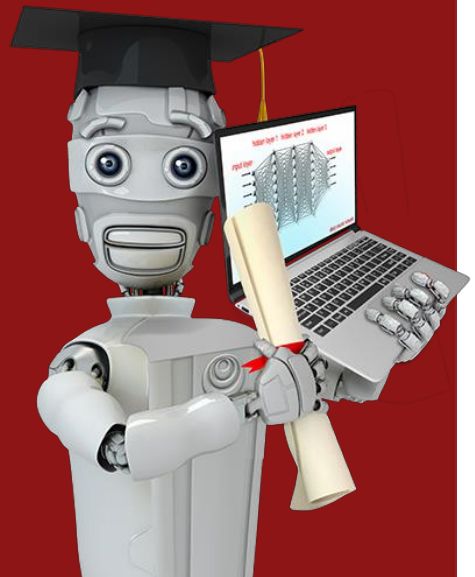

Vectorization (optional)

How neural networks are implemented efficiently

For loops vs. vectorization

```
x = np.array([200, 17])

W = np.array([[1, -3, 5],
              [-2, 4, -6]])

b = np.array([-1, 1, 2])

def dense(a_in,W,b):
    a_out = np.zeros(units)
    for j in range(units):
      w = W[:,j]
      z = np.dot(w,x) + b[j]
      a[j] = g(z)
  return a
```

[1,0,1]

Vectorized

```
X = np.array([[200, 17]])

W = np.array([[1, -3, 5],
              [-2, 4, -6]])

B = np.array([[-1, 1, 2]])

def dense(A_in,W,B):
    Z = np.matmul(A_in,W) + B
    A_out = g(Z)
    return A_out
```

[[1,0,1]]

2D array

same

1x3 2D array

all 2D arrays

matrix multiplication

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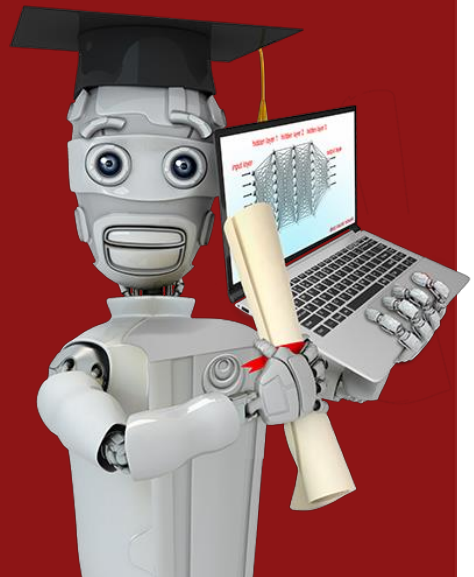

Vectorization (optional)

Matrix multiplication

Dot products

example

$\begin{bmatrix}1\\2\end{bmatrix} \cdot \begin{bmatrix}3\\4\end{bmatrix}$

$z = (1 \times 3) + (2 \times 4)$

$3 + 8$

11

in general

$\begin{bmatrix}\uparrow\\ \vec{a} \\ \downarrow\end{bmatrix} \cdot \begin{bmatrix}\uparrow\\ \vec{w} \\ \downarrow\end{bmatrix}$

$z = \vec{a} \cdot \vec{w}$

transpose

$\vec{a} = \begin{bmatrix}1\\2\end{bmatrix}$

$\vec{a}^T = [1 \quad 2]$

vector vector multiplication

$[\leftarrow \ \vec{a}^T \ \rightarrow] \begin{bmatrix}\uparrow\\ \vec{w} \\ \downarrow\end{bmatrix}$

$1 \times 2 \qquad 2 \times 1$

$z = \vec{a}^T \vec{w}$

equivalent

useful for understanding matrix multiplication

Vector matrix multiplication

$$\vec{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad \vec{a}^T = \begin{bmatrix} 1 & 2 \end{bmatrix} \qquad W = \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}$$

$$Z = \vec{a}^T W \quad \begin{bmatrix} \leftarrow & \vec{a}^T & \rightarrow \end{bmatrix} \begin{bmatrix} \uparrow & \uparrow \\ \vec{w}_1 & \vec{w}_2 \\ \downarrow & \downarrow \end{bmatrix}$$

1 by 2

$$Z = \begin{bmatrix} \vec{a}^T \vec{w}_1 & a^T \vec{w}_2 \end{bmatrix}$$

$$(1 * 3) + (2 * 4) \qquad (1 * 5) + (2 * 6)$$

$3 + 8$ $\quad$ $5 + 12$

11 $\quad$ 17

$$Z = [11 \quad 17]$$

matrix matrix multiplication

$$A = \begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix} \quad W = \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix} \quad Z = A^T W = \begin{bmatrix} \leftarrow & \vec{a}_1^T & \rightarrow \\ \leftarrow & \vec{a}_2^T & \rightarrow \end{bmatrix} \begin{bmatrix} \uparrow & \uparrow \\ \vec{w}_1 & \vec{w}_2 \\ \downarrow & \downarrow \end{bmatrix}$$

rows

columns

$$= \begin{bmatrix} \vec{a}_1^T \vec{w}_1 & \vec{a}_1^T \vec{w}_2 \\ \vec{a}_2^T \vec{w}_1 & \vec{a}_2^T \vec{w}_2 \end{bmatrix}$$

row 1 col 1

row 2 col 1

row 1 col 2

row 2 col 2

$(-1 \times 3) + (-2 \times 4)$

$-3 + -8$

-11

$(-1 \times 5) + (-2 \times 6)$

$-5 + -12$

-17

$$= \begin{bmatrix} 11 & 17 \\ -11 & -17 \end{bmatrix}$$

general rules for matrix multiplication
↳ next video!

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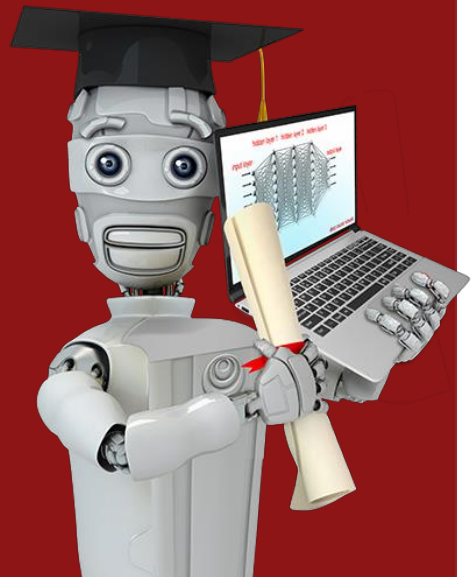

Vectorization (optional)

Matrix multiplication rules

Matrix multiplication rules

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0.1 \\ 2 & -2 & 0.2 \end{bmatrix} \quad \mathbf{A}^T = \begin{bmatrix} 1 & 2 \\ -1 & -2 \\ 0.1 & 0.2 \end{bmatrix} \quad \mathbf{W} = \begin{bmatrix} 3 & 5 & 7 & 9 \\ 4 & 6 & 8 & 0 \end{bmatrix}$$

$\vec{a}_1^T$ $\vec{a}_2^T$ $\vec{a}_3^T$

$\vec{w}_1$ $\vec{w}_2$ $\vec{w}_3$ $\vec{w}_4$

$$Z = \mathbf{A}^T\mathbf{W} = \begin{bmatrix} & & & \\ & & & \\ & & & \end{bmatrix}$$

Matrix multiplication rules

$$A = \begin{bmatrix} 1 & -1 & 0.1 \\ 2 & -2 & 0.2 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 2 \\ -1 & -2 \\ 0.1 & 0.2 \end{bmatrix} \quad W = \begin{bmatrix} 3 & 5 & 7 & 9 \\ 4 & 6 & 8 & 0 \end{bmatrix} \quad Z = A^T W = \begin{bmatrix} & & & \\ & & & \\ & & & \end{bmatrix}$$

3 by 4 matrix

$$\vec{a}_1^T \vec{w}_1 = (1 \times 3) + (2 \times 4) = 11$$

row 3 column 2

$$\vec{a}_3^T \vec{w}_2 = (0.1 \times 5) + (0.2 \times 6) = 1.7$$

0.5 + 1.2

row 2 column 3?

$$\vec{a}_2^T \vec{w}_3 = (-1 \times 7) + (-2 + 8) = -23$$

-7 + -16

Matrix multiplication rules

$$A = \begin{bmatrix} 1 & -1 & 0.1 \\ 2 & -2 & 0.2 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 2 \\ -1 & -2 \\ 0.1 & 0.2 \end{bmatrix} \quad W = \begin{bmatrix} 3 & 5 & 7 & 9 \\ 4 & 6 & 8 & 0 \end{bmatrix} \quad Z = A^T W = \begin{bmatrix} 11 & 17 & 23 & 9 \\ -11 & -17 & -23 & -9 \\ 1.1 & 1.7 & 2.3 & 0.9 \end{bmatrix}$$

3 by 4 matrix

$\vec{a}_1^T \vec{w}_1 = (1 \times 3) + (2 \times 4) = 11$

row 3 column 2

$\vec{a}_3^T \vec{w}_2 = (0.1 \times 5) + (0.2 \times 6) = 1.7$

$0.5 + 1.2$

row 2 column 3?

$\vec{a}_2^T \vec{w}_3 = (-1 \times 7) + (-2 + 8) = -23$

$-7 + -16$

Matrix multiplication rules

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0.1 \\ 2 & -2 & 0.2 \end{bmatrix} \quad \mathbf{A}^T = \begin{bmatrix} 1 & 2 \\ -1 & -2 \\ 0.1 & 0.2 \end{bmatrix} \quad \mathbf{W} = \begin{bmatrix} 3 & 5 & 7 & 9 \\ 4 & 6 & 8 & 0 \end{bmatrix} \quad \mathbf{Z} = \mathbf{A}^T\mathbf{W} = \begin{bmatrix} 11 & 17 & 23 & 9 \\ -11 & -17 & -23 & -9 \\ 1.1 & 1.7 & 2.3 & 0.9 \end{bmatrix}$$

3 x 2 2 x 4

can only take dot products of vectors that are same length

[0.1 0.2] length 2

$\begin{bmatrix} 5 \\ 6 \end{bmatrix}$ length 2

3 by 4 matrix

↳ same # rows as A^T

same # columns as W

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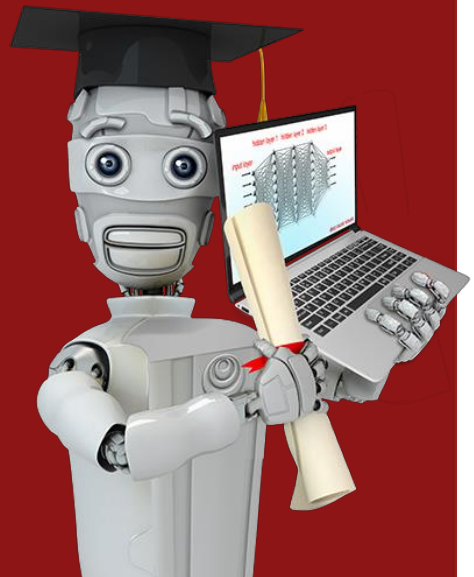


Vectorization (optional)

Matrix multiplication code

Matrix multiplication in NumPy

$$A = \begin{bmatrix} 1 & -1 & 0.1 \\ 2 & -2 & 0.2 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 2 \\ -1 & -2 \\ 0.1 & 0.2 \end{bmatrix} \quad W = \begin{bmatrix} 3 & 5 & 7 & 9 \\ 4 & 6 & 8 & 0 \end{bmatrix} \quad Z = A^T W = \begin{bmatrix} 11 & 17 & 23 & 9 \\ -11 & -17 & -23 & -9 \\ 1.1 & 1.7 & 2.3 & 0.9 \end{bmatrix}$$

```
A=np.array([1,-1,0.1],
           [2,-2,0.2]])
```

```
W=np.array([3,5,7,9],
           [4,6,8,0])
```

```
Z = np.matmul(AT,W)
```

or

```
Z = AT @ W
```

```
AT=np.array([1,2],
            [-1,-2],
            [0.1,0.2])
```

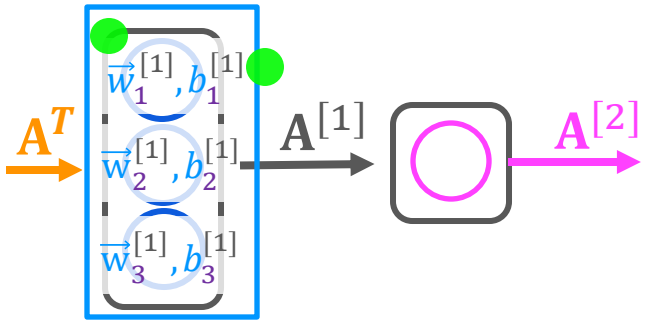
result

```
[[11,17,23,9],
 [-11,-17,-23,-9],
 [1.1,1.7,2.3,0.9]
]
```

```
AT=A.T
```

transpose

Dense layer vectorized



$$\mathbf{A}^T = [200 \quad 17] \qquad 1 \times 2$$

$$\mathbf{W} = \begin{bmatrix} 1 & -3 & 5 \\ -2 & 4 & -6 \end{bmatrix} \qquad 2 \times 3$$

$$\vec{\mathbf{b}} = [-1 \quad 1 \quad 2] \qquad 1 \times 3$$

$$\mathbf{Z} = \mathbf{A}^T\mathbf{W} + \mathbf{B}$$

$$[165 \quad -531 \quad 900]$$

$$z_1^{[1]} \quad z_2^{[1]} \quad z_3^{[1]}$$

$$\mathbf{A} = g(\mathbf{Z})$$

$$[1 \quad 0 \quad 1]$$

A

```
AT = np.array([[200, 17]])
W = np.array([[1, -3, 5],
              [-2, 4, -6]])
b = np.array([[-1, 1, 2]])
```

a_in

```
def dense(AT,W,b,g):
  z = np.matmul(AT,W) + b
  a_out = g(z)
  return a_out
```

a_in

```
[[1,0,1]]
```

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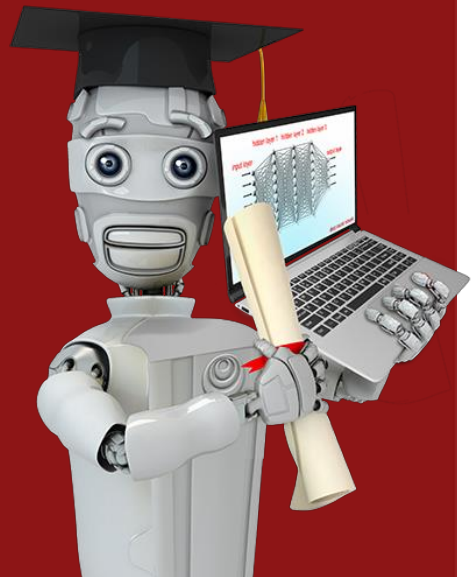
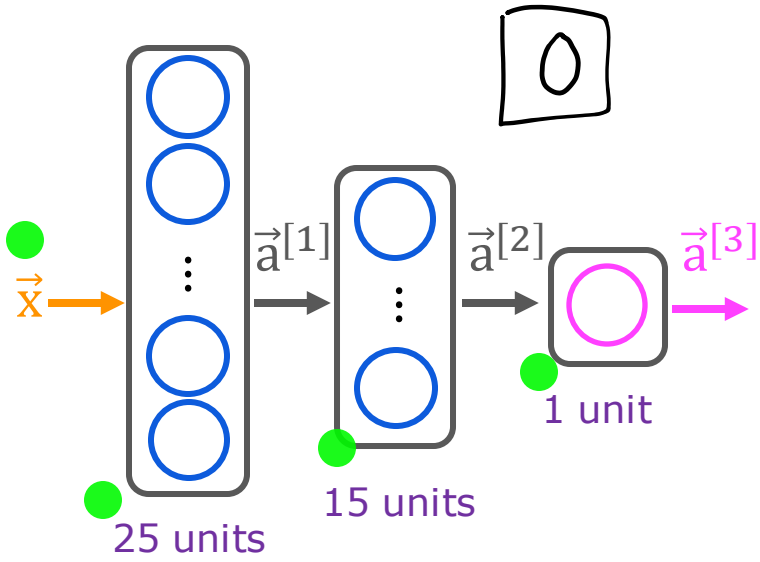


Neural Network Training

TensorFlow implementation

Train a Neural Network in TensorFlow




```
import tensorflow as tf
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense
   model = Sequential([
     Dense(units=25, activation='sigmoid')
     Dense(units=15, activation='sigmoid')
     Dense(units=1, activation='sigmoid')
                              )]
from tensorflow.keras.losses import
BinaryCrossentropy
   model.compile(loss=BinaryCrossentropy())

   model.fit(X,Y,epochs=100)
```

0 1 2 3

epochs: number of steps in gradient descent

Given set of (x,y) examples

How to build and train this in code?

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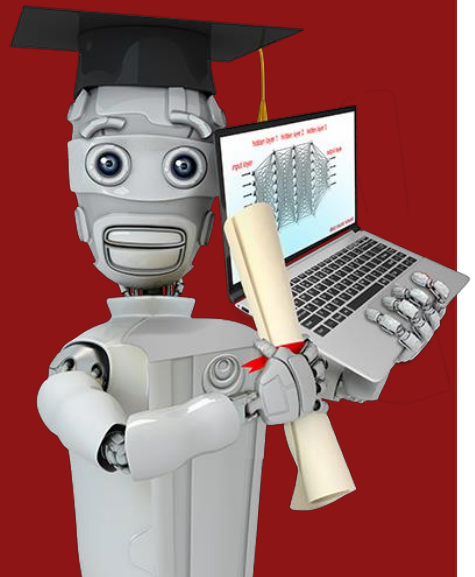


Neural Network Training

Training Details

Model Training Steps

TensorFlow

1. specify how to compute output given input x and parameters w,b (define model)

$f_{\vec{w},b}(\vec{x}) = ?$

2. specify loss and cost

$L\left(f_{\vec{w},b}(\vec{x}), y\right)$ 1 example

$$J(\vec{w}, b) = \frac{1}{m}\sum_{i=1}^{m} L\left(f_{\vec{w},b}(\vec{x}^{(i)}), y^{(i)}\right)$$

3. Train on data to minimize $J(\vec{w}, b)$

logistic regression

```
z = np.dot(w,x) + b

f_x = 1/(1+np.exp(-z))
```

logistic loss

```
loss = -y * np.log(f_x)
   -(1-y) * np.log(1-f_x)
```

```
w = w - alpha * dj_dw
b = b - alpha * dj_db
```

neural network

```
model = Sequential([
          Dense(...)
          Dense(...)
          Dense(...)
                    ])
```

binary cross entropy

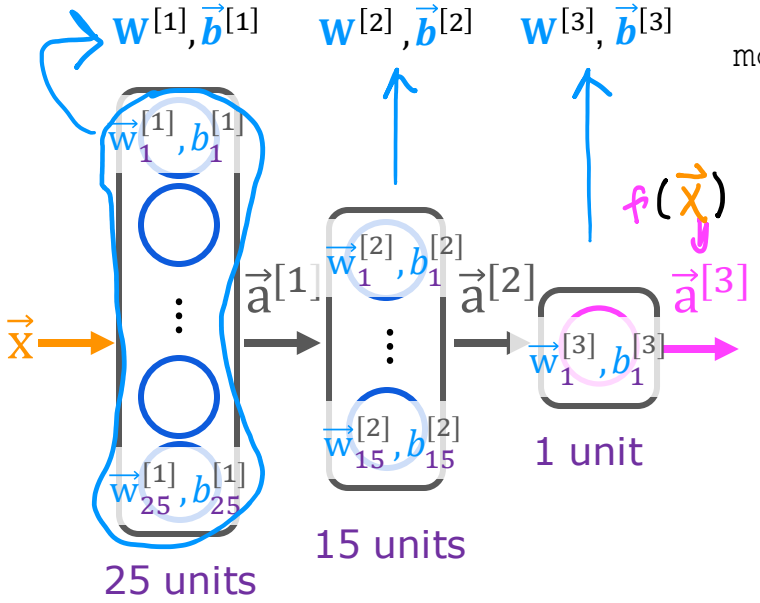
```
model.compile(
loss=BinaryCrossentropy())
```

```
model.fit(X,y,epochs=100)
```

1. Create the model

define the model

$f(\vec{x}) = ?$




```
import tensorflow as tf
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense

model = Sequential([
    Dense(units=25, activation='sigmoid')
    Dense(units=15, activation='sigmoid')
    Dense(units=1, activation='sigmoid')
                    ])
```

2. Loss and cost functions

Mnist digit classification problem

binary classification

$$L(f(\vec{x}), y) = -y\log(f(\vec{x})) - (1-y)\log(1-f(\vec{x}))$$

Compare prediction vs. target

logistic loss
also known as binary cross entropy

$$J(\mathbf{W}, \mathbf{B}) = \frac{1}{m}\sum_{i=1}^{m} L(f(\vec{x}^{(i)}), y^{(i)})$$

$\mathbf{W}^{[1]}, \mathbf{W}^{[2]}, \mathbf{W}^{[3]}$ $\vec{b}^{[1]}, \vec{b}^{[2]}, \vec{b}^{[3]}$

$f_{W,B}(\vec{x})$

```
model.compile(loss= BinaryCrossentropy())
```

```
from tensorflow.keras.losses import
  BinaryCrossentropy
```

K Keras

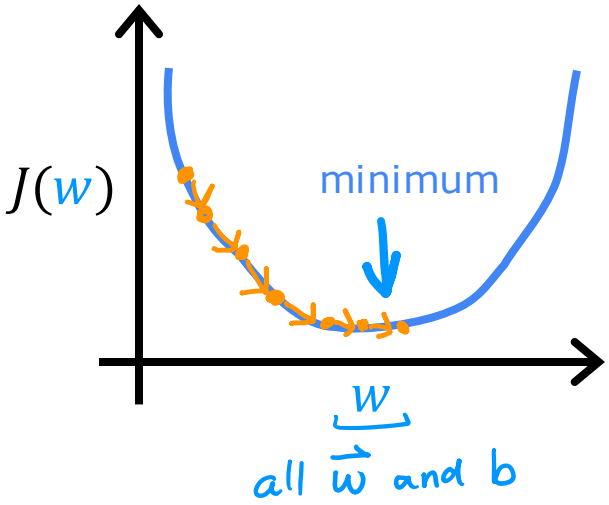
regression
(predicting numbers
and not categories)

mean squared error

```
model.compile(loss= MeanSquaredError())
```

```
from tensorflow.keras.losses import
  MeanSquaredError
```

3. Gradient descent



repeat {

$$w_j^{[l]} = w_j^{[l]} - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$$

$$b_j^{[l]} = b_j^{[l]} - \alpha \frac{\partial}{\partial b} J(\vec{w}, b)$$

}

Compute derivatives for gradient descent using "back propagation"

```
model.fit(X,y,epochs=100)
```

Neural network libraries

Use code libraries instead of coding ”from scratch”

Good to understand the implementation
(for tuning and debugging).

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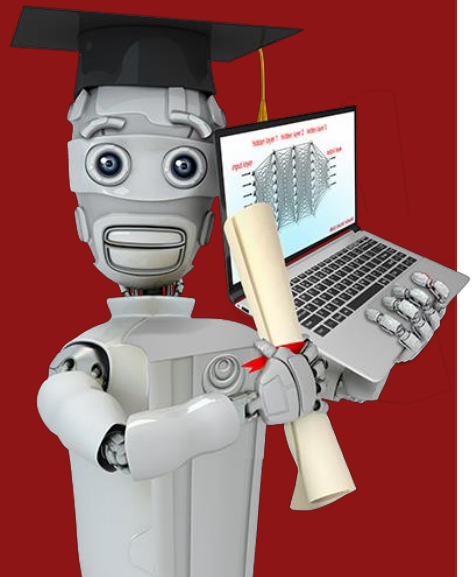
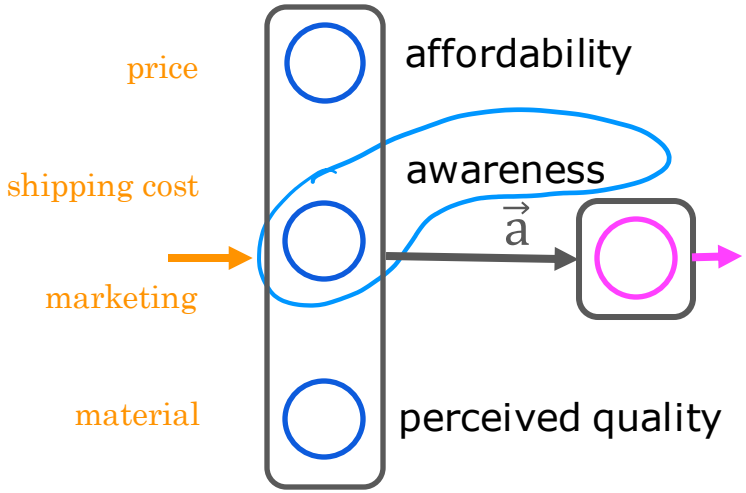


Activation Functions

Alternatives to the sigmoid activation

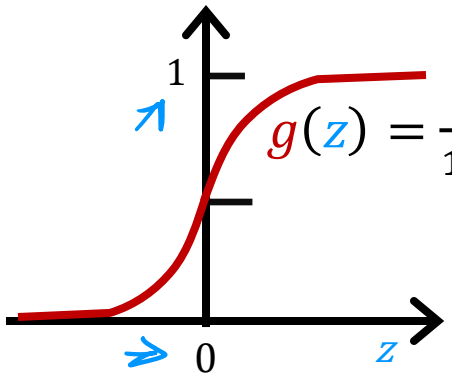
Demand Prediction Example



$$a_2^{[1]} = g(\vec{w}_2^{[1]} \cdot \vec{x} + b_2^{[1]})$$

z

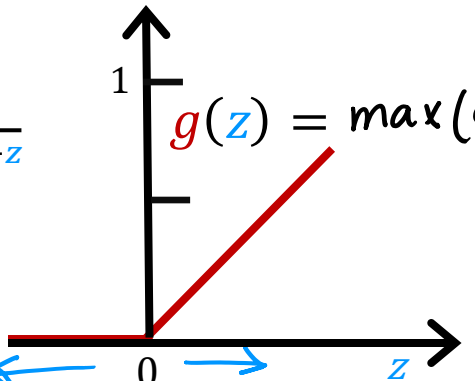
Sigmoid



$$g(z) = \frac{1}{1+e^{-z}}$$

$$0 < g(z) < 1$$

ReLU Rectified Linear Unit



$$g(z) = \max(0, z)$$

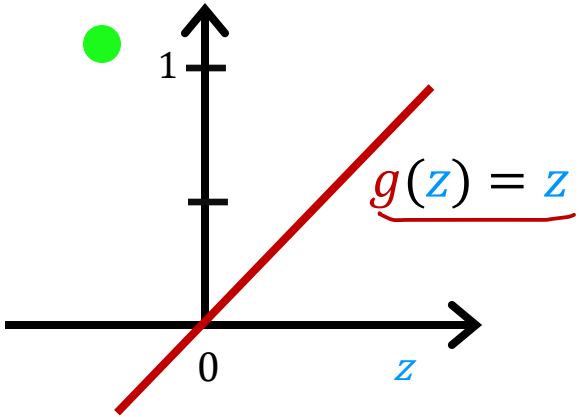
if $z < 0$, $g(z)$ is 0

if $z \geq 0$, $g(z)$ is z

Examples of Activation Functions

"No activation function"

Linear activation function



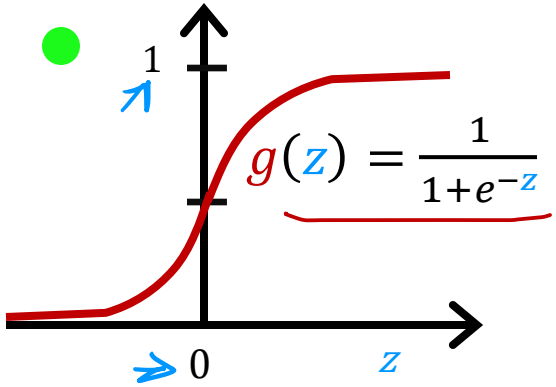
$a = g(z) = \vec{w} \cdot \vec{x} + b$

z

$a_2^{[1]} = g(\vec{w}_2^{[1]} \cdot \vec{x} + b_2^{[1]})$

z

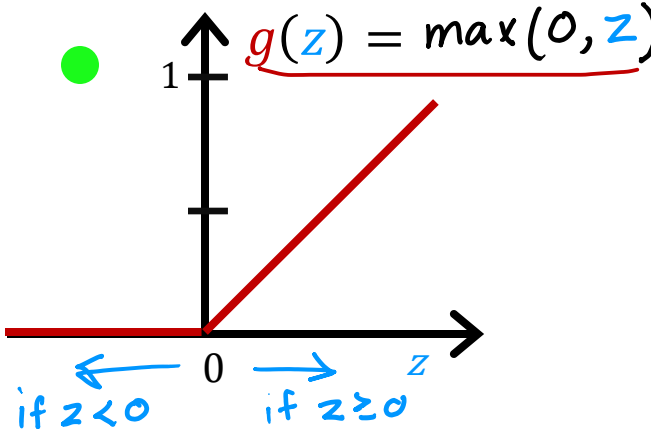
Sigmoid



$0 < g(z) < 1$

Later: softmax activation

ReLU Rectified Linear Unit



if $z < 0$ $g(z)$ is 0

if $z \geq 0$ $g(z)$ is z

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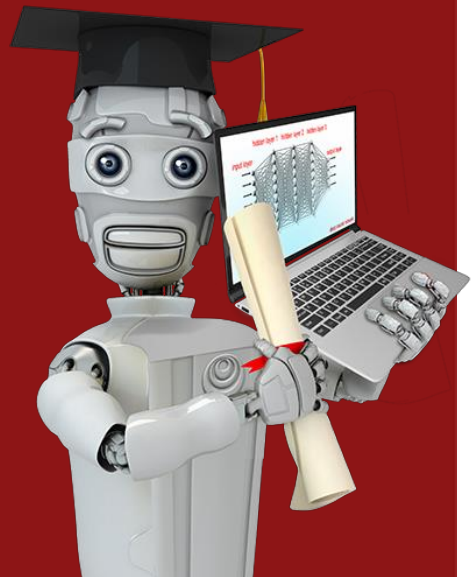
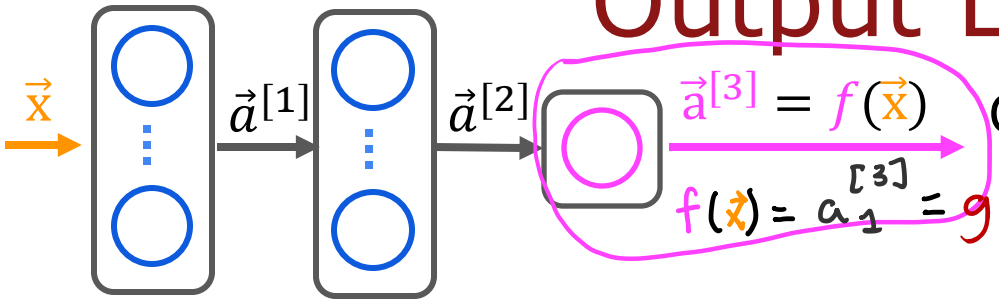


Activation Functions

Choosing activation functions

Output Layer

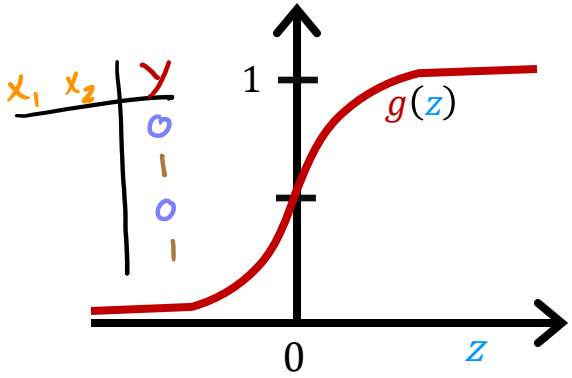


Choosing $g(z)$ for output layer?

Binary classification

Sigmoid

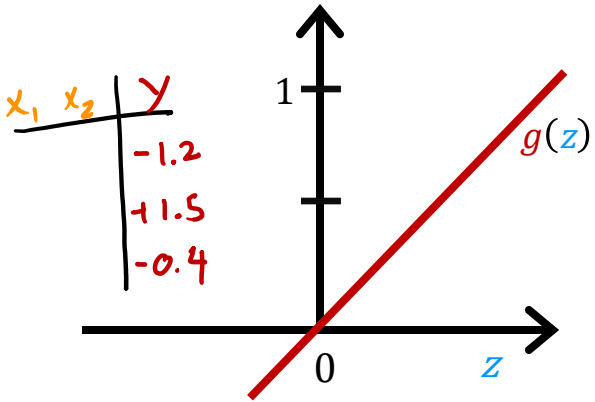
y=0/1



Regression

Linear activation function

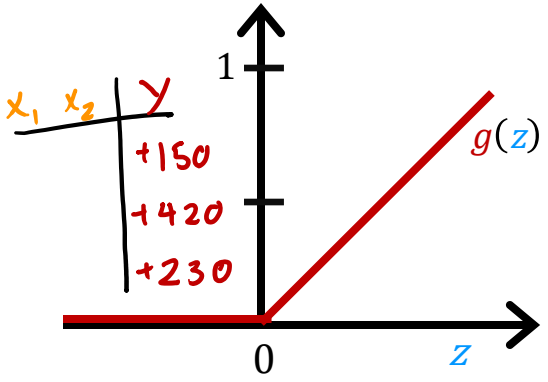
y = +/-



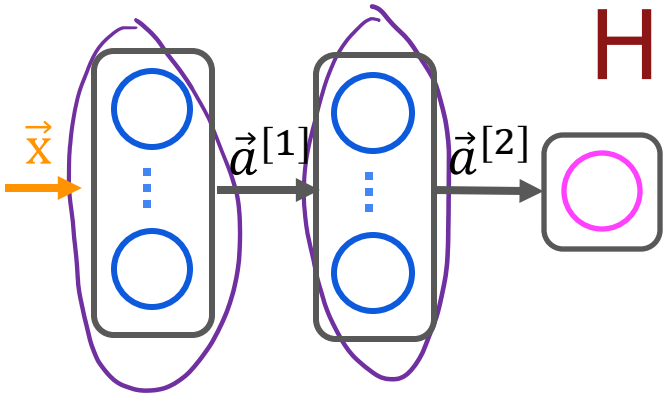
Regression

ReLU

Y = 0 or +



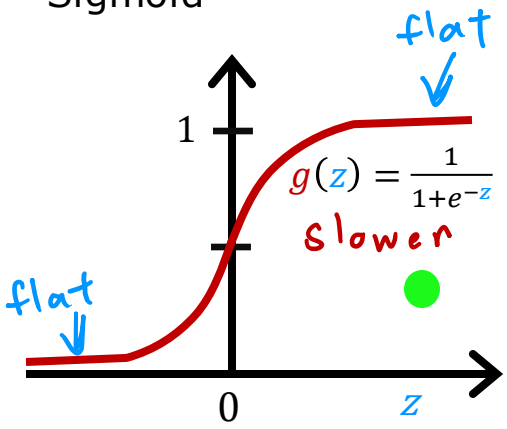
Hidden Layer

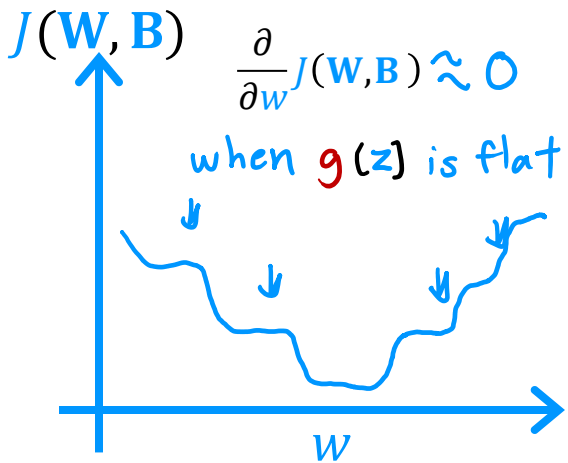


$\vec{a}^{[3]} = f(\vec{x})$

Choosing $g(z)$ for hidden layer

Sigmoid

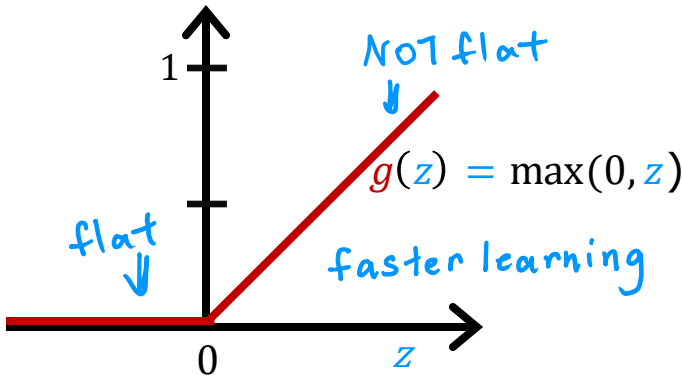




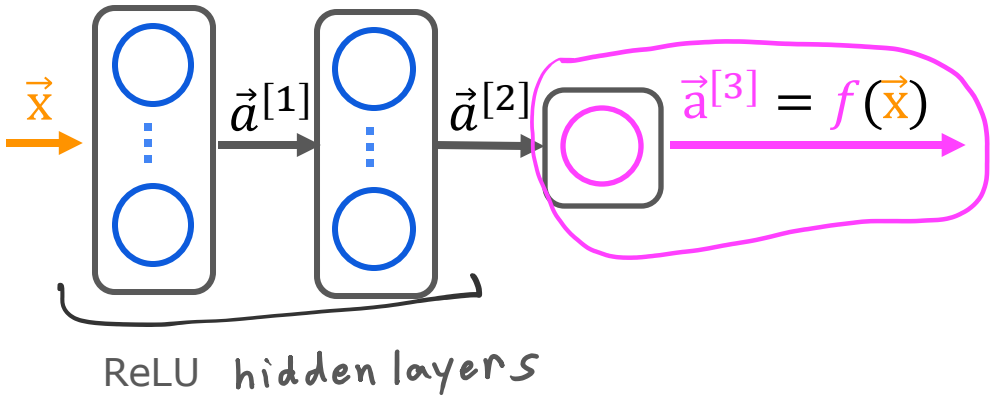
most common choice

ReLU

faster



Choosing Activation Summary



ReLU hidden layers

binary classification

activation='sigmoid'

regression y negative/positive

activation='linear'

regression $y \geq 0$

activation='relu'

```
from tf.keras.layers import Dense
model = Sequential([
  Dense(units=25, activation='relu'),    layer1
  Dense(units=15, activation='relu'),    layer2
  Dense(units=1,  activation='sigmoid')  layer3
])
```

or 'linear'
or 'relu'

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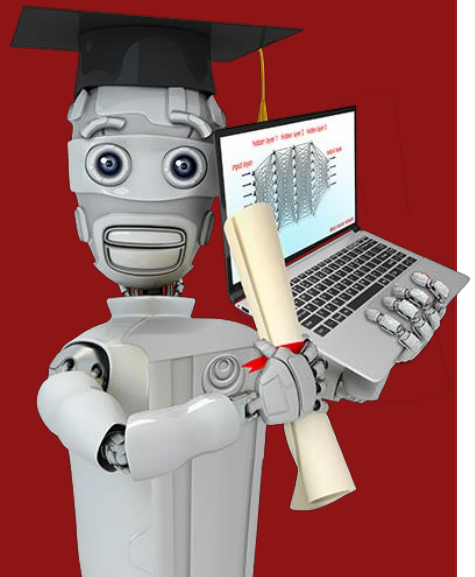
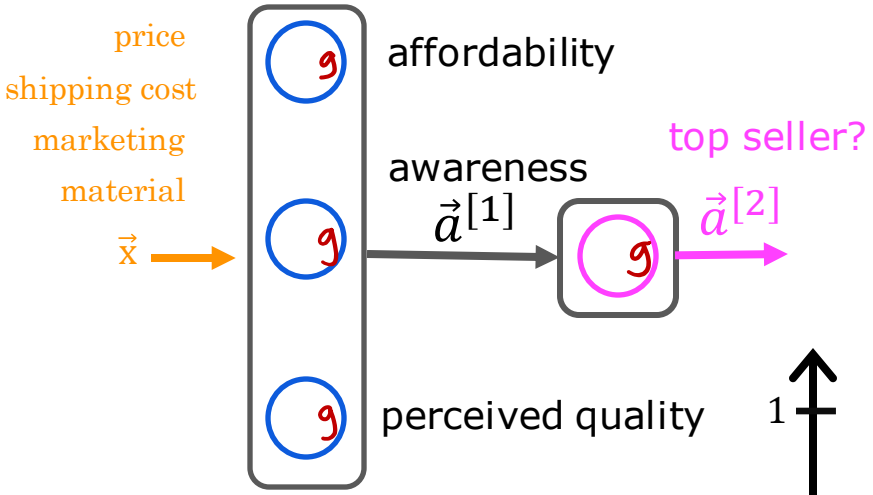
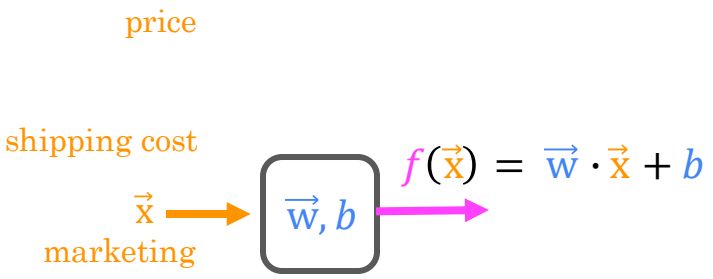


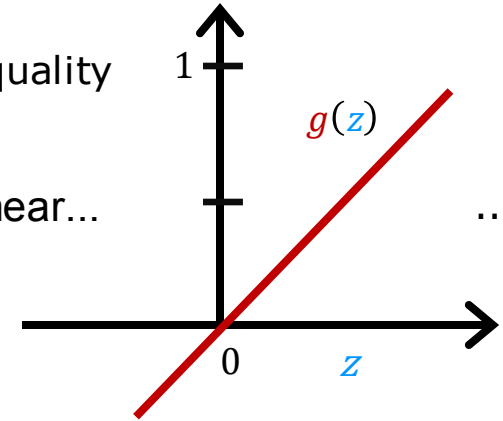
Activation Functions

Why do we need activation functions?

Why do we need activation functions?



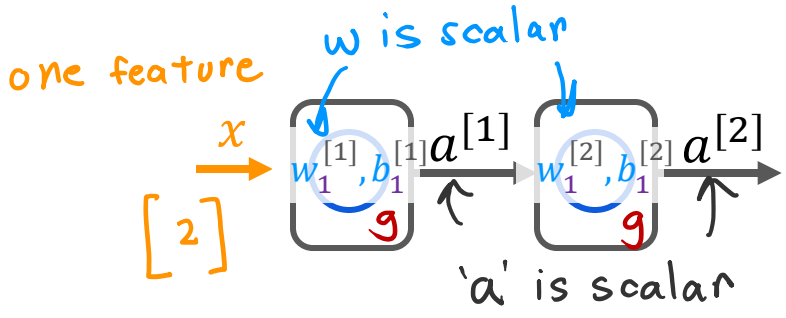




if all $g(z)$ are linear...

...no different than linear regression

Linear Example



$$g(z) = z$$

$$a^{[1]} = w_1^{[1]} x + b_1^{[1]}$$

$$a^{[2]} = w_1^{[2]} a^{[1]} + b_1^{[2]}$$

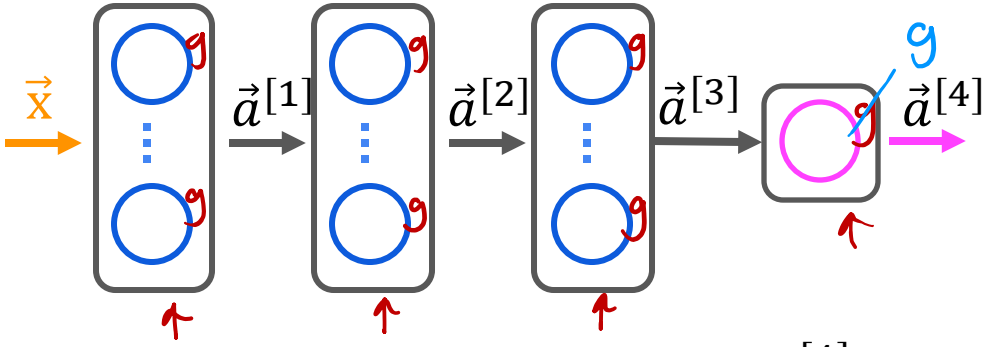
$$= w_1^{[2]} (w_1^{[1]} x + b_1^{[1]}) + b_1^{[2]}$$

$$\vec{a}^{[2]} = \underbrace{(\vec{w}_1^{[2]} \vec{w}_1^{[1]})}_{w} x + \underbrace{w_1^{[2]} b_1^{[1]} + b_1^{[2]}}_{b}$$

$$\vec{a}^{[2]} = w x + b$$

$f(x) = wx + b$ linear regression

Example



$g(z) = z$

$$\vec{a}^{[4]} = \vec{w}_1^{[4]} \cdot \vec{a}^{[3]} + b_1^{[4]}$$

all linear (including output)
↳ equivalent to linear regression

$$\vec{a}^{[4]} = \frac{1}{1+e^{-(\vec{w}_1^{[4]} \cdot \vec{a}^{[3]} + b_1^{[4]})}}$$

output activation is sigmoid
(hidden layers still linear)
↳ equivalent to logistic regression

Don't use linear activations in hidden layers (use ReLU)

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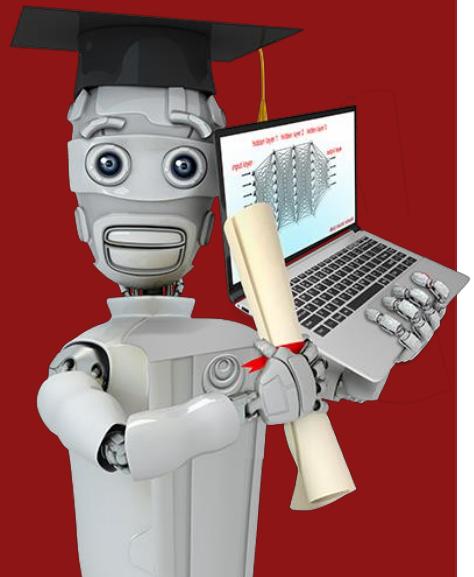
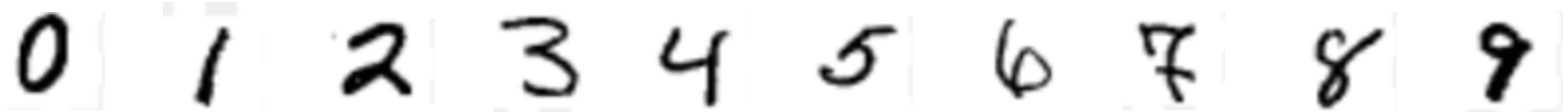


Multiclass Classification

Multiclass

MNIST example

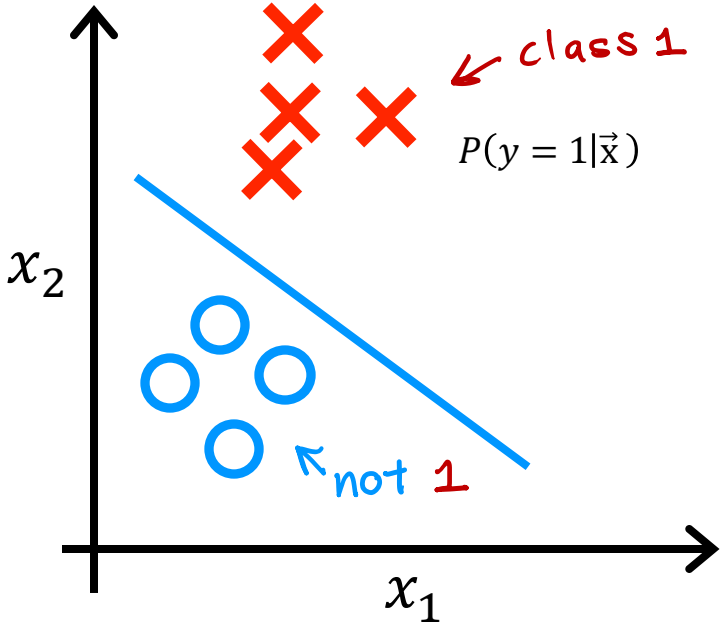


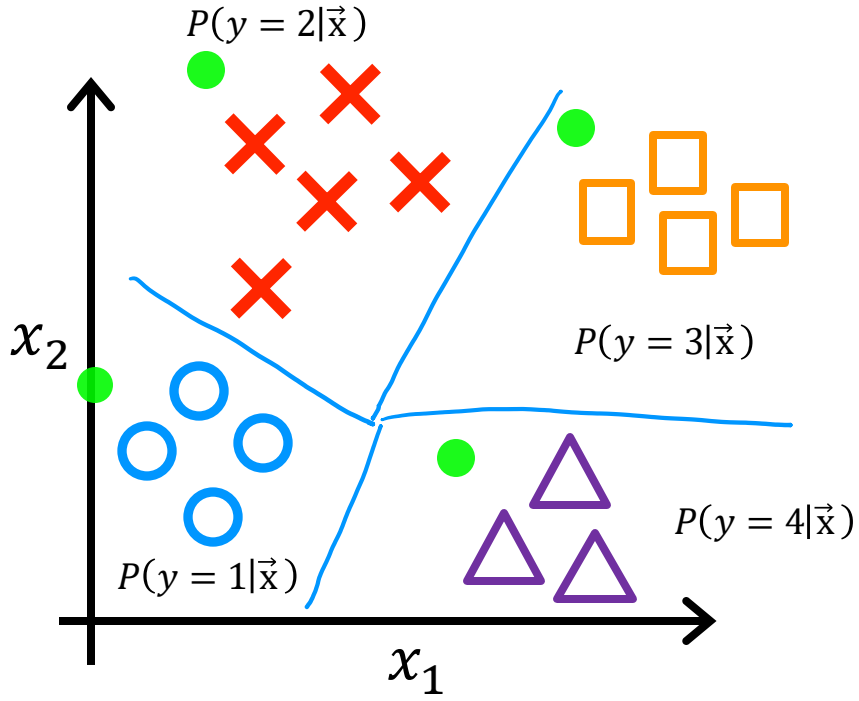
$y = 0$ 1 2 3 4 5 6 7 8 9

$y = 7$

multiclass classification problem:
target y can take on more than two possible values

Multiclass classification example





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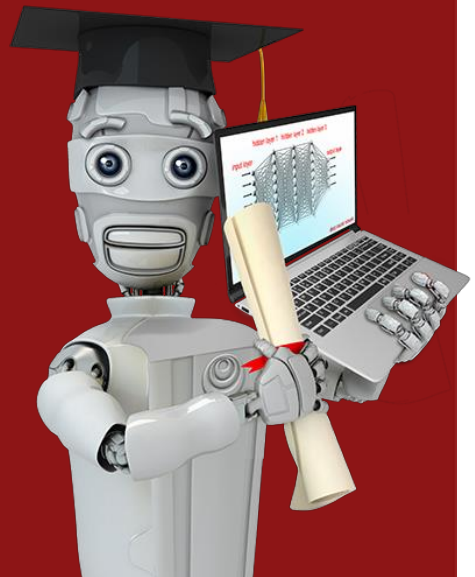


Multiclass Classification

Softmax

Logistic regression (2 possible output values)

$z = \vec{w} \cdot \vec{x} + b$

$a_1 = g(z) = \frac{1}{1+e^{-z}} = P(y=1|\vec{x})$ 0.71

$a_2 = 1 - a_1 = P(y=0|\vec{x})$ 0.29

Softmax regression (N possible outputs) $y = 1, 2, 3, \ldots, N$

$z_j = \vec{w}_j \cdot \vec{x} + b_j \quad j = 1, \ldots, N$

parameters $w_1, w_2, \ldots, w_N$ $b_1, b_2, \ldots, b_N$

$a_j = \frac{e^{z_j}}{\sum_{k=1}^{N} e^{z_k}} = P(y=j|\vec{x})$

note: $a_1 + a_2 + \ldots + a_N = 1$

Softmax regression (4 possible outputs) $y = 1, 2, 3, 4$

$z_1 = \vec{w}_1 \cdot \vec{x} + b_1$

$a_1 = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3} + e^{z_4}} = P(y=1|\vec{x})$ 0.30

$z_2 = \vec{w}_2 \cdot \vec{x} + b_2$

$a_2 = \frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3} + e^{z_4}} = P(y=2|\vec{x})$ 0.20

$z_3 = \vec{w}_3 \cdot \vec{x} + b_3$

$a_3 = \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3} + e^{z_4}} = P(y=3|\vec{x})$ 0.15

$z_4 = \vec{w}_4 \cdot \vec{x} + b_4$

$a_4 = \frac{e^{z_4}}{e^{z_1} + e^{z_2} + e^{z_3} + e^{z_4}} = P(y=4|\vec{x})$ 0.35

Cost

Logistic regression

$$z = \vec{w} \cdot \vec{x} + b$$

$$a_1 = g(z) = \frac{1}{1 + e^{-z}} = P(y = 1|\vec{x})$$

$$a_2 = 1 - a_1 = P(y = 0|\vec{x})$$

$$loss = -y \log a_1 - (1 - y) \log(1 - a_1)$$

if $y = 1$ if $y = 0$

$$J(\vec{w}, b) = \text{average loss}$$

Softmax regression

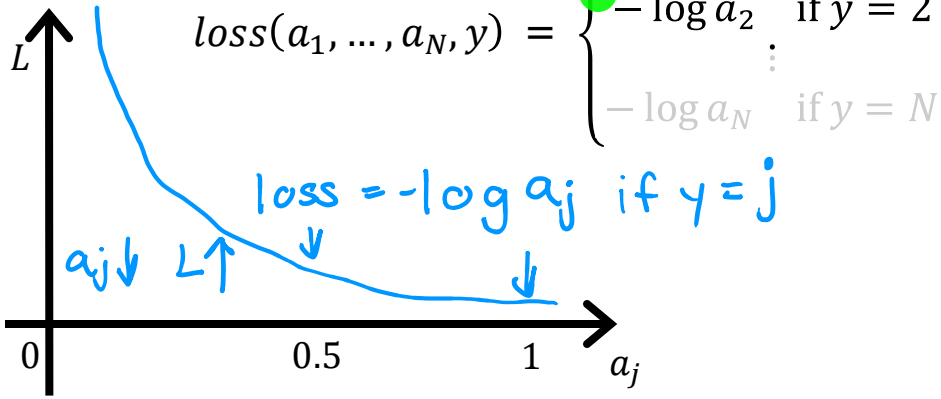
$$a_1 = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + \cdots + e^{z_N}} = P(y = 1|\vec{x})$$

$$\vdots$$

$$a_N = \frac{e^{z_N}}{e^{z_1} + e^{z_2} + \cdots + e^{z_N}} = P(y = N|\vec{x})$$

Crossentropy loss

$$loss(a_1, \ldots, a_N, y) = \begin{cases} -\log a_1 & \text{if } y = 1 \\ -\log a_2 & \text{if } y = 2 \\ \vdots & \\ -\log a_N & \text{if } y = N \end{cases}$$



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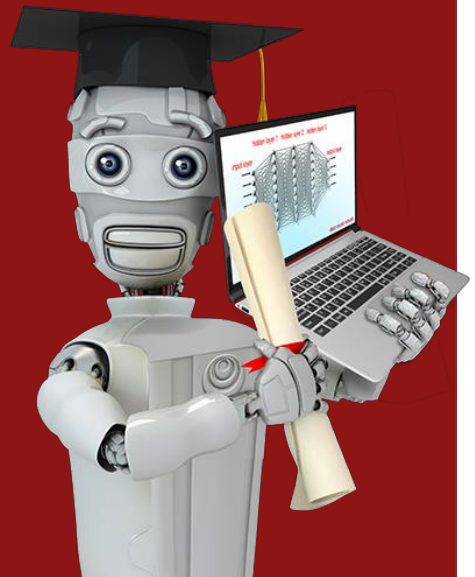
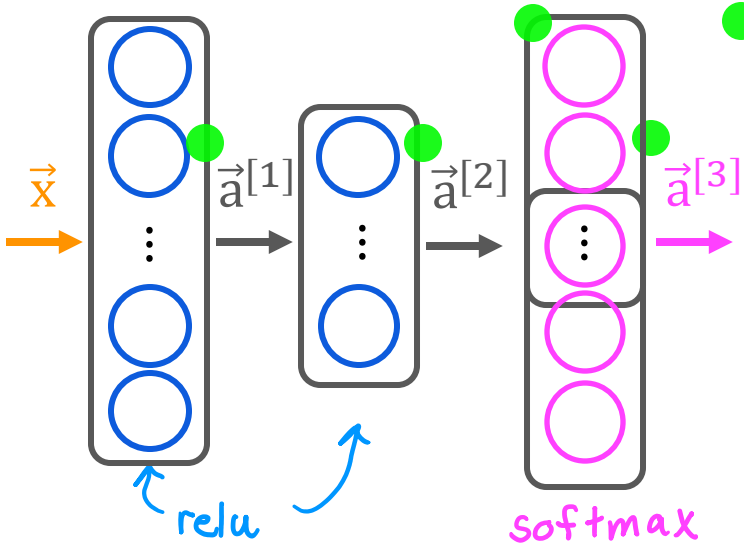


Multiclass Classification

Neural Network with Softmax output

Neural Network with Softmax output



$$z_1 = \vec{w}_1 \cdot \vec{a}^{[2]} + b_1 \qquad a_1 = \frac{e^{z_1}}{e^{z_1} + \cdots + e^{z_{10}}} = P(y = 1|\vec{x})$$

$$\vdots$$

$$z_{10} = \vec{w}_{10} \cdot \vec{a}^{[2]} + b_{10} \qquad a_{10} = \frac{e^{z_{10}}}{e^{z_1} + \cdots + e^{z_{10}}} = P(y = 10|\vec{x})$$

logistic regression

$$a_1^{[3]} = g\left(z_1^{[3]}\right) \quad a_2^{[3]} = g\left(z_2^{[3]}\right)$$

softmax

$$\vec{a}^{[3]} = \left(a_1^{[3]}, \ldots a_{10}^{[3]}\right) = g\left(z_1^{[3]}, \ldots, z_{10}^{[3]}\right)$$

MNIST with softmax

1. specify the model

$f_{\vec{w},b}(\vec{x}) = ?$

```
import tensorflow as tf
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense
model = Sequential([
  Dense(units=25, activation='relu')
  Dense(units=15, activation='relu')
  Dense(units=10, activation='softmax')
                  )]
```

2. specify loss and cost

$L\left(f_{\vec{w},b}(\vec{x}), y\right)$

```
from tensorflow.keras.losses import
  SparseCategoricalCrossentropy
model.compile(loss= SparseCategoricalCrossentropy() )
```

3. Train on data to minimize $J(\vec{w}, b)$

```
model.fit(X,Y,epochs=100)
```

Note: better (recommended) version later.

Don't use the version shown here!

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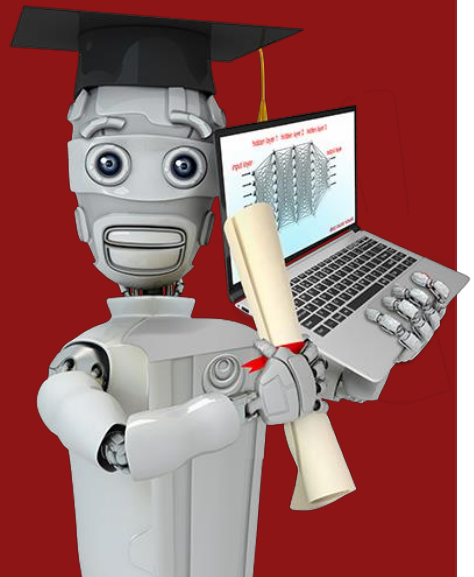

Multiclass Classification

Improved implementation of softmax

Numerical Roundoff Errors

option 1

$$x = \frac{2}{10{,}000}$$

option 2

$$x = \left(1 + \frac{1}{10{,}000}\right) - \left(1 - \frac{1}{10{,}000}\right) =$$

Numerical Roundoff Errors

More numerically accurate implementation of logistic loss:

Logistic regression:

$$a = g(z) = \frac{1}{1+e^{-z}}$$

Original loss

$$loss = -y\log(a) - (1-y)\log(1-a)$$

More accurate loss (in code)

$$loss = -y\log\left(\frac{1}{1+e^{-z}}\right) - (1-y)\log\left(1-\frac{1}{1+e^{-z}}\right)$$

logit: z

```
model = Sequential([
    Dense(units=25, activation='relu')
    Dense(units=15, activation='relu')
    Dense(units=10, activation='sigmoid')
])
model.compile(loss=BinaryCrossEntropy() )
```

'linear'

```
model.compile(loss=BinaryCrossEntropy(from_logits=True) )
```

More numerically accurate implementation of softmax

Softmax regression

$$(a_1, \ldots, a_{10}) = g(z_1, \ldots, z_{10})$$

$$\text{Loss} = L(\vec{a}, y) = \begin{cases} -\log a_1 & \text{if } y = 1 \\ \vdots & \\ -\log a_{10} & \text{if } y = 10 \end{cases}$$

```
model = Sequential([
    Dense(units=25, activation='relu')
    Dense(units=15, activation='relu')
    Dense(units=10, activation='softmax')
```

'linear'

```
model.compile(loss=SparseCategoricalCrossEntropy() )
```

More Accurate

$$L(\vec{a}, y) = \begin{cases} -\log \dfrac{e^{z_1}}{e^{z_1} + \cdots + e^{z_{10}}} & \text{if } y = 1 \\ \vdots & \\ -\log \dfrac{e^{z_{10}}}{e^{z_1} + \cdots + e^{z_{10}}} & \text{if } y = 10 \end{cases}$$

```
model.compile(loss=SparseCrossEntropy(from_logits=True) )
```

MNIST (more numerically accurate)

model

```
import tensorflow as tf
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense
model = Sequential([
  Dense(units=25, activation='relu')
  Dense(units=15, activation='relu')
  Dense(units=10, activation='linear') )]
```

loss

```
from tensorflow.keras.losses import
  SparseCategoricalCrossentropy
model.compile(...,loss=SparseCategoricalCrossentropy(from_logits=True) )
```

fit

```
model.fit(X,Y,epochs=100)
```

predict

```
logits = model(X)
f_x = tf.nn.softmax(logits)
```

not $a_1 \ldots a_{10}$ is $z_1 \ldots z_{10}$

logistic regression (more numerically accurate)

model

```
model = Sequential([
  Dense(units=25, activation='sigmoid')
  Dense(units=15, activation='sigmoid')
  Dense(units=1, activation='linear')
from tensorflow.keras.losses import
  BinaryCrossentropy
```

loss

```
model.compile(..., BinaryCrossentropy(from_logits=True)) )
```

```
model.fit(X,Y,epochs=100)
```

fit

```
logit = model(X)
```

predict

```
f_x = tf.nn.sigmoid(logit)
```

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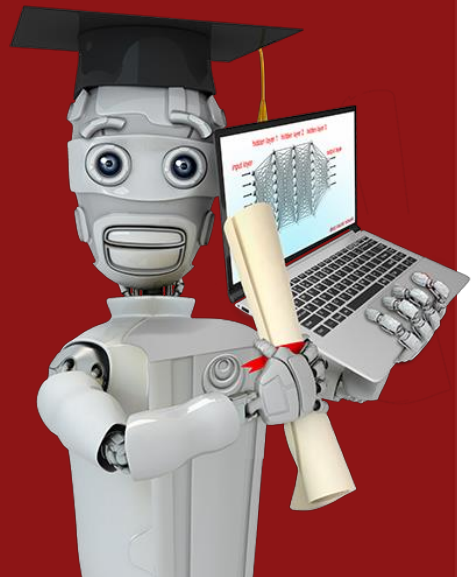


Multi-label Classification

Classification with multiple outputs (Optional)

Multi-label Classification

x

Is there a car?
Is there a bus?
Is there a pedestrian

yes
no
yes

$y = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

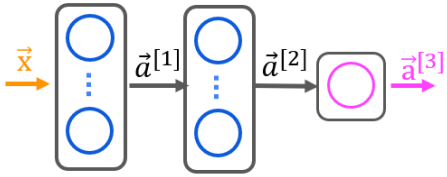
no
no
yes

$y = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

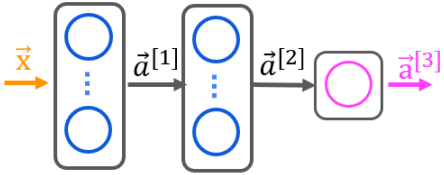
yes
yes
no

$y = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

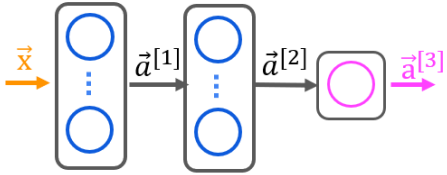
Multiple classes



car

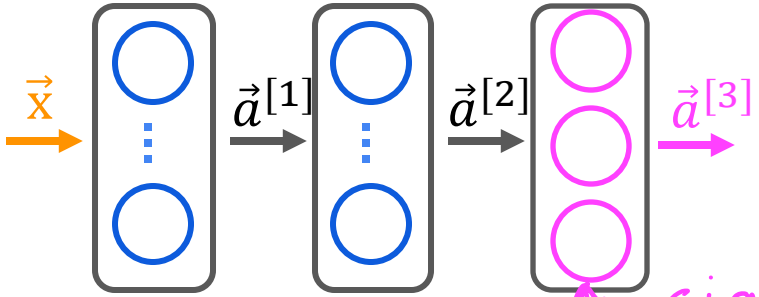


bus



pedestrian

Alternatively, train one neural network with three outputs



$$\vec{a}^{[3]} = \begin{bmatrix} a_1^{[3]} \\ a_2^{[3]} \\ a_3^{[3]} \end{bmatrix}$$

car
bus
pedestrian

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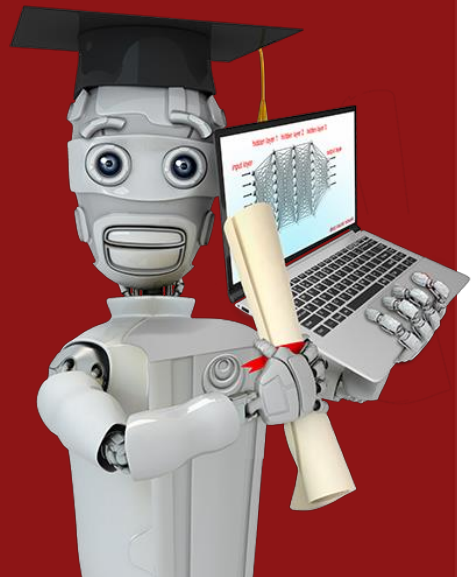


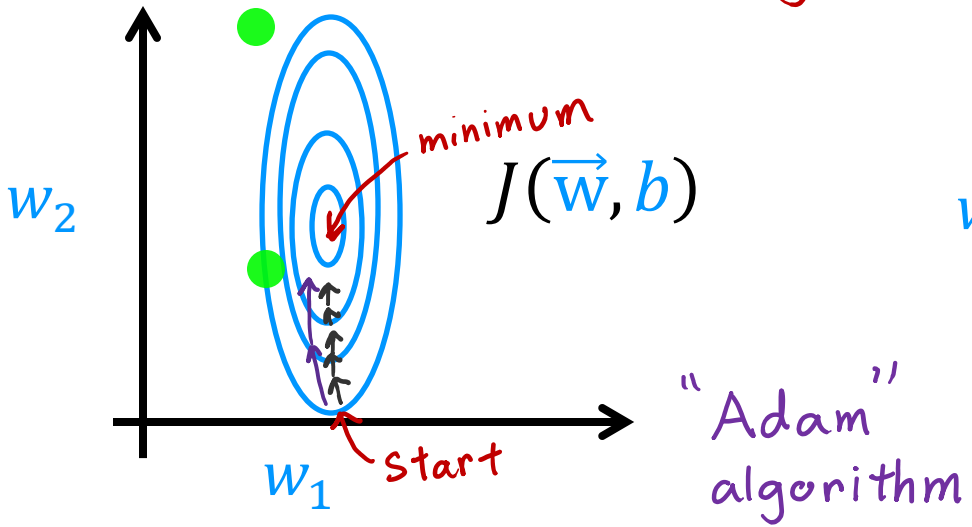
Additional Neural Network Concepts

Advanced Optimization

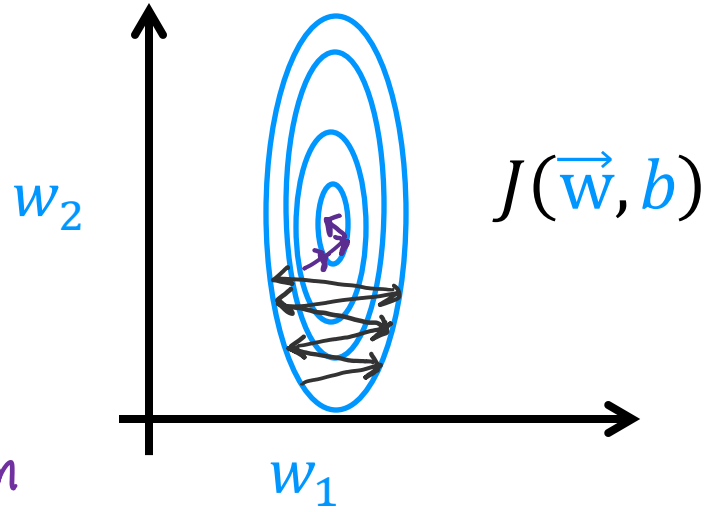
Gradient Descent

$$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$$

learning rate



"Adam" algorithm



Go faster – increase α

Go slower – decrease α

Adam Algorithm Intuition

Adam: Adaptive Moment estimation

not just one α

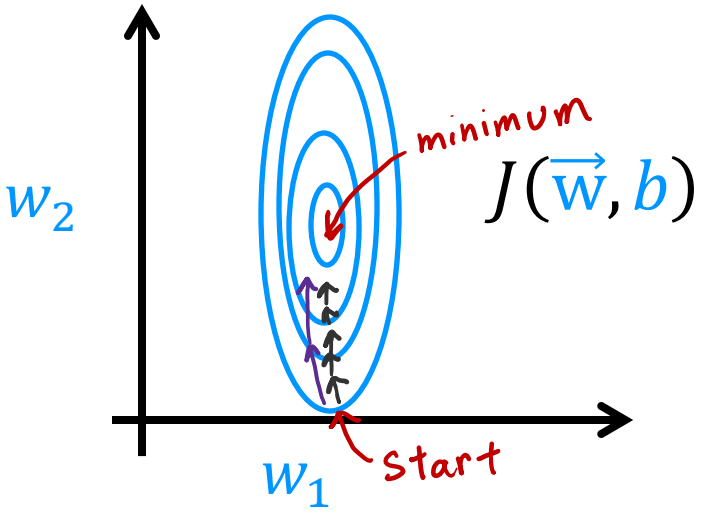
$$w_1 = w_1 - \alpha_1 \frac{\partial}{\partial w_1} J(\vec{w}, b)$$

$$\vdots$$

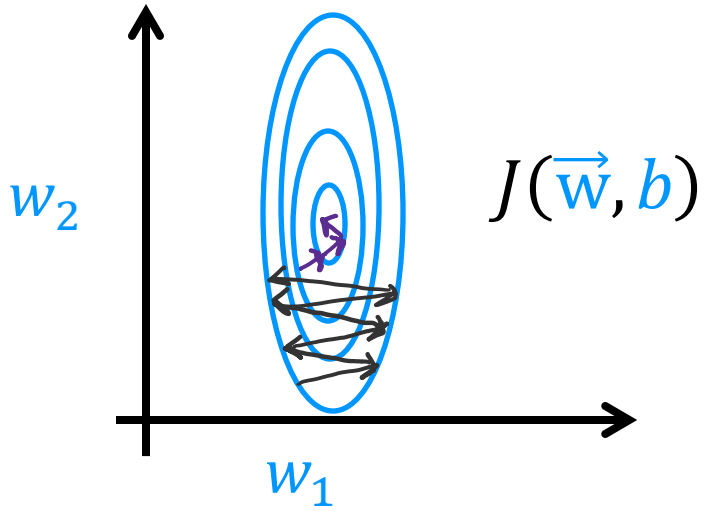
$$w_{10} = w_{10} - \alpha_{10} \frac{\partial}{\partial w_{10}} J(\vec{w}, b)$$

$$b = b - \alpha_{11} \frac{\partial}{\partial b} J(\vec{w}, b)$$

Adam Algorithm Intuition



If w_j (or b) keeps moving in same direction, increase α_j.



If w_j (or b) keeps oscillating, reduce α_j.

MNIST Adam

model

```
model = Sequential([
          tf.keras.layers.Dense(units=25, activation='sigmoid')
          tf.keras.layers.Dense(units=15, activation='sigmoid')
          tf.keras.layers.Dense(units=10, activation='linear')
])
```

compile

$\alpha = 10^{-3} = 0.001$

```
model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=1e-3),
  loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=True))
```

fit

```
model.fit(X,Y,epochs=100)
```

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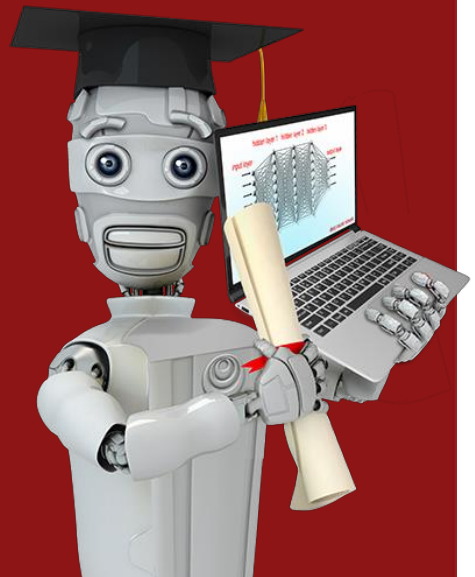
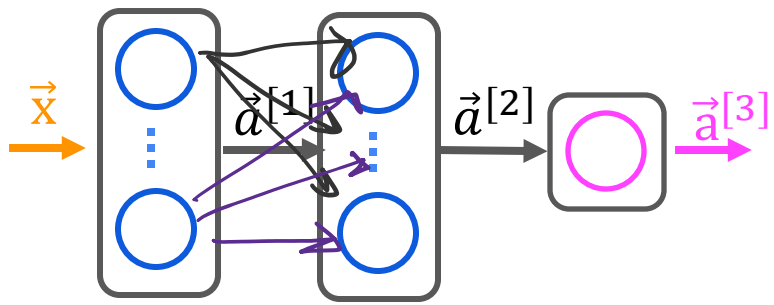


Additional Neural Network Concepts

Additional Layer Types

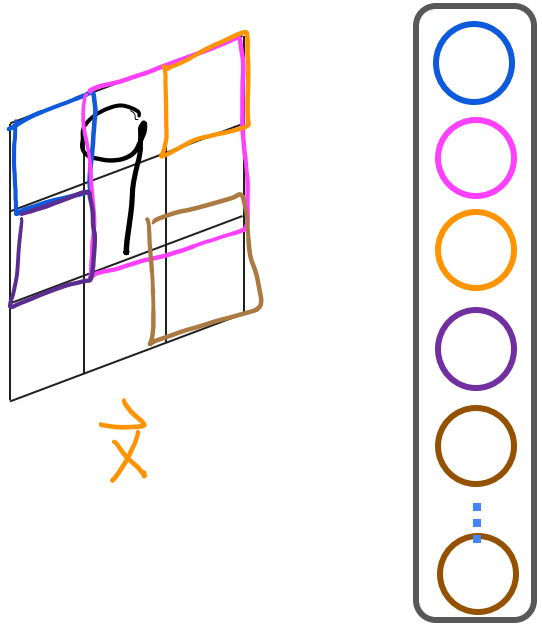
Dense Layer



Each neuron output is a function of all the activation outputs of the previous layer.

$$\vec{a}_1^{[2]} = g\left(\vec{w}_1^{[2]} \cdot \vec{a}^{[1]} + b_1^{[2]}\right)$$

Convolutional Layer

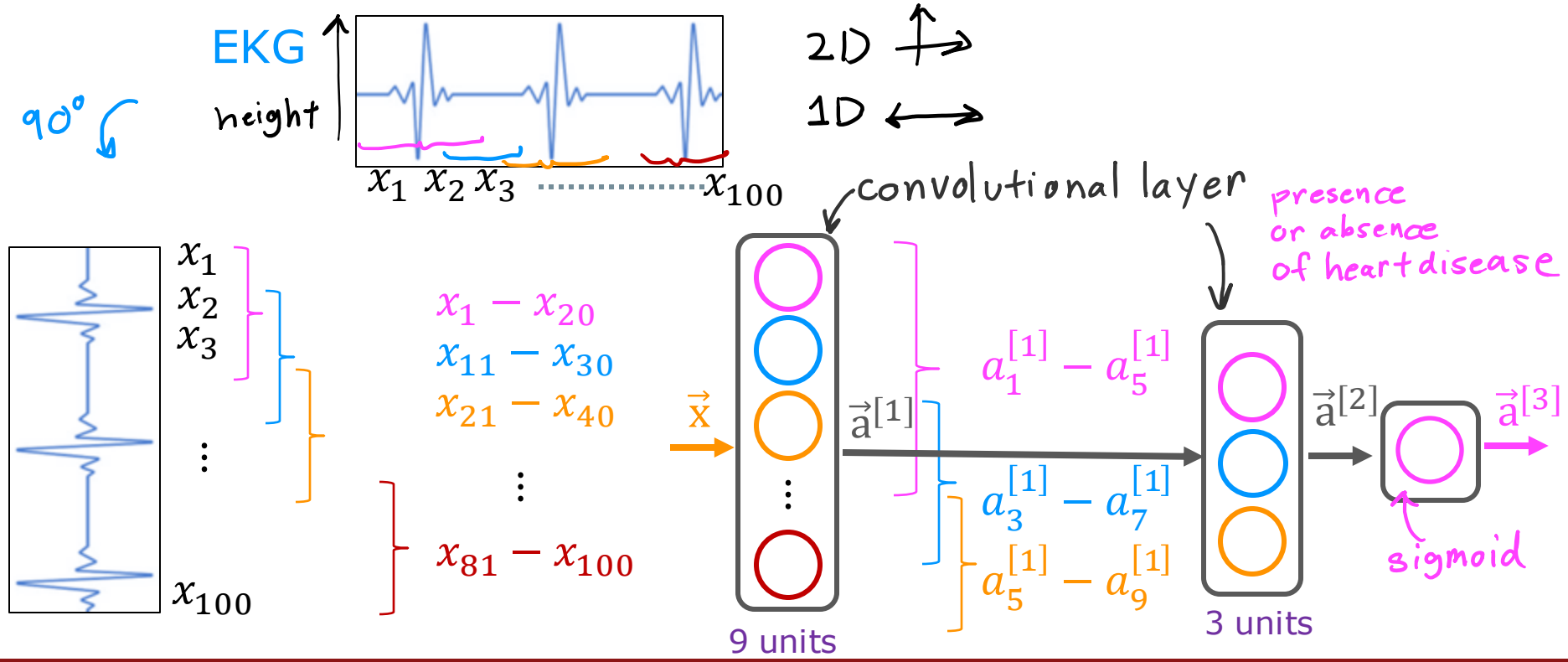


Each Neuron only looks at part of the previous layer's inputs.

Why?

- Faster computation
- Need less training data (less prone to overfitting)

Convolutional Neural Network



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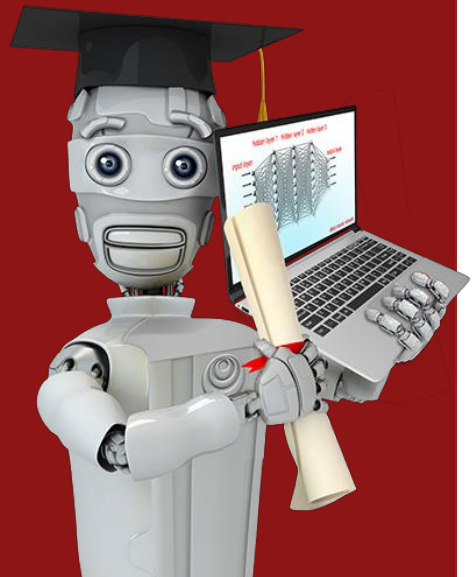

Advice for applying machine learning

Deciding what to try next

Debugging a learning algorithm

You've implemented regularized linear regression on housing prices

$$J(\vec{w}, b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right) - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$

But it makes unacceptably large errors in predictions. What do you try next?

- Get more training examples
- Try smaller sets of features
- Try getting additional features
- Try adding polynomial features $(x_1^2, x_2^2, x_1x_2, etc)$
- Try decreasing λ
- Try increasing λ

Machine learning diagnostic

Diagnostic: A test that you run to gain insight into what is/isn't working with a learning algorithm, to gain guidance into improving its performance.

Diagnostics can take time to implement but doing so can be a very good use of your time.

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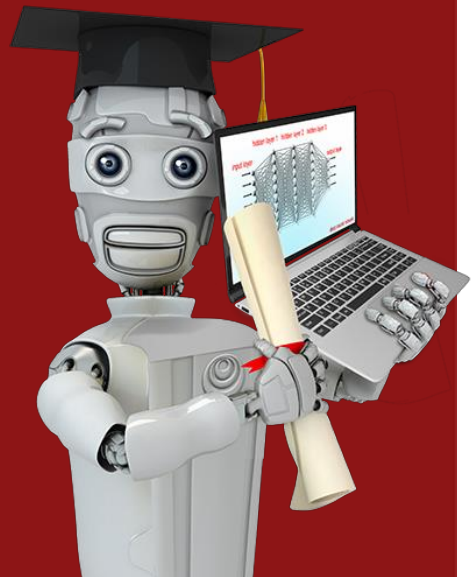
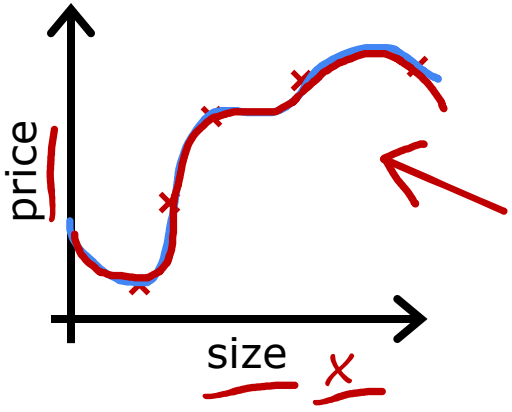


Evaluating and choosing models

Evaluating a model

Evaluating your model



→ Model fits the training data well but will fail to generalize to new examples not in the training set.

$x = x_1$

$$f_{\vec{w},b}(\vec{x}) = w_1 x + w_2 x^2 + \cdots + w_n x^n + b$$

just x

$w_n x^4$

- x_1 = size in feet^2
- x_2 = no. of bedrooms
- x_3 = no. of floors
- x_4 = age of home in years

$f(\vec{x})$

Evaluating your model

Dataset:

size	price
2104	400
1600	330
2400	369
1416	232
3000	540
1985	300
1534	315
1427	199
1380	212
1494	243

70%

30%

training set

test set

$(x^{(1)}, y^{(1)})$
$(x^{(2)}, y^{(2)})$
$\vdots$
$(x^{(m_{train})}, y^{m_{train}})$

m_{train} = no. training examples = 7

$(x_{test}^{(1)}, y_{test}^{(1)})$
$\vdots$
$(x_{test}^{(m_{test})}, y_{test}^{(m_{test})})$

m_{test} = no. test examples = 3

Train/test procedure for linear regression (with squared error cost)

Fit parameters by minimizing cost function $J(\vec{w}, b)$

$$J(\vec{w}, b) = \min_{\vec{w}, b} \left[\frac{1}{2m_{train}} \sum_{i=1}^{m_{train}} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)} \right)^2 + \frac{\lambda}{2m_{train}} \sum_{j=1}^{n} w_j^2 \right]$$

Compute test error:

$$J_{test}(\vec{w}, b) = \frac{1}{2m_{test}} \left[\sum_{i=1}^{m_{test}} \left(f_{\vec{w},b}(\vec{x}^{(i)}_{test}) - y^{(i)}_{test} \right)^2 \right]$$

Compute training error:

$$J_{train}(\vec{w}, b) = \frac{1}{2m_{train}} \left[\sum_{i=1}^{m_{train}} \left(f_{\vec{w},b}(\vec{x}^{(i)}_{train}) - y^{(i)}_{train} \right)^2 \right]$$

Train/test procedure for linear regression (with squared error cost)

Fit parameters

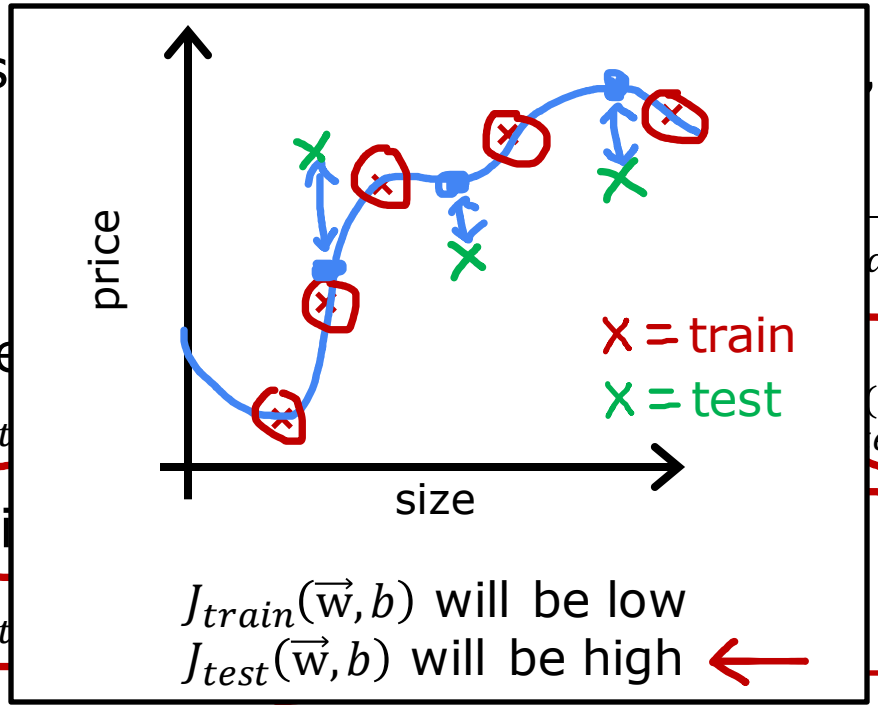
$J(\vec{w}, b) = \min_{\vec{w}, b}$

Compute test e

J_t

Compute traini

J_t



$, b)$

$\text{ain} \sum_{j=1}^{n} w_j^2 \Big]$

$^{(i)}_{est}\Big)^2 \Big]$

$+ \sum w^2$

$- y^{(i)}_{train}\Big)^2 \Big]$

Train/test procedure for classification problem

0/1

Fit parameters by minimizing $J(\vec{w}, b)$ to find $\vec{w}, b$

E.g.,

$$J(\vec{w}, b) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) + \left(1-y^{(i)}\right)\log\left(1-f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right)\right] + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$

Compute test error:

$$J_{test}(\vec{w}, b) = -\frac{1}{m_{test}}\sum_{i=1}^{m_{test}}\left[y_{test}^{(i)}\log\left(f_{\vec{w},b}\left(\vec{x}_{test}^{(i)}\right)\right) + \left(1-y_{test}^{(i)}\right)\log\left(1-f_{\vec{w},b}\left(\vec{x}_{test}^{(i)}\right)\right)\right]$$

Compute train error:

$$J_{train}(\vec{w}, b) = -\frac{1}{m_{train}}\sum_{i=1}^{m_{train}}\left[y_{train}^{(i)}\log\left(f_{\vec{w},b}\left(\vec{x}_{train}^{(i)}\right)\right) + \left(1-y_{train}^{(i)}\right)\log\left(1-f_{\vec{w},b}\left(\vec{x}_{train}^{(i)}\right)\right)\right]$$

Train/test procedure for classification problem

0/1

Fit parameters by minimizing $J(\vec{w}, b)$ to find $\vec{w}, b$

E.g.,

$$J(\vec{w}, b) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(f_{\vec{w},b}(\right.\right.$$

Compute test error:

$$J_{test}(\vec{w}, b) = -\frac{1}{m_{test}}\sum_{i=1}^{m_{test}}\left[y_{\right.}$$

Compute train error:

$$J_{train}(\vec{w}, b) = -\frac{1}{m_{train}}\sum_{i=1}^{m_{train}}$$

fraction of the test set and the fraction of the train set that the algorithm has misclassified.

$$\hat{y} = \begin{cases} 1 \text{ if } f_{\vec{w},b}(\vec{x}^{(i)}) \geq 0.5 \\ 0 \text{ if } f_{\vec{w},b}(\vec{x}^{(i)}) < 0.5 \end{cases}$$

count $\hat{y} \neq y$

$J_{test}(\vec{w}, b)$ is the fraction of the test set that has been misclassified.

$J_{train}(\vec{w}, b)$ is the fraction of the train set that has been misclassified.

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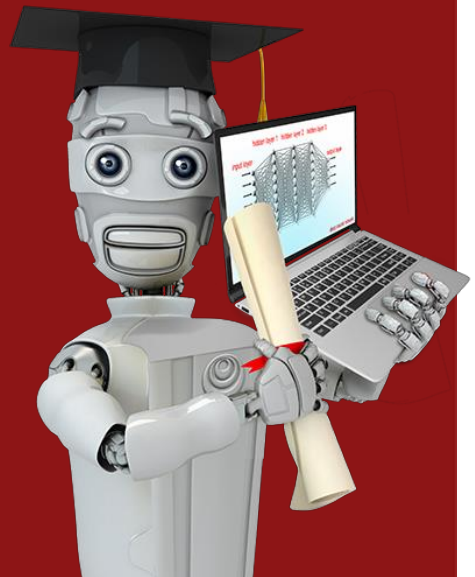
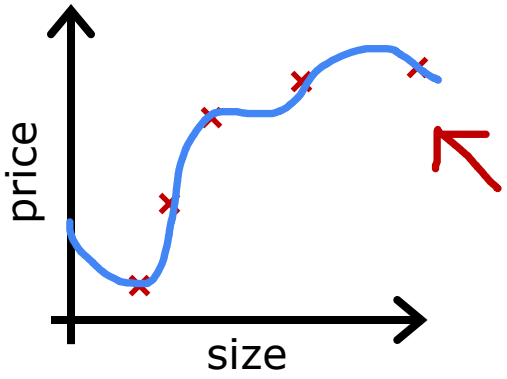


Evaluating and choosing models

Model selection and training/cross validation/test sets

Model selection (choosing a model)



$$f_{\vec{w},b}(\vec{x}) = w_1x_1 + w_2x^2 + w_3x^3 + w_4x^4 + b$$

Once parameters $\vec{w}, b$ are fit to the training set, the training error $J_{train}(\vec{w}, b)$ is likely lower than the actual generalization error.

$J_{test}(\vec{w}, b)$ is better estimate of how well the model will generalize to new data than $J_{train}(\vec{w}, b)$.

Model selection (choosing a model)

d=1 1. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + b$ → $w^{<1>}, b^{<1>}$ → $J_{test}(w^{<1>}, b^{<1>})$

d=2 2. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x^2 + b$ → $w^{<2>}, b^{<2>}$ → $J_{test}(w^{<2>}, b^{<2>})$

d=3 3. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x^2 + w_3 x^3 + b$ → $w^{<3>}, b^{<3>}$ → $J_{test}(w^{<3>}, b^{<3>})$

$\vdots$

d=10 10. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x^2 + \cdots + w_{10} x^{10} + b$ → $J_{test}(w^{<10>}, b^{<10>})$

Choose $w_1 x_1 + \cdots + w_5 x^5 + b$ d=5 $J_{test}(w^{<5>}, b^{<5>})$

How well does the model perform? Report test set error $J_{test}(w^{<5>}, b^{<5>})$?

The problem is $J_{test}(w^{<5>}, b^{<5>})$ is likely to be an optimistic estimate of generalization error. Ie: An extra parameter d (degree of polynomial) was chosen using the test set.

w,b

Training/cross validation/test set

size	price
2104	400
1600	330
2400	369
1416	232
3000	540
1985	300
1534	315
1427	199
1380	212
1494	243

validation set
development set
dev set

training set
60%

cross validation
20%

test set
20%

$(x^{(1)}, y^{(1)})$
$\vdots$
$(x^{(m_{train})}, y^{(m_{train})})$
$m_{train} = 6$

$(x_{cv}^{(1)}, y_{cv}^{(1)})$
$\vdots$
$(x_{cv}^{(m_{cv})}, y_{cv}^{(m_{cv})})$
$m_{cv} = 2$

$(x_{test}^{(1)}, y_{test}^{(1)})$
$\vdots$
$(x_{test}^{(m_{test})}, y_{test}^{(m_{test})})$
$m_{test} = 2$

Training/cross validation/test set

Training error: $$J_{train}(\vec{w}, b) = \frac{1}{2m_{train}} \left[\sum_{i=1}^{m_{train}} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)} \right)^2 \right]$$

Cross validation error: $$J_{cv}(\vec{w}, b) = \frac{1}{2m_{cv}} \left[\sum_{i=1}^{m_{cv}} \left(f_{\vec{w},b}\left(\vec{x}_{cv}^{(i)}\right) - y_{cv}^{(i)} \right)^2 \right]$$ (validation error, dev error)

Test error: $$J_{test}(\vec{w}, b) = \frac{1}{2m_{test}} \left[\sum_{i=1}^{m_{test}} \left(f_{\vec{w},b}\left(\vec{x}_{test}^{(i)}\right) - y_{test}^{(i)} \right)^2 \right]$$

Model selection

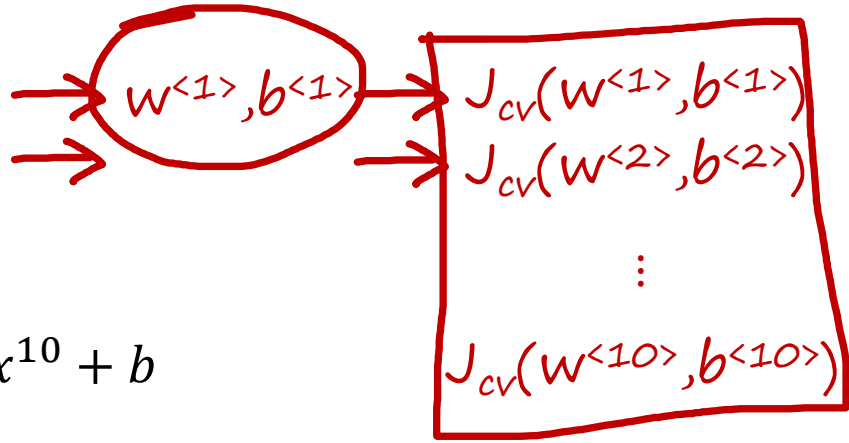
d=1 1. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + b$

d=2 2. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x^2 + b$

d=3 3. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x^2 + w_3 x^3 + b$

⋮ ⋮

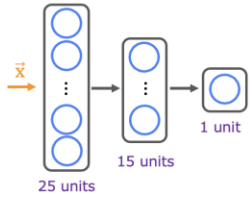
d=10 10. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x^2 + \cdots + w_{10} x^{10} + b$



→ Pick $w_1 x_1 + \cdots + w_4 x^4 + b$ $(J_{cv}(w^{<4>}, b^{<4>}))$

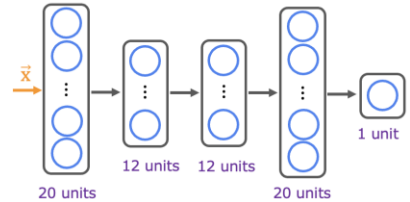
Estimate generalization error using test the set: $J_{test}(w^{<4>}, b^{<4>})$

Model selection – choosing a neural network architecture

1. 

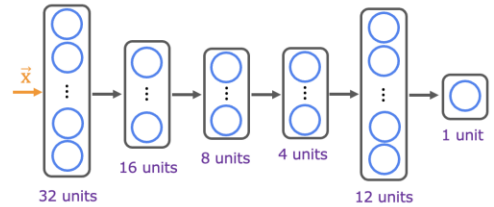
$w^{(1)}, b^{(1)}$

$J_{cv}(\mathbf{W}^{(1)}, \mathbf{B}^{(1)})$

2. 

$w^{(2)}, b^{(2)}$

$J_{cv}(\mathbf{W}^{(2)}, \mathbf{B}^{(2)})$

3. 

$w^{(3)}, b^{(3)}$

$J_{cv}(\mathbf{W}^{(3)}, \mathbf{B}^{(3)})$

Train, CV

Pick $\mathbf{W}^{(2)}, \mathbf{B}^{(2)}$

Estimate generalization error using the test set: $J_{test}(\mathbf{W}^{(2)}, \mathbf{B}^{(2)})$

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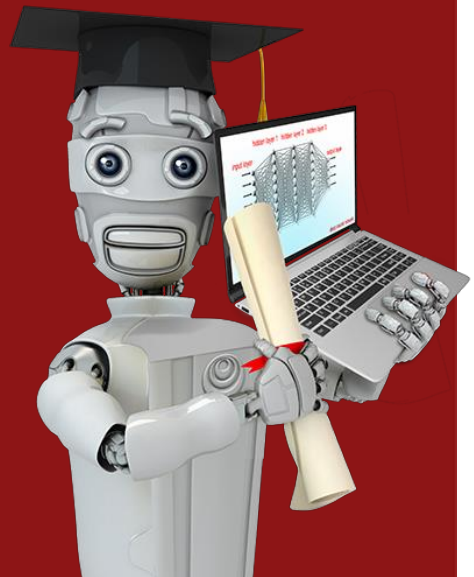
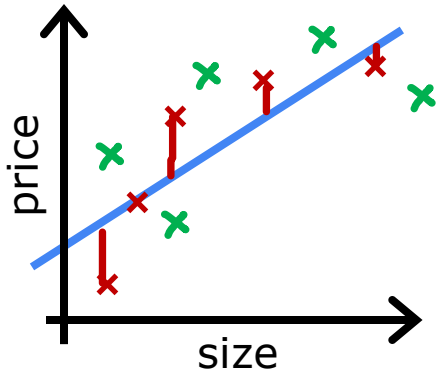


Bias and variance

Diagnosing bias and variance

Bias/variance



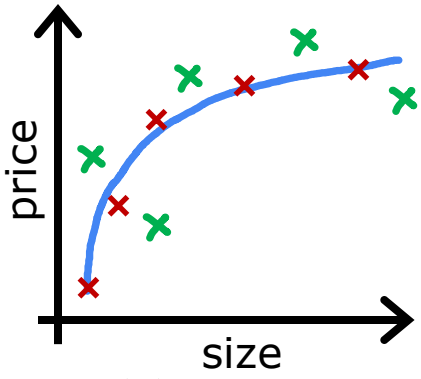
$f_{\vec{w},b}(x) = w_1 x + b$

High bias (underfit)

$d = 1$

J_{train} is high

J_{cv} is high



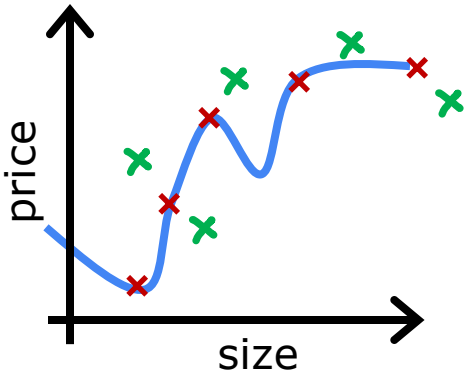
$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + b$

"Just right"

$d = 2$

J_{train} is low

J_{cv} is low



$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b$

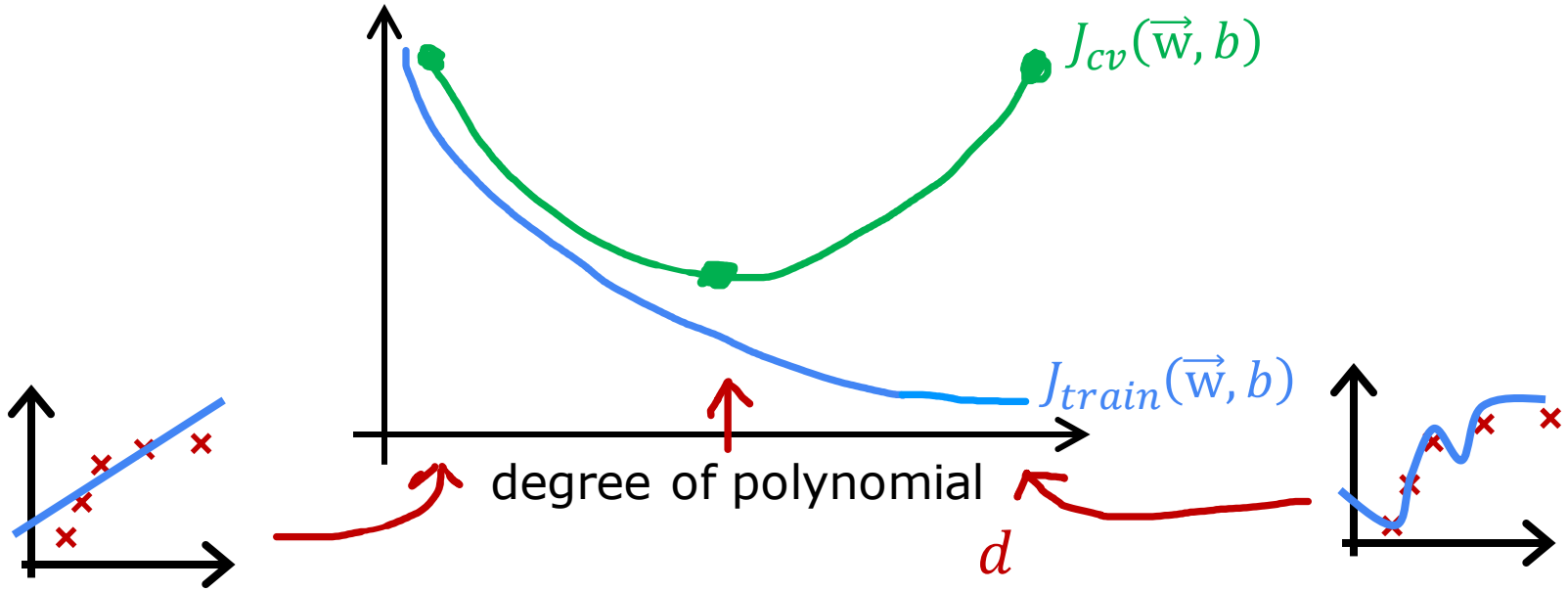
High variance (overfit)

$d = 4$

J_{train} is low

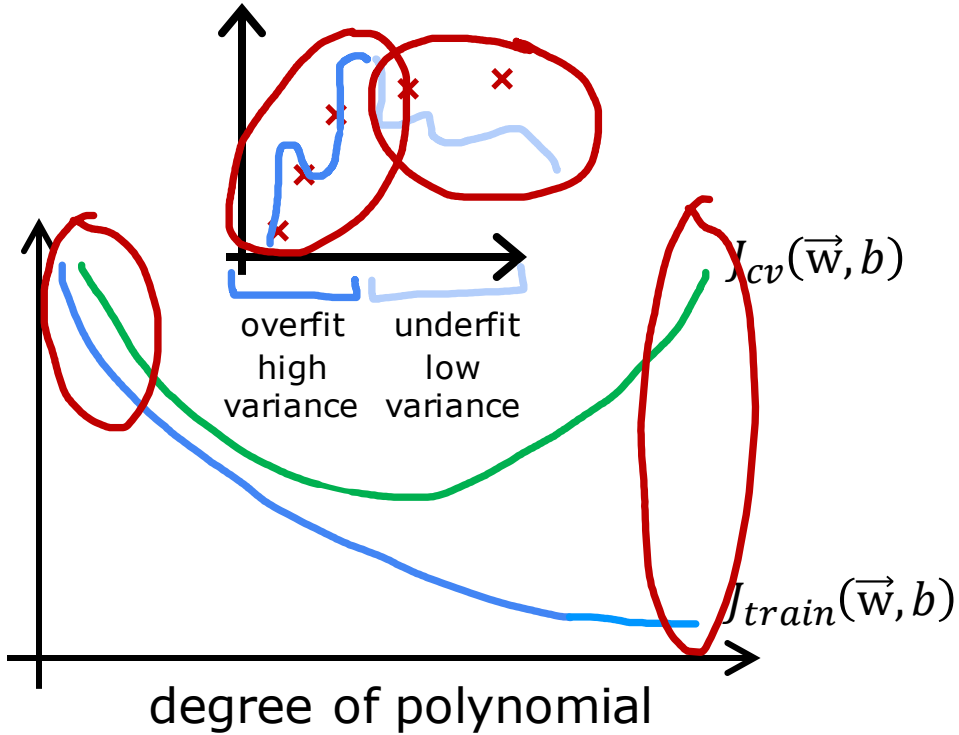
J_{cv} is high

Understanding bias and variance



Diagnosing bias and variance

How do you tell if your algorithm has a bias or variance problem?



High bias (underfit)

J_{train} will be high

($J_{train} \approx J_{cv}$)

High variance (overfit)

$J_{cv} \gg J_{train}$

(J_{train} may be be low)

High bias and high variance

J_{train} will be high

and $J_{cv} \gg J_{train}$

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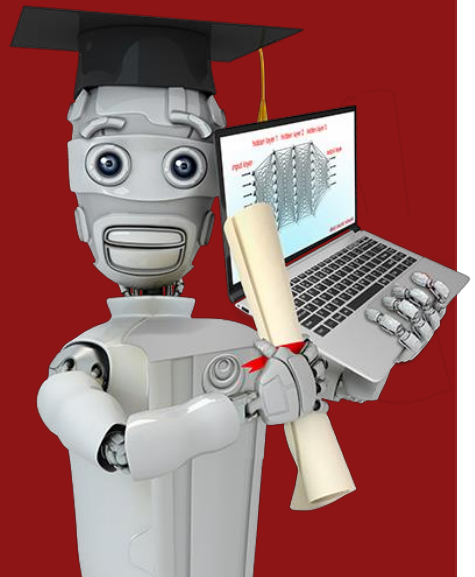


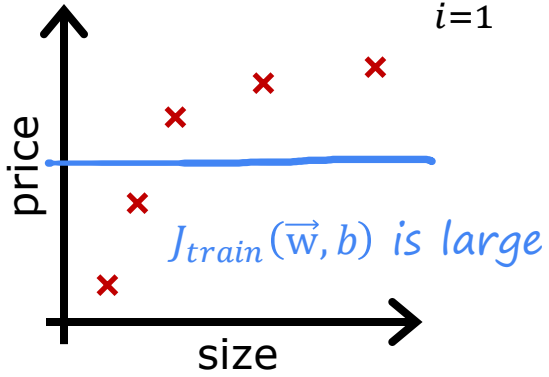
Bias and variance

Regularization and bias/variance

Linear regression with regularization

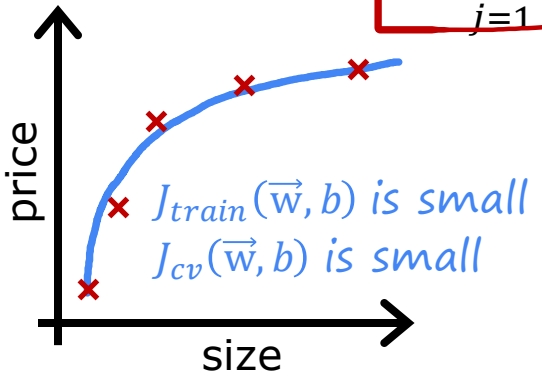
Model: $f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b$

$$J(\vec{w}, b) = \frac{1}{2m} \sum_{i=1}^{m} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)} \right)^2 + \frac{\lambda}{2m} \sum_{j=1}^{n} w_j^2$$

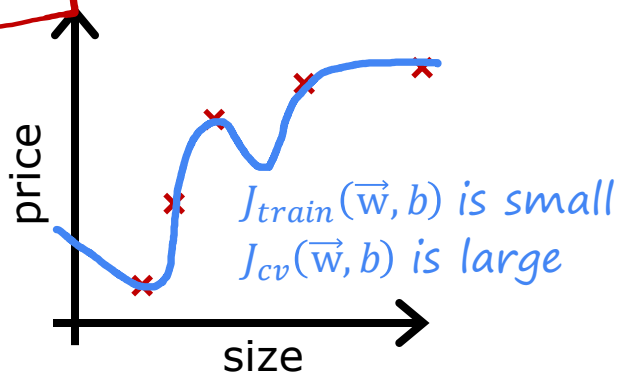


Large λ
High bias (underfit)

$\lambda = 10{,}000$ $w_1 \approx 0, w_2 \approx 0$
$f_{\vec{w},b}(\vec{x}) \approx b$



Intermediate λ



Small λ
High variance (overfit)
$\lambda = 0$

λ

Choosing the regularization parameter λ

Model: $f_{\vec{w},b}(x) = w_1x + w_2x^2 + w_3x^3 + w_4x^4 + b$

1. Try $\lambda = 0$ → $\min_{\vec{w},b} J(\vec{w},b)$ → $w^{<1>}, b^{<1>}$ → $J_{cv}(w^{<1>}, b^{<1>})$
2. Try $\lambda = 0.01$ → $w^{<2>}, b^{<2>}$ → $J_{cv}(w^{<2>}, b^{<2>})$
3. Try $\lambda = 0.02$ → $J_{cv}(w^{<3>}, b^{<3>})$
4. Try $\lambda = 0.04$
5. Try $\lambda = 0.08$ → $J_{cv}(w^{<5>}, b^{<5>})$

⋮

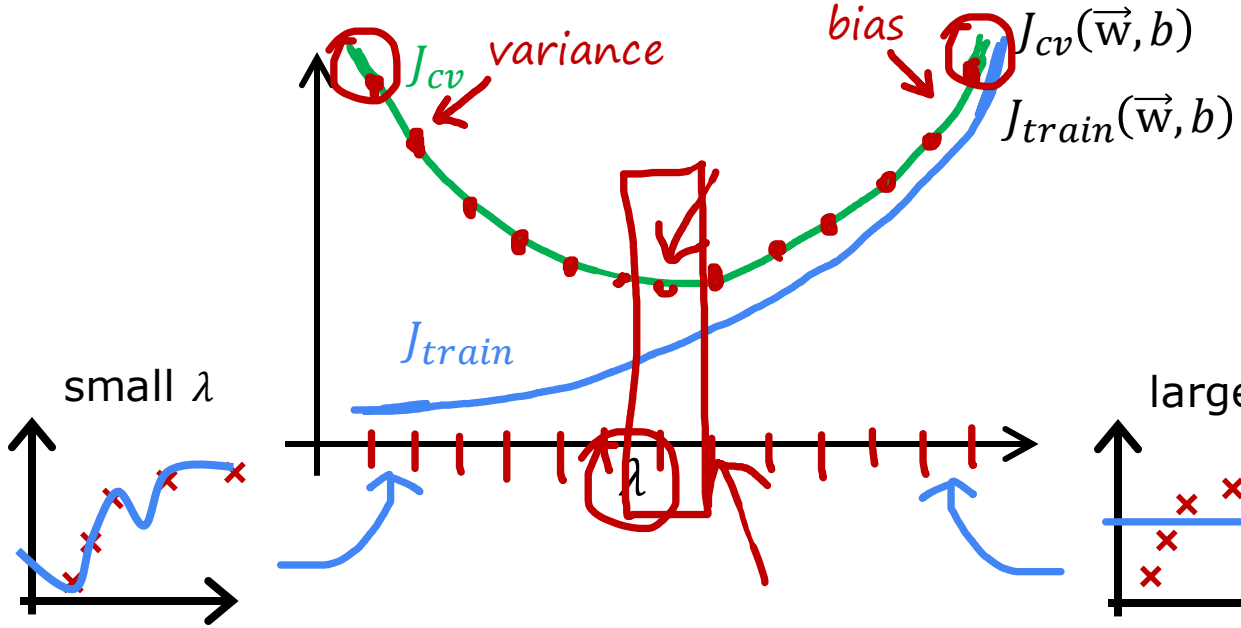
12. Try $\lambda \approx 10$ → $w^{<12>}, b^{<12>}$ → $J_{cv}(w^{<12>}, b^{<12>})$

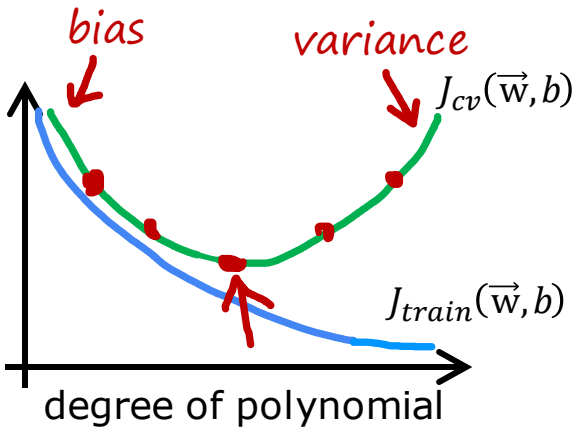
Pick $w^{<5>}, b^{<5>}$

Report test error: $J_{test}(w^{<5>}, b^{<5>})$

Bias and variance as a function of regularization parameter λ

$$J(\vec{w}, b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$





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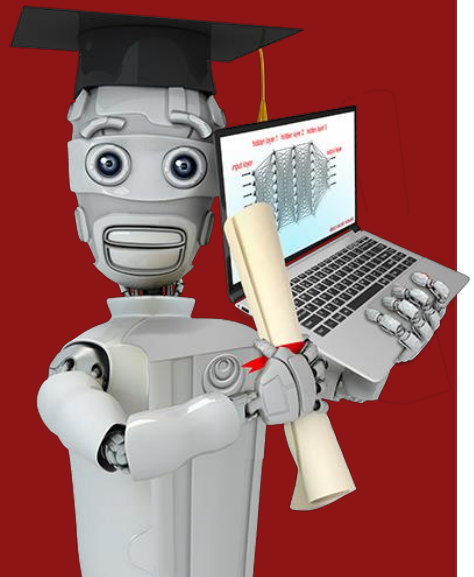


Bias and variance

Establishing a baseline level of performance

Speech recognition example

Human level performance : 10.6%

Training error J_{train} : 10.8%

Cross validation error J_{cv} : 14.8%

0.2%

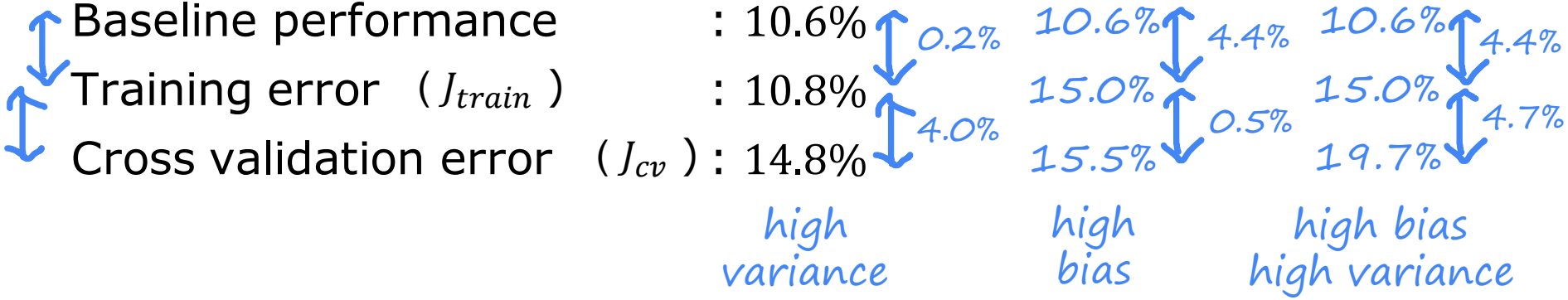
4.0%

Establishing a baseline level of performance

What is the level of error you can reasonably hope to get to?

- Human level performance
- Competing algorithms performance
- Guess based on experience

Bias/variance examples



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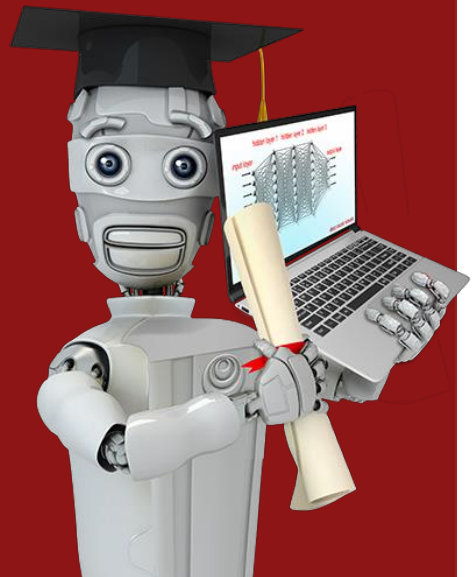


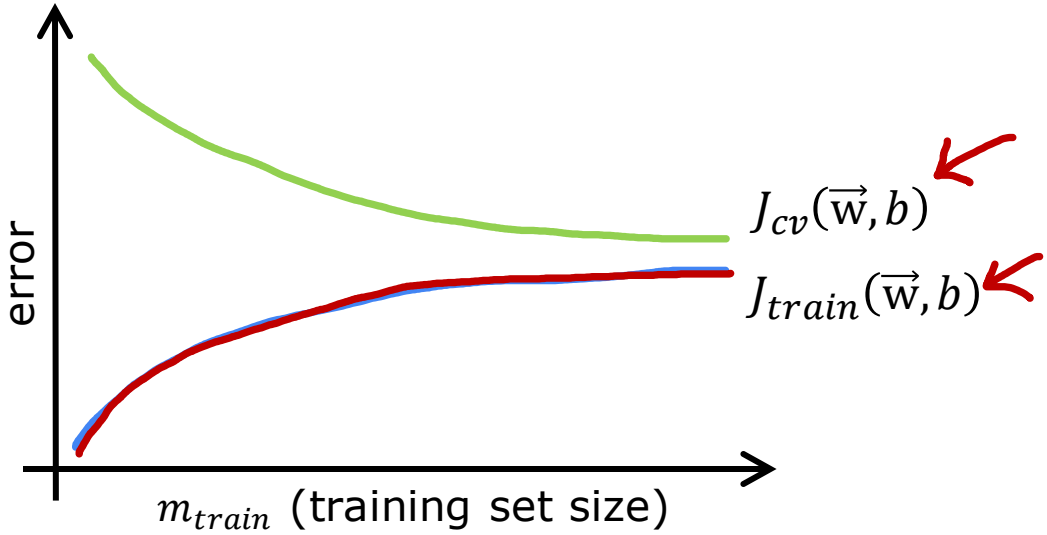
Bias and variance

Learning curves

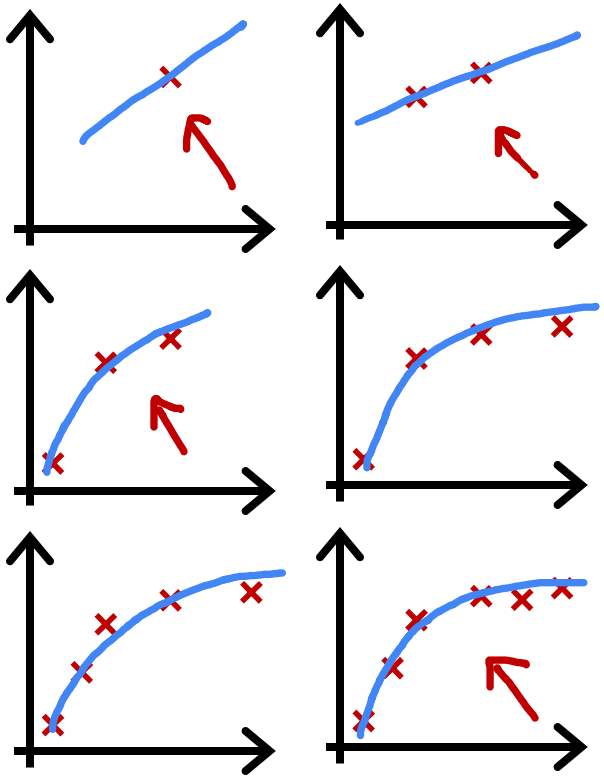
Learning curves

J_{train} = training error

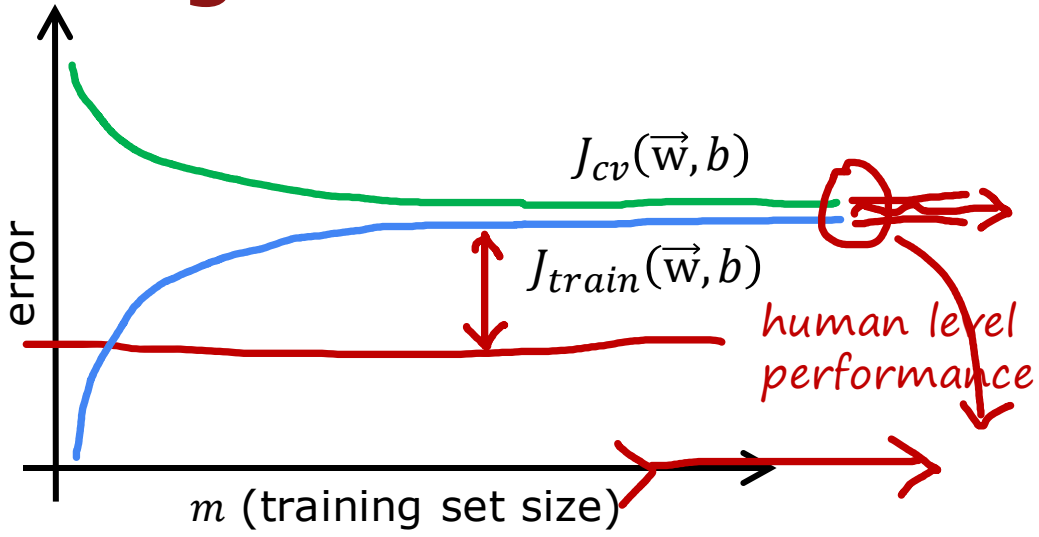
J_{cv} = cross validation error

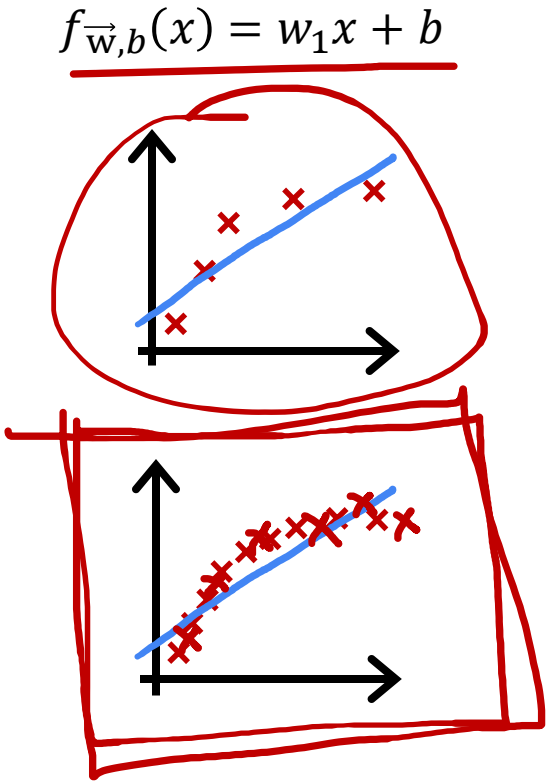


$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + b$



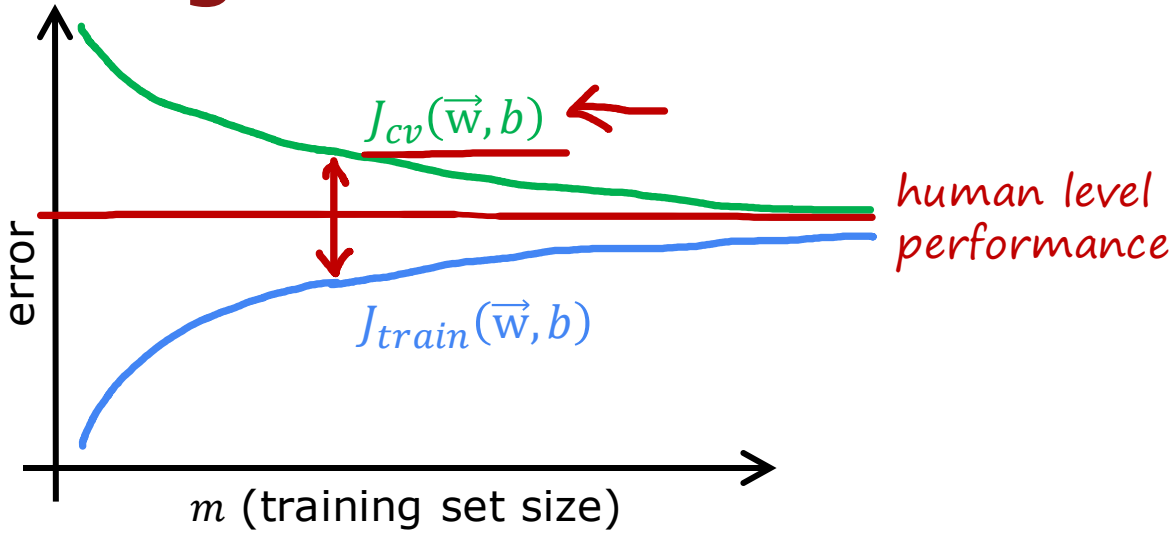
High bias





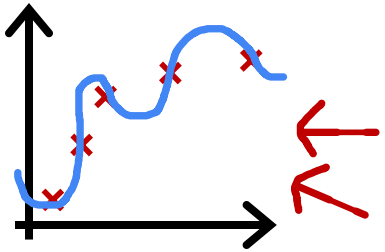
if a learning algorithm suffers from high bias, getting more training data will not (by itself) help much.

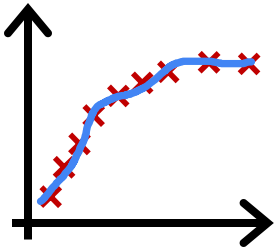
High variance



$$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b$$

(with small λ)





if a learning algorithm suffers from high variance, getting more training data is likely to help.

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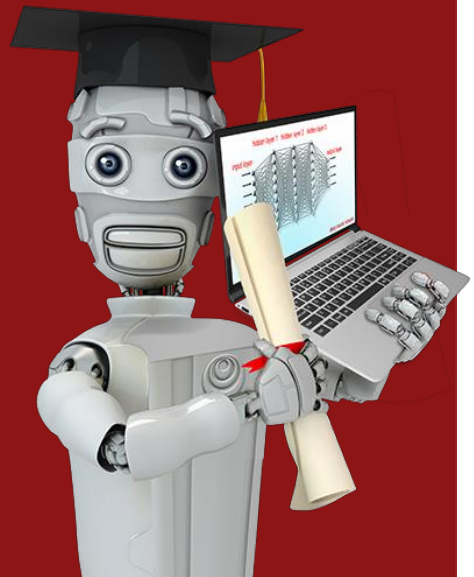

Bias and variance

Deciding what to try next revisited

Debugging a learning algorithm

You've implemented regularized linear regression on housing prices

$$J(\vec{w}, b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$

But it makes unacceptably large errors in predictions. What do you try next?

- Get more training examples — fixes high variance
- Try smaller sets of features $x, x^2, x^3, x^4, x^5 \ldots$ — fixes high variance
- Try getting additional features — fixes high bias
- Try adding polynomial features $(x_1^2, x_2^2, x_1x_2, etc)$ — fixes high bias
- Try decreasing λ — fixes high bias
- Try increasing λ — fixes high variance

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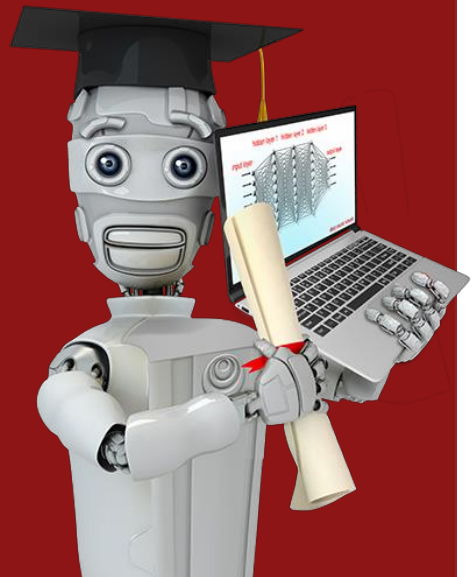


Bias and variance

Bias/variance and neural networks

The bias variance tradeoff

$f_{\vec{w},b}(x) = w_1 x + b$

$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + b$

$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b$

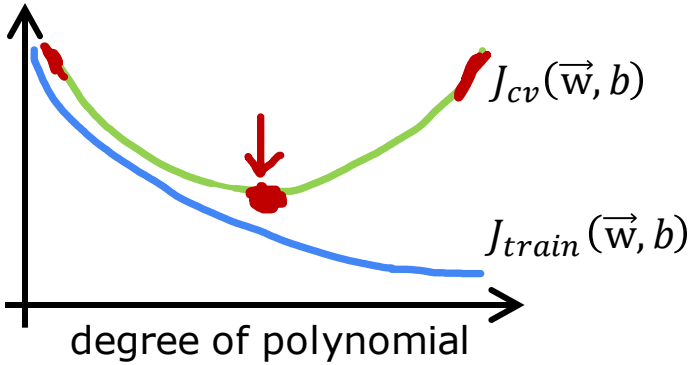
Simple model

High bias

tradeoff

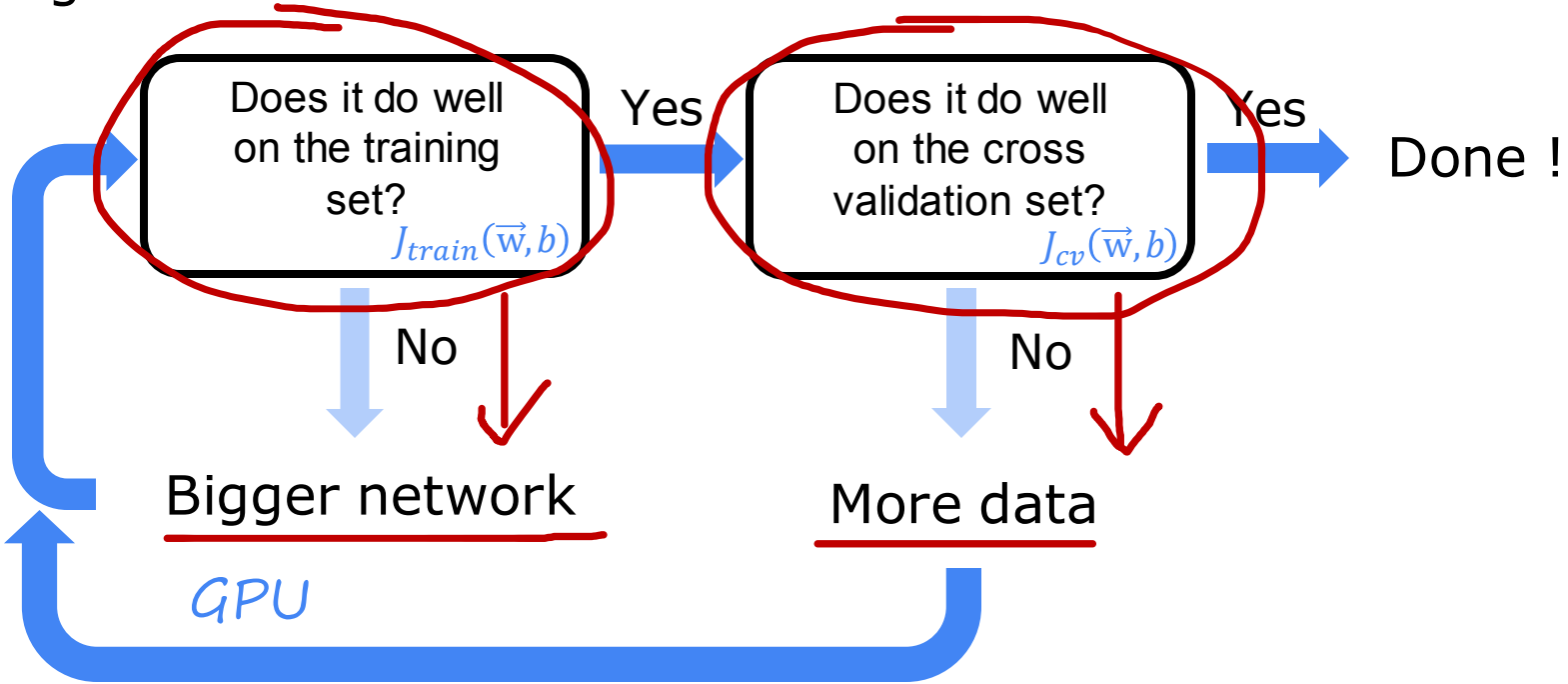
Complex model

High variance

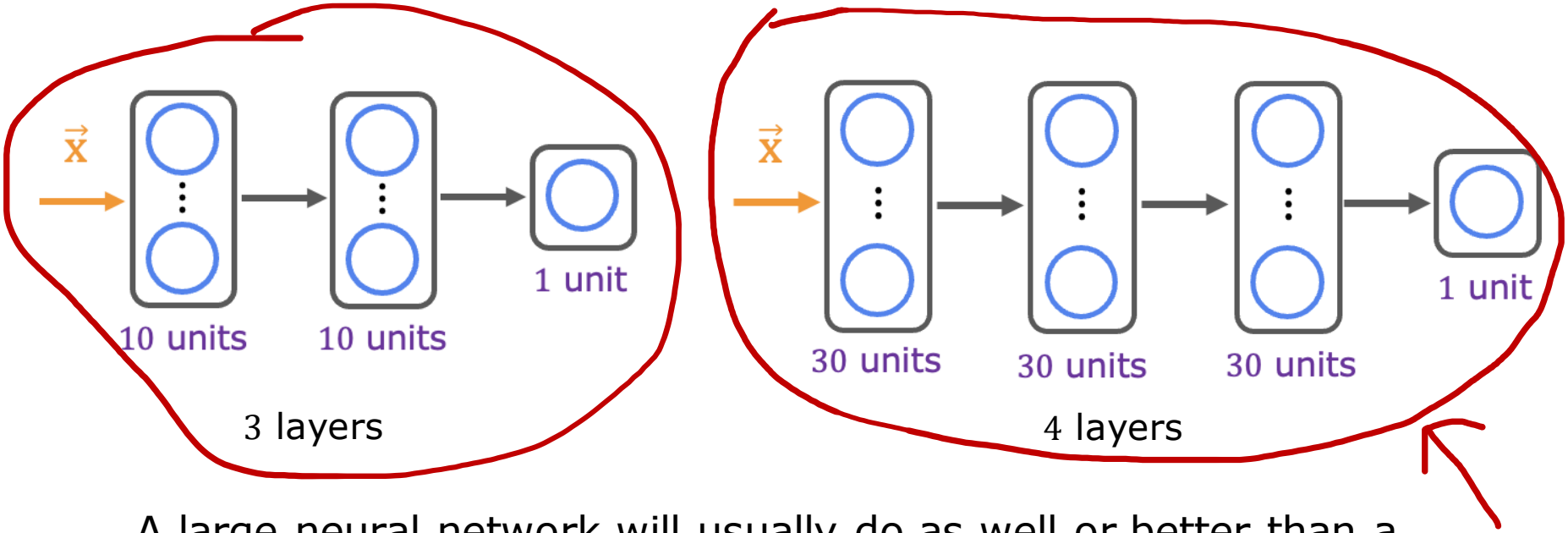


Neural networks and bias variance

Large neural networks are low bias machines



Neural networks and regularization



A large neural network will usually do as well or better than a smaller one so long as regularization is chosen appropriately.

Neural network regularization

$$J(\mathbf{W},\mathbf{B}) = \frac{1}{m}\sum_{i=1}^{m} L\left(f\left(\vec{x}^{(i)}\right), y^{(i)}\right) + \frac{\lambda}{2m}\sum_{\text{all weights } \mathbf{W}} \left(w^2\right)$$

Unregularized MNIST model

```
layer_1 = Dense(units=25, activation="relu")
layer_2 = Dense(units=15, activation="relu")
layer_3 = Dense(units=1, activation="sigmoid")
model = Sequential([layer_1, layer_2, layer_3])
```

Regularized MNIST model

```
layer_1 = Dense(units=25, activation="relu", kernel_regularizer=L2(0.01))
layer_2 = Dense(units=15, activation="relu", kernel_regularizer=L2(0.01))
layer_3 = Dense(units=1, activation="sigmoid", kernel_regularizer=L2(0.01))
model = Sequential([layer_1, layer_2, layer_3])
```

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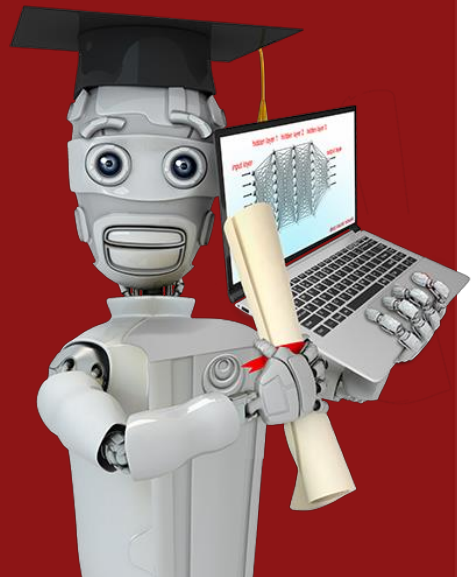
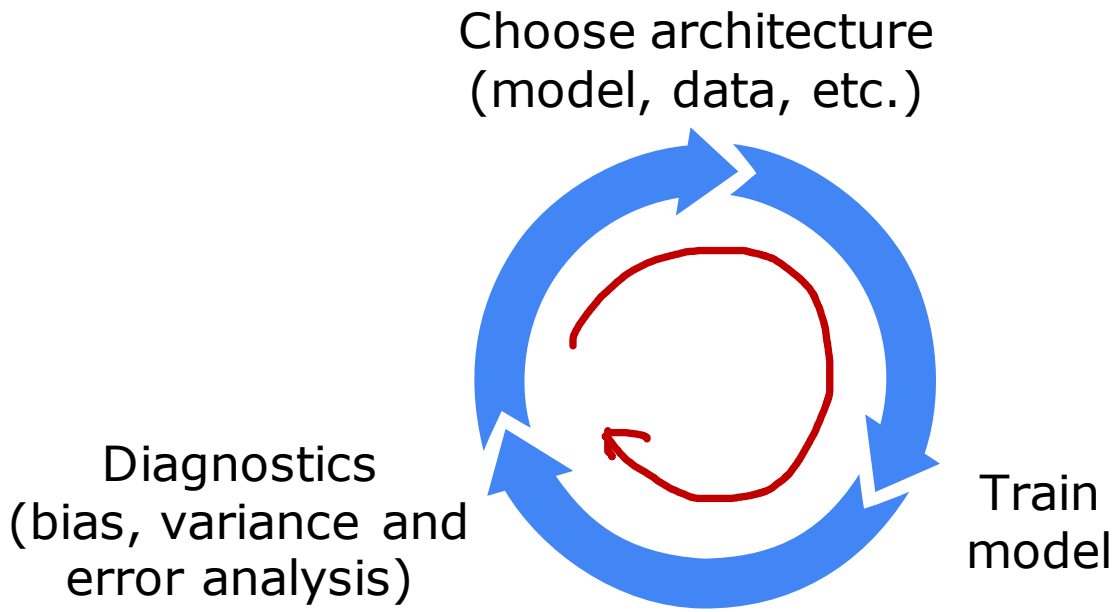


Machine learning development process

Iterative loop of ML development

Iterative loop of ML development



Spam classification example

From: cheapsales@buystufffromme.com
To: Andrew Ng
Subject: Buy now!

Deal of the week! Buy now!
Rolex w4tchs - $100
Med1cine (any kind) - £50
Also low cost M0rgages
available.

From: Alfred Ng
To: Andrew Ng
Subject: Christmas dates?

Hey Andrew,
Was talking to Mom about plans for Xmas. When do you get off work. Meet Dec 22?
Alf

Building a spam classifier

Supervised learning: $\vec{x}$ = features of email
y = spam (1) or not spam (0)

Features: list the top 10,000 words to compute $x_1, x_2, \cdots, x_{10,000}$

$$\vec{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 0 \\ \vdots \end{bmatrix} \begin{matrix} a \\ andrew \\ buy \\ deal \\ discount \\ \vdots \end{matrix}$$

From: cheapsales@buystufffromme.com
To: Andrew Ng
Subject: Buy now!

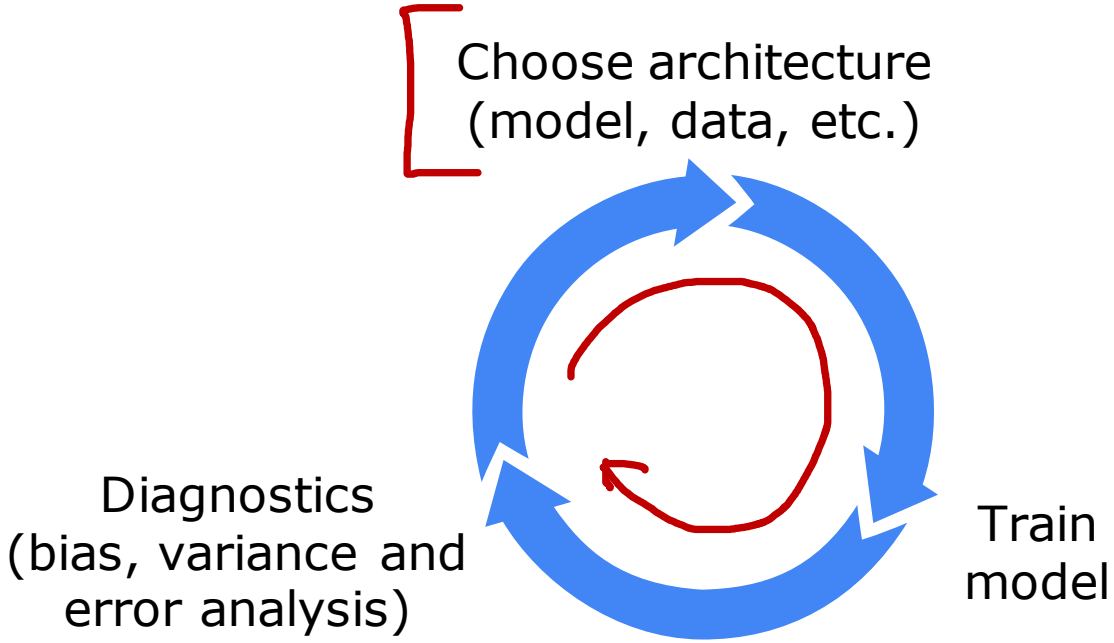
Deal of the week! Buy now!
Rolex w4tchs - $100
Med1cine (any kind) - £50
Also low cost M0rgages available.

Building a spam classifier

How to try to reduce your spam classifier's error?

- Collect more data. E.g., "Honeypot" project.
- Develop sophisticated features based on email routing (from email header).
- Define sophisticated features from email body. E.g., should "discounting" and "discount" be treated as the same word.
- Design algorithms to detect misspellings. E.g., w4tches, med1cine, m0rtgage.

Iterative loop of ML development



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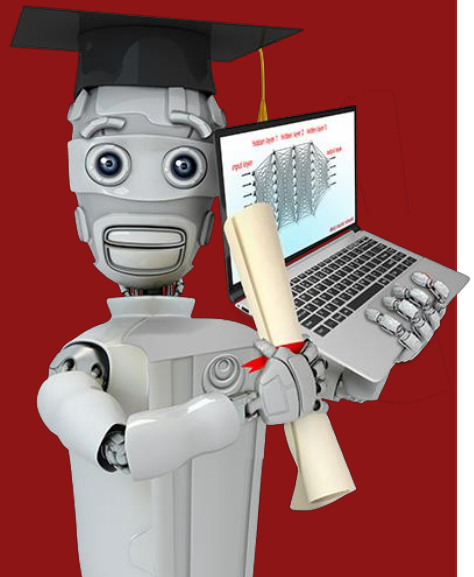


Machine learning development process

Error analysis

Error analysis

m_{cv} = ~~500~~ 5000 examples in cross validation set.

Algorithm misclassifies ~~100~~ 1000 of them.

Manually examine 100 examples and categorize them based on common traits.

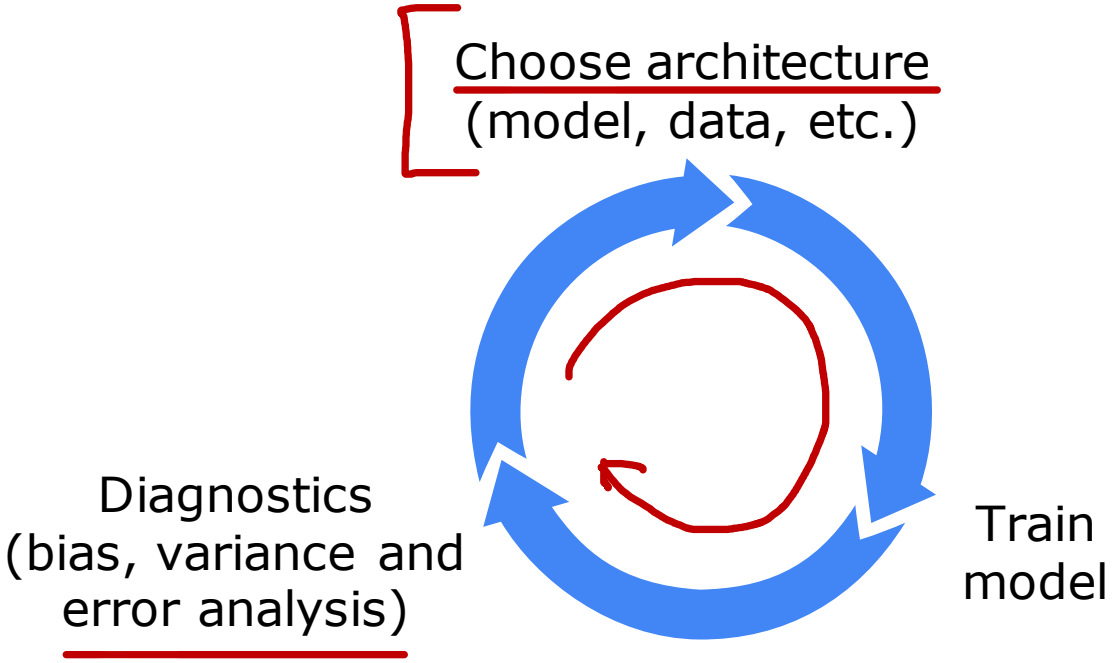
- Pharma: 21 → more data features
- Deliberate misspellings (w4tches, med1cine): 3
- Unusual email routing: 7
- Steal passwords (phishing): 18 → more data features
- Spam message in embedded image: 5

Building a spam classifier

How to try to reduce your spam classifier's error?

- Collect more data. E.g., "Honeypot" project.
- Develop sophisticated features based on email routing (from email header).
- Define sophisticated features from email body. E.g., should "discounting" and "discount" be treated as the same word.
- Design algorithms to detect misspellings. E.g., w4tches, med1cine, m0rtgage.

Iterative loop of ML development



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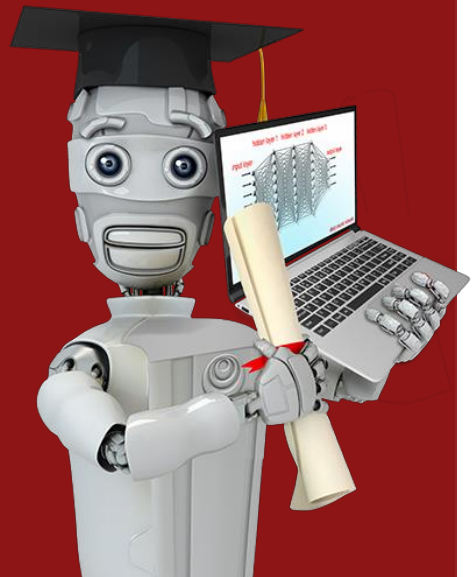


Machine learning development process

Adding data

Adding data

- Add more data of everything. E.g., "Honeypot" project.
- Add more data of the types where error analysis has indicated it might help.

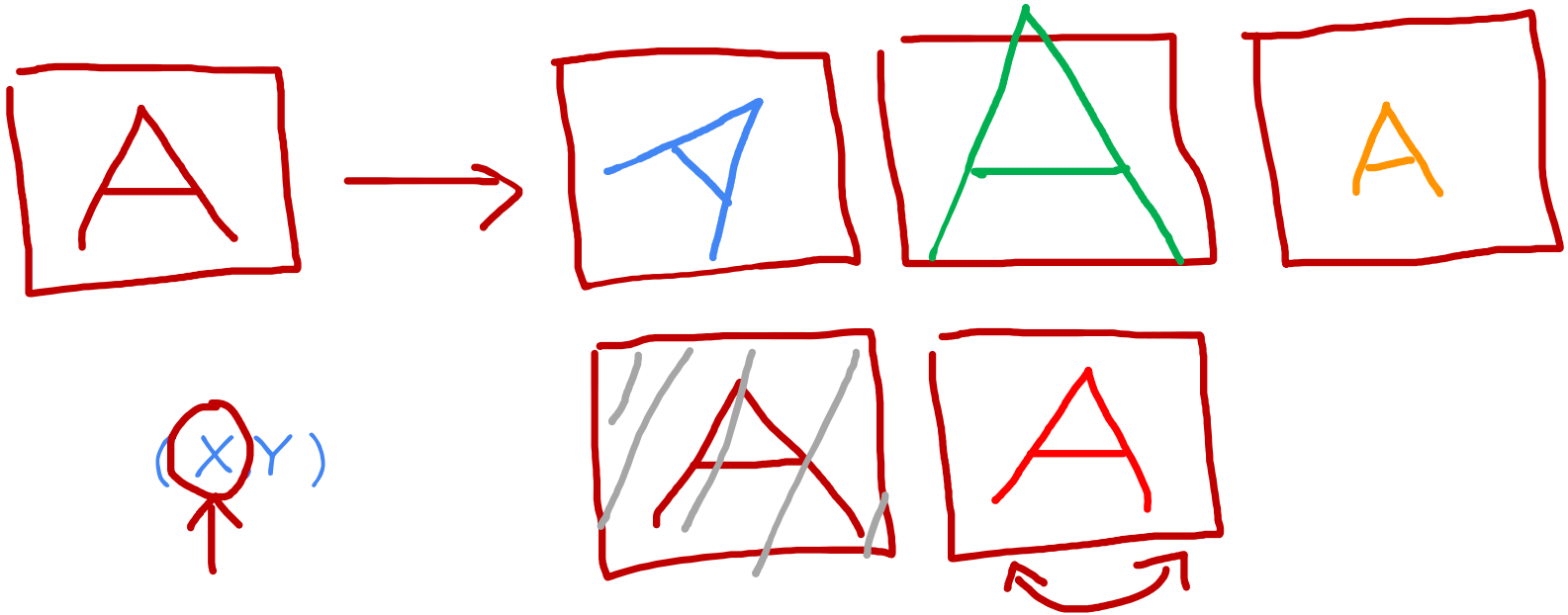
Pharma spam

E.g., Go to unlabeled data and find more examples of Pharma related spam.

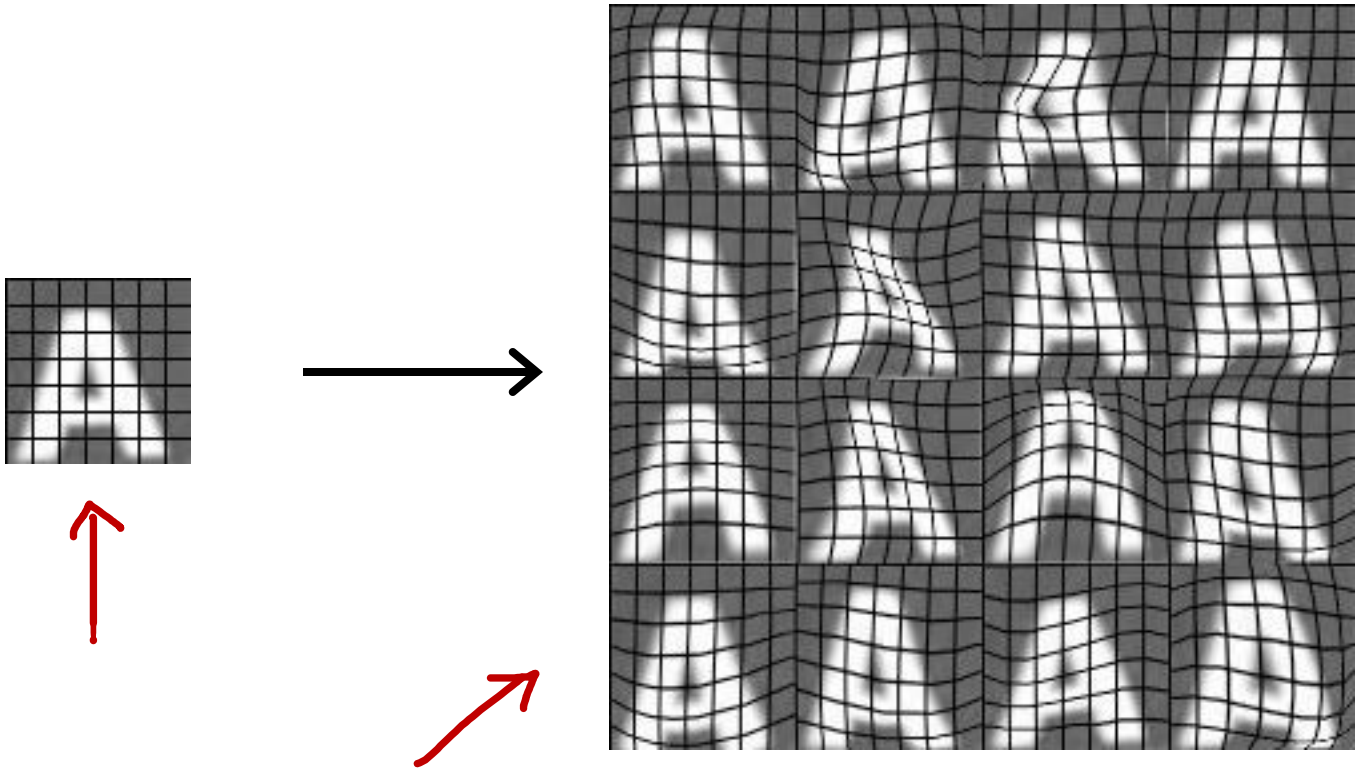
(x, y)

Data augmentation

Augmentation: modifying an existing training example to create a new training example.



Data augmentation by introducing distortions



Data augmentation for speech

Speech recognition example

Original audio (voice search: ”What is today’s weather?”)

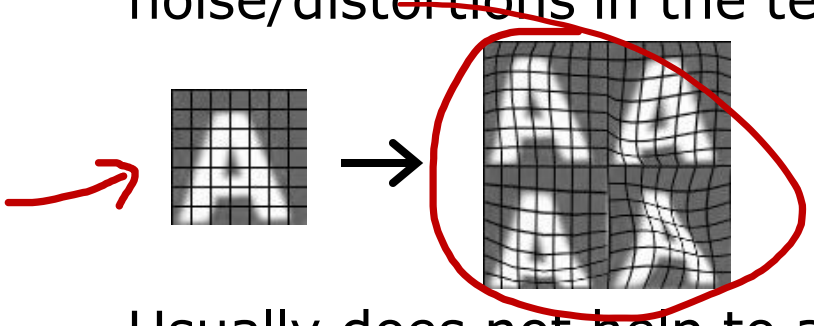
+ Noisy background: Crowd

+ Noisy background: Car

+ Audio on bad cellphone connection

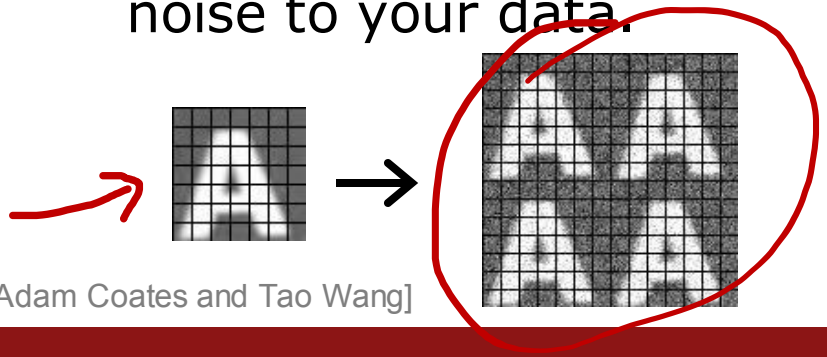
Data augmentation by introducing distortions

Distortion introduced should be representation of the type of noise/distortions in the test set.



Audio:
Background noise,
bad cellphone connection

Usually does not help to add purely random/meaningless noise to your data.



[Adam Coates and Tao Wang]

x_i=intensity (brightness) of pixel i

$x_i \leftarrow x_i +$ random noise

Data synthesis

Synthesis: using artificial data inputs to create a new training example.

Artificial data synthesis for photo OCR

[http://www.publicdomainpictures.net/view-image.php?image=5745&picture=times-square]

Artificial data synthesis for photo OCR

Real data

[Adam Coates and Tao Wang]

Artificial data synthesis for photo OCR

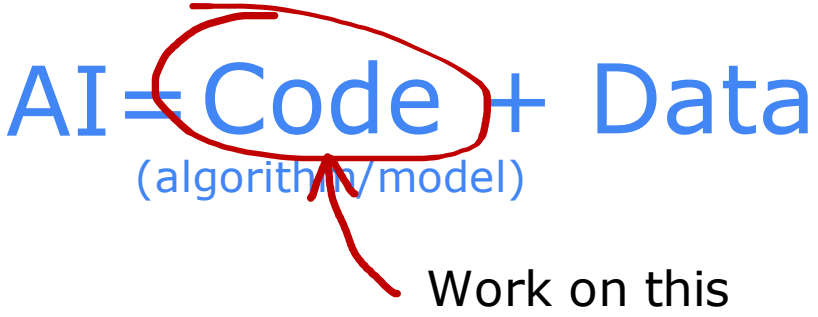
Real data

Synthetic data

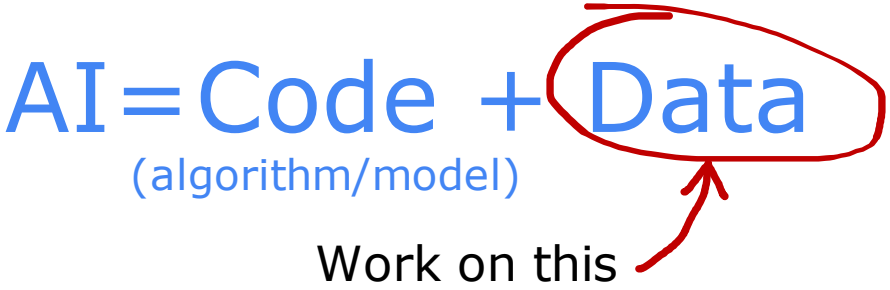
[Adam Coates and Tao Wang]

Engineering the data used by your system

Conventional model-centric approach:



Data-centric approach:



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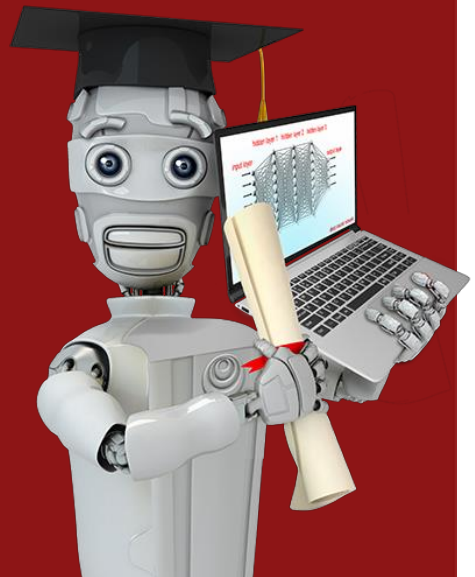
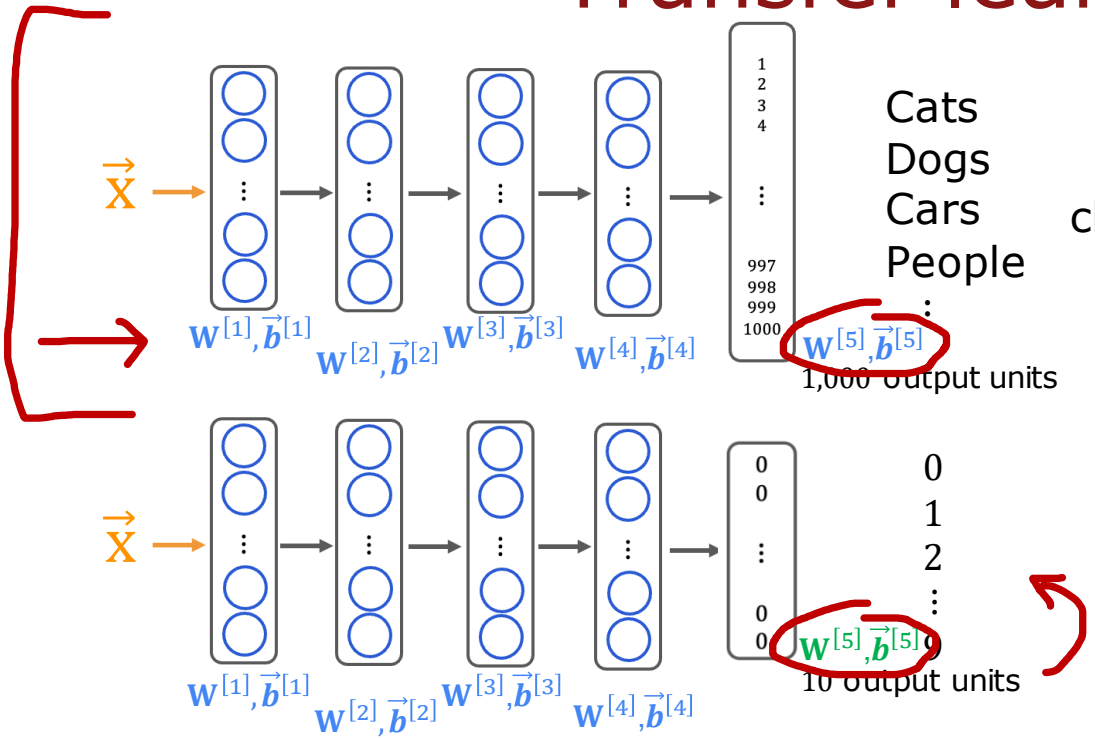


Machine learning development process

Transfer learning: using data from a different task

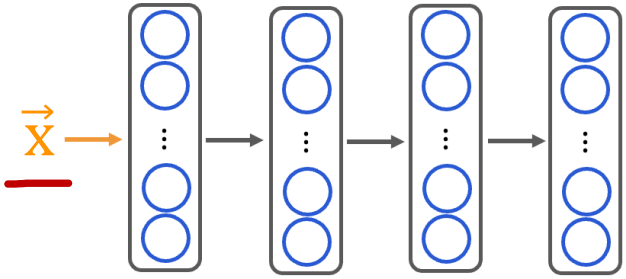
Transfer learning



Option 1: only train output layers parameters.

Option 2: train all parameters.

Why does transfer learning work?

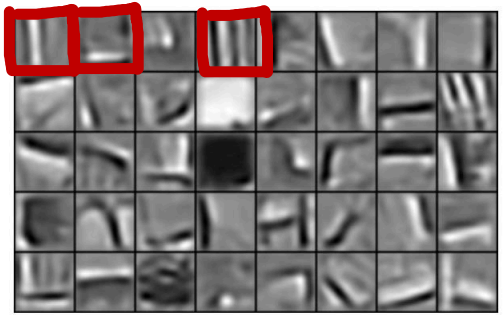


detects edges

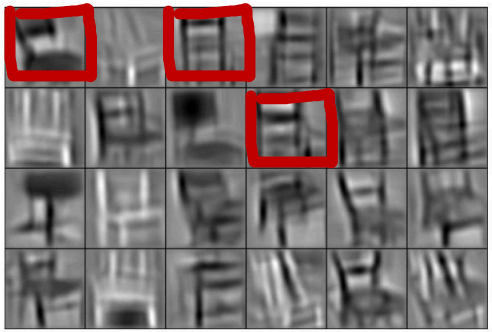
detects corners

detects curves/basic shapes

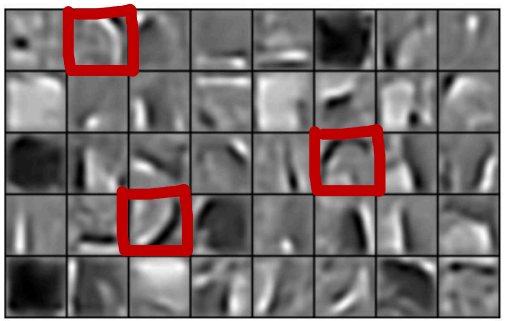
use the same input type



Edges



Corners



Curves / basic shapes

Transfer learning summary

1. Download neural network parameters pretrained on a large dataset with same input type (e.g., images, audio, text) as your application (or train your own).

1M

2. Further train (fine tune) the network on your own data.

1000

50

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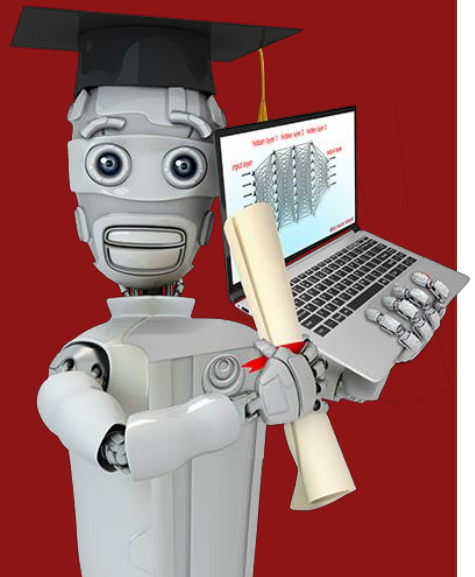
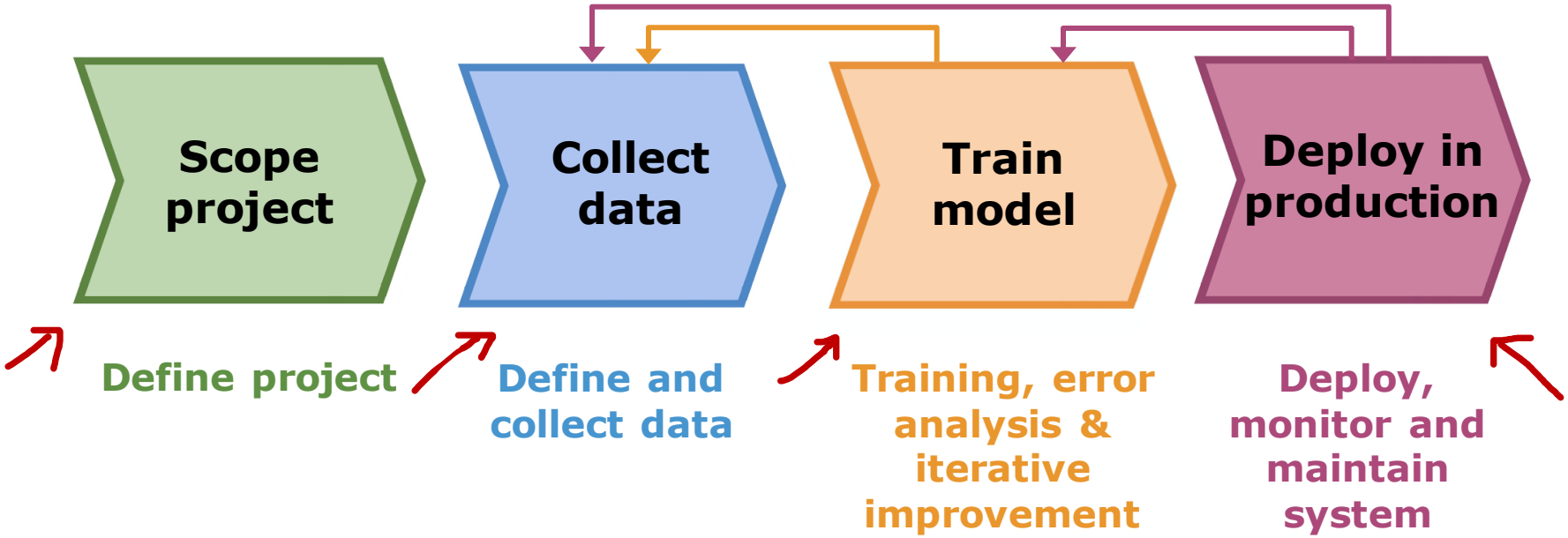


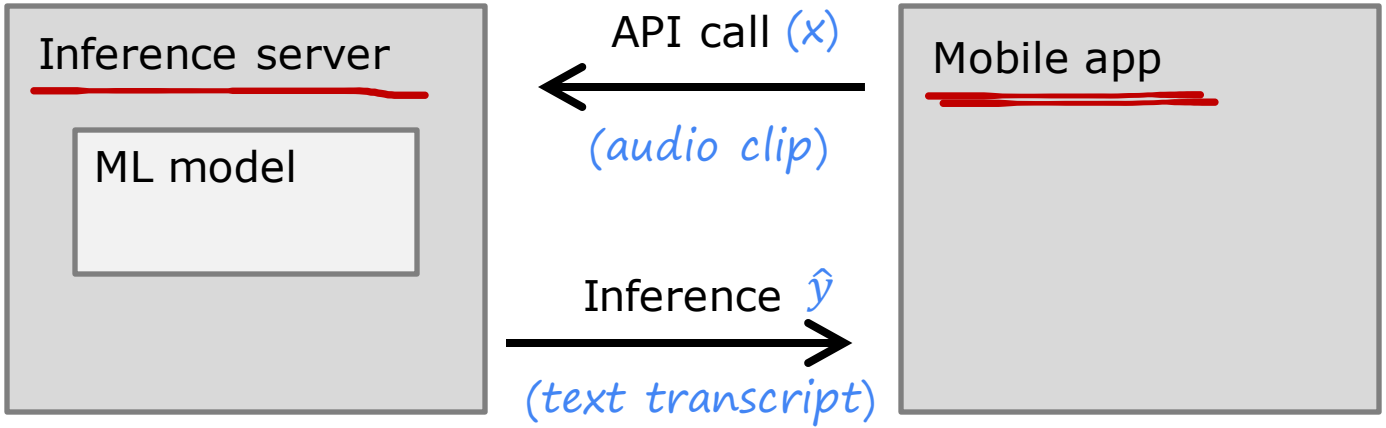
Machine learning development process

Full cycle of a machine learning project

Full cycle of a machine learning project



Deployment



→ Software engineering may be needed for:

- Ensure reliable and efficient predictions
- Scaling
- Logging
- System monitoring
- Model updates

MLOps
machine learning operations

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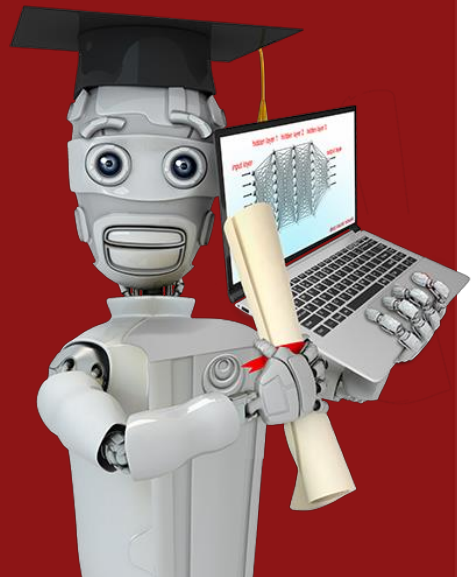


Machine learning development process

Fairness, bias, and ethics

Bias

Hiring tool that discriminates against women.

Facial recognition system matching dark skinned individuals to criminal mugshots.

Biased bank loan approvals.

Toxic effect of reinforcing negative stereotypes.

Adverse use cases

Deepfakes

Spreading toxic/incendiary speech through optimizing for engagement.

Generating fake content for commercial or political purposes.

Using ML to build harmful products, commit fraud etc.

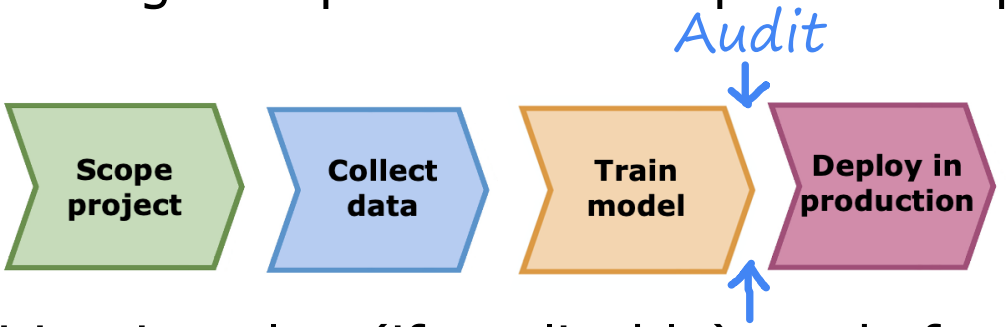
Spam vs anti-spam : fraud vs anti-fraud.

Guidelines

Get a diverse team to brainstorm things that might go wrong, with emphasis on possible harm to vulnerable groups.

Carry out literature search on standards/guidelines for your industry.

Audit systems against possible harm prior to deployment.



Develop mitigation plan (if applicable), and after deployment, monitor for possible harm.

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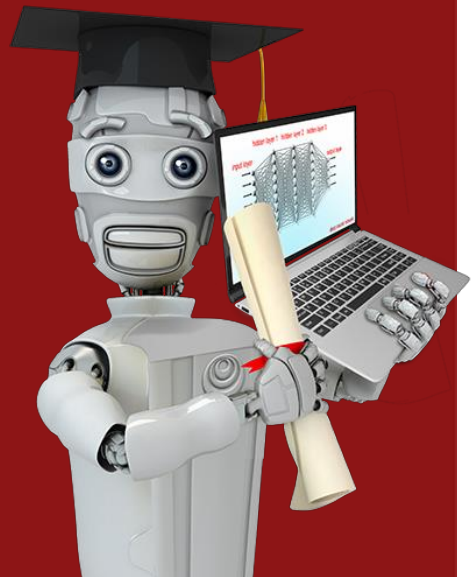


Skewed datasets (optional)

Error metrics for skewed datasets

Rare disease classification example

Train classifier $f_{\vec{w},b}(\vec{x})$

($y = 1$ if disease present,
$y = 0$ otherwise)

Find that you've got 1% error on test set
(99% correct diagnoses)

Only 0.5% of patients have the disease

```
print("y=0")
```

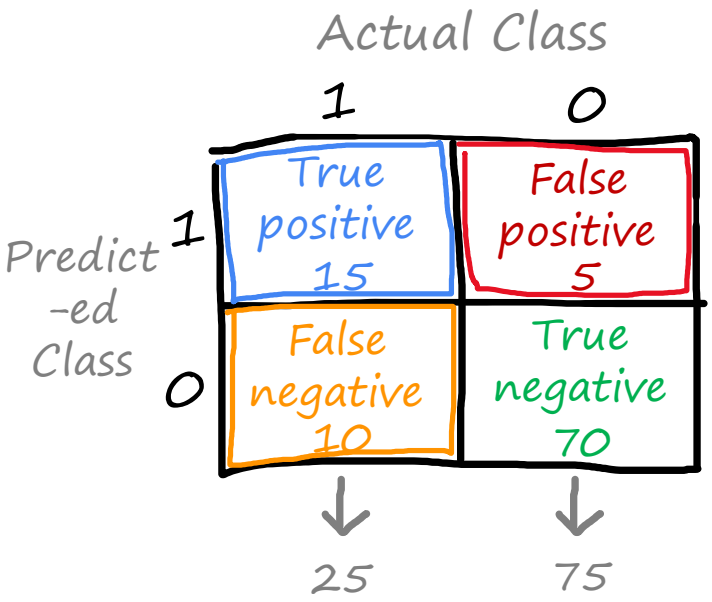
99.5% accuracy, 0.5% error

1%

1.2%

Precision/recall

$y = 1$ in presence of rare class we want to detect.



Precision:
(of all patients where we predicted $y = 1$, what fraction actually have the rare disease?)

$$\frac{\text{True positives}}{\#\text{predicted positive}} = \frac{\text{True positives}}{\text{True pos} + \text{False pos}} = \frac{15}{15+5} = 0.75$$

Recall:
(of all patients that actually have the rare disease, what fraction did we correctly detect as having it?)

$$\frac{\text{True positives}}{\#\text{actual positive}} = \frac{\text{True positives}}{\text{True pos} + \text{False neg}} = \frac{15}{15+10} = 0.6$$

```
print("y=0")
```

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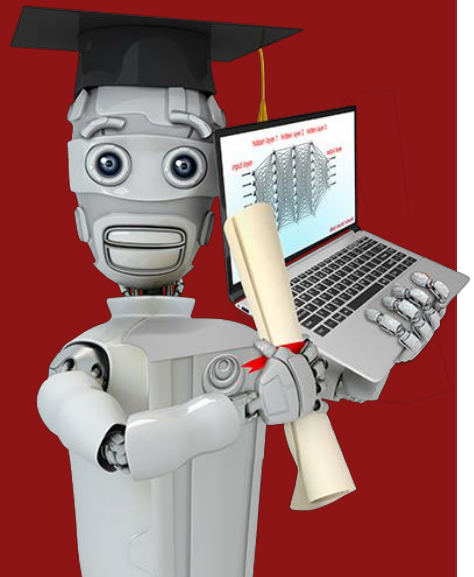


Skewed datasets (optional)

Trading off precision and recall

Trading off precision and recall

Logistic regression: $0 < f_{\vec{w},b}(\vec{x}) < 1$

Predict 1 if $f_{\vec{w},b}(\vec{x}) \geq 0.5$ ~~0.7~~ ~~0.9~~ 0.3

Predict 0 if $f_{\vec{w},b}(\vec{x}) < 0.5$ ~~0.7~~ ~~0.9~~ 0.3

$$\text{precision} = \frac{\text{true positives}}{\text{total predicted positive}}$$

$$\text{recall} = \frac{\text{true positives}}{\text{total actual positive}}$$

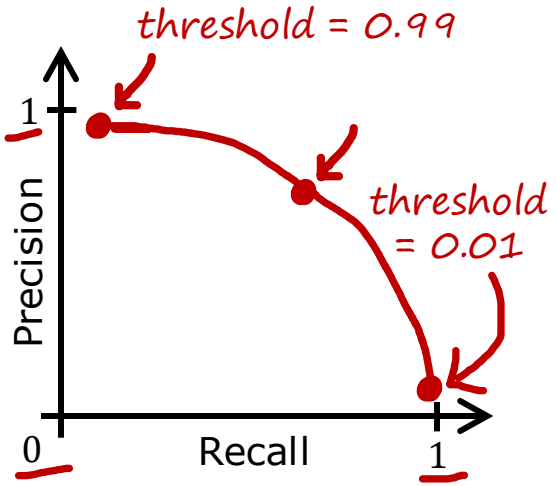
Suppose we want to predict $y = 1$ (rare disease) only if very confident.

→ higher precision, lower recall.

Suppose we want to avoid missing too many cases of rare disease (when in doubt predict $y = 1$)

→ lower precision, higher recall.

More generally predict 1 if: $f_{\vec{w},b}(\vec{x}) \geq$ threshold.



F1 score

How to compare precision/recall numbers?

	Precision (P)	Recall (R)	Average	F_1 score
Algorithm 1	0.5	0.4	0.45	0.444
Algorithm 2	0.7	0.1	0.4	0.175
Algorithm 3	0.02	1.0	0.501	0.0392

```
print("y=1")
```

Average $= \frac{P+R}{2}$

$$F1\ score = \frac{1}{\frac{1}{2}\left(\frac{1}{P}+\frac{1}{R}\right)} = 2\frac{PR}{P+R}$$

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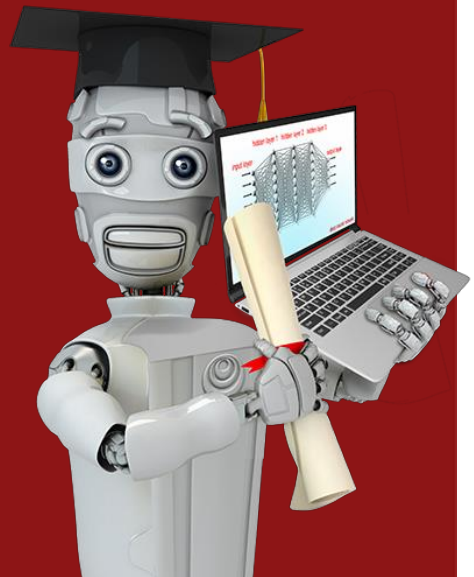


Decision Trees

Decision Tree Model

Cat classification example

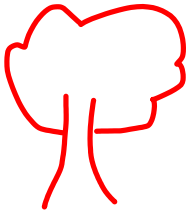
	Ear shape (x_1)	Face shape (x_2)	Whiskers (x_3)	Cat
	Pointy	Round	Present	1
	Floppy	Not round	Present	1
	Floppy	Round	Absent	0
	Pointy	Not round	Present	0
	Pointy	Round	Present	1
	Pointy	Round	Absent	1
	Floppy	Not round	Absent	0
	Pointy	Round	Absent	1
	Floppy	Round	Absent	0
	Floppy	Round	Absent	0

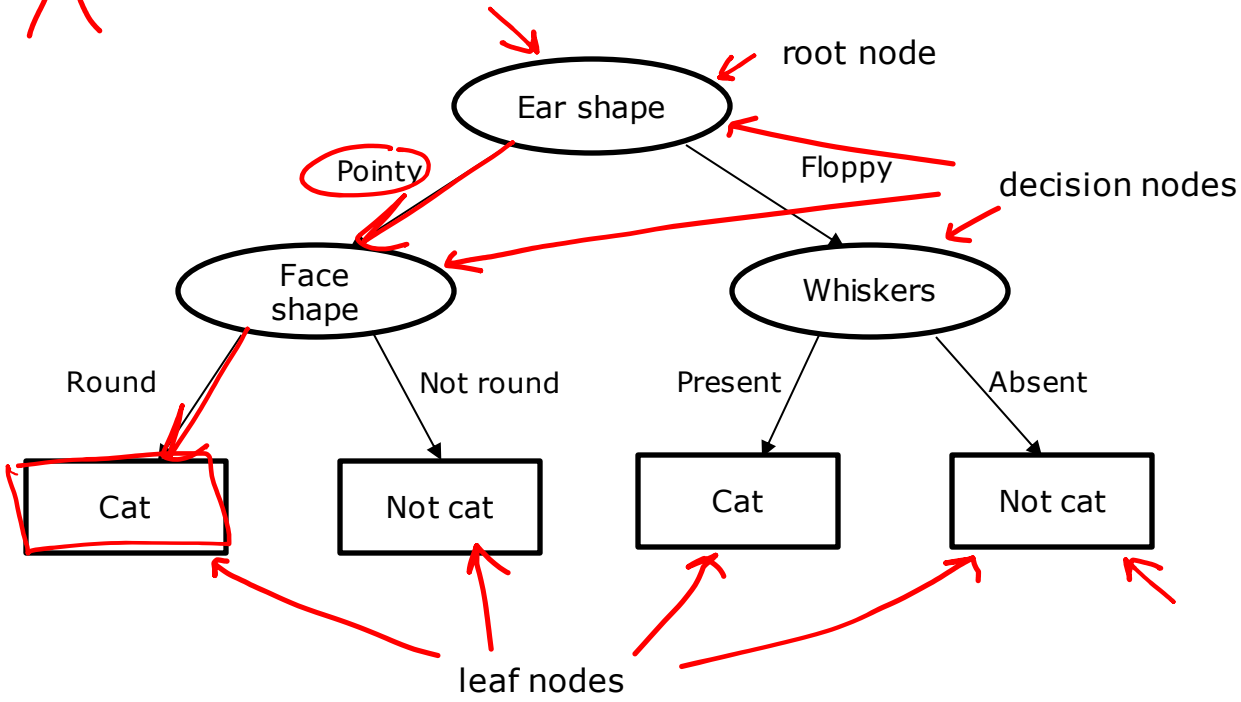
Categorical (discrete values)

X

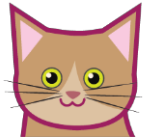
y

Decision Tree



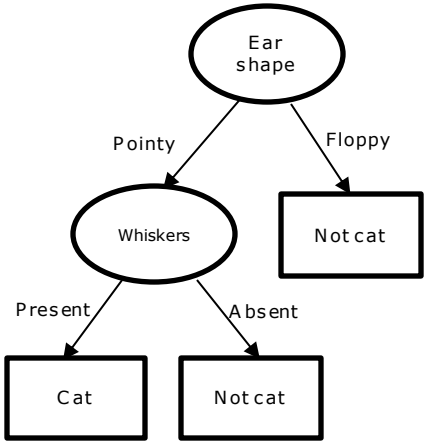


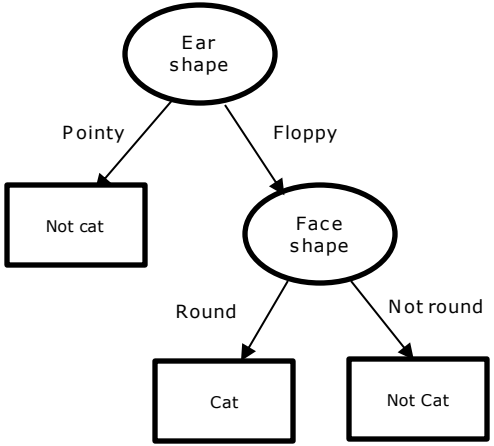
New test example

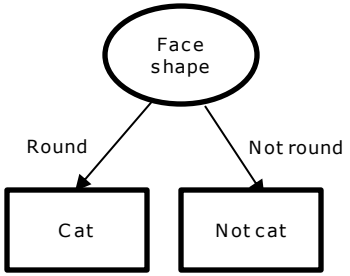


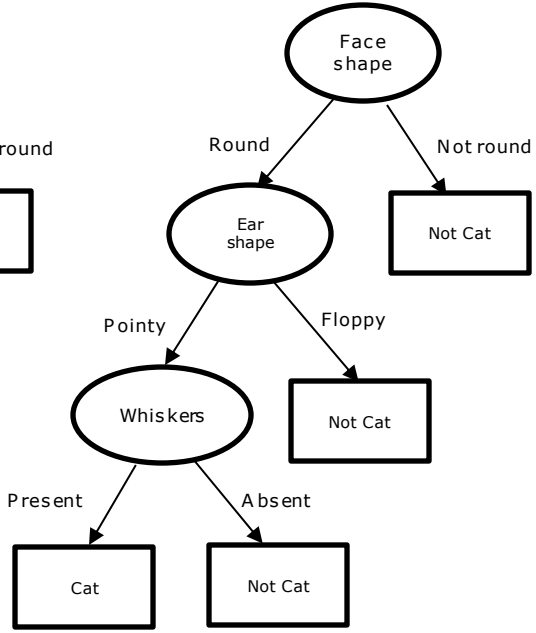
Ear shape: Pointy
Face shape: Round
Whiskers: Present

Decision Tree









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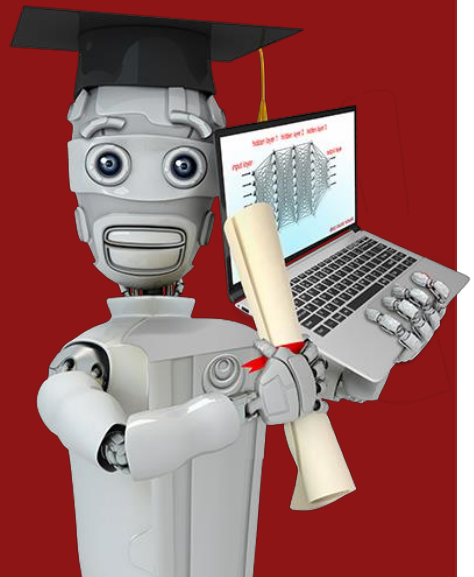
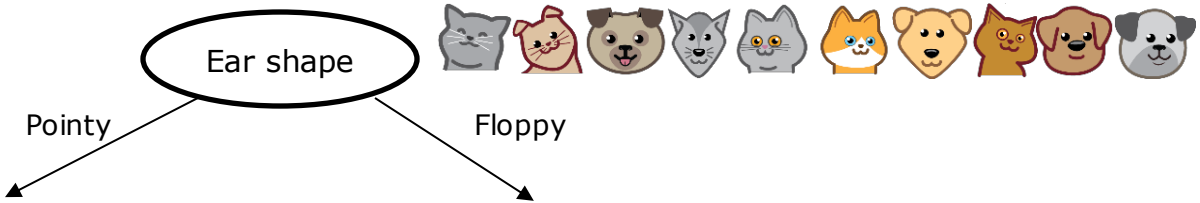


Decision Trees

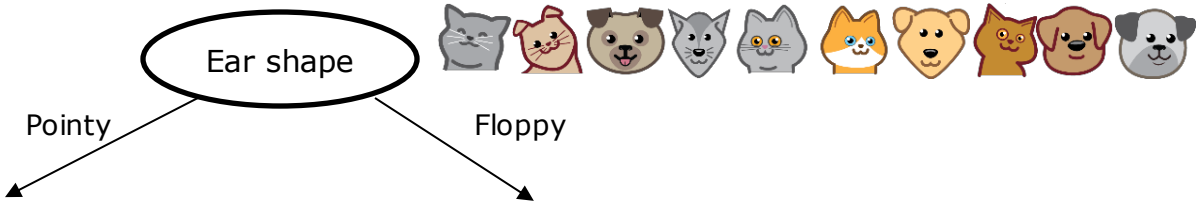
Learning Process

Decision Tree Learning

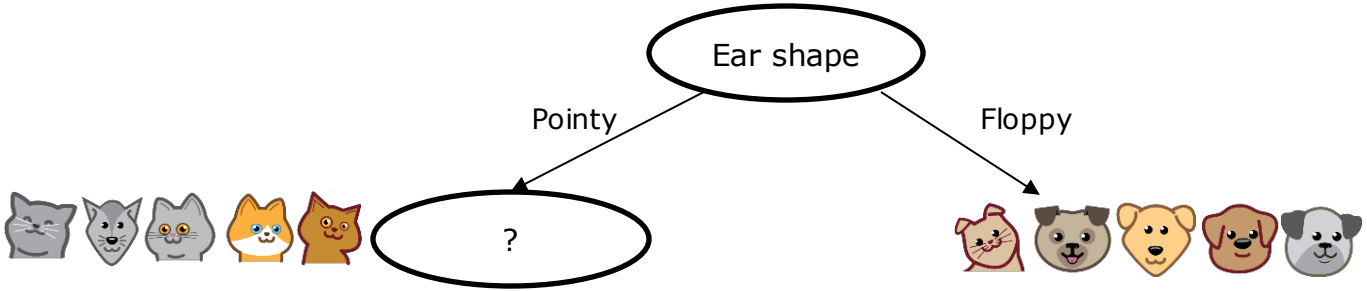
Decision Tree Learning



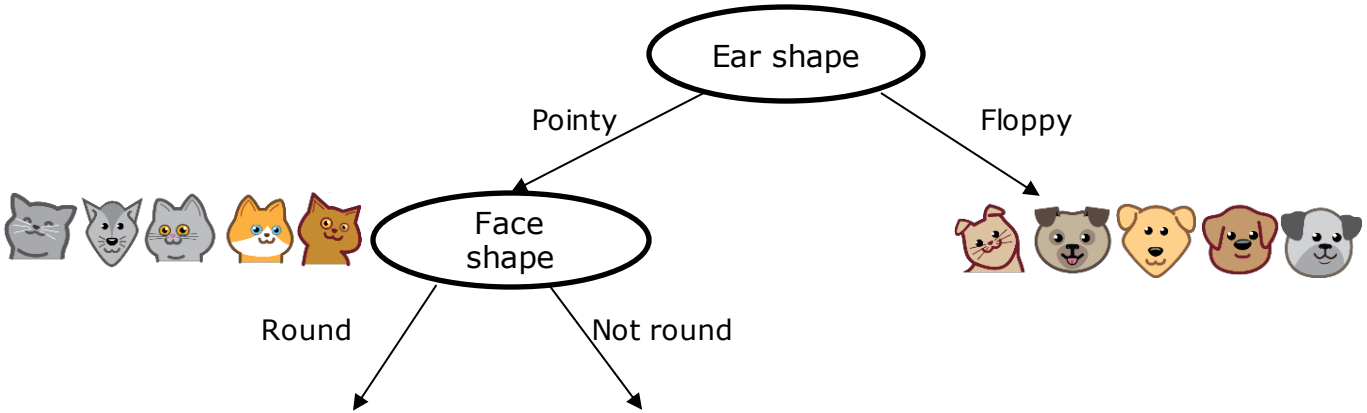
Decision Tree Learning



Decision Tree Learning

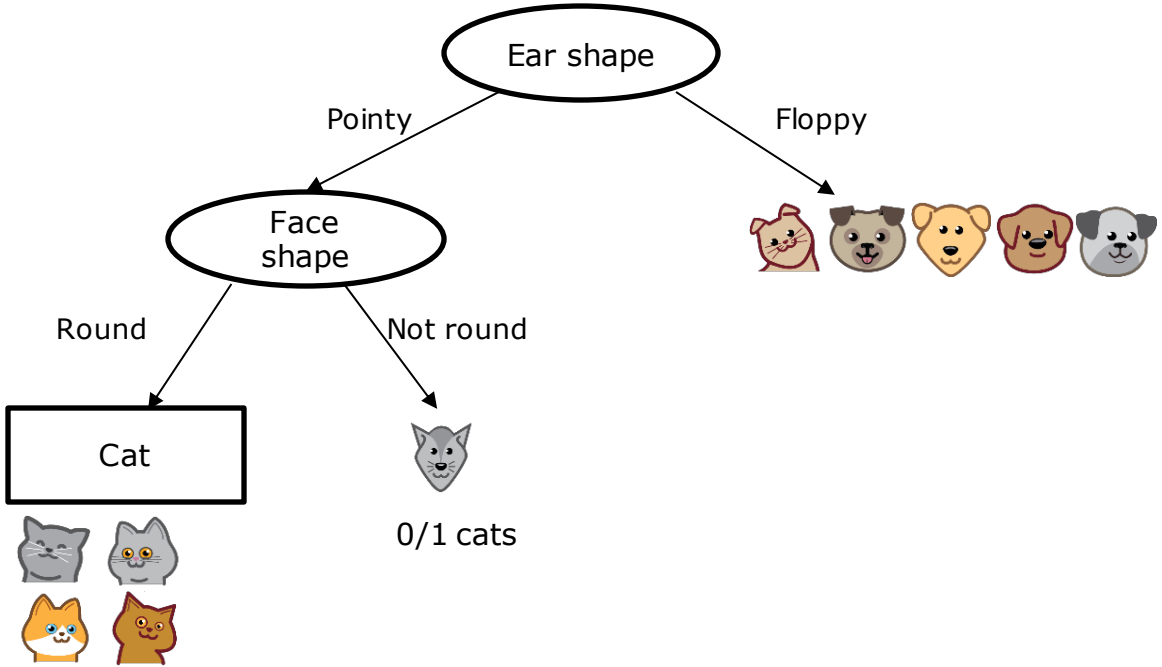


Decision Tree Learning

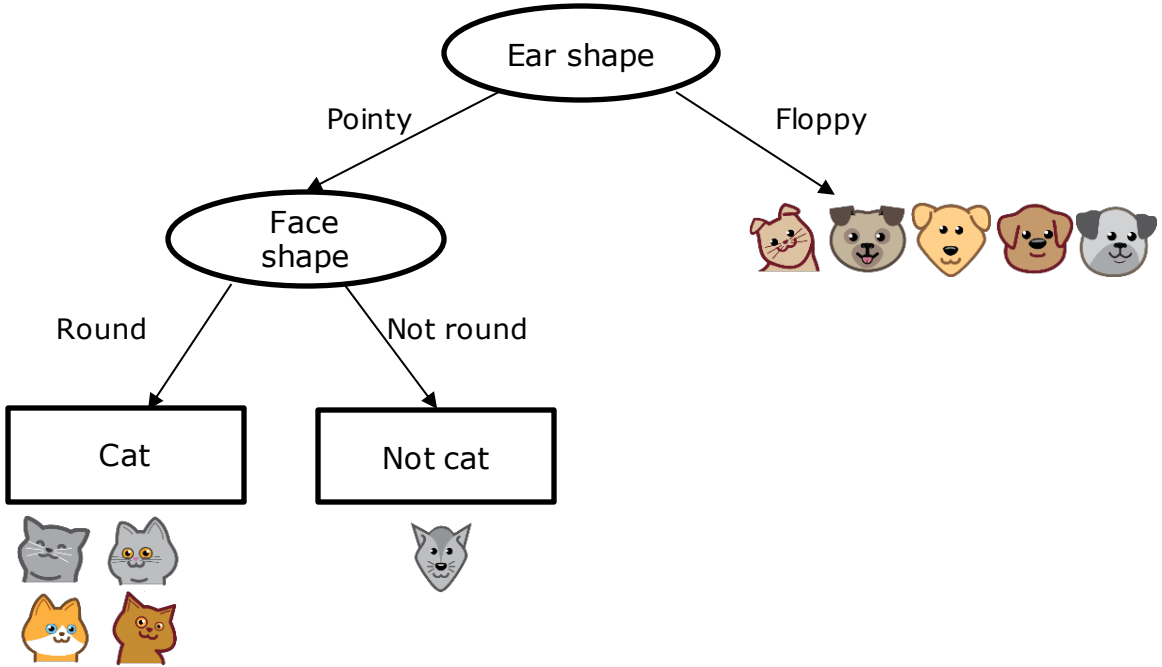


4/4 cats

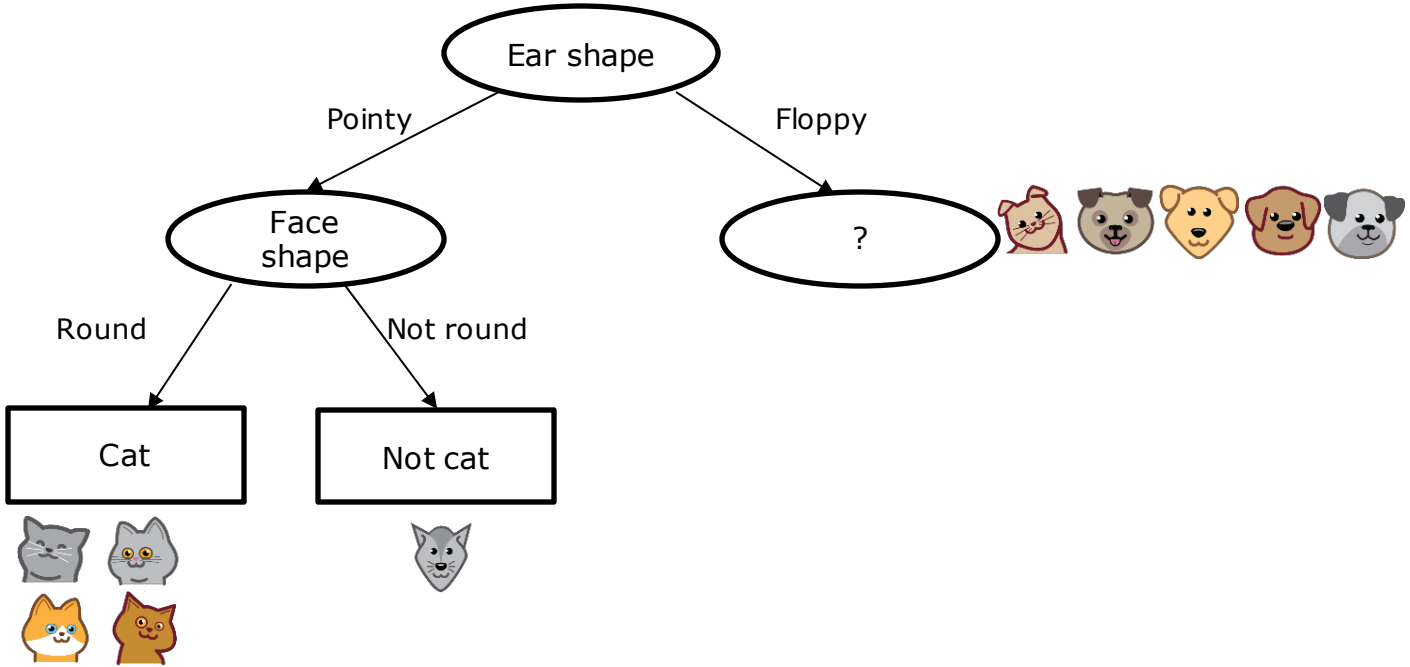
Decision Tree Learning



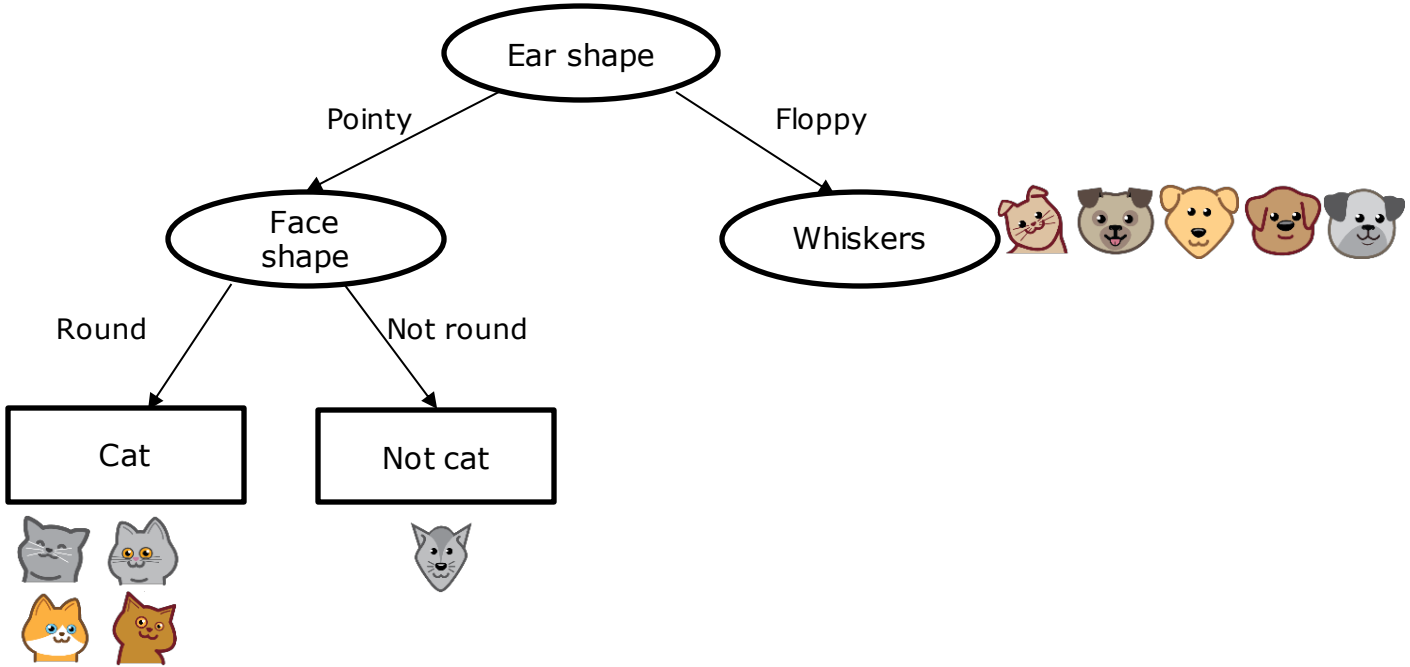
Decision Tree Learning



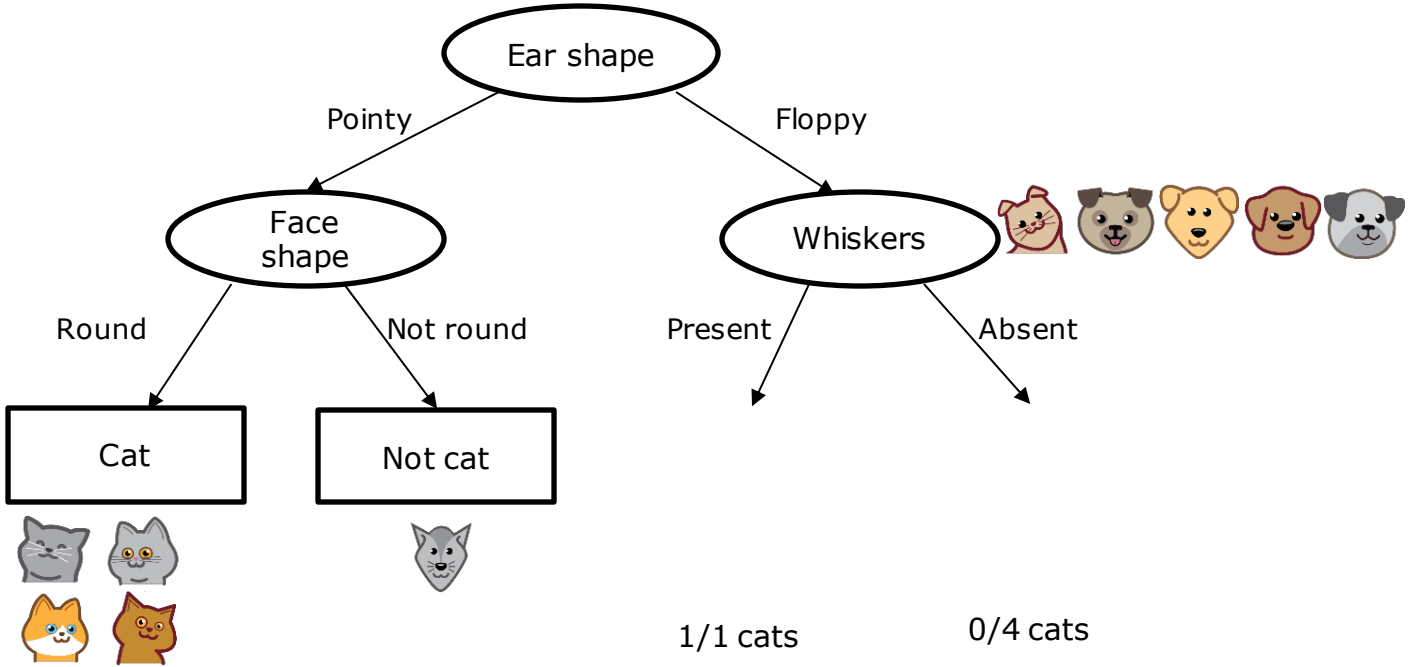
Decision Tree Learning



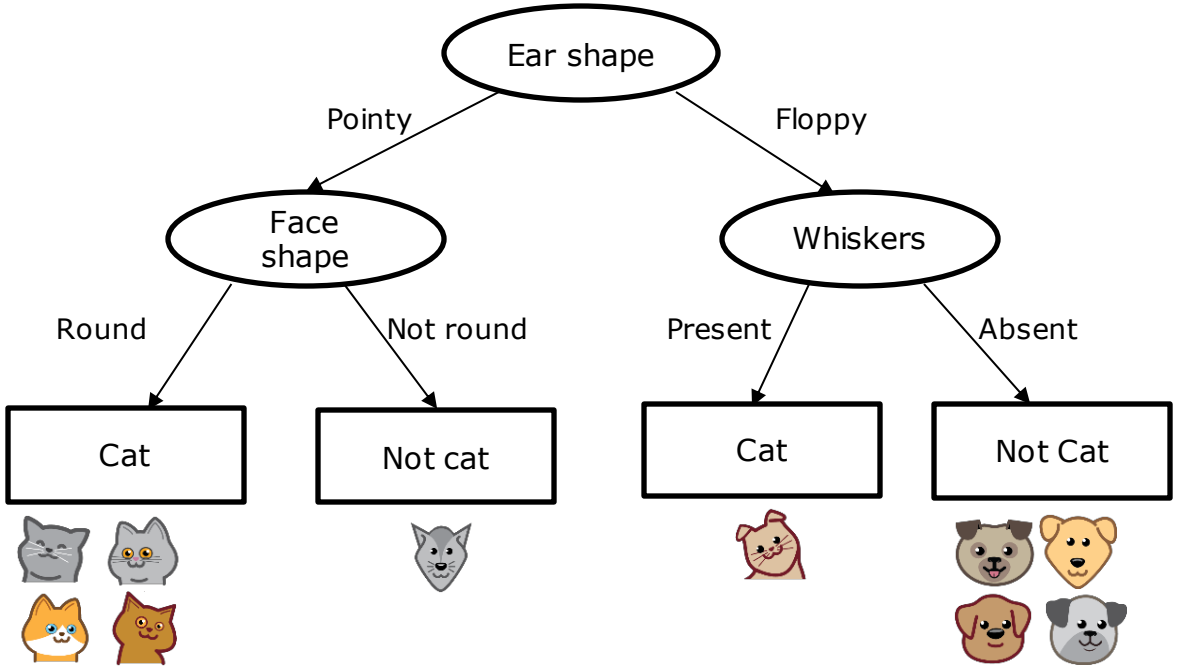
Decision Tree Learning



Decision Tree Learning



Decision Tree Learning



Decision Tree Learning

Decision 1: How to choose what feature to split on at each node?

Maximize purity (or minimize impurity)

Decision Tree Learning

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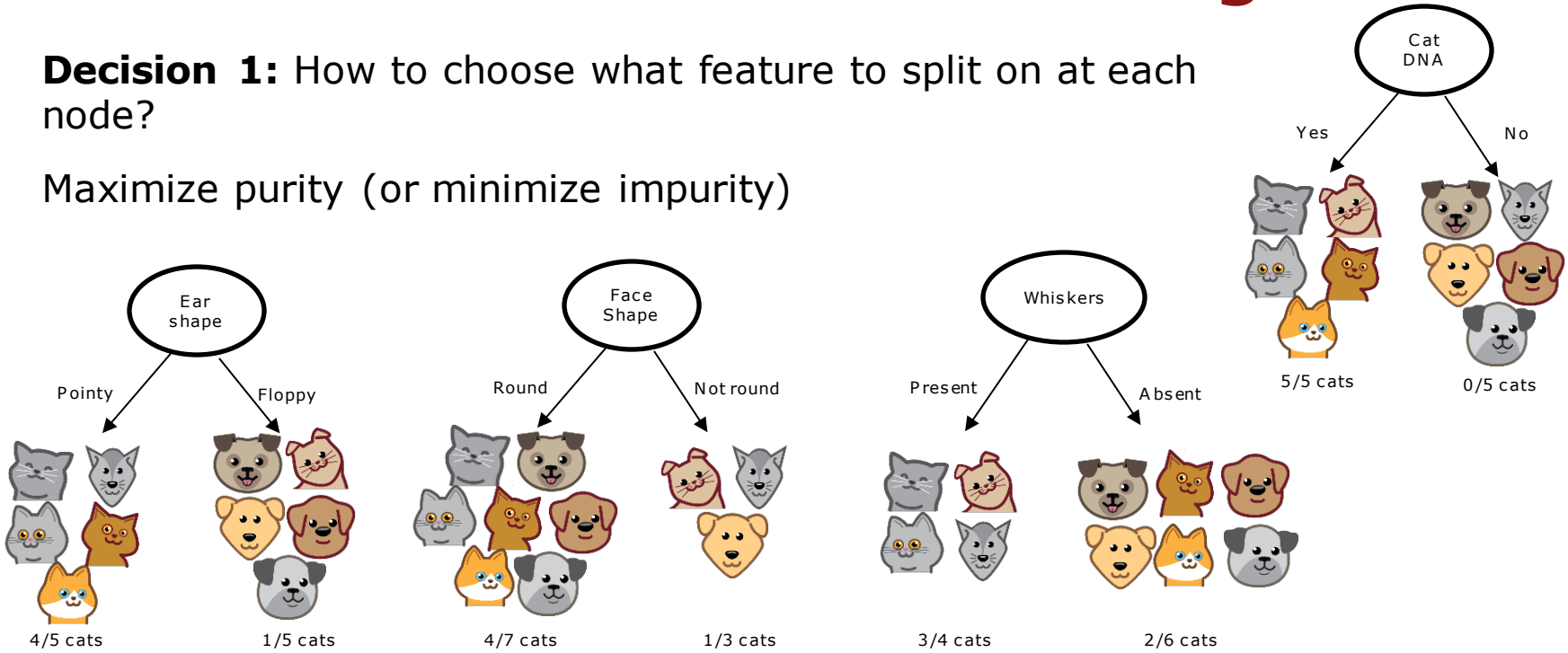
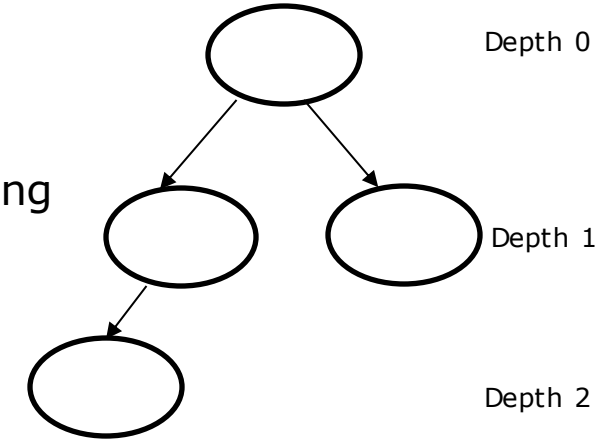
Maximize purity (or minimize impurity)



Decision Tree Learning

Decision 2: When do you stop splitting?

- When a node is 100% one class
- When splitting a node will result in the tree exceeding a maximum depth



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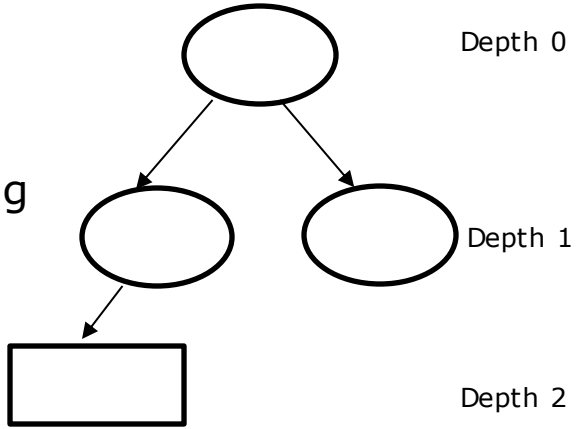
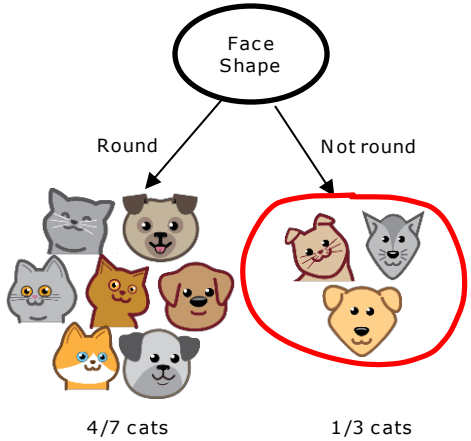


Decision Tree Learning

Decision 2: When do you stop splitting?

- When a node is 100% one class
- When splitting a node will result in the tree exceeding a maximum depth
- When improvements in purity score are below a threshold
- When number of examples in a node is below a threshold



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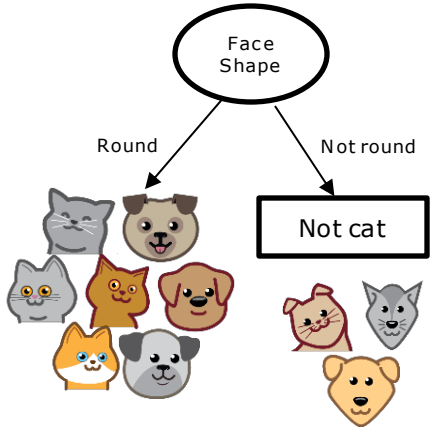

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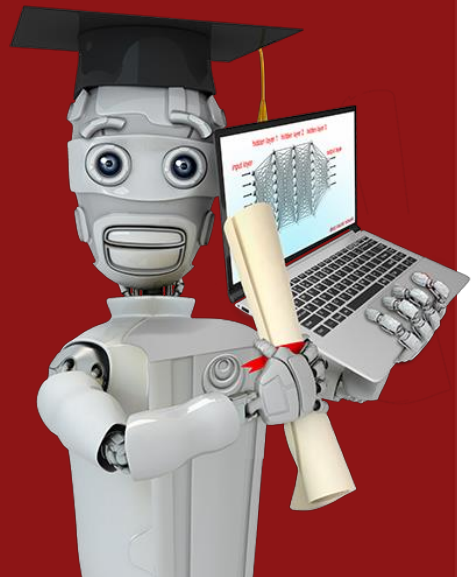
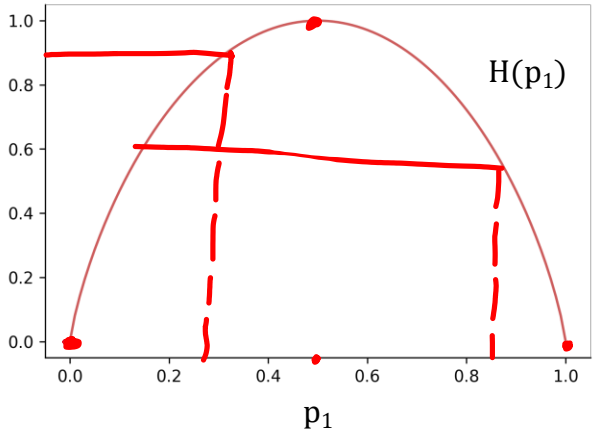


Decision Tree Learning

Measuring purity

Entropy as a measure of impurity

p_1 = fraction of examples that are cats



$p_1 = 0 \quad H(p_1) = 0$

$p_1 = 2/6 \quad H(p_1) = 0.92$

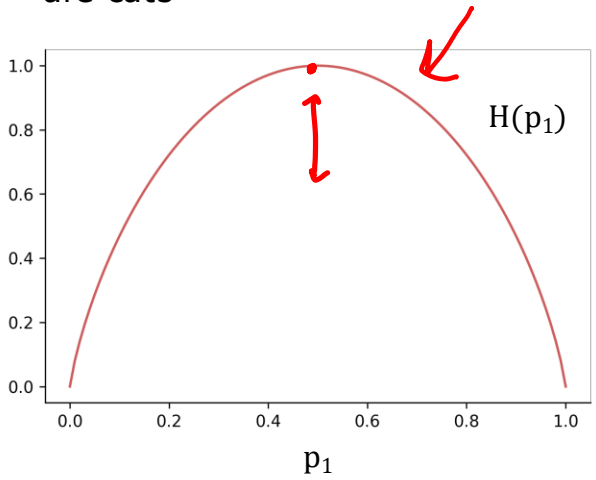
$p_1 = 3/6 \quad H(p_1) = 1$

$p_1 = 5/6 \quad H(p_1) = 0.65$

$p_1 = 6/6 \quad H(p_1) = 0$

Entropy as a measure of impurity

p_1 = fraction of examples that are cats



$$p_0 = 1 - p_1$$

$$H(p_1) = -p_1 log_2(p_1) - p_0 log_2(p_0)$$

$$= -p_1 \ log_2(p_1) - (1 - p_1) log_2(1 - p_1)$$

Note: "$0 \log(0)$" $= 0$

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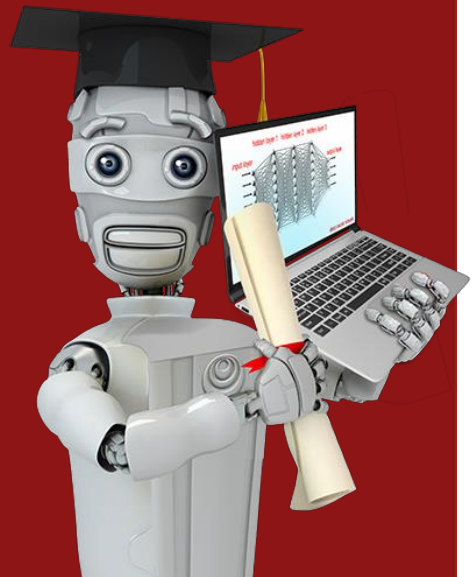


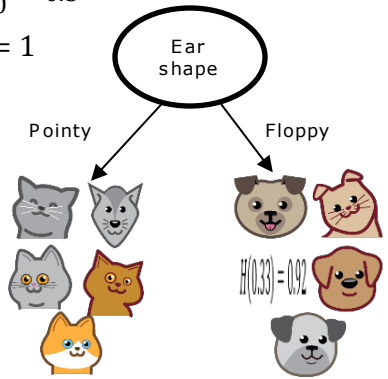
Decision Tree Learning

Choosing a split: Information Gain

Choosing a split

$p_1 = {}^5/_{10} = 0.5$

$H(0.5) = 1$



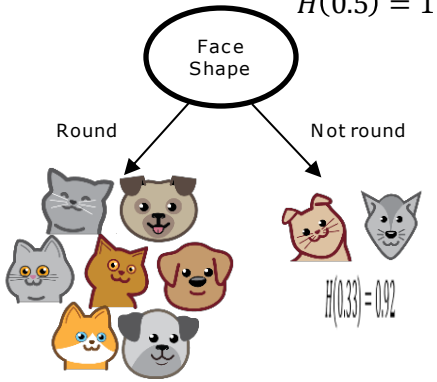
$p_1 = {}^4/_5 = 0.8 \quad p_1 = {}^1/_5 = 0.2$

$H(0.8) = 0.72 \quad H(0.2) = 0.72$

$$H(0.5) - \left(\frac{5}{10}H(0.8) + \frac{5}{10}H(0.2)\right)$$

$= 0.28$

$H(0.5) = 1$



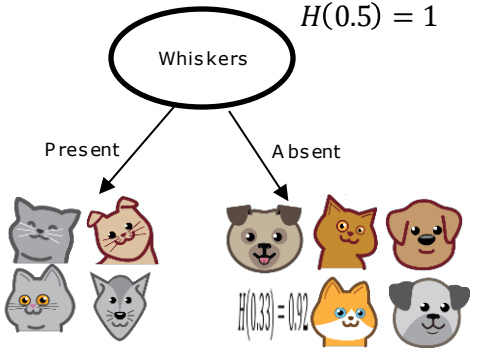
$p_1 = {}^4/_7 = 0.57 \quad p_1 = {}^1/_3 = 0.33$

$H(0.57) = 0.99 \quad H(0.33) = 0.92$

$$H(0.5) - \left(\frac{7}{10}H(0.57) + \frac{3}{10}H(0.33)\right)$$

$= 0.03$

$H(0.5) = 1$



$p_1 = {}^3/_4 = 0.75 \quad p_1 = {}^2/_6 = 0.33$

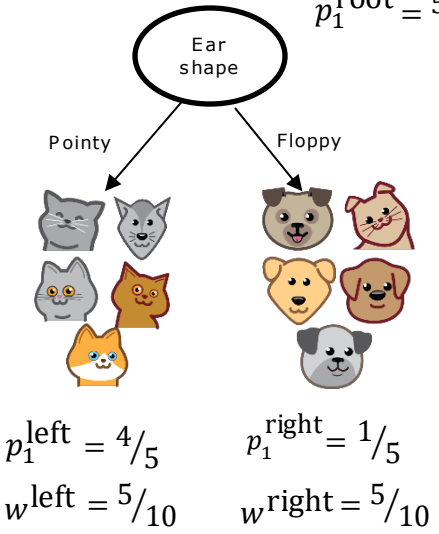
$H(0.75) = 0.81 \quad H(0.33) = 0.92$

$$H(0.5) - \left(\frac{4}{10}H(0.75) + \frac{6}{10}H(0.33)\right)$$

$= 0.12$

Information Gain

$p_1^{\text{root}} = 5/10 = 0.5$



$p_1^{\text{left}} = 4/5$

$w^{\text{left}} = 5/10$

$p_1^{\text{right}} = 1/5$

$w^{\text{right}} = 5/10$

Information gain

$$= \mathrm{H}(\mathrm{p}_1^{\text{root}}) - \left(w^{\text{left}} H\left(p_1^{\text{left}}\right) + w^{\text{right}} H\left(p_1^{\text{right}}\right) \right)$$

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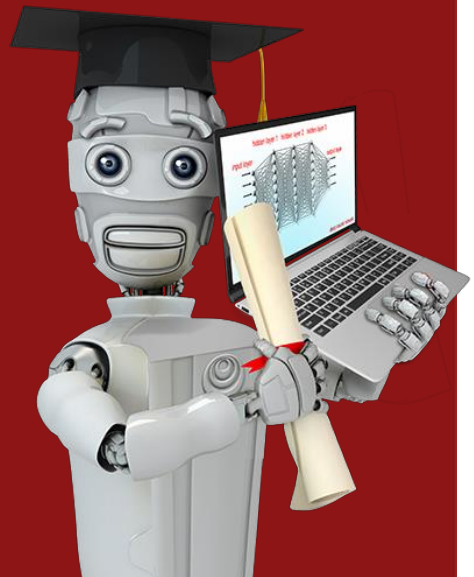


Decision Tree Learning

Putting it together

Decision Tree Learning

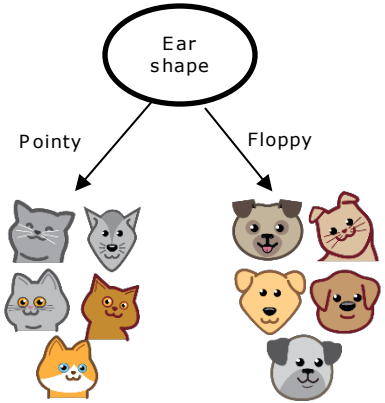
- Start with all examples at the root node
- Calculate information gain for all possible features, and pick the one with the highest information gain
- Split dataset according to selected feature, and create left and right branches of the tree
- Keep repeating splitting process until stopping criteria is met:
    - When a node is 100% one class
    - When splitting a node will result in the tree exceeding a maximum depth
    - Information gain from additional splits is less than threshold
    - When number of examples in a node is below a threshold

Recursive splitting

Recursive splitting

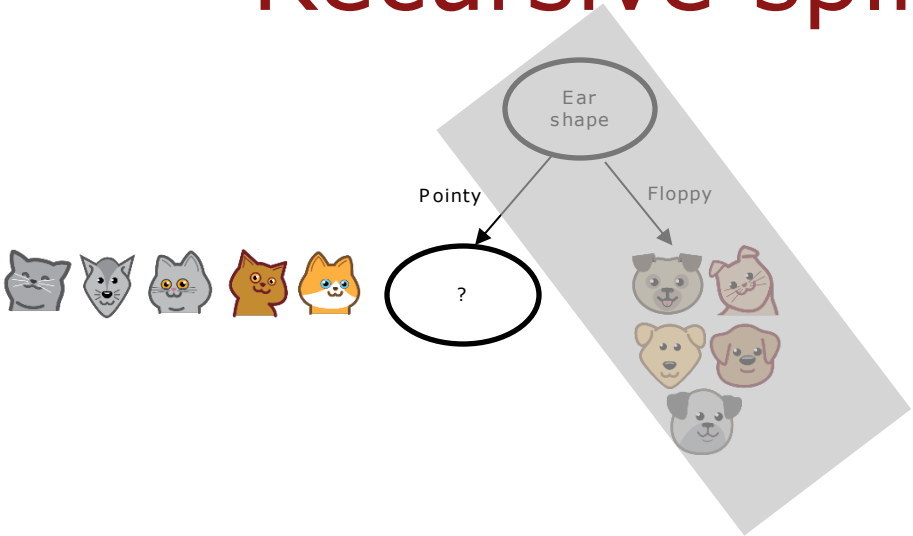
Recursive splitting

Recursive splitting

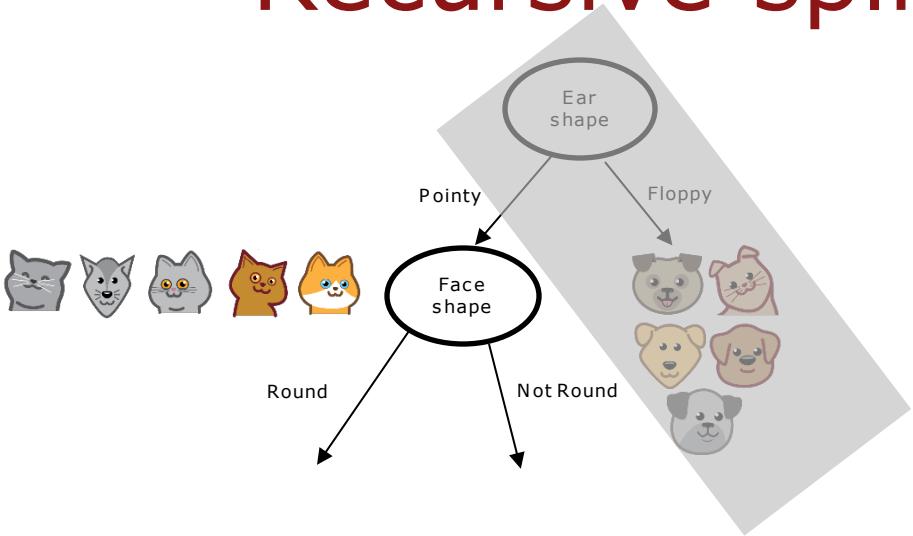


Recursive splitting

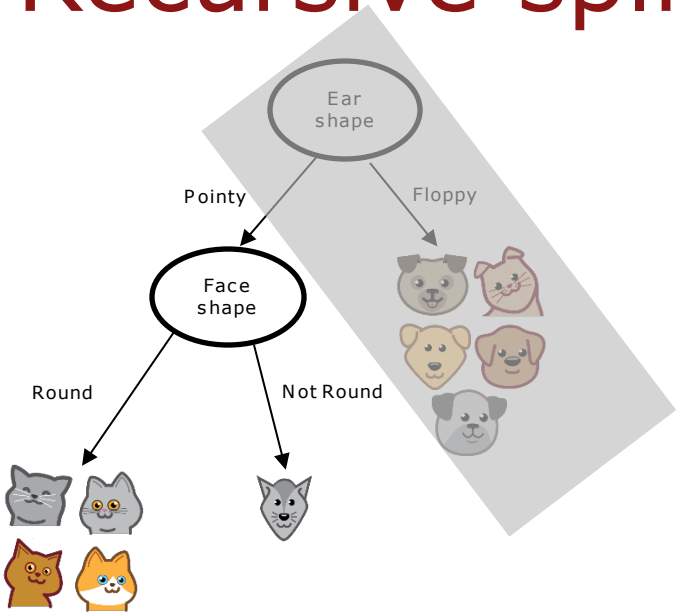
Recursive splitting



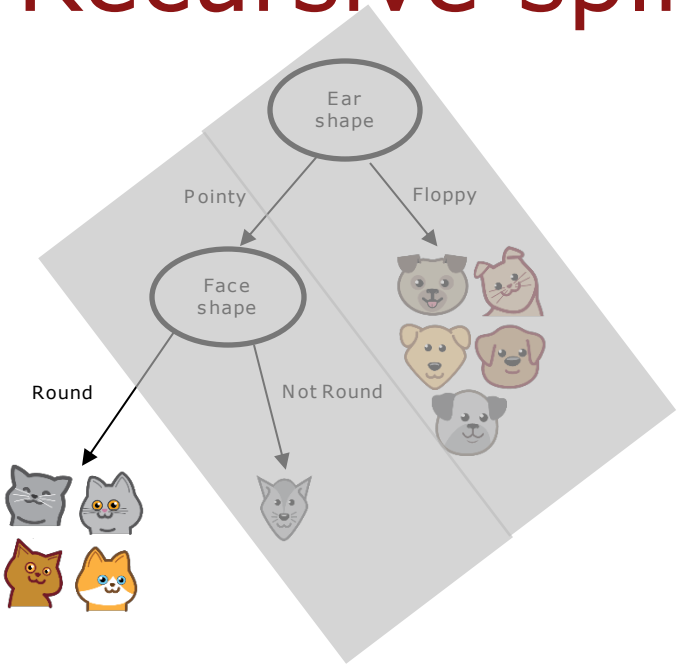
Recursive splitting



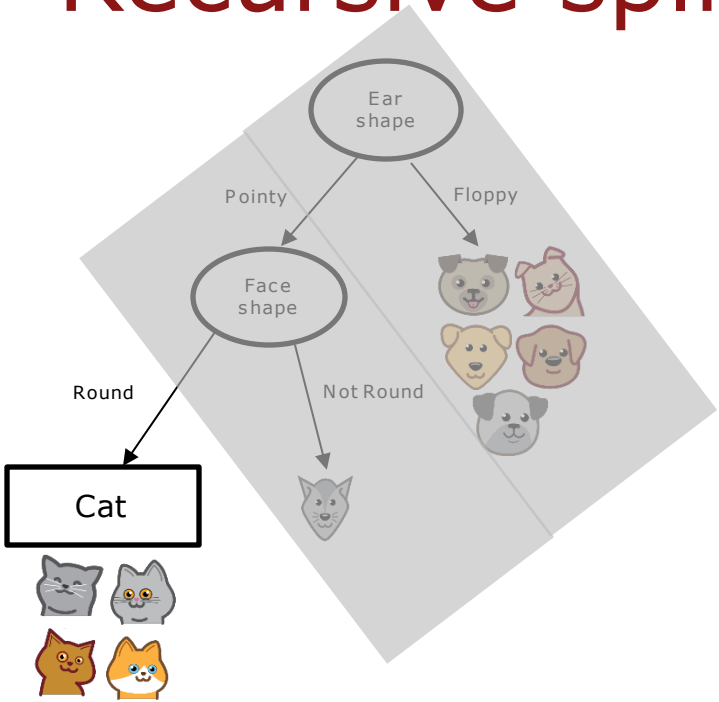
Recursive splitting



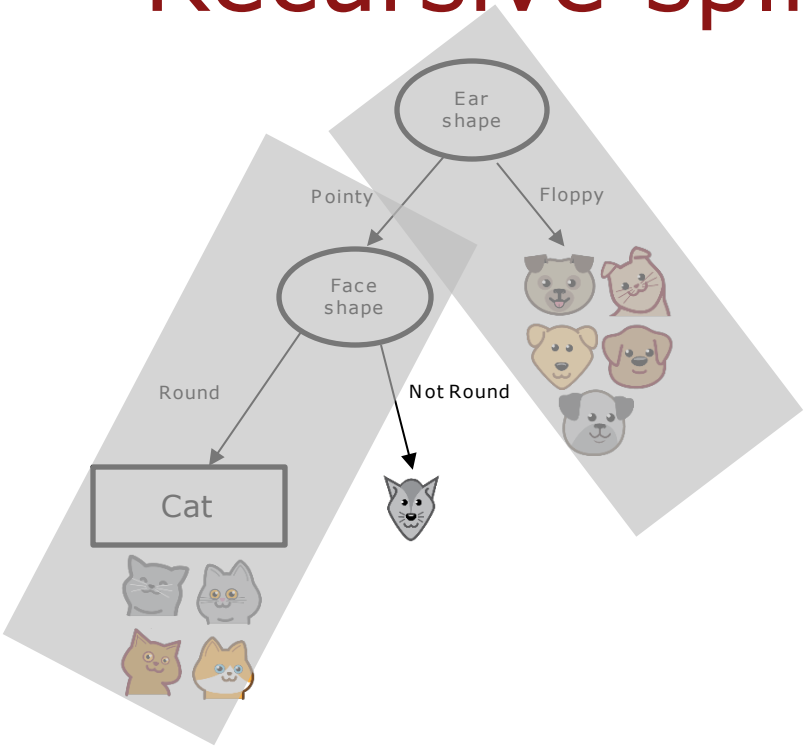
Recursive splitting



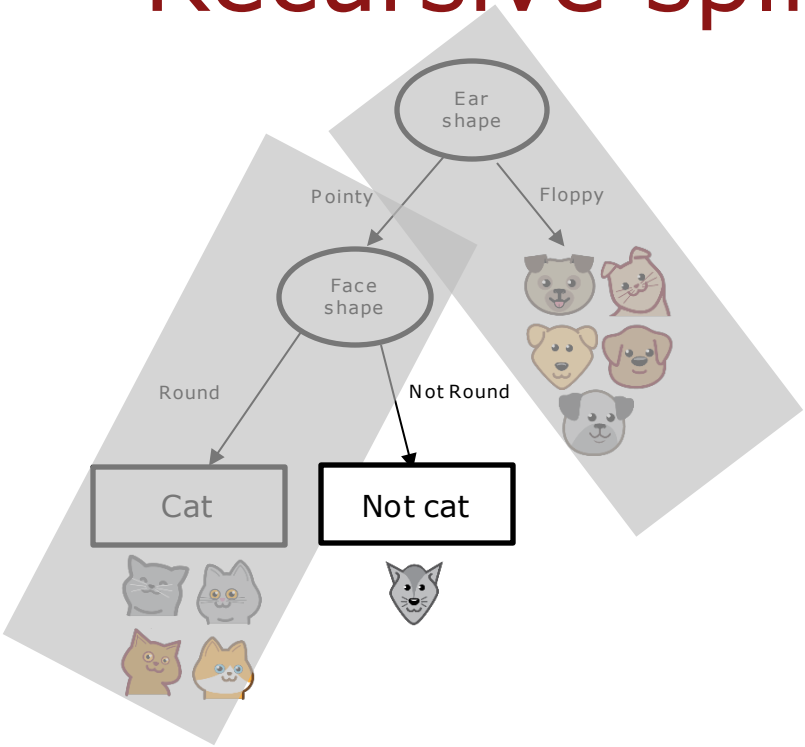
Recursive splitting



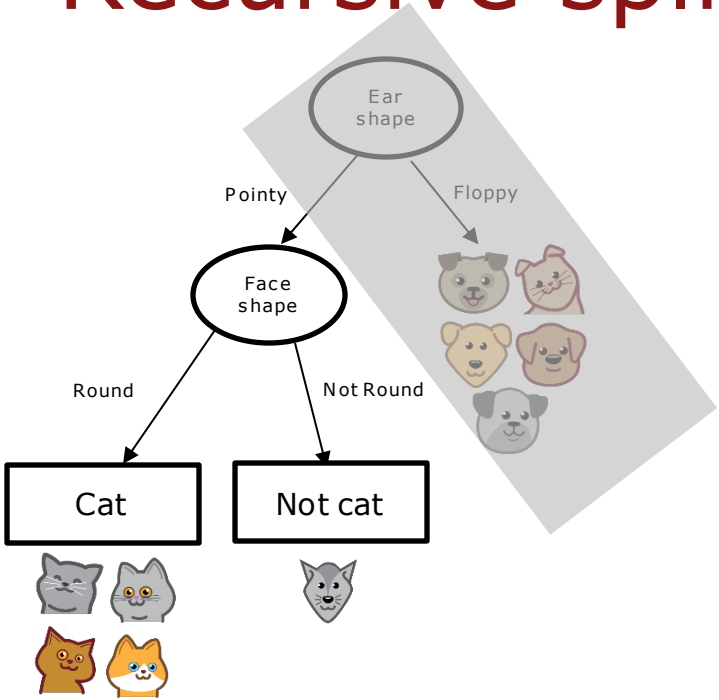
Recursive splitting



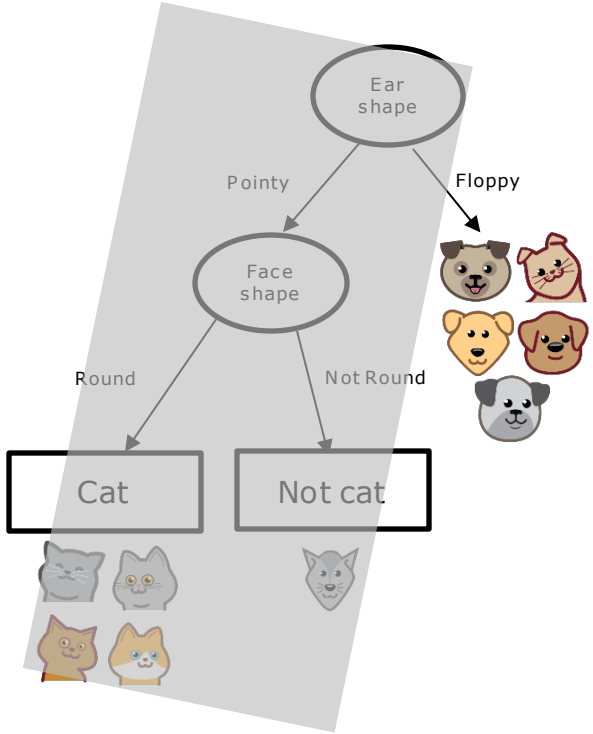
Recursive splitting



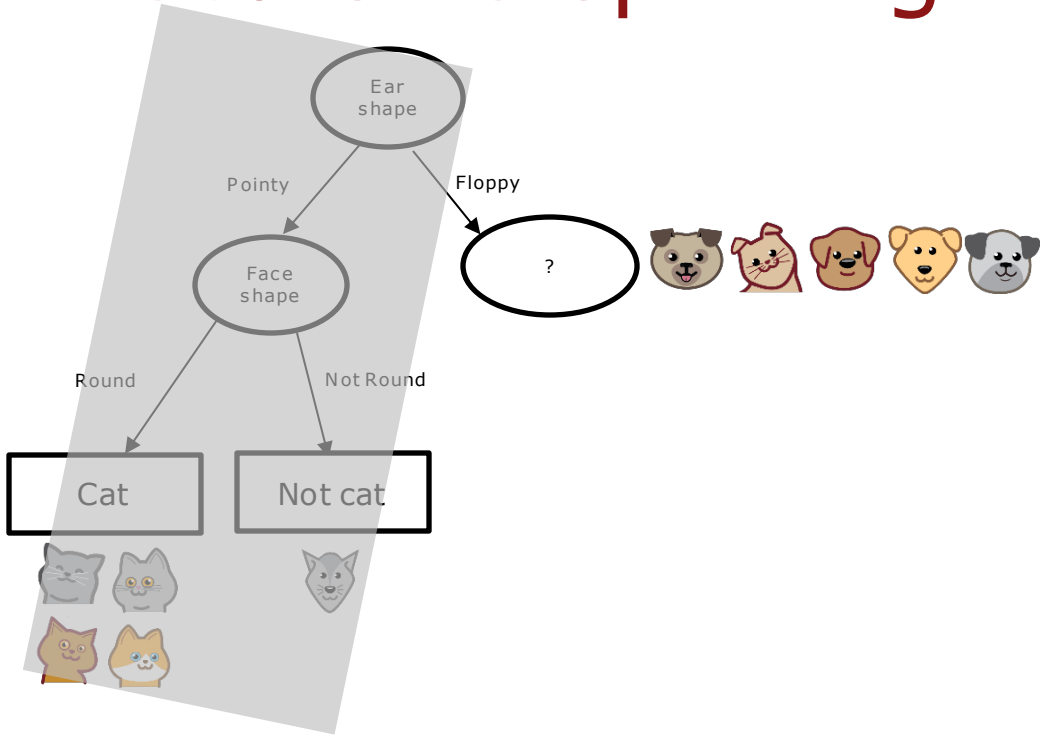
Recursive splitting



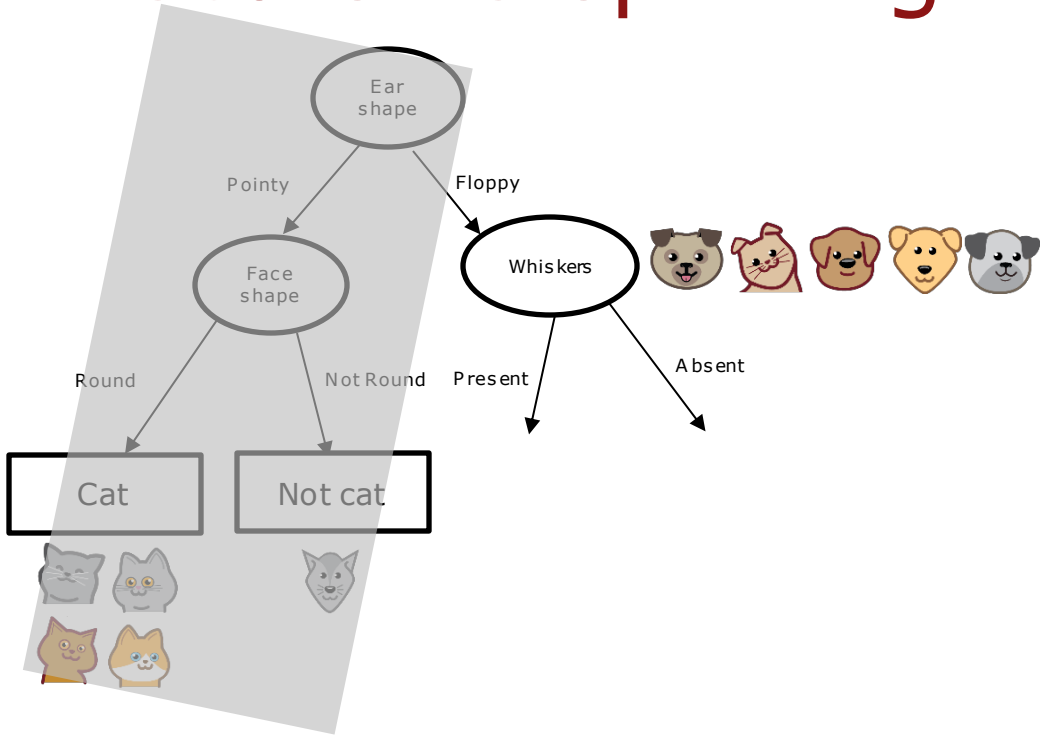
Recursive splitting



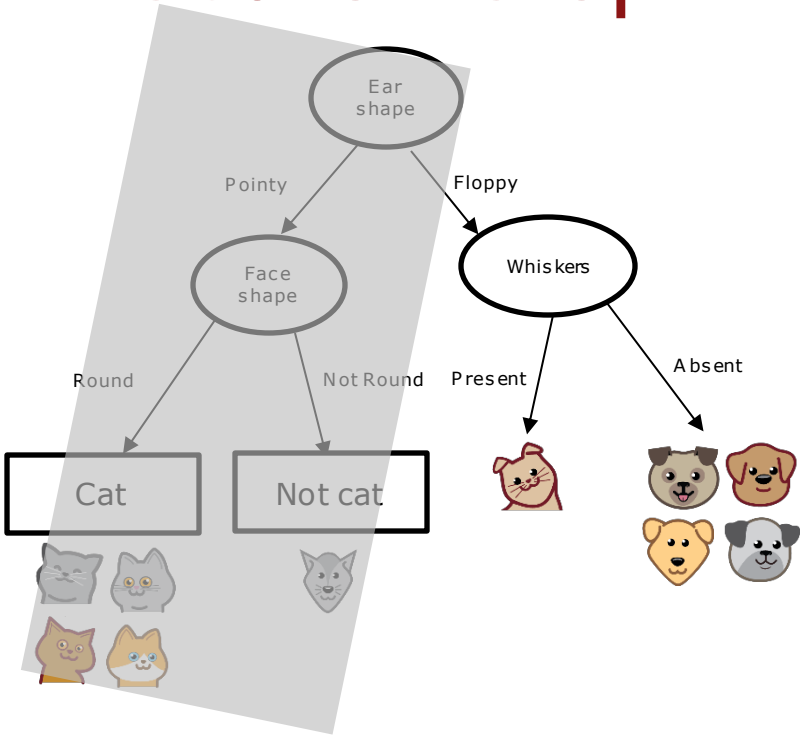
Recursive splitting



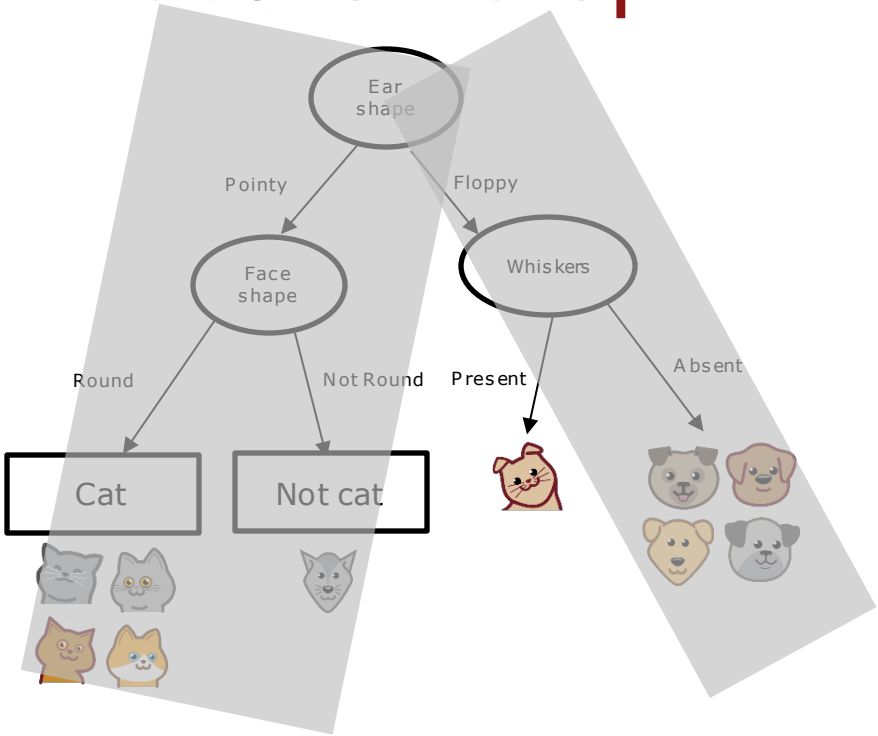
Recursive splitting



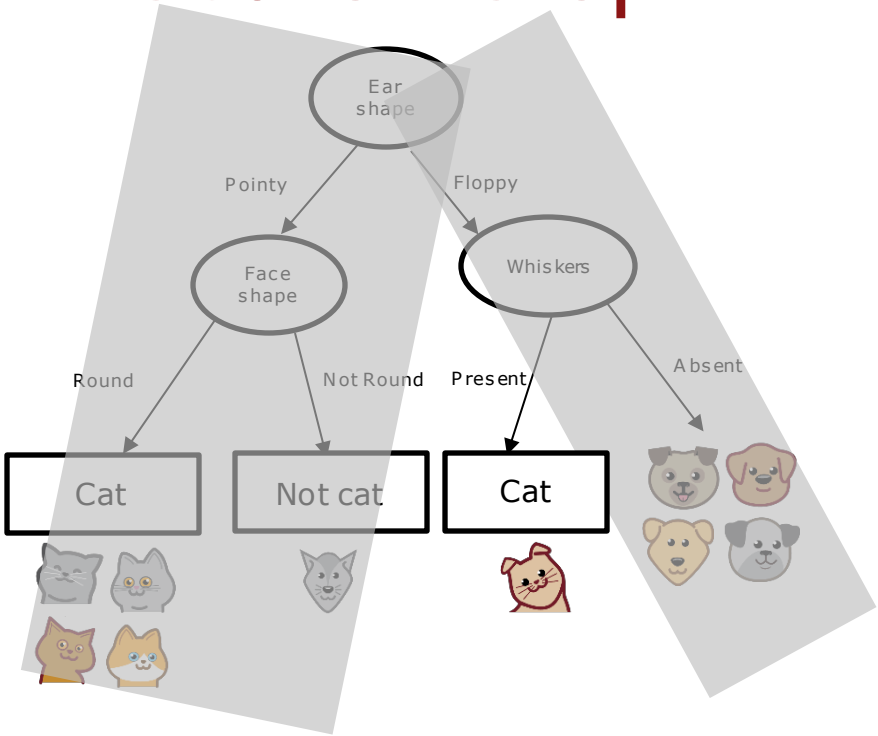
Recursive splitting



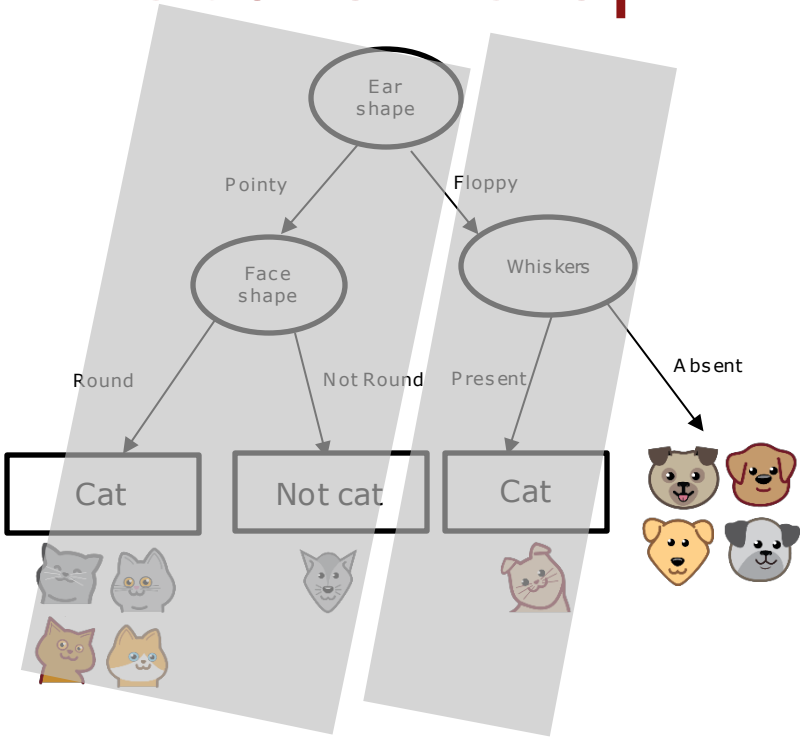
Recursive splitting



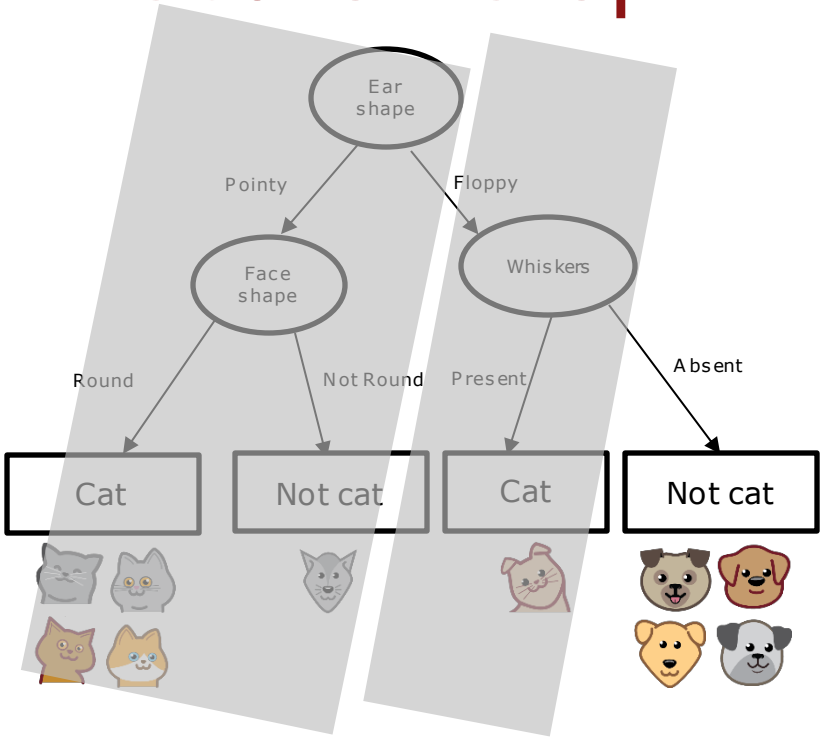
Recursive splitting



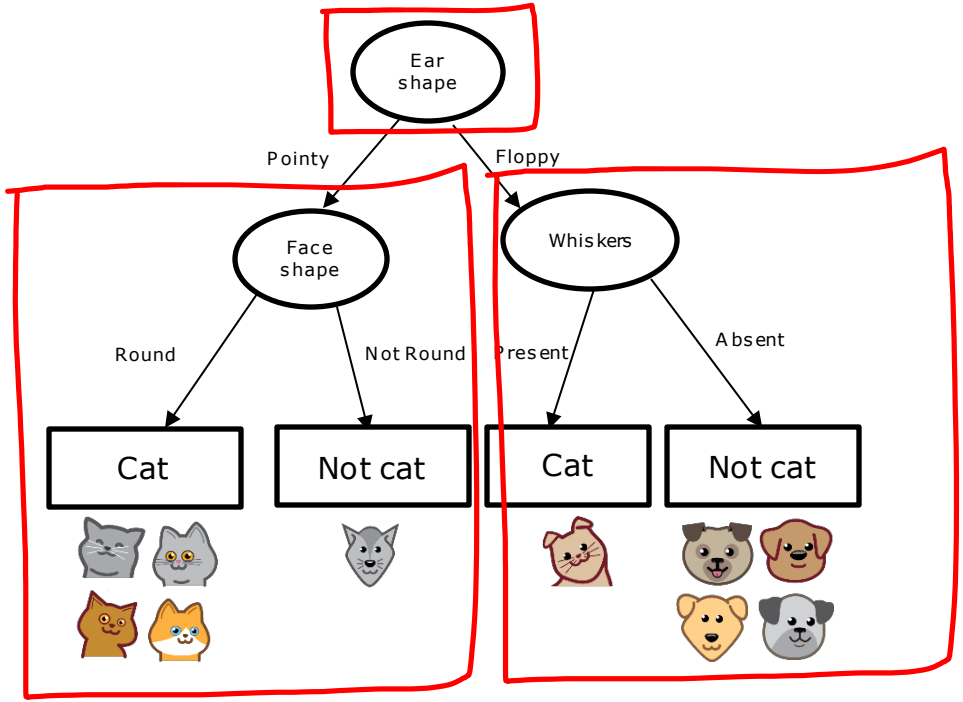
Recursive splitting



Recursive splitting



Recursive splitting



Recursive algorithm

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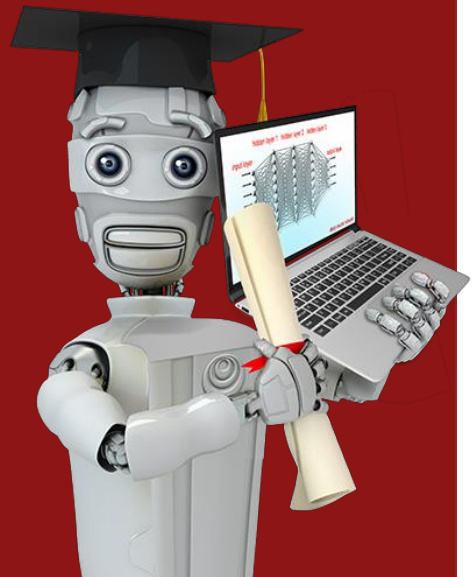
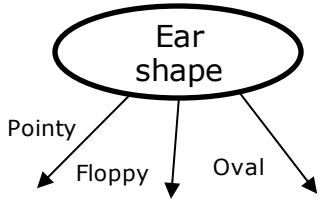


Decision Tree Learning

Using one-hot encoding of categorical features

Features with three possible values

	Ear shape (x_1)	Face shape (x_2)	Whiskers (x_3)	Cat (y)
	Pointy	Round	Present	1
	Oval	Not round	Present	1
	Oval	Round	Absent	0
	Pointy	Not round	Present	0
	Oval	Round	Present	1
	Pointy	Round	Absent	1
	Floppy	Not round	Absent	0
	Oval	Round	Absent	1
	Floppy	Round	Absent	0
	Floppy	Round	Absent	0



3 possible values

One hot encoding

	~~Ear shape~~	Pointy ears	Floppy ears	Oval ears	Face shape	Whiskers	Cat
	~~Pointy~~	1	0	0	Round	Present	1
	~~Oval~~	0	0	1	Not round	Present	1
	~~Oval~~	0	0	1	Round	Absent	0
	~~Pointy~~	1	0	0	Not round	Present	0
	~~Oval~~	0	0	1	Round	Present	1
	~~Pointy~~	1	0	0	Round	Absent	1
	~~Floppy~~	0	1	0	Not round	Absent	0
	~~Oval~~	0	0	1	Round	Absent	1
	~~Floppy~~	0	1	0	Round	Absent	0
	~~Floppy~~	0	1	0	Round	Absent	0

One hot encoding

If a categorical feature can take on k values, create k binary features (0 or 1 valued).

One hot encoding

	~~Ear shape~~	Pointy ears	Floppy ears	Oval ears	Face shape	Whiskers	Cat
	~~Pointy~~	1	0	0	Round	Present	1
	~~Oval~~	0	0	1	Not round	Present	1
	~~Oval~~	0	0	1	Round	Absent	0
	~~Pointy~~	1	0	0	Not round	Present	0
	~~Oval~~	0	0	1	Round	Present	1
	~~Pointy~~	1	0	0	Round	Absent	1
	~~Floppy~~	0	1	0	Not round	Absent	0
	~~Oval~~	0	0	1	Round	Absent	1
	~~Floppy~~	0	1	0	Round	Absent	0
	~~Floppy~~	0	1	0	Round	Absent	0

One hot encoding and neural networks

	Pointy ears	Floppy ears	Round ears	Face shape	Whiskers	Cat
	1	0	0	~~Round~~ 1	~~Present~~ 1	1
	0	0	1	~~Not round~~ 0	~~Present~~ 1	1
	0	0	1	~~Round~~ 1	~~Absent~~ 0	0
	1	0	0	~~Not round~~ 0	~~Present~~ 1	0
	0	0	1	~~Round~~ 1	~~Present~~ 1	1
	1	0	0	~~Round~~ 1	~~Absent~~ 0	1
	0	1	0	~~Not round~~ 0	~~Absent~~ 0	1
	0	0	1	~~Round~~ 1	~~Absent~~ 0	1
	0	1	0	~~Round~~ 1	~~Absent~~ 0	1
	0	1	0	~~Round~~ 1	~~Absent~~ 0	1

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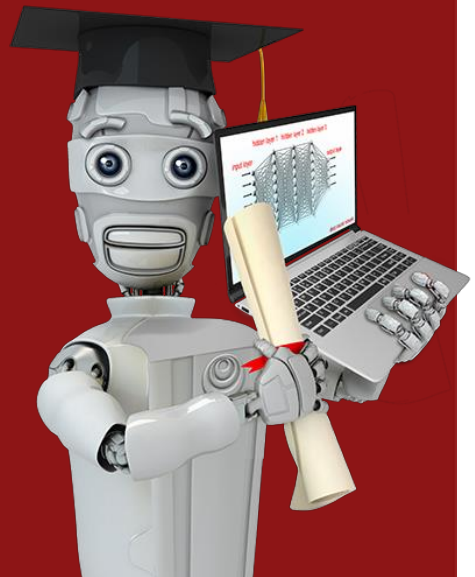


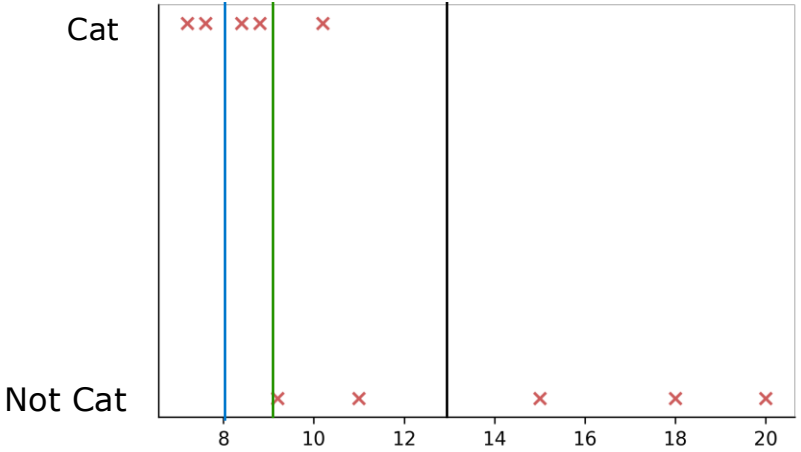
Decision Tree Learning

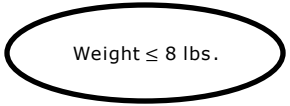
Continuous valued features

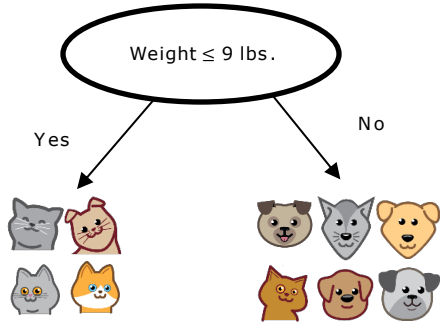
Continuous features

	Ear shape	Face shape	Whiskers	Weight (lbs.)	Cat
	Pointy	Round	Present	7.2	1
	Floppy	Not round	Present	8.8	1
	Floppy	Round	Absent	15	0
	Pointy	Not round	Present	9.2	0
	Pointy	Round	Present	8.4	1
	Pointy	Round	Absent	7.6	1
	Floppy	Not round	Absent	11	0
	Pointy	Round	Absent	10.2	1
	Floppy	Round	Absent	18	0
	Floppy	Round	Absent	20	0

Splitting on a continuous variable







$$\mathrm{H}(0.5) - \left(\frac{2}{10} H\left(\frac{2}{2}\right) + \frac{8}{10} H\left(\frac{3}{8}\right)\right) = 0.24$$

$$\mathrm{H}(0.5) - \left(\frac{4}{10} H\left(\frac{4}{4}\right) + \frac{6}{10} H\left(\frac{1}{6}\right)\right) = 0.61$$

$$\mathrm{H}(0.5) - \left(\frac{7}{10} H\left(\frac{5}{7}\right) + \frac{3}{10} H\left(\frac{0}{3}\right)\right) = 0.40$$

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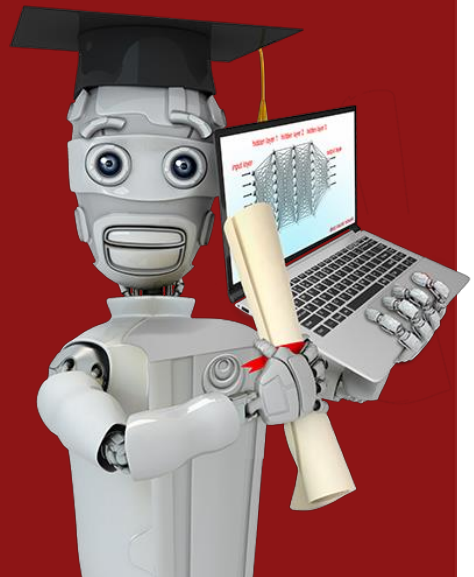


Decision Tree Learning

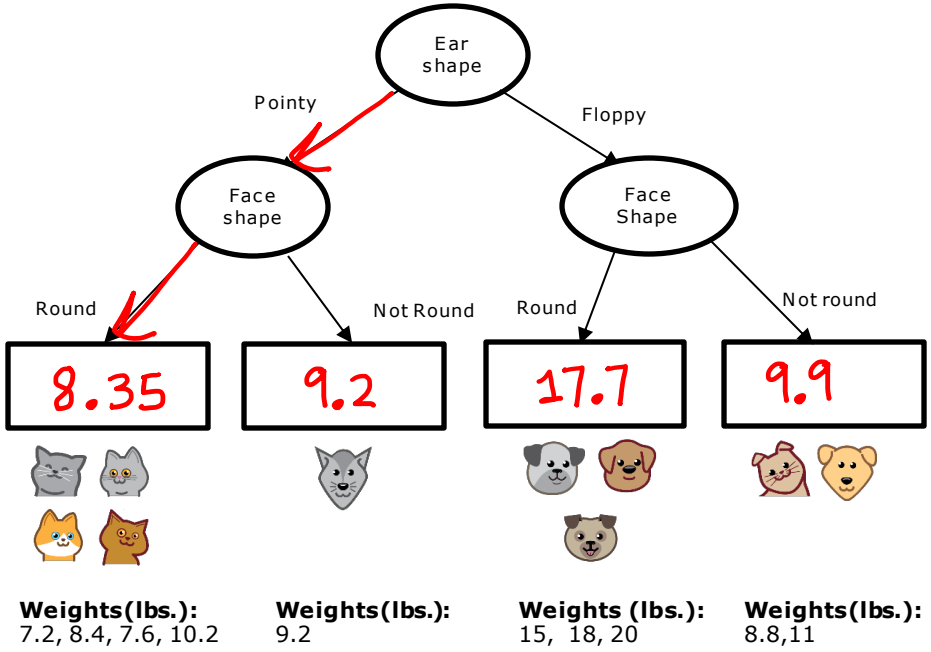
Regression Trees (optional)

Regression with Decision Trees: Predicting a number

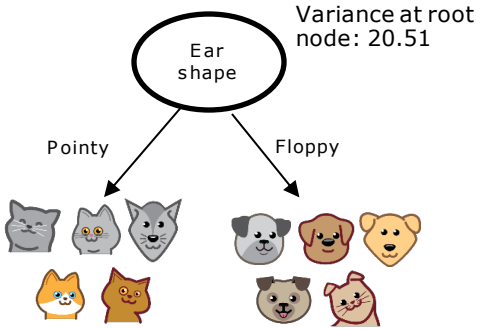
	Ear shape	Face shape	Whiskers	Weight (lbs.)
	Pointy	Round	Present	7.2
	Floppy	Not round	Present	8.8
	Floppy	Round	Absent	15
	Pointy	Not round	Present	9.2
	Pointy	Round	Present	8.4
	Pointy	Round	Absent	7.6
	Floppy	Not round	Absent	11
	Pointy	Round	Absent	10.2
	Floppy	Round	Absent	18
	Floppy	Round	Absent	20

X y

Regression with Decision Trees



Choosing a split



Variance at root node: 20.51

Weights: 7.2, 9.2, 8.4,7.6, 10.2

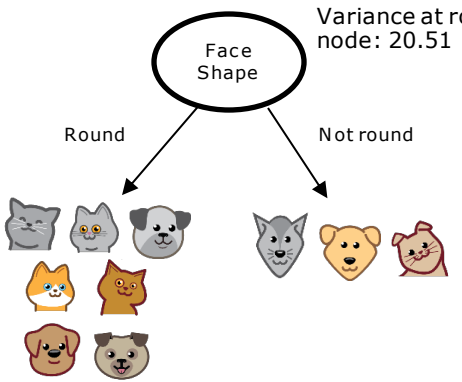
Variance: 1.47

Weights: 8.8, 15, 11, 18, 20

Variance: 21.87

$w^{\text{left}} = 5/10$ $w^{\text{right}} = 5/10$

$$20.51 - \left(\frac{5}{10} * 1.47 + \frac{5}{10} * 21.87\right)$$
$$= 8.84$$



Variance at root node: 20.51

Weights: 7.2, 15, 8.4, 7.6,10.2, 18, 20

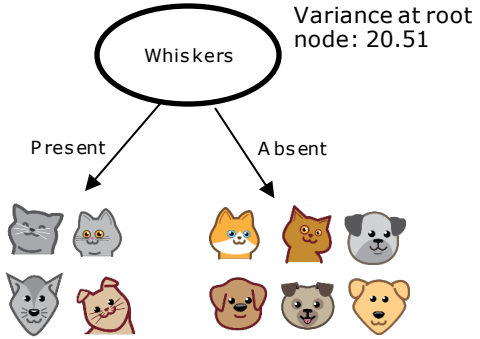
Variance: 27.80

Weights: 8.8,9.2,11

Variance: 1.37

$w^{\text{left}} = 7/10$ $w^{\text{right}} = 3/10$

$$20.51 - \left(\frac{7}{10} * 27.80 + \frac{3}{10} * 1.37\right)$$
$$= 0.64$$



Variance at root node: 20.51

Weights: 7.2, 8.8, 9.2, 8.4

Variance: 0.75

Weights: 15, 7.6, 11, 10.2, 18, 20

Variance: 23.32

$w^{\text{left}} = 4/10$ $w^{\text{right}} = 6/10$

$$20.51 - \left(\frac{4}{10} * 0.75 + \frac{6}{10} * 23.32\right)$$
$$= 6.22$$

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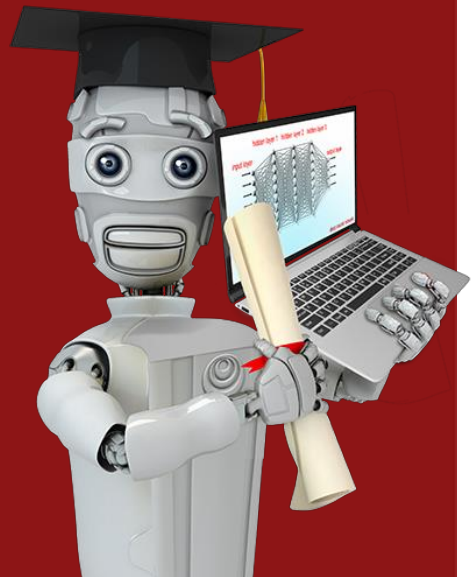
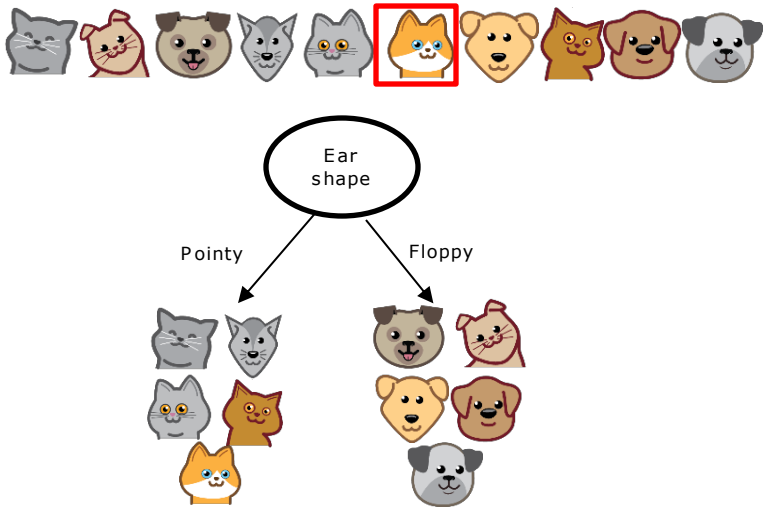
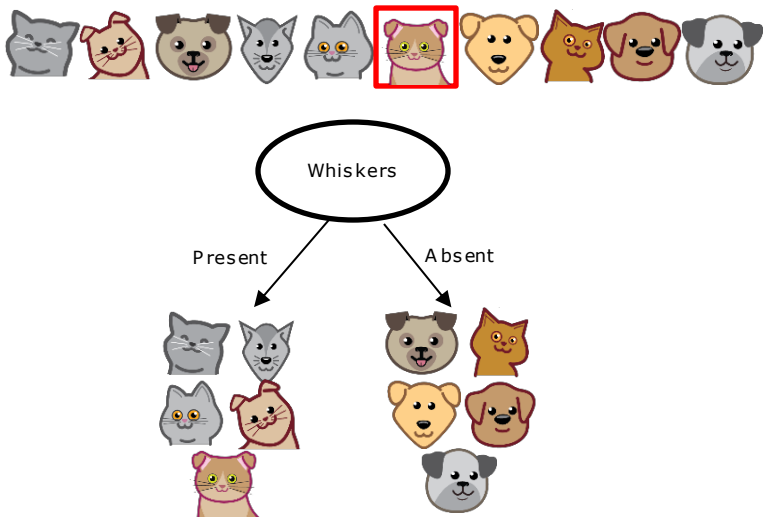


Tree ensembles

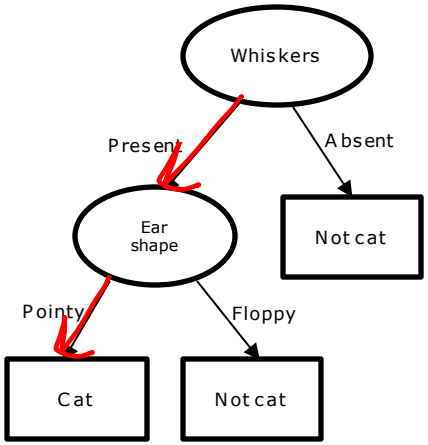
Using multiple decision trees

Trees are highly sensitive to small changes of the data

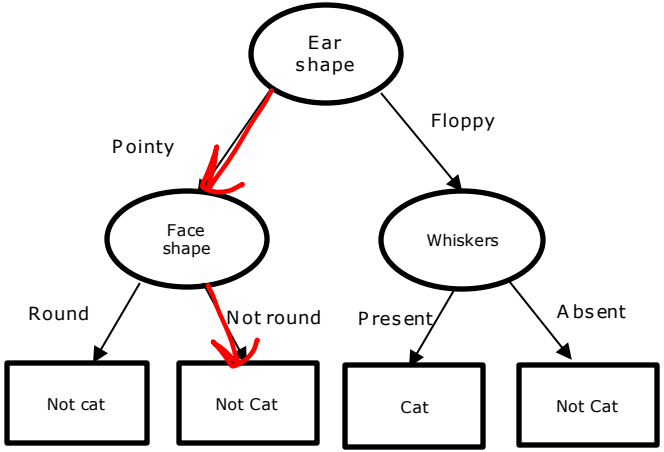




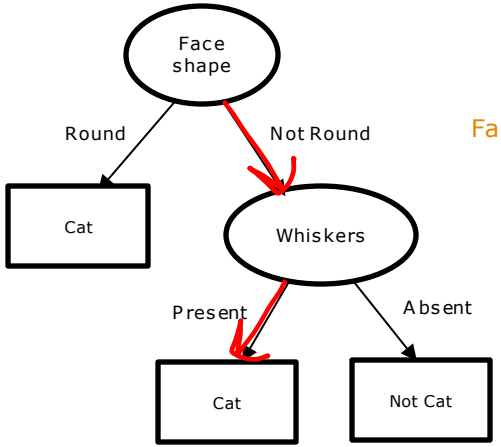
Tree ensemble



Prediction: Cat



Prediction: Not cat



Prediction: Cat

New test example

Ear shape: Pointy
Face shape: Not Round
Whiskers: Present

Final prediction: Cat

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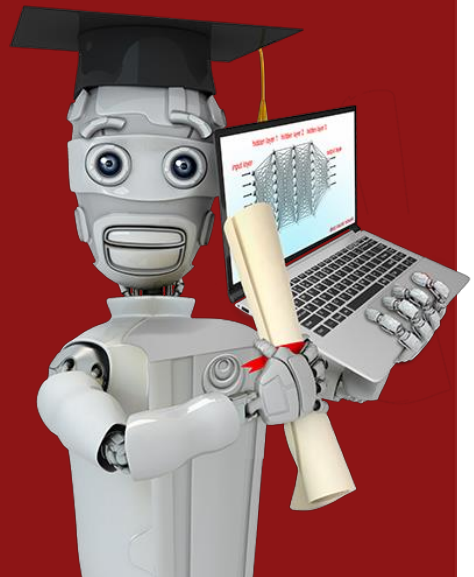


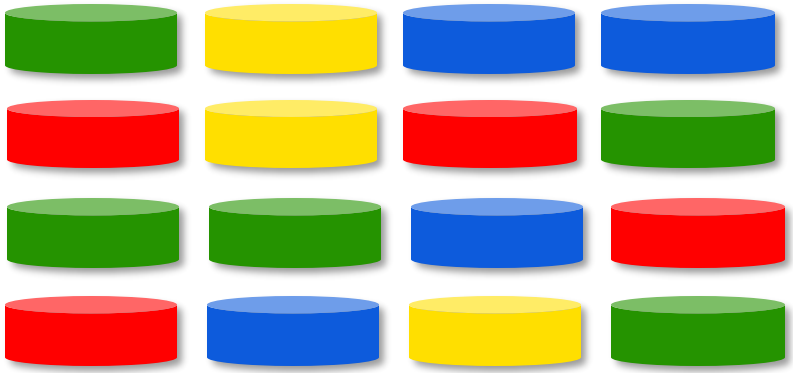
Tree ensembles

Sampling with replacement

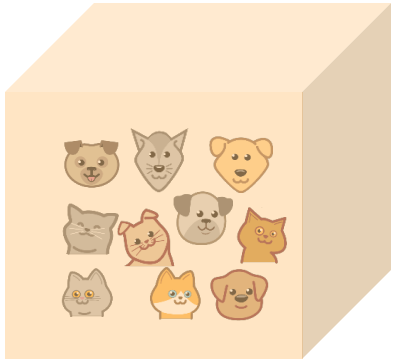
Sampling with replacement

Tokens

Sampling with replacement:



Sampling with replacement



	Ear shape	Face shape	Whiskers	Cat
	Pointy	Round	Present	1
	Floppy	Not round	Absent	0
	Pointy	Round	Absent	1
	Pointy	Not round	Present	0
	Floppy	Not round	Absent	0
	Pointy	Round	Absent	1
	Pointy	Round	Present	1
	Floppy	Not round	Present	1
	Floppy	Round	Absent	0
	Pointy	Round	Absent	1

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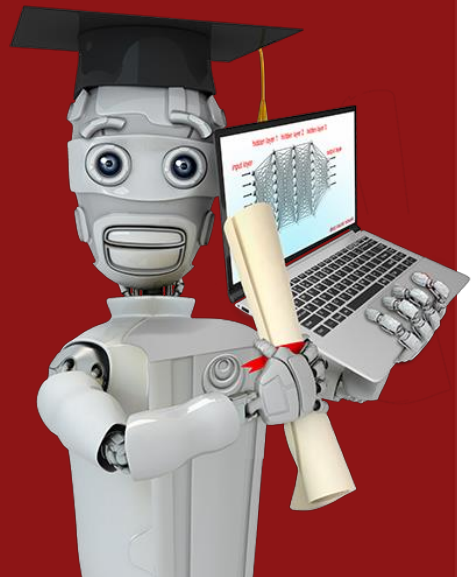


Tree ensembles

Random forest algorithm

Generating a tree sample

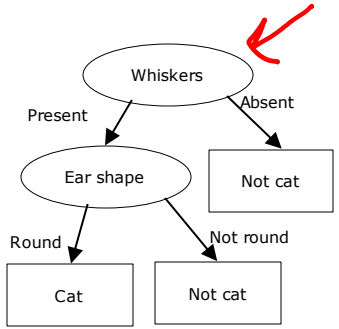
Given training set of size m

For $b = 1$ to B:

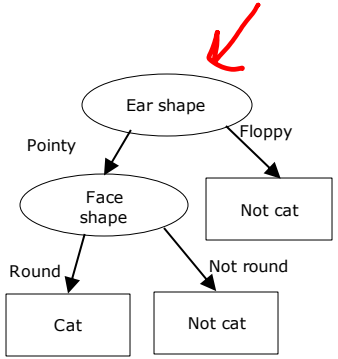
Use sampling with replacement to create a new training set of size m

Train a decision tree on the new dataset

Ear shape	Face shape	Whiskers	Cat
Pointy	Round	Present	Yes
Floppy	Round	Absent	No
Floppy	Round	Absent	No
Pointy	Round	Present	Yes
Pointy	Not Round	Present	Yes
Floppy	Round	Absent	No
Floppy	Round	Present	Yes
Pointy	Not Round	Absent	No
Pointy	Not Round	Absent	No
Pointy	Not Round	Present	Yes



Ear shape	Face shape	Whiskers	Cat
Pointy	Round	Present	Yes
Pointy	Round	Absent	Yes
Floppy	Not Round	Absent	No
Floppy	Not Round	Absent	No
Pointy	Round	Absent	Yes
Floppy	Round	Absent	No
Floppy	Round	Absent	No
Floppy	Round	Absent	No
Pointy	Not Round	Absent	No
Pointy	Round	Present	Yes



...

Bagged decision tree

Randomizing the feature choice

At each node, when choosing a feature to use to split, if n features are available, pick a random subset of $k < n$ features and allow the algorithm to only choose from that subset of features.

$k = \sqrt{n}$

Random forest algorithm

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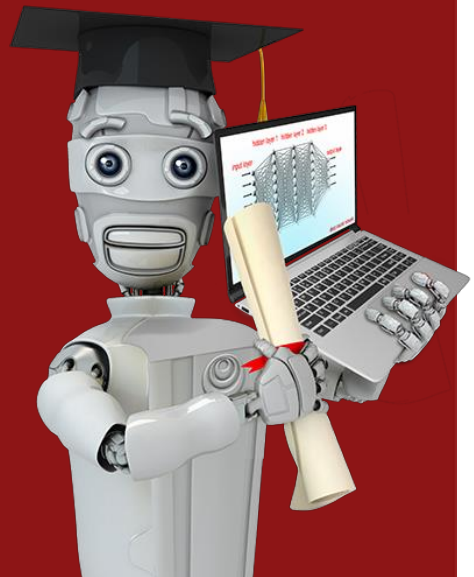


Tree ensembles

XGBoost

Boosted trees intuition

Given training set of size m

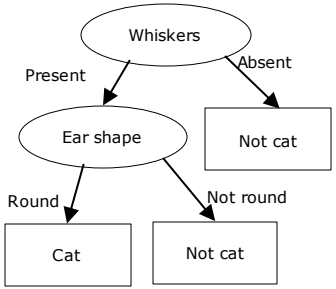
For $b = 1$ to B:

Use sampling with replacement to create a new training set of size m

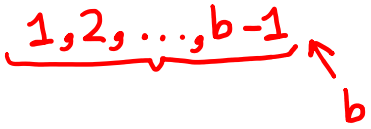
But instead of picking from all examples with equal (1/m) probability, make it more likely to pick examples that the previously trained trees misclassify

Train a decision tree on the new dataset

Ear shape	Face shape	Whiskers	Cat
Pointy	Round	Present	Yes
Floppy	Round	Absent	No
Floppy	Round	Absent	No
Pointy	Round	Present	Yes
Pointy	Not Round	Present	Yes
Floppy	Round	Absent	No
Floppy	Round	Present	Yes
Pointy	Not Round	Absent	No
Pointy	Not Round	Absent	No
Pointy	Not Round	Present	Yes



Ear shape	Face shape	Whiskers	Prediction
Pointy	Round	Present	Cat ✓
Floppy	Not Round	Present	Not cat ✗
Floppy	Round	Absent	Not cat ✓
Pointy	Not Round	Present	Not cat ✓
Pointy	Round	Present	Cat ✓
Pointy	Round	Absent	Not cat ✗
Floppy	Not Round	Absent	Not cat ✓
Pointy	Round	Absent	Not cat ✗
Floppy	Round	Absent	Not cat ✓
Floppy	Round	Absent	Not cat ✓



XGBoost (eXtreme Gradient Boosting)

- Open source implementation of boosted trees
- Fast efficient implementation
- Good choice of default splitting criteria and criteria for when to stop splitting
- Built in regularization to prevent overfitting
- Highly competitive algorithm for machine learning competitions (eg: Kaggle competitions)

Using XGBoost

Classification

```
from xgboost import XGBClassifier

model = XGBClassifier()

model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Regression

```
from xgboost import XGBRegressor

model = XGBRegressor()

model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

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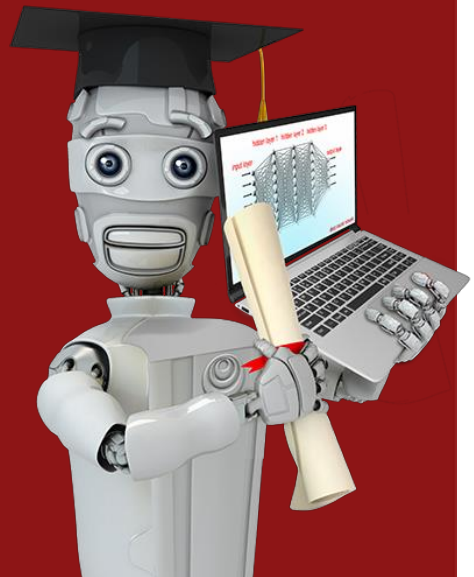


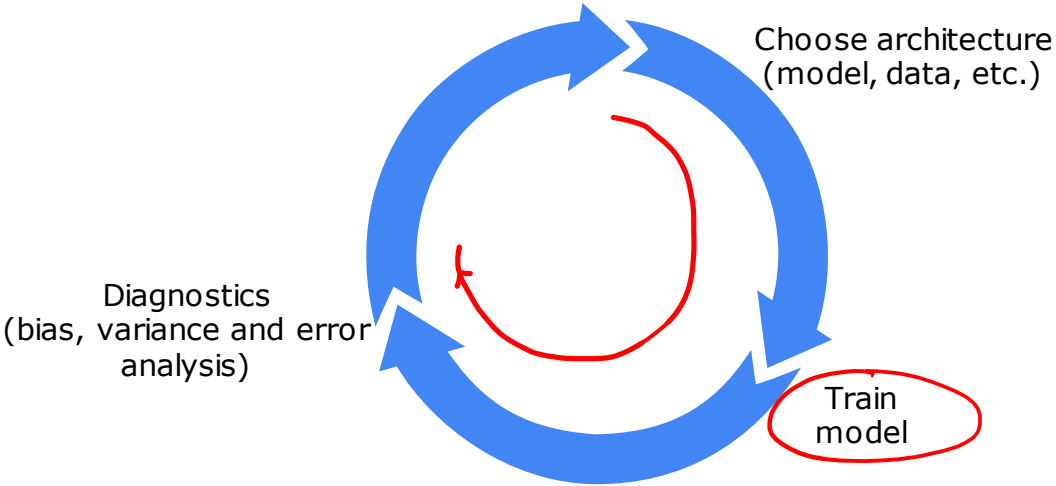
Conclusion

When to use decision trees

Decision Trees vs Neural Networks

Decision Trees and Tree ensembles

- Works well on tabular (structured) data
- Not recommended for unstructured data (images, audio, text)
- Fast



Decision Trees vs Neural Networks

Decision Trees and Tree ensembles

- Works well on tabular (structured) data
- Not recommended for unstructured data (images, audio, text)
- Fast
- Small decision trees may be human interpretable

Neural Networks

- Works well on all types of data, including tabular (structured) and unstructured data
- May be slower than a decision tree
- Works with transfer learning
- When building a system of multiple models working together, it might be easier to string together multiple neural networks

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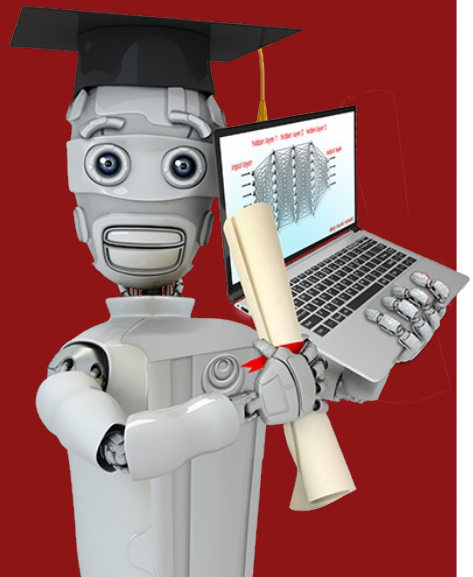

Unsupervised learning, recommender systems and reinforcement learning

Welcome!

Beyond Supervised Learning

- Unsupervised Learning
  - Clustering
  - Anomaly detection
- Recommender Systems
- Reinforcement Learning

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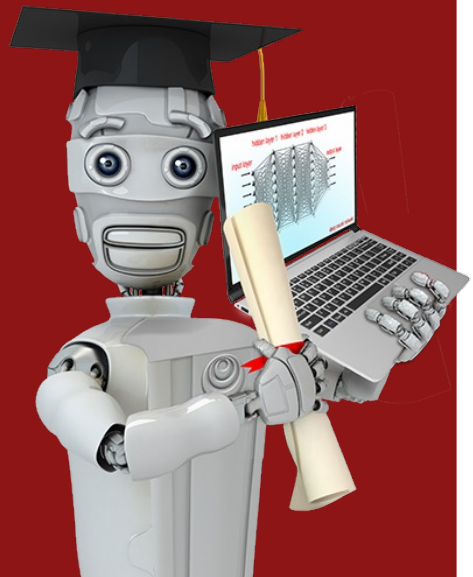
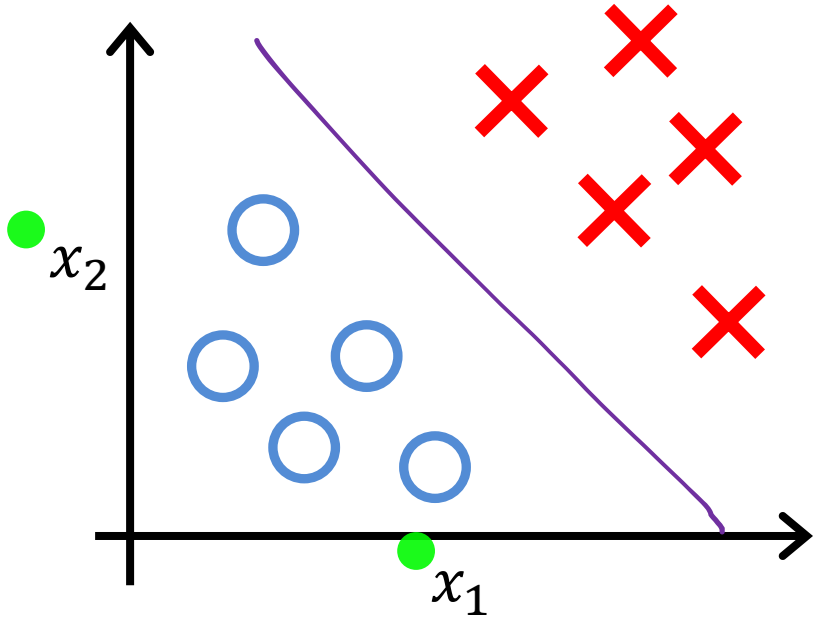


Clustering

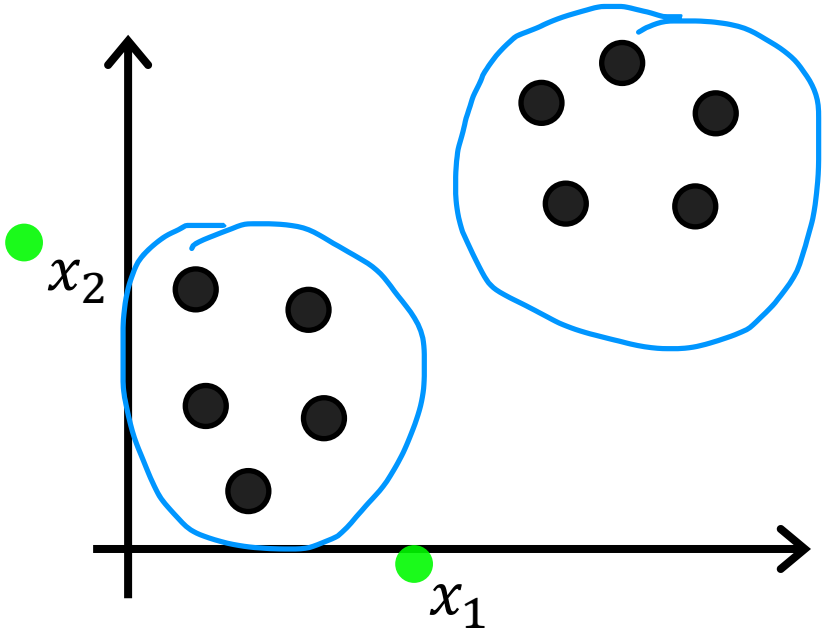
What is clustering?

Supervised learning



Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)}), \ldots, (x^{(m)}, y^{(m)})\}$?

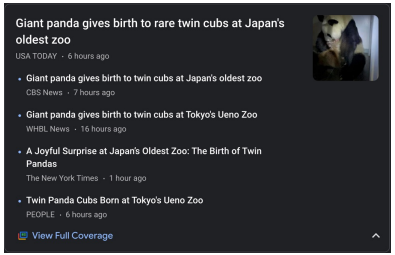
Unsupervised learning



Clustering

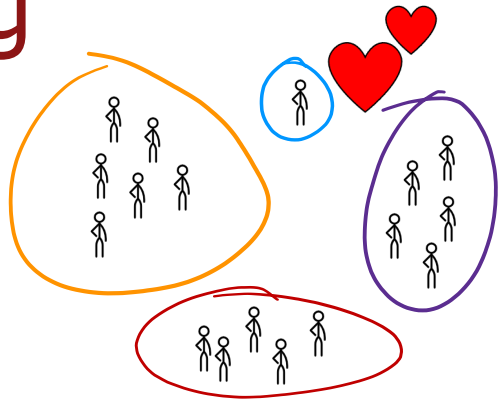
Training set: $\{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(m)}\}$

Applications of clustering

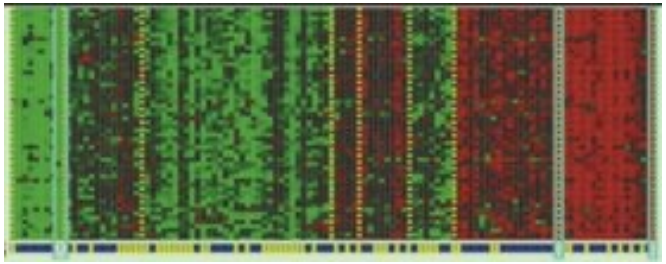


Grouping similar news

- Growing skills
- Develop career
- Stay updated with AI, understand how it affects your field of work



Market segmentation



DNA analysis

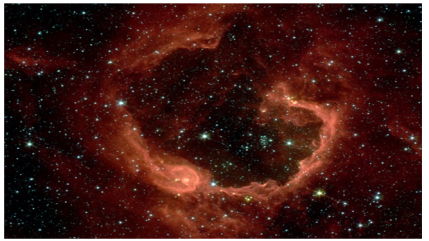


Image credit: NASA/JPL-Calte
ch/E. Churchwell (Univ. of Wisconsin, Madison)

Astronomical data analysis

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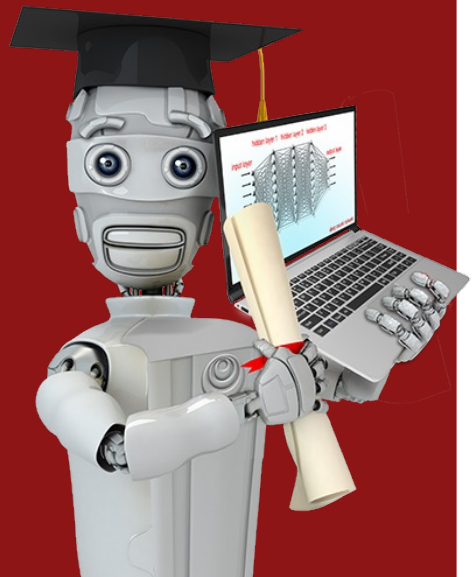
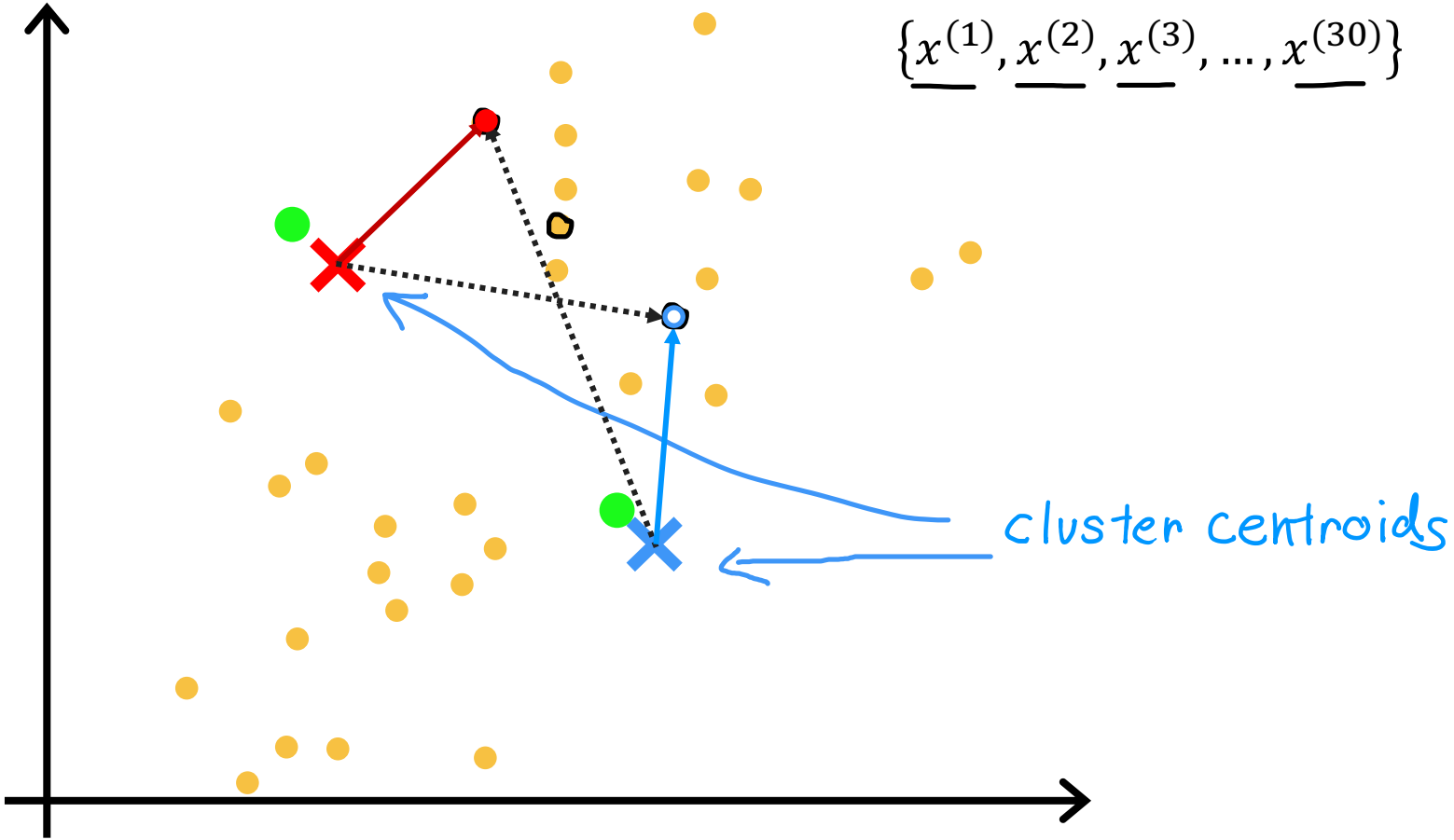


Clustering

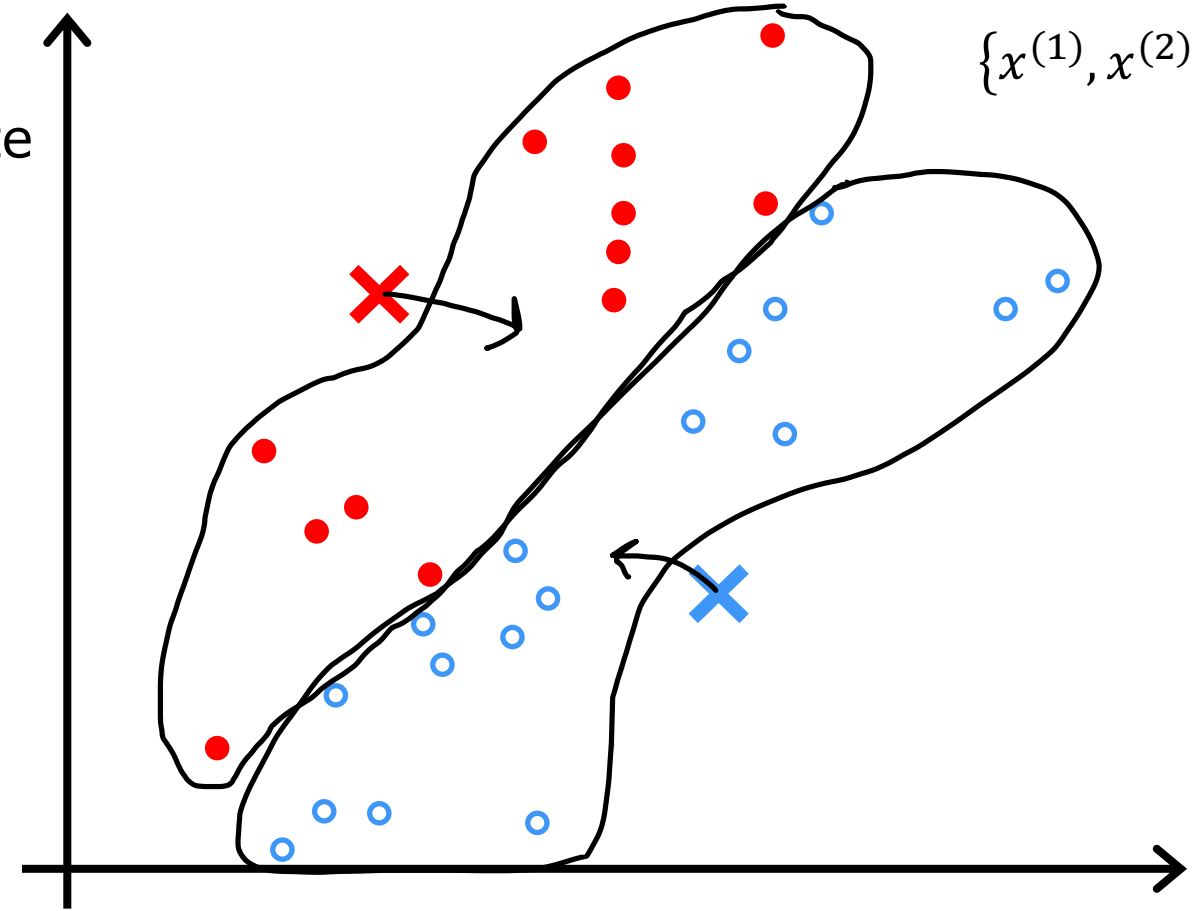
K-means intuition

Step 1:
Assign each point to its closest centroid

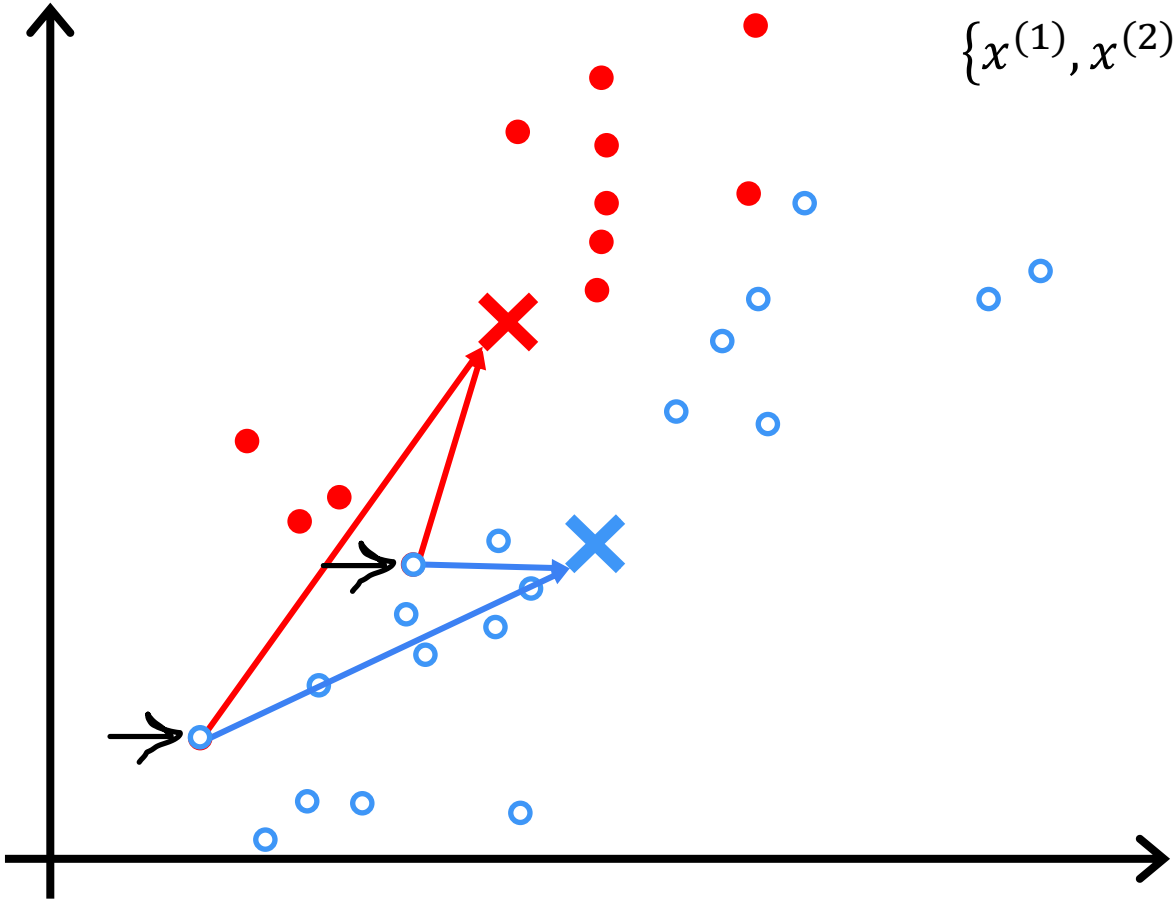


Step 2:
Recompute the centroids

$\{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(30)}\}$

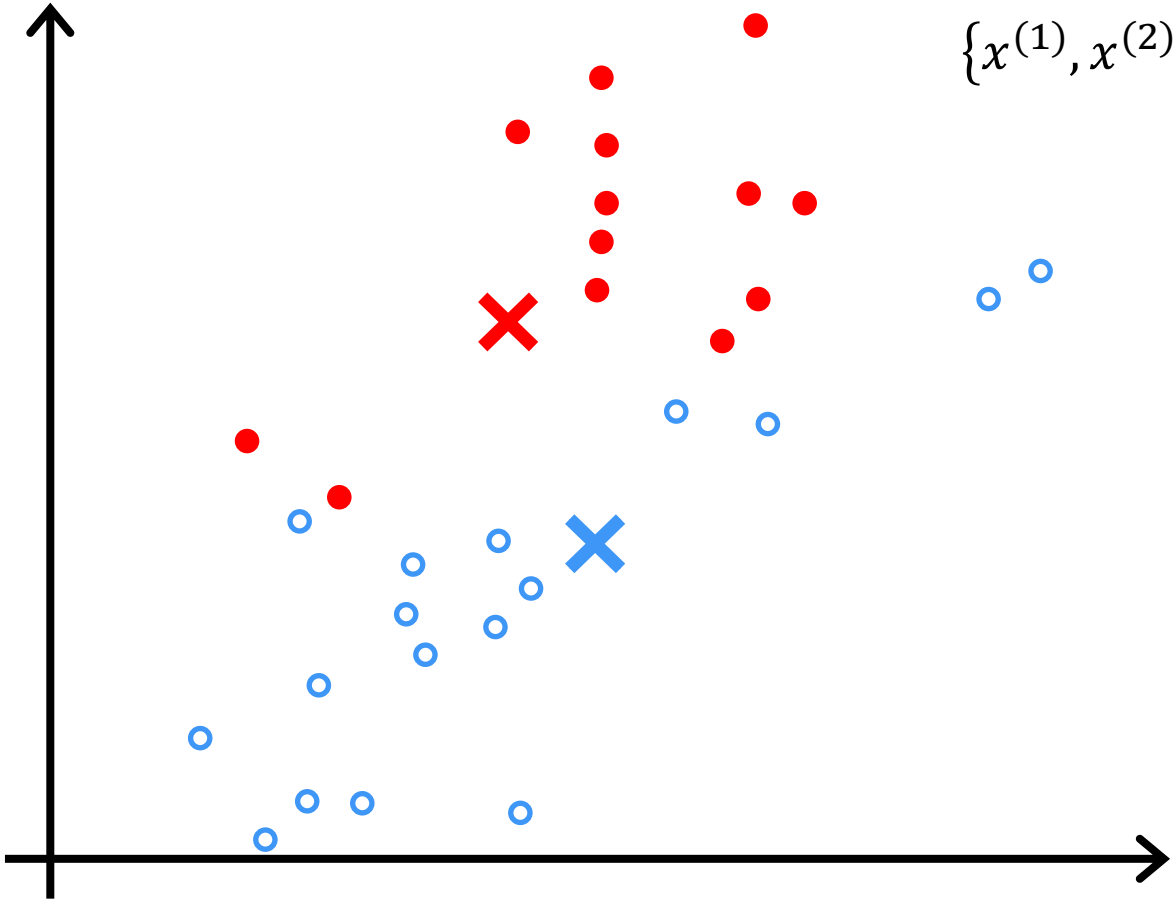


Step 1: Assign each point to its closest centroid



$\{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(30)}\}$

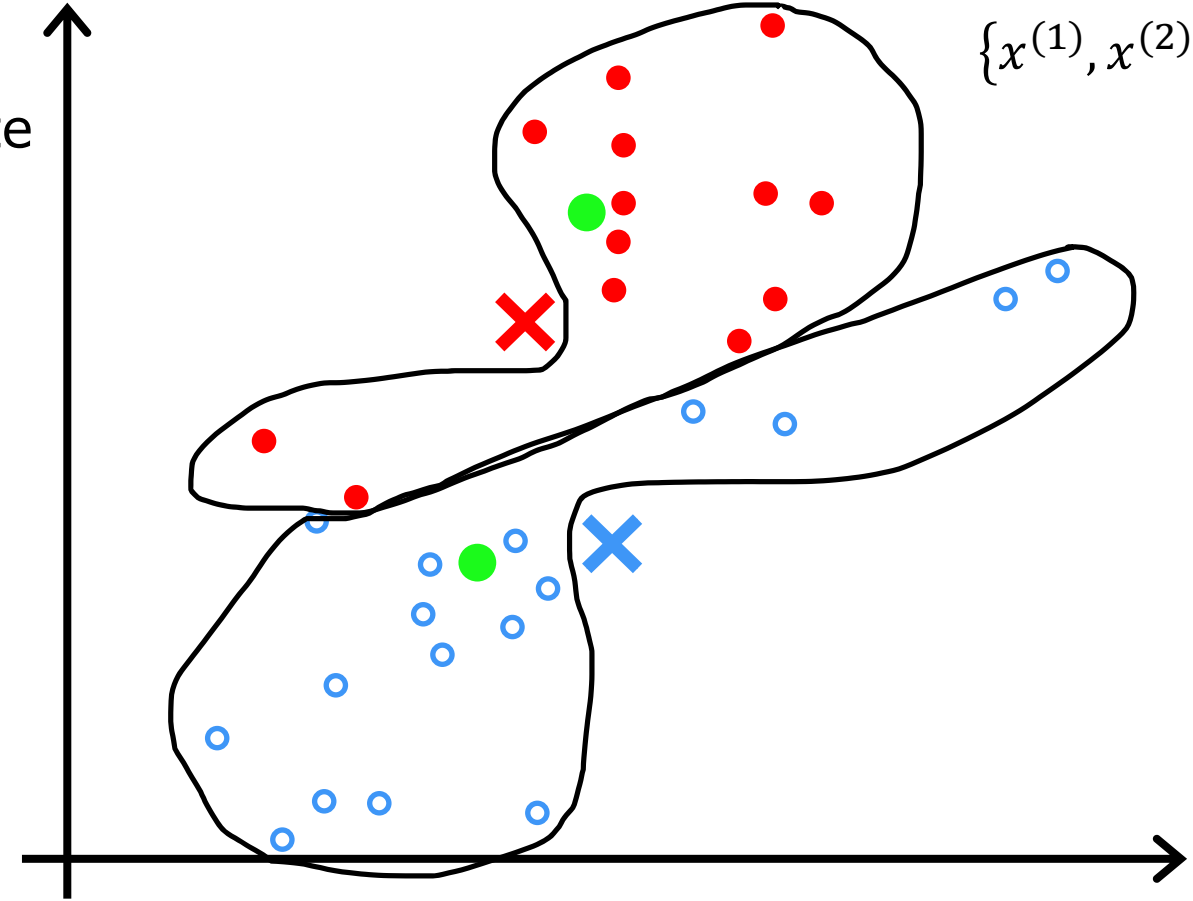
Step 1: Assign each point to its closest centroid



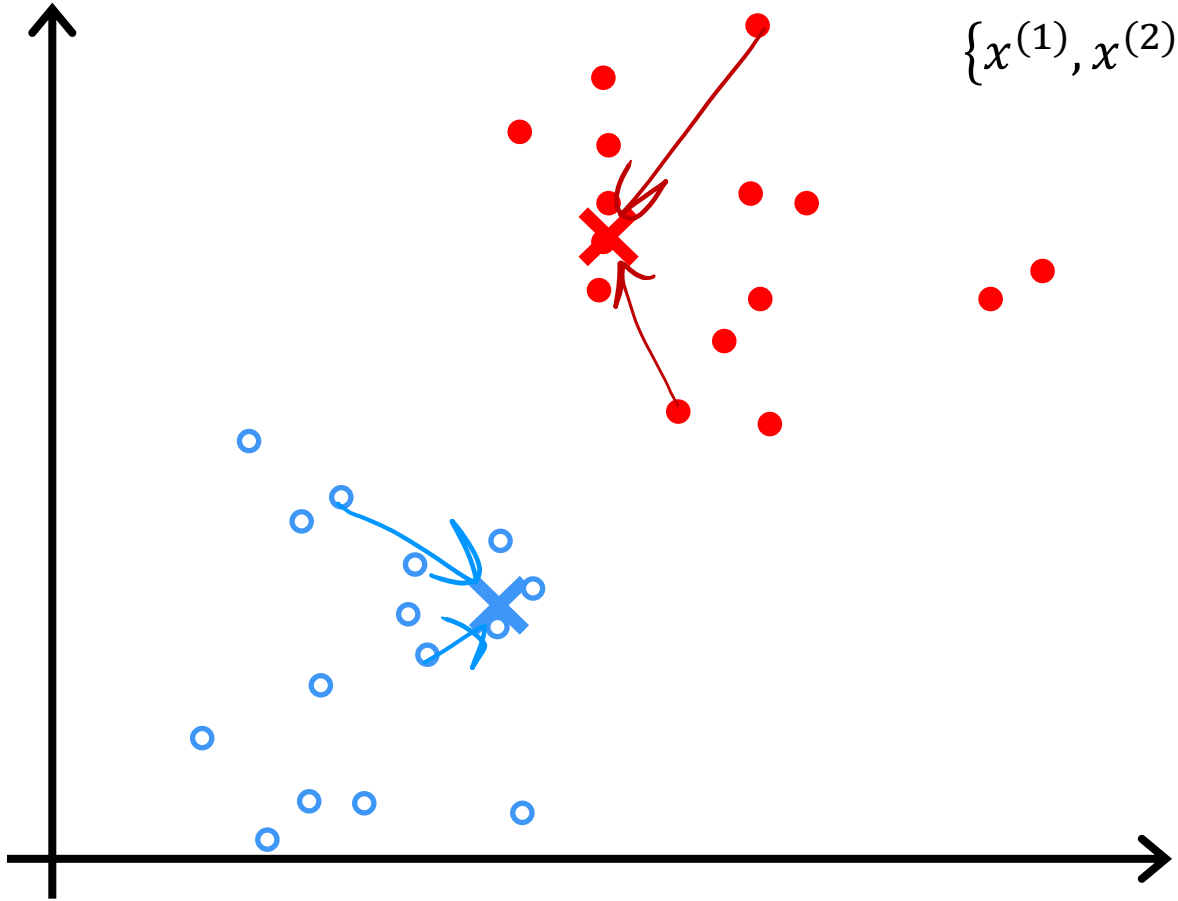
$\{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(30)}\}$

Step 2:
Recompute the centroids

$\{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(30)}\}$

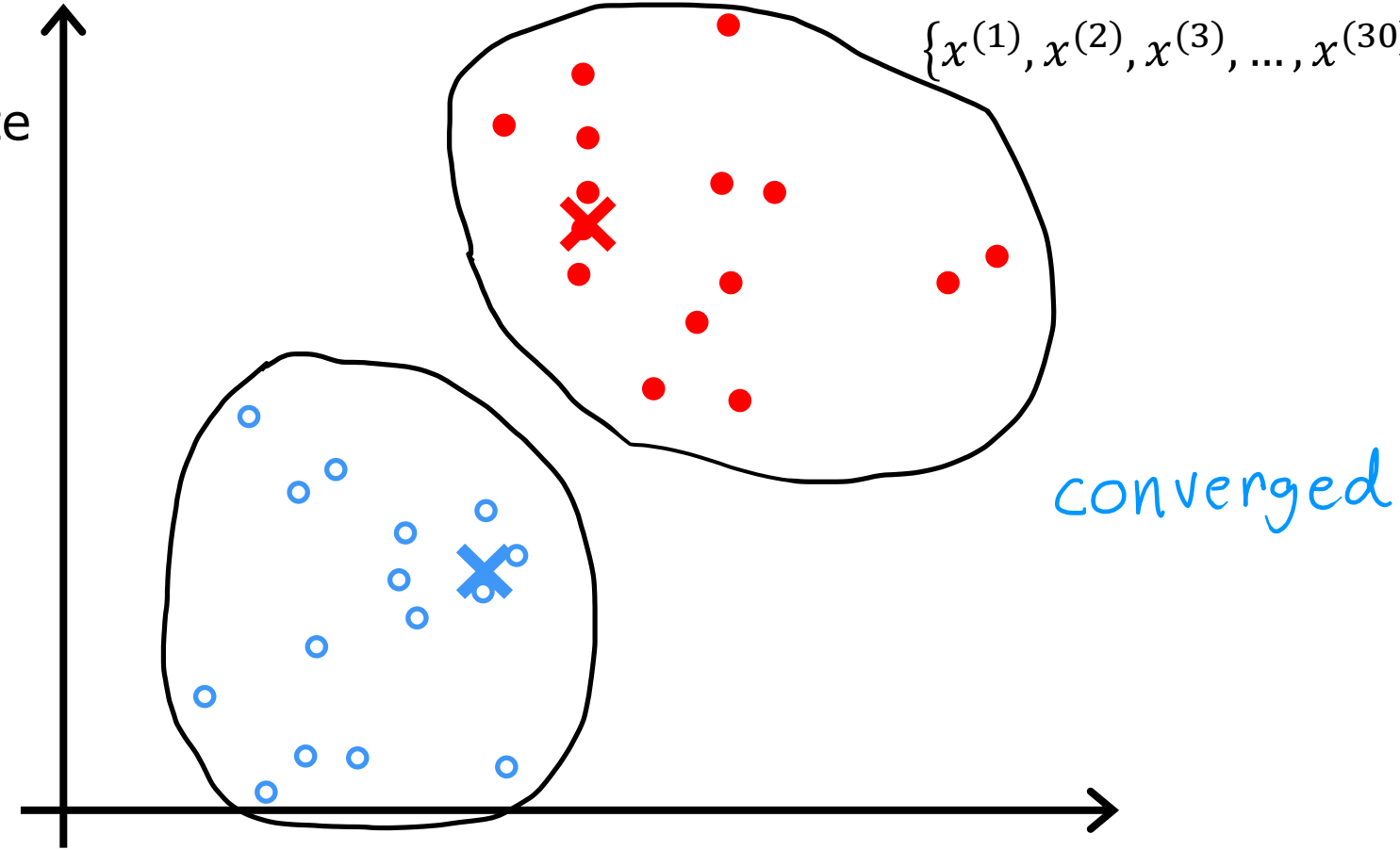


Step 1: Assign each point to its closest centroid



$\{x^{(1)}, x^{(2)}, x^{(3)}, \ldots, x^{(30)}\}$

Step 2:
Recompute the centroids



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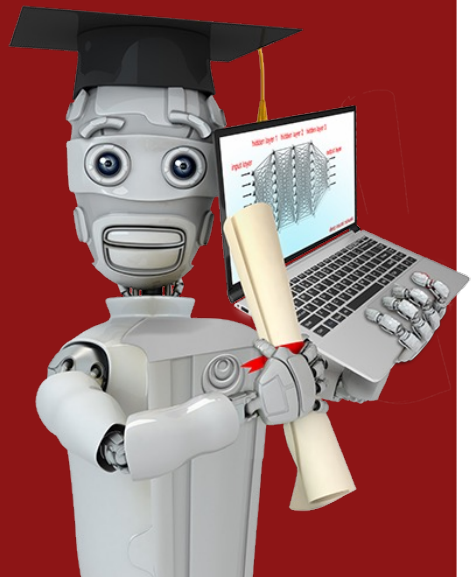


Clustering

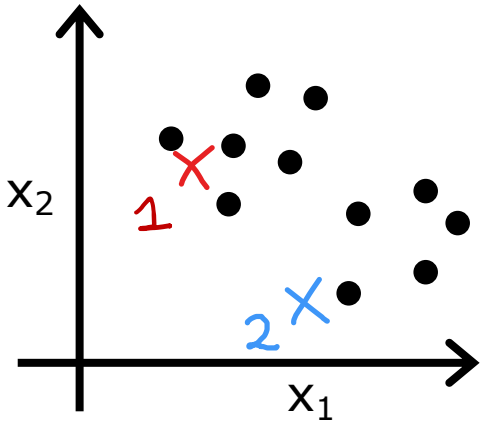
K-means algorithm

K-means algorithm

Randomly initialize K cluster centroids $\mu_1, \mu_2, \ldots, \mu_K$

μ_1 μ_2 $x^{(1)}, x^{(2)}, \ldots, x^{(30)}$

$n=2$ $K=2$



K-means algorithm

μ_1 μ_2 $x^{(1)}, x^{(2)}, \ldots, x^{(30)}$
$n=2$ $K=2$

Randomly initialize K cluster centroids $\mu_1, \mu_2, \ldots, \mu_K$

Repeat {

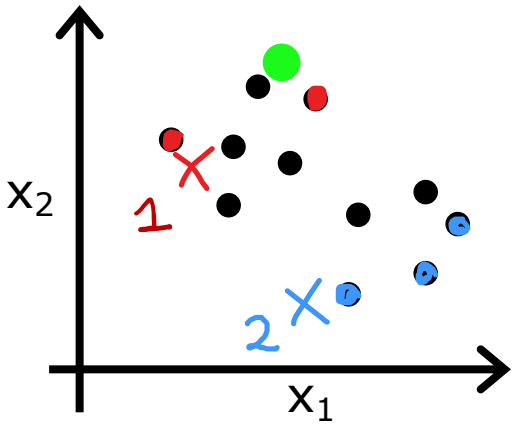
 # Assign points to cluster centroids

for i = 1 to m

$c^{(i)}$:= index (from 1 to K) of cluster centroid closest to $x^{(i)}$

L2 norm

$\min_K \|x^{(i)} - \mu_K\|^2$



K-means algorithm

Randomly initialize K cluster centroids $\mu_1, \mu_2, \ldots, \mu_K$

Repeat {

 # Assign points to cluster centroids

for i = 1 to m

$c^{(i)}$:= index (from 1 to K) of cluster centroid closest to $x^{(i)}$

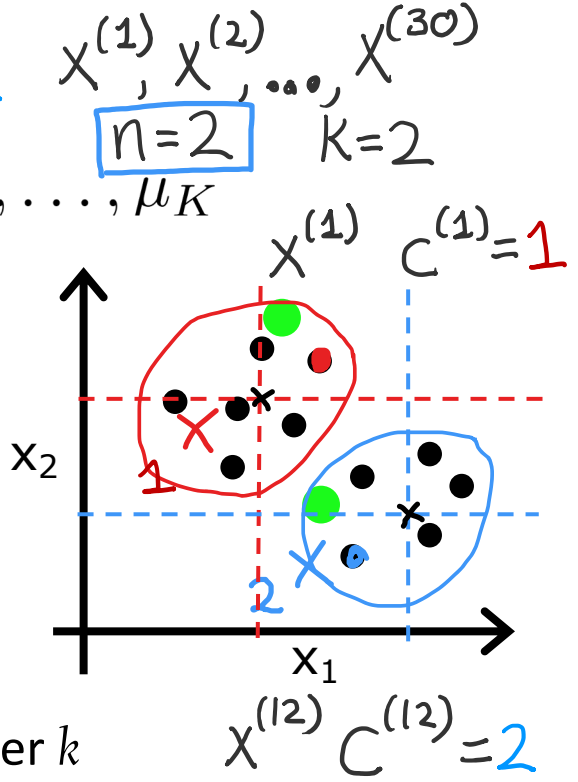
 # Move cluster centroids

for k = 1 to K

μ_k := average (mean) of points assigned to cluster k

}

$$\mu_1 = \frac{1}{4}\left[x^{(1)} + x^{(5)} + x^{(6)} + x^{(10)}\right]$$



K-means algorithm

$\mu_1 \mu_2$ $x^{(1)}, x^{(2)}, \ldots, x^{(30)}$

$n = 2$ ~~$k = 2$~~

$K = K - 1$ ✓

Randomly initialize k cluster centroids, $\mu_1, \mu_1, \ldots \mu_k$

Repeat {

 # Assign points to cluster centroids

for i = 1 to m

$c^{(i)}$:= index (from 1 to K) of cluster centroid closest to $x^{(i)}$

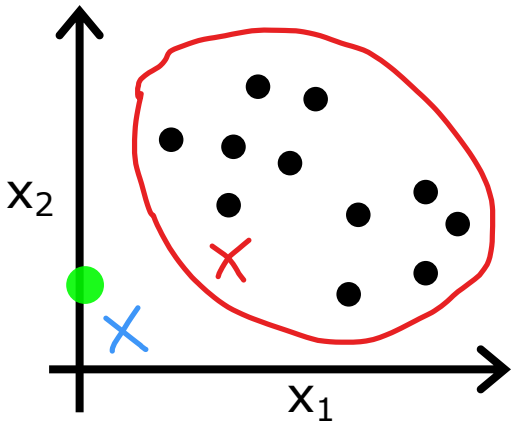
$\min_K \|x^{(i)} - \mu_K\|^2$

 # Move cluster centroids

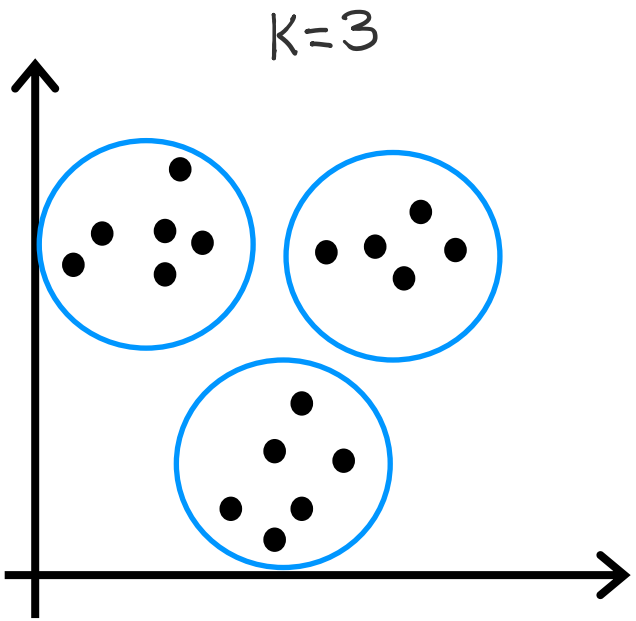
for k = 1 to K

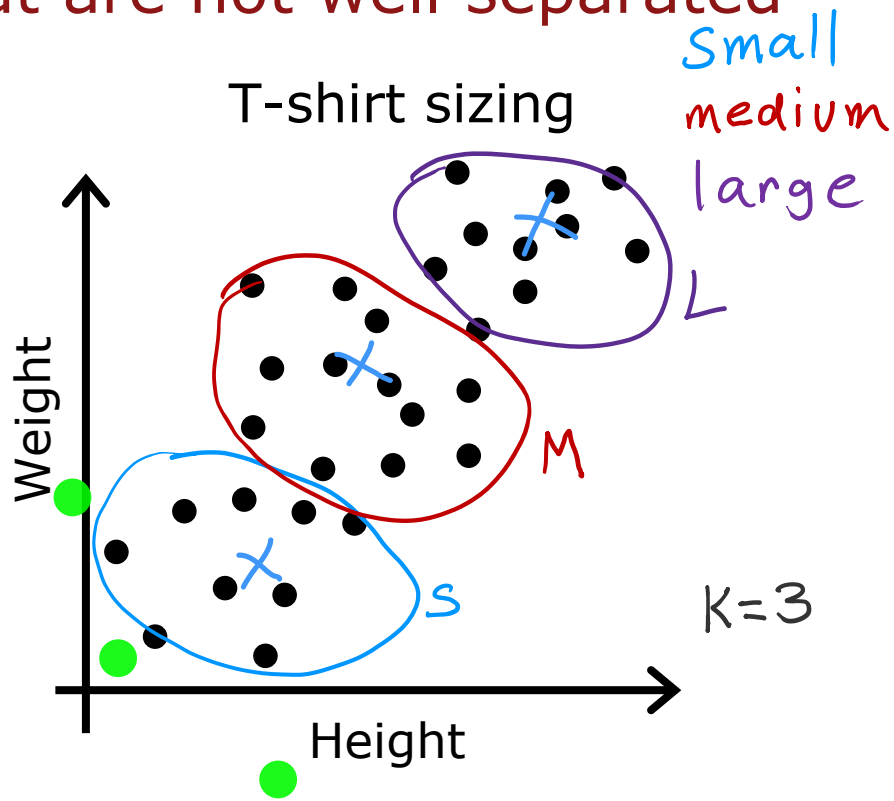
μ_k := average (mean) of points assigned to cluster k

}



K-means for clusters that are not well separated





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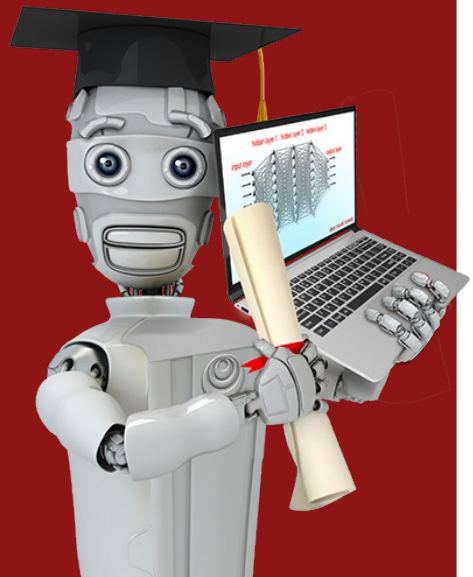


Clustering

Optimization objective

K-means optimization objective

$c^{(i)}$ = index of cluster $(1, 2, \ldots, K)$ to which example $x^{(i)}$ is currently assigned

μ_k = cluster centroid k

$\mu_{c^{(i)}}$ = cluster centroid of cluster to which example $x^{(i)}$ has been assigned

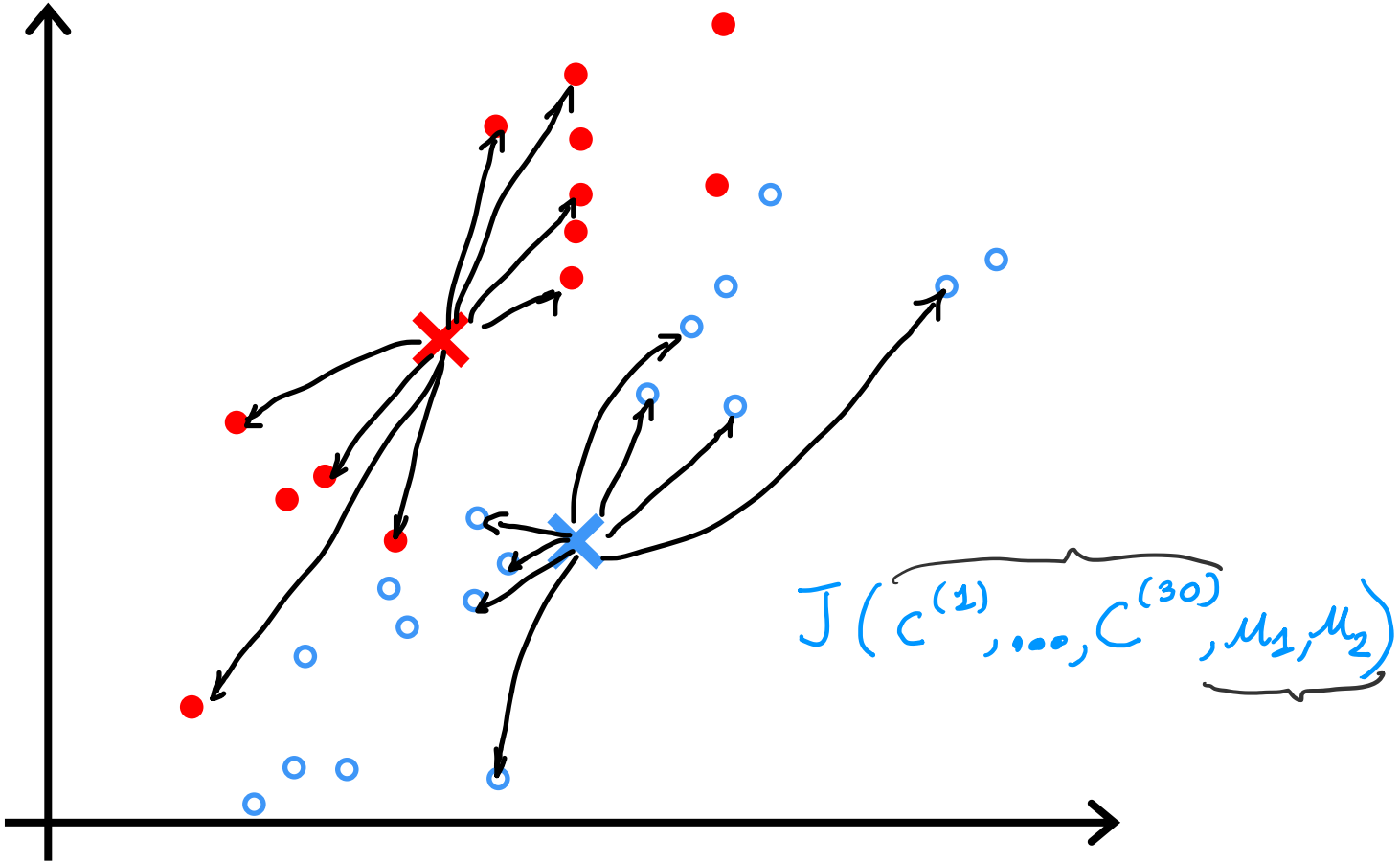
$x^{(10)}$ $c^{(10)}$ $\mu_c^{(10)}$

Cost function

$$J(c^{(1)}, \ldots, c^{(m)}, \mu_1, .., \mu_K) = \frac{1}{m} \sum_{i=1}^{m} \| x^{(i)} - \mu_{c^{(i)}} \|^2$$

$$\min_{\substack{c^{(1)}, \ldots, c^{(m)} \\ \mu_1, \ldots, \mu_K}} J(c^{(1)}, \ldots, c^{(m)}, \mu_1, .., \mu_K)$$

distortion


$J(c^{(1)}, \ldots, c^{(30)}, \mu_1, \mu_2)$

Cost function for K-means

$$J\left(c^{(1)}, \ldots, c^{(m)}, \mu_1, \ldots, \mu_K\right) = \frac{1}{m} \sum_{i=1}^{m} \left\| x^{(i)} - \mu_{c^{(i)}} \right\|^2$$

Repeat {

 # Assign points to cluster centroids

for i = 1 to m:

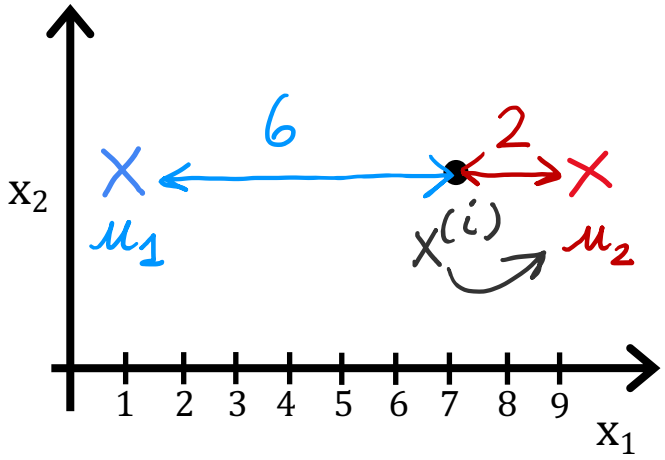
$c^{(i)}$ = index of cluster centroid closest to $x^{(i)}$

 # Move cluster centroids

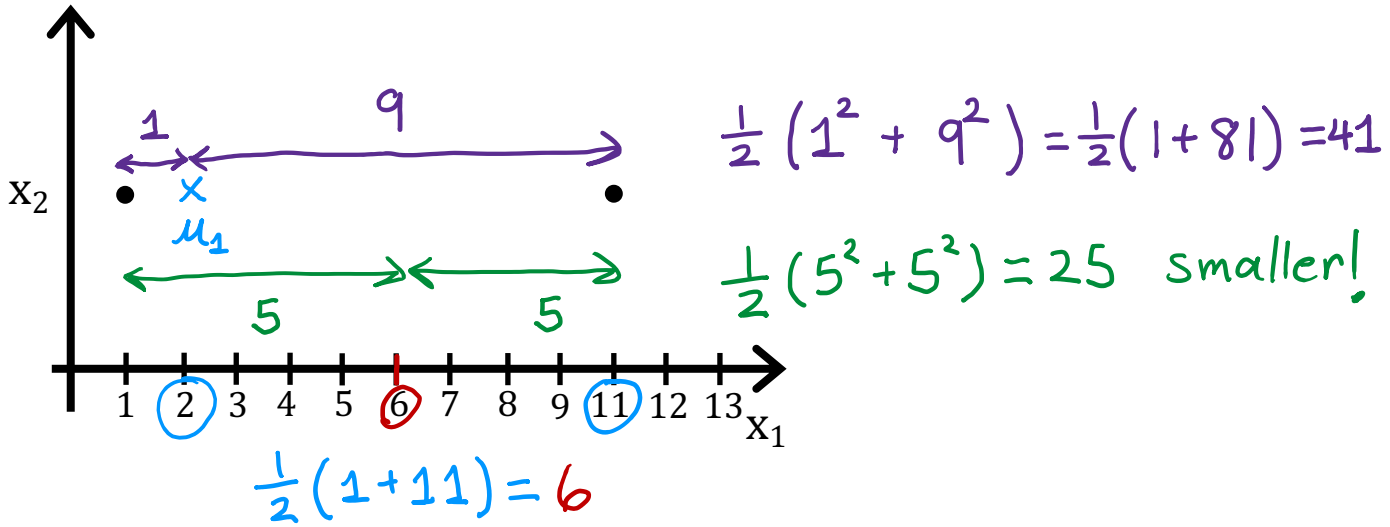
for i = 1 to K:

μ_k = average of points in cluster

}



Moving the centroid



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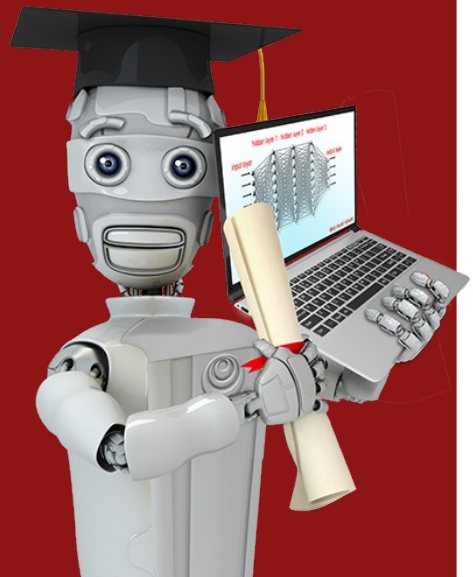


Clustering

Initializing K-means

K-means algorithm

- Step 0: Randomly initialize K cluster centroids $\mu_1, \mu_1, \ldots, \mu_k$

Repeat {

Step 1: Assign points to cluster centroids

Step 2: Move cluster centroids

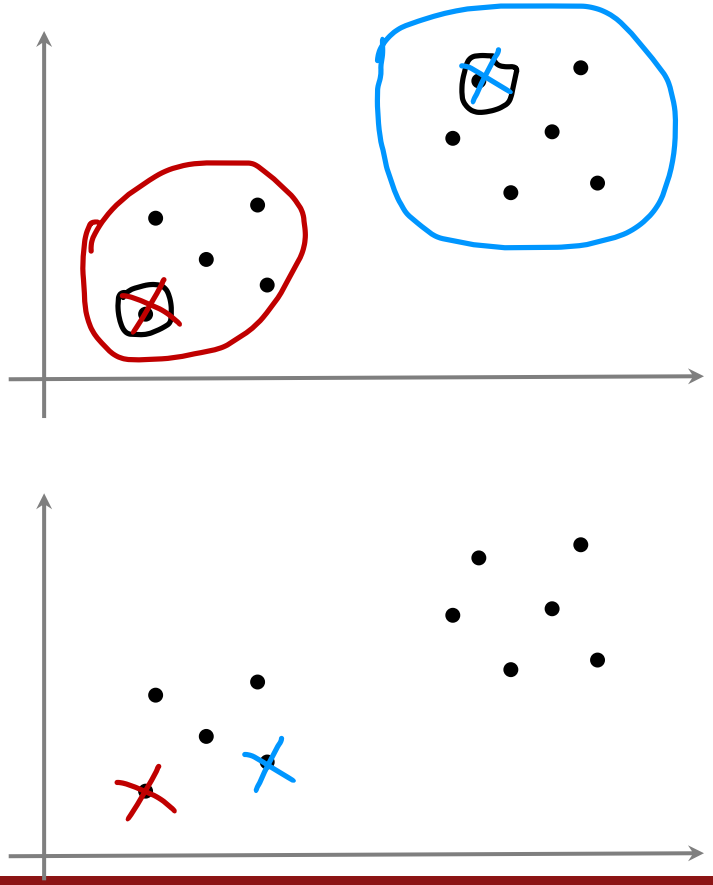
}

Random initialization

Choose $K < m$

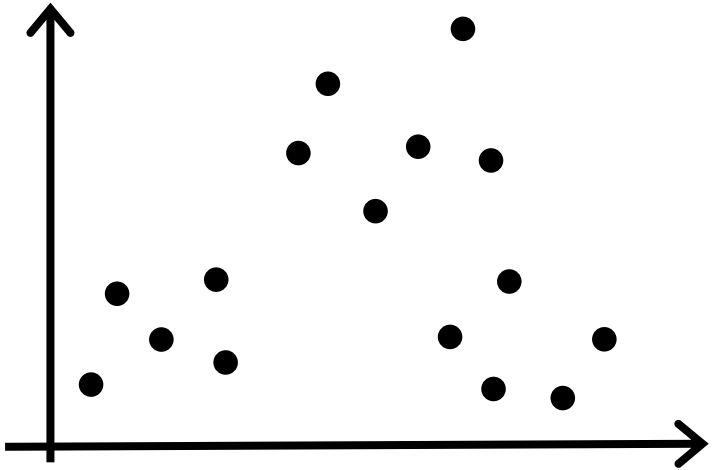
Randomly pick K training examples.

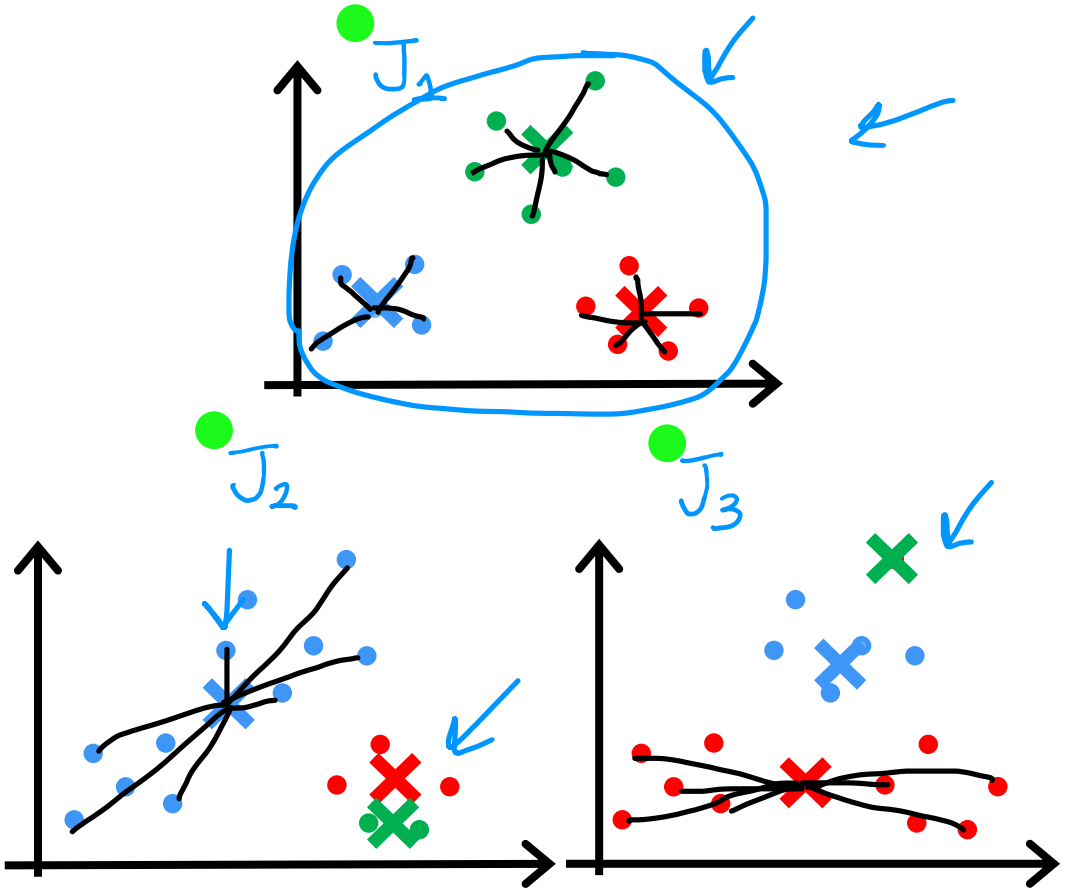
Set μ_1, μ_1,…, μ_k equal to these K examples.



K=3

$$J(c^{(1)}, \ldots, c^{(m)}, \mu_1, .., \mu_K)$$





Random initialization

better than running K-means once

50-1000

For $i = 1$ to 100 {

pick K random examples
set as cluster centroids

Randomly initialize K-means.
Run K-means. Get $c^{(1)}, \ldots, c^{(m)}, \mu_1, \mu_1, \ldots, \mu_k$
Compute $J(c^{(1)}, \ldots, c^{(m)}, \mu_1, \mu_1, \ldots, \mu_k)$

distortion
(cost function)

}

Pick set of clusters that gave lowest cost J

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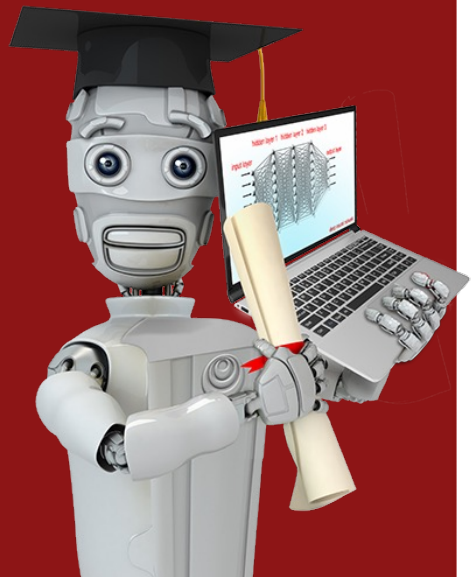
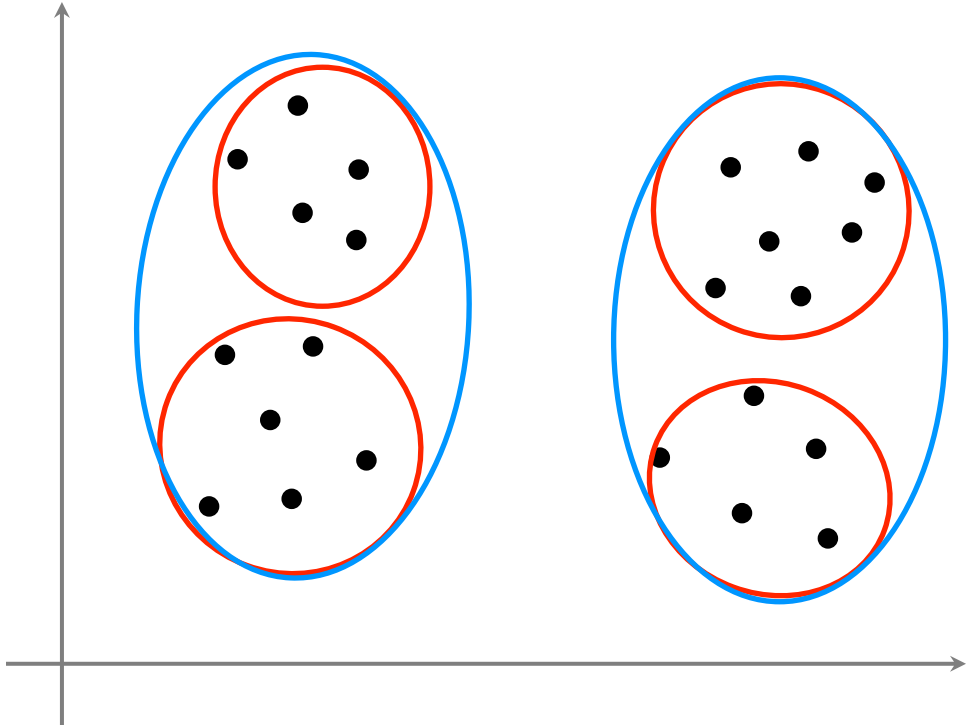


Clustering

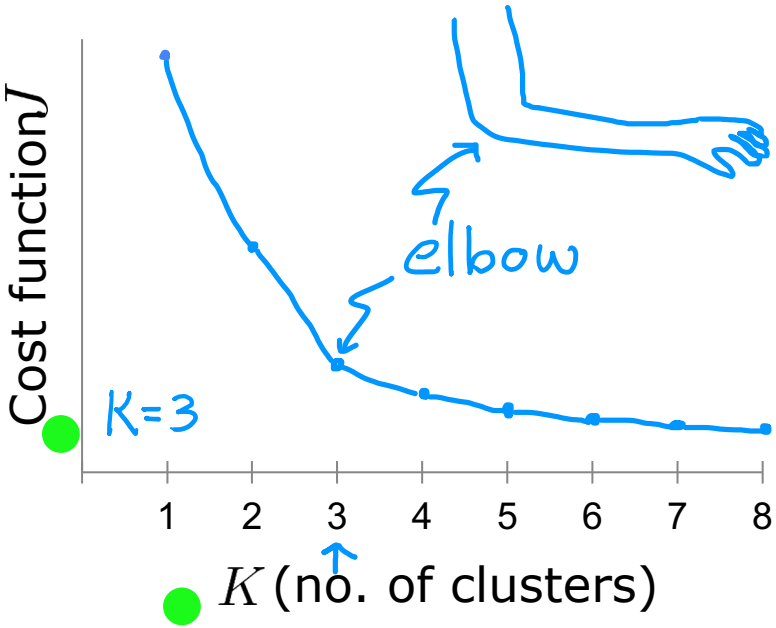
Choosing the Number of Clusters

What is the right value of K?



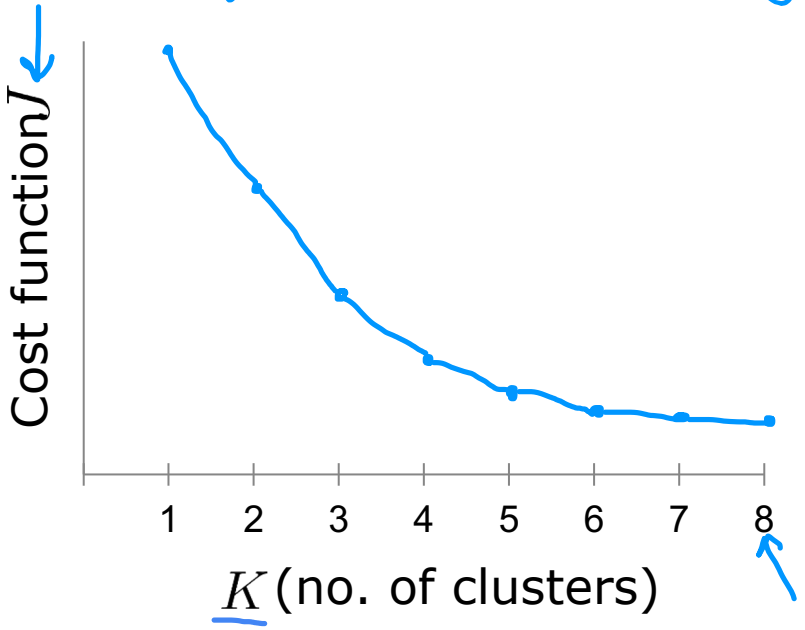
Choosing the value of K

Elbow method



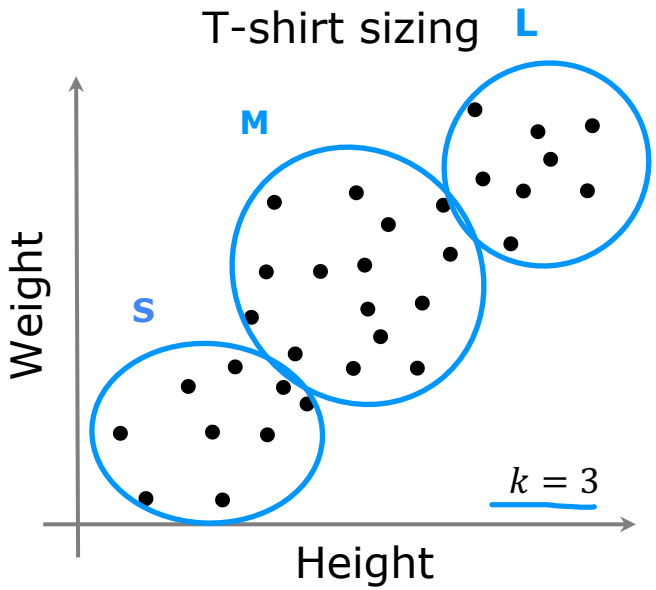
the right "K" is often ambiguous

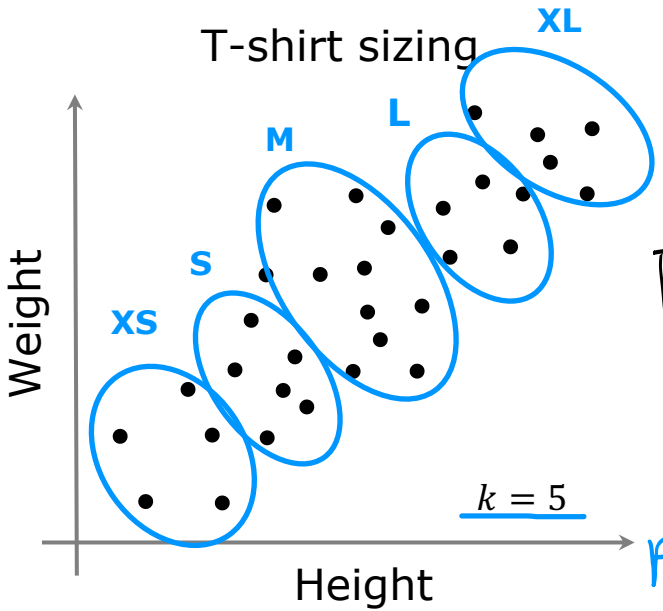
Don't choose K just to minimize cost J

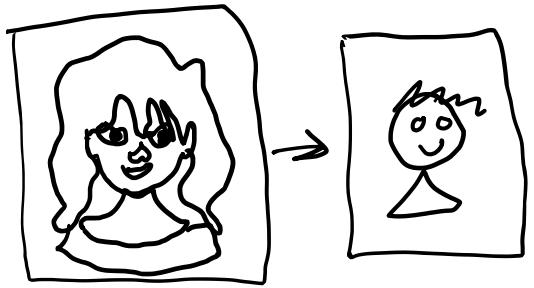


Choosing the value of K

Often, you want to get clusters for some later (downstream) purpose.
Evaluate K-means based on how well it performs on that later purpose.







practice lab:
image compression!

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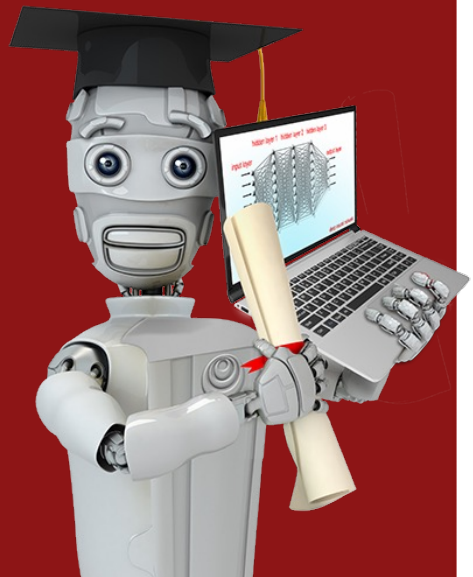


Anomaly Detection

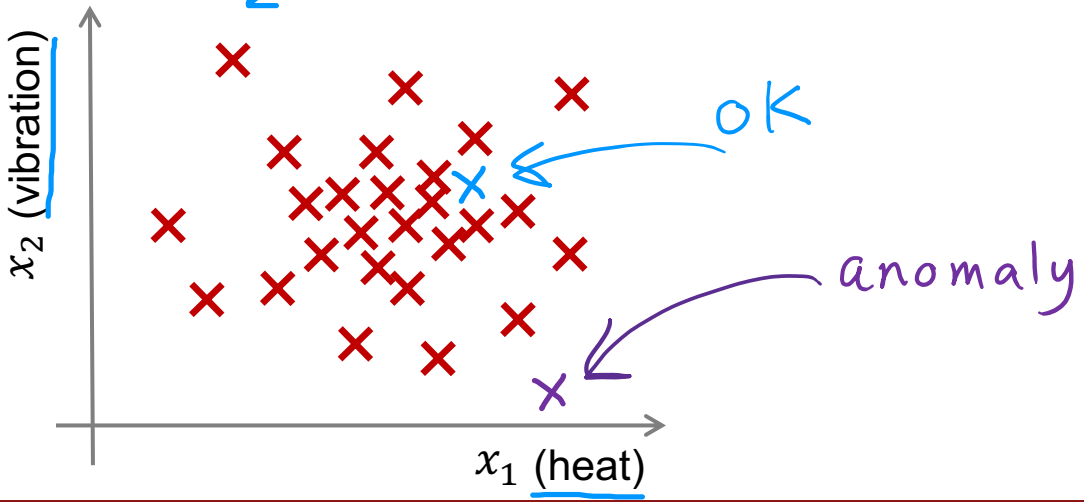
Finding unusual events

Anomaly detection example

Aircraft engine features:
x_1 = heat generated
x_2 = vibration intensity
…

Dataset: $\{x^{(1)}, x^{(2)}, \ldots, x^{(m)}\}$

New engine: x_{test}



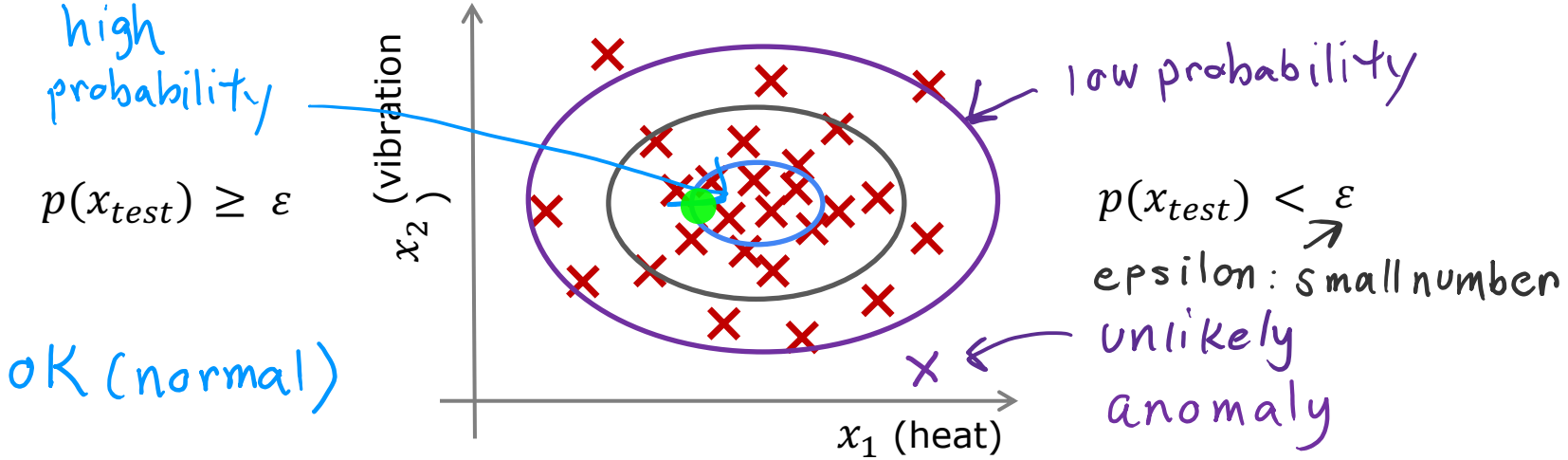
Density estimation

Dataset: $\{x^{(1)}, x^{(2)}, \ldots, x^{(m)}\}$

probability of x being seen in dataset

Model $p(x)$

Is x_{test} anomalous?



Anomaly detection example

how often log in?
how many web pages visited?
transactions?
posts? typing speed?

Fraud detection:

- $x^{(i)}$ = features of user i's activities
- Model $p(x)$ from data.
- Identify unusual users by checking which have $p(x) < \varepsilon$

perform additional checks to identify real fraud vs. false alarms

Manufacturing:

$x^{(i)}$= features of product i

airplane engine
circuit board
smartphone

ratios

Monitoring computers in a data center:

$x^{(i)}$= features of machine i

- x_1 = memory use,
- x_2= number of disk accesses/sec,
- x_3 = CPU load,
- x_4= CPU load/network traffic.

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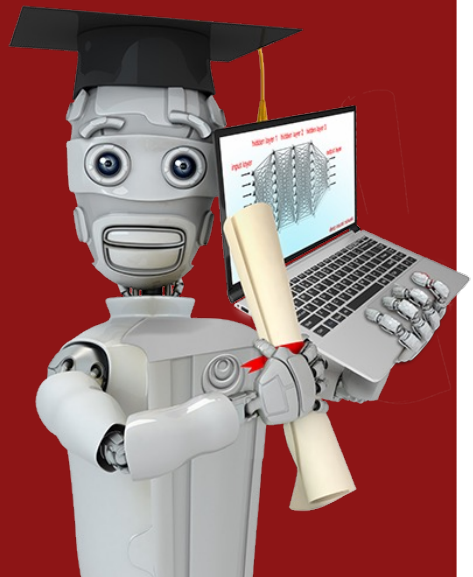


Anomaly Detection

Gaussian (Normal) Distribution

Gaussian (Normal) distribution

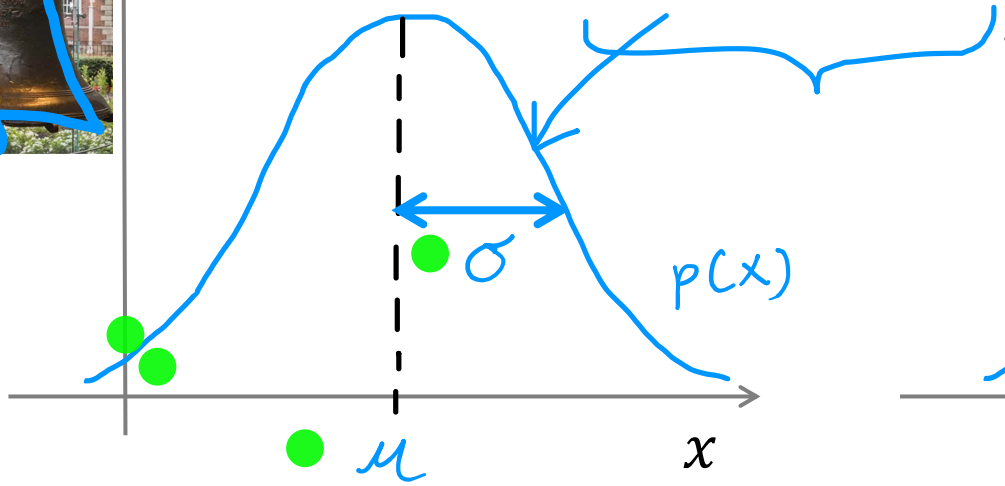
σ standard deviation

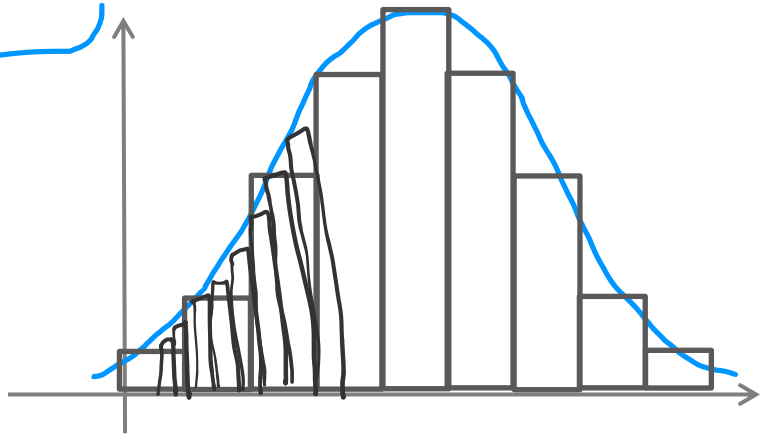
σ^2 variance

Say x is a number.
Probability of x is determined by a Gaussian with mean μ, variance σ^2.

$\pi = 3.14$

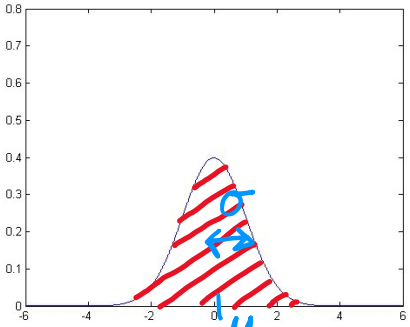
$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$



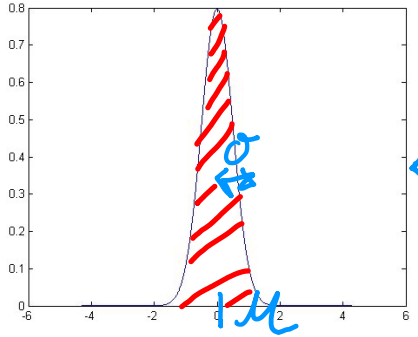


Gaussian distribution example

$\mu = 0, \sigma = 1$

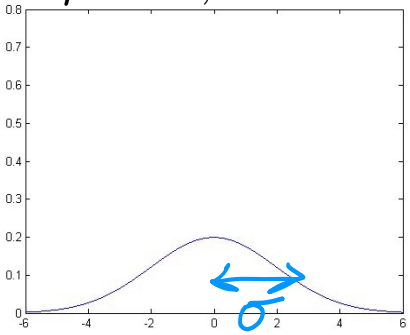


$\mu = 0, \sigma = 0.5$



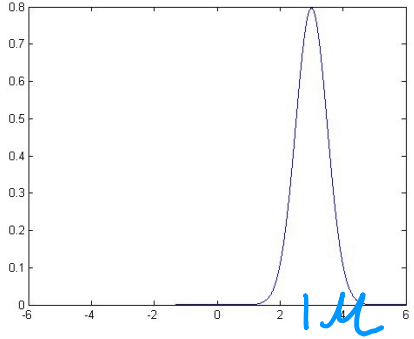
$\sigma^2 = 0.25$

$\mu = 0, \sigma = 2$



$\sigma^2 = 4$

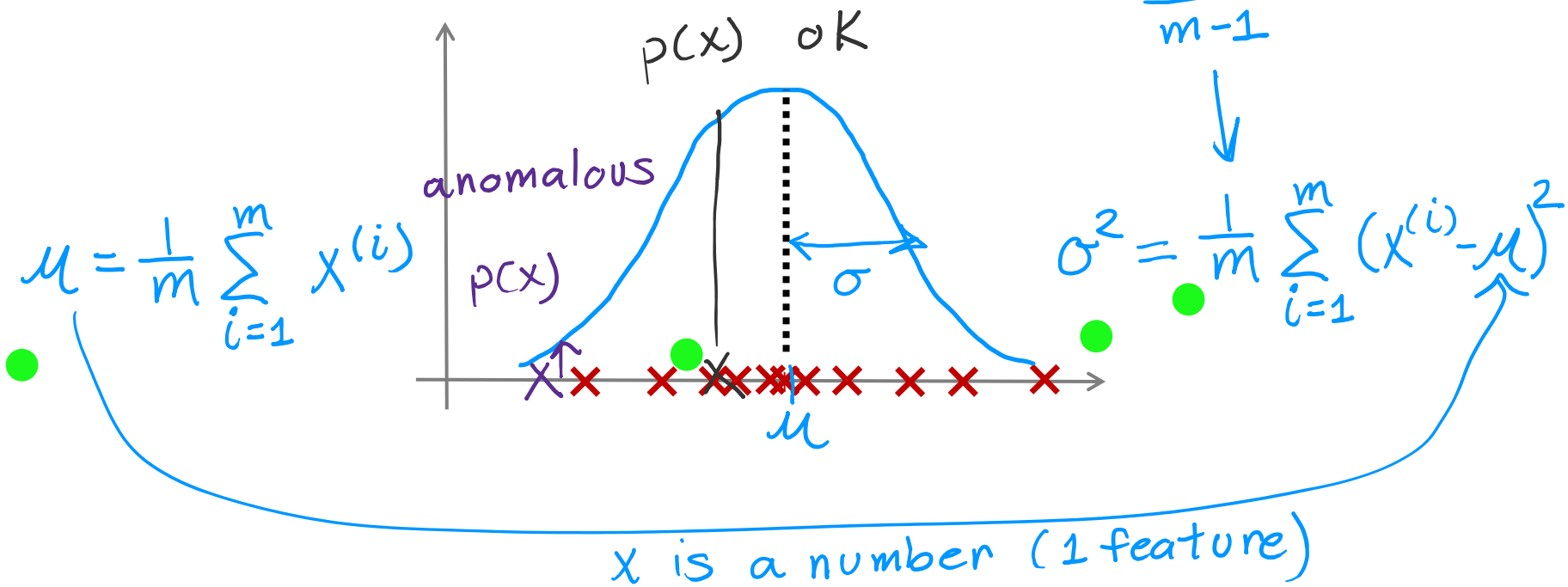
$\mu = 3, \sigma = 0.5$



Parameter estimation

Dataset: $\{x^{(1)}, x^{(2)}, \ldots, x^{(m)}\}$

maximum likelihood for μ, σ



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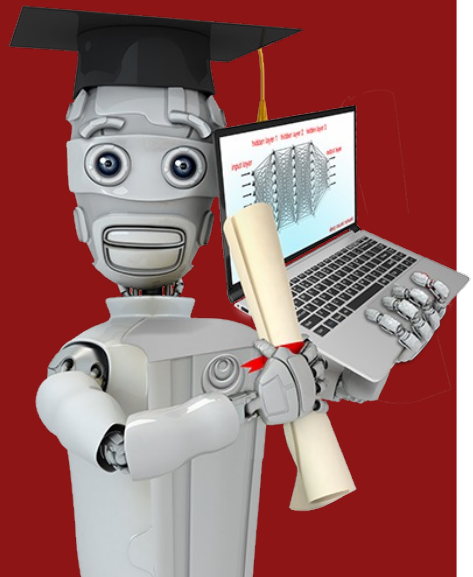


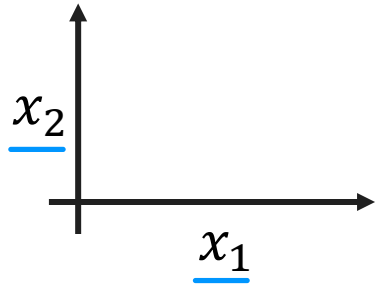
Anomaly Detection

Algorithm

Density estimation

Training set: $\{\vec{x}^{(1)}, \vec{x}^{(2)}, \ldots, \vec{x}^{(m)}\}$

Each example $\vec{x}^{(i)}$ has n features



$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$p(\vec{x}) = p(x_1; \mu_1, \sigma_1^2) * p(x_2; \mu_2, \sigma_2^2) * p(x_3; \mu_3, \sigma_3^2) * \cdots * p(x_n; \mu_n, \sigma_n^2)$$

$$= \prod_{j=1}^{n} p(x_j; \mu_j, \sigma_j^2)$$

$\sum$ "add" $\prod$ "multiply"

$p(x_1 = \text{high temp}) = 1/10$

$p(x_2 = \text{high vibra}) = 1/20$

$p(x_1, x_1) = p(x_1) * p(x_2)$

$= \frac{1}{10} \times \frac{1}{20} = \frac{1}{200}$

Anomaly detection algorithm

1. Choose n features x_i that you think might be indicative of anomalous examples.
2. Fit parameters $\mu_1, \ldots, \mu_n, \sigma_1^2, \ldots, \sigma_n^2$

$$\mu_j = \frac{1}{m}\sum_{i=1}^{m} x_j^{(i)} \qquad \sigma_j^2 = \frac{1}{m}\sum_{i=1}^{m}(x_j^{(i)} - \mu_j)^2$$

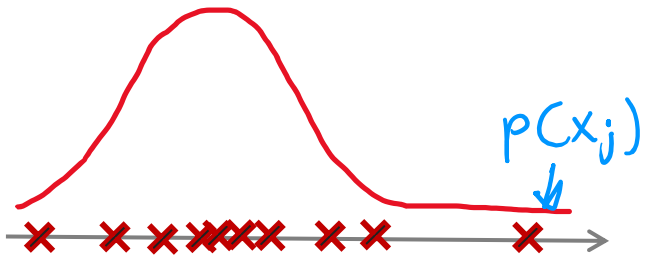
Vectorized formula

$$\vec{\mu} = \frac{1}{m}\sum_{i=1}^{m} \vec{x}^{(i)} \qquad \vec{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \cdots \\ \mu_n \end{bmatrix}$$

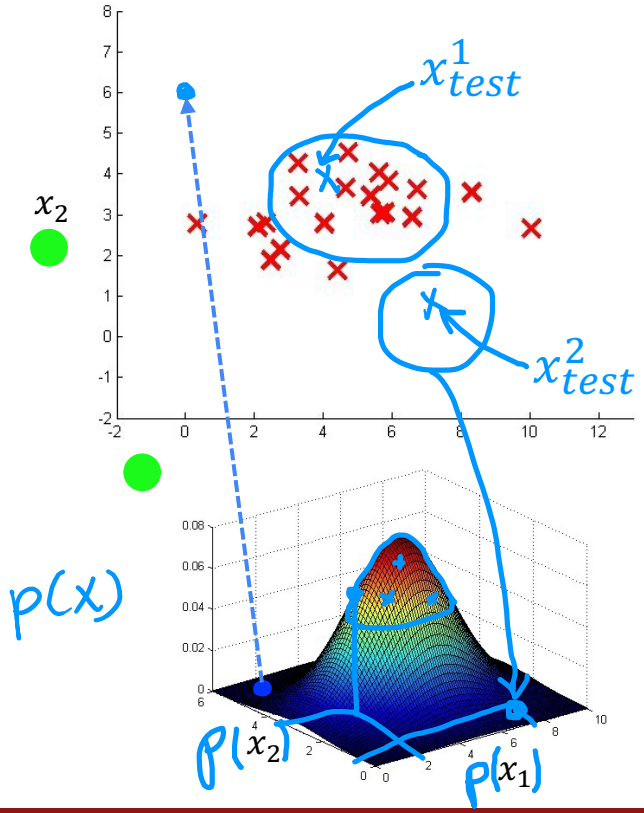
3. Given new example x, compute $p(x)$:

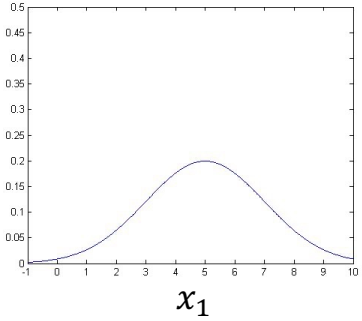
$$p(x) = \prod_{j=1}^{n} p(x_j; \mu_j, \sigma_j^2) = \prod_{j=1}^{n} \frac{1}{\sqrt{2\pi}\sigma_j} exp(-\frac{(x_j - \mu_j)^2}{2\sigma_j^2})$$

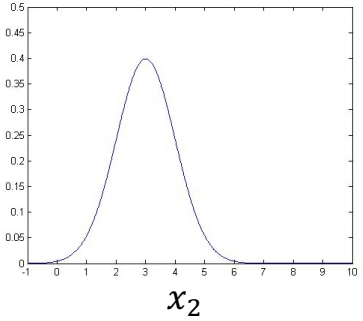
Anomaly if $p(x) < \varepsilon$



Anomaly detection example







$\mu_1 = 5,\ \sigma_1 = 2$

$p(x_1; \mu_1, \sigma_1^2)$

$\mu_2 = 3,\ \sigma_2 = 1$

$p(x_2; \mu_2, \sigma_2^2)$

$\varepsilon = 0.02$

$p\left(x_{test}^{(1)}\right) = 0.0426$ → "ok"

$p\left(x_{test}^{(2)}\right) = 0.0021$ → anomaly

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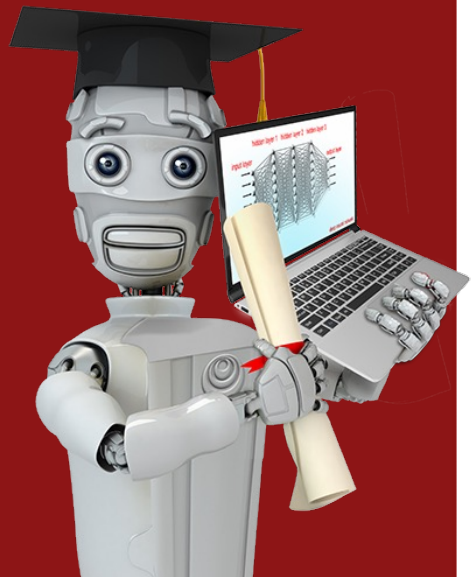

Anomaly Detection

Developing and evaluating an anomaly detection system

The importance of real-number evaluation

When developing a learning algorithm (choosing features, etc.),
making decisions is much easier if we have a way of evaluating our learning algorithm.

Assume we have some labeled data, of anomalous and non-anomalous examples.

$y = 1$ $y = 0$

Training set: $x^{(1)}, x^{(2)}, \ldots, x^{(m)}$ (assume normal examples/not anomalous)

$y = 0$ for all training examples

Cross validation set: $\left(x_{cv}^{(1)}, y_{cv}^{(1)}\right), \ldots, \left(x_{cv}^{(m_{cv})}, y_{cv}^{(m_{cv})}\right)$

Test set: $\left(x_{test}^{(1)}, y_{test}^{(1)}\right), \ldots, \left(x_{test}^{(m_{test})}, y_{test}^{(m_{test})}\right)$

include a few anomalous examples $y = 1$

mostly normal examples $y = 0$

Aircraft engines monitoring example

10000 good (normal) engines
20 flawed engines (anomalous)

2 to 50

y=1

2

y=0

Training set: 6000 good engines

train algorithm on training set

CV: 2000 good engines ($y = 0$) 10 anomalous ($y = 1$)

use cross validation set

tune ε tune x_j

Test: 2000 good engines ($y = 0$), 10 anomalous ($y = 1$)

Alternative: No test set

Use if very few labeled anomalous examples

higher risk of overfitting

Training set: 6000 good engines

2

CV: 4000 good engines ($y = 0$), 20 anomalous ($y = 1$)

tune ε tune x_j

Algorithm evaluation

course 2 week 3

skewed datasets

Fit model $p(x)$ on training set $x^{(1)}, x^{(2)}, \ldots, x^{(m)}$

On a cross validation/test example x, predict

$$y = \begin{cases} 1 & if\ p(x) < \varepsilon\ (anomaly) \\ 0 & if\ p(x) \geq \varepsilon\ (normal) \end{cases}$$

10

2000

Possible evaluation metrics:

- True positive, false positive, false negative, true negative
- Precision/Recall
- F_1-score

Can also use cross validation set to choose parameter ε

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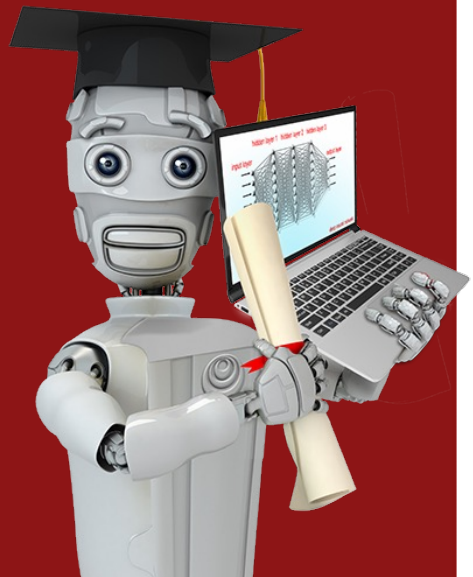

Anomaly Detection

Anomaly detection vs. supervised learning

Anomaly detection vs. Supervised learning

Very small number (0 to 20) of positive examples $y = 1$,
large number of negative examples ($y = 0$);
Model $p(x)$ with just negative examples.
Use positive examples for cv and test sets

Many different "types" of anomalies.
Hard for any algorithm to learn (from just positive examples) what the anomalies look like.
Future anomalies may look nothing like any of the anomalous examples seen so far.

financial fraud

Large number of positive and negative examples.

≥ 20 positive examples

Enough positive examples for algorithm to get a sense of what positive examples are like.
Future positive examples likely to be similar to ones in training set.

email spam

Anomaly detection vs. Supervised learning

- Fraud detection

- Manufacturing:
Finding new previously unseen defects.
(e.g. aircraft engines)

- Monitoring machines in a data center
- Security applications

- Email spam classification

- Manufacturing:
Finding known, previously seen defects
(e.g. scratches on smartphones, $y = 1$)

- Weather prediction (sunny/rainy/etc.)

- Diseases classification

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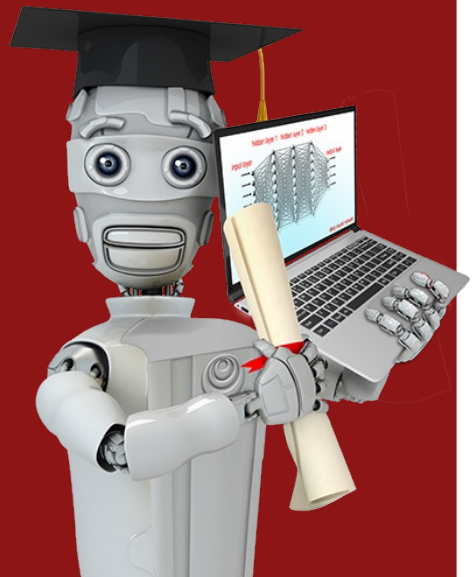
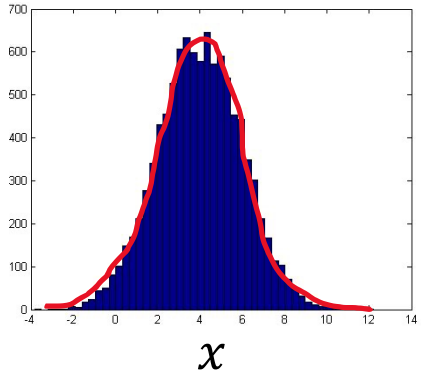


Anomaly Detection

Choosing what features to use

Non-gaussian features



$p(x_1; \mu_1, \sigma_1^2)$

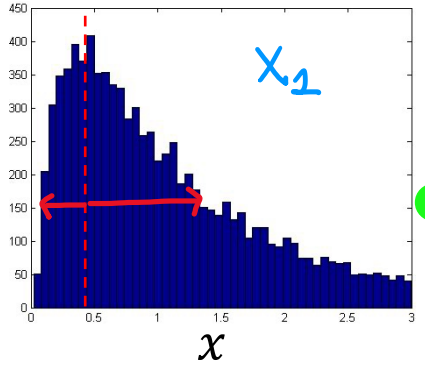
`plt.hist(x)`

$x_1 \leftarrow \log(x_1)$

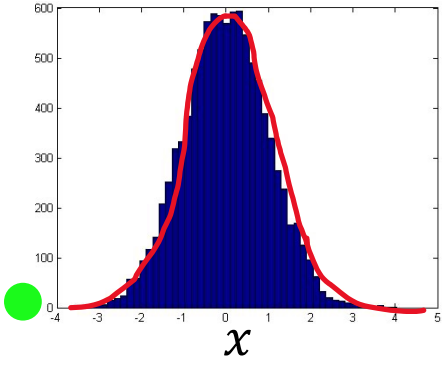
$x_2 \leftarrow \log(x_2 + 1)$ $\log(x_2 + c)$

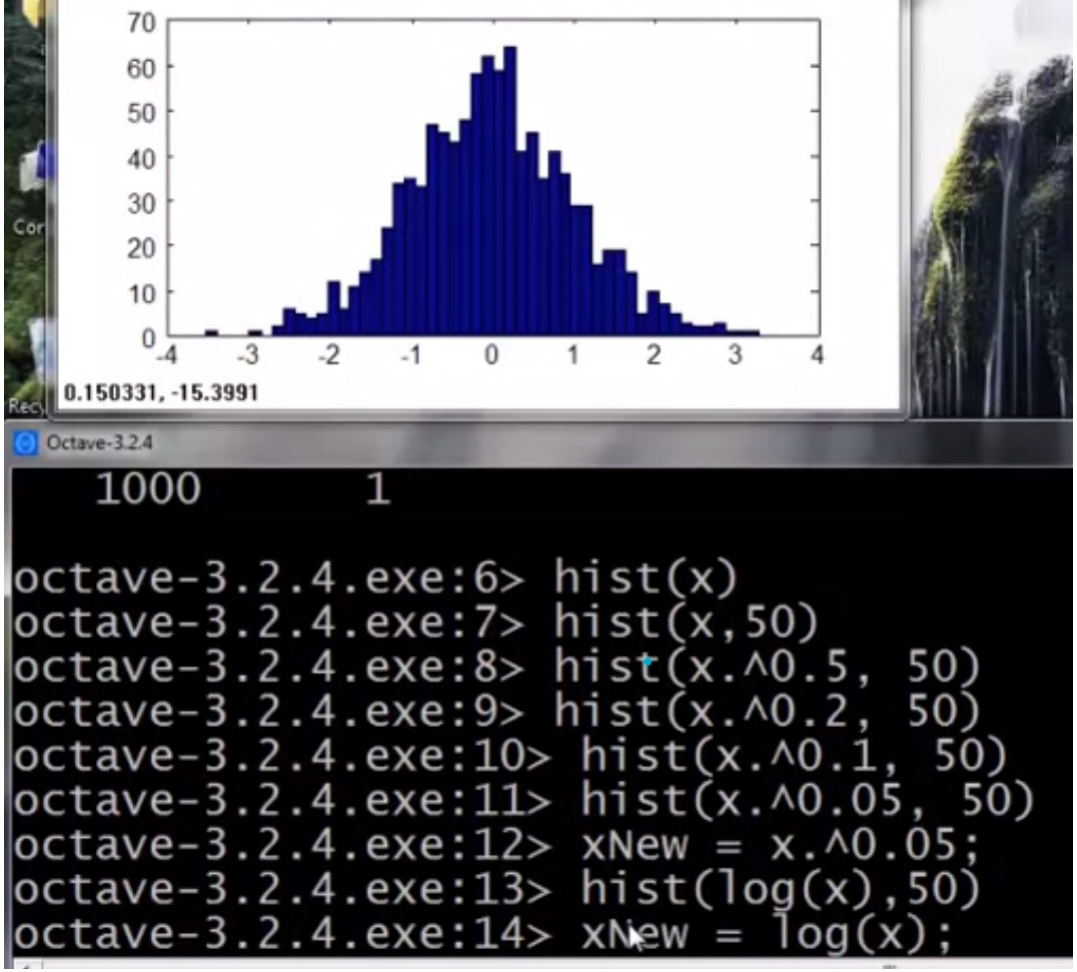
$x_3 \leftarrow \sqrt{x_3} = x_3^{1/2}$

$x_4 \leftarrow x_4^{1/3}$



`np.log(x)`

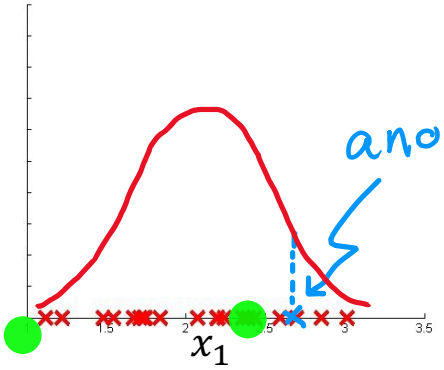


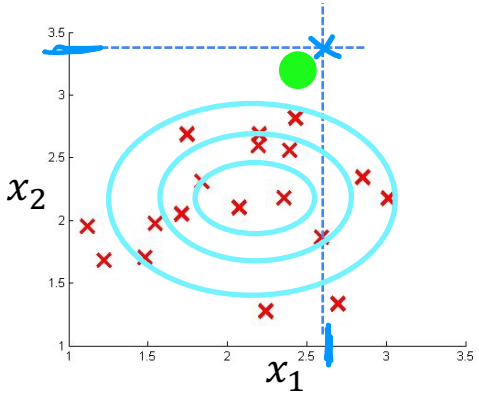

70
60
50
40
30
20
10
0
-4
-3
-2
-1
0
1
2
3
4
0.150331, -15.3991
Octave-3.2.4
1000 1
octave-3.2.4.exe:6> hist(x)
octave-3.2.4.exe:7> hist(x,50)
octave-3.2.4.exe:8> hist(x.^0.5, 50)
octave-3.2.4.exe:9> hist(x.^0.2, 50)
octave-3.2.4.exe:10> hist(x.^0.1, 50)
octave-3.2.4.exe:11> hist(x.^0.05, 50)
octave-3.2.4.exe:12> xNew = x.^0.05;
octave-3.2.4.exe:13> hist(log(x),50)
octave-3.2.4.exe:14> xNew = log(x);

Error analysis for anomaly detection

Want $p(x) \geq \varepsilon$ large for normal examples x.
$p(x) < \varepsilon$ small for anomalous examples x.

Most common problem:
$p(x)$ is comparable for normal and anomalous examples.
($p(x)$ is large for both)





x_1 num transactions x_2 typing speed

Monitoring computers in a data center

Choose features that might take on unusually large or small values in the event of an anomaly.

x_1 = memory use of computer

x_2 = number of disk accesses/sec

x_3 = CPU load

x_4 = network traffic

high low

not unusual

$$x_5 = \frac{\text{CPU load}}{\text{network traffic}}$$

$$x_6 = \frac{(\text{CPU load})^2}{\text{network traffic}}$$

Deciding feature choice based on $p(x)$

Large for normal examples;

Becomes small for anomaly in the cross validation set

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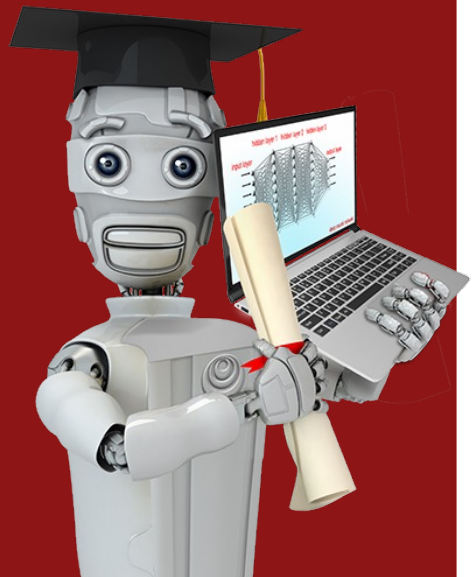

Recommender Systems

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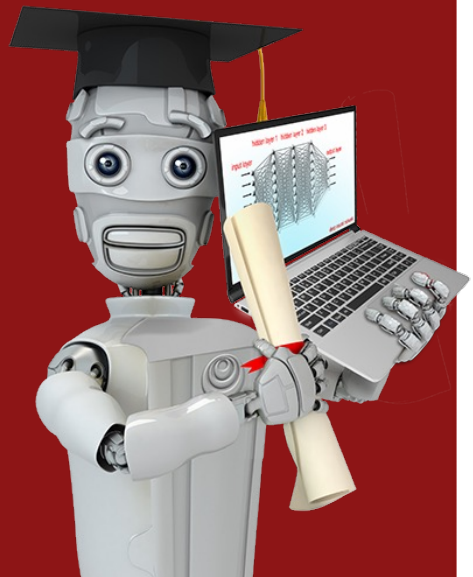
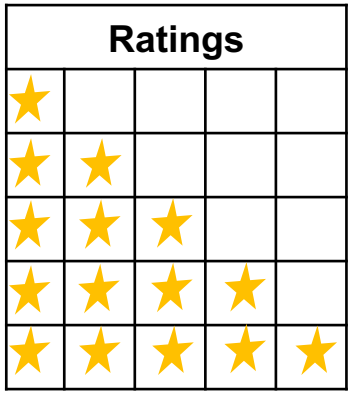


Recommender System

Making recommendations

Predicting movie ratings

User rates movies using ~~one~~ zero to five stars



Movie	Alice(1)	Bob(2)	Carol(3)	Dave(4)
Love at last	5	5	0	0
Romance forever	5	?	?	0
Cute puppies of love	?	4	0	?
Nonstop car chases	0	0	5	4
Swords vs. karate	0	0	5	?

n_u = no. of users

n_m = no. of movies

$r(i,j)=1$ if user j has rated movie i

$n_u = 4$ $\quad r(1,1) = 1$

$n_m = 5$ $\quad r(3,1) = 0$

$y^{(3,2)} = 4$

$y^{(i,j)}$ = rating given by user j to movie i (defined only if $r(i,j)=1$)

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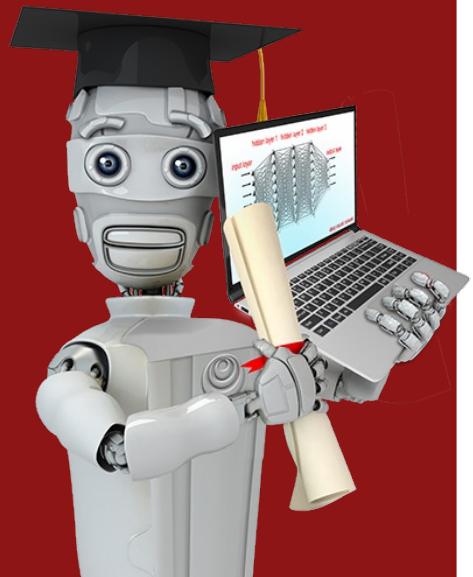

Collaborative Filtering

Using per-item features

What if we have features of the movies?

$n_u = 4$
$n_m = 5$
$n = 2$

Movie	Alice(1)	Bob(2)	Carol(3)	Dave(4)	x_1 (romance)	x_2 (action)
Love at last	5	5	0	0	0.9	0
Romance forever	5	?	?	0	1.0	0.01
Cute puppies of love	?	4	0	?	0.99	0
Nonstop car chases	0	0	5	4	0.1	1.0
Swords vs. karate	0	0	5	?	0	0.9

$x^{(1)} = \begin{bmatrix} 0.9 \\ 0 \end{bmatrix}$

$x^{(3)} = \begin{bmatrix} 0.99 \\ 0 \end{bmatrix}$

For user 1: Predict rating for movie *i* as: $w^{(1)} \cdot x^{(i)} + b^{(1)}$ — just linear regression

$w^{(1)} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$ $b^{(1)} = 0$ $x^{(3)} = \begin{bmatrix} 0.9 \\ 0 \end{bmatrix}$

$w^{(1)} \cdot x^{(3)} + b^{(1)} = 4.95$

For user *j*: Predict user j's rating for movie *i* as $w^{(j)} \cdot x^{(i)} + b^{(j)}$

Cost function

Notation:

- $r(i,j) = 1$ if user j has rated movie i (0 otherwise)
- $y^{(i,j)}$ = rating given by user j on movie i (if defined)
- $w^{(j)}$, $b^{(j)}$ = parameters for user j
- $x^{(i)}$ = feature vector for movie i

For user j and movie i, predict rating: $w^{(j)} \cdot x^{(i)} + b^{(j)}$

$m^{(j)}$ = no. of movies rated by user j

To learn $w^{(j)}$, $b^{(j)}$

$$\min_{w^{(j)} b^{(j)}} J\left(w^{(j)}, b^{(j)}\right) = \frac{1}{2m^{(j)}} \sum_{i:r(i,j)=1} \left(w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)}\right)^2 + \frac{\lambda}{2m^{(j)}} \sum_{k=1}^{n} \left(w_k^{(j)}\right)^2$$

number of features

Cost function

To learn parameters $w^{(j)}, b^{(j)}$ for user j :

$$J\left(w^{(j)}, b^{(j)}\right) = \frac{1}{2} \sum_{i:r(i,j)} \left(w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)}\right)^2 + \frac{\lambda}{2} \sum_{k=1}^{n} \left(w_k^{(j)}\right)^2$$

To learn parameters $w^{(1)}, b^{(1)}, \; w^{(2)}, b^{(2)}, \cdots w^{(n_u)}, b^{(n_u)}$ for all users :

$$J\begin{pmatrix} w^{(1)}, & \ldots, w^{(n_u)} \\ b^{(1)}, & \ldots, b^{(n_u)} \end{pmatrix} = \frac{1}{2} \sum_{j=1}^{n_u} \sum_{i:r(i,j)=1} \left(\underbrace{w^{(j)} \cdot x^{(i)} + b^{(j)}}_{f(x)} - y^{(i,j)}\right)^2 + \frac{\lambda}{2} \sum_{j=1}^{n_u} \sum_{k=1}^{n} \left(w_k^{(j)}\right)^2$$

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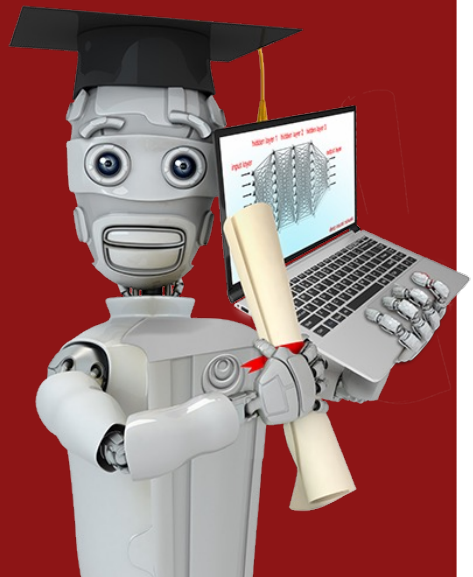

Collaborative Filtering

Collaborative filtering algorithm

Problem motivation

Movie	Alice (1)	Bob (2)	Carol (3)	Dave (4)	x_1 (romance)	x_2 (action)
Love at last	5	5	0	0	0.9	0
Romance forever	5	?	?	0	1.0	0.01
Cute puppies of love	?	4	0	?	0.99	0
Nonstop car chases	0	0	5	4	0.1	1.0
Swords vs. karate	0	0	5	?	0	0.9

Problem motivation

Movie	Alice (1)	Bob (2)	Carol (3)	Dave (4)	x_1 (romance)	x_2 (action)
Love at last	5	5	0	0	?	?
Romance forever	5	?	?	0 ←	?	? $x^{(2)}$
Cute puppies of love	?	4	0	? ←	?	? $x^{(3)}$
Nonstop car chases	0	0	5	4 ←	?	?
Swords vs. karate	0	0	5	? ←	?	?

$w^{(1)} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$, $w^{(2)} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$, $w^{(3)} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$, $w^{(4)} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$

$b^{(1)} = 0$, $b^{(2)} = 0$, $b^{(3)} = 0$, $b^{(4)} = 0$ ←

using $w^{(j)} \cdot x^{(i)} + b^{(j)}$

$w^{(1)} \cdot x^{(1)} \approx 5$

$w^{(2)} \cdot x^{(1)} \approx 5$

$w^{(3)} \cdot x^{(1)} \approx 0$

$w^{(4)} \cdot x^{(1)} \approx 0$

$\rightarrow \quad x^{(1)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Cost function

Given $w^{(1)}, b^{(1)}, w^{(2)}, b^{(2)}, \cdots, w^{(n_u)}, b^{(n_u)}$

to learn $x^{(i)}$:

$$J\left(x^{(i)}\right) = \frac{1}{2} \sum_{j:r(i,j)=1} \left(w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)}\right)^2 + \frac{\lambda}{2} \sum_{k=1}^{n} \left(x_k^{(i)}\right)^2$$

→ To learn $x^{(1)}, x^{(2)}, \cdots, x^{(n_m)}$:

$$J\left(x^{(1)}, x^{(2)}, \ldots, x^{(n_m)}\right) = \frac{1}{2} \sum_{i=1}^{n_m} \sum_{j:r(i,j)=1} \left(w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)}\right)^2 + \frac{\lambda}{2} \sum_{i=1}^{n_m} \sum_{k=1}^{n} \left(x_k^{(i)}\right)^2$$

Collaborative filtering

	$j=1$ Alice	$j=2$ Bob	$j=3$ Carol
$i=1$ Movie1	5	5	?
$i=2$ Movie2	?	2	3

Cost function to learn $w^{(1)}, b^{(1)}, \cdots w^{(n_u)}, b^{(n_u)}$:

$$\min_{w^{(1)},b^{(1)},\cdots,w^{(n_u)},b^{(n_u)}} \frac{1}{2}\sum_{j=1}^{n_u}\sum_{i:r(i,j)=1}\left(w^{(j)}\cdot x^{(i)}+b^{(j)}-y^{(i,j)}\right)^2 + \frac{\lambda}{2}\sum_{j=1}^{n_u}\sum_{k=1}^{n}\left(w_k^{(j)}\right)^2$$

Cost function to learn $x^{(1)},\cdots,x^{(n_m)}$:

$$\min_{x^{(1)},\cdots,x^{(n_m)}} \frac{1}{2}\sum_{i=1}^{n_m}\sum_{j:r(i,j)=1}\left(w^{(j)}\cdot x^{(i)}+b^{(j)}-y^{(i,j)}\right)^2 + \frac{\lambda}{2}\sum_{i=1}^{n_m}\sum_{k=1}^{n}\left(x_k^{(i)}\right)^2$$

Put them together:

$$\min_{\substack{w^{(1)},\ldots,w^{(n_u)}\\ b^{(1)},\ldots,b^{(n_u)}\\ x^{(1)},\ldots,x^{(n_m)}}} J(w,b,x) = \frac{1}{2}\sum_{(i,j):r(i,j)=1}\left(w^{(j)}\cdot x^{(i)}+b^{(j)}-y^{(i,j)}\right)^2 + \frac{\lambda}{2}\sum_{j=1}^{n_u}\sum_{k=1}^{n}\left(w_k^{(j)}\right)^2 + \frac{\lambda}{2}\sum_{i=1}^{n_m}\sum_{k=1}^{n}\left(x_k^{(i)}\right)^2$$

Gradient Descent

collaborative filtering

Linear regression (course 1)

repeat {

$$w_i = w_i - \alpha \frac{\partial}{\partial w_i} J(w,b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w,b)$$

$$w_i^{(j)} = w_i^{(j)} - \alpha \frac{\partial}{\partial w_i^{(j)}} J(w,b,x)$$

$$b^{(j)} = b^{(j)} - \alpha \frac{\partial}{\partial b^{(j)}} J(w,b,x)$$

$$x_k^{(i)} = x_k^{(i)} - \alpha \frac{\partial}{\partial x_k^{(i)}} J(w,b,x)$$

}

parameters w, b, x

X is also a parameter

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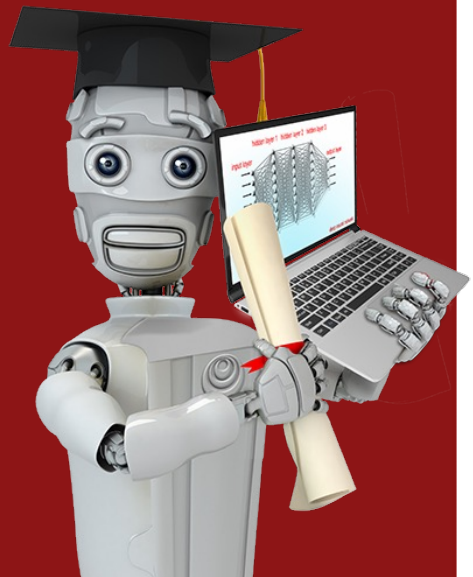

Collaborative Filtering

Binary labels:
favs,
likes and clicks

Binary labels

Movie	Alice(1)	Bob(2)	Carol(3)	Dave(4)
Love at last	1	1	0	0
Romance forever	1	?	?	0
Cute puppies of love	?	1	0	?
Nonstop car chases	0	0	1	1
Swords vs. karate	0	0	1	?

1
0
?

Example applications

1. Did user j purchase an item after being shown? 1, 0, ?
2. Did user j fav/like an item? 1, 0, ?
3. Did user j spend at least 30sec with an item? 1, 0, ?
4. Did user j click on an item? 1, 0, ?

Meaning of ratings:
- 1 - engaged after being shown item
- 0 - did not engage after being shown item
- ? - item not yet shown

From regression to binary classification

- Previously:
  - Predict $y^{(i,j)}$ as $w^{(j)} \cdot x^{(i)} + b^{(j)}$
- For binary labels:
  - Predict that the probability of $y^{(i,j)} = 1$ is given by $g\left(w^{(j)} \cdot x^{(i)} + b^{(j)}\right)$

where $g(z) = \frac{1}{1+e^{-z}}$

Cost function for binary application

Previous cost function:

$$\frac{1}{2}\sum_{(i,j):r(i,j)=1}\left(w^{(j)}\cdot x^{(i)}+b^{(j)}-y^{(i,j)}\right)^2+\frac{\lambda}{2}\sum_{i=1}^{n_m}\sum_{k=1}^{n}\left(x_k^{(i)}\right)^2+\frac{\lambda}{2}\sum_{j=1}^{n_u}\sum_{k=1}^{n}\left(w_k^{(j)}\right)^2$$

$w^{(j)}\cdot x^{(i)}+b^{(j)}$: $f(x)$

Loss for binary labels $y^{(i,j)}$: $f_{(w,b,x)}(x)=g(w^{(j)}\cdot x^{(i)}+b^{(j)})$

$$L\left(f_{(w,b,x)}(x),y^{(i,j)}\right)=-y^{(i,j)}\log\left(f_{(w,b,x)}(x)\right)-\left(1-y^{(i,j)}\right)\log\left(1-f_{(w,b,x)}(x)\right)$$

Loss for single example

$$J(w,b,x)=\sum_{(i,j):r(i,j)=1}L\left(f_{(w,b,x)}(x),y^{(i,j)}\right)$$

$g(w^{(j)}\cdot x^{(i)}+b^{(j)})$

cost for all examples

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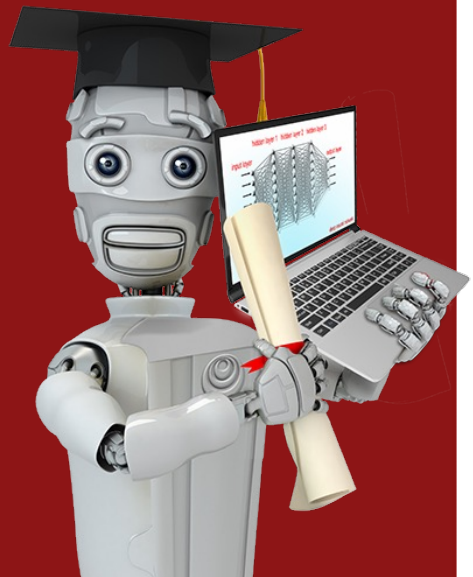

Recommender Systems implementation

Mean normalization

Users who have not rated any movies

Movie	Alice(1)	Bob (2)	Carol (3)	Dave (4)	Eve (5)	
Love at last	5	5	0	0	?	$r(i,1)$
Romance forever	5	?	?	0	?	$r(i,2)$
Cute puppies of love	?	4	0	?	?	$r(i,3)$
Nonstop car chases	0	0	5	4	?	$r(i,4)$
Swords vs. karate	0	0	5	?	?	$r(i,5)$

$$\begin{bmatrix} 5 & 5 & 0 & 0 & ? \\ 5 & ? & ? & 0 & ? \\ ? & 4 & 0 & ? & ? \\ 0 & 0 & 5 & 4 & ? \\ 0 & 0 & 5 & 0 & ? \end{bmatrix}$$

$$\min_{\substack{w^{(1)},\ldots w^{(n_u)} \\ b^{(1)},\ldots b^{(n_u)} \\ x^{(1)},\ldots x^{(n_m)}}} \frac{1}{2} \sum_{(i,j):r(i,j)=1} \left(w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)}\right)^2 + \frac{\lambda}{2} \sum_{j=1}^{n_u} \sum_{k=1}^{n} \left(w_k^{(j)}\right)^2 + \frac{\lambda}{2} \sum_{i=1}^{n_m} \sum_{k=1}^{n} \left(x_k^{(i)}\right)^2$$

$w^{(5)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $b^{(5)} = 0$ $w^{(5)} \cdot x^{(i)} + b^{(5)}$

Mean Normalization

$$\begin{bmatrix} 5 & 5 & 0 & 0 & ? \\ 5 & ? & ? & 0 & ? \\ ? & 4 & 0 & ? & ? \\ 0 & 0 & 5 & 4 & ? \\ 0 & 0 & 5 & 0 & ? \end{bmatrix} \quad \begin{matrix} 2.5 \\ 2.5 \\ 2 \\ 2.25 \\ 1.25 \end{matrix}$$

$$\mu = \begin{bmatrix} 2.5 \\ 2.5 \\ 2 \\ 2.25 \\ 1.25 \end{bmatrix}$$

$$\begin{bmatrix} 2.5 & 2.5 & -2.5 & -2.5 & ? \\ 2.5 & ? & ? & -2.5 & ? \\ ? & 2 & -2 & ? & ? \\ -2.25 & -2.25 & 2.75 & 1.75 & ? \\ -1.25 & -1.25 & 3.75 & -1.25 & ? \end{bmatrix}$$

For user j, on movie i predict:

$$w^{(j)} \cdot x^{(i)} + b^{(j)} + \mu_i$$

$y^{(i,j)}$

$$w^{(j)} \cdot b^{(j)} \cdot x^{(i)}$$

User 5 (Eve):

$$w^{(5)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad b^{(5)} = 0$$

$$\underbrace{w^{(5)} \cdot x^{(1)} + b^{(5)}}_{0} + \mu_1 = 2.5$$

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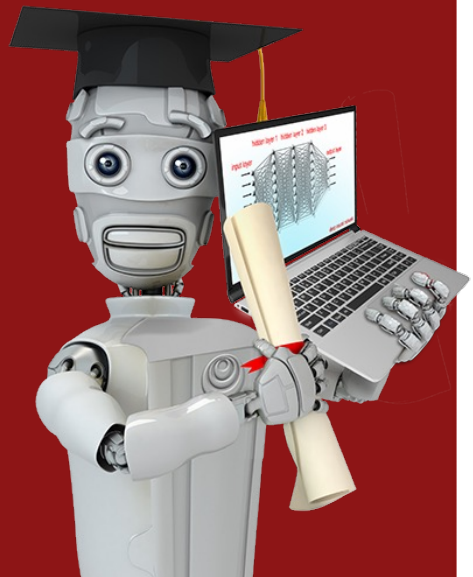


Recommender Systems implementational detail

TensorFlow implementation

Derivatives in ML

Gradient descent algorithm

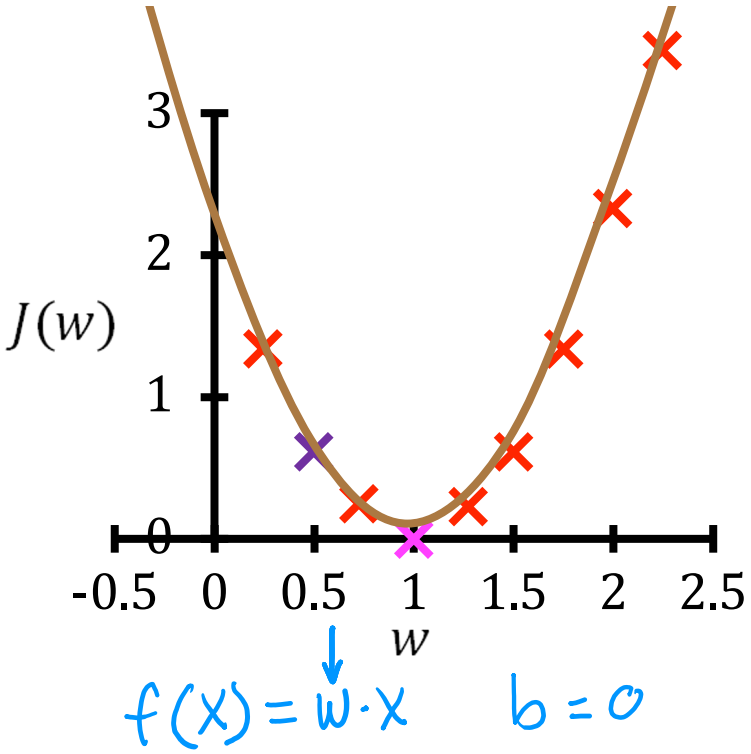
Repeat until convergence

$w = w - \alpha \frac{d}{dw} J(w,b)$

$b = b - \alpha \frac{d}{db} J(w,b)$ ← $b = 0$

Learning rate

Derivative



Custom Training Loop

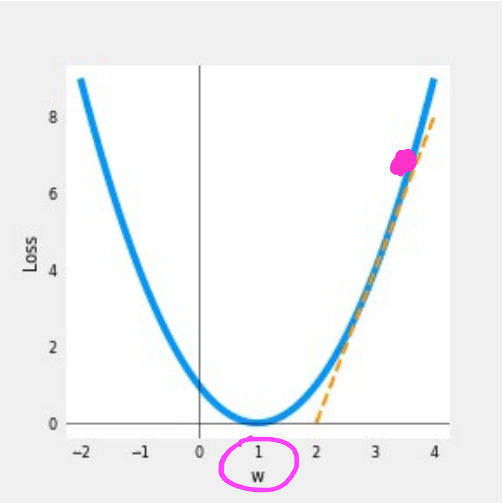
$$J = (\underbrace{wx}_{f(x)} - \underbrace{1}_{y})^2$$

Gradient descent algorithm

Repeat until convergence

$$w = w - \alpha \frac{d}{dw} J(w,b)$$

Fix b = 0 for this example




```
w = tf.Variable(3.0)
x = 1.0
y = 1.0 # target value
alpha = 0.01

iterations = 30
for iter in range(iterations):
    # Use TensorFlow's Gradient tape to record the steps
    # used to compute the cost J, to enable auto differentiation.
    with tf.GradientTape() as tape:
        fwb = w*x
        costJ = (fwb - y)**2

    # Use the gradient tape to calculate the gradients
    # of the cost with respect to the parameter w.
    [dJdw] = tape.gradient( costJ, [w] )

    # Run one step of gradient descent by updating
    # the value of w to reduce the cost.
    w.assign_add(-alpha * dJdw)
```

f(x)

Tf.variables are the parameters we want to optimize

Auto Diff

Auto Grad

$$\frac{\partial}{\partial w} J(w)$$

tf.variables require special function to modify

Implementation in TensorFlow

Gradient descent algorithm

Repeat until convergence

$$w = w - \alpha \frac{\partial}{\partial w} J(w,b,X)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w,b,X)$$

$$X = X - \alpha \frac{\partial}{\partial X} J(w,b,X)$$

```
# Instantiate an optimizer.
optimizer = keras.optimizers.Adam(learning_rate=1e-1)

iterations = 200
for iter in range(iterations):
    # Use TensorFlow's GradientTape
    # to record the operations used to compute the cost
    with tf.GradientTape() as tape:

      # Compute the cost (forward pass is included in cost)
      cost_value = cofiCostFuncV(X, W, b, Ynorm, R,
          num_users, num_movies, lambda)
                   # n_u        # n_m
    # Use the gradient tape to automatically retrieve
    # the gradients of the trainable variables with respect to
          the loss
    grads = tape.gradient( cost_value, [X,W,b] )

    # Run one step of gradient descent by updating
    # the value of the variables to minimize the loss.
    optimizer.apply_gradients( zip(grads, [X,W,b]) )
```

Dataset credit: Harper and Konstan. 2015. The MovieLens Datasets: History and Context

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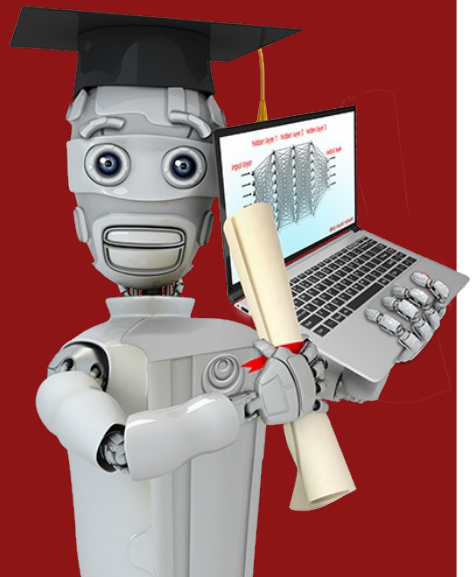

Collaborative Filtering

Finding related items

Finding related items

The features $x^{(i)}$ of item i are quite hard to interpret.

To find other items related to it,
find item k with $x^{(k)}$ similar to $x^{(i)}$

romance
action

x_1, x_2, x_3
n

i.e. with smallest distance

$$\sum_{l=1}^{n} \left(x_l^{(k)} - x_l^{(i)} \right)^2$$

$x^{(k)}$ $x^{(i)}$

$$\left\| x^{(k)} - x^{(i)} \right\|^2$$

Limitations of Collaborative Filtering

Cold start problem. How to

- rank new items that few users have rated?
- show something reasonable to new users who have rated few items?

Use side information about items or users:

- Item: Genre, movie stars, studio, ….
- User: Demographics (age, gender, location), expressed preferences, …

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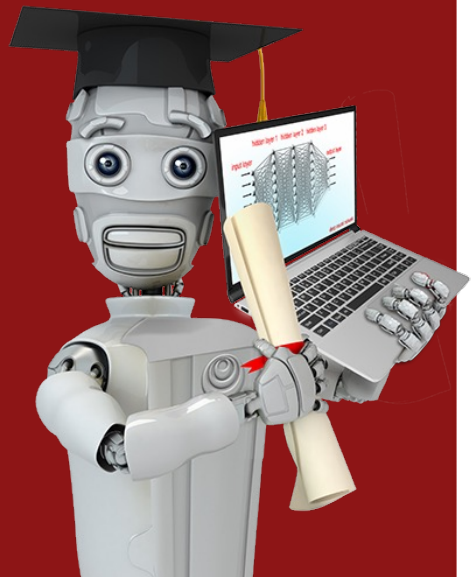


Content-based Filtering

Collaborative filtering
vs
Content-based filtering

Collaborative filtering vs Content-based filtering

- Collaborative filtering:
  Recommend items to you based on rating of users who gave similar ratings as you

- Content-based filtering:
  Recommend items to you based on features of user and item to find good match

$r(i,j)=1$ if user j has rated item i

$y^{(i,j)}$ rating given by user j on item i (if defined)

Examples of user and item features

User features:

- Age
- Gender (1 hot)
- Country (1 hot, 200)
- Movies watched (1000)
- Average rating per genre
- …

Movie features:

- Year
- Genre/Genres
- Reviews
- Average rating
- …

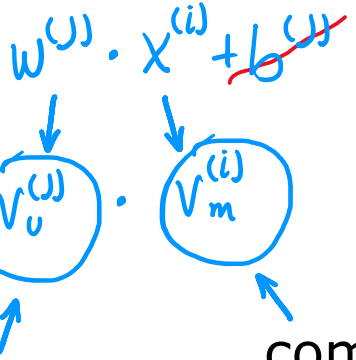
$x_u^{(j)}$ for user j

$x_m^{(i)}$ for movie i

Vector size could be different

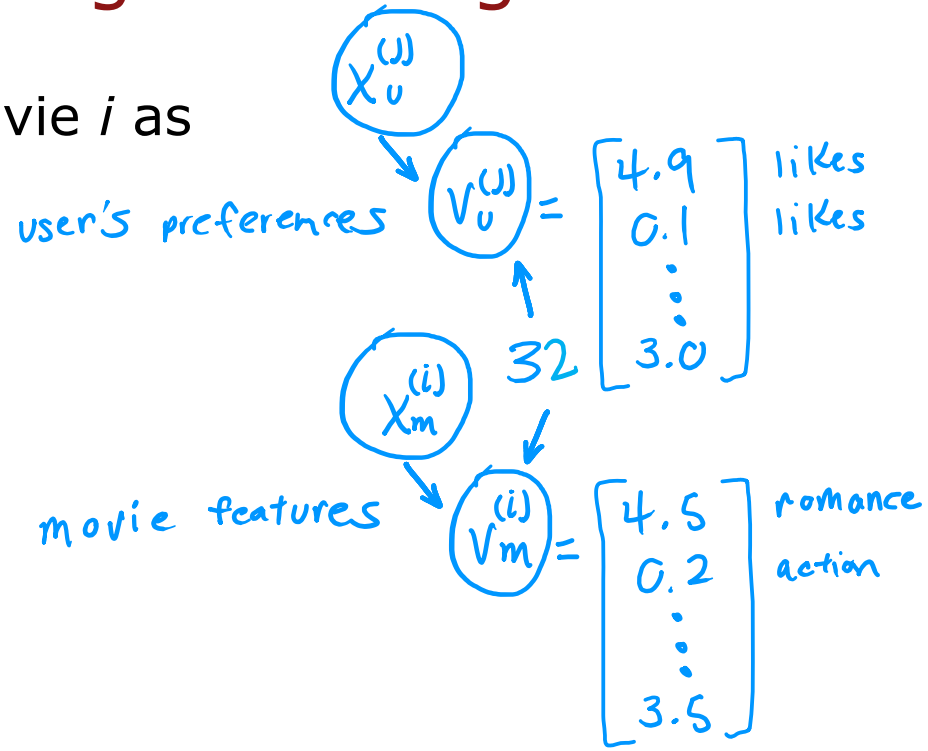
Content-based filtering: Learning to match

Predict rating of user *j* on movie *i* as



computed from $\boldsymbol{x}_{\boldsymbol{u}}^{(\boldsymbol{j})}$

computed from $\boldsymbol{x}_{\boldsymbol{m}}^{(\boldsymbol{i})}$



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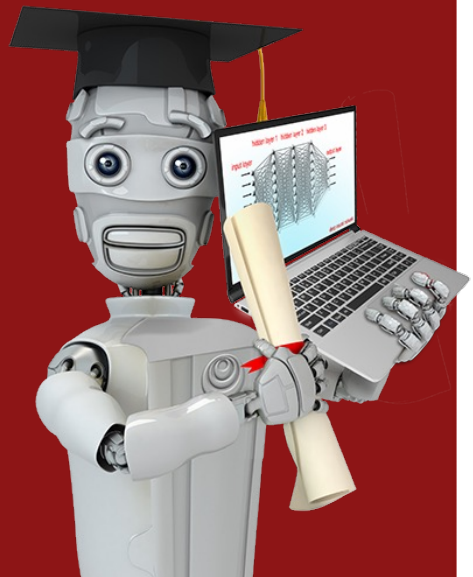
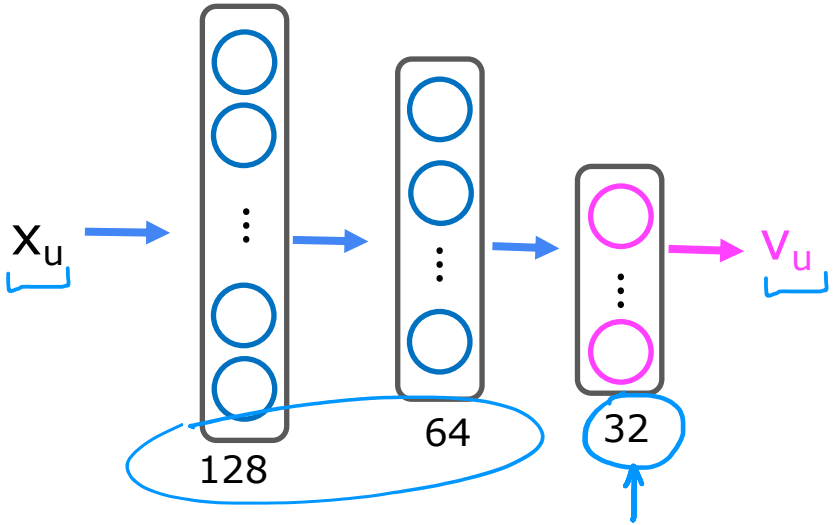


Content-based Filtering

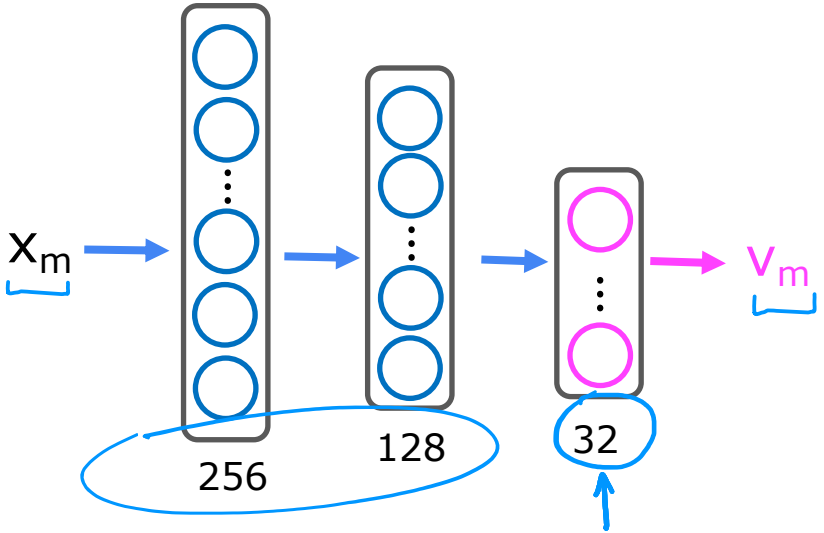
Deep learning for content-based filtering

Neural network architecture

$x_u \rightarrow v_u$ User network



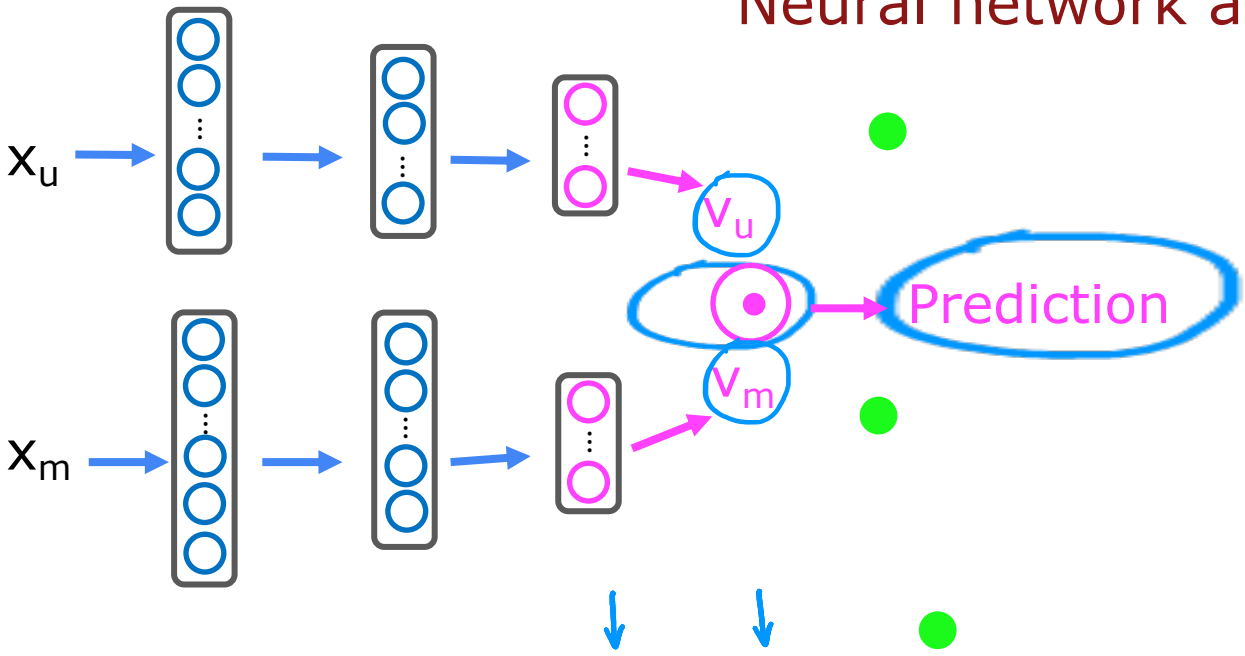
$x_m \rightarrow v_m$ Movie network



Prediction : $v_u^{(j)} \cdot v_m^{(i)}$

$g(v_u^{(j)} \cdot v_m^{(i)})$ ***to predict the probability that*** $y^{(i,j)}$ ***is*** 1

Neural network architecture



Cost function

$$J = \sum_{(i,j):r(i,j)=1} \left(v_u{}^{(j)} \cdot v_m{}^{(i)} - y^{(i,j)}\right)^2 + \text{NN regularization term}$$

Learned user and item vectors:

- $v_u^{(j)}$ is a vector of length 32 that describes user j with features $x_u^{(j)}$
- $v_m^{(i)}$ is a vector of length 32 that describes movie i with features $x_m^{(i)}$

To find movies similar to movie i:

$\| v_m^{(k)} - v_m^{(i)} \|^2$ small

$\| x^{(k)} - x^{(i)} \|^2$

- Note: This can be pre-computed ahead of time

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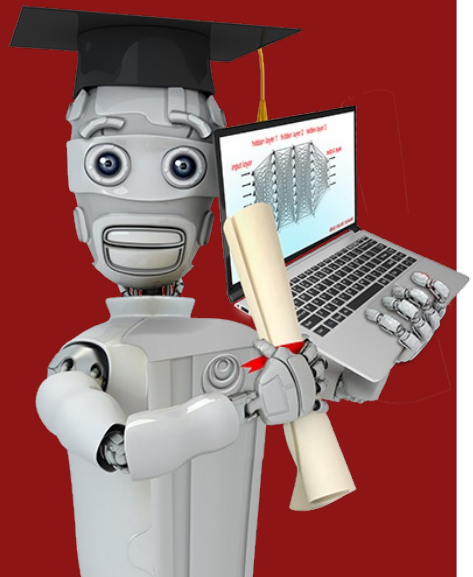
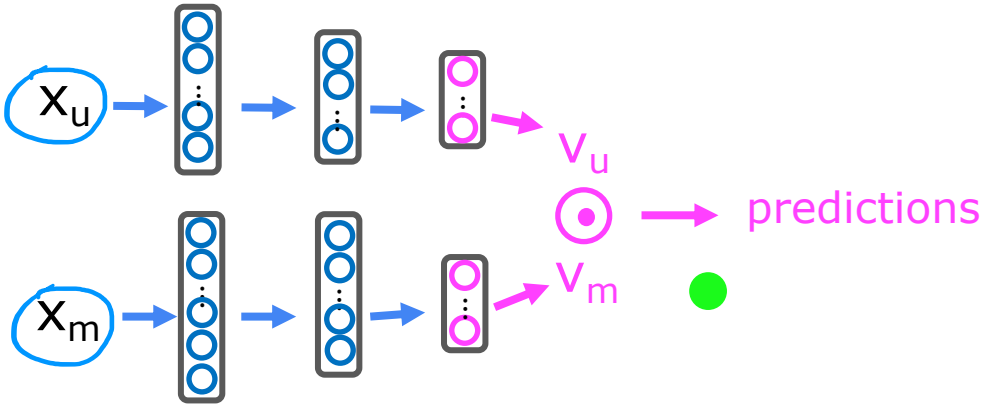


Advanced implementation

Recommending from a large catalogue

How to efficiently find recommendation from a large set of items?

- Movies 1000+
- Ads 1m+
- Songs 10m+
- Products 10m+



Two steps: Retrieval & Ranking

Retrieval:

- Generate large list of plausible item candidates ~100s

  e.g.

  1) For each of the last 10 movies watched by the user, find 10 most similar movies

  $\| v_m^{(k)} - v_m^{(i)} \|^2$

  2) For most viewed 3 genres, find the top 10 movies

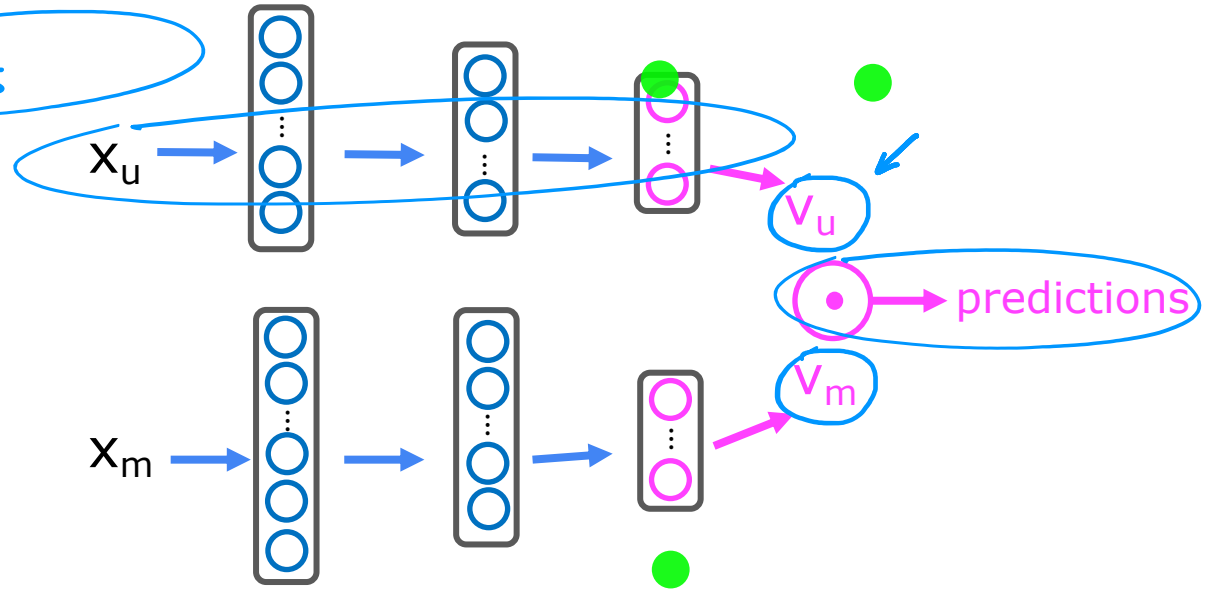
  3) Top 20 movies in the country

- Combine retrieved items into list, removing duplicates and items already watched/purchased

Two steps: Retrieval & ranking

Ranking:

- Take list retrieved and rank using learned model



- Display ranked items to user

Retrieval step

- Retrieving more items results in better performance, but slower recommendations.
- To analyse/optimize the trade-off, carry out offline experiments to see if retrieving additional items results in more relevant recommendations (i.e., $p(y^{(i,j)}) = 1$ of items displayed to user are higher).

100 500

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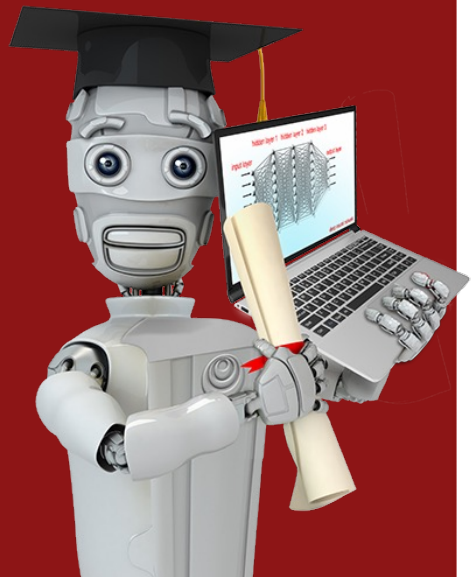


Advanced implementation

Ethical use of recommender systems

What is the goal of the recommender system?

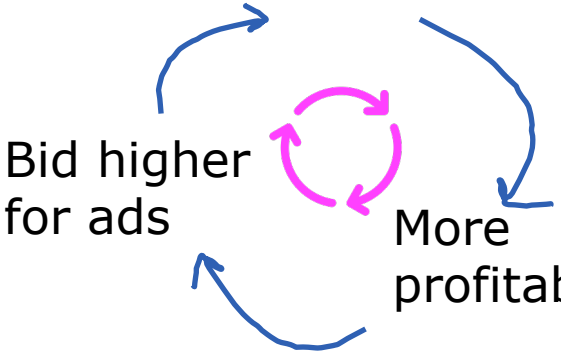
Recommend:

- Movies most likely to be rated 5 stars by user
- Products most likely to be purchased
- Ads most likely to be clicked on +high bid
- Products generating the largest profit
- Video leading to maximum watch time

Ethical considerations with recommender systems

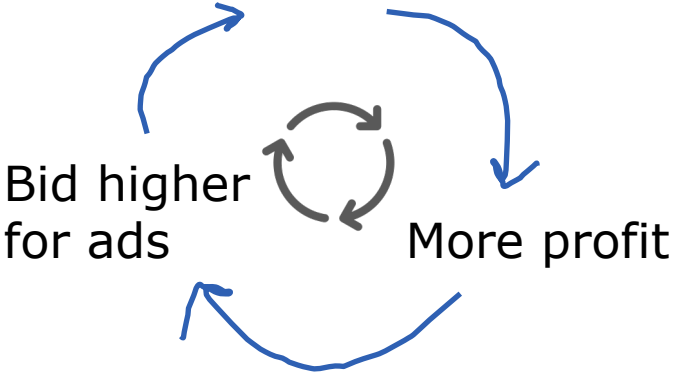
Travel industry

Good travel experience to more users



Payday loans

Squeeze customers more



Amelioration: Do not accept ads from exploitative businesses

Other problematic cases:

- Maximizing user engagement (e.g. watch time) has led to large social media/video sharing sites to amplify conspiracy theories and hate/toxicity

  → Amelioration : Filter out problematic content such as hate speech, fraud, scams and violent content

- Can a ranking system maximize your profit rather than users' welfare be presented in a transparent way?

  → Amelioration : Be transparent with users

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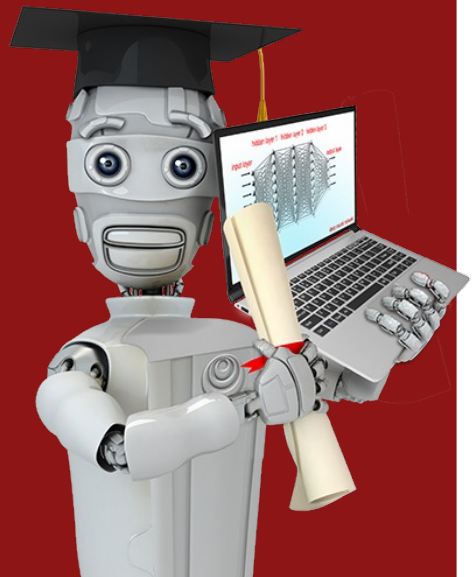
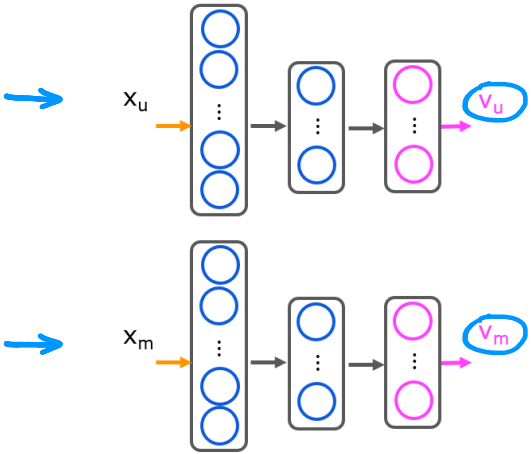


Content-based Filtering

TensorFlow Implementation




```
user_NN = tf.keras.models.Sequential([
    tf.keras.layers.Dense(256, activation='relu'),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(32)
])
```

```
item_NN = tf.keras.models.Sequential([
    tf.keras.layers.Dense(256, activation='relu'),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(32 )
])
```

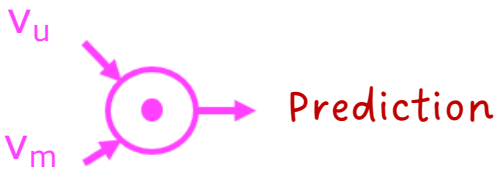
```
# create the user input and point to the base network
input_user = tf.keras.layers.Input(shape=(num_user_features))
vu = user_NN(input_user)
vu = tf.linalg.l2_normalize(vu, axis=1)

# create the item input and point to the base network
input_item = tf.keras.layers.Input(shape=(num_item_features))
vm = item_NN(input_item)
vm = tf.linalg.l2_normalize(vm, axis=1)

# measure the similarity of the two vector outputs
output = tf.keras.layers.Dot(axes=1)([vu, vm])

# specify the inputs and output of the model
model = Model([input_user, input_item], output)

# Specify the cost function
cost_fn = tf.keras.losses.MeanSquaredError()
```



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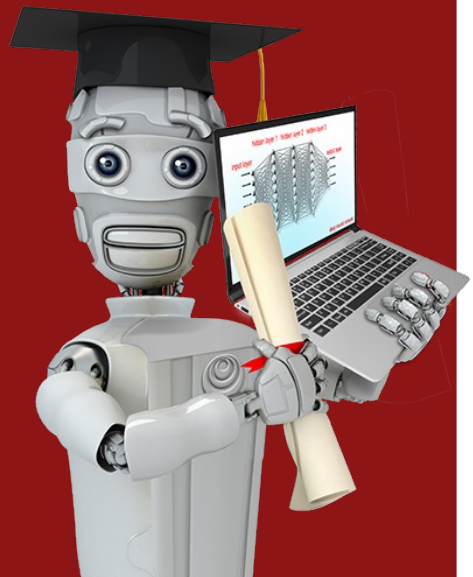
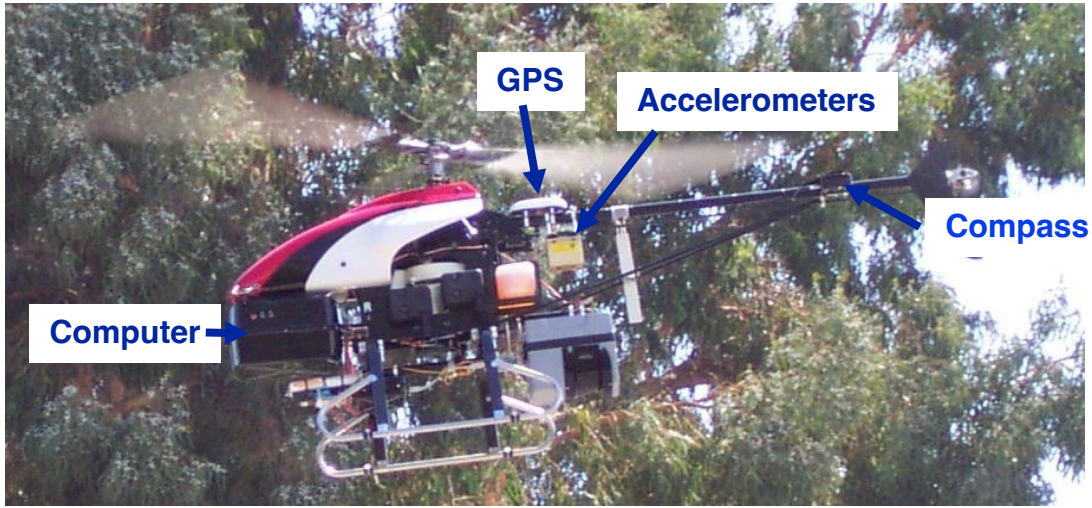


Reinforcement Learning Introduction

What is Reinforcement Learning?

Autonomous Helicopter



How to fly it?

Autonomous Helicopter

[Thanks to Pieter Abbeel, Adam Coates and Morgan Quigley]

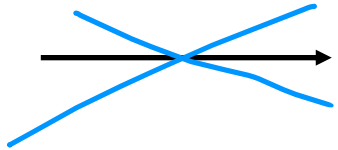
→ For more videos: http://heli.stanford.edu.

Reinforcement Learning

position of helicopter → how to move control sticks

state s → action a

x → y



X, Y

reward function

positive reward : helicopter flying well +1

negative reward : helicopter flying poorly −1000

Robotic Dog Example

[Thanks to Zico Kolter]

Applications

- Controlling robots
- Factory optimization
- Financial (stock) trading
- Playing games (including video games)

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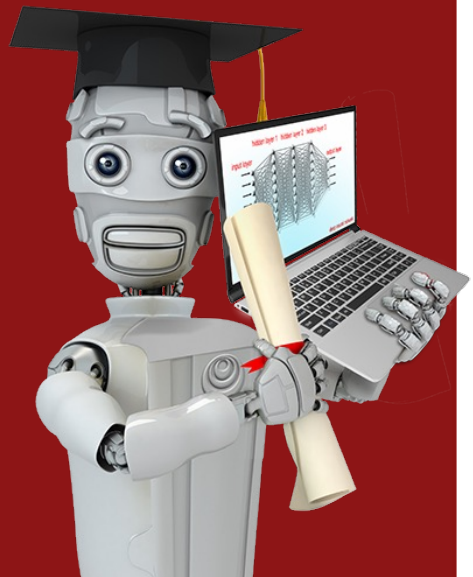
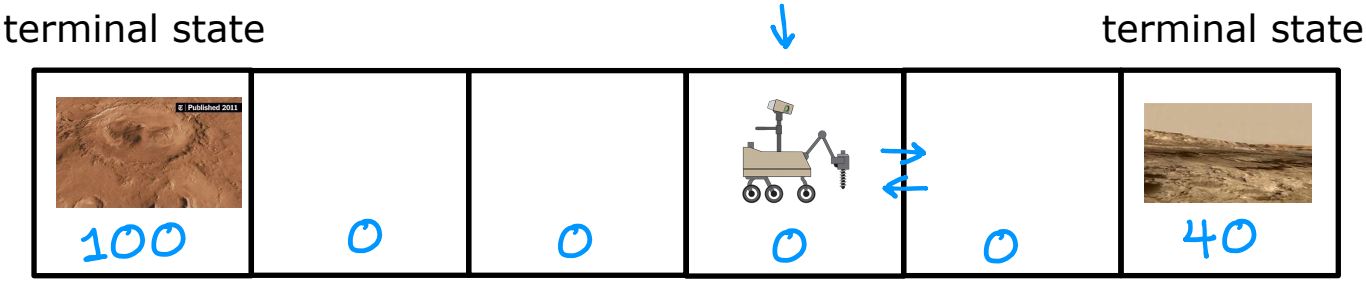
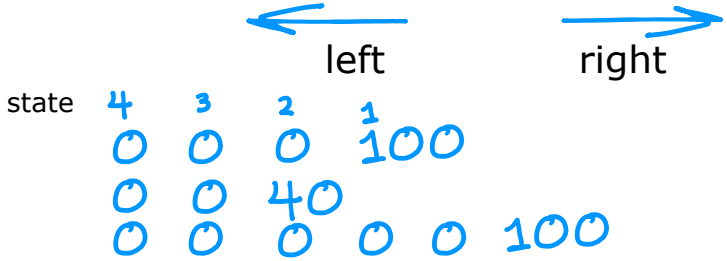


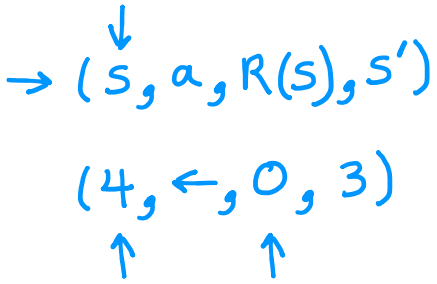
Reinforcement Learning formalism

Mars rover example

Mars Rover Example







[Credit: Jagriti Agrawal, Emma Brunskill]

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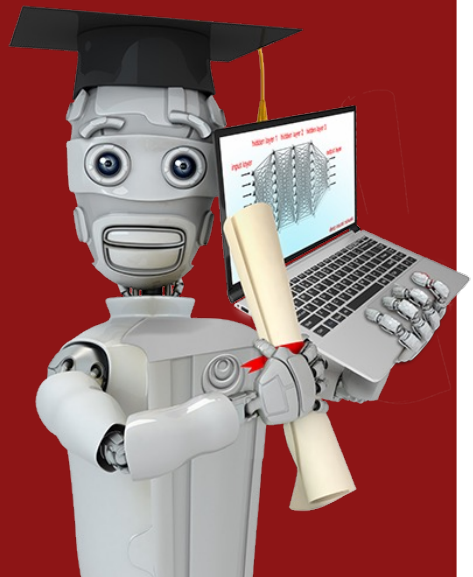
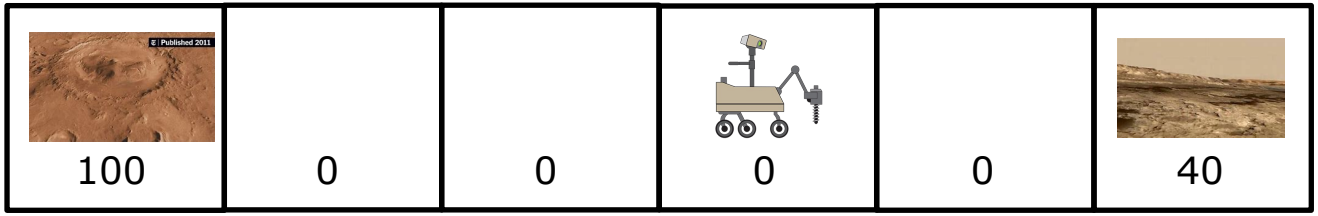


Reinforcement Learning formalism

The Return in reinforcement learning

Return



Return $= 0 + (0.9)0 + (0.9)^2 0 + (0.9)^3 100 = 0.729 \times 100 = 72.9$

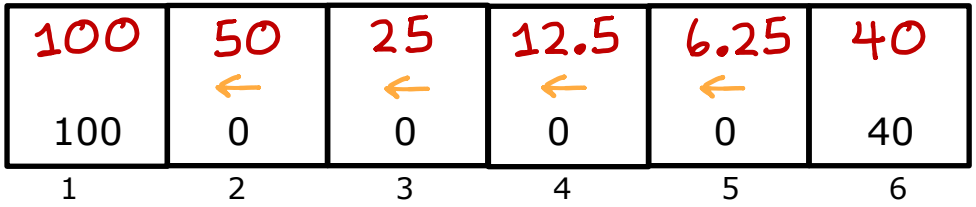
Return $= R_1 + \gamma R_2 + \gamma^2 R_3 + \cdots$ (until terminal state)

Discount Factor $\gamma = 0.9 \quad 0.99 \quad 0.999$

$\gamma = 0.5$

Return $= 0 + (0.5)0 + (0.5)^2 0 + (0.5)^3 100 = 12.5$

Example of Return

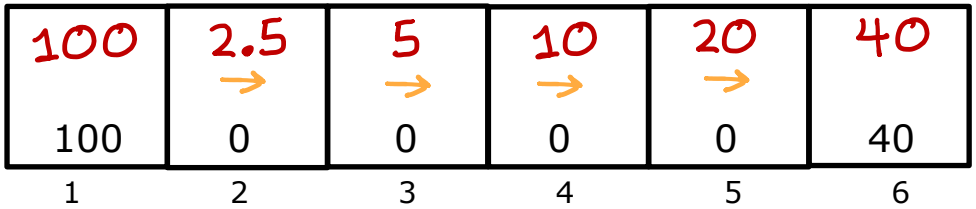


← return

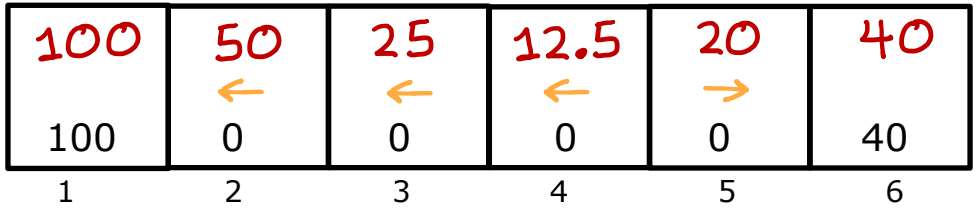
← reward

$\gamma = 0.5$

The return depends on the actions you take.



$0 + (0.5)0 + (0.5)^2 40 = 10$



$0 + (0.5)40 = 20$

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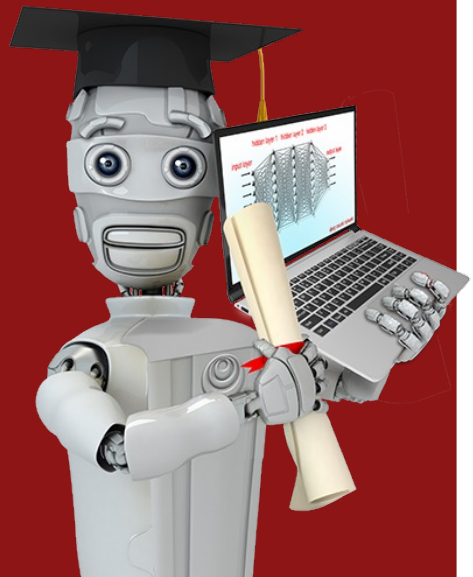
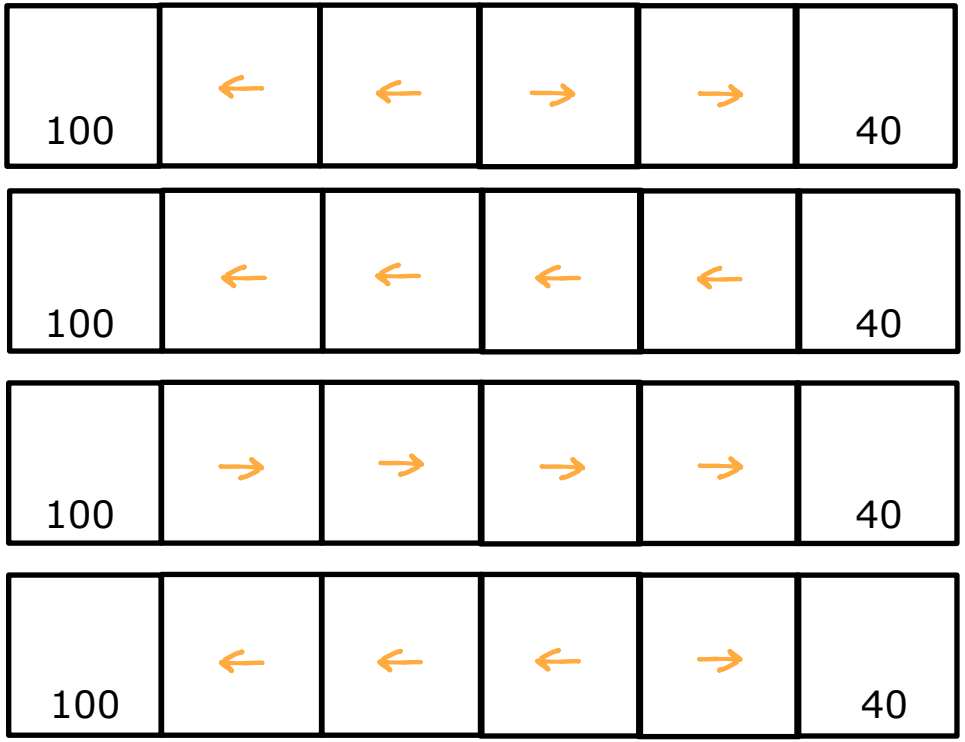


Reinforcement Learning formalism

Making decisions: Policies in reinforcement learning

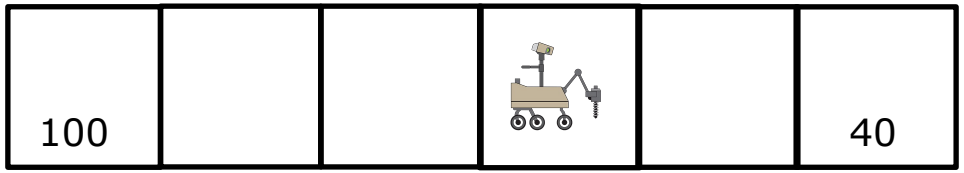
Policy



$\pi(5) = \rightarrow$
$\pi(4) = \leftarrow$
$\pi(3) = \leftarrow$
$\pi(2) = \leftarrow$
$\pi(s) = a$

A policy is a function $\pi(s) = a$ mapping from states to actions, that tells you what action *a* to take in a given state *s*.

The goal of reinforcement learning



Find a policy π that tells you what action (a = $\pi(s)$) to take in every state (s) so as to maximize the return.

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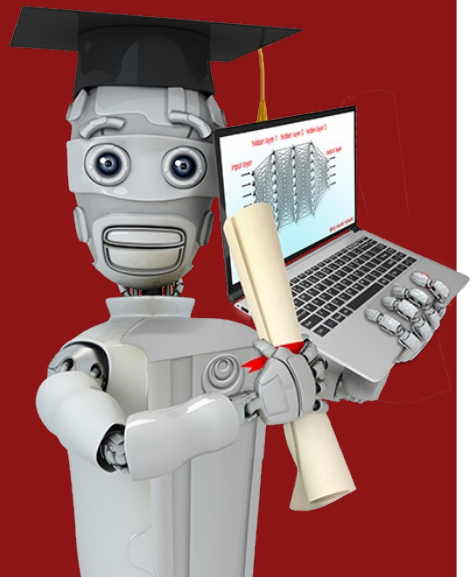
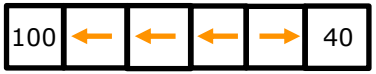
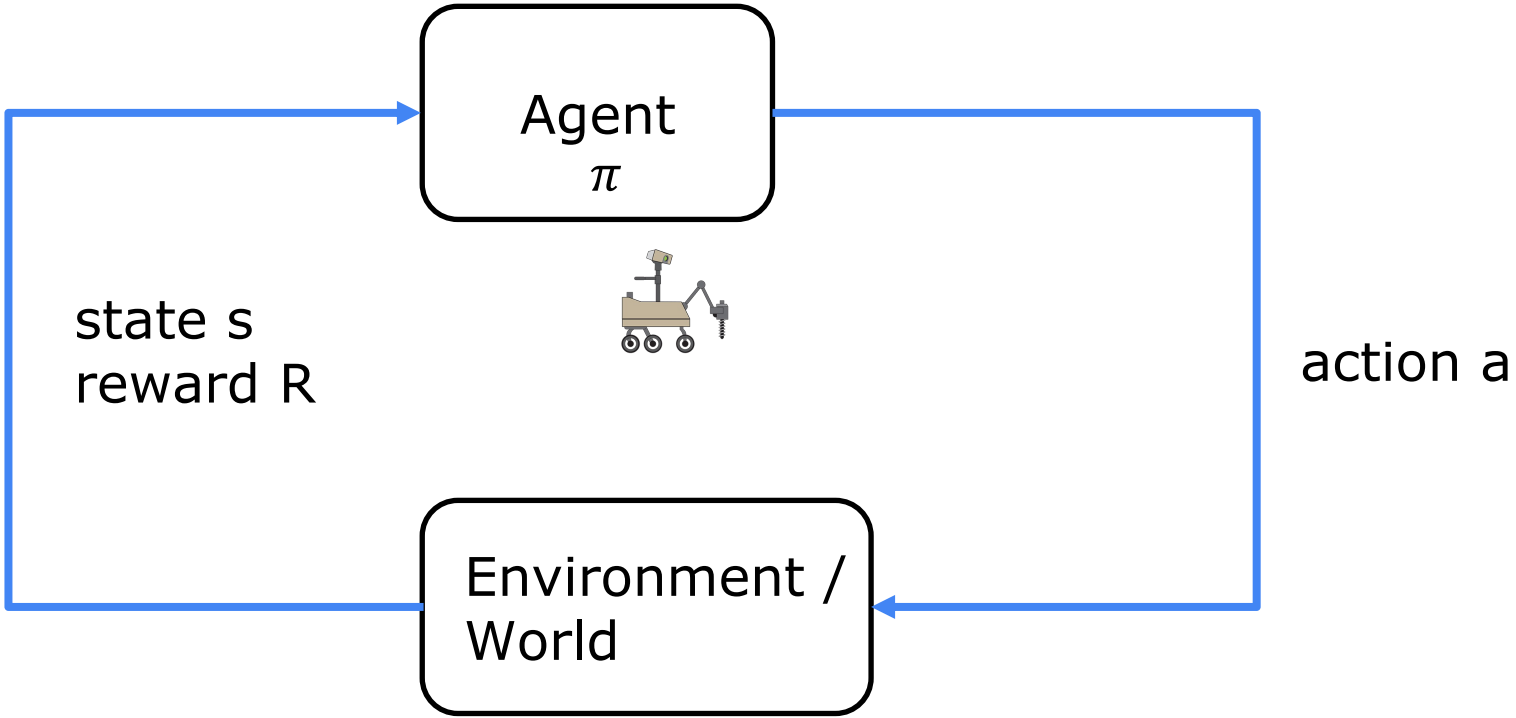


Reinforcement Learning formalism

Review of key concepts

	Mars rover	Helicopter	Chess
states	6 states	position of helicopter	pieces on board
actions	← →	how to move control stick	possible move
rewards	100, 0, 40	+1, −1000	+1, 0, −1
discount factor γ	0.5	0.99	0.995
return	$R_1 + \gamma R_2 + \gamma^2 R_3 + \cdots$	$R_1 + \gamma R_2 + \gamma^2 R_3 + \cdots$	$R_1 + \gamma R_2 + \gamma^2 R_3 + \cdots$
policy π		Find $\pi(s) = a$	Find $\pi(s) = a$

Markov Decision Process (MDP)



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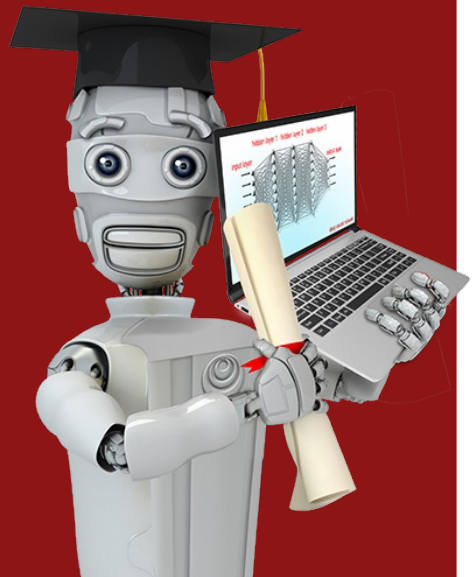


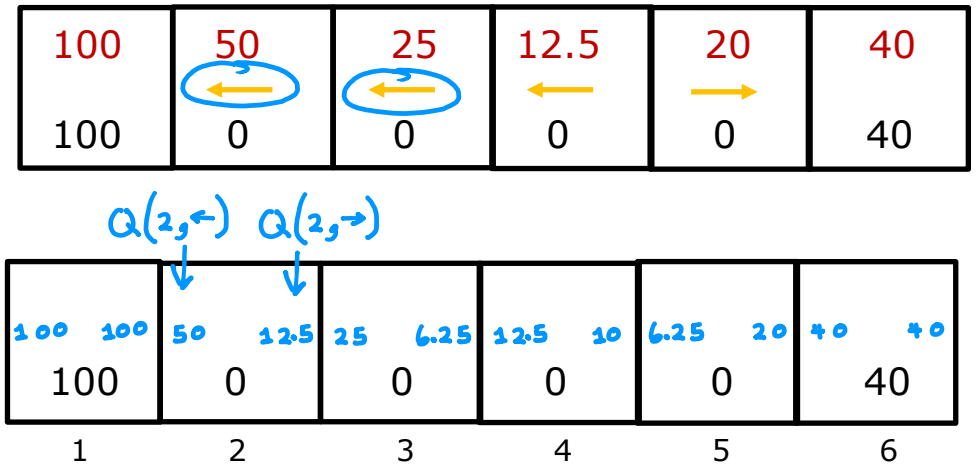
State-action value function

State-action value function definition

State action value function (Q-function)

$Q(s,a)$ = Return if you
- start in state s.
- take action a (once).
- then behave optimally after that.

$Q(s,a)$



← return
← action
← reward

$Q(2,\rightarrow) = 12.5$

$0+(0.5)0+(0.5)^2 0+(0.5)^3 100$

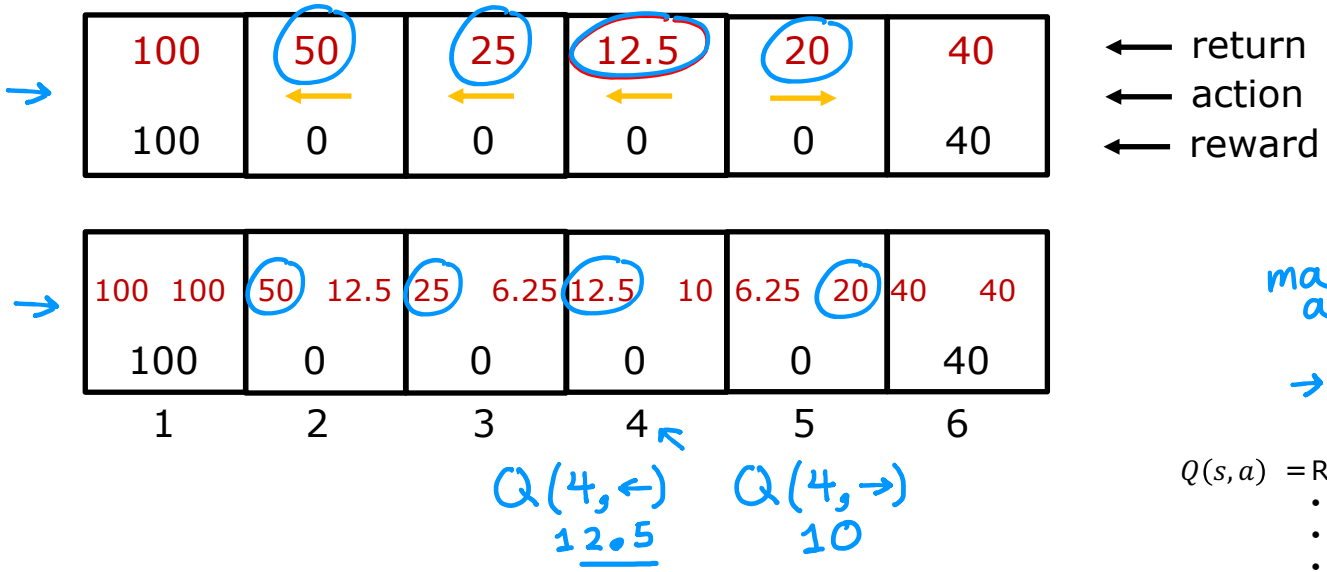
$Q(2,\leftarrow) = 50$

$0+(0.5)100$

$Q(4,\leftarrow) = 12.5$

$0+(0.5)0+(0.5)^2 0+(0.5)^3 100$

Picking actions



← return
← action
← reward

$\max_a Q(s, a)$

$\rightarrow \pi(s) = a$

$Q(s, a)$ = Return if you
- start in state s.
- take action a (once).
- then behave optimally after that.

The best possible return from state s is $\max_a Q(s, a)$.

The best possible action in state s is the action a that gives $\max_a Q(s, a)$.

Q^*

Optimal Q function

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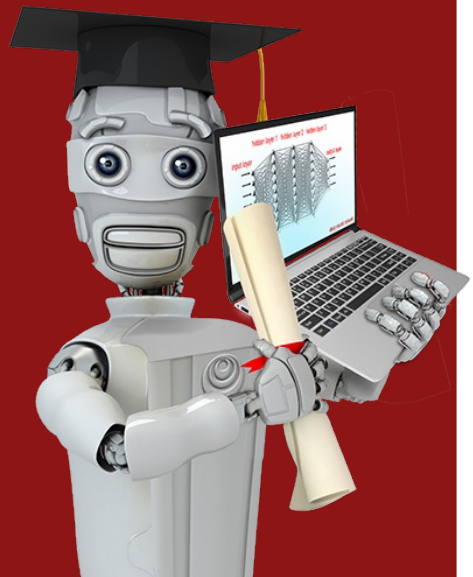

State-action value function

State-action value function example

Jupyter Notebook

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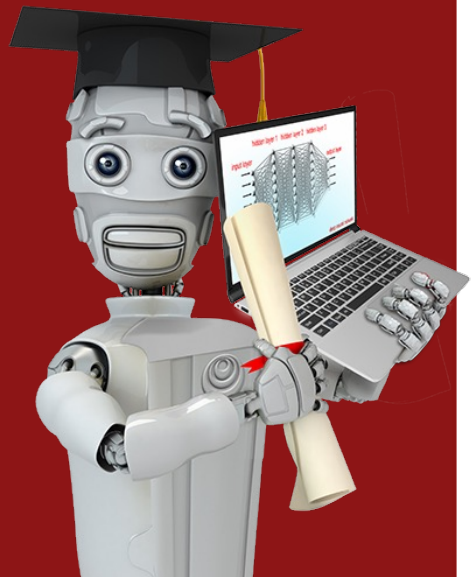
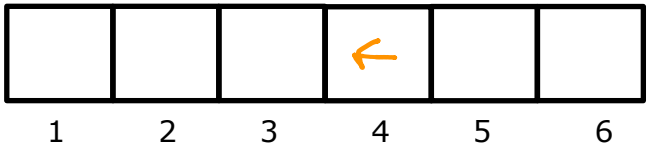


State-action value function

Bellman Equation

Bellman Equation

$Q(s, a) =$ Return if you
- start in state s.
- take action a (once).
- then behave optimally after that.



s : current state
a : current action

$R(s)$ = reward of current state

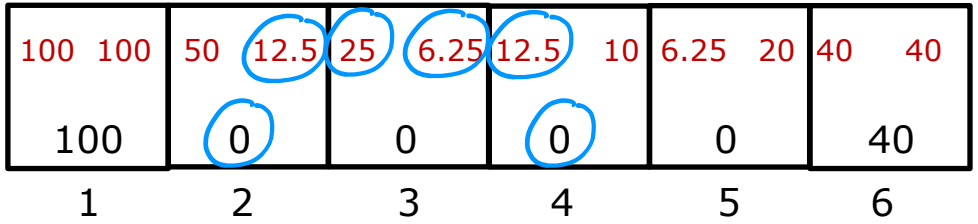
s' : state you get to after taking action a
a' : action that you take in state s'

$$Q(s,a) = R(s) + \gamma \max_{a'} Q(s',a')$$

Bellman Equation

$$Q(s,a) = R(s) + \gamma \max_{a'} Q(s',a')$$

$Q(s,a) = R(s)$



$s = 2$
$a = \rightarrow$
$s' = 3$

$$Q(2,\rightarrow) = R(2) + 0.5 \max_{a'} Q(3,a')$$
$$= 0 + (0.5)25 = 12.5$$

$s = 4$
$a = \leftarrow$
$s' = 3$

$$Q(4,\leftarrow) = R(4) + 0.5 \max_{a'} Q(3,a')$$
$$= 0 + (0.5)25 = 12.5$$

Explanation of Bellman Equation

$Q(s, a)$ = Return if you
- start in state s.
- take action a (once).
- then behave optimally after that.

$s \rightarrow s'$

The best possible return from state s' is $\max_{a'} Q(s', a')$

$$Q(s,a) = R(s) + \gamma \max_{a'} Q(s', a')$$

Reward you get right away

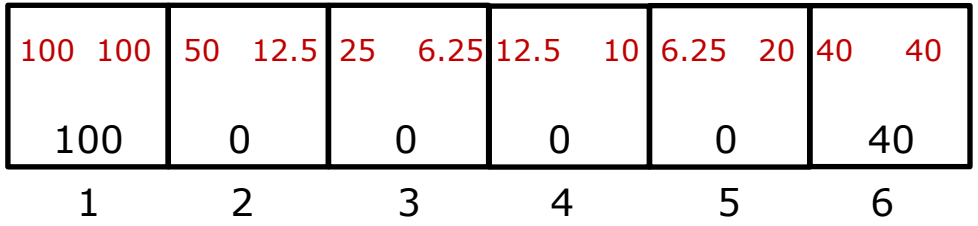
Return from behaving optimally starting from state s'.

$$R_1 + \gamma R_2 + \gamma^2 R_3 + \gamma^3 R_4 + \cdots$$

$$Q(s,a) = R_1 + \gamma [R_2 + \gamma R_3 + \gamma^2 R_4 + \cdots]$$

Explanation of Bellman Equation

$$Q(s,a) = R(s) + \gamma \max_{a'} Q(s',a')$$



$Q(4,\leftarrow)$

$= 0 + (0.5)0 + (0.5)^2 0 + (0.5)^3 100$

$= R(4) + (0.5)[0 + (0.5)0 + (0.5)^2 100]$

$= R(4) + (0.5)\max_{a'} Q(3,a')$

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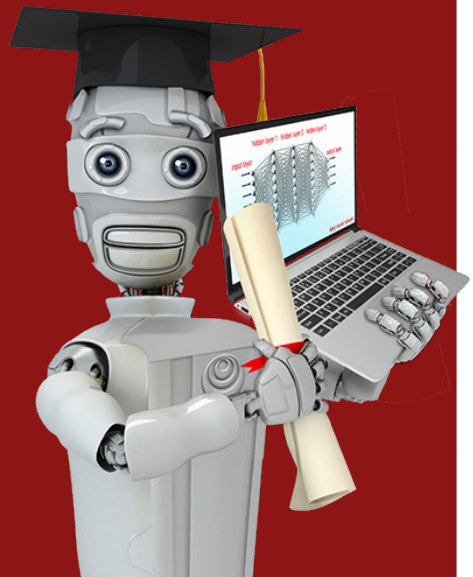
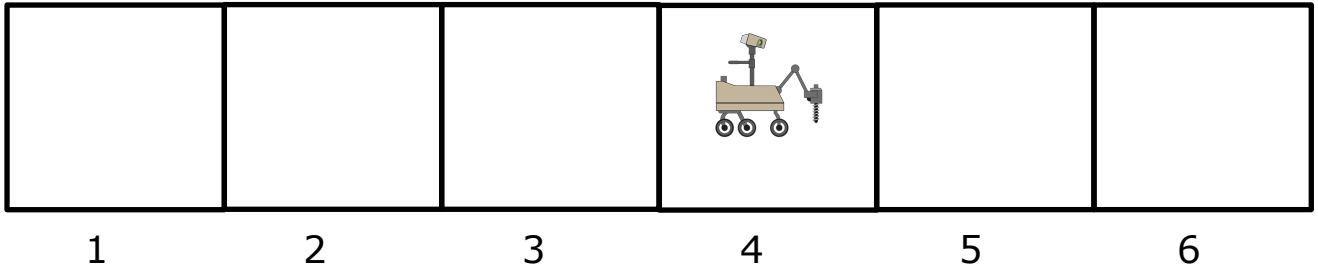


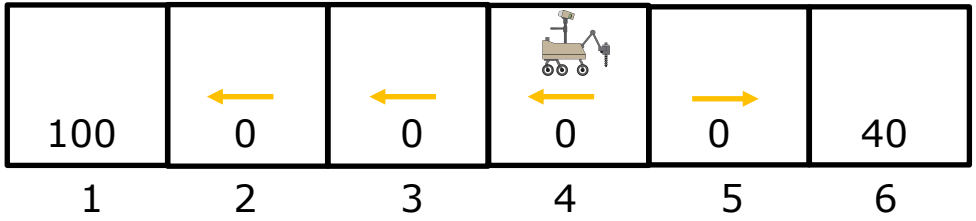
State-action value function

Random (stochastic) environment (Optional)

Stochastic Environment



Expected Return



0 0 0 100 0 100
0 0 0 0
0 0 40

Expected Return = Average($R_1 + \gamma R_2 + \gamma^2 R_3 + \gamma^3 R_4 + \cdots$)

$= \mathrm{E}[R_1 + \gamma R_2 + \gamma^2 R_3 + \gamma^3 R_4 + \cdots]$

Expected Return

Goal of Reinforcement Learning:

Choose a policy $\pi(s) = a$ that will tell us what action a to take in state s so as to maximize the expected return.

Bellman Equation:

$$Q(s,a) = R(s) + \gamma E[\max_{a'} Q(s',a')]$$

3 ←

2 or 4

Jupyter Notebook

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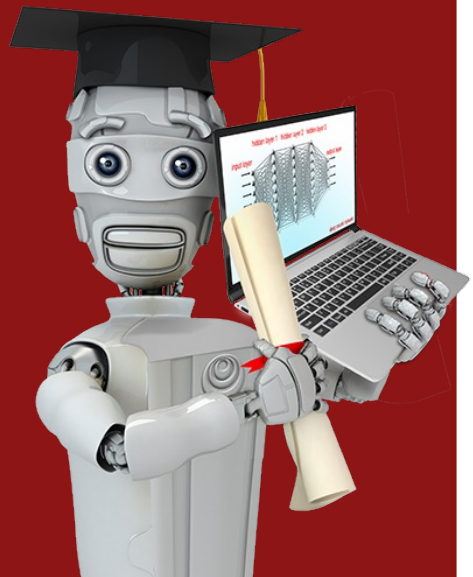
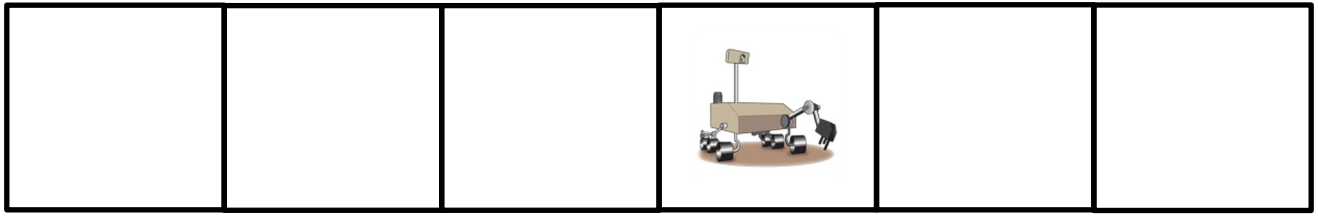


Continuous State Spaces

Example of continuous state applications

Discrete vs Continuous State

Discrete State: $S = 1, 2, 3, 4, 5, 6$



Continuous State:

$$S = \begin{bmatrix} x \\ y \\ \theta \\ \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} \leftarrow 0\text{-}360^\circ$$

Autonomous Helicopter

$$s = \begin{bmatrix} x \\ y \\ z \\ \phi \\ \theta \\ \omega \\ \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\phi} \\ \dot{\theta} \\ \dot{\omega} \end{bmatrix}$$

roll
pitch
yaw

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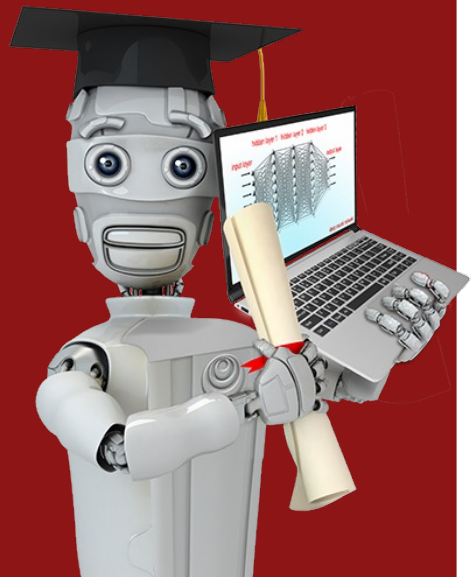
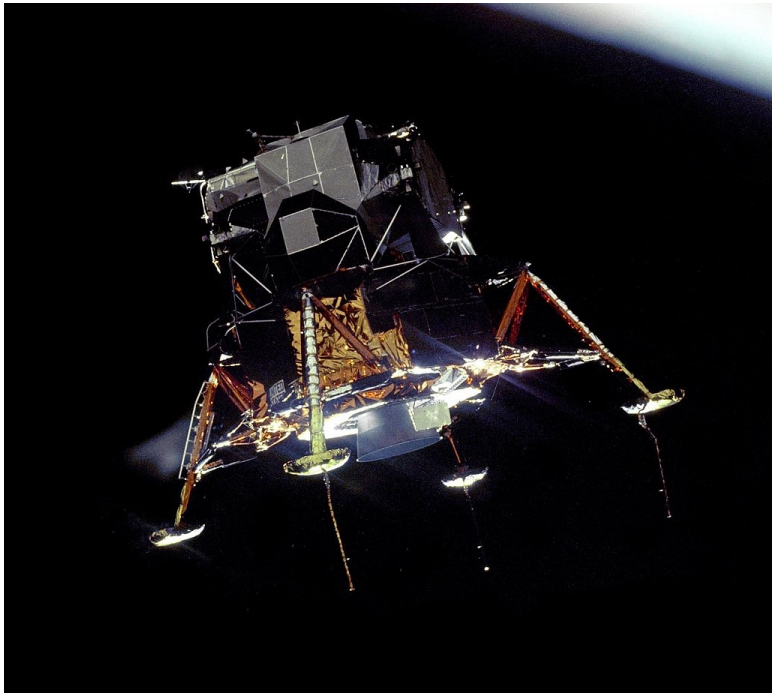
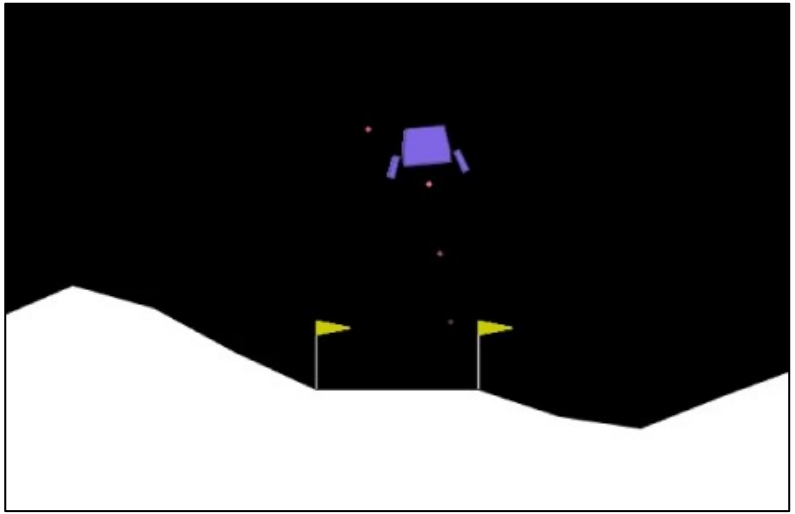


Continuous State Spaces

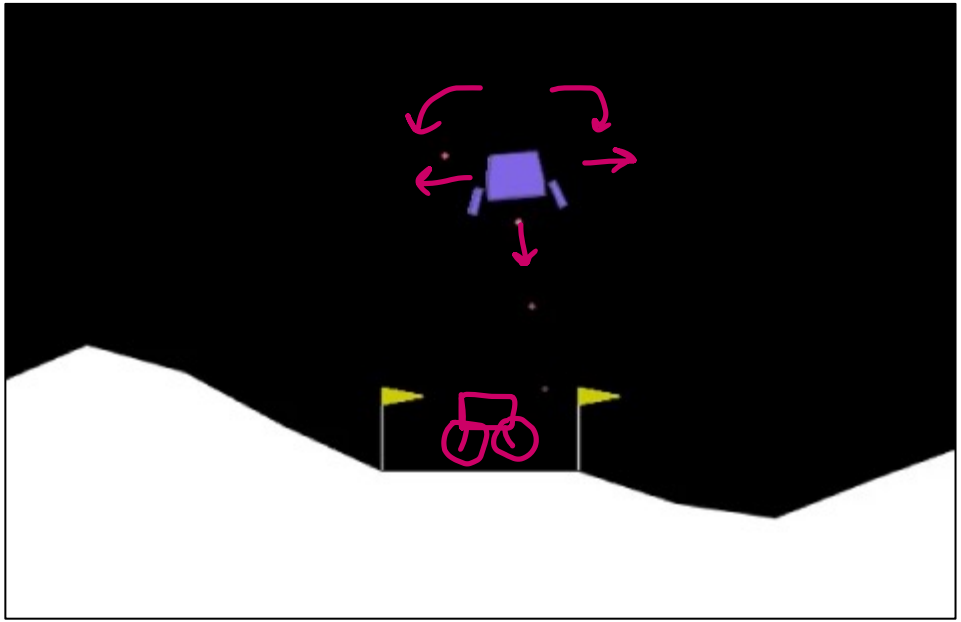
Lunar Lander

Lunar Lander





Lunar Lander



actions:

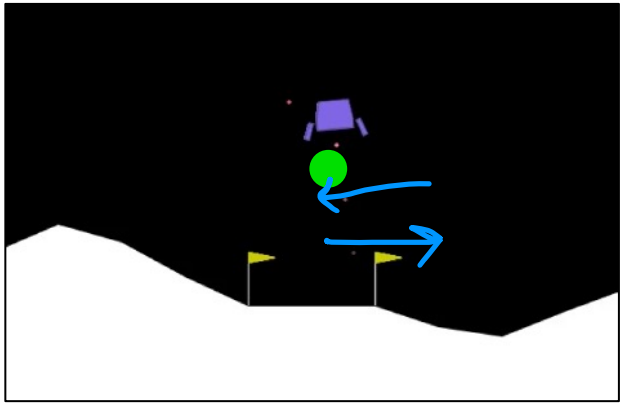
- do nothing
- left thruster
- main thruster
- right thruster

$$s = \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \\ \theta \\ \dot{\theta} \\ l \\ r \end{bmatrix}$$

0 or 1

Reward Function

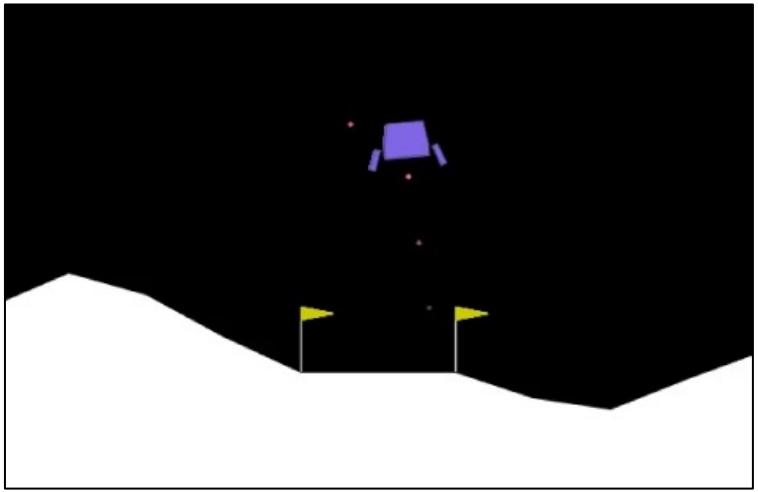
- Getting to landing pad: 100 – 140
- Additional reward for moving toward/away from pad.
- Crash: -100
- Soft landing: +100
- Leg grounded: +10
- Fire main engine: -0.3
- Fire side thruster: -0.03



Lunar Lander Problem

Learn a policy π that, given

$$s = \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \\ \theta \\ \dot{\theta} \\ l \\ r \end{bmatrix}$$



picks action $a = \pi(s)$ so as to maximize the return.

$$\gamma = 0.985$$

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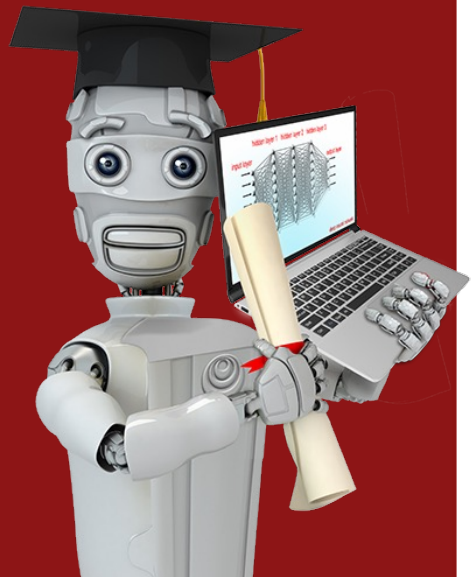
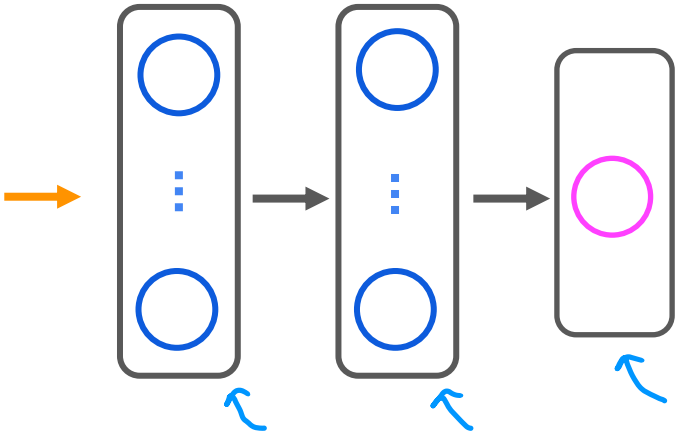


Continuous State Spaces

Learning the state-value function

Deep Reinforcement Learning

$$\vec{x} = \begin{bmatrix} s \\ a \end{bmatrix} \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \\ \theta \\ \dot{\theta} \\ l \\ r \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$



$Q(s,a)$

y

$x \rightarrow y$

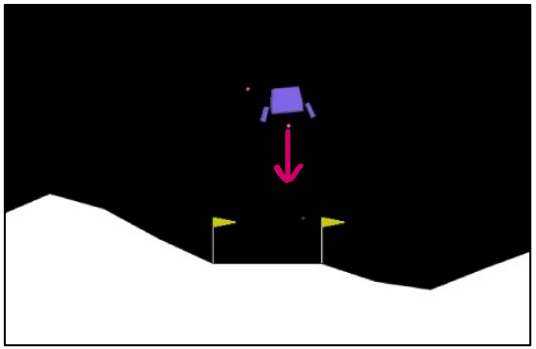
12 inputs

64 units

64 units

1 unit

- In a state s, use neural network to compute
  $Q(s, \text{nothing})$, $Q(s, \text{left})$, $Q(s, \text{main})$, $Q(s, \text{right})$
- Pick the action a that maximizes $Q(s,a)$



Bellman Equation

$$Q(s,a) = R(s) + \gamma \max_{a'} Q(s',a')$$

x $\quad$ y

$$f_{w,b}(x) \approx y$$

$$y^{(1)} = R(s^{(1)}) + \gamma \max_{a'} Q(s'^{(1)}, a')$$

$$y^{(2)} = R(s^{(2)}) + \gamma \max_{a'} Q(s'^{(2)}, a')$$

$(s, a, R(s), s')$

$(s^{(1)}, a^{(1)}, R(s^{(1)}), s'^{(1)})$ ←

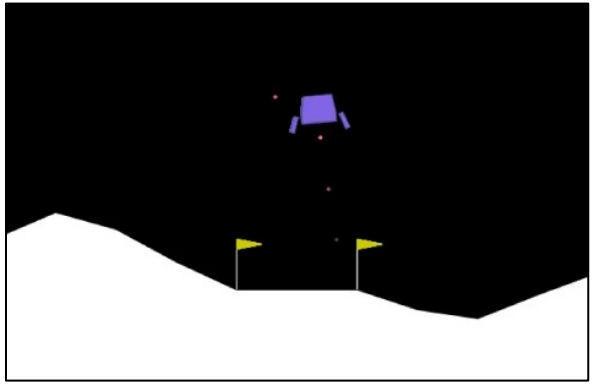
$(s^{(2)}, a^{(2)}, R(s^{(2)}), s'^{(2)})$ ←

$(s^{(3)}, a^{(3)}, R(s^{(3)}), s'^{(3)})$ ←

x	y
$x^{(1)} = (s^{(1)}, a^{(1)})$	$y^{(1)}$
$x^{(2)} = (s^{(2)}, a^{(2)})$	$y^{(2)}$
$x^{(10,000)}$	$y^{(10,000)}$

8 $\quad$ 4

Learning Algorithm



Initialize neural network randomly as guess of $Q(s,a)$.

Repeat {

Take actions in the lunar lander. Get $(s,a,R(s),s')$.

Store 10,000 most recent $(s,a,R(s),s')$ tuples.

replay buffer

Train neural network:

Create training set of 10,000 examples using

$x = (s,a)$ and $y = R(s) + \gamma \max_{a'} Q(s',a')$

Train Q_{new} such that $Q_{new}(s,a) \approx y$.

$f_{w,b}(x) \approx y$

Set $Q = Q_{new}$.

$x^{(1)}, y^{(1)}$

x, y

$x^{(100000)}, y^{(100000)}$

[Mnih et al., 2015, Human-level control through deep reinforcement learning]

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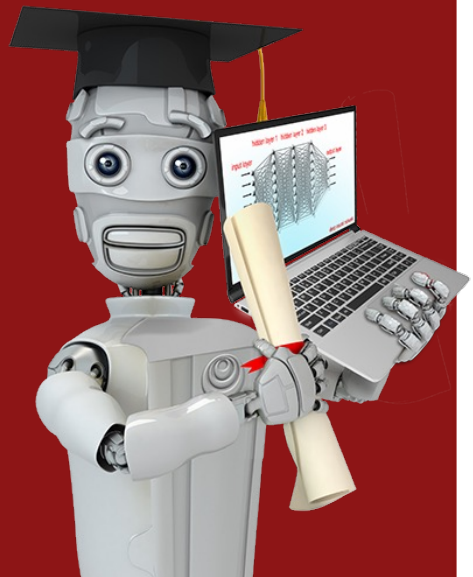
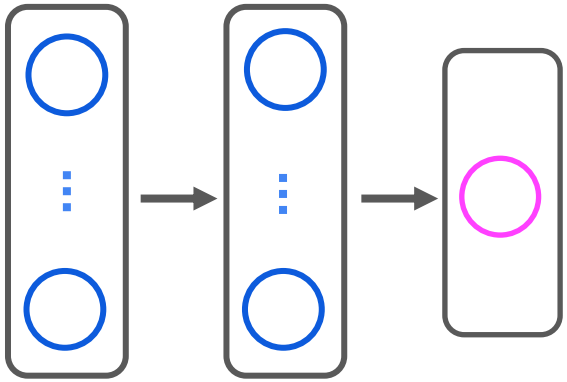


Continuous State Spaces

Algorithm refinement: Improved neural network architecture

Deep Reinforcement Learning

$$\vec{x} = \begin{bmatrix} s \\ a \end{bmatrix} \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \\ \theta \\ \dot{\theta} \\ l \\ r \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$



$Q(s, a)$

12 inputs

64 units

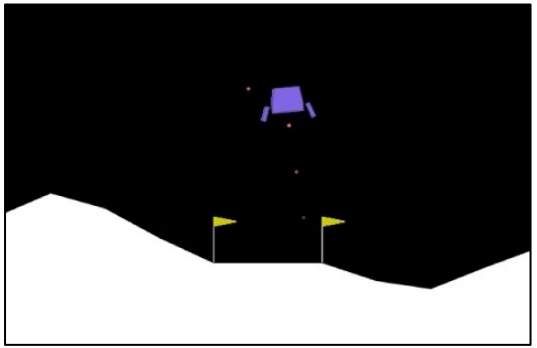
64 units

1 unit

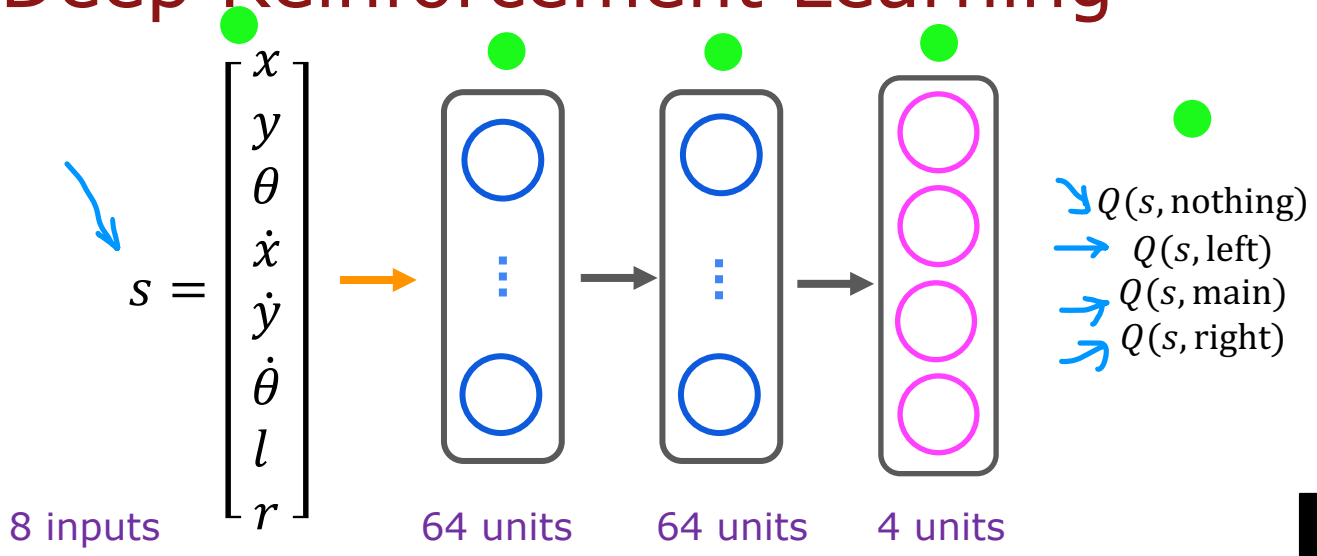
In a state s, use neural network to compute

$Q(s, \text{nothing})$, $Q(s, \text{left})$, $Q(s, \text{main})$, $Q(s, \text{right})$

Pick the action a that maximizes $Q(s, a)$

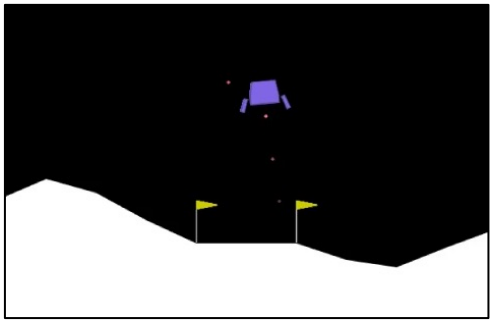


Deep Reinforcement Learning



In a state s, input s to neural network.

Pick the action a that maximizes $Q(s,a)$. $R(s) + \gamma \max_{a'} Q(s', a')$



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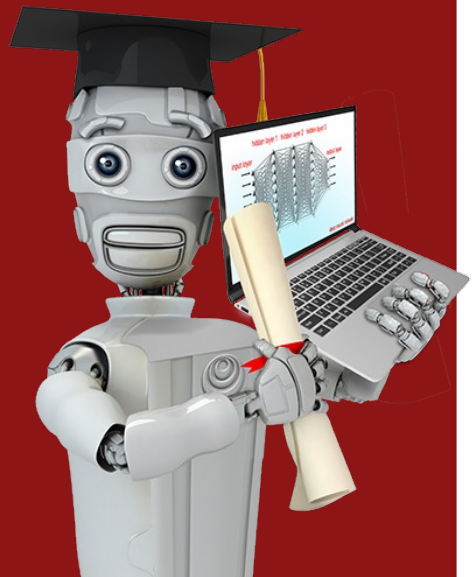
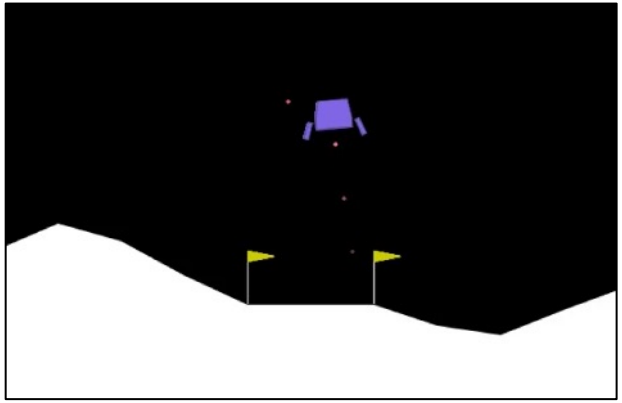


Continuous State Spaces

Algorithm refinement: ε-greedy policy

Learning Algorithm



Initialize neural network randomly as guess of $Q(s,a)$.

Repeat {

Take actions in the lunar lander. Get $(s,a,R(s),s')$.

Store 10,000 most recent $(s,a,R(s),s')$ tuples.

Train model:

Create training set of 10,000 examples using

$x = (s,a)$ and $y = R(s) + \gamma \max_{a'} Q(s',a')$.

Train Q_{new} such that $Q_{new}(s,a) \approx y$. $f_{w,b}(x) \approx y$

Set $Q = Q_{new}$.

How to choose actions while still learning?

In some state s

what if init $Q(s, \text{main})$ to be low?

Option 1: → will never try other actions ↳ will never fire main thrusters!

Pick the action a that maximizes $Q(s,a)$.

Option 2:

With probability 0.95, pick the action a that maximizes $Q(s,a)$. greedy exploitation

With probability 0.05, pick an action a randomly. exploration

ε-greedy policy ($\varepsilon = 0.05$)

greedy 95% of the time

exploring 5% of the time

start ε high

gradually decrease

1.0 → 0.01

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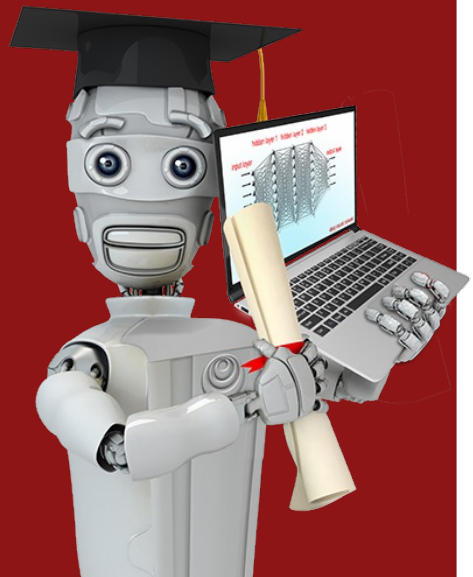


Continuous State Spaces

Algorithm refinement: Mini-batch and soft update (optional)

How to choose actions while still learning?

x	y
2104	400
1416	232
1534	315
852	178
...	...
3210	870

100,000,000

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{w,b}\left(x^{(i)}\right) - y^{(i)}\right)^2$$

$m = 100,000,000$

$m' = 1,000$

repeat {

$$w = w - \alpha \frac{\partial}{\partial w} \frac{1}{2m'}\sum_{i=1}^{m'}\left(f_{w,b}\left(x^{(i)}\right) - y^{(i)}\right)^2$$

$$b = b - \alpha \frac{\partial}{\partial b} \frac{1}{2m'}\sum_{i=1}^{m'}\left(f_{w,b}\left(x^{(i)}\right) - y^{(i)}\right)^2$$

}

Mini-batch

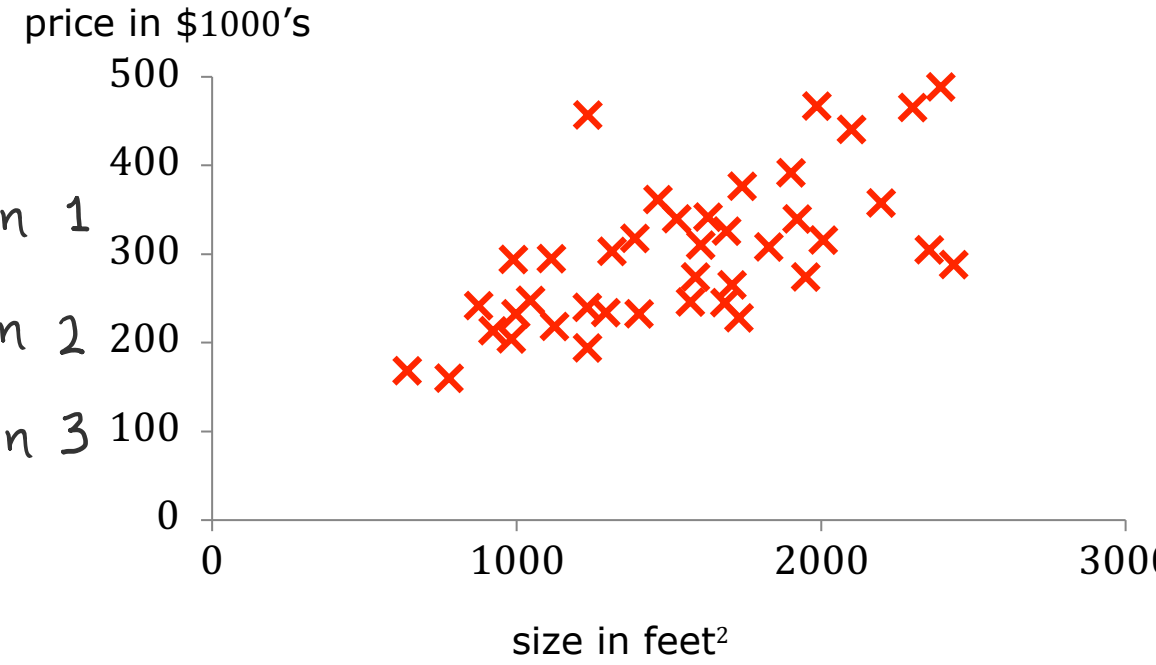
x	y
2104	400
1416	232
1534	315
852	178
...	...
3210	870

batch

iteration 1

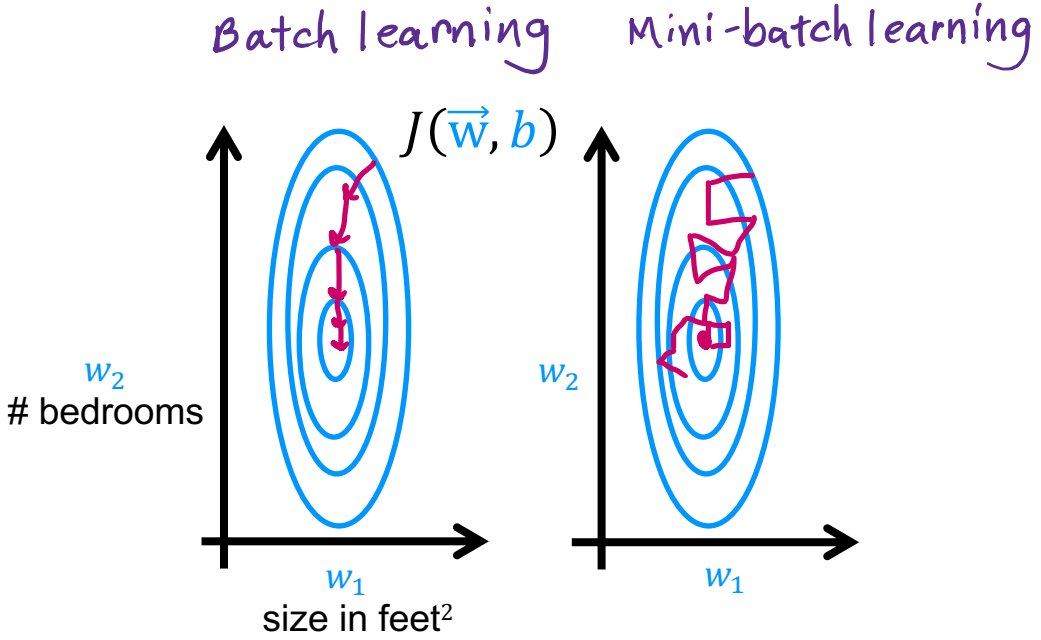
iteration 2

iteration 3

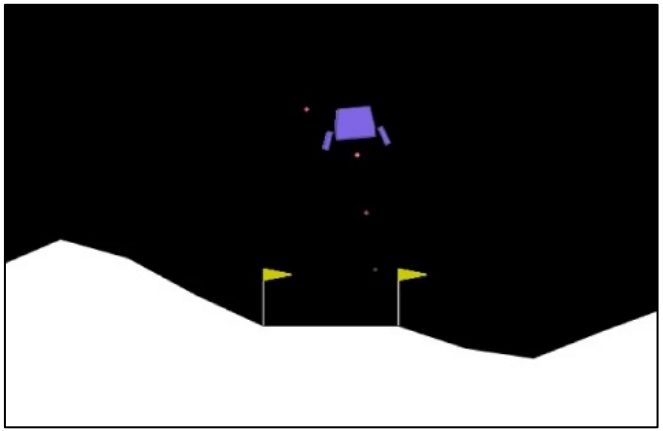


Mini-batch

x	y
2104	400
1416	232
1534	315
852	178
...	...
3210	870



Learning Algorithm



Initialize neural network randomly as guess of $Q(s,a)$

Repeat {

Take actions in the lunar lander. Get $(s, a, R(s), s')$.

Store 10,000 most recent $(s, a, R(s), s')$ tuples.

Replay Buffer

Train model:

Create training set of ~~10,000~~ 1,000 examples using

$x = (s, a)$ and $y = R(s) + \gamma \max_{a'} Q(s', a')$

Train Q_{new} such that $Q_{new}(s, a) \approx y$.

Set $Q = Q_{new}$.

$x^{(1)}, y^{(1)}$

$x^{(1000)}, y^{(1000)}$

Soft Update

Set $Q = Q_{new}$.

w, B $\quad$ w_{new}, B_{new}

$Q(s,a)$

$W = 0.01 W_{new} + 0.99 W$

$B = 0.01 B_{new} + 0.99 B$

$w = 1 w_{new} + 0 w$

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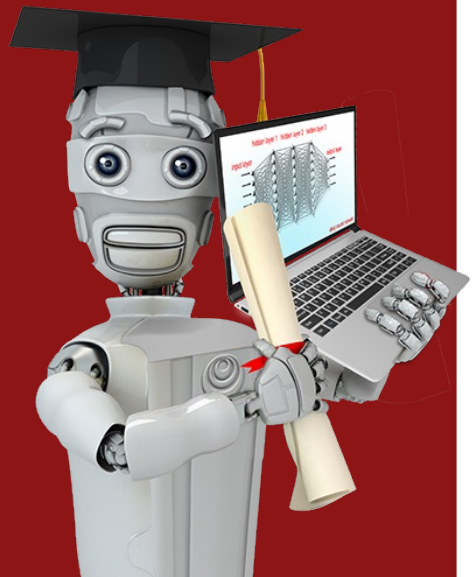

Continuous State Spaces

The state of reinforcement learning

Limitations of Reinforcement Learning

- Much easier to get to work in a simulation than a real robot!
- Far fewer applications than supervised and unsupervised learning.
- But … exciting research direction with potential for future applications.

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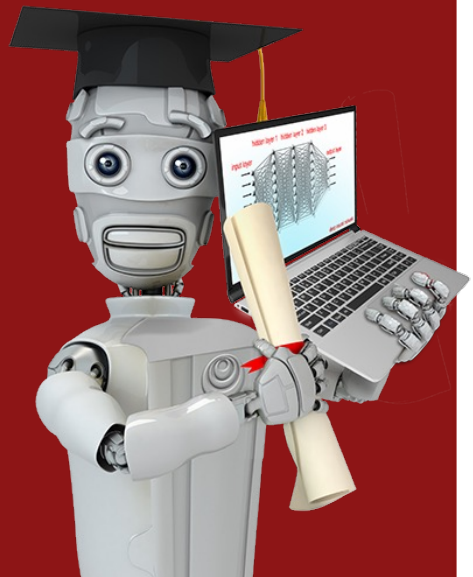


Conclusion

Summary and
Thank you

Courses

- Supervised Machine Learning: Regression and Classification

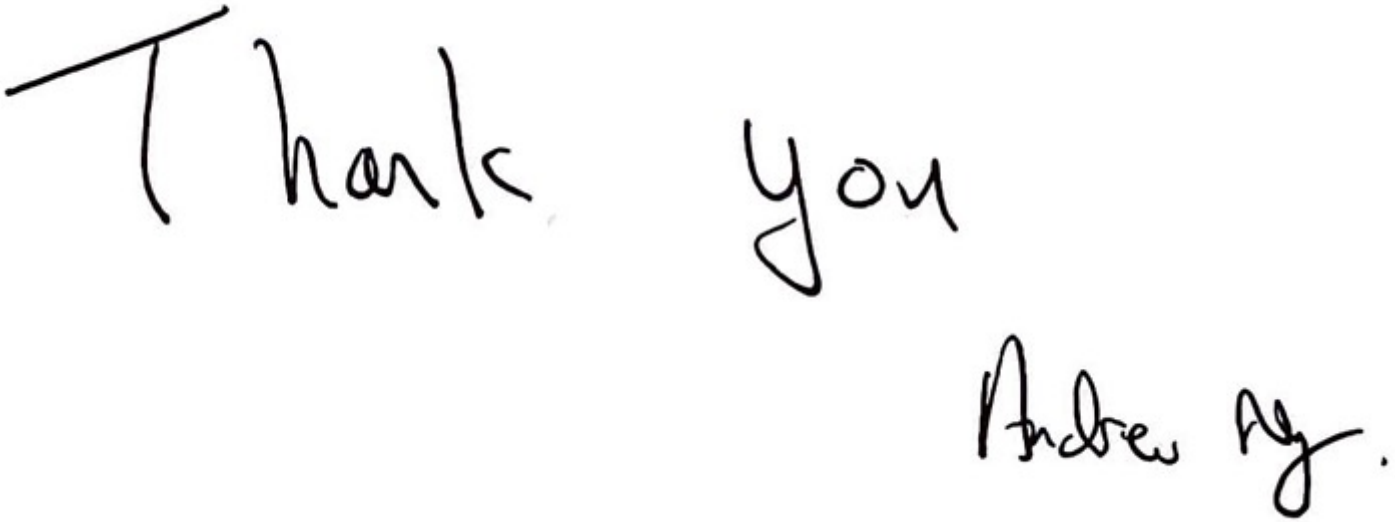
  Linear regression, logistic regression, gradient descent

- Advanced Learning Algorithms

  Neural networks, decision trees, advice for ML

- Unsupervised Learning, Recommenders, Reinforcement Learning

  Clustering, anomaly detection, collaborative filtering, content-based filtering, reinforcement learning


Thank you
Andrew Ng.